Studies in Computational Intelligence 235

Reiner Onken
Axel Schulte

System-Ergonomic Design of Cognitive Automation

Dual-Mode Cognitive Design of Vehicle Guidance and Control Work Systems



Springer

System-Ergonomic Design of Cognitive Automation

Studies in Computational Intelligence, Volume 235

Editor-in-Chief
Prof. Janusz Kacprzyk
Systems Research Institute
Polish Academy of Sciences
ul. Newelska 6
01-447 Warsaw
Poland
E-mail: kacprzyk@ibspan.waw.pl

Further volumes of this series can be found on our homepage: springer.com

Vol. 213. Stefano Cagnoni (Ed.)
Evolutionary Image Analysis and Signal Processing, 2009
ISBN 978-3-642-01635-6

Vol. 214. Been-Chian Chien and Tzung-Pei Hong (Eds.)
Opportunities and Challenges for Next-Generation Applied Intelligence, 2009
ISBN 978-3-540-92813-3

Vol. 215. Habib M. Ammari
Opportunities and Challenges of Connected k-Covered Wireless Sensor Networks, 2009
ISBN 978-3-642-01876-3

Vol. 216. Matthew Taylor
Transfer in Reinforcement Learning Domains, 2009
ISBN 978-3-642-01881-7

Vol. 217. Horia-Nicolai Teodorescu, Junzo Watada, and Lakhmi C. Jain (Eds.)
Intelligent Systems and Technologies, 2009
ISBN 978-3-642-01884-8

Vol. 218. Maria do Carmo Nicoletti and Lakhmi C. Jain (Eds.)
Computational Intelligence Techniques for Bioprocess Modelling, Supervision and Control, 2009
ISBN 978-3-642-01887-9

Vol. 219. Maja Hadzic, Elizabeth Chang, Pornpit Wongthongtham, and Tharam Dillon
Ontology-Based Multi-Agent Systems, 2009
ISBN 978-3-642-01903-6

Vol. 220. Bettina Berendt, Dunja Mladenic, Marco de de Gemmis, Giovanni Semeraro, Myra Spiliopoulou, Gerd Stumme, Vojtech Svatek, and Filip Zelezny (Eds.)
Knowledge Discovery Enhanced with Semantic and Social Information, 2009
ISBN 978-3-642-01890-9

Vol. 221. Tassilo Pellegrini, Sören Auer, Klaus Tochtermann, and Sebastian Schaffert (Eds.)
Networked Knowledge - Networked Media, 2009
ISBN 978-3-642-02183-1

Vol. 222. Elisabeth Rakus-Andersson, Ronald R. Yager, Nikhil Ichalkaranje, and Lakhmi C. Jain (Eds.)
Recent Advances in Decision Making, 2009
ISBN 978-3-642-02186-2

Vol. 223. Zbigniew W. Ras and Agnieszka Dardzinska (Eds.)
Advances in Data Management, 2009
ISBN 978-3-642-02189-3

Vol. 224. Amandeep S. Sidhu and Tharam S. Dillon (Eds.)
Biomedical Data and Applications, 2009
ISBN 978-3-642-02192-3

Vol. 225. Danuta Zakrzewska, Ernestina Menasalvas, and Liliana Byczkowska-Lipinska (Eds.)
Methods and Supporting Technologies for Data Analysis, 2009
ISBN 978-3-642-02195-4

Vol. 226. Ernesto Damiani, Jechang Jeong, Robert J. Howlett, and Lakhmi C. Jain (Eds.)
New Directions in Intelligent Interactive Multimedia Systems and Services - 2, 2009
ISBN 978-3-642-02936-3

Vol. 227. Jeng-Shyang Pan, Hsiang-Cheh Huang, and Lakhmi C. Jain (Eds.)
Information Hiding and Applications, 2009
ISBN 978-3-642-02334-7

Vol. 228. Lidia Ogiela and Marek R. Ogiela
Cognitive Techniques in Visual Data Interpretation, 2009
ISBN 978-3-642-02692-8

Vol. 229. Giovanna Castellano, Lakhmi C. Jain, and Anna Maria Fanelli (Eds.)
Web Personalization in Intelligent Environments, 2009
ISBN 978-3-642-02793-2

Vol. 230. Uday K. Chakraborty (Ed.)
Computational Intelligence in Flow Shop and Job Shop Scheduling, 2009
ISBN 978-3-642-02835-9

Vol. 231. Mislav Grgic, Kresimir Delac, and Mohammed Ghanbari (Eds.)
Recent Advances in Multimedia Signal Processing and Communications, 2009
ISBN 978-3-642-02899-1

Vol. 232. Feng-Hsing Wang, Jeng-Shyang Pan, and Lakhmi C. Jain
Innovations in Digital Watermarking Techniques, 2009
ISBN 978-3-642-03186-1

Vol. 233. Takayuki Ito, Minjie Zhang, Valentin Robu, and Shaheen Fatima, Tokuro Matsuo (Eds.)
Advances in Agent-Based Complex Automated Negotiations, 2009
ISBN 978-3-642-03189-2

Vol. 234. Aruna Chakraborty and Amit Konar
Emotional Intelligence, 2009
ISBN 978-3-540-68606-4

Vol. 235. Reiner Onken and Axel Schulte
System-Ergonomic Design of Cognitive Automation, 2010
ISBN 978-3-642-03134-2

Reiner Onken and Axel Schulte

System-Ergonomic Design of Cognitive Automation

Dual-Mode Cognitive Design of Vehicle Guidance and Control Work Systems

Springer

Prof. Dr. Ing. Reiner Onken
Universität der Bundeswehr München
Institut für Flugsysteme
Werner-Heisenberg-Weg 39
85577 Neubiberg
Germany
E-mail: reiner.onken@unibw.de

Prof. Dr. Ing. Axel Schulte
Universität der Bundeswehr München
Institut für Flugsysteme
Werner-Heisenberg-Weg 39
85577 Neubiberg
Germany
E-mail: axel schulte@unibw.de

ISBN 978-3-642-03134-2 e-ISBN 978-3-642-03135-9

DOI 10.1007/978-3-642-03135-9

Studies in Computational Intelligence ISSN 1860-949X

Library of Congress Control Number: Applied for

© 2010 Springer-Verlag Berlin Heidelberg

This work is subject to copyright. All rights are reserved, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilm or in any other way, and storage in data banks. Duplication of this publication or parts thereof is permitted only under the provisions of the German Copyright Law of September 9, 1965, in its current version, and permission for use must always be obtained from Springer. Violations are liable to prosecution under the German Copyright Law.

The use of general descriptive names, registered names, trademarks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.

Typeset & *Cover Design:* Scientific Publishing Services Pvt. Ltd., Chennai, India.

Printed in acid-free paper

9 8 7 6 5 4 3 2 1

springer.com

We can only see a short distance ahead,
but we can see plenty there that needs to be done.
(Alan Turing)

Acknowledgements

This book came about when there was the latest transition in office for the chair of Flight Dynamics and Flight Guidance at our Universität der Bundeswehr München (Munich University of German Armed Forces). The two of us were personally involved in that transition, i.e. one of us came in office and the other retired. It was the younger one who eventually stimulated the project to write this book on a research and development topic by which our institute became known worldwide.

Since about two decades the institute has been putting feet on new grounds regarding the introduction of knowledge-based systems and cognitive automation into the work process of guidance and control of air and road vehicles for the sake of performance enhancement and safety. A great number of publications accumulated and groundbreaking experiments were conducted during that time, including complex simulator and field trials.

There are always good reasons why such success comes about. The one to be mentioned in the first place are the resources of our university, thereby offering best provisions for successful research. The basic financial support as to the laboratory equipment and experimental facilities has been outstanding throughout. Additional funding mainly was granted by the German Departments of Defence (BMVg) and Research and Technology (BMFT), by the European Union, as well as by companies like Dornier Luftfahrt, EADS (former DASA), Daimler-Benz, MAN, and Opel. This made it possible that a rather small group of people could come to some amazing achievements.

In addition, the organisational setup of the faculty lends itself to initiate fruitful co-operations. This cannot be appreciated highly enough. For a long period of time the institute was run together with the colleague Prof. Dr. Ernst-Dieter Dickmanns who was holding the chair of Control Engineering. During that time, benefit was mutually gained from common participations in research projects like for instance the notably funded Prometheus project Pro-Driver and the CAMA (Crew Assistant Military Aircraft) project. Now, as part of a recent reorganisation a new promising formation, the Institute of Flight Systems, is established together with the colleague Prof. Dr. Peter Stütz who now holds the chair of Aeronautical Engineering.

The content of our book is substantially based on the work being outlined in the doctoral dissertations which were completed during the last two decades, resulting from enthusiastic engagement in a fascinating research field. Without them, there would be hardly anything to write on an authentic basis. All started with the dissertation of Heinz-Leo Dudek on the ASPIO cockpit assistant system. This achievement was a kind of foundation stone for the following work. He also was

significantly involved in the CASSY project as actively participating representative of the company Dornier Luftfahrt GmbH. The following dissertations which we like to mention here are those of Matthias Kopf (the first dissertation on the automotive application, resulting in the driver assistant system DAISY), Thomas Wittig, Thomas Prévôt, Axel Schulte, Marc Gerlach, Johann Peter Feraric, Wilhelm Ruckdeschel, Michael Strohal, Frank Schreiner, Stephan Grashey, Peter Stütz, Frank Ole Flemisch, Anton Walsdorf, Udo von Garrel, Henrik Putzer, Andreas Frey, Hans-Jörg Otto, and lately Claudia Meitinger, the first researcher of the new generation after the aforementioned transition. At this point, also the work of André Lenz should be especially appreciated. Diana Donath and Michael Kriegel shall be named as further representatives for the new generation, not enumerating those, at present more than ten members of the research staff, who were only involved for a relatively short period of time or have not yet finished up their research work. Through all the years Dr. Werner Fohrer supported the research team as enthusiastic expert of classical flight mechanics.

Undoubtedly, the experimental work would not have been possible without the technical staff helping to build up and maintaining the laboratory and computing equipment as well as the simulators as larger experimental facilities of the institute. The work of these staff people was always highly appreciated, even if we are not citing them here by name. One other name should be mentioned, though. That is our secretary Madeleine Gabler. She can be considered as the living form of the good spirit of the institute during all the years. She was also involved in the editing work for shaping up the formal layout of our book.

Having continuously been active in scientific discourse with the international research community, we received innumerable valuable suggestions of many colleagues for our work when discussing the results of our research. This was also the case in the context of preparing our book. Representative for all of them, we would like to mention by name the German colleagues Prof. Dr. H. Bubb, Prof. Dr. B. Döring, Prof. Dr. B. Färber, Prof. Dr. K.-P. Gärtner, Prof. Dr. P. Hecker, Prof. K.-F. Kraiss, Prof. Dr. U. Krogmann, Prof. Dr. H.-P. Krüger, Dr. E. Rödig, Prof. Dr. C. Schlick, Prof. Dr. F. Thielecke, Prof. Dr. K.-P. Timpe, Prof. Dr. U. Völckers, and Prof. Dr. H. Winter, and from abroad Prof. R. Amalberti, G. Champigneux, Prof. Dr. M. Cummings, Dr. M. Draper, Prof. Dr. P. Fabiani, Dr. R. Frampton, Prof. K. Funk, Prof. Dr. E. Hollnagel, S. Howitt, C.A. Miller, E. Palmer, Dr. J.T. Platts, J. Ramage, J. Reising, R.M. Taylor, Prof. C.D. Wickens, Dr. S. Wood, Prof. D.D. Woods. Certainly, there are many more who have contributed to our work in one or the other way. We apologize, if we might accidentally have missed someone who might also deserve mention. Nevertheless, we hope that it will be forgiven and accepted at this point that we hereby thank all of them across the board.

Neubiberg Reiner Onken
May 2009 Axel Schulte

Contents

1 Motivation and Purpose of This Book……... 1

2 Introductory Survey on Operational Guidance and Control Systems…….. 7
2.1 Aircraft Systems…… 7
2.2 Systems in Automotive Vehicles……..... 14

3 Basics about Work and Human Cognition……..... 17
3.1 Concept of Work……. 17
3.1.1 Work Process…… 17
3.1.2 Work System…… 19
3.1.3 Systems of Work Systems…… 28
3.1.4 Engineering Potentials and Challenges of Work System Development……..... 30
3.2 Conceptual Essentials about Human Cognition…….. 32
3.2.1 Implementation Principles of Human Cognition…… 35
3.2.1.1 Connectionistic Information Processing…… 36
3.2.1.2 Structural Division of Cognitive Functions into Distinctive Brain Areas…….. 37
3.2.1.3 Principles of Human Memory…… 43
3.2.1.4 The Two Modes of Information Processing……... 49
3.2.1.5 The Limbic Censorship…….. 57
3.2.1.6 Conclusions for the Work System Designer…….. 60
3.2.2 Framework of Functional and Behavioural Aspects of Cognition…….. 63
3.2.2.1 The Three Functional Levels of Human Cognition…… 64
3.2.2.2 The Three Levels of Human Cognitive Behaviour…….. 66
3.2.2.3 A Simple Framework of Cognition……..... 76

4 Dual-Mode Cognitive Automation in Work Systems..... 79
4.1 Conventional Automation 80
4.2 Experience with Conventional Automation..... 85
4.3 Cognitive Automation..... 91
4.3.1 Cognitive Automation: Mode 1..... 92
4.3.2 Cognitive Automation: Mode 2..... 95
4.3.3 Conceptual Conclusions..... 98
4.4 Cognitive Teaming in the Work System..... 104
4.4.1 Forms of Co-operation..... 105
4.4.2 Team Structuring..... 110
4.4.3 Team Management..... 112
4.4.3.1 Co-ordination..... 112
4.4.3.2 Communication..... 117
4.5 Engineering Approach for Artificial Cognitive Units..... 119
4.5.1 Background Aspects..... 119
4.5.2 The Cognitive Process of an Artificial Cognitive Unit (ACU)..... 122

5 Examples of Realisations of Cognitive Automation in Work Systems..... 129
5.1 Realisations of Supporting Cognitive Units..... 129
5.1.1 ACC System..... 130
5.1.2 Support by Co-operating UAVs..... 132
5.2 Operating Cognitive Units (Assistant Systems)..... 142
5.2.1 Associative Assistance..... 143
5.2.2 Alerting Assistance..... 145
5.2.3 Substituting Assistance..... 147
5.2.4 Summary on Characteristic Styles of Assistants..... 152
5.2.5 Conclusions for General Design Guidelines..... 153
5.3 OCU Prototypes..... 159
5.3.1 Prototypes of Cockpit Assistant Systems..... 159
5.3.1.1 PA and RPA (USA)..... 160
5.3.1.2 Copilote Electronique (France)..... 165
5.3.1.3 COGPIT (United Kingdom)..... 166
5.3.1.4 ASPIO, CASSY, CAMA, and TIMMS (Germany)..... 168
5.3.2 Prototypes of Driver Assistant Systems..... 189
5.3.2.1 Generic Intelligent Driver Support (GIDS)..... 190
5.3.2.2 Driver Assistant System (DAISY)..... 192
5.3.3 Assistant for Driver Tutoring (ADT)..... 206

6 Implementation Examples of Crucial Functional Components of Cognitive Automation..... 213
6.1 Knowledge Management..... 214
6.1.1 General Discussion on Knowledge Representation..... 215

6.1.2 Management of Explicit Knowledge…….. 220
6.1.2.1 Some Milestones Concerning Management of Explicit Knowledge…….. 221
6.1.2.2 Semantic Vector Space…… 239
6.1.2.3 Concluding Remarks…… 247
6.1.3 Management of Implicit Knowledge…….. 247
6.2 A-Priori Knowledge Components in ACUs…….. 250
6.2.1 Components with Emphasis on Skill-Based Behaviour……. 250
6.2.1.1 A Classical Approach……..… 251
6.2.1.2 A Soft-Computing Approach…… 255
6.2.2 Components with Emphasis on Procedure-Based Behaviour…… 269
6.2.2.1 Piloting Expert Implementation in CASSYand CAMA…… 269
6.2.2.2 Implementation of Rule Bases for DAISY……... 290
6.2.3 Components with Emphasis on Concept-Based Behaviour…… 293
6.2.3.1 Co-operation of UAVs…… 293
6.2.3.2 Example for the Identification Function: Pilot Intent and Error Recognition…… 297

7 Operationalisation of Cognitive Automation in Work Systems…….. 311
7.1 Cognitive System Architecture (COSA)…… 311
7.1.1 Design Goals of COSA……..... 312
7.1.2 Architecture…… 313
7.1.2.1 Overview…….. 313
7.1.2.2 Kernel…… 314
7.1.2.3 Distribution Layer……. 315
7.1.2.4 Front-End…….. 316
7.1.3 Implementation……... 320
7.1.4 Conclusion and Perspectives…….. 321
7.2 Integrity Amendments through Cognitive Automation……..... 321
7.2.1 Metafunction for Online Monitoring and Control of System Performance (Application Example)…… 322
7.2.1.1 The Technical Concept…… 323
7.2.1.2 A Functional Prototype……. 324
7.2.1.3 The Scenario……. 324
7.2.1.4 Conclusion…… 327
7.2.2 Identification of Performance Deficits of Work Systems…... 329

8 Abbreviations……..... 333

Appendix…….. 339
1 Supplementary Useful Facts about Human Factors……... 339
1.1 Main Brain Structures…… 339

1.1.1 Anatomical Aspects…… 339
1.2 Perceptual Processing..... 345
1.3 Motor Processing……... 346
1.4 Language-Based Communication…….. 346
2 Input/Output Modalities……. 347
2.1 Sensing Modalities…… 347
2.2 Effecting Modalities…….. 349

References……. 351

Author Index……... 369

Subject Index…… 375

Video demonstration of CASSY (DVD)

Chapter 1
Motivation and Purpose of This Book

Why this book? Simply because it is due. Cognitive automation and its system-ergonomic introduction into work systems have been advanced in the meantime to such a degree that already applications for operational work systems are slowly becoming reality. This interdisciplinary book is meant for designers of work systems and associated machines who are interested in this modern approach and its implementation.

Cognition as the functioning of the human brain is no more anything mysterious. Although [Singer, 2007] starts with his article on understanding the brain with: *"People find it difficult to get into their heads what goes on in their heads…"* he makes pretty evident, jointly with other scientists, that in the meantime cognition can be considered as fairly well understood. There is no more any question about that fact. It makes great sense to exploit the findings on cognition in system design, since it is definitely representing the most outstanding capability of the human species. The success story of mankind would not have been possible without it. As humans, cognition puts us in a position to reach out for extraordinary challenges and to succeed with it. Typical examples are the achievements in science, i.e. exploring, understanding and manipulating nature or making use of natural resources wherever it makes sense. Other examples are creating abstract tools like languages as well as powerful physical tools like tiny nano-chips and sensors to explore the micro worlds on the one side, and on the other side to achieve gigantic engineering accomplishments like huge bridges or skyscrapers, or airplanes for almost thousand passengers. The examples can be enumerated further and further. Above all, besides these achievements, human cognition enables us to cope with the demanding complexity of every day's life as seen from an individual's point of view, as well as to succeed after all as a breed in the process of biological evolution.

Undoubtedly, all the fundamental findings, inventions and developments are great achievements, however we are more and more becoming aware of the fact that the achievements as such and to make effective use of them are two different things. Oftentimes, exploration of what can be done is driving new products rather than what is really demanded by potential users as technical aid. In that sense, it is characteristic, for instance, that we as engineers are building highly sophisticated devices like mobile phones or computers with lots of functions, but most of them are by far too complex for the average user. If we want to use all these functions, we need much proficient assistance. Mostly, users take it as it is, since they do not have an alternative.

This book shall contribute to give system designers some more guidelines about effective work system design, in particular for those related to vehicle guidance

R. Onken and A. Schulte: System-Ergonomic Design of Cognitive Automation, SCI 235, pp. 1–6.
springerlink.com © Springer-Verlag Berlin Heidelberg 2010

and control. The issue is that the findings on cognition have to become sufficient commonsense for all from the various disciplines involved in system design, and that guidelines are given how to make use of it in an appropriate and systematic manner. These guidelines are to account for both the needs of the human operator in the work process and the use of technical potentials to make the work system a really most effective one. In other words, this book is meant to provide guidelines for the organisational and technical design of work systems. Therefore, this book is an interdisciplinary one. Findings in individual disciplines are not the main issue. It is rather the combination of these findings for the sake of the performance of work systems which makes this book a useful one.

Having worked in this field for more than twenty years, a great deal of experience has been accumulated on work system design for work domains like the execution of civil and military flight missions, driving on the road, and driver tutoring. Particular emphasis was placed on issues like the introduction of artificial cognition into the work process and to take advantage of artificial cognition by enhancing human-machine interaction on the cognitive level for the sake of work system performance and safety. A number of prototype systems have been realised and field-tested. Among them the work system design for civil flight mission accomplishment by introducing a cockpit assistant system CASSY has probably got the greatest recognition. It was the first system of that kind worldwide which was successfully flight-tested. This took place in the year of 1994.

When we started to write this book, the first plan was, regarding the introduction of cognitive assistant systems, to provide a survey about how cognitive assistant systems are placed in the full picture of automation. Then, considering that automation is part of the design of work systems, we discovered that there is also not a common understanding of the place of artificial cognitive systems in the full picture of work system design. Thus, the scope of this book was extended correspondingly.

The crucial and most challenging design hurdle in this respect is the task to harmonise humans and machines on a cognitive level while they are acting in a work system. It is the *cognitive design*, as we call it, which stands for the endeavour to overcome this hurdle. [Billings, 1997] has created the term *human-centred design* in that context, because the needs of the human operator have to be always accounted for by the engineers when they are looking for technical solutions of work system enhancement. In essence, this postulates that technical means, mainly automated functions, should take care of work tasks which are not suited to be carried out by the human operator, but should be withdrawn, where the human operator has his particular strengths. At the first glance this seems to be quite easy, but the complexity of human behaviour made this to become a gigantic task of interdisciplinary design work.

There are three basic observations which finally led to our decision to undertake the project of writing this book. They can be easily identified when discussing with people, who are involved in the process of human-machine system layout and design. In short, these observations are

1. Still there are many people being involved in ergonomics of human-machine systems, who are simply too much tied up with former views

about pros and cons of automation. For good reasons, they are critiquing the way automation is designed in human-machine systems in use ([Wiener & Curry, 1980] [Bainbridge, 1983] [Wiener, 1989] [Woods et al., 1994] [Woods, 1996] [Billings, 1997]). They formulate certain general recommendations for better human-machine interaction and human operator training, but they do not follow up technical means to systematically avoid the problems they have revealed. The prevailing thinking is that supervisory control (see [Sheridan, 1987]) is the only way to introduce interactive automation. As a consequence, they take it as a given fact that with increased deployment of computerised automation the resulting increase of software complexity leads to systems which are opaque and brittle, i.e. being too sensitive to changes of the environment which were not anticipated by the system designer.

2. On the other side, another type of pre-occupation can be observed among engineers, in particular among those who have been involved in the design of *operational systems* as part of engineering teams for a long time. They are so used to certain routine ways of working off the design contracts that they hardly realise that with the advent of powerful computers and the feasibility of artificial cognition in one or the other way the interactive capabilities of machines have been considerably extended beyond that what they apply in their day-to-day work. They seem to ignore that these technical advances offer new opportunities for the deployment of machines in work processes, in particular concerning new modes of interactive forms of automation with human-like performance, including effective co-operation between the human operator and the assisting automation.
3. The third observation concerns a group of design people who acknowledge most of the technical evolution in artificial cognition, but who still lack confidence that the findings so far are sufficient to offer the solutions they are looking for. This group needs more encouragement not to get stuck with the acknowledgement only. Apparently, they have not realised that we can achieve a lot by use of cognitive systems in work processes without having to wait for the one hundred percent solution. In the context of robot design, [Brooks, 1989] has stated the fundamental point of view in this context:

   > *"Note that we are not saying we should build creatures in simple worlds and then gradually increase the complexity of the worlds. Rather we are arguing for building simple creatures in the most complex world we can imagine and gradually increase the creatures."* [Brooks, 1989]

   In fact, even simple machines can do a lot of work assistance, if they fit well in the complex work environment, including the *human operator*. For instance, we can quite easily build a relatively simple machine for driver assistance when parking our car in a specific parking lot of a specific city garage in our hometown as compared to building a machine for the same purpose of parking, now in *any* garage and *any* parking lot.

> We can, of course, imagine further improvements in coping with higher environmental complexity by a corresponding increase of machine complexity. The problem is, however, that the range of the operation of the machine function has to be clearly delineated regarding its interaction with the real environment and that this delineation is never obscured for the human operator when anything unanticipated occurs (see [Kopf, 1997]).

Typically, the different groups of people with different background, which are representative for each of these observed pre-occupations, are used to work separately from each other. As a consequence, there are inherent difficulties to understand the other's motivations. From our own experience, readiness to learn from each other and to co-operate among the different schools of psychology, computer science, organisational science, and engineering can be a magic recipe to create new interdisciplinary ideas. No question, it takes more than conferences and journal papers to come to that point, in particular, if safety-critical work processes are concerned. A more comprehensive elaboration in a book of the "essential stuff", i.e. the underlying theoretical lines of thought combined with illustrative exemplifications may do a better job. That is what we try to do in the following.

From our own experience we know that research and development programs of great prominence can push things forward, too. We belonged to the third group at the time when we started with the ASPIO (Assistant for the Single Pilot IFR Operation) project [Dudek, 1990]. At that time, about twenty years ago, it was the contemporary existence of the prominent *Pilot's Associate* program in the US [Lizza et al., 1990] which created a lot of encouragement and which inspired us, to start pushing things forward on our own. From that on, we have learned a lot about work systems and the substantial role of cognitive systems as part of them, and we are still learning.

Only rather recently we eventually have fully realised with all its consequences that the basic structuring of human-machine systems as known so far needs further refinement regarding the way automation can be applied. With the advent of artificial cognitive systems, the design of operator-controlled automation has to account for the following fact:

There are two significantly distinct ways of operator-controlled automation which can be applied in human-machine system design with correspondingly two distinct types of human-machine interaction.

One of these ways of automation corresponds to what everybody has been used to since long, no matter whether the automated sub-systems concerned are highly intelligent or less. This way of automation is what corresponds to the type of human-machine interaction where the human operator is placed over the automation in the so-called *supervisory control* [Sheridan, 1992]. We all are aquainted to the automatic speed control in our cars as a typical example. It is the human operator who pursues an objective which is the one of the underlying work

process he is involved in. The automation sub-systems, which are at the disposal of the human operator in supervisory control, do not have any commitment to make sure on their own that they comply with the objective the human operator is pursueing. They do not know about it such that they can solely do what they are instructed to do within the range of their capabilties. If we instruct the automatic speed control to hold the speed at a certain value this will be carried out no matter whether there is something in the way which should give rise to slow down. This is also true if the speed control would have got cognitive capabilities like fancy robots as long as the instruction of the human operator is not specifying this exemption.

In distinction to supervisory control, the other possible way of automation can be designated as *co-operative control* [Schulte et al., 2008]. Usually, designers are still hardly familiar with it, because it has only become possible since artificial cognitive systems have proved technically feasible. They were accustomed to supervisory control as only option for too long. This second way of automation is characterised by the fact that both the human operator and artificial cognitive systems are co-operatively pursuing a common objective which is pertinent to the underlying work process they are operating in. Similar to the human operator these systems may be able to assess what might be the consequences and necessary actions in case of not foreseen events. With the human operator as team leader commitments on certain tasks are co-ordinated between both the operator and the co-operating cognitive system and are executed accordingly very similar to a human team.

Apparently, these two ways of automation show great differences regarding design rational and resulting features and capabilities. Correspondingly, they are very different when we consider what the human operator can and will expect from them and what the interaction with the human operator looks like.

Although supervisory control was a very good means if not the only means for enhancement of the human-machine system performance over a long period of time when nobody was thinking of cognitive systems, it has also shortcomings. It is assumed that the human operator compensates for the weaknesses. This is accounted for by a great effort on permanently presenting all information about the controlled sub-system to the human operator which might be potentially necessary for him to ensure that it works as he intended it to do. This works fine for rather simple automated functions like the automatic speed control in our cars. However, in case of highly complex subsystems to be controlled, there is too little concern about overtaxing the human operator with an overflowing amount of information where most of it is in fact not relevant at a certain point of time in the course of work. There is also little concern about the fact that the human operator might develop certain expectancies about the performance of these sub-systems which are not the same as intended by the designer. This is a kind of dilemma. How can the design cope with the requirement that the human operator should not overlook any important information when needed at a critical situation under high workload. In summary, the design of this way of automation, also in case of cognitive sub-systems, is usually considered to be almost purely a task of technical engineering. It mostly led to a deliberate separation of the sub-system

design from the overall human-machine system design. For instance, designing a system of supervisory control oftentimes has led to a more or less independent development of the automated functions as such to be supervised on the one hand and the human-machine interface functions on the other hand. Thereby, the designers of the human-machine interface are heavily constrained by the mostly given design of the automated function as such. Under these circumstances an all-embracing system-ergonomic consideration is contaminated. Human factors, if at all considered, become too much confined on barely the man-machine interface design. In that sense, it supports the separation of the participating groups of designers of different disciplines and as a consequence also the biases as mentioned above. Despite these drawbacks, though, there is still sufficient reason to make use of supervisory control as long as the complexity of the corresponding functions is kept on a manageable level.

On the other hand, the communicative interaction on a cognitive level between the human operator and a co-operating cognitive system can take place as it is known from the communication within a human team. This might also include initiatives on the side of the co-operative cognitive system to channel the operator's attention towards something he might not have been aware of before. In the extreme, there might be even an intervention in order to avoid an accident. In summary, this way of automation brings with it a new way of human-machine interaction. This interaction is like that of co-operating humans which is characterised by taking into account the other's strengths and weaknesses and assisting eachother based on this knowledge and the knowledge of the common objective. Systems of that capacity are becoming technically feasible now. There are no examples yet of operative systems for our day-to-day life, but there are already fielded prototypes which clearly demonstrate the achievable gain in performance. Of course, this kind of cognitive design is quite a challenge. Therefore, the change to make simultaneous use of both ways of controlled automation will not come over night. Most designers have still to become acquainted with it.

Summing up, the governing question is, how these two distinct ways of automation can be implemented to the best of the human operator's work performance. How should the systems be structured, correspondingly? How to tackle this problem for work processes of vehicle guidance and control can be considered as the central topic of this book. Many future system designs cannot do without. Some of the successfully fielded prototype systems will be briefly described in this book as an illustration and an incentive for also making use of these potentials in future design of work systems.

Chapter 2
Introductory Survey on Operational Guidance and Control Systems

If we talk about vehicle guidance and control to people who have their own experience in being often directly involved in such a process like you and us as car drivers or like airplane pilots, they may at first think about what have been the challenges and problems when driving a car or flying an aircraft, and how they mastered the demanding situations. If we talk about vehicle guidance and control with control engineers, they may look at the same thing from a different perspective. At first they may think about technical systems which are or can be installed in the vehicle in order to support the vehicle operator, usually to make his job an easier one. Either view does not cover the whole issue, but both are important ones on their own right. Before we put both these views together into a single comprehensive one, this chapter will deal just with the engineer's view with the focus on operational systems in aircraft and automotive vehicles.

2.1 Aircraft Systems

Although even today aircraft, e.g. in general aviation, aerobatics or gliding are still operated in pure manual control mode, the pilots' flight guidance tasks are in general dominated by the use of automated systems of various kinds. Major reasons for the early and ongoing introduction of automation in aircraft cockpits were limitations of human

- sensing capabilities for aircraft state parameters and the aircraft environment,
- bandwidth in manual control tasks, and
- endurance in continuous and fatiguing tasks. [Brockhaus, 2001]

In addition to this list, the demand for information and communication may be added, especially in networked scenarios. Billings [Billings, 1997] classifies the various types of automation serving the various requirements into

1. control automation
2. information automation
3. management automation.

Concerning sensing capabilities humans perform poorly regarding the direct perception of e.g. the flight altitude, velocity or heading. Being essential parameters for flight guidance, these have to be measured and displayed to the pilot by technical means. Human bandwidth limitations have to be taken into account

R. Onken and A. Schulte: System-Ergonomic Design of Cognitive Automation, SCI 235, pp. 7–16.
springerlink.com © Springer-Verlag Berlin Heidelberg 2010

when it is demanded to stabilise highly agile aircraft, where the given dynamic frequencies are beyond human cut-off frequency and the given stability margin of the aircraft is not sufficient. In these cases usually stability augmentation systems ranging from simple damping feedback contol loops to highly complex flight control automation will be used. As a very simple example, the pitch damper, which mainly improves the stability of the short-period mode of the longitudinal aircraft dynamics might be mentioned. Furthermore, autopilot and automated flight management functions relieve the pilot from tedious manual control tasks associated with the maintenance or acquisition of certain flight parameters such as a desired altitude, speed or heading. Thinking of an intercontinental airline flight there might be the need to keep these parameters within certain narrow limits over hours with only very infrequent adjustments of set points. By use of modern Flight Management Systems (FMS) even these flight guidance tasks may be automated. As a result, at least in civil air transport, it is possible to perform a full flight mission from take off to landing concerning its navigational aspects in a more or less fully automated manner. About the only input needed is a flight plan containing the geographical and some additional information about the desired flight, e.g. the vertical profile.

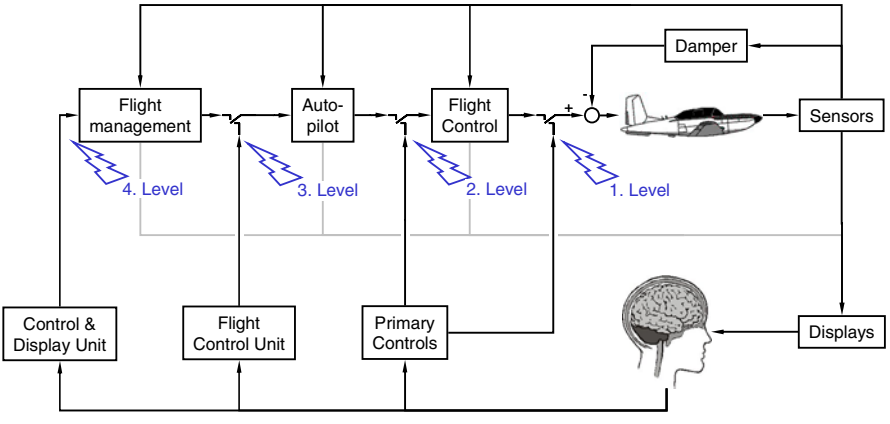


Fig. 1 Hierarchy of automation functions in flight guidance

Figure 1 depicts a simplified setup of these primary functions of automation in flight guidance indicating the four most important levels of possible pilot's interaction with the system.

On the first level of interaction the pilot directly operates on the control inputs of the aircraft, i.e. aerodynamic surfaces and engine throttle by use of dedicated primary flight control devices such as a control stick or wheel. Characteristic to this interaction level is the continuous control of the aircraft in 3D space and time. Thus, the pilot continuously has to compute the control law in order to achieve the desired aircraft movements as according to the given aircraft dynamics. The pilot observes the aircraft movement by use of primary flight instruments, above all the artificial horizon. Related sensors are gyro instruments and atmospheric probes. In

flight mechanics the aircraft dynamics commonly is modelled by use of a set of differential equations describing the development of the aircraft state variables over time as effect of the external forces and moments. The movement of the aircraft seen as rigid body can be described sufficiently accurate for the most principle considerations by twelve degrees of freedom, i.e. the three rotational attitude angles (yaw, pitch, roll), the related angular velocities, the three spatial positions (longitude, latitude, altitude) and the related translational velocities. While the three position parameters and the rotation angle against north in the lateral plane characterise the navigational movement of the aircraft against the geographic frame of reference, the remaining eight degrees of freedom allow describing the dynamic characteristics of the aircraft movement including stability and control issues. These are the important ones for the first and second interaction level of the pilot with the aircraft, since the dynamic characteristics have to satisfy certain standardised requirements (e.g. "Military Standard – Flying Qualities of Piloted Aircraft MIL-STD-1797", or "Federal Aviation Regulations – FAR, Part 23 an 25"). It might be interesting to mention that these handling quality specifications belong to about the earliest human factors related requirements in aeronautical engineering. The natural flight characteristics of a given aircraft can be modified to some very limited extent by controllers feeding back angular rate information to the aerodynamic control surfaces, thereby, generating damping moments.

When there is need for some more radical changes of the dynamic behaviour, either for the sake of stability or for any other control requirement coming out of the application domain of the aircraft, further automation functions are necessary. In this case automated flight control systems (AFCS) are put into place. Although still being operated by the pilot in continuous manual control mode through the conventional primary flight control elements, i.e. control stick or control wheel, there is at least one fundamental difference to the first interaction level. The pilot no longer determines the deflection of aerodynamic surfaces, but commands desired demand values for certain aircraft state parameters, such as accelerations or angular rates. A very common example for level 2 interaction is the so-called *rate command / attitude hold* mode. Here the angular attitude of the aircraft will be kept constant, while there is no pilot input on the stick. A pilot's pitch input to the control stick will be interpreted by the AFCS as a proportional pitch rate command to the aircraft and the actual pitch rate will be adjusted accordingly by the automatic control. In order to technically achieve this, the classical principle of a mechanical connection between the control stick and the aerodynamic surfaces had to be given up. Because of its technical solution mainly based upon electrical signals, this automation technology is often referred to as *fly-by-wire* (FBW). Figure 2 depicts the technical principle of a fly-by-wire system implementing a vertical acceleration command.

On the third level of interaction the pilot now is relieved from the task of continuous input of demand values for kinematic parameters such as accelerations or angular rates. On this level the pilot has to enter certain flight trajectory related set points whenever a change is demanded. This may occur rather infrequently. In between these adjustments there is no further control operation required by the pilot, though, except for the monitoring of the proper execution of the function.

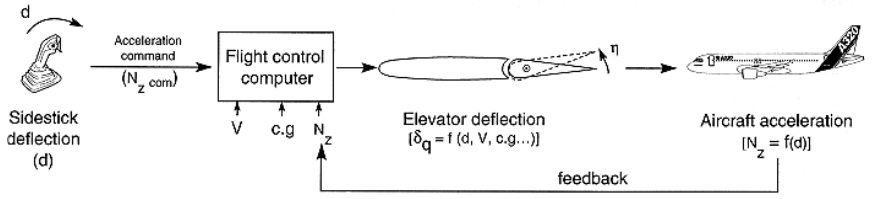


Fig. 2 Principle of fly-by-wire system

Therefore, operating on the third interaction level can be seen as a typical supervisory control task. The family of very diverse systems being in use on this interaction level can be summarised under the term *autopilot*. Simple autopilots are capable of functions such as ALTITUDE (or SPEED, or HEADING) HOLD, i.e. keeping a desired barometric flight altitude (or airspeed, or aircraft heading) at a constant value. In fact, realised autopilot systems, e.g. in airliners are usually some more complex concerning modes and operation. Operation of an autopilot of transport aircraft is done via a dedicated Flight Control Unit (FCU) (see Figure 3 for an example).



Fig. 3 Flight Control Unit of an Airbus 320

In military high performance aircraft the concept of operation might be considerably different, e.g. the control of the autopilot modes and settings integrated in a HOTAS (Hands on Throttle and Stick) concept.

Before the advent of FBW technology autopilots were in use temporarily displacing the pilot in operating on the primary flight control organs by meshing into the mechanical linkages, causing the control stick to move according to the nested outer and inner loop commands of the autopilot. Today, autopilots may be integrated into aircraft equipped with an AFCS in FBW technology. In this case the autopilot will only generate so called outer-loop-guidance-commands (OLG) which will be processed by the AFCS inner control loops.

Finally, on the fourth level the interaction of the pilot with the automated system is again changed in a radical manner. On this level so called Flight Management Systems (FMS) integrate various navigation, guidance and management functions. Traditionally, flight navigation was supported by various automated

sub-systems covering very specific functionalities used to determine the position of the own aircraft. Mainly these were inertial navigation and radio navigation. More recently satellite navigation technologies came along. While in former days until the 1970ies it was the duty of a dedicated person in the cockpit, i.e. the navigation officer, to integrate the information provided by the various navigation aids, the early Navigation Management Systems integrated these functions into one single system entity, functioning mostly automatic. Since then, especially with the advent of digital computer technology in avionics systems Flight Management Systems evolved rapidly. Main functional characteristics of modern FMS can be classified into three main categories:

1. Position determination
   - integration of navigation information aiming at providing a navigational solution of high precision and integrity
2. Route planning
   - storage of the flight plan and optionally a secondary flight plan
   - support for manipulating the flight plan, i.e. navigational databases, automated planning functions
   - calculation of trajectory predictions, e.g. for precise flight timing
   - providing access to database information, e.g. routes, waypoints, navaids, airfields
   - performance and resources calculations, e.g. optimal speed, distances, arrival times, fuel, top-of-decent
3. Guidance computation
   - generation of outer-loop-guidance-commands for automatic flight plan following,
   - monitoring of the flight progress
   - management and provision of information for display on the Control and Display Unit (CDU) and the Electronic Flight Instrumentation System (EFIS)

Figure 4 shows the Control and Display Unit (CDU) of an Airbus 320 representing the traditional main human-machine interface to the FMS. The various sub-systems can be accessed via moding keys. Data shall be entered via an alphanumerical keypad. Line select keys allow the direct ineraction with software menus displayed on the screen. Suchlike single-box solutions, sometimes even the flight management computer was hosted by the CDU, were strongly motivated by the fact that when FMS first entered flight-decks, they were usually introduced as retrofit and therefore had a rather low degree of integration with other systems. In most recent flight-deck designs, e.g. the A380, the FMS is controlled in a more GUI-style (Graphical User Interface) interaction, using multifunctional head-down displays, trackball and standard QWERTY keyboard. This approach allows e.g. an easy geometrical manipulation of flight plan information on a map display.

Another feature of modern FMS is the integration into information networks such as e.g. ADS-B (Automatic Dependent Surveillance-Broadcast), TIS-B (Traffic information services-broadcast) or CPDLC (Controller Pilot Data Link Communications) to the main purpose of air traffic management and deconfliction.



Fig. 4 Control and Display Unit of an Airbus 320

The variety of information provided by the diverse means of control automation including the communication systems described above will be displayed on an Electronic Flight Instrument System (EFIS) replacing the traditional electromechanical cockpit instruments and gauges of former times in modern flight-decks. An EFIS usually consists of several Multi Function Displays (MFDs). These glass cockpit displays being capable of displaying and interacting with various display formats, the most common of which are the Primary Flight Display (PFD) and the Navigation Display (ND) (see Figure 5).

Furthermore, many additional display formats are in use conveying information about various aircraft systems, such as propulsion, fuel or electrical systems. Systems like ECAM (Electronic Centralised Aircraft Monitor) or EICAS (Engine Indicating and Crew Alerting System) go beyond the pure display of status information by also providing corrective action to be taken by the flight crew in case of system failures, as well as system limitations resulting from these failures.

While ECAM or EICAS only consider the internal status of the basic aircraft systems and the engine, providing advice and warning to the flight-deck crew, there further exist dedicated warning systems concerning the external situation of the aircraft such as traffic and ground proximity.

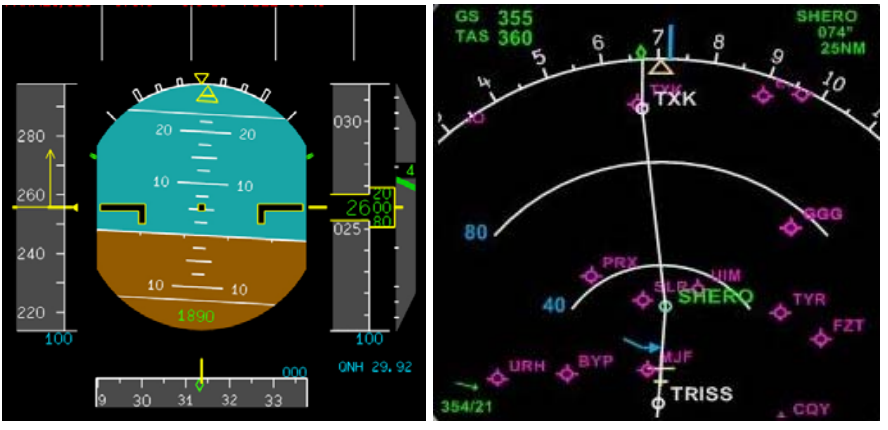


Fig. 5 Primary Flight and Navigation Display

The Traffic Alert and Collision Avoidance System (TCAS) is designed to reduce the incidence of mid-air collisions between aircraft. It monitors the airspace around an aircraft for other aircraft independent of air traffic control. It warns the flight-deck crew of the presence of other aircraft which may present a threat of mid-air collision. TCAS is a cooperative system in a sense that only transponder-equipped aircraft can in principle participate. TCAS is an implementation of the Airborne Collision Avoidance System (ACAS) standard mandated by the International Civil Aviation Organization (ICAO). Depending on the implementation version TCAS provides traffic information (TA, Traffic Advisory) and direct instructions to minimise the danger of collision (RA, Resolution Advisory) by vertical manoeuvres. Future implementations will include advices for lateral evasive manoeuvering.

In order to counteract the occurrence of CFIT (Controlled Flight into Terrain) accidents, so-called Terrain Awareness and Warning Systems (TAWS) are in use in civil aircraft. The first-generation of these TAWS systems is known as Ground Proximity Warning System (GPWS), which uses the radar altimeter information to determine terrain closure. This system has now been further improved by use of digital terrain elevation databases. The TAWS calculates interferences of the anticipated flight trajectory with the digital terrain by use of integrated navigation information. In case of hazardous approach to terrain, warnings will be issued to the flight-deck crew. The availablility of digital terrain elevation data allows the display of terrain information and proximity on the navigation display or even as three-dimensional perspective view (see Figure 6 taken from [Brämer & Schulte, 2002]).

In summary, aircraft guidance and control systems are probably the most comprehensive ones at the time being compared to corresponding systems in other categories of vehicles. They evolved in the meantime from a design tradition of more than hundred years and cover about all tasks which might occur during flight. Up to now, though, there are very few exemptions in operational aircraft to

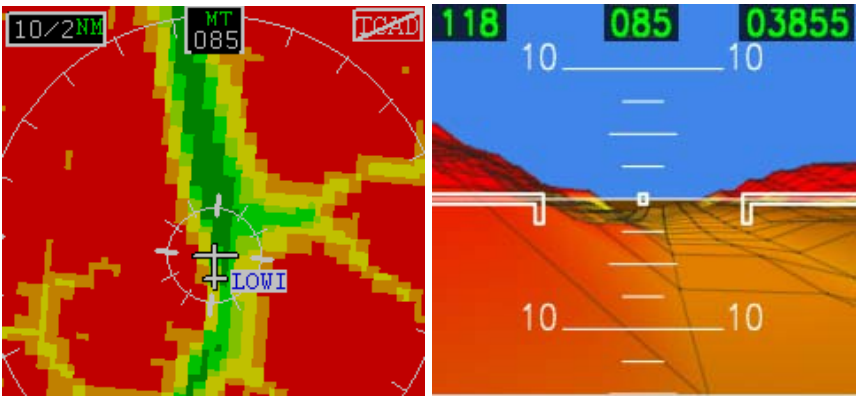


Fig. 6 TAWS displays (left: map display, right: perspective view)

the traditional design principle of exclusively letting the pilot decide which task has to be carried out at whatever flight situation. One exemption is the terrain avoidance mode in some fighter aircraft, where an automatic pull-up manoeuvre takes place in case of being too close to the terrain at high speed. With the ongoing advent of unmanned aerial vehicles this will change as can easily be concluded from the following of this book. In turn, the experience with unmanned aircraft surely will eventually have an impact on the design of guidance and control systems for future manned aircraft, too.

2.2 Systems in Automotive Vehicles

In distinction to the domain of aviation it took some time that automatic systems for guidance and control were introduced in road vehicles. There are many reasons for that. One of them is the fact that any malfunction might rather immediately lead to an accident before the driver can intervene. Another aspect is the cost relative to the total cost of the vehicle. The first and only automatic support for the driver for a long time was the automatic gear. Later, in addition to that the function of cruise control could be installed. The driver activated this rather simple control function to have it keep the actual speed of the vehicle until the driver intervenes by stepping on the gas pedal or using the brakes. Only since some years we experience that the automotive industry endeavours more and more to support the driver by a greater variety of automatic functions, built-in ones or other systems the driver can activate himself/herself in modern cars (see [Wallentowitz & Reif, 2006]). The built-in ones usually are for warning purposes or for automatic enhancement of the vehicle's dynamic behaviour. A typical example for a warning system is that which guides the driver's attention to a vehicle on the adjacent lane close enough that the started lane change would lead to a collision. Systems for automatic enhancement of the vehicle's dynamic behaviour are for instance the ESC (Electronic Stability Control) which combines the ABS (Anti-lock Braking

System) and traction control. Systems of that kind are automatically turned on when the engine is started. The driver might even not notice them as special automatic systems in the car.

The systems mentioned so far by name, are of the procedural type of control systems we are used to since long in all kinds of work domains. These are characterised by feedback loops (like ABS etc.), possibly triggered by command inputs (like cruise control). The control laws of these loops work on sensor signals for relevant physical quantities pertinent to the controlled element. The procedural knowledge involved in these conrol systems implicitly lies in the control laws.

Since in the meantime computers and mass storage devices became cheaper and cheaper we have become acquainted since several years to the so-called navigation systems in our cars. Thereby, the position of the vehicle is determined by means of a GPS receiver. These systems offer a new quality. They support the driver for the task of route planning and of correctly following the selected route, making use of explicit declarative knowledge about geographical maps in terms of data bases. These systems are of similar supporting effect than the flight management systems in airplanes.

A further step forward is the kind of automation as exhibited for instance by systems like the parking aid for lateral control and the ACC (Adaptive Cruise Control) system for longitudinal control (see Figure 7). The parking aid, for instance, takes care of the lateral control during an automatic parking manoeuvre in between cars in front and ahead. Thereby, driver executes longitudinal control through gas pedal and brakes.

The ACC probably is the best known system of that kind so far. It holds a set speed, as known from the former systems for cruise control, and automatically decelerates and holds a safe distance to vehicles ahead as long as they are moving at a lower speed than that which was set by the driver. The basic components of ACC systems are shown in Figure 7.

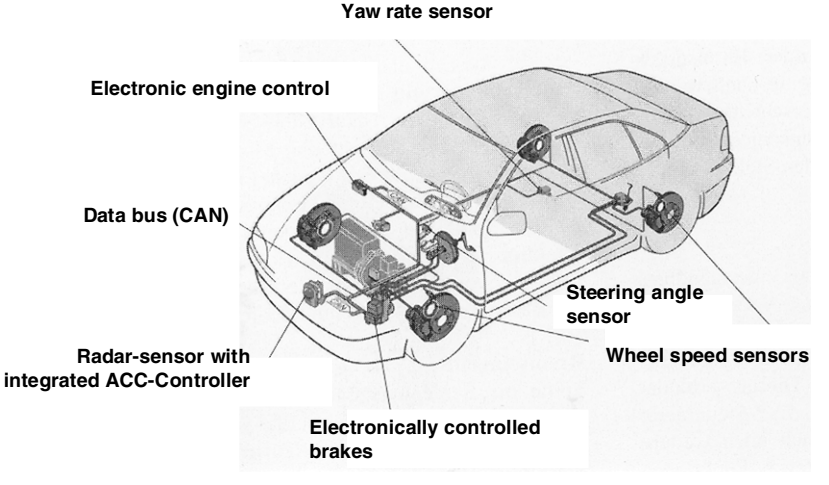


Fig. 7 Basic structure of a generic ACC system [Wallentowitz & Reif, 2006]

By now, ACC systems work within a speed range from 0 to about 200 km/h. Thus, these also include automatic longitudinal control for stop and go in congested traffic situations (see [Venhovens et al., 2000]). The performance of ACC systems is steadily increasing. Besides radar sensors, infrared and camera-based sensing is introduced to ensure as much information about the relevant environment of the vehicle as possible. So far, camera-based sensors are in operational use to obtain information about the position of the own vehicle within the road lane or about passing traffic from behind. Also data bases about stationary environmental objects together with those in current navigation systems will be used for relevant information in future systems not far from now. This will be the time when we can also expect further coupling of systems, also with the navigation system, which is now stand-alone.

The current featuring of the ACC system makes it pretty obvious, that it exhibits a new characteristic: It functions on the basis of explicitly built-in knowledge. This is the crucial step to introduce artificial cognition. In that sense, the advancement in automotive systems supporting the driver has already gone much further than that of the aircraft systems. Later on in this book, this will become much clearer.

Chapter 3
Basics about Work and Human Cognition

In order to initiate new developments of vehicle guidance and control systems which go beyond the state of the art, it appears to be worthwhile to put in front some basic facts and considerations. In particular, it makes sense to clarify how a system is structured which enables the process of vehicle guidance and control and what we can learn from human cognition in order to make use of it for the enhancement of vehicle guidance and control. In essence, this leads to an understanding of two general concepts, that of *work* and pertinent work systems, and that of *cognition*. These two concepts compose the basis for the considerations of system design which will be outlined thereafter. This chapter therefore will start with a discourse on the concept of work and its implications, if we consider the process of vehicle guidance and control as a work process. Subsequently, the main features of (human) cognition are outlined in order to capture its design potentials, if one makes use of introducing artificial cognition in the work process.

3.1 Concept of Work

Similar to what has been described in [Bubb, 2006] as the nature of work, we consider any private or professional human teleological activity serving a certain purpose as covered by the concept of work. As a consequence, this concept of work holds no matter whether the activity is done as a necessity for living or for fun, whether it is mainly a physical activity like playing tennis or an intellectual one like reading in a journal on Sunday night. In this book, the focus is on work associated with the activity of vehicle guidance and control. Certainly, professional work will play a key role in the following discussions, but by far not an exclusive one.

3.1.1 Work Process

Work is a dynamic process, the so-called *work process* (see Figure 8). The work process follows a certain agenda of procedures and actions based on assessments of the current world situation and, if necessary or explicitly wanted, on incidental deliberations. More than two third of our lifetime being awake we are involved in work processes. A natural and very plain example with only a single person involved is the process of the purposeful activity to swim in a lake while being on vacation. The purpose might be a combination of carrying out an exercise, or having some refreshment, or just having fun. Another example, which is

R. Onken and A. Schulte: System-Ergonomic Design of Cognitive Automation, SCI 235, pp. 17–77.
springerlink.com © Springer-Verlag Berlin Heidelberg 2010

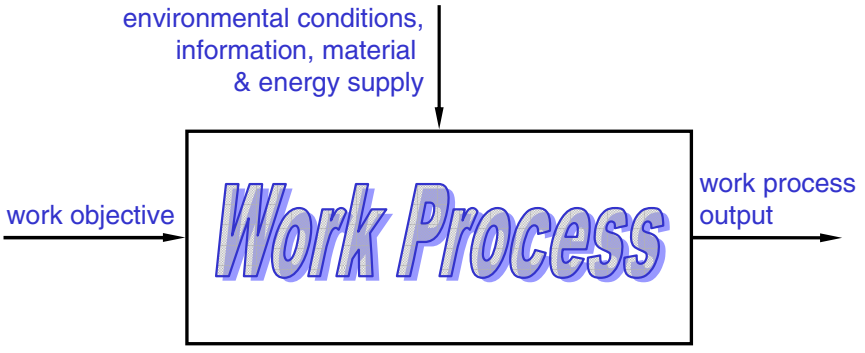


Fig. 8 The work process

associated to the more complex application area of vehicle guidance and control as being considered more closely in this book, driving a car to get to a certain destination in order to meet some friends or flying an airplane for the purpose of transportation of goods or passengers from one location to another.

The purpose of a work process in explicit terms is what we call the *work objective*. There is no work process without a work objective. In professional life, the work objective mostly comes as an *instruction*, *order* or *command* from a different supervising agency with its own work processes, preceding the work process being considered. In certain cases, though, in particular in the private domain, the human who is carrying out the work to accomplish the work objective specifies it by himself.

Besides the work objective as main input, other inputs to the work process are the

- *environmental conditions,*
- *information, material and energy supply*.

Environmental conditions can play an important role, in particular regarding the performance of the work process. They might affect the work process in one or the other way. For instance, in case of flying an aircraft (work process), subject to a certain mission order (work objective) atmospheric processes generate environmental conditions with physical effects on the work process which cannot be ignored.

The work process also has got certain supply inputs. A typical supply input is the energy supply. This might include associated material supply. Another important supply input the work process is drawing on is information in addition to that about the work objective. This includes all available information of relevance for the work process about the surrounding real world.

The accomplishment of the work objective is that what really matters in a work process. The physically effecting measures are showing up at the *work process output*. They are to produce the corresponding changes of the real world subject to the work objective. In that sense, the work process output can also be an act of information processing and transmission.

3.1.2 Work System

Describing a work process means to state *how* it is performing, i.e. which functions are being carried out in order to comply with the work objective. Another dimension of representing an operative work scenario is the physical one, dealing with the real world entities which are supposed to be involved in the work process. We call a real world entity a *work system* (see [Karg & Staehle, 1982], [REFA, 1984], [Winter, 1995], and [Döring, 2007]), if it performs a work process specified by a work objective. In essence, the work system represents the components which accommodate the functions of the work process. Thus, a work processes needs a corresponding work system. Hence, the person who is swimming in a lake as alluded to earlier in the context of a simple example for a work process represents the corresponding work system. There is always a one-to-one relationship between individual work systems, work processes and work objectives.

In Figure 9, it is shown how a work system is embedded in the real world. The work system has got the same inputs and outputs as the corresponding work process (see Figure 8). There are certain parts of the real world from which inputs are provided for the work system and also certain parts which are being affected by the work system. They form the relevant work environment which is of interest for design investigations of the work system.

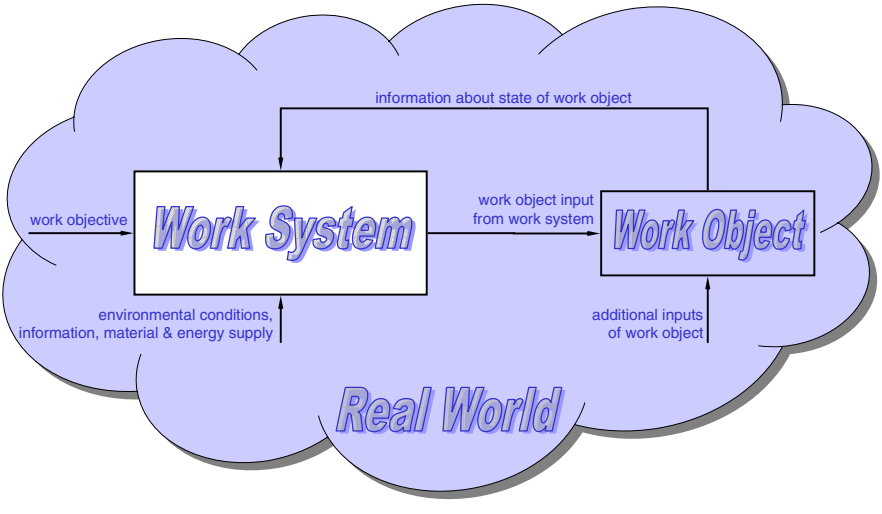


Fig. 9 Work system and the surrounding real world, including the work object

In Figure 9, the *work object* is explicitly shown as the most relevant part of the surrounding real world. It is the purpose of the work process to achieve a certain change of the work object subject to the work objective. The work object exhibits this change as a change of the state of the real world. The *work object input from the work system* as the intended productive work system output is effecting these

changes of the real world. There might be other work system outputs as side-effects on the real world in the course of the work process which are not explicitly intended but inevitably generated in order to produce the intended one. These are not depicted in Figure 9. They cause changes of the real world, too, apart from the work object, and are just part of the game when endeavouring for the accomplishment of the work objective. As long as these are not compromising the success of the work process, one might tolerate them. It should be noted at this point that the work system is part of the real world as well and that parts of the work system or even the whole thing can be part of the work object or even the work object as such. Therefore, in order to make a clear distinction between work system and work object, we depict the work object in separation of the work system (see Figure 9), although it might be part of the work system at the same time. Think of the earlier mentioned example of a person swimming in a lake. In this case, that person is representing both the work system and the work object. That person simply generates the input on himself/herself as the work object. Hence, in conclusion, work objects may also represent any kind of active system like a work system with respective types of inputs.

Referring to the earlier mentioned example where a person is operating a car for the purpose to move to a certain destination in order to meet some friends, the work object is the driving person as part of the real world whose location is being changed by the car ride. The change of the work object being brought about by the work process is always being brought to light by the *state of the work object* which may consist of a number of state variables. Here, the state variable "location" is being changed by the work process. In case of the other example mentioned earlier of flying an airplane for the purpose of transportation of goods from one location to another, the work object is nothing else but the transported goods.

The state of the work object indicates the essential part of the work state as such. For instance, state variables indicate whether the work objective is accomplished or what is the difference of the current state of the work object from what is wanted to meet the work objective. In case of the car ride it is possibly just the information about the position of the car (coinciding with the location of the car driver), which for instance might also be of interest for other work systems in the environment, then being an information supply to them. This information would be passed on to the other work systems via other work system outputs to communicate with the external world. As already mentioned, other work systems, in turn, might as well transmit information as inputs into the work system concerned (information and energy supply) and the associated work object (*additional inputs of the work object*).

At the time the work process has been finished, the work object becomes the *work end product* of the work system concerned. The work end product might comply with what is wanted in the sense of the work objective. It also might not comply for any reason. Oftentimes, the work end product as a physical entity is passed on to another work system, for instance as material input.

The design of work systems is the main issue of this book. It is the work domain of system engineers when defining the so-called user requirements and,

based on that, the system requirements. They have to deal with the peculiarity that there are human operators as core component of work systems, usually interacting with technical devices. Taking this into account, the basic structure of work system components is shown in Figure 10. When dealing in the following with a work system in general as intended in Figure 10, we make things more concrete by exemplifying the work process according to the application domain of main interest in this book (vehicle guidance and control) as that of flying an aircraft. As being depicted in Figure 10, the work system comprises two main components, the

- *operating force* and as an additional, usually existing optional main component the
- *operation-supporting means*.

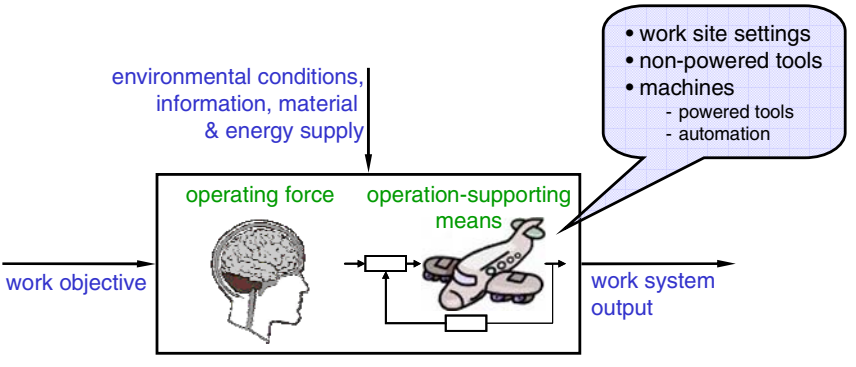


Fig. 10 Basic structure of a work system

The effectiveness of the work system is completely determined by the functional contents of these components. Both the operating force and the operation-supporting means, here in particular the actively supporting equipment, have got the necessary receptor and effector devices. Thereby, the work system can interact with the physical environment including the work object and other work systems. As to the operation-supporting means receptors are typically sensors or communication receivers providing work system inputs, effectors may include communication transmitters. There is also internal interaction between the operating force and the operation-supporting means through receptors and effectors on both sides.

Operation-supporting means are not mandatory for all work systems one could imagine. In the extreme case, it takes just a single human being to form a work system. Humans possess senses and means to receive the work process inputs and they also possess motor systems (effectors) to produce outputs in terms of body motions. Think of the work process of swimming in a lake. The senses of the swimming person provide the data needed to determine her/his position or to check for obstacles, for instance, and his motor systems let him move, if the energy supply is warranted.

The operating force and the operation-supporting means will be described in more detail in the following:

Operating force (OF): Besides carrying out certain tasks, if not all, to comply with the work objective, the operating force is the high-end decision component of the work system. This component determines and supervises what will happen in the course of the work process and which operation-supporting means will be deployed at which time. In other words, this work system component has got the highest authority level. It is only this component where the complete knowledge about the work objective is placed as well as the knowledge needed to understand and pursue the work objective effectively. Therefore, the operating force exclusively consists of operating units which have got the necessary cognitive capabilities to do this job or at least, in case of multiple operating units, to contribute to the pursuit of the work objective with it in view. Thereby they exhibit the indispensable behavioural feature based on the knowledge mentioned to monitor the work process as a whole from their own perspective, subject to the work objective and the work plan agreed upon.

By definition, the operating force is the only component in the work system containing a human element. It consists of at least one human individual, i.e. the human operator. Already this makes this component very special. Traditionally, the operating force has been exclusively human. The human element as an autonomous entity warrants that the operating force can make use of this basic capacity to be principally capable, if wanted, to create or modify the work objective independent from any external agency. Therefore, **we can claim that a work system is an autonomous system, in principal.** This is not in contradiction with the fact that the work objective can come from a supervising external agency, too. For instance, without any further comment on the earlier work system example of a person swimming in a lake we would not hesitate to presume that this person has made up the work objective independently as an autonomous agent. However, although this is probably true, he or she could also have been instructed by a supervising agency, and even in this case the individual still would have the capacity, in principle, to modify the instructed objective or to create a new objective based on its own value judgement. This is one of the great strengths of the autonomous human operator, keeping in mind, though, that it can be a crucial weakness, too, in certain rare cases. Everything has its price. The Titanic accident is a spectacular example to illustrate that the natural autonomy of the human operator in the work process can lead to a catastrophe. It was not a matter of loosing awareness of possible environmental threats. It was rather a matter of the governing personal desires of the people in charge. Thus, unfortunately, the work objective and the desires of the operating units have not to be automatically in coincidence in all cases.

The operating force does not necessarily consist of just a single person. It might also be formed by a human operating team. The necessary condition to operate as a member of the operating team is to know enough about the work objective in order to understand and pursue it effectively, regardless of the fact that the

individual capabilities may be different and may lie in different fields of expertise. Think of a medical team being the operating force in the surgery room. All team members as the operating units are human. All of them know about the work objective, but the individual capabilities of the surgeon compared to those of a surgery nurse are obviously very different. Still the whole team is engaged in collaboration, each member with a certain role, to meet the common work objective. This is an imperative and satisfactory property of any ordinary team member of the operating force.

Traditionally, improvements of work system effectiveness and efficiency as far as the human operating force is concerned are so far only subject to selection, education, training, motivation and other issues related to human resources and performance improvement. However, at this point the question is reasonable whether to consider not only humans as team members. One main purpose of this book is to deal with that question that the operating force might be supplemented by artificial team members as a special case of co-operative automation. One step further, one could theoretically think of an autonomous artificial system by replacing the human team members by artificial ones. With advancing technology, in theory a situation is imaginable where the human operator will completely be substituted by some automation, probably with human-like cognitive capabilities. In Figure 11, right, this kind of artificial cognitive unit is depicted by a pictogram like a robot head being used throughout the book in similar contexts. The reduction of the human flight-deck crew of airliners about two decades ago is a good example for this tendency of substituting human manpower, although that was provided only by automation for task execution (e.g. flight management system). Regardless of the discussion, what kind of capabilities such automation might provide in the future, there is one crucial requirement to be fulfilled to refer to such a system as an *autonomous system*: It has to have the capability to independently assign the work objective on its own, just as the human operator in a work system is principally capable to do so. Such a system we call an *autonomous system* (see Figure 11, right with robot head), having the capacity to operate in **complete** independence of other agencies, in particular independent of human ones. Theoretically, this seems to be a principally feasible engineering challenge.

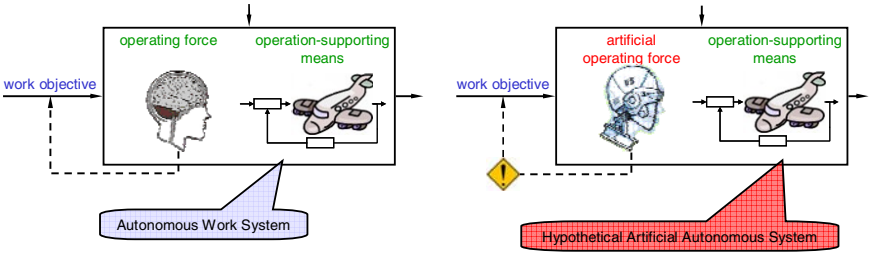


Fig. 11 Hypothetical conversion of an ordinary autonomous work system with human operator into an artificial autonomous system

However, considering such an artificial autonomous system there are two good reasons, why it is not desirable to create a machine like this:

1. From an *ethical* point of view we want to refuse building machines, which potentially could self-assign a work objective, as long as this implies the possibility of unforeseeable, harmful consequences for us humans. This statement is pretty much in line with the philosophical discourse on the principle of responsibility in [Jonas, 1989].
2. From a *pragmatic* point of view we do not want machines not being subjected to human authority. Such technological artefacts for themselves are of no use, since they are no longer serving the human for his work in its broadest sense.

As a consequence, technology may provide very powerful automation capabilities, also in terms of co-operating cognitive units as part of the operating force in the work system, or in replacement of human performance if useful, unless it is not entitled to the authority of self-assigning a work objective. We call the resulting artefact a(n) (artificial) semi-autonomous system, the structure of which is very similar to that of a work system. Figure 12 shows such a semi-autonomous system being part of the operation-supporting means of the work system for the application as shown in Figure 12. In this set-up, this operation-supporting semi-autonomous system will not be allowed to self-define an objective. It will receive it from the operating force of the associated work system which at least comprises one human operator. Surely, such a semi-autonomous system may work very independently, without continuous guidance by the operating force.

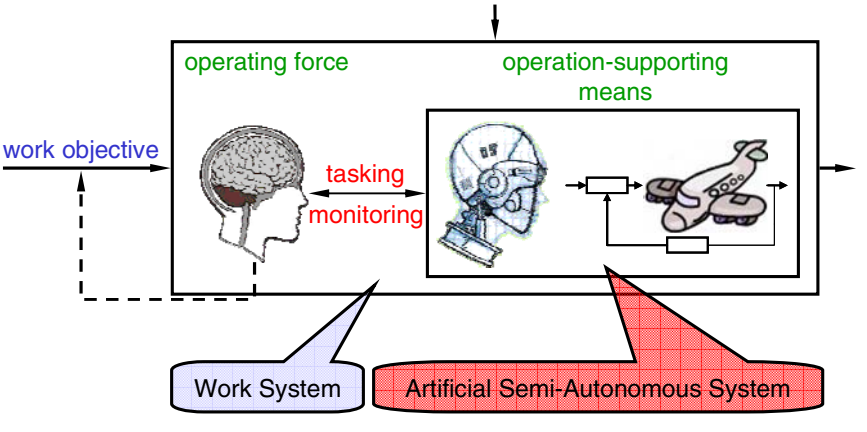


Fig. 12 Semi-autonomous system as part of a work system

In conclusion, the definition of a work system excludes the existence of a purely artificial one. There will be no artificial autonomous system addressed in this book as a work system. There will be no artificial system addressed which would have got the capacity to generate a work objective on its own. Instead, we

claim the necessary condition that there will be at least one human operator as part of the operating force of the work system. In other words, every technological artefact, no matter how high its "level of autonomy" might be advertised by its creators, will finally either support human work (as part of a work system) in its broadest sense or be not of any use at all.

Finally, the operating force has not necessarily to be positioned at one single work site. If there is more than one operating unit, these might be easily distributed over different locations, if appropriate communication is warranted between these units. One might think in this context of any data link connection, e.g. via internet. Figure 13 sums up the main characteristics of the operating force.

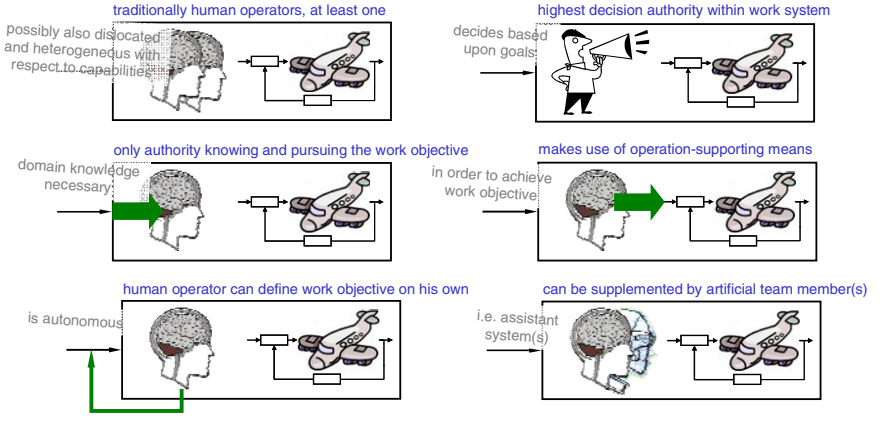


Fig. 13 Main characteristics of the operating force in a work system

Operation-supporting means (OSM): Taking into account that capabilities of the human operator cannot be improved beyond certain sometimes rather narrow limits, system designers focus on the enhancement of the operation-supporting means, i.e. to design artificial means by exploiting technology to support the human operator for the sake of a human-centred [Billings, 1997] and efficient work system. If there are several sub-processes in the work process each sub-process may have its own set of operation-supporting means. All vehicle guidance and control systems, like the autopilot in the aircraft or the cruise control in our car, which are presently in use in the aerial and automotive domain as described in the previous Chapter 2, belong to the operation-supporting means of whatever work system they are part of. Therefore, what is known about human-machine systems is closely related to work systems. When we talk about a human-machine system, this intrinsically corresponds to or is part of a work system. This will become more obvious when we talk about applications. At this point, [Kirchner & Rohmert, 1974], [Wickens, 1992], [Vicente & Rasmussen, 1992], [Johannsen, 1993], [Helander et al., 1997], [Salvendy, 1997], [Sheridan, 2002], [Harris, 2002], and [Timpe et al., 2002] may be recommended as literature on basics about human-machine systems. The machines so far have been exclusively part of

operation-supporting means in the work system. Our proposition is that machines may even be part of the operating force as was alluded to earlier and will be dealt with later on in more detail.

In principal, as already mentioned, operation-supporting means are not a necessary component of a work system. There may be work systems without any operation-supporting means. Consider the example of the swimming human or, for instance, to work in a purely natural environment without any equipment. If the performance of these work processes is not a matter of concern or if there is a deliberate abandonment of artificial means for support there will be no engineering effort, too. This kind of work system, though, will not be further considered in this book.

The operation-supporting means include the following optional components:

- work site settings
- non-powered tools
- machines
  - powered tools
  - automation

The basic *work site settings* are passive means in the sense that they are not used as a tool which is part of the active productive work process. Work site settings include artificial means for accommodation, furnishing and accessories like e.g. illumination or air conditioning.

Non-powered tools at the disposal of the operator are tools like books, maps, screw drivers as workshop tool and other tool box items, bicycles as vehicles and others.

All *powered tools* belong to the category of *machines*. For the work objective "to move for a visit to relatives at some other place in town", for instance, the technical means of transportation like an automobile is a typical powered tool as part of the operation-supporting means of the corresponding work system. This is true as long as all changes of the state of this machine is totally controlled by the operating force of the work system it belongs to (in this case by me). Otherwise it would belong to another entailed work system. Therefore, both, whether your friend gave you a courtesy ride, or if you were using public transportation, this would not be an operation-supporting means of the work system you are operating while being transported. In fact, you would have been the work object, i.e. a part of the material supply input of another entailed work system with an operating force you did not belong to. To offer another example for a machine as powered tool apart of any motorised vehicles, let us think of a drilling machine which our dentists use. This machine has in common with a motorised vehicle that it is totally controlled by the human operator.

Opposed to the powered tools there are other machines as sub-systems which provide so-called automated functions. They may stand for themselves like a robot or a multi-agent system, or are built into a powered tool like modern car equipment such as ABS (Anti-lock Braking System) or automatic speed control. All this belongs to the special sub-category of machinery which is called *automation*. Regarding the scope of this book, automation within a work process

stands for any technical function, which partially or fully carries out a control or information manipulation task which, in the last place, is defined by an autonomous entity like the human operator, and which otherwise a human operator would have to accomplish. As already mentioned in Chapter 2.1 corresponding functions of automation (see [Billings, 1997]) can be subsumed under

- automation of information acquisition and delivery,
- automation of system control, and
- automation of system management.

Automation has become a major playground for engineers in order to enhance the work system performance. In the context of this book sub-systems of automated functions in vehicle guidance and control are in our primary focus. Thereby, these sub-systems, on the one hand, can appear as built-in systems which are not necessarily recognisable by the human operator as such, because they are working on a self-contained basis, once the work process is initiated. Their impact on the work system output can be a direct or an indirect one. The ABS in our cars, for example, which was mentioned above, directly impinges on both the work product as the way the car is moving as well as the work state as the state of progress of the transportation task. An automatic display function, however, changes only the internal work state without any immediate effect on the work system output. The authority level of these systems may range from low to full authority, depending on the reliability and the certainty that it cannot violate the work objective by any means.

On the other hand, there are sub-systems of automation at the disposal of the operating force which need to be activated each time when required, like the automatic speed control as mentioned above, too. These sub-systems often work with full authority, when activated, but they might be overdriven or turned off by the operating force at any time. Apparently, it is up to the operating force to decide whether to allocate a task or sub-task execution at this sub-system. Depending on the functions these sub-systems represent, their impact on the work process output can also be both, a direct or an indirect one. [Sheridan, 2002] is offering scales of *levels of automation* (LoA) in different dimensions, like for

- the degree of specifity required of the human for inputting requests to the machine,
- the degree of specifity with which the machine communicates decision alternatives or action recommendations to the human,
- the degree to which the human has responsibility for initiating implementation of action, and
- the timing and detail of feedback to the human after machine action is taken.

These days, operational systems go as far as incorporating robots as part of the operation-supporting means which exhibit some kind of artificial cognition. Automation has probably become the most emphasised area in system design during the past decades. The effects of automation, as they were experienced so far, were satisfying to a great extent, but there were severe disappointments, too, as will be discussed in more detail in Chapter 4.2.

We consider operation-supporting means exclusively as artefacts. This does not exclude external manpower supporting the work process. In that sense we rather deal with supporting manpower as separate supporting work systems instead of embedded ones as part of the operation-supporting means. This does not make any functional difference but it provides clear-cut entities with respect to autonomy allocation. System engineers need to break down large-scale systems into clear-cut entities in order to keep them manageable in the design process as well as during operation. The autonomy of the operating force should not be contaminated by any other autonomous party as part of the operation-supporting means. With the increase of automation there are many operation-supporting means, which originally were work systems having been replaced by an artificial sub-system. One can say in general that a work system, originally being operated by a human operator, looses its status of being a work system and converts into an operation-supporting means, once its objective will be accomplished by an artefact without any human operator involved. The bounds and the performance of the operation-supporting means of a work system are crucial design characteristics. They depend on the layout of the work objective, the operating force, and the other interacting work systems and the technology used. The work process performance crucially depends on the interplay of the operating force and the operation-supporting means and the characteristics of the operation-supporting means.

3.1.3 Systems of Work Systems

The charm of the concept of the work system lies in the fact that a complex work environment can be broken down into convenient partitions, handy enough for investigations. This partition can be easily and unambiguously described and delineated by corresponding work objectives and pertinent work systems and work objects. This is the platform, from where it can be investigated how the performance of the respective work process can be enhanced, for instance by exploiting new potentials of technology.

Thus, the work environment usually comprises a system of interacting work systems. In Chapter 3.1.2 was already mentioned a configuration of two interacting work systems where the work system concerned is manipulating a work object as an active system which in turn is a work system by itself. The latter work system, here also considered as the work object, is for instance manipulated by the instruction of a certain work objective as generated by the other work system. All work systems correspond to respective work objectives in a one-to-one relationship. The whole of process knowledge embedded in the operating force of a work system decides about its delineation from interacting work systems in other levels. Although it is possible that the work objective of an upper-level work system is known, merging with that one is not reasonable, if there is too much lack of process knowledge about the upper level work process. There are different ways of interaction of work systems, depending on the inputs or outputs through which the interaction links are established. One category of systems of work systems is that one structured by a work *hierarchy*. This is typical for the work environment of organisations like a company. In a work hierarchy of

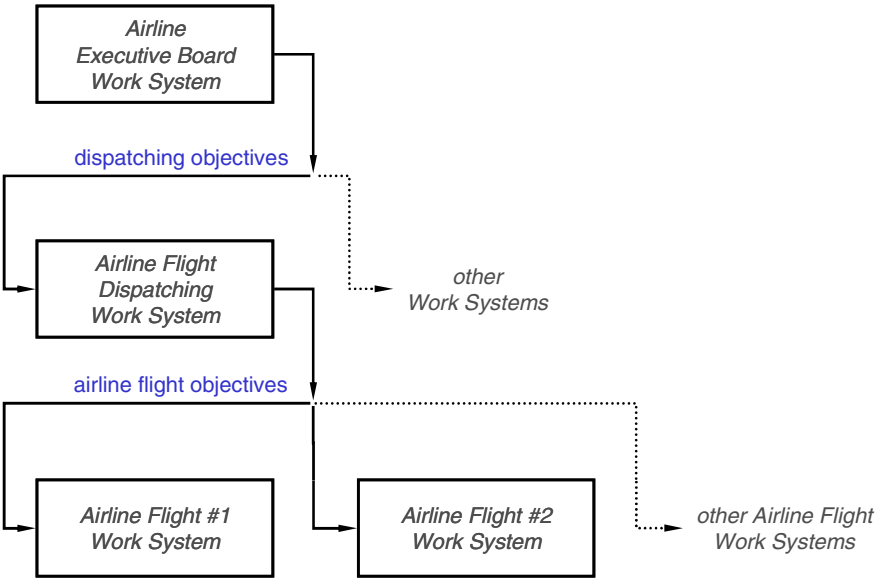


Fig. 14 Hierarchical decomposition into work systems in the airline flight operation work domain

work systems these systems are linked by the work objective forwarded as work system output on the higher hierarchy level and as input of the work system on the lower work hierarchy level. Figure 14 shows a simple example for a section of the hierarchical work structure of the airline flight operation. The work system, which is operated by the airline flight operation executive board, provides the objectives for the airline flight dispatching work system, and the work system operating the flight dispatching provides the flight objectives of the various airline flights, which are to be met by the corresponding work systems with the pilots as operating force.

The other possible category of systems of work systems is characterised by a *network* work structure. A network of work systems is characterised by links between them which are providing any inputs other than work objectives. Thus, there are supplying work systems and receiving ones. It may also happen that the supplier in this case is one of the work systems which were previously a receiver. Therefore, in general, these *work system networks* are omni-directional. Figure 15 shows a simple example for a work system network of the flight operation domain. From this figure it becomes obvious that work system networks include concurrent activity of participating work systems.

The work systems of the various airline flights receive air space clearance information from an ATC work system, and the ATC work system receives information about the state of the airline flights from the respective flight work systems. Take also notice of the fact that Figure 14 and Figure 15 can be

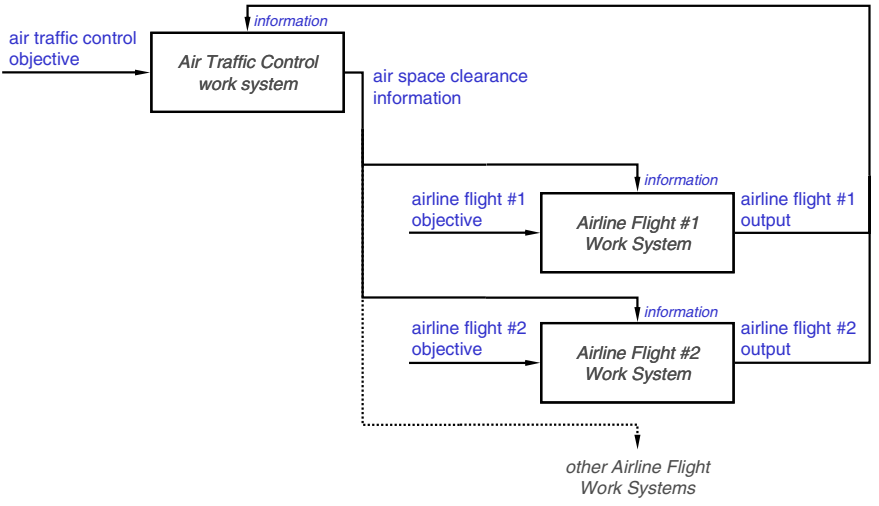


Fig. 15 Work system network between an ATC work system and work systems of individual flights

combined into a structure of work systems which contains both categories of systems of work systems the hierarchical work structure and the network structure.

3.1.4 Engineering Potentials and Challenges of Work System Development

Now we know how work systems are structured and how the work system components are interacting in the work process. Usually the work system design is not started from scratch. In most cases, there already exists a so-called system of work systems. Usually, the design objective is to achieve an enhancement of only one of the member work systems, which is considered to provide the best prospect of cost reduction in whatever sense, e.g. economic gain, safety enhancement, more convenience for the human operator(s), and so on.

There are certain given facts concerning the work system which are not a matter of change in the design process. One fact is that the operating force has to consist of at least one human operator. Usually, another given fact as the starting point for the development of a work system is the work objective. In fact, it is the common systems engineering practice to start a development from the operational user requirements which will be condensed in the work objective and the environmental conditions under which the work system is supposed to operate (see also [INCOSE, 2007]). Thus, the remaining potential items for design enhancements are the

- supply,
- operation-supporting means, and
- augmentation of the operating force.

Enhancements concerning the supply will not be investigated in this book. There is a continuous process of checking for advantageous changes in this work

process input, which usually is of no consequence for the internal work system design. Although currently energy-saving human behaviour for the minimisation of energy supply (and in turn the minimisation of contaminative emissions) is for good reasons a hot topic, this is not within the scope of this book.

Changes of the operation-supporting means are considered in about all cases of work system design we are interested in. As mentioned earlier, this is the main area of concern for engineers when a new design project for a work system enhancement is launched. So far, this was the work system component, and only this one, where new technology (automation, microcomputers, etc.) has been applied. Augmentations of the operating force were only considered regarding the necessary amount of manpower, the screening and selection of candidate operators, and the subsequent and continuous training of personnel. The more was done on the side of the operation-supporting means, the less or less skilled manpower was assumed to be needed. A typical example is the reduction of the cockpit crew for airline flights from four down to two during the past forty years. This caused great changes in the work processes through the years, though, in particular the way pilots share their job, and how they are making use of the new developments of operation-supporting means they find in their cockpits. Caused by a steady increase of novel operation-supporting means in our cars, we experience as drivers a similar tendency of work process changes, although we still kept our position as active driver, so far.

The engineering challenge in work system development is twofold:

- to develop, as we have become used to, more powerful operation-supporting means, and
- to find the right proportion about what tasks are left to the operating force in the work process and which functions are allocated to the operation-supporting means.

Not everything which is technically possible on the side of operation-supporting means must necessarily lead to an overall enhancement of the work system considered. Does automation always do a good job? And what, if not? What is needed for good interaction within the operating force and between the operating force and the operation-supporting means? Have human operators all automated functions among the operation-supporting means fully under control or can it be warranted by the designer that all automated functions are in full compliance with the work process objective, whatever it will be? Can it be warranted that there are no interferences between different tasks which are simultaneously operated by one and the same human operator? These are the crucial questions to be answered in the process of work system design.

We know from accident analysis how difficult it is to warrant a system design positively answering these questions, in particular for work systems of great complexity like those airline or military pilots have to deal with. One has to admit that this is the plain unpleasant reality the designer is confronted with today and that it is not satisfyingly solvable by the means we had at our disposal so far. This almost compulsorily leads to the demand to make use of a higher level potential of

automation technology which happens to become available these days: The artificial cognitive systems being part of the work system.

3.2 Conceptual Essentials about Human Cognition

As been elaborated in the previous chapter, the concept of the work system is based on the fact that the human operator is its central element. There is no work system without a human as main part of the operating force. The most obvious reason for this absolute necessity is the fact that all work activity is done in order to comply with certain human desires whatsoever. Undoubtedly, as part of all work processes there is a human operator who represents the highest operating authority.

The human's prominent role of being the highest authority in the work process would not be conceivable without his capabilities based on what we call *cognition*. Originally, the term cognition was used in disciplines which address the questions of how human brains work and how human behaviours could be explained, like physiology and psychology. Based on these disciplines, other interdisciplinary approaches evolved like neuroscience and cognitive psychology dealing with cognition as main research subject. Meanwhile, all this research work on human cognition tends to be subsumed collectively under a combined discipline called *cognitive science*. Cognitive science has made enormous progress in the course of the last two decades. It should not be overlooked here that there is also other use of the notion of cognition in recent literature like in the context of *distributed cognition* ([Hutchins, 1995] [Hollan, et al., 1995]) and *joint cognitive systems* [Hollnagel & Woods, 2005]. These publications take cognition in a broader sense. They emphasise that cognition is not solely located in a single human operator's mind when operating in a work process but that it is distributed over the work system components and environment, whether the pertinent objects are human-made artefacts or other humans involved. This view can even be extended by adding the organisational dimension of system design and realisation to become operative. Although this is a valid approach we will on purpose stay with the original way of interpretation, building up work systems with emphasis on the individual entities which incorporate cognition.

Today, the term and concept of cognition is no longer limited to natural cognition only, like human cognition. Along with the tremendous ongoing technological advances in computer science, concerning computation power and software design, artificial cognition increasingly comes into focus. This leads to the question how the kinds of functions of human cognition can also apply for artificial cognition. It turns out that, in principal, these kinds of functions are valid for both natural and artificial cognition, however the realisation of these functions may be very different, and indeed are.

Both, natural and artificial cognition have their particular scientific community with different views on which aspects of cognition are to be studied and for which purpose. There is the world of psychology and neuroscience considering human cognition, and on the other hand, there are the disciplines like computer science and to some extent engineering dealing with artificial cognition. Both sides,

pursuing not exactly the same objectives, though, are making use of the other's research results with great benefit. Since cognitive scientists provide more and more quantifiable results about these characteristics, the technology-related community is making use of these findings. On the other hand, cognitive scientists follow up the achievements in information processing artefacts capable of representing certain cognitive characteristics from which they can learn exactly how these are working in comparison to human information processing, thereby possibly gaining new insights in human cognition.

Therefore, the contributions on the analysis and realisation of cognition in both the cognitive scientists' and the engineers' realm will be discussed in this book. Emphasis will be laid on what has to be learned from the cognitive scientists and what are the main lines of thought to actually derive a concept of artificial cognition.

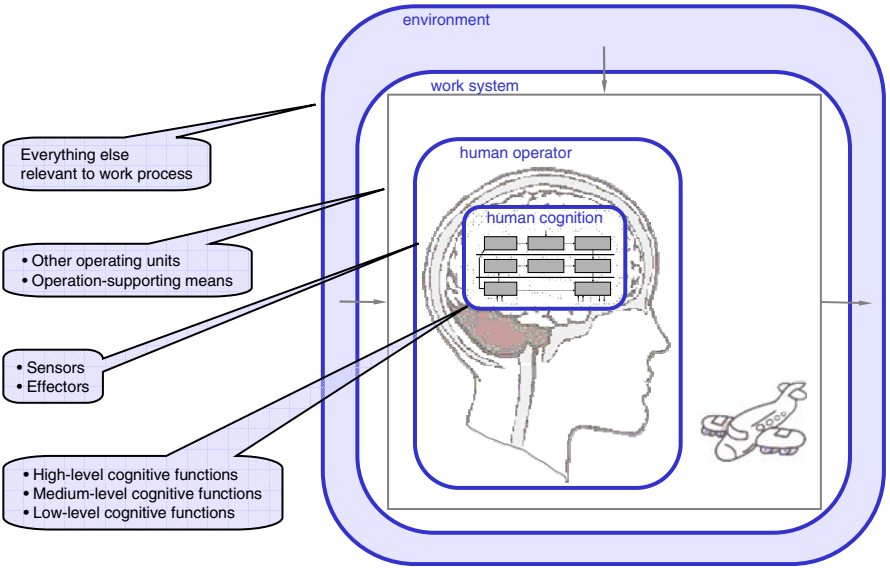


Fig. 16 Human cognition in the work system context

From the point of view of the work system designer *human cognition* is seen as a functional component, implemented on the tissue of the human brain and embedded into some kind of work environment. Figure 16 shows the environment in a stepwise inside-out arrangement. The *human operator* provides the most inner environmental layer of the brain, embodying this functional component and making use of it in the context of the work process. In the first place, this inner environmental layer provides sensors (receptors) and effectors without which no work at all can be done. It is not an easy task to separate between the sensing and effecting systems in the human body on the one hand and the human cognition on the other hand. Despite this fact, however, a separation seems adequate and very

useful for the purpose of the organisation of this chapter. The sensors are represented by the various human sensory modalities (visual, auditory, haptic, vestibular and olfactory) and their physiological properties. Effectors are represented by the muscle system of the human body (e.g. muscles to move arm or/and leg). This includes also the muscles for the articulation of spoken language and gestures. Sensing and effecting modalities are described in some more detail in Appendix 2. Remaining body systems of the human operator, like for instance those basic ones concerning circulation, respiration, and metabolism, are not in our focus here and shall not be further discussed in this book.

Both, the human sensing and effecting systems represent the interface to the world outside the human operator with two further environmental layers, first the *work system* layer and finally the *external environment* layer outside the work system. For the work system layer, all existing work system components are to be considered except the human operator. This layer is optional depending on which components are definitely involved in the work system (see also Chapter 3.1.2). In Figure 16, this is symbolised by the aircraft, in particular representing the diversity of possible operation-supporting means. The external environment represents everything beyond the work system bounds which is of relevance to the work process.

Since the work system has already been described in more depth, the focus of this chapter shall lie on some insight into the subject of human cognition as the most inner layer in Figure 16. Many readers may wonder at this point, in particular those who have to design technical equipment, why sound knowledge about human cognition can be of great benefit for the design of technical support systems as part of a work system. However, this insight is in fact of great significance for the work system designer, since the knowledge about human cognition lends itself to more systematic approach for enhancing the effectiveness of work systems. This is subsumed under the discipline of *human engineering*. Unfortunately, designers often have become used to live with (too) little feedback of the users and what could have been achieved, if one would rigorously apply what is known about human cognition and resulting needs for the effective human-machine interaction in the work process. They are used to intuitively imagine how the human operator will make effective use of what they are designing, not realising that it is about impossible to comprehensively match with the real needs. From our experience we know that taking some effort of studying the findings about human cognition will pay off with great yield. Along these lines, knowledge about human cognition can be of great advantage for the work system designer in at least the following three ways:

1. The **design of artificial components supporting the human operator**, in particular artificial cognition, can be specified **on a more rational basis**, because it is better known where the human operator really needs support, when artificial cognition makes sense, and for which function. If we want to come to a decision on how the function allocation between the human operator and artificial cognitive components shall be designed, knowledge about human cognition is crucial.

2. **In order to make sure that the artificial cognitive functions are being in compliance with the actual needs of the human operator, the embodiment of knowledge about human cognitive behaviour into artificial systems as part of work system is very useful. Only then effective co-operation on a cognitive level between artificial systems (automation) and the human operator can be made a design characteristic.** This can be achieved by quantitative models of human behaviour like those we have in mind ourselves when we are supporting any co-workers in our daily life or are co-operating with somebody.
3. **The use of the knowledge about the capabilities and the architecture of human cognition as a reference is a prudent way to achieve better designs**. This does not mean that a designer should feel compelled to copy the design of the human brain. He rather can make use of his knowledge to benefit from assessments about on which functional basis the outstanding capabilities of human cognition are founded. This might lead to the conclusion to simply take over design features, if it makes sense, or to be stimulated by the unique realisations in the brain in both physical and functional structuring, like the mechanisms of learning, attention control and consciousness and to try for similar effects features in the artificial system. Moreover, the use of knowledge about human cognition might lead the designer to solutions where artificial cognition may have a chance to do better.

In summary, it is the human cognitive system the designer indeed can learn from, keeping in mind, though, that there are certain basic features which cannot be copied easily and that there are certain new options different from human cognition when starting to design an artificial cognitive system.

Consequently, this chapter is structured in the following way: First, we highlight key principles concerning the implementation of human cognition, before we establish a framework of functional and behavioural aspects of cognition which may serve as a framework for the transition from human cognition to artificial cognition. Further useful findings about peculiarities in human factors, like certain ones of human sensing and sensori-motor modalities, perception etc. are delt with in Appendix 2.

3.2.1 Implementation Principles of Human Cognition

It was the process of evolution of mankind which has determined what we can consider as the current design plan of the human species. Undoubtedly, the evolution has generated an outstanding excellence in cognitive performance, thereby adapting to the environment encountered. This is unsurpassed so far by anything we can think of. Nevertheless, there is a certain sub-optimality concerning our performance in modern times. The slow pace of natural evolution lets us assume that the human race of today mirrors the predominating needs of survival at times about hundreds of thousand years ago, but not those of today. This evolutionary adaptation process has not been able to catch up to all the demands

humans are facing in modern times with rapidly changing work environments corresponding to the steadily advancing technical equipment. As a simple example, deliberately not considering latest technical developments, the vehicles humans are operating generate speed levels which are by far no more comparable with the running speed the human can accomplish without any supporting technical means. An equivalent decrease of human reaction time necessary to master sudden encounters of threats has not evolved simultaneously. Thus, there is a price to pay for the fact that the environment of human operators has changed, mostly by humans themselves. Therefore, the work system designer has to account for both the excellences of human cognition and the weaknesses as they exist here and now. Thus, let us move to more details about the features which are of interest for the work system designer.

We know that the topic of human cognition is at least worth a book for itself, like those which are available (see for instance [Roth & Prinz, 1996], [Birbaumer & Schmidt, 2003], and [Pinel, 2007]). Nevertheless, it makes sense to introduce this topic here by selectively highlighting what is of fundamental interest for the designer of artificial cognitive components in work systems. That is the discussion on

- connectionistic information processing,
- division of cognitive functions in distinctive network structures,
- the two modes of information processing,
- principles of human memory, and
- the ventral loop.

In Appendix 1.1 the main brain structures are described in some depth from the anatomical and functional point of view for those who want some more biopsychological background for what will be outlined in the following.

3.2.1.1 Connectionistic Information Processing

Since biological systems are principally based on chemo-physical processes, the solution for the implementation of cognition in the human brain presumably cannot be but a connectionistic one. The functional elementary building blocks for cognition are nerve cells (neurons) in a highest possible packing density. The neocortex with its two hemispheres, for instance, the largest of the main subdivisions of the brain (90% of the cerebral cortex, also called isocortex because of its more or less continuously layered networking structure of neurons) comprises about 10^{10} neurons [Schüz, 2001] [Roth & Wullimann, 2001]. In anatomical terms these are packed in a flat sheet with a thickness of about 3 millimeters. It is considerably convoluted in order to fit into the skull, including all neural interconnections. The main furrows (fissures) of this folded sheet spreading out over an area of about 0.2 squaremeters give a good indication about cortex divisions, resulting in four main lobes of each of the two hemispheres: frontal lobe, parietal lobe, temporal lobe and occipital lobe (see Appendix 1.1).

The neurons are interconnected to a very high degree (about ten thousand interconnections per neuron) and mainly organised in columnar arrays. There is the principal property that these structures are not invariant in the course of the human individual's life. The brain adapts to changing demands by increasing or decreasing the number of neurons and/or interconnections to a certain degree. The interconnections take care of the transmission of the output signals of neurons to pertinent destination neurons which in turn are processing the incoming signals and are producing their own output which may be widely spread out. It is noteworthy at this point that despite the enormous performance we demand from this neural supercomputer the energy consumption is extremely low, only at about 15 Watts. There are certain mechanisms located at the connection points for incoming signals into the nerve cell, the so-called synapses, which determine the contribution of these signals to the output of the neuron. The synaptical parameter setting represents the status of embodied knowledge available for a human individual. This knowledge is the main determinant of human behaviour. The synaptical setting is modifiable through the process of learning.

The connectionisitic implementation lends itself to integrate the ability to learn. Our cognitive performance would be rather miserable, if we could not rely on our learning capabilities. This is fundamentally different to artificial cognition. Knowledge cannot easily be "implanted" into the human brain, at least not at the time being. Therefore, we are permanently learning in our life. We never arrive at a stationary state of accumulated knowledge. We can push the learning process by conscious training, which usually is a very tedious process, otherwise we learn by in-built automatic learning mechanisms which rely on a great number of repetitons of presentations in the course of normal operations. Learning not only relies on changes of parameter setting of the synapses by these mechanisms, but also on a certain amount of plasticity of the neural structures by adding and closing down interconnections. In addition, changing demands might also lead to increase or decrease of the number of neurons. This accounts for the so-called plasticity versus stability dilemma by increased limitation of the plasticity with the individual's age. Some more details on human learning mechanisms are made in the context of memory management in Chapter 3.2.1.3.

Furthermore, the connectionisitic implementation by virtue takes care of semantic coding of embodied knowledge. This principle of knowledge representation is highly effective in avoiding any ontology problems. Further comments on this issue can be found in Chapter 3.2.1.3 and Chapter 3.2.1.6.

3.2.1.2 Structural Division of Cognitive Functions into Distinctive Brain Areas

The interconnection of the nerve cells is not arbitrary. There is irresistible evidence from about all known studies in biopsychology on subjects, whether through non-invasive visualising methods of investigation like positron emission tomography (PET) and functional magnetic resonance tomography (fMRT) or through findings about behavioural deficits as a result from surgical operations and trials with animals, that cognition is established by the interplay of distinctive

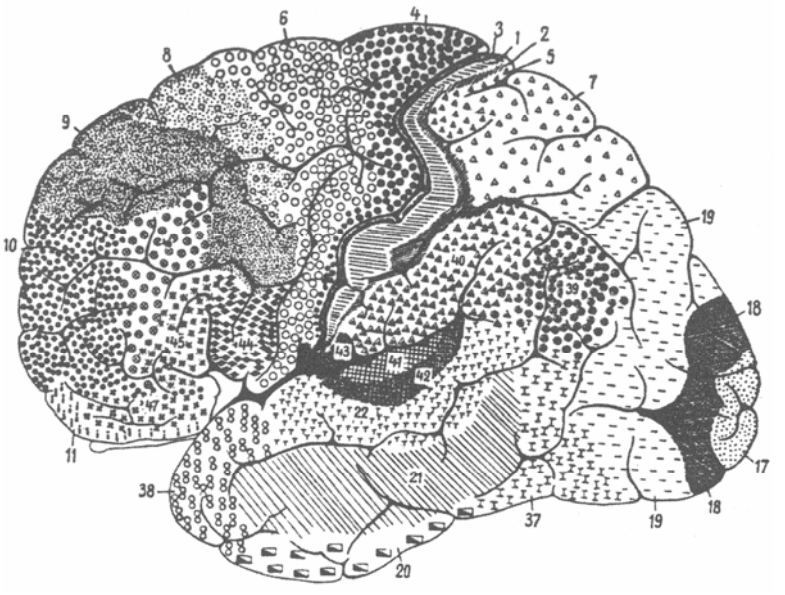


Fig. 17 Structure of neocortex and functional localisation based on differences of neural formations. ([Brodmann, 1909])

neural network structures which, depending on the functions considered, can be associated to certain brain regions:

> *"Thus, a physiological map of the different brain functions including voluntary action appears feasible. Based on the results presented, brain functions can be conceived as being organised in systems of concertedly working areas, each of which performs its role in functional subsystems." [Seitz, 2003]*

The well-known picture from [Brodmann, 1909], as shown in Figure 17, depicting the distribution of neurons across the neocortex in differing configurations, already gives an indication, although a very coarse one, on the fact that the different regions of the neocortex are associated to different cognitive functions. We can distinguish between functional modules for perceptual processing, functional subsystems for the generation of voluntary actions, and sensori-motor processing. As an example we will briefly give some insight into the function of perceptual processing of the visual modality in the following.

A significant part of the necortex is devoted to visual perceptual processing. Once the visual receptor information enters neural information processing, various functionally specialised areas in the brain are involved at the different levels of the hierarchical functional structure of perceptual processing. Each of these specialised functional assemblies at the different hierarchical levels plays a certain role. It appears that the human perceptual processing is mainly organised as a one-way pathway for sensory inputs from the receptors to the association cortex via

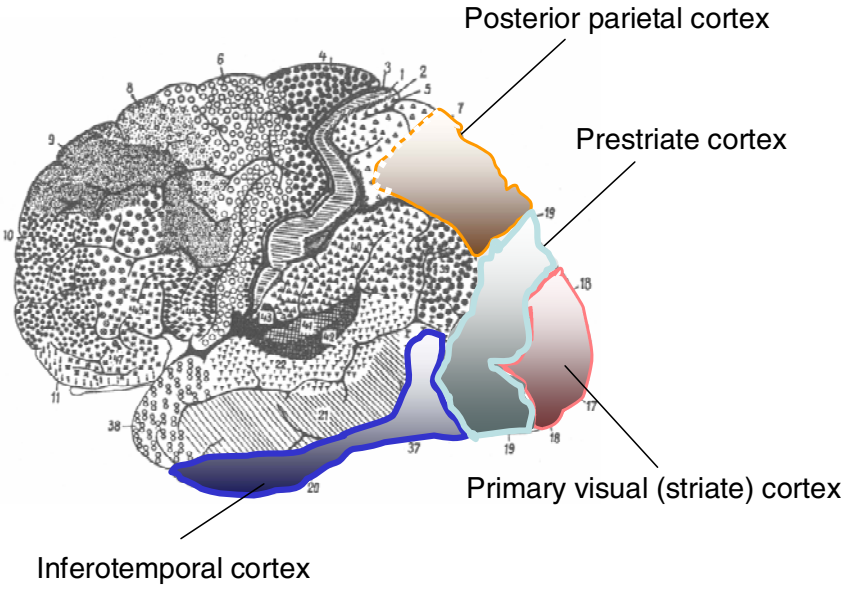


Fig. 18 Main areas of human visual processing in the cerebral cortex (roughly indicated in colour up to those of association cortex)

the thalamus, the primary sensory cortex, and the secondary sensory cortex. Descending interconnections are much less in numbers, although important, too. Also the interconnections between functional assemblies within each hierarchy level of processing are of great importance for the process of perception.

It is the posterior region of the occipital lobe where the primary visual cortex is located (see Figure 18). The secondary visual cortex lies in the adjacent areas of the inferior temporal lobes and the anterior region of the occipital lobe surrounding the primary visual cortex, the so-called prestriate cortex. The main stream of visual information goes along the pathway from the primary visual cortex to the areas of the secondary visual cortex, and from there to those parts of the association cortex which receive visual inputs. The largest single area of of association cortex regarding visual inputs is located in the posterior parietal cortex (see Figure 18). It signifies the so-called dorsal visual stream which stands for the perception of *where* the perceived object is located in the environment. The association cortex as part of the inferotemporal cortex is also known to receive visual input. It signifies the so-called ventral visual stream which stands for the perception of *what* object is perceived.

It is the activation of these areas when the human is having conscious experiences about what he/she sees, and it is on these pathways where the combined activity of different interconnected cortical areas, the so-called *binding* takes place. Because of the parallel processing of stimuli there have to be certain mechanisms in order to selectively combine responses on elementary features into neural activation patterns at higher processing levels which represent feature

formations, e.g. objects. The synaptical setting (unconscious processing knowledge) along these pathways determines what is recognised as objects and what is not. The number of features to account for has to be sufficient to avoid confusion. The features will be increasingly fine-grained the more knowledge is accumulated about possible objects to be recognised. There is experimental evidence that neurons not only report on the elementary feature they encode but that they also indicate with which other neurons they are currently related for the purpose of representing the object the feature belongs to (see [Bickhard & Terveen, 1995] and [Singer 2003]. These findings suggest that the participating neurons are temporally synchronised in concert with mechanisms like

- inhibiting and excluding unrelated responses from further processing and
- enhancing of discharge frequency of the seleted responses.

As pointed out by Singer,

> *"synchronization can be used: to define, with high temporal precision and flexibility, the relationships between distributed processes and to bind them together for further joint processing, to select responses for further processing, to support the selective routing from sender to receiver within distributed networks, to bind responses from different sensory subsystems into coherent reprsentations, to establish connections between sensory and executive structures, to maintain contents in working memory* (see also Chapter 3.2.1.3)*, to strengthen associations between between synchronously active cell assemblies by synaptic plasticity, and to support the formation of activity patterns that have access to conscious processing."[Singer, 2007]*

These mechanisms, which are crucial for the perception process as well as others of the total process of cognition, are amazingly efficient and by far better than any technical device we know of so far. They matter not only for the perception of tangible objects, but also for representations of abstract entities like for instance feelings and intended actions. If looking at the neural processing based on binding in more abstract terms, one can state that

> *"every such representation corresponds to one of a vast number of possible states, or, to put it differently, the cerebral cortex system continusly moves from one point to the next in an inconceivable multidimensional space … During its procession through this multidimensional space, the system continuously changes because its functional architecture is constantly altered by the experience it gains along the way. Therefore, it can never return to the same location … The second time we see a certain object, … the new state also reflects the fact that we have seen it before." [Singer, 2007]*

This is done in a way exhibiting an amazing combination of tremendous generalisation capability and equally high perception reliability. This makes the



Fig. 19 Different appearances of the capital letter T as semantically identical objects (source unknown)

human perceptual processing a highly exceptional one. It can reliably spell out when two identical objects are recognised, or what the dissimilarities are if there are two objects which are not identical. In addition, it can identify, if objects are semantically identical, although being of quite different appearance (see Figure 19). Not only neural structures of the sensory cortex in different modalities might be involved but also other brain structures and corresponding functions, including the attention control (see also Chapter 3.2.1.4). Certainly, the process will be somewhat different for recognising known objects, at least when thinking about what we consciously experience as individual features of the object, as opposed to the perception of unknown objects.

One of the fundamental outcomes of the binding process is the way in which we automatically seggregate the viewed scenery into the so-called figure (one or more objects of predominant concern) as a result of the binding process which we try to take into central focus of fovea vision, and everything else, the background. In that respect, the extremely small central area on the retina of high-resolution vision in combination with the ability of fast eye movement control helps a lot to keep the data stream in a manageable size. The figure has been called the *Gestalt* by the community of researchers who worked on this phenomenon. They proposed a number of simple laws of perceptual organisation (e.g. grouping together visual elements with common color, motion, or alignment which are illustrated in Figure 20 by some simple examples from [Engel, 1996] and in Figure 21 a figure-background example from [Thurston & Carraher, 1966].

Although there is also a lot of criticism about the Gestalt laws they help a lot to make good use in the design process of display presentations in work systems. This research also stimulated the finding that complex stimuli are normally perceived as integrated wholes, not as combinations of independent attributes. That means that the overall Gestalt may be perceived before the parts of the object stimulus which make up the whole or that the whole is identified consciously before the individual features are. There has not been found an area in the cortex, though, which receives all partial sensory information in order to integrate it to the complete figure what refers to the binding problem as mentioned earlier. The neuropsychological explanation for that phenomenon, as it has been evidenced so far to some degree, is that various neural areas are working in parallel at different locations in the cortex on the perception of an object. They separately analyse the features of the object, thereby forming a so-called assembly for that object which is bound by a certain common activation pattern of all neurons involved. Thus, it

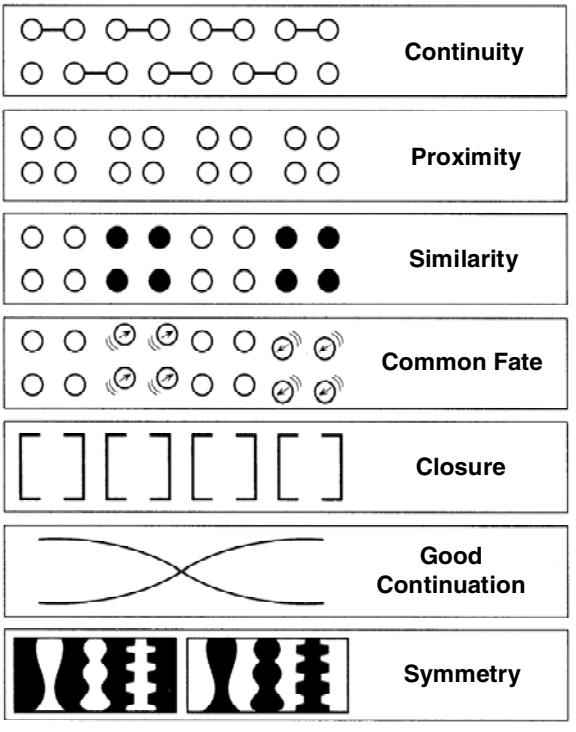


Fig. 20 Perceptual organisation by Gestalt criteria. [Engel, 1996]

is this assembly which represents the object considered. This is a great concept, exploiting the given multitude of neural interconnections in a clever way in order to be able to be responsive to the tremendous variety of objects encountered in real life and to warrant efficient processing. Artificial cognition might make use of this concept in future developments. This conception is not confined to the visual system. Also the other sensory systems as well as object perception based on parallel sensing through more than one modality show similar phenomena.

Finally, referring to the thalamus (lateral geniculate nuclei), it represents a kind of relay for all information on its way into the cortex. Obviously, many other brain structures project into the thalamus through more than 80% of its afferences. Presumably, thereby the information is selectively modulated before it enters the cortex. The thalamus is also involved in the process of attention control, and it decides which of the information coming from sensing organs and other parts of the body is important enough to be forwarded to respective cortical areas. This kind of function of the thalamus, by the way, is true for all senses, except the olfactory one.

In many work systems human perception is confined to a great extent to information gathering from displays. This information is provided by technical sensing devices and communication links which take care of all information



Fig. 21 Demonstration of amazing human visual performance with application of Gestalt laws (separation of figure and background). This picture is also a demonstration of how different cortical areas are working in parallel on the features to produce the perception result. We are sure the reader is able to recognise the snuffling dog. (From [Thurston & Carraher, 1966])

needed about the environment. Artificial cognitive systems as opposed to humans might even manage with no sensing at all, if there are appropriate communication links to other agents which have got the sensing capability needed. Nevertheless, it is unquestionable that future artificial cognitive systems in work systems will most likely not do without sensing. Great progress has already been made in the field of artificial perception, also making use of the findings about human perception. Just to mention one invention of nature regarding visual perception, for instance, which must not be missing in work systems for computational efficiency reasons, that is dynamic camera alignment control in analogy to human eye movements [Dickmanns, 2007].

3.2.1.3 Principles of Human Memory

The human memory is the storage for all types of knowledge which is made use of by a person working as operating force in a work system. About hundred years ago more serious research has begun on the human memory functionality (see [Ebbinghaus, 1885] and [James, 1890]). It is from thereon that a dualistic view on the memory phenomena has been pursued. It has been found that the memorising performance depends heavily on the recency of the information presented or how often it has been repeatedly experienced. The more refreshment is happening by

repetitions of deliberately induced or unintended activations on the same neural traces the more permanent memory is warranted.

It seems to be commonsense that there are three categories of memory in the human brain with evident differences in size and retention capacity. These are the sensory, short-term, and long-term memories.

We start with that one of the three memory categories, the *long-term memory*, which contains all permanently available knowledge what is learned by the human operator up to the time being. This knowledge determines the human operator's behaviour in a given situation in the course of a work process. We can distinguish between two types of long-term memories, the so-called *explicit* (*declarative*) ones the contents of which can by virtue become conscious, and the so-called *implicit* ones which are involved in automatic procedures and which are only becoming globally conscious by way of obvious impingement on something we can observe consciously. We identify our implicit knowledge, if we cannot accurately explain our behaviour in explicit terms. For instance, I can only give a rather global statement that I lift my arm in order to put some food into my mouth. I cannot provide much more detail on the precise kinematics and kinetics of the arm motion or how much force by which muscles is involved in all phases of that action. Implicit knowledge by way of automatic procedures is an important feature, though, for the sake of processing efficiency. The implicit memories are oftentimes also called *procedural*. This is in particular the case in the context of the motor systems, i.e. for instance the memories for sensori-motor patterns in the cerebellum. Both the explicit and the implicit memories impress with an almost unlimited memory capacity never exhausting during life time and no significant decay rate.

The contents of the explicit memories are about facts (semantic memory) and autobiographical episodic events and experiences (episodic memory) [Tulving, 1993]. The storage into the episodic memory is achieved almost effortless as opposed to the storage into the semantic memory of facts like concepts, rules or abstract ideas [Solso, 2001]. On the other hand, the reliability of retention in the episodic memory is relatively poor compared to that of the semantic one. It can be stated, though, that both of these memory systems are heavily interacting. A model of the semantically coded representation can be thought of as a high-dimensional semantic space. This might explain that the explicit memories will never be exhausted. As long as explicit knowledge is not verbally represented, individuals often fail, too, to exactly describe anything in verbal terms what is semantically represented. This is usually due to the imperfect linguistic capabilty of accurately expressing semantic contents, to exactly describe anything in explicit verbal terms what is semantically represented in their brain as explicit knowledge. The representation of explicit knowledge might be in various forms when passing through the sequence of memorising stages, depending on the kind of stimuli. No matter whether the knowledge representation might be visually or acoustically verbalised, provided through images or through feelings like pain, for instance, there is always represented an explicit meaning. Thus, human cognition based on the long-term memories works on a semantic knowledge representation. Therefore, reading a word which can have two different meanings (for example the adjective

"light", which can mean the opposite of both dark and heavy), will only lead to ambiguity, if no additional contextual information is available what only very rarely might be the case.

Consequence for approaches of artificial cognition

To know the way knowledge is coded and represented in the brain is of great importance for the designer of work systems. This is so important, because both the human operator and an artificial cognitive system in a work system should be able to understand each other on the cognitive level. Basically, the knowledge representation in the human brain is semantically coded. The neuronal processing leads to an activation pattern which stands for a meaning, not necessarily for a symbolic expression. Therefore, there usually is no difficulty for the human operator to understand the artificial system, no matter how the knowledge in this system is represented. Although there might be ambuigity of vocabulary, this will not lead to misunderstandings on the side of the human operator. The semantics of the vocabulary used can be considered as known by the human operator. Therefore, we strongly appreciate that the human representation of explicit knowledge is semantic. This is a feature of the human mind which can be considered as a spin-off of the biological design principle of connectionism. It makes human cognition a very reliable one and avoids errors when ambuigious symbolic expressions are to be processed. It is probably one of the great strengths of humans when operating a work system. It should be noted, though, that the price paid for that strength is the relative weakness in human computing power.

Thinking of an artificial cognitive system with symbolic knowledge representation only which is usually the case these days, this system does by far not understand the incoming information the same way as the human operator does. There might come up serious misunderstandings and errors. This is because it cannot be inserted all by way of symbolic knowledge representation what the designer has semantically in mind as real world entities when he is coding the artificial cognitive system. As a typical example, when a designer has in mind a "car", its representation takes into account innumerable features and relations which go with this entity as semantic determinants, thereby making it unambigously distinct from other similar concepts. Although the sematically coded representation of "car" is also not exactly the same among humans, there usually is still sufficient overlap to avoid a misunderstanding. In that sense, it becomes obvious that symbolic representation alone in artificial systems has its disadvantages and that semantic coding in artificial cognitive systems is a design feature to strive for. It seems that semantic knowledge representation is one of the human strengths being not yet sufficiently appreciated by the designers of artificial cognitive systems. **This characteristic of human explicit memory also lends itself to have co-ordinated knowledge representations of all kinds of application domains which are encountered by us during life. We suspect that, besides the capability of information binding as described in Chapter 3.2.1.2 for the human perceptual processing, this might be one of the main reasons that there is still more confidence in humans than in cognitive machines to do the main job as part of the operating force in work systems.**

There still remains the question about the localisation of explicit knowledge in the cortex. Although the hippocampus in concert with other adjacent neural formations (entorhinal, perirhinal, and parahippocampal cortex (EPPC)) is responsible of the process of storing explicit knowledge, it is not the place where the knowledge is located. It is assumed that it is stored in the associative areas of the neocortex which are pertinent to the modality, quality, and function at question [Menzel & Roth, 1996]. Thus, the memory of visually experienced objects, for instance, is supposed to be located in the associatve areas of the visual processing system and so on.

The process of recalling an item of the explicit memory is surprisingly efficient. This is an important fact regarding the human role in a work process. Usually, the access to the memory is very fast, in particular in the case of recognition (as opposed to reproduction). The so-called retrieval cues certainly play a significant role in this process. Therefore, the context out of which the memory access is activated is of great influence on the resulting retrieval. Here, it becomes evident that the retrieval process heavily depends on the encoding in the learning process.

The contents of the long-term memory are surprisingly stable once they are stored. Continuously, new pieces of knowledge are integrated through the process of learning, but there are almost no losses due to interferences in the course of the learning process. The corresponding neural activation patterns might change, but the semantic code they hold remains the same. In summary, the human long-term memory as a whole is in average operating as a fast-read, slow-write system [Card et al., 1983].

While the long-term memory contains the knowledge, which has already been stored in the past, the sensory memory and the *short-term memory* store new incoming information. The *sensory memory* is a kind of transient memory for the overwhelming amount of external stimuli with a high decay rate. It exhibits a retention performance of just a few seconds of stimulus-specific activation, then giving room for new incoming information. Within this time span associations between external stimuli or internal projections are taking place, if attention is directed on them. Otherwise the information is lost.

The short-term memory has a much longer decay time than the sensory memory, enough to serve as the transient memory needed for the process of learning. This process consolidates the contents of the short-term memory as a part of the long-term memory. All pieces of information which are attended to and adopted by the short-term memory must temporarily have become a part of the working memory.

The storage capacity of the short-term memory is very limited with seven (plus/minus two) so-called chunks. A chunk constitutes a meaningful unit of information which can correspond to any kind of concepts no matter how many bits and bytes it would consume in an artificial memory like that we are used to in a computer. If one would try to desribe a chunk in terms of explicit features and sub-features (attributes), one could possibly succeed in certain cases by use of a fixed number of them, possibly a very small one (or just a single feature). However, this is soly true for only a part of the explicit knowledge which accounts for chunks as objects (like for instance well defined geometrical figures) which are perfectly defined in semantics by symbolic coding. In other cases, though, the

number of explicit features and sub-features would shoot up to become countless, if a precise description is intended. These chunks usually consist of implicit features and therefore are implicitly representing explicit knowledge entities.

The decay time of the short-term memory depends heavily on the contents and on the occurrence of disturbing events. It can be less than 10 seconds in case of rather unfavourable circumstances, but it can also last for very much longer under certain accommodating circumstances. More recent studies come to conclusions that there is not one unitary short-term memory but rather several ones. They may be located in different areas of the cortex depending on the contents whether they are modality-specific or associated to the central executive function of voluntary action, for instance. It turns out as a consequence that the capacity limitation is not always quite as strict as it was outlined before.

Anyway, the capacity limitation of the short-term memory is one of the greatest drawbacks of human cognition. This is one of the top issues to be accounted for in the work system design. Concerning the memory drawback the introduction of external memories in terms of displays can help a lot (adaptive display presentations). Therefore, the layout of displays for the human operator has been a major realm in work system engineering regarding operation-supporting means (see [Wickens, 1992]). As an example, the techniques of *virtual reality* can be of great effect by presenting the 3D environment which often cannot be looked at in completeness by the human operator (see [Stanney, 2002]).

The capacity drawback is impressively illustrated by [Newell & Simon, 1972] with the simple brain-twister of considering the following cryptarithmetic problem:

```
   DONALD
+  GERALD
   ROBERT
```

Even, if the letter D is given as digit 5 at the start of the experiment, subjects usually will not succeed to solve this problem, if they are not allowed to make use of any external memory as, for instance, by use of paper and pencil. By the way, the solution of this little problem is: T = 0; G = 1; O = 2; B = 3; A = 4; D = 5; N = 6; R = 7; L = 8, and E = 9.

We have just learned that display design for external memory support is a valuable recommendation for the work system engineer to cope with the deficiencies of the short-term memory of the human operator. The way the long-term memory is organised, though, might be even more exciting, in particular the semantic coding as was already alluded to.

The learning process of explicit knowledge, which consolidates contents of the short-term memory as part of the long-term memory, modifies, simply speaking, the parameter setting at the neural synapses. In [Schüz, 2001] we can read the following:

> *"A mechanism for the storage of correlations was proposed by Hebb (1949) in his theory of cell assemblies, a concept that was taken up by Legendy (1971) and further elaborated by Braitenberg (1978). According to this theory, learning is achieved by strengthening the connections between those neurons that are often*

> *active together. When a child is exposed to a new object, the various properties of the object, such as its particular colour, shape, consistency, and so forth, will be projected onto the cortex by the sensory input systems and will excite a certain set of neurons there. This set will be more or less the same, whenever this or a similar object reappears. If there happen to be connections between members of this set, these connections might be strengthened whenever the neurons fire together. After a while, the connections will have become strong enough to ignite the whole set when some of its members are activated from outside. By then, the new object will have a representation in the brain, and the concept of it can be evoked relatively independently from the input. Such "cell assemblies" might constitute the units of cognition." [Schüz, 2001]*

Learning always includes what has been experienced before in similar situations, also taking into account whether strong feelings like satisfaction, delight, or fear went along with these experiences [Menzel & Roth, 1996] as an involvement of neural structures like the amygdala and the mesolimbic system. For instance, regarding an accident which has been survived, there most likely takes place immediate permanent storing in the episodic memory by way of the hippocampus and implicit storing of certain aspects in the amygdala.

We can conclude that the way knowledge is built up and consolidated is presumably different for different categories of memories. Thus, the limbic system plays a major role in the process of storing knowledge, in particular by the interaction of amygdala and hippocampus with other cortical areas. The hippocampus is deeply involved in the so-called long-term-potentiation (LTP) which leads to a memory consolidation. We can imagine the following about learning and memory. There are certain functionally relevant neural formations which determine what is new or already known, important or not important about the current perceptions. If there is a situation which is considered as new and important, then certain neural formations pertinent to the category of knowledge concerned are being made ready for a reorganization of the synaptic connections.

Learning usually takes time. This is true for both the explicit and the implicit knowledge. It does not help, too, that short-term memory rehearsal is compulsorily connected with the process of sequential conscious experience. [Card et al., 1983] stated:

> *"This asymmetry (of the long-term memory being a fast-read, slow write system) puts great importance on the limited capacity of the short-term memory, since it is not possible in tasks of short duration to transfer very much knowledge to long-term memory as a working convenience." [Card et al., 1983]*

Learning of implicit knowledge is different from learning of explicit knowledge. Patients with lesions of the hippocampus on both sides have no noticeable deficiencies in learning senso-motoric skills, for instance. Implicit learning happens straightforward through repeated experience, hence with involvement of attention and consciousness. PET-studies of practising senso-motoric tasks show that

activation is increased in typical motoric brain components like the motor cortex, thalamus, basal ganglia, and the cerebellum [Grafton et al., 1992]. Implicit learning of humans has great similarities to the training process of artificial neural nets. Therefore, for the sake of work system design, modelling of human implicit learning by approaches using artificial neural net techniques has been already quite successful.

Although learning is not an obligatory capability for artificial cognition, it provides great potentials for performance enhancements, also with regard to the modelling of the human operator's behaviour as operating force in the work system as exemplified in Chapters 6.1.2 and 6.2.1. Most of the techniques applied for knowledge acquisition are trying to mimick human learning behaviour, although not all facets of human learning are known so far.

3.2.1.4 The Two Modes of Information Processing

Human cognition works on the basis of two modes of information processing. By far the most part of cognition works on the basis of the mode which is characterised by mainly parallel and therefore swift processing. This kind of mode is no surprise, because the connectionistic neural net implementation of human cognition is best suited for this mode. Theoretically, its capacity can hardly be exhausted, but the processing lacks flexibility because of fixed, stored processing patterns (automatisms) and is therefore unqualified for the purpose of creative reasoning. Moreover, it is hardly possible to accurately describe any process of this kind in verbal terms, because the main part of it works unconsciously. Have you thought about what are we really conscious of about our steering wheel actions for lane keeping when driving on the freeway? Automatic activation of this mode can be considered as the standard procedure, although it can be activated consciously, too.

The other mode can be characterised by conscious, mainly serial, and therefore slow processing of explicit entities (symbols). This processing mode we would not expect at the first glance in case of a neural net implementation. Therefore, we will dwell on it in some more detail in the following. This process is limited in capacity, rather tedious, but well-suited for reasoning purposes. It is in the loop, if we are facing complex problems with no solutions at hand. Then, a great deal of data from different memory contents have usually to be put in context in order to consciously figure out how to proceed. It also provides the crucial capability which supports communication by means of languages and which we owe the spectacular progress of science mankind has achieved. For the processing in this mode, two enabling mechanisms play a central role:

- the attention control mechanism and
- the so-called working memory,

which will be discussed in the following.

Attention control

Attention control is an effective mechanism to concentrate the existing resource of cognitive processing on those few external or internal stimuli which are the

relevant ones in the light of the motivational contexts and which we therefore ought to look at (see [Posner, 1989], [Posner & DiGirolamo, 1998], [Pashler, 1998], and [Driver, 1998]). The attended matter can be a content item of the working memory or is just becoming one, originated by external stimuli. We can activate attention control to focus on a particular matter of voluntary choice, subject to ongoing intentions. However, it is also possible that attention is automatically drawn to external stimuli, subject to contexts like

- motivational contexts,
- conceptual contexts,
- cultural contexts, and
- perceptual contexts.

A mechanism like this is a necessity for the sake of efficiency, because we have only a limited capacity of information processing. [Anderson, 2000] concludes:

> *"The brain consists of a number of parallel-processing systems for the various perceptual systems, motor systems, and central cognition. Each of these parallel systems seems to suffer bottlenecks at which it must focus its processing on a single thing. Attention is best conceived as the process by which each of these systems is allocated to potentially competing information-processing demands. The amount of interference one gets among tasks is a function of the overlap in the demands that these tasks make on the same system." [Anderson, 2000]*

Attention control is one of the most complex mechanisms of the brain being still a matter of intense biopsychological research. It can be taken for granted in the meantime that top-down connections onto the neural pathways which are responsible for the processing of sensory inputs, are selectively amplifying and reducing the activation levels. There is some evidence that at least the pre-frontal cortex and the limbic system are partially involved in attention control. This makes sense, because these brain structures are mainly involved in central executive functions like the motivational evaluation of the overall situation concerning the work process, the goal setting, and action planning based on this evaluation. [Corbetta et al. 1990] have shown in cognitive neuroscience experiments that attention to colour and shape produced amplified activity in areas of the ventral visual pathway, and attention to movement produced amplified activity in an area of the dorsal pathway. It is still rather unclear, though, how the detailed implementation looks like. At least, it seems to be verified that the following structures like the reticular formation, thalamus, basal ganglia, gyrus cingulated, parietal lobe and frontal lobe of the neocortex (see Appendix 1.1) are participating to make happen the phenomenon of attention control [Birbaumer & Schmidt, 2003]. There are still debates going on about theories of either early-selection or late-selection along the perception pathways which indicates a lack of a widely accepted view on the cortical processes involved. Certainly, there is a pre-attentive phase of external stimuli processing, as clearly accentuated by [Treisman & Gelade, 1980]. In summary, the explanation still relies to a great extent on phenomenological behavioural evidences.

When moving on with phenomenological behavioural evidences as a basis, attention is very much associated to the sensory systems, although one has to separate attention in perceptual processes from that in (voluntary) action generation. There is *overt* and *covert attention* regarding the process of selecting external stimuli for attention. Overt attention is the act to direct our senses towards a source of relevant stimuli. Usually, the rule applies that the more peripheral a stimulus on the retina, the less the chance to be captured by the attention mechanism. Still, through covert attention we mentally focus on a stimulus which is not simultaneously in the focus of our senses. This brings to light that attention control is a selective mechanism. We cannot simultaneously attend to all external stimuli which are for instance included in what is received on the retina. This also leads to the definition of *selective* or *focused* attention, when talking about stimuli processing of one modality. The term focused attention means that we can only focus at one item at a time, whereas selective attention is used in the context of necessary priority rules to be applied in the light of the existence of various alternatives to attend to, possibly distracting from what primarily has to be attended to.

This limitation of the attention resource is not only true for the perceptual processes, but also for the other cognitive processes where attention is involved. In particular, selective and focused attention can become a critical weakness in certain situations. To a great extent, erroneous behaviour of the human operator can be attributed to this weakness. There is the so-called *strong-but-wrong error* form, for instance. This sometimes fatal type of error can happen when attention is primarily focused on a limited amount of features which, as a result of the phenomenon of habituation regarding context preconceptions (see [Birbaumer & Schmidt, 2003]), seem to exclusively characterise the encountered situation. The pertinent tasks are expected to be easily mastered by existing preconceptions. It may happen, though, that the expectations might be proven wrong, if all features of that situation were taken into account. This can lead to great surprises of the operator and to catastrophic failures at the worst. Although attention is immediately shifted to this mismatch, the knowledge about alternative corrective procedures, although existing, might still not be promptly accessible, since the attention is not necessarily immediately directed towards the conceptual lead-in cues which are crucial to adequately identify the situation. In summary, attention control can be misled by existing prevailing preconceptions. A typical situation of that kind can arise when a fire breaks out in a building crowded by assembling people. The crowd is rushing to the door in panic, and the person being first at the door might just be able to open it before the crowd is pressing hard from behind. If this person is being preoccupied by a usual concept that *pushing* the door is the way to open it, it might just do the wrong thing in this particular case. The door remains closed and, as a consequence, this person and possibly others would be crushed to death before any alternative action can be carried out. Being strongly minded, even further attempts to open the door the same wrong way before eventually looking for possible alternatives is somewhat typical of strong-but-wrong errors. Another kind of error due to the limitations of attention control, but without any acute problem solving demand, is the so-called *skill-based slip* which

might occur, if the focus of attention is distracted for some reason from the present task. Probably all of us have experienced it, for instance when driving home from work at night, but having planned to make a little detour to a shop in order to buy something on the way home. It might happen that we think of our family and intentions for the next day, not recognising, that we have passed the intersection, where we should have turned off instead of directly driving home as we do almost every day. [Reason, 1988] describes these errors in a framework for human error forms as a generic error modelling system (GEMS).

In summary, the limitations in attention control probably represent the most serious weakness of human cognition. Many dangerous incidents and even accidents in vehicle guidance and control can be attributed to these types of errors in attention control. In particular, when monitoring of complex work contexts with a great amount of tasks to be performed in a short period of time the human operator might miss important relevant cues. If it comes to the worst, the operator is closing in towards a catastrophe without having the slightest idea about it until it is too late. **Therefore, particularly for safety-critical work systems, it has to be taken care to avoid this kind of human overtaxing by design.**

On the other hand, it is possible under certain circumstances, though, to keep attention on different matters in parallel by performing on a time sharing basis. In this case one is speaking of *divided* attention. For tasks demanded in parallel, time sharing might be very successful and almost not measurable, if there is little resource competition between the cognitive processes needed for these tasks. This is the case, for instance, when we time share driving and verbal conversation on an open freeway. On the other hand, when driving in a city with heavy traffic and lots of distractions our conversation may die because of processing resource conflict. [Wickens, 1984] proposed the so-called multiple-resource theory as a result of dual-task studies, breaking down resources in the following dimensions: sensing modalities (visual and auditory), processing codes (spatial and verbal), and processing stages (perceptual and central processing on the one side and responding on the other side). If separate resources of the three dimensions are demanded when carrying out the tasks, time sharing performance is more efficient. Also increase in difficulty in one of the tasks is less likely having any interfering effect on the performance of carrying out the other task. A typical example is the fact that talking and listening to the co-driver while driving on the freeway (auditory, verbal processing stages and vocal response) usually has no interfering effect on the performance of the driver of simultaneously keeping the lane and controlling speed (visual, spatial processing stages and manual response).

A prerequisite for attention is the state of vigilance. Vigilance might be compromised by fatigue ([Gaillard, 2003], [Parasuraman, 1986]). We have to accept that for working humans a decline of vigilance over time is inevitable. This kind of principle deficiency regarding the inevitable decline of vigilance with increased duration of the work process, accompanied by a loss in performance, is referred to by a parameter called vigilance decrement. In particular, the vigilance decline has to be accounted for in case of tiring inspection tasks (for more information (see [Wickens, 1992]). Awakening into a vigilant state correlates with

progressive increase of blood flow across the brain, starting with the brain stem region, subsequently the thalamus, and eventually followed by the cortex [Dehaene et al., 2006]. The reticular formation (see Appendix 1.1) plays an important role in this respect. It is the only brain structure a breakdown of which leads to the loss of consciousness [Roth, 2001a].

It has become obvious from the preceding paragraphs that attention control is fundamental for human cognition. Designers of artificial cognitive systems are recommended to follow closely what will be further discovered by neuro scientists about this issue in the coming years in order to make use of it in one or the other way. The evidence that the limbic system as the main evaluation authority and the pre-frontal cortex as a main site of processes of goal-directed planning are identified as important players in attention control, may trigger ideas for similar mechanisms in artificial cognitive systems for the purpose of filtering relevant information out of the whole of environmental stimuli.

Working memory

As to the working memory, [Baddeley, 1990] described and investigated this mechanism as a psychological construct for temporary assignments of particular memory items to be kept ready by means of rehearsing loops for further processing in the context of personal behaviour control [Kimberg et al., 1998] (see Figure 22).

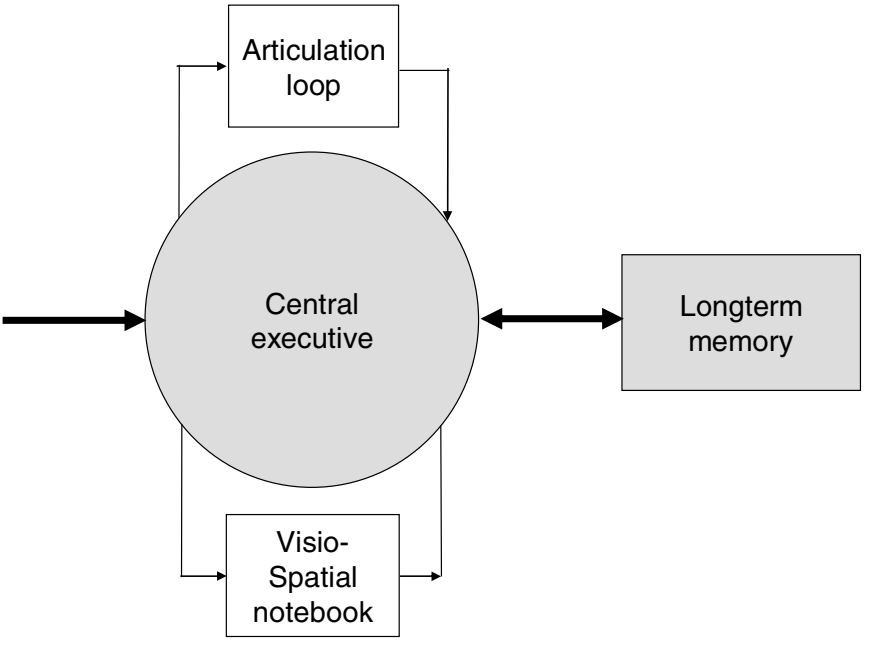


Fig. 22 Concept of working memory (cf. [Baddeley, 1990])

Table 1 Cognitive processing rate [Card et al., 1983]

Rate at which an item can be matched against working memory			
digits	33 [27~39]	msec/item	[Cavanaugh, 1972]
colors	38	msec/item	[Cavanaugh, 1972]
letters	40 [24~65]	msec/item	[Cavanaugh, 1972]
words	33 [36~52]	msec/item	[Cavanaugh, 1972]
geometrical shapes	50	msec/item	[Cavanaugh, 1972]
random forms	68 [42~93]	msec/item	[Cavanaugh, 1972]
nonsense syllables	73	msec/item	[Cavanaugh, 1972]
Rate at which four or fewer objects can be counted			
dot patterns	46	msec/item	[Chi & Klahr, 1975]
3D shapes	94 [40~172]	msec/item	[Akin & Chase, 1978]
Perceptual judgements	92	msec/inspection	[Welford, 1973]
Choice reaction time	92	msec/inspection	[Welford, 1973]
	153	msec/bit	[Hyman, 1953]
Silent counting rate	167	msec/digit	[Landauer, 1962]

There is great evidence that the central part of it is located in the dorsolateral pre-frontal cortex [Petrides, 2000]. The content of the working memory changes in a serial manner. Only a single entry of the working memory can be processed at a time. This warrants that causality is preserved for ensueing processes in brain circuits which make use of the working memory content. The entry may be a perceptual one (externally activated cortex) and one of “inner senses” like inner speech or imagery (internally activated sensory cortex). The serial processing of the working memory entries leads to a rather low processing power compared to other processing in the brain which is running in parallel. Some experimental results (Table 1) provide some data about processing rates we can expect of working memory operations. In order to change the content of the working memory as needed for a next step of processing, an assignment process for pertinent long-term memory items has to be made (also called matching of long-term memory against working memory [Card et al., 1983]). It is not surprising that this serial process is rather time-consuming.

It becomes immediately obvious from the Table 1 that there might come up problems, if for instance, as a common experience in the work realm of automotive guidance and control, tasks have to be performed under narrow time constraints which demand for certain more complex deliberations. This is the reason for the automatic mode of unconscious parallel information processing with direct access to stored (learned) time-dependent action patterns which can be matched in only one processing cycle to fulfil a task (automatic behaviour).

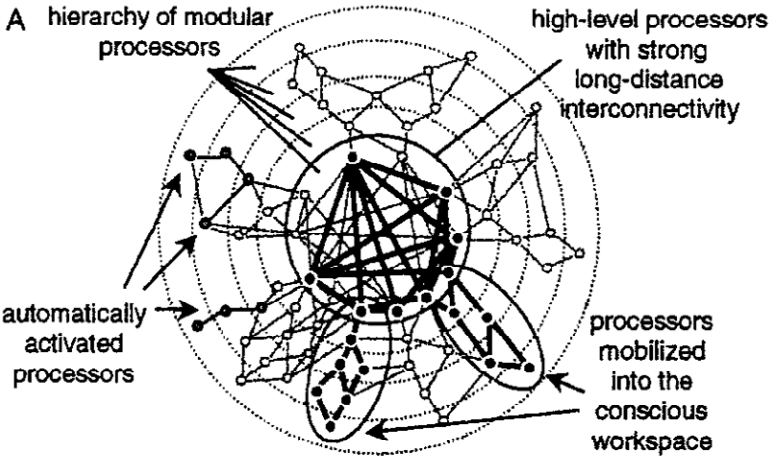


Fig. 23 Neural structure as hypothesized for gobal workspace with amplified state of workspace activity (black circles), bringing together several peripheral processors in coherent brain-scale activation pattern [Dehaene & Naccache, 2001]

In the meantime, strongly supported by brain-imaging investigations in the pursuit of revealing the mechanisms of human consciousness, the working memory model has been extended by way of the so-called *global workspace hypothesis* (see for instance [Dehaene et al., 1998] [Dehaene & Naccache, 2001] [Varela et al., 2001], and publications by Baars, starting with [Baars, 1983]). The global workspace hypothesis can be considered as a framework for human consciousness. As illustrated in Figure 23, this hypothesis postulates that

- there are two main computational spaces in the brain, a processing network of distributed, modular subsystems working in parallel as highly specialised processors, and a so-called global workspace consisting of a distributed set of cortical neurons, located in cortex layers 2 and 3 and characterised by their ability to receive from and send back horizontal projections to homologous neurons in other cortical areas with modular processors through long-range excitatory axons [Dehaene et al., 1998],
- the global neural workspace is formed by the higher levels of a hierarchy of connections between brain processors (each symbolised by a circle in Figure 23) where these levels are assumed to be widely interconnected by strong long-distance interconnections [Dehaene & Naccache, 2001],
- there are processors which do not directly interchange information in an automatic mode, but which can nevertheless gain access to each other's content in a co-ordinated, though variable manner, if they become part of the global workspace [Dehaene & Naccache, 2001],
- the global workspace is characterised by the spontaneous activation of a subset of workspace neurons in a sudden, coherent, and exclusive manner with the rest of workspace neurons being inhibited, thereby exhibiting a

particular kind of "brain-scale" activity state; there can only be one such activity state at any given time [Dehaene et al., 1998],

- because of its integrative function the global workspace theory accounts for the ability to reach a wide range of specialised processors so that there knowledge sources can be related to each other in an unpredictable manner and in consciousness, therefore the global workspace is the arena of a limited capacity stream of sequential conscious experiences [Baars, 1993],
- an amplified state of workspace activity, bringing together several peripheral processors in a coherent brain-scale activation pattern (black circles in Figure 23) can coexist with the automatic activation of multiple local chains of processors outside the workspace (gray circles in Figure 23) [Dehaene & Naccache, 2001],
- the underlying mechanism of selective gating in the global workspace is a top-down attentional amplification of the activity of certain peripheral processors (see also attention control in Chapter 3.2.1.4), by which these processes can be temporily mobilised, i.e. being maintained for a limited duration of time and being made available to the global workspace, and eventually to consciousness [Dehaene & Naccache, 2001]; this reveals that attention control is intimately involved in what we consider as consciousness; without dynamic mobilisation a process may still contribute to cognitive performance, but only unconsciously [Dehaene et al., 1998],
- the maintained activities of processors attached to the global workspace constitute the working memory, also called the "stage" in the *theatre metaphor* for consciousness of [Baars, 1997]; only one of the processor activities may enter consciousness at a given time, depending on various types of contexts residing in evaluation processors which are also connected to the gobal workspace,
- the conscious experience is represented by a chunk of *explicit meaning* (semantic coding); this representation is much richer than any representation which consists of a set of *explicit features* only (see Chapter 6.1.2.2). The chunk can represent a set of explicit features, too, though, or a single one, but the meaning of these features is also explicit,
- bidirectional connections must exist between workspace neurons and a peripheral processor which is in an active state, so that a sustained amplification loop can be established,
- at least the following network systems participate in the workspace: perceptual circuits that inform about the present state of the environment, motor circuits that allow the preparation and controlled execution of actions, long-term memory that can reinstate past workspace states, evaluation circuits that attribute them a valence in relation to previous experience, and attentional or top-down circuits that selectively gate the focus of interest.

What can we consider as neural substrates of the contents of consciousness? Great evidence has been achieved by brain-imaging studies for a common characteristic in case of human performance when carrying out effortful cognitive

tasks which presumably can only be carried out consciously: there is always intense activation of prefrontal cortex areas and the anterior cingulates, thus conferring those areas a dominant role regarding the flow of conscious experiences. Others may contribute in addition, depending on the specific task. Studies like those of [Petrides, 2000], [Rämä et al., 2001], and [Collete & van Linden, 2002] underline this aspect as well as the fact that other regions of the prefrontal cortex contribute to underlying processes like attention control and entailed amplification and mobilisation acts. A conscious experience signifies the mobilisation of a particular content in the global workspace. As a consequence, it also provides the ability to be conscious about the self.

The mechanisms of the gobal workspace, in particular the memorising ability to maintain explicit information for a sufficient amount of time by means of attentional control and entailed amplification, and the ability to make available this information for distant circuits, renders consciousness to represent a crucially integrative function. This is the basis for us (see [Dehaene & Naccache, 2001]) to be able

- to combine several mental operations in order to perform a novel or unusual task and
- to spontaneously generate intentional behaviour.

This makes human cognition very powerful. Therefore, we can sum up by returning to the issue of the two modes of human information processing, in that we conclude that nature in fact has achieved to combine conscious behaviour of *voluntary* action preparation with unconscious, automatic cognitive processing of low-level functions for efficiency reasons on the basis of neural network design.

3.2.1.5 The Limbic Censorship

The limbic system (see Appendix 1.1) is often called the centre of feelings. Affective/emotional experiences are originated in the limbic system. It is considered to be the central unconscious evaluator of the information entering the brain and is in charge to provide the top motivational contexts and to deal with enabling mechanisms like attention control, associations, and memory management subject to the motivational contexts. It more or less dictates the first cognitive reaction, and after loop-like activations sweeping through numbers of brain structures, it also has the final judgement. This can lead to strongly emotional behaviour, for instance panic behaviour in the extreme case.

The ventral loop (also called limbic loop) represents a number of pathways between certain regions of the cortex and certain subcortical structures. Most of the neuronal formations involved belong to the limbic system. This is the reason why the ventral loop represents our top executive level which decides what we do and what we will not do in order to comply with the motivational contexts. By means of the ventral loop the unconsciously working limbic centres shape our conscious experiences as to feelings (positive and negative), goals and the intensity of our desire to bring them to reality.

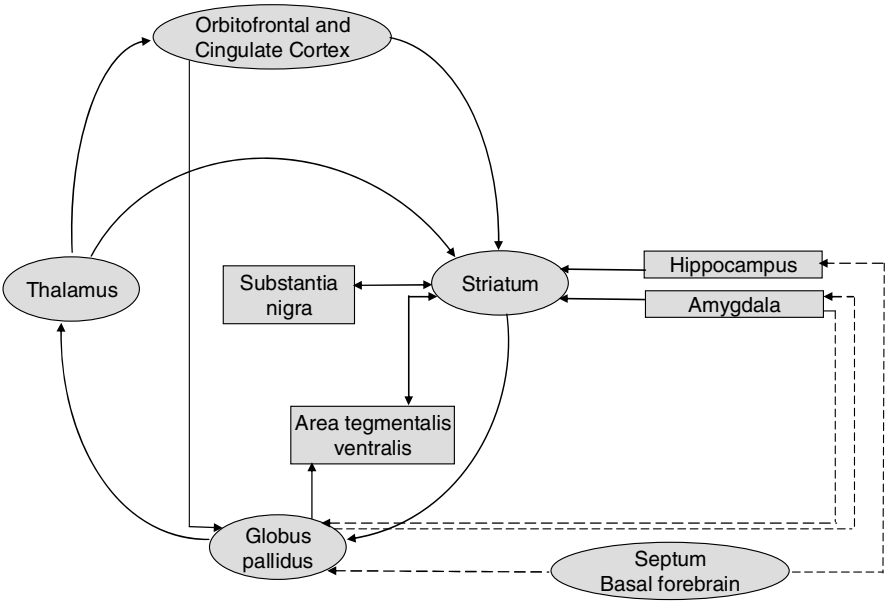


Fig. 24 Schematic structure of the ventral (limbic) loop [Roth, 2001b]

Figure 24 shows the main components of the ventral loop and where the main connecting pathways go [Roth, 2001b]. The loop projects from the limbic cortex, i.e. the orbitofrontal and the cingulate cortex, to the ventral striatum, from there to the ventral pallidum, and finally, being relayed by thalamic nucleus mediodorsalis, back to the limbic cortex. Both the ventral striatum and the pallidum are in close connection with the substantia nigra. The centres of emotional implicit memory, i.e. the amygdala and the ventral tegmental area (mesolimbic system), both representative of the human negative and positive value system respectively, directly affect the loop through the striatum as shown in Figure 24. Also the hippocampus projects to the striatum taking care that the experiences about context get involved, while the basal forebrain takes care of the attention control necessary in that process.

Whereas the dorsolateral prefrontal cortex accommodates what can be considered as the human problem solving capability, including the working memory, it is the orbitofrontal cortex as part of the pre-frontal cortex which provides in full consciousness the functional means that rational human cognitive behaviour can be generated. Rational behaviour abides by given behavioural standards or boundaries which are generally accepted. The orbitofrontal cortex is also the place where voluntary goals (short-, mid-, and long-term) are being developed, for instance according to motivational contexts with expectance of reward (mesolimbic system), and corresponding actions are being prepared. Also later consequences of emotional short-term demands of the limbic system are assessed and adjusted by taking account of certain constraints, for instance social

ones. Thereby, exaggerated emotional reactions (for instance from the amygdala) might be mitigated or held down. It seems that the orbitofrontal cortex accommodates what we name human prudence. At this point it also becomes obvious that human cognition not only stands for feedforward input–output information processing as known from invesitigations of the school of behaviourism in psychology but also for the individual's self-initialised behaviour of goal setting, and entailed actions. In that sense, the orbitofrontal cortex is the consciously experiencing rational counterpart of the unconsciously working limbic system, unconsciously taking the emotional limbic demands for granted, but in consciousness trying to make them compliant with the real world conditions and requirements, if these are known and if there is sufficient time which can be spent for necessary reasoning activity. [Roth, 2003] summarises, based on [Libet et al., 1983] and other work:

> *"… it seems that internally-guided voluntary actions, which are not sufficiently automatized, the pre-frontal and orbito-frontal cortex (in co-operation with other cortical areas) plans and prepares a certain action. This already happens under strong influence of the unconscious limbic emotional centres (above all amygdala and meso-limbic/meso-cortical system) via the limbic loop. Interestingly, our conscious brain is unaware of the subcortical origin of many, if not most our wishes; they seem to come from "nowhere".*
>
> *Before being eventually realized, these wishes and plans must pass through the dorsal loop for a second subcortical "censorship" before the cortex can influence the pre-motor and motor centres of the cerebral cortex, so that these determine the actions in detail and induce them. This censorship of the basal ganglia is, in turn, controlled via the ventral loop by experiential declarative (explicit) and emotional memory contained in the limbic system. It appears that the censorship of the basal ganglia is essentially concerned with two questions:*
>
> 1. *Whether in the light of previous experience the planned action should really be carried out rather than another action.*
> 2. *Whether the intended action is appropriate to the situation.*
>
> *Only when both questions are answered affirmatively do the basal ganglia activate the pre-motor and motor cortex via the thalamus (more precisely, they reduce or abolish the inhibition of the thalamic nuclei). This happens via the "dopamine signal", which in turn is under control of the unconscious emotional-experiential memory.*
>
> *This means that the "final word" about whether voluntary actions are actually carried out comes from brain centres that are inaccessible to consciousness, namely the basal ganglia, the amygdala, the meso-limbic system, and the limbic thalamic nuclei."*

3.2.1.6 Conclusions for the Work System Designer

The preceding sections indeed reveal useful findings which can be exploited in the work system design for the sake of work performance. All these findings can be used by the designer according to the ways mentioned at the beginning of this chapter, i.e.

- to take the capabilities and the pertinent functionality of human cognition as a performance reference, but not necessarily as an implementation model to be copied,
- to effectively design for the real needs of the human operator in the work process based on the knowledge about the strengths and weaknesses in human cognition,
- to generate models of human behaviour which can be incorporated in artificial components as knowledge components to enable human-like capabilities of effective co-operation with the human operator, and
- in summary, to design work systems by use of artificial cognition on a more systematic way.

Some of the findings have already been mentioned right along with the description of the particular implementation principles. The conclusions in this section, though, do not dwell now on others for the sake of completeness. Instead, they highlight just a few findings, no matter whether they are already mentioned or not, which appear to us as those of most potential effect on the work system design. The vast amount of accumulated valuable knowledge about human factors in addition to that cannot be further commented here anyway, although it is strongly recommended to study these findings, too, if they are not kown yet. Here, we only focus on the following six conclusions:

(1) The finding of probably best yield for the designer of artificial cognitive systems is that about the the two modes of human information processing with one of them being

- standing for the widely distributed unconscious specialised processing subsystems which are essentially working in parallel, and

the other one being

- standing for a process which is working on the basis of sequential processing steps (accompanied by conscious experiences) for the purpose to selectively (goal-relevant) drawing information from widely distributed unconscious specialised processing subsystems it can communicate with by means of the so-called global workspace and to integrate this information and to broadcast results among the unconscious specialised processing subsystems.

The unconscious cognition in processing subsystems, which is often called the *automatic* one, is extremely efficient, if routine behaviour with quick reactions is demanded. It is taking advantage of massive parallel processing. This relates to the by far largest part of cognitive processing activity going on in our brain at all

times. The conscious cognitive processing, which is often called the *controlled* one, is time-consuming but very effective, too, and the only effective one, if higher level behaviour of deliberations is demanded which cannot be provided by the automatic one. This includes *"durable information maintainance, novel combinations of operations, or spontaneous generation of intentional (voluntary) behaviour [Dehaene & Naccache, 2001]"*. As a consequence, [Sloman, 1999] is proposing that artificial software agents should be furnished with hybrid architectures of that kind layering two kinds of agents, reactive subsystems forming a basic layer for the automatic cognitive processing and deliberative agents in a layer on top for the controlled cognitive processing. The latter may also include agents for meta-management, providing self-assessing, self-monitoring and self-modifying capabilities.

According to these two modes of information processing human cognition features two main principles of memory implementation, explicit and implicit memory. Unconscious processing for routine behaviour takes advantage of implicit memory. Deliberative behaviour, though, which copes with situations demanding for conscious decisions and more effortful deliberations, is characterised by making use of explicit memory.

(2) The content of the explicit memory is semantically coded, i.e. by explicit meaning. These semantically coded explicit memory items (so-called chunks) are represented sub-symbolically, possibly including an abundance of implicitly represented features, which offers a representation much richer as other representations which consist of a limited set of *explicit features* only. Moreover, semantic coding does not exclude the representation of explicit features either which provides our capability to deal with symbols, too. We might also realise, perhaps as a surprise, that many features, which are also part of the implicitly represented ones of a content item of explicit memory, can be selectively evoked as explicit ones by the global workspace. Thus, human cognition demonstrates how to take advantage of the benefits of both connectionism and symbolism. This kind of knowledge representation

- avoids ontology problems (ambiguities etc.),
- ensures a fine-grained representation in the feature space along with outstanding generalisation capabilities, and
- enables coexisting knowledge representations of almost an arbitrary number domains, independent and dependent ones.

As we think, this is not being sufficiently appreciated by the community of designers of artificial cognition, so far. Unfortunately, the corresponding mechanisms in human cognition are not yet fully explored, but the theory of global workspace as a well-accepted hypothesis might yield a valid basis to generate ideas for how to implement a similar approach in artificial cognition. There is very little undertaking to our knowledge, so far, like that of [Franklin & Graesser, 2001] for instance, which seriously is pursueing this aspect, but we are sure this is the way to go for future designs.

In order to come to the point that machines deserve to be trusted to the same degree as we trust in human operators, it seems indispensable to provide something equivalent to semantic coding as this characterstic of human cognition.

(3) It is also of great interest to the designer of cognitive systems how learning is integrated in the overall system architecture of human cognition by connecting perception with learning, where attention control in the context of the issue of relevance of environmental stimuli plays a role, too. The way of achieving tremendous generalisation capability and equally high reliability as accomplished for human perception has to be taken into account in that context. This is directly pointing to the so-called plasticity versus stability dilemma.

The functional component of learning was neglected in most cases of existing artificial cognitive systems, because knowledge can principally be implemented in artificial systems as a given thing, which might have been acquired offline by knowledge acquisition methods. This admittedly pragmatic approach of system implementation will not necessarily be a leading one for future applications, because online learning will be a valuable design feature for the performance of artificial systems, too, when online adaptation capability plays an important role.

(4) The processes of human voluntary action are pivotal and very complex. The design of these processes indeed account for control dilemmas in a satisfying way [Goschke, 2003] like the already mentioned plasticity versus stability dilemma, and in addition the

- maintenance versus switching dilemma and
- selection versus monitoring dilemma.

This addresses the following: Although human behaviour is goal-driven in the first place instead of being data-driven, persistence in the pursuit of long-term goals is demanded and at the same time enough flexibility is demanded to interrupt this pursuit in favour of new dispositions which should be immediately carried out in response to certain urging changes of the situation. In addition, the selection of goal-relevant information has to be designed that it does not suppress information, which is relevant to another competitive goal of higher level. To design for these kinds of dilemmas is a great challenge for the designer of artificial cognitive systems. Further findings in neuropsychological investigations on these processes can be expected in the coming years which might be helpful. These should be followed up as well as findings on the other mental processes, in order to be able to make use of them.

(5) It should be realised by work system designers that the amount of findings about human cognition has been developed to an extent that qualitative and quantitative models of human behaviour are no longer something one only dreams of. Modelling of about all high-level functions concerning human situation-dependent action generation based on explicit and implicit knowledge can be accomplished today. Examples will be presented in Chapter 6. Some of the

prototype systems of artificial cognition in work systems are significantly taking advantage by making use of these kinds of models.

(6) Finally, some words about limitations of human cognition. An often neglected design recommendation for work systems is the rule to bring the human operators in full action when there are tasks which match ideally with their excellences, and to alleviate him from tasks or support him effectively where his weaknesses could play a shabby trick. For instance, if visual perception as such is a central task, there is still no artificial means which generally can do better, but if this is coupled with specific demands on attention, for instance, compliance with these demands is not warranted as we learned from Chapter 3.2.1.4. Here, for instance, support by means of artificial systems can be very effective. Artificial cognition is always vigilant and attentive. There is also no fatigue problem. Also consciousness is not an issue for the design of artificial cognition, because everything which goes along with consciousness like working memory and attention control can be warranted by design. These are typical properties of artificial cognition which can help to avoid violations of human limitations in the work system design.

In artificial cognition motivational contexts are also governing the cognitive behaviour, but usually it is not wanted to design a system with the kind of emotions which are insufficiently controlled. The motivational contexts in artificial cognition are defined on a rational basis. However, more than one level of motivational contexts as it is realised with human cognition by the level of limbic motivational contexts and the cortical ones (orbitofrontal cortex) is also for artificial cognition an interesting approach.

3.2.2 Framework of Functional and Behavioural Aspects of Cognition

In the preceding Chapter 3.2.1 an introductory survey was given on findings about human cognition. This survey is by far not comprehensive, but it is sufficient for an assessment of how sound knowledge about human cognition may be useful for the work system designer and where it might make sense to have a closer look at the literature to take advantage of the findings. In particular, this survey can help to decide about which features of human cognition are essential to be saved when heading for artificial cognition in a certain application. It is very useful for the designer of artificial cognition to know about the given facts regarding human cognition as the most prominent case of cognition in nature, but the designer's most vital task before actually endeavouring to implement an artificial cognitive system is to set up structured requirements for the behaviour of the system to be designed. In the following of this chapter, a framework of cognition will be established where both human and artificial cognition can be captured in a common basic structure of functional and behavioural levels from a top view, thereby embracing the functional aspects of human cognition as presented in Chapter 3.2.1, but looking at them from the point of view that these functions are to generate certain behavioural components. Therefore, in addition to the definition of the functional levels as dwelt on in Chapter 3.2.1, we define behavioural levels, thereby following an existing qualitative model of human

cognitive behaviour which seems well-suited for this purpose. This framework provides a good basis for the functional demands which have to be satisfied in artificial cognitive systems. The other way around, it can be used to show where blank spots are still left in the design of artificial cognitive systems. This framework will be used, too, in this book to indicate what is covered by the implementation examples in Chapter 6.

3.2.2.1 The Three Functional Levels of Human Cognition

We ascertain that cognition is the functional basis of intelligent behaviour as it is exhibited by human beings. Admittedly, the biopsychologists still do not know all details of how human cognition works, but those essential cognitive functions in our brain, which enable the outstanding human capabilities we are so proud of, can yet be described to a quite satisfying extent. We have become acquainted with them to a sufficient extent in the preceding chapter that we can exploit it for work system design. For this purpose, this section will sum up by suggesting in the following a structure of three cognitive functional levels.

High-level cognitive functions: These functions represent the chain of cognitive functions which stand for the mainstream subfunctions to generate any kind of voluntary action in a work process and thereby to impinge on the real world environment. These subfunctions primarily are in our focus. It is the chain of these subfunctions which take care to determine the situational beliefs, the adjusted goals and constraints, a task agenda, the correspondingly due task (current intent), the resulting action instructions, and the control of these actions. Thus, these are the functions, which provide the visible behaviour of the human operator, being solely based on relevant relevant a-priori knowledge which is innate or might have been acquired beforehand by any learning process. This knowledge can be partially accessed consciously. The most part, though, of a-priori knowledge about motivational contexts which are the drivers for this chain of functions does not become conscious. [Roth & Rudnick, 2008] identified four levels of a-priori knowledge in the human brain which become involved for decision-making in any work situation:

- *level of innate nature*: This level is the very basic one which can hardly be affected by any experiences or education. The pertinent a-priori knowledge may settle personality traits like patience, endurance, openness to experience etc. There is no conscious access to anything explicit of this knowledge.
- *emotional level*: Together with the level of innate nature this level constitutes the unconscious kernel of our personality, the a-priori knowledge of motivational contexts which are guiding us (most of it located in the amygdala and the mesolimbic system). It develops during the first years of our life and can only be affected by extreme emotional experiences or long lasting influence. There is also no conscious access to anything explicit of this knowledge.
- *level of social mind set*: This level consists of the a-priori knowledge which helps to make our behaviour socially compliant (predominantly located in the orbitofrontal cortex). The reasoning process and the resulting decisions

are being consciously experienced. In essence, the knowledge becomes encoded by continuously learning from social experiences in the course of our life. This level considers the long-term consequences of our voluntary actions.

- *level of know-how to communicate*: The pertinent a-priori knowledge of this level guides our ways of communication to be in compliance with other a-priori knowledge available in the prefrontal cortex. Again, this knowledge develops by a continuous learning process during our life and any reasoning activity in this context and the behavioural decisions are being consciously experienced.

The a-priori knowledge of the latter three levels and the corresponding subfunctions are highly application-oriented, e.g. dependent on the concrete work demands (work objective) in an application domain. Some of these subfunctions include knowledge processing in the sense of reasoning, thereby taking into account the constraints of limited mental resources. It should be noted that these functions also include voluntary actions for deliberate active acquisition of additional knowledge which may be pursued to be utilised from there on in the work process. Deliberate acquisition of knowledge cannot be done without voluntary actions like reading, for instance, and might become a work process for itself.

Medium-level cognitive functions: These are the functions needed to enable the high-level cognitive functions. They are carried out unconsciously, like learning, if it is not deliberately pursued (learning by doing like we have learned our mother language) and therefore taking place as a concurrent data-driven background processes, or the memory mechanisms including the mechanisms of knowledge representation, retention and retrieval, and the mechanisms of consciousness including those of working memory and attention control.

Low-level cognitive functions: These functions are those of the different kinds of basic information processing elements (neurons) and the structure and internal processing organisation of the neural networking in the human brain. These functions are the basic enablers of the functions of the higher levels.

For the discussion of human cognitive functions the focus will mainly lie on the first two higher functional levels of cognition as mentioned above. It would be nice to also cover the third level comprehensively, but this is not possible at the time being. The functions of human cognition pertinent to this level, basic brain elements of information processing and, in particular, the functional organisation of these elements in terms of neural networks as functional units with millions of interacting neurons are still not sufficiently understood in the neuroscience community. This is still a fact, although there are great advances, thanks to the techniques available in combination of electroencephalography (EEG), magneto-encephalography (MEG), positron emission tomography (PET) and functional magnetic resonance tomography (fMRT) [Pinel, 2007]. These techniques allow the localisation of the time-dependent activation of neural areas associated to certain cognitive tasks with rather good accuracy and in real time. Other

techniques allow the investigation of the processes within and around single neurons in great detail, which enables the formulation of computer models. For further informative introduction in what has been achieved see for instance [Roth & Prinz, 1996], [Pinel, 2007], and [Maasen et al., 2003].

The great gap, though, lies in the fact that there is alarmingly little knowledge of how the billions of neurons actually work together to provide the unique performance of human cognition. For the purpose of this book this gap will be overcome by covering it on a phenomenological basis, as it is common practise in cognitive science. Therefore, we treat the innermost part of low-level functions more loosely. Looking at the engineers' realm of artificial cognition, it is much easier to carry out the corresponding discussion for the basic functional level of artificial cognition. There, we deal with the technology available as part of the enabling foundations for artificial cognition, keeping in mind that this is a matter of permanent progress of the state of the art with steadily increasing performance of the basic functional elements. This difference in the low-level functions between human and artificial cognition is of great importance, since they essentially provide the explanation for the observable differences in the two higher functional levels between human and machine cognition: The differences mainly lie in the performance of the basic functional elements and the structuring of these elements. This is the reason, too, for the fact that at the time being a one-to-one identity in the functional performance of natural and artificial cognitive systems is not within reach. Consequently, performance differences between human and artificial cognition within the functional levels of cognition have to be carefully accounted for in cognitive system design for the sake of overall system performance.

In the context of high-level functions we have discussed *perception, deliberate learning, voluntary action, sensorimotor systems, and communication* to some extent. As medium- level functions we have dwelt on the basics of *attention control, vigilance, consciousness, unconscious learning, and memory functions*. The low-level functions are considered to a certain extent when describing the higher-level functions from a neuropsychological point of view.

3.2.2.2 The Three Levels of Human Cognitive Behaviour

For the work systems considered in this book, which are associated with vehicle guidance and control, it can be assumed that the human operator works on the basis of rational cognitive behaviour. In that sense, the complexity of the limbic loop will not become much apparent. The motivational contexts comprise only those which are sensible ones in the light of the work objective. There are no motivations (emotions) which lead to a situation such that they have to be "socialised" in the sense of the limbic loop (see Chapter 3.2.1.5). Hence, when viewing on the subject of human cognitive behaviour we settle down in this book on a qualitative model developed by [Rasmussen, 1983], which is well-suited as an introduction in the topic of behavioural models and as a specification of the main behavioural functions of human cognition. This model has been derived to

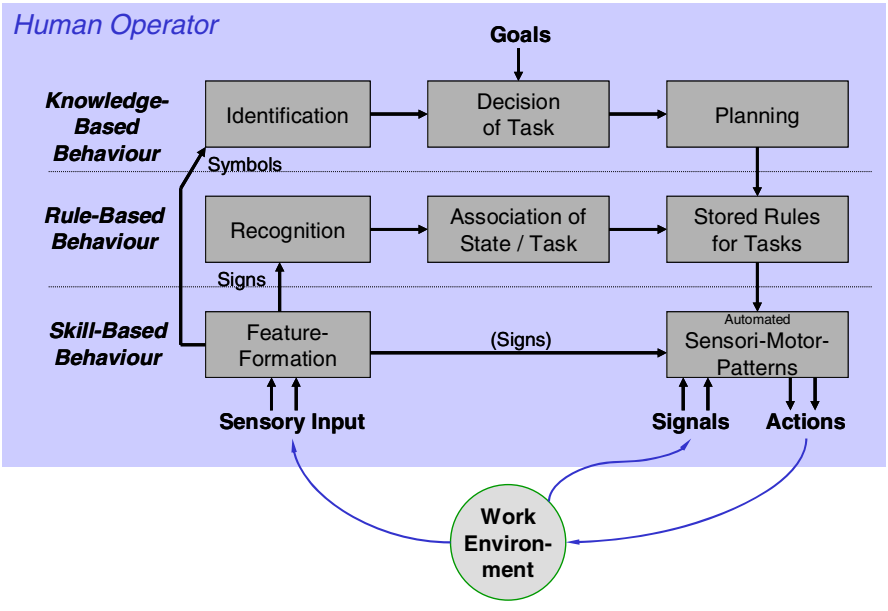


Fig. 25 Rasmussen's model of human operator's performance levels linked to work environment [Rasmussen, 1983]

describe the behavioural structure of the human when working on control processes. Rasmussen's model appears to be a rather grainy but generally well-accepted one, and it has the advantage of nicely offering a reasonable break down of the main high level cognitive functions and their interrelations. The simplicity and intelligibility made this model, originally having its seeds in ergonomics research, quite popular in the circles of cognitive psychologists as well as amongst human factors minded engineers. In fact, Rasmussen's work became the probably most commonly known psychological model within the engineering community.

In the following sub-sections Rasmussens's model of human behavioural levels shall be described in some more detail. According to the scope of this book there will be derived an interpretation of the model, keeping the basic model structure, but incorporating some details with respect to a computational use of the model.

Rasmussen's model of human cognitive behaviour

Without going into too much detail here (for a detailed discussion it is referred to the original literature [Rasmussen, 1983]), the model distinguishes between three levels of human performance, the skill-based, the rule-based, and the knowledge-based behaviour (see Figure 25). Thereby, it captures the two modes of human information processing for voluntary actions where one is working consciously-explicit (rule-based and knowledge-based behaviour) and the other unconsciously-implicit (skill-based behaviour). In this context, [Krüger et al, 2001] is alluding to the model of [Sanders, 1983], distinguishing between controlled actions and automated action respectively.

On the skill-based level highly automated control tasks are being performed, with hardly any mental effort or consciousness. This is the level representing the unconscious/implicit information processing of humans. Typical for this level is the continuous perception and manipulation of the physical environment in order to control processes in three-dimensional space and time. Most of this performance is carried out in feed-forward control mode by configuring and triggering sensori-motor patterns (stored as compact behavioural chunks) on the basis of task specific features. Typical behaviour on this level, like keeping one's car in a street lane while driving, is generated by running a sequence of parameterised templates with some feedback control ratio for precision enhancement.

On the rule-based level most of the everyday *conscious* behaviour takes place in a strict feed-forward control manner. Here, humans follow pre-stored procedures (explicit a-priori knowledge about rules and scripts) in order to activate the appropriate sensori-motor patterns on the basis of the presence of clearly recognised objects characterising the actual situation. Rule-based performance is goal-oriented, although goals are not explicit, but encoded in the pre-conditions of the applicable rules.

The knowledge-based level will be entered in situations, where there are no applicable rules. This requires dealing with a problem without pre-stored solution. In this case matching mental concepts have to be figured out in order to identify the situation. If this identification process succeeded, goals derived from any motivational contexts (goals) explicitly direct the entailing planning process. Planning, often used as a synonym for problem-solving, will be deployed in order to generate new procedures, which can be executed on the rule-based level. In general, planning can be considered as a highly versatile process, incorporating strategies such as *difference reduction* and *means-ends analysis* [Anderson, 2000] as well as *search in problem space* [Newell, 1990].

It can be postulated that in the course of the evolutionary design process of the human brain this structure of behaviour levels evolved chiefly in order to optimise information processing efficiency by accounting for a very large memory capacity for the storage of a-priori knowledge, while being stuck with principal limitations in computation power. In essence, the corresponding a-apriori knowledge for the skill-based and rule-based level is being developed by a process which forms more compact chunks pertinent to situations which usually demand quick reaction and were encountered very frequently. This training process gradually reformulates behavioural knowledge in that way which was derived on the knowledge-base level at an earlier stage, first shifting behavioural capablities from the knowledge-based level down to the rule-based level and, if appropriate, further down to the skill-based level.

Interpretation of Rasmussen's model

Rasmussen's model is very common and of great usefulness to the human factors minded engineering community because of its appealing structure of clear-cut blocks and their interrelations. In spite of having stated that, as to the purpose of

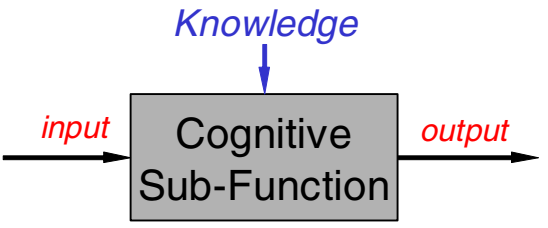


Fig. 26 Formal description of cognitive sub-function and related knowledge

this book, we are going to work in the following with an accustomed interpretation of this model. Similar ways of dealing with the model, depending on the application they are focusing on, have been embarked on by other authors, too (see for instance [Sheridan, 1992]). Here, it shall be done from an information technology point of view, in particular with respect to the implementation of knowledge and its processing within the model. In Chapter 4.5, when we develop a model of an artificial cognitive process, we will come back to this.

To begin with, the layered arrangement of behavioural levels, and the different functional characteristics of these levels remain to be those of [Rasmussen, 1983]. The labelling of the levels is modified, though, in a way which is more consistent with terms used in other disciplines.

Most of the blocks in Rasmussen's diagram as shown in Figure 25 represent dedicated cognitive sub-functions (e.g. 'recognition', 'planning'). Two particular blocks on the other hand, i.e. 'stored rules for tasks' and 'sensori-motor patterns' represent knowledge, without having the pertinent functions specified. Only one block in Rasmussen's notion (i.e. 'decision of task') makes use of an explicit knowledge basis (i.e. 'goals'). From an information technology point of view it would be desirable to modify the depiction according to the following convention (see Figure 26), e.g.

- using blocks only for the representation of cognitive sub-functions,
- identifying the *knowledge* which is made use of in each block, and
- naming all inputs and outputs of the functional blocks representing states of conscious experience.

The knowledge of each functional block shall be considered as static data generated prior to the actual performance of the function. We will refer to this kind of knowledge as *a-priori knowledge*. This a-priori knowledge is equivalent to the persistent content of human long-term memory, i.e. it consists of semantically coded implicit and explicit knowledge (see Chapter 3.2.1.4). The explicit knowledge consists of chunks as also mentioned earlier in Chapter 3.2.1.4. A chunk constitutes a meaningful unit of information. This is also true for the implicit knowledge, which is exclusively implicit, though. For instance, the implicit knowledge comprises the sensori-motor patterns to be applied for skill-based reactive control behaviour. This is an important design characteristic to augment the reaction time (computation time), albeit at the expense of less certainty of correctness.

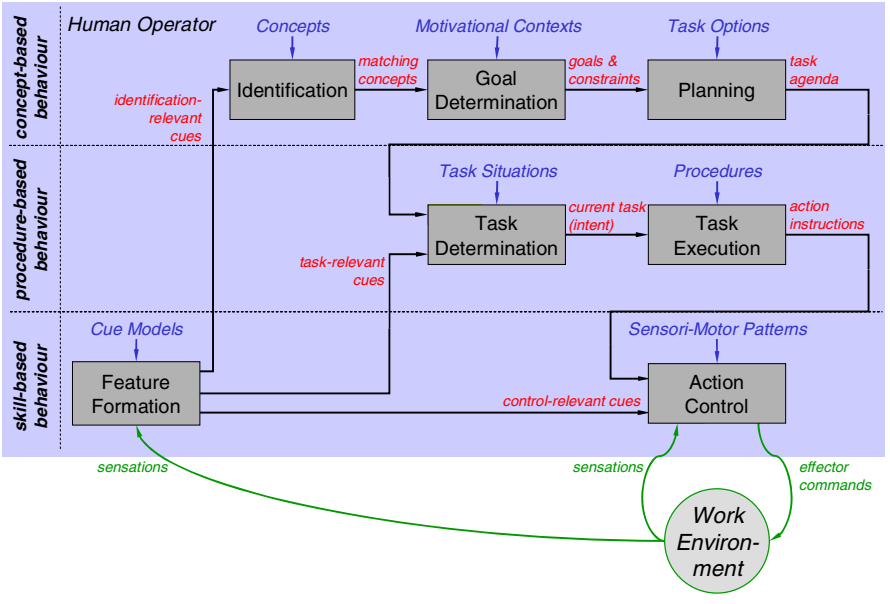


Fig. 27 Interpretation of Rasmussen's model of human performance incorporating an information technology approach

At this point, it becomes immediately obvious that knowledge is fundamental for cognitive processes and that the human capability of knowledge acquisition for all levels of behaviour is crucial. It is a typical feature of implicit knowledge acquisition for skill-based behaviour, that it is acquired (learned) by doing. With increased practise the representation changes from simple rules with single discrete actions to combined and streamlined ones with condensed action patterns. This is the result of a behavioural shift through the behavioural levels from conscious behaviour in the knowledge-based level down to the subconscious behaviour in the skill-based level, mainly for the sake of processing efficiency. Acquisition of explicit knowledge usually is a data-driven process as a consequence of experiences when monitoring what is taking place in the course of the work process at hand. This way of knowledge acquisition is dependent on sensing capabilities.

It should be noted though, too, that active knowledge acquisition might be dispensable in certain cases as long as there are other ways to make available the necessary knowledge. For instance, knowledge might be stored a-priori before the cognitive process is started. Remember, new-borns not having learned anything yet, immediately know how to behave to get to the food they need. Furthermore, knowledge might be acquired by direct communication links and direct storage of communicated information. That might take place without having any perceptual encoding involved. Of course, this cannot be the case for humans, as long as we do not consider any devices to directly transmit data into the brain. For artificial

cognitive systems, though, this is a valid alternative. In that sense, artificial cognition is less dependent on learning than humans. However, the capability of self-reliant knowledge acquisition via observing the environment through whatever senses is an extremely valuable feature in any case.

On the basis of this approach our interpretation of Rasmussen's model of human performance results in some modifications, as for instance to the way of presentation as shown in Figure 27. As being immediately obvious, Rasmussen's original terms of knowledge-based and rule-based are replaced in this figure by the notions of *concept-based* and *procedure-based behaviour*. This modification will be used throughout the following of this book and is done to highlight the kind of knowledge format which seems to be most characteristic for the respective behavioural level. Since the knowledge about motivational contexts is anyway penetrating all three levels in one or the other way we take the knowledge of concepts and procedures as the most significant one in the respective levels. For the procedure-based level one could possibly argue that the knowledge about task situations is at least similarly significant compared to the procedure knowledge. With the decision made, though, we think to adhere best with the choice made by Rasmussen by designating this behavioural level as rule-based. Since we want to carry out the crossover between human cognition and the computer implementation for the main aspects of cognition, it also offers a better chance to be understood by either research community. Computer scientists, for instance, speak of knowledge-based systems to emphasize the separation between explicitly represented knowledge and its processing, i.e. inference mechanisms. They use the term rule-based systems, when they want to address applications which build up upon a knowledge representation using rules, i.e. productions.

On the skill-based behaviour level the *feature formation* is performed as a subconscious perceptual process by transforming sensations from various sensual modalities into spatio-temporal *cues* characterising the present situation relevant to the current control task. This function covers the range from pre-attentively (bottom-up) mechanised functions of perception [Treisman & Gelade, 1980] up to a-priori-knowledge-driven (top-down) pattern matching functions for (situation-independent) object recognition. This corresponds to the cognitive functions of the primary and secondary sensory cortex, of the dorsal and ventral pathway into the posterior parietal and the inferotemporal cortex, including participating areas of the association cortex, and those structures which are participating in attention control in addition to those mentioned, including the frontal lobe and subcortical structures. These also indicate the areas where the a-priori knowledge is situated (see also Chapter 3.2.1). All imaginable kinds of models (also dynamic world models) for situation-relevant cues can be used as *cue models* to be part of the a-priori knowledge. The kind of cues provided by the function of *feature formation* mainly depends on their further use on the three behavioural levels which are renamed by the way according to Figure 27 for the sake of consistency with other parts of this book. These formations can also be considered as chunks, in general, as was outlined earlier.

The three modes of feature formation

As an example for the mode of *feature formation* on the **skill-based level** may serve a situation, where the curvature of the street ahead changes from straight to turning while driving along that street. This may trigger the *action control* function to utilise a different sensori-motor pattern to follow the changing gradient of the curvature. The features which describe the curvature are possibly extremely manifold. They are by far not just explicit features as could be used in a symbolic representation. Instead, they are embracing an abundance of implicit features in an integral formation which make up a semantic entity of a cue describing the curvature as perceived by the driver. Rasmussen refers to it by the notion of *signs* to refer to the typical semantically coded representation. There is typically only one *control-relevant cue* at a time which might consist of both explicit and implicit features as indicated by the example mentioned.

On the **procedure-based level** an example for the pertinent mode of feature formation (Rasmussen again refers to the notion of signs here) might be a traffic situation which points to an opportunity for a takeover manoeuvre. Again, there will be only one relevant compact cue as a chunk which semantically matches a pre-stored *task situation* and triggers the *current task* as output of the *task determination* function. The task determination function takes into account that there are two kinds of possibly matching task situations. For one hand there is that one (type 1) already expected for the initialisation of the next task when following the task agenda, or on the other hand there is that one (type 2) which is caused by intervening enviromental players, not foreseeable but essentially familiar and possibly more compelling. When considering artificial cognition later in this book, the type 2 of task determination will play a central role for the delineation between cognitive automation and conventional automation. If the situation is of higher complexity, explicit featuring might not be sufficient. Therefore, also in case of procedure-based behaviour an implicit or an explicit *task-relevant cue* can be the relevant one. The resulting *intent* to "stay behind", for instance will trigger the *task execution* to generate the *action instruction* "decelerate" as input to the *action control* function.

On the **concept-based level** the pertinent mode of feature formation also deals with both explicit and implicit cueing (Rasmussen refers to the notion of *symbols* here), which lend themselves to be used in a reasoning process about the identification of a complex situation. As opposed to the other behaviour levels, though, there may be considered several *identification-relevant cues* by the concept-based function of *identification*. This may be enlightened by an anecdote, which one of the authors of this book experienced at the age of 19 when he was by no means prepared for a scary situation like this: "driving in my car", "red light flashes on the instrument panel", "little oilcan symbol turns on next to it". Consequently, in our example, the *task determination* on the basis of known *task situations* failed, because there weren't any appropriate pre-stored task situations matching. Meanwhile, the *identification* of various related *concepts* proceeded, e.g. "red light – technical vehicle problem", "oilcan – engine", red light & oilcan – engine failure". The *determination* of also-ran *motivational contexts* and the generation of an, as turned out later, catastrophic *task agenda* that followed deserve no further memorandum here. (Next time I'll do better.)

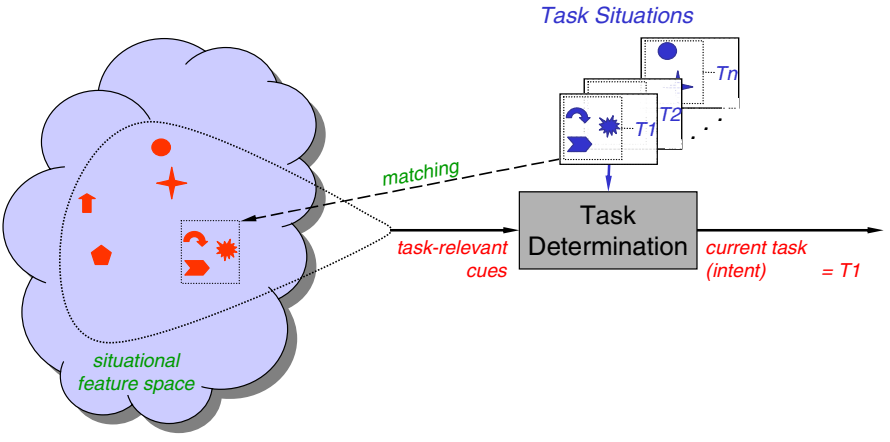


Fig. 28 Cognitive sub-function of task determination on the procedure-based performance level

Stepping up to the procedure-based level, the *task determination* will be performed by matching task-relevant cues with a-priori known *task situations*. The knowledge on task situations comprises the entirety of models of perceivable aspects the world exhibits in certain situations which are implicitly related to the motivational contexts. They are derived from the individual's experience. The *task determination* process automatically decides on just one task situation, if there is a matching one. This will be consciously perceived. Figure 28 shows this basic idea of task determination as well as the fact that the matching situation is identified by a corresponding task-relevant cue. Only the result of the task determination will trigger further decisions how to proceed, i.e. how to work on the *current task* or *intent*. Those links, being the immediate mapping between familiar situations and what to do in these situations, are based on records made earlier (e.g. "to stop in certain road traffic situations", as one learns in driving school).

This kind of matching process is not without risk. There might be situations, in particular if they are of type 2, where additional features and corresponding cues do have relevance, however may not be considered, either because of attention or other resource-related limitations (e.g. "overlooked car close behind me while heavily breaking on red traffic light", see also [Eriksen & Yeh, 1985], [Wickens, 1992]), or because of incomplete knowledge on *task situations* (e.g. the earlier mentioned scary situation when driving, recognising a flashing red light and the oilcan symbol next to it turning on).

In order to proceed on the procedure-based performance level with the decision made for the current intent to be worked on, this particular task will be passed on to the function of *instruction generation.* Here, the knowledge of *procedures* related to the current task or intent will be instantiated. These procedures incorporate certain pre-conditions concerning the current situation and the task to be worked on, and the consequential *instructions* to be given to the action control function for further pursuit. These procedures usually encode task related behaviour rules or "recipes" as sequence of instructions in order to cope with the

variety of potential tasks. The nature of the instructions heavily depends on what sensori-motor patterns are available to the individual. Performing a complex mechanical assembly, being trained regularly, certainly becomes a highly automated operation for a skilled worker. In this case this operation might be triggered by giving just one single instruction.

Procedure-based performance covers those situations, where the pre-recorded knowledge facilitates a satisfactory way to proceed. It directly maps familiar situations into intents to be worked on using standard behaviour particular for these situations. During normal procedure-based performance the concept-based behaviour level is not passive, although in most cases not interfering. Here, the concept-based level is responsible for the monitoring of the compliancy of the procedure-based processes with the motivational contexts and respective goals and task agenda. This monitoring is for instance rather efficient in the detection of *slips*, *lapses*, and mistakes, i.e. possible errors on the skill-based and procedure-based performance level [Reason, 1988]. But it might as well fail due to lack of attention allocation (see also [Cacciabue, 2004]. The concept-based performance level eventually actively interferes, if the understanding of the situation is not in sufficient compliance with what has been expected according to the actual goals, constraints, or task agenda and if these deviations are unfamiliar. Not only adverse situations in conflict with the motivational contexts are identified by the performance level of concept-based behaviour. It also offers the possibility to identify welcomed opportunities and to exploit them for the sake of better compliance with the individual motivational contexts.

Example for engagement of concept-based performance level

There can be identified several types of situations where the concept-based level will be engaged for the sake of performance optimisation. According to our understanding of the cognitive functions the most obvious trigger is the lack of a-priori knowledge on the procedure-based performance level (i.e. missing knowledge about task situations, related intents, or behaviour rules). An example might illustrate this: After the re-unification in Germany the "green arrow plate" allowing the right turn on red traffic light has been introduced as heritage from the former East-German road traffic system. Imagine a foreigner, familiar with the pre-1990 West-German road signs, returning after 1990 and not knowing about this innovation. This person will stop on every red traffic light despite the presence of that green arrow even with the wish to turn right. Certainly there is a-priori knowledge available on the procedure-based level responsible for this behaviour, but there is no model of a task situation comprising a red traffic light with a green arrow plate and the resulting task. How can now an engagement of concept-based behaviour for performance enhancement be achieved? Probably the green arrow, which now is an additional *identification-relevant cue*, might be associated with the concept of the phenotype of a traffic sign (concerning e.g. green colour, directional information, and mounted on a pylon). Together with other *matching concepts* and based on the *motivational context* not wanting to loiter away too much time on red traffic lights (still considering safety motives), one might develop the plan to drive on. It might as well be necessary to have this process be pushed by some feedback over the physical world, i.e. the guy in the car behind

honking the horn, or the guy in front just doing it and driving on. Thereby, further cues will be added encouraging concept-based considerations.

As an additional aspect, even if the a-priori knowledge about the *task situation* and the corresponding rules are available, concept-based behaviour might nevertheless be required, because one crucial feature (green arrow) might not be perceived due to resource-related bottlenecks (e.g. visual scanning of the scene can be disturbed by something more appealing) or any psychological pre-occupation effects [Solso, 2001] (e.g. "Sure, they don't have this green arrow here in Germany!").

On the concept-based behaviour level situation assessment takes place by the *identification* of relevant *concepts*. This will be performed by matching a-priori known *concepts* with incoming *identification-relevant cues*. These concepts comprise the highly networked entirety of the individual's knowledge about the world. Thereby, the connotation of further related knowledge will be facilitated. Figure 29 shows the basic idea of the cognitive sub-function of identification.

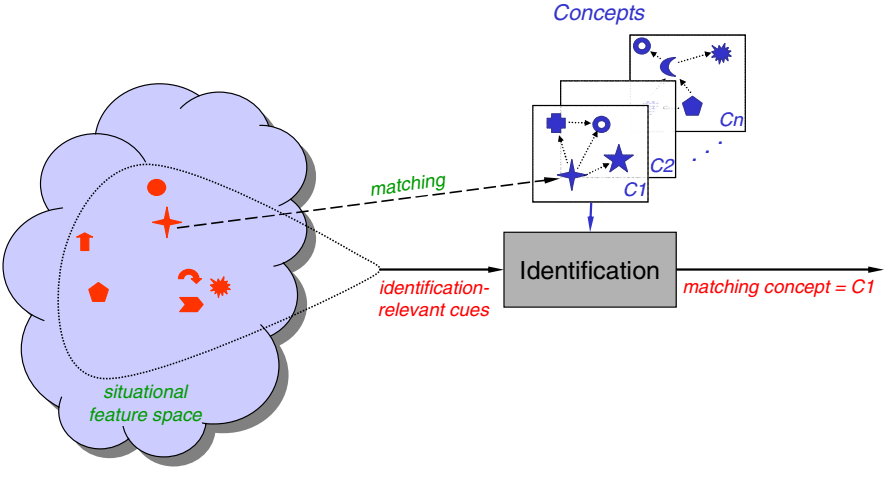


Fig. 29 Cognitive sub-function of identification on the concept-based performance level

Because of the identification of certain violations of the a-priori known motivational contexts, the cognitive function of *goal determination* derives *goals and constraints* delimiting future action to comply with the *motivational contexts* in the course of further performance on the concept-based level. They are deduced from wanted work rewards and can be considered as part of our a-priori mind setting about values which we make use of in all situations we encounter in life, including all kinds of situations while we are working. There is a kind of mental automatism which we are not quite aware of, and which has established the explicit a-priori knowledge of motives in harmony to an implicit value system we are conditioned to (see Chapter 3.2.1.5). As a consequence of an actual situational context some of the existing motives will mentally be activated and

transformed into actually relevant new tangible goals. In that sense, goals must not be mixed up with work objectives. The goals and constraints constitute the state of the world currently aimed at, on the basis of which the cognitive function of *planning* will be initiated. Planning or problem solving as a synonym tries to concatenate several intentional steps in order to minimise the difference between the current situation and the desired state, when executed. For that purpose planning selects from alternatives how to proceed, taken from the knowledge of *task options*. Hopefully, this leads to with the accomplishment of the actual goals. The chosen options result in a *task agenda* to be pursued in the further course of work, i.e. the plan for action.

As one possible step towards further refinement one could also think of an involvement of the concept-based behaviour level without the necessity of a new planning process is not needed. This is the case in a situation which cannot be immediately associated to a known task situation. Then, the existing cues are processed by the identification function, possibly resulting in a concept match which might nevertheless trigger the task determination function. For the sake of simplicity at this point, we will not pursue this further in this book, though. A typical example would be a little error of the human operator which nonetheless needs more complex deliberations to have it identified, but which can immediately be counteracted only interrupting the given task agenda in the sense of a task situation of the aforementioned type 2.

The behavioural functions just described which comprise the task determination and the task execution on the procedure-based level as well as the goal determination and planning on the concept-based level, correspond to the cognitive functions of voluntary action as described in Chapter 3.2.1.5. Here, the pre-frontal, the orbito-frontal cortex, and the cingulate cortex play a major role, in connection with the limbic system. This is not a straight forward pathway of action determination as indicated in Figure 27 in a simplified fashion. The simple block diagram in Figure 27 neither indicates the emotional behaviour corresponding to the limbic loops. Rather there are several interplaying loops via the working memory. From a work system engineer's point of view, who is working on the design of cognitive units as part of work systems of vehicle guidance and control, one can still use this scheme as a model of artificial cognition. Modelling of emotional behaviour is not of primary interest for him.

3.2.2.3 A Simple Framework of Cognition

In the framework to be sketched here the biopsychological findings on human cognition and those of the behaviorists shall be combined. The relations between both the functional levels from the biopsychological point of view and the behavioural ones as discussed in the preceding section can be illustrated in a two-dimensional array as a framework, indicating which functions cover which macro types of behaviour. This may serve as a vehicle for the transition from human cognition to artificial cognition. In this framework we find both the functional labels as used in Chapter 3.2.1 and those as mentioned in Chapter 3.2.2. As mentioned earlier, it provides a good basis for the functional demands which have

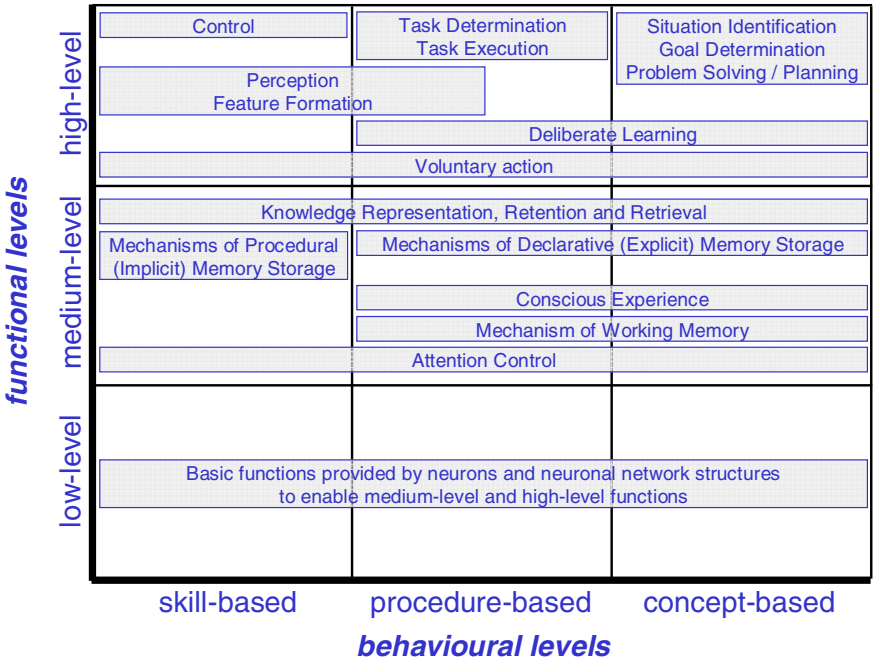


Fig. 30 Framework for functional and behavioural aspects of human cognition

to be satisfied in artificial cognitive systems in order to generate certain behaviour. The most favourable artificial cognitive system is supposed to address each cell of the array. The other way around, the framework can be used to show where the blank spots are left in the design of an artificial cognitive system. This framework will be used, too, in this book to indicate what is covered by the implementation examples which are described in Chapter 6.

The top row of array cells in Figure 30 contains all behavioural functions as high-level functions which we know from the block diagram in Figure 27. In addition, there is the high-level function of deliberate learning which is not accounted for in the Rasmussen diagram. The second row identifies the functions as mainly described in Chapters 3.2.1.3 and 3.2.1.4. The functions in the bottom row have been coarsely described in Chapter 3.2.1.1 and partially in Chapter 3.2.1.2. Here, we only know about the macro structures and the nerve cells as building elements, but still we know too little about how the nerve cells within the macro structures and the macro structures are interacting that we are in a position to describe the functional brain “hardware” in detail.

Depending on the design requirements, the designer of artificial cognition may make his decisions from hereon, to match the behavioural requirements with what the technology offers as bulding blocks.

Chapter 4
Dual-Mode Cognitive Automation in Work Systems

Considering the system design in work environments, engineers as opposed to psychologists are interested to build adequate tools and support systems for human operators for the sake of good, satisfying work results. The engineers' focus lies on the issue of system synthesis in the work environment rather than system analysis, thereby working around the human operator, who is considered as a given system element. As to the human component, work system designers can only make a good job, if they do not ignore what was already found about human factors, including human cognition. These findings are indeed of great value for the work system design, i.e. to define when and how support for the human operator by artificial means, in particular automation, is to be considered for the sake of both work process performance and that there is a good feeling on the human's side that he has everything well under control.

Technical systems do not turn out from a lengthy random process of evolution like biological systems. Their design can be immediately adjusted, if new task requirements have to be coped with and possibilities for design improvements become obvious. This is also true for potentials which arise from the introduction of cognitive automation, to be realised by means of artificial cognitive systems which deploy in part or as a whole those kinds of cognitive functions as known from human cognition. This might involve the kinds of high-level cognitive functions which are depicted in Figure 27 as an interpretation of the qualitative Rasmussen model. However, it will not necessarily be pursued in system design to make artificial cognition perform precisely the same as humans do. Rather, solely the requirements for the work system performance and costs will provide the criteria for the decision about which cognitive function will be designed and how similar it should be to human cognition.

Cognitive automation can be introduced into work systems in many ways and in varying degrees. Referring to the format of defining *levels of automation* in different dimensions [Sheridan, 2002], we can add another dimension indicating the degree of cognition by taking the framework of functional and behavioural levels of human cognition as a basis (see Figure 30 in Chapter 3.2.2.3). In the following, we will define only two major behavioural levels under this dimension. Level 1 stands for what we designate as *conventional automation* and what will be discussed in more detail in the following section. Level 2 stands for what we call *cognitive automation* which comprises the work system components which include functional aspects of cognitive behaviour as depicted in Figure 27 which go beyond

R. Onken and A. Schulte: System-Ergonomic Design of Cognitive Automation, SCI 235, pp. 79–127.
springerlink.com © Springer-Verlag Berlin Heidelberg 2010

those of conventional automation. Cognitive automation will be introduced as a new design option for enhancements of work systems in *two* distinctive possible modes (dual-mode) as will be also described later in more detail.

4.1 Conventional Automation

Conventional automation is the employment of automated functions in work systems we are used to so far. Many people think it is the only way to automate. One fundamental characteristic of conventional automation is the fact that it is focused on functions of *execution* as part of the operation-supporting means. Regarding vehicle guidance and control work systems, for instance, one would think in this context of autopilot and flight management systems for the cockpit work site of pilots or of cruise control and the anti-lock braking system (ABS) in our cars (see also Chapter 2). These automated execution functions include at one end subtask functions like those, for instance, to alleviate the operator's access to important information (automation of information acquisition and display, see [Billings, 1997]). At the other end, execution functions for system control (automation of system control) and higher prefixed functions for problem solving and planning tasks like those of the flight management system and the navigation system in our cars (automation of system management) are included.

In addition, if we consider the cognitive elements of a conventionally automated function, the following characteristic has to be accounted for, too:

Conventional automation is never driven by any own explicit or implicit motivational contexts as depicted in Figure 27. There are no own initiatives subject to own motivational contexts to change the current task in case of any unforeseen situational interference. This is exclusively left to the human part.

Therefore, only the motivational contexts of the human operator and the work system designer are the driving mechanisms to operate the work system. The designer, based on his/her motivational contexts, tries to imagine the motivational contexts of the human operator while operating the work system subject to the work objective. Having this in mind the designer tries to envision the associated task situations, to figure out the tasks (see also Figure 27) to be automated, and possibly to envision goals to be associated to an automated planning function, if this would have to be designed, too. That means: **all situational options have to be considered in that design process**. Obviously, this is not an easy job if not an impossible one. The more complex the work process is the more it becomes uncertain in a particular situation that the automated functions are being in sufficient compliance with the motivational contexts of the human operator. The crucial fact is that the automated function cannot verify on its own here and now in the course of the work process, whether it really acts in compliance with the motivational contexts of the human operator. One has to live with the fact that mismatches might occur and might have fatal consequences in extreme cases.

A very simple example might illustrate this. If a pilot activates the autopilot for the subtask “altitude hold”, the system is doing its best to comply with the assigned altitude, even if a high mountain might be in the way. The prime goal to warrant safety in this case which the pilot has in his mind as motivational context, i.e. to explicitly avoid a crash into the mountain, is not known by the autopilot. Therefore, it does not care. Here, the deficiency of the conventionally automated autopilot function of only complying with a task command of holding an assigned altitude and not accounting for any more possible superiror goals makes the work system vulnerable to human errors in the course of the work process. Admittedly, it is hard to believe that the pilot does not notice the threat of the mountain popping up, at least if flying under VMC (Visual Meteorological Conditions). But think of the accident of the American Airlines flight 965 from Miami to Cali (Columbia) in 1995 which happened in poor visibility conditions (see also [Baberg, 2000]). Being inbound to Cali about 35 miles from the destination on a southerly heading the air traffic controller offered a straight in approach with a much shorter distance to touchdown than planned. The crew accepted this clearance, although the airplane was too high and too fast. Immediately, there was a high workload on the crew to keep everything under control. Power was reduced and speed brakes extended to acieve a high descent rate. Approach charts had to be changed and the flight management system (FMS) had to be reprogrammed. When the controller cleared the Boeing 757 to fly directly to Rozo which is a radio beacon close to the runway, the copilot keyed into the FMS its identifier “**R**” as being looked up from the map and executed the entry right away. The plausibility of the entry was not checked, though, due to high workload. Now, being under automatic control by means of the FMS the airplane was immediately leaving the course for the straight in approach and made a left turn towards a mountainous area, still with a high descent rate. The crew did not notice this considerable change of the flight path until the aircraft had turned up to about 90^0. Even sharply turning back the plane towards the approach centreline when noticing the wrong heading, could not prevent the accident. The airplane crashed into a mountain about 20 miles short of its destination. This controlled flight into terrain (CFIT) mainly happened because the crew was kept up with too many activities to manage the landing approach. Therefore, they did not sufficiently abide to the standard operating procedures (SOPs) to pay attention to important cues for situation awareness, apparently not knowing of the ambuigity problem when entering the identifier “**R**”. In fact, “R” is the correct code for Rozo as far as the approach chart is concerned; but the FMS database was programmed according to the ARINC standard. This makes a terrible difference, because in this very specific case the identifier “R” in truth stands for a so-called **R**omeo beacon close to Bogota just not too far away. Typically, the FMS, although a rather complex operation-supporting means of conventional automation for flight planning and execution has no idea about what the motivational contexts of the piloting crew really were. It could not do anything to avoid the accident. It lacked the necessary cognitive capabilities which were exclusively reserved to the human crew.

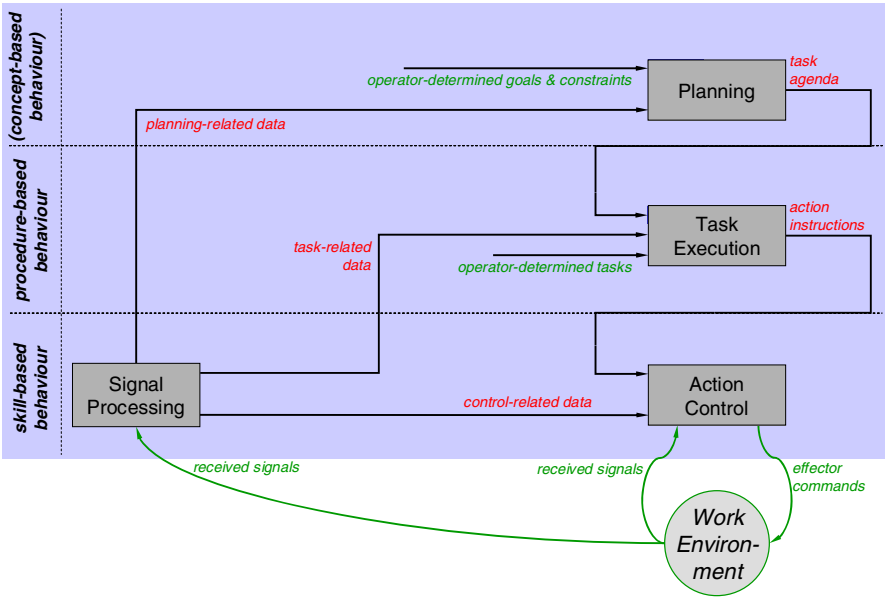


Fig. 31 Potential coverage of cognitive functions by conventional automation Reference: cognitive functions according to Figure 27

This example illustrates that we have not named conventional automation that way in order to delineate between past and future developments of automation only, although this is probably the case to a very high degree. We rather intend thereby to mark the principle difference in design philosophy when talking about conventional automation in distinction to what we shall describe later on (see Chapter 4.3) under cognitive automation.

We have to account for two different ways of introducing automation into the work system design. On the one hand, there is the concept of *operator-controlled automation* which can be considered as what is known under supervisory control in a wider sense [Sheridan, 2002]. In this case, the automated function is activated, monitored, and turned off by the operator in the course of the work process. By experiencing the behaviour of the automated function the operator forms a model of its behaviour and is using it correspondingly subject to his/her motivational contexts. On the other hand, there are automated functions which are to be dealt with as an intrinsic part of the vehicle or other components of the operation-supporting means, not leaving an option for the operator of turning off these functions or directly controlling them. The operator might even be totally unaware of the fact that there exists an automated function of that kind in the system. At one end, for example, the main part of information displays we are used to in vehicles and the means to derive the information presented belong to that category. At the other end the ABS in our car is also a typical example. In the following, we shall call this concept *built-in automation*. It is entirely up to the work system designer to decide whether a function of that kind is to be implemented as well as when and

how it is to be activated in the course of the work process. Therefore, the design of conventionally automated functions of this kind has to be even more careful. Automation in work systems has usually been started by introducing built-in automation. Built-in automation and operator-controlled automation often are present in work systems working in parallel. In the following, the main focus will be on operator-controlled automation (supervisory control).

Looking at the actual cognitive capabilties of conventional automation when taking Figure 27 as a reference, one can only figure out functions which are part of low-level human cognitive behaviour. Therefore, one might even hesitate to call them cognitive functions. In Figure 31 this is depicted for the case of operator-controlled automation, thereby hooking up as much as possible on how the human cognitive functions were presented in Figure 27. Accordingly, considering the function of goal determination and the pertinent knowledge about motivational contexts, this remains with the human operator and may lead to *operator-determined* inputs about goals and constraints, presuming that there exists an automatic planning function at all. The same is true for any operator input determining a new task in distinction to those of the pre-planned task agenda. Since the operator exclusively represents the motivational contexts for the work process, his/her task inputs principally are of higher priority than the automatised ones as part of the expected preset sequence of tasks according to the task agenda. In that case this sequence of planned tasks is interrupted, whether to directly initiate a new task when reacting on an unexpected situation or to reinitiate the planning for a new task agenda. In Chapter 3.2.2.2 the corresponding task situation was already addressed as one of type 2 in distinction to those of type 1 for the tasks being carried out in compliance with the task agenda. **Hence, a crucial part of the task determination function is that of type 2 to be able to react on an unexpected situation and to decide about a new task not being part of the current task agenda. This is not automated in case of conventional automation.** Therefore, as depicted in Figure 31, it makes sense when considering conventional automation to merge the remaining part of the type 1 task determination function with the task execution function, since underlying motivational contexts of the human operator are not involved in the task determination process. The type 1 task determination function is simply controlled by task-related situational data which indicate the completion of a task and trigger the subsequent task according to the task agenda. If there are functions of built-in automation in addition to the operator-controlled ones, they can be thought as an always automatically engaged task as part of the task agenda, working in parallel and even being able in extreme cases of functional conflict as *predefined* by the designer to override effector commands which would be expected from the execution of the sequence of planned tasks in the task agenda. For instance, while an automatic descent is taking place when approaching an airfield, this function might be made ineffective by a built-in function of executing a terrain avoidance task. There is *no deliberation* taking place on whether this is really in sufficient compliance with the motivational contexts of the human operator.

We also recognise immediately from Figure 31 that, as a consequence, some of the cognitive functions in Figure 27 are missing in case of conventional automation. Since the goal determination is exclusively the responsibility of the human operator, also the function of identification is missing in Figure 31. In addition, the function of feature formation is hardly realised in a way equivalent to human cognition. Therefore, this function is better relabelled as *signal processing*. It is only providing the time-varying input data the automatic task execution is usually tied to, whether it is directly determined by the operator or whether it results from the task agenda based on the operator-determined goal(s). The planning function, if we refer to the concept-based cognitive human behaviour level as depicted in Figure 27, cannot truly be considered as concept-based for itself, since the genuine functions of the concept-based behaviour level like identification and goal determination lie in the responsibility of the human operator. In the context of conventional automation the planning function rather becomes a higher level kind of execution task, assigned by the human operator, who has got the motivational contexts to define the goals to be planned for. All this essentially results from the already mentioned central characteristic that it cannot be perfectly ensured that conventional automation is by design satisfying the motivational contexts of the operator.

Figure 31 is also illustrating that explicit a-priori knowledge similar to that of human cognition according to Figure 27 is usually not present in ordinary conventional automation. Since there is no function of goal and task determination, it often does not make much sense for the sake of computing efficiency to explicitly represent a-priori knowledge.

That does not mean that there is not made use of any knowledge at all, but this is usually implicitly represented as task-specific knowledge and, therefore, not available for use outside the dedicated context given by the function which contains it. Rather certain data bases of explicit data might be available for conventionally automated functions like a data basis of the navigation system in our cars for geographical data or the digital terrain data base of a ground proximity warning system in an airliner. However, this kind of explicit knowledge usually does not show any logical relations between these data. Referring to the digital terrain data base, any knowledge about logically connected facts like “where a valley is and where a mountain is” is usually not explicitly represented in that system.

At this point, we also have to mention a specific class of conventional automation which indeed is working on the basis of explicit a-priori knowledge. This class can be subsumed under the label of *expert systems* as knowledge-based systems, which we are already used to since some time, mainly as systems for planning or problem solving in more general terms (see [Shortliffe, 1976]). These systems suffer from the same crucial limitations of conventional automation like the lack of

- knowledge about the motivational contexts of the human operator and of
- cognitive functions such as goal determination and the crucial part of type 2 task determination which are dealt with by the human operator alone.

The data processing usually is much more elaborated compared to that of ordinary conventional automation. The resulting output typically is the problem solution in terms of a decision proposal. It can be a task agenda as depicted in Figure 31 but also simply just a proposal about “yes” or “no”. Further entailing functions like task execution and action control as depicted in Figure 31 and as we are used to for vehicle guidance and control are not part of these systems. From the historical point of view, expert systems emerged at a time when the so-called artificial intelligence developments made available more efficient techniques for the processing of explicit knowledge. This was a great advancement. By considerably extending the potentials of automation that way, it can probably be considered as the main achievement to enable the transition to cognitive automation. Insofar, it is not unreasonable to dedicate an intermediate level to these systems between conventional (level 1) and cognitive automation (level 2). For the sake of not letting things become too complex in this respect, we are reluctant, though, to follow that viewpoint in this book.

4.2 Experience with Conventional Automation

The concept of having conventionally automated functions in the work process as low authority artificial specialists of being good for a particular execution support but lacking the full perspective, was very convenient as long as only simple automated functions were used for simple tasks. It was of no harm that the responsibility load not to violate the prime goals of the work process was exclusively on the human operator’s shoulders. However, in highly automated work systems for complex and demanding work objectives as we are used to now, like, for instance, a demanding mission of a modern fighter aircraft. Conventionally automated functions of high degree of complexity are accommodated as operation-supporting functions. They are considered as necessary to keep the human operator freed from a sufficiently high number of tasks which otherwise could lead to overloads. This might become critical, though, since with increasing conventional automation complexity, the task of the operator as supervisor becomes more and more complex, too. Instead of the intended unloading, danger of overcharge of the operator may inevitably arise in certain situations, resulting in a loss of situation awareness and control performance, i.e. lacking to comply with the prime goals. The operator might be unaware of discrepancies between sub-task activities of automated functions or/and the prime goal necessities or even of his/her insufficiency to adapt to this inadequacy. This is just the opposite of what was intended by the introduction of automation.

Figure 32 is illustrating the principle effects for a pilot’s work site, although not based on real data. In fact, experiences with conventional automation show that it reduces the demand on the pilot’s resources on the average most of the time, but it might also generate, rarely though, excessive load which would not have happened without automation. Many accidents can be attributed to that phenomenon. [Hollnagel, 1995] is talking about the erosion of operator responsibilities in this respect, but still sees the situation-dependent goal-setting as

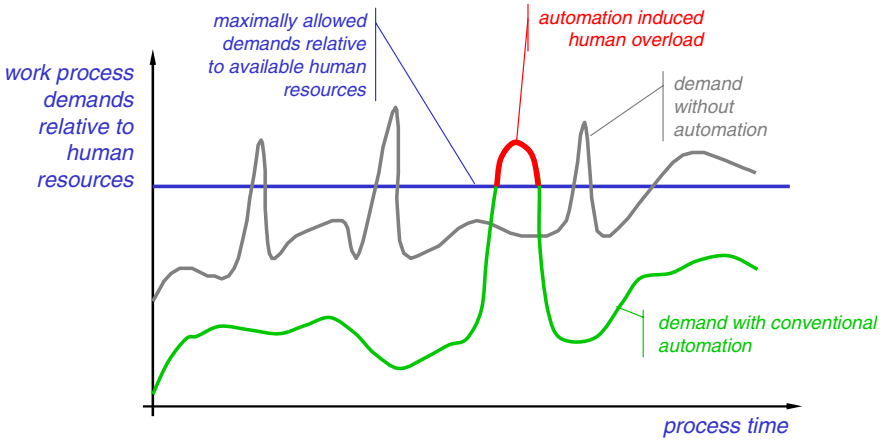


Fig. 32 Effect of conventional automation on relative pilot resource demands during flight mission

the responsibility of the human operator. Indeed, there has been taking place a steadily increasing encroachment of automation upon human work activities up to a very high degree of automation complexity, all as part of the operation-supporting means.

[Sarter & Woods, 1995] for instance brought into light the fact that pilots of modern airliners have to deal with a proliferation of flight system modes (see Table 2). These authors also claim:

> *"As technology allows for the proliferation of more automated modes of operation…human supervisory control faces new challenges. The flexibility achieved through these mode-rich systems has a price: it increases the need for mode awareness – human supervisory controllers tracking what their machine counterparts are doing, what they will do next, and why they are doing it…While we understand a great deal about mode problems, the research to examine specific classes of countermeasures in more detail and to determine what is required to use them effectively, singly or in combination, is just beginning." [Sarter & Woods, 1994]*

In order to illustrate the problem of mode awareness, due to the proliferation of modes which usually goes with intrinsic mode coupling, an example for an accident is given caused by this effect. The accident occurred at Nagoya (Japan) with an Airbus 300-600R of the China Airline in 1994 (see [Billings, 1997]):

> *"The airplane was in a normal approach to landing at the Nagoya runway 34 in visual meteorological conditions. The speed was automatically controlled by the autothrottle system. The copilot flying apparently triggered the autopilot TOGA (takeoff-go-around) switch (possibly inadvertently), whereupon the automation added power and commanded a pitch-up. The captain warned the copilot*

> *of the mode change and the copilot deactivated the go-around mode. Everything seemed to be recovered, but when the copilot continued to attempt to guide the aircraft down the glide slope the automation countered his inputs with nose-up elevator trim. Ultimately, with stabilizer trim in an extreme nose-up position, the copilot was unable to counteract the trim with nose-down elevator and turned off the autopilot. However, it was too late. The aircraft had already nosed up to an attitude in excess of 50°, stalled and slid backward to the ground, 264 people were killed in the crash."* [Billings, 1997]

In the past, system engineers misunderstood mishaps of that kind by vigorously pushing further increased use of this type of (conventional) automation, not carefully enough looking for possible negative effects on the work system as a whole. This fits well the characteristics of a vicious circle. This is depicted in Figure 33 for the application field of UAV (Unmanned Aerial Vehicle) guidance and control.

Table 2 Flight Management System (FMS) and Autoflight modes in the Airbus A320 ([Sarter & Woods, 1995], Human Factors, 37(1), 5-9, Copyright 1995 Human Factors and Ergonomics Society. All rights reserved)

Autothrust Modes	**Vertical Modes**	**Lateral Modes**
TOGA	SRS	RWY
FLX 42	CLB	NAV
MCT	DES	HDG/TRK
CLB	OPEN CLB	LOC*
IDLE	OPEN DES	LOC/APP NAV
THR	EXPEDITE	LAND
SPD/MACH	ALT	ROLLOUT
ALPHA FLOOR	V/S-FPA	
TOGA LK	G/S-FINAL	
	FLARE	

As (historical) starting point of the vicious circle, work systems with none or only little automation might be taken. Here, manual control was the predominant mode of operation, while the supervision of automated systems could be neglected. However, as characteristic of reactive designs to inescapably occurring human error, automation and its further extention is taken as the solution. Designers transfer the authority over more and more comprehensive work process tasks from

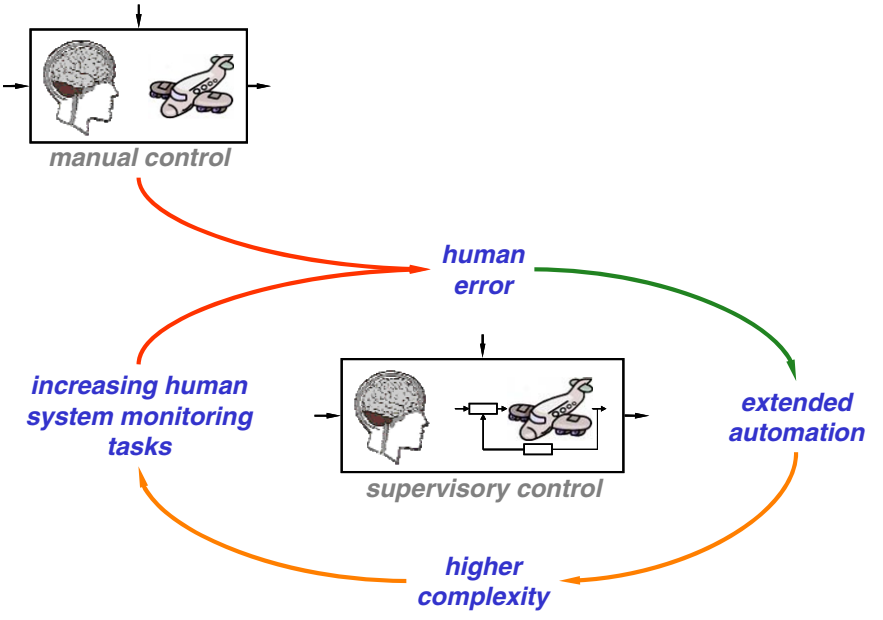


Fig. 33 The vicious circle of extending conventional automation

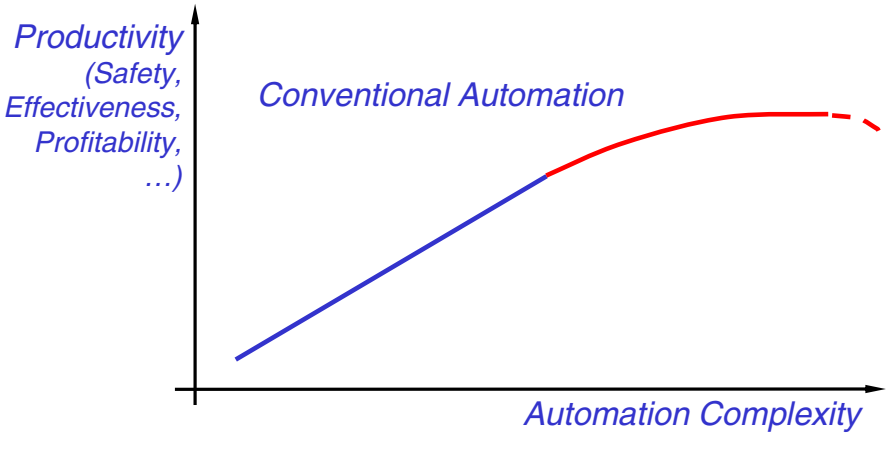


Fig. 34 Productivity of conventional automation as a function of system complexity

the operating force i.e. the human operator to increasingly complex automated functions being part of the operation-supporting means. Although, being relieved from workload in the first place, the human operator will be burdened with extra demands from increasing system monitoring and management tasks. At a certain point the human operator cannot but fail in the supervision of suchlike automated systems. In reaction, as being used to, typically the only solution considered is to

further extend the share of automation as part of the operation supporting means in conjunction with further increase of complexity. In conclusion, as we are interested in a design with gains in productivity of a work system, we experience that the increase of system complexity due to further addition of conventional automation results in less and less increase of productivity gain and might eventually even end up with decreases (see Figure 34).

Obviously, the limits of growth of productivity through increase of conventional automation are already attained in certain domains like aviation. Apparently, there were lacking well-founded methods and theoretical frameworks which thoroughly account for the overall performance demand on a work system as a whole (including the human operator). The FAA report [FAA, 1996] was an attempt to promote endeavours in that direction. Unfortunately, the predominantly drawn conclusion, however, was to pursue only the probably simplest one to realise, i.e. to intensify the training of the human operator instead of intensifying research on a work system design with the potential to effectively counteract the deficiencies of conventional automation. Intensified training does not do any harm, but it cannot be very effective as to the deficiencies explicitly mentioned about implicit automation.

The deficiencies of conventional automation as being experienced (see [Sarter & Woods, 1995] and [Billings, 1997]) are well-captured in explicit terms in the meantime by the phenomena known under automation

- brittleness
- opacity (entailing increased demands on monitoring the automation status)
- literalism
- clumsiness, and
- data overload,

all of them being consequences of too high system complexity.

Brittleness is describing inadequacies of conventional automation due to the fact that it is almost impossible to verify during the development process of highly complex functions that everything is working allright in all possibly encountered situations. Therefore, brittleness refers to the characteristic feature of conventional automation to perform well, i.e. according to specification, within clear-cut operational limits, but close or even beyond these limits usually quit service rather rapidly. These limitations of proper operation are usually not known by the human operator and there will always be a situation number “n+1” which could not be foreseen during the design process.

Opacity typically refers to situations when the human operator gets surprised by what turns out as the output of the automated function, although there is no system error. The human operator simply cannot know all about the complex functionality he is tying to use and has no chance to understand what is going on. [Wiener, 1989] condensed this phenomenon under phrasing the questions the operator may ask in situations like that without getting answers:

- What is it doing?
- Why is it doing that?
- What’s it going to do next?

This is why [Sarter & Woods, 1994] also refer to these questions in their statement as cited above.

Literalism simply is the kind of inflexibility of a computer program of conventional automation which stubbornly carries out how it is instructed by the human operator. It does not necessarily account for the objectively given demands in the context of certain situations in order to comply with the pertinent work objective. In other words, literalism describes the character of a conventionally automated system, strictly to follow the instructions given by the human operator no matter whether they are correct or might be wrong. Conventional automation does not question or check control operations concerning their ability to make sense in the given context.

Finally, *clumsiness* (see [Wiener, 1989]) stands for the type of automation which decreases the operator's workload in situations when it is anyway low, and which does not help or even increases the operator's workload when it is already high. This mirrors the fact that conventional automation usually does not help in unanticipated situations when it is urgently needed. A cause for clumsiness can also be the presentation of excessive amounts of information to the human operator, overloading him with data which are only partially needed. Clumsiness can also be attributed to cases which lead to behavioural traits of the human operator to "trick" the automated function concerned. [Billings, 1997] gives an example of the possibly automated vertical navigation in the aviation domain. The automatic system calculates an optimal point when to start the descent for the landing approach. The algorithm often denies the pilot's desire of a smooth approach starting at a point where the plane is "not too high and not too fast". This causes the pilot to insert wind data as important input for the algorithm which indicate a higher tailwind component than what is really communicated such that the start of the descent is calculated to be at an earlier point of time.

These intrinsic characteristics of conventional automation are being confirmed as shortcomings by a number of studies like those already mentioned from Sarter and Woods of the aviation realm as forrunner domain, or the FAA study published by [Funk et al., 1996]. The have become concrete by the multitude of mishaps and and catastrophes which occurred over the years. Most of them are caused through human failure, whether it happened in the course of work system design or in the course of work system operation. We know that nothing is perfect in this world, but we have to take care that built-in barriers avoid that imperfections lead to disaster (see also [Hollnagel, 2004]). In [Billings, 1997] some representative accidents are discussed in more detail. They reveal why the potential of conventional automation as part of the operation-supporting means has been overestimated for a long time.

At least, certain principles for the so-called *human-centred automation* were formulated by [Billings, 1997] to counteract these drawbacks. In this context, he derived some general recommendations. He claimed that all acting components of the work system must

- be actively involved,
- keep each other adequately informed,
- monitor each other, and therefore should be predictable, and
- know the intent(s) of all other acting components.

These requirements demand for various capabilities of automation such as comparing expected and observed activities of the human operator in order to be able to monitor him in terms of his intents.

4.3 Cognitive Automation

As has become evident from Chapter 4.2, automation often has been implemented in work systems in the past without use of well-founded theoretical frameworks which thoroughly account for the overall performance demand that a work system (including the human operator) as a whole should be designed for, i.e. to comply best with the work objective. The lack of a theoretically-founded systematic approach is the reason why the potential of conventional automation has been overestimated for a long time by system engineers (see Chapter 4.2) and that, on the other hand, ergonomists often underestimated the potential of automation as such (see [Bainbridge, 1983]).

We hypothesise at this point that the crucial technological step to avoid the drawbacks of conventional automation is the introduction of artificial cognition in terms of *cognitive automation*. This allows to deal with higher degrees of complexity the same way as humans can brilliantly do, based on motivational contexts. Higher automation complexity certainly is the price to pay for making possible significant gains in productivity of work systems, but we will realise in the following of this book that in fact cognitive automation can pave the way towards these gains without loosing much as it is the case for conventional automation. Therefore, it is a crucial issue in work system design to develop a methodological approach for exploiting the automation potential in the most beneficial manner and therefore making use of artificial cognition by means of cognitive automation in addition to conventional automation. By the way, this can be considered as the major issue of this book.

How this can be achieved by a more systematic design strategy will become evident in the following. First, though, we shall clarify what cognitive automation means in distinction to conventional automation before describing the benefits of it and the design requirements to make use of it in a work system. Also in case of cognitive automation, we also have to deal with both ways of automation, ACUs as operator-controlled or built-in automation.

Taking again the functional model of human cognitive behaviour in Figure 27 as a reference, everything what includes a functional component of feature formation as opposed to the pure data processing of conventional automation and what goes otherwise beyond conventional automation in terms of replacing the human operator for the cognitive functions of goal and task determination, will be accounted for in the following as cognitive automation. Thereby, the goal determination is based on a-priori knowledge of motivational contexts and the input of the identification function, whereas the task determination relies on a-priori knowledge of task situations according to Figure 27. At the same time a corresponding functional component of feature formation is evolving. At the low end, only a task determination function is involved. At its best, cognitive automation comprises the whole set of functions as shown in Figure 27. The more

knowledge is involved, the more comprehensive are the cognitive capabilities. This indicates that theoretically there is an unlimited potential for increase of performance. In summary, cognitive automation stands for artificial capabilities

- to understand the situation in case of unforeseen events and to independently interpret it in the light of the known *motivational contexts* as drivers for voluntary actions,
- to develop an understanding of the necessary sequence of actions best-suited to accomplish the desired result according to the assignment, thereby distinguishing between important and unimportant information, urgent and less urgently needed actions,
- to perform those actions which are authorised by its assignment, and
- to effectively initiate the necessary communication to other units of the pertinent work environment, thereby evening up how to proceed in case of conflicts and opportunities.

These are the cognitive capabilities, in particular that of independently interpreting the work situation in the light of known own motivational contexts, whose potential for the work system design will be investigated further in the following. In that sense, we consider in the following of this book cognitive automation as being realised in terms of a particular kind of agents which have got these capabilities and which we call ***artificial cognitive units*** **(ACUs)**. Comparing ACUs with humans, we make use of the option in the following that ACUs are merely components analogue to what is meant by the human cognition as the inner functional layer of the human operator in Figure 16. Of course, also the outer layer with sensing and effecting means, as indicated in Figure 16, is necessary for the ACU, too, in order to make the ACU an effective player in the work system. However, in distinction to humans, these could typically also be provided by corresponding separate operation-supporting means as being true for many operational systems. In order to make appropriate use of these means the ACU has got detailed knowledge about their capabilities and functionality.

4.3.1 Cognitive Automation: Mode 1

This section deals with ACUs as part of the operation-supporting means. As a consequence, these ACUs usually work under supervision of the human operator, i.e. they usually are instructed and monitored by the human operator. The main advancement is the fact that it is driven by explicitly represented motivational contexts. Besides others their motivational contexts include the desire to carry out the instructions of the human operator as perfectly as possible. This is the crucial precondition to cope with encounters which have not been envisioned in the course of the design process. Thereby, ACUs make the best use they can of their cognitive capabilities. **However, it cannot be assumed that they have got any knowledge about the work objective the human operator is engaged in**. If the corresponding work system design is a good one, this lack of work objective knowledge is for good reasons.

This is similar to cases when human "specialists" are instructed to work on functions which are needed in a hierarchically higher-level work system with its respective work objective. Normally, forming an own work system with its pertinent work objective being equivalent to the instruction of the higher-level work system, the specialist endeavours to make best use of available cognitive capabilities to comply with the instruction. However, while doing this, he or she is not necessarily also doing the best to achieve the work objective of the hierarchically higher-level work system. This is because the insufficient knowledge about that work objective might just be inadequate.

Now, think of the possibility that this human operator is replaced by an ACU which performs quite similarly. According to the work system definition (see Chapter 3.1.2), the subordinated specialist work system converts into an automated sub-system which automatically becomes part of the supervised operation-supporting means of the higher-level work system. Sure enough, although now belonging to the higher-level work system, it is again not warranted that it complies with the work objective of the higher-level work system. Again, it only complies, if the supervisor's instruction is a sensible one, i.e. being in fact in full compliance with the work objective. In that sense, an SCU in place of the conventional FMS apparently would not have avoided the Cali accident (see Chapter 4.1) either. Its own knowledge of that work objective is still insufficient, although compliance with the instruction is much better warranted.

By the way, be aware of the fact that a separate work system with a human specialist and the replacement of this work system by an ACU as part of the operation-supporting means will never be functionally equivalent, because of the basic human property to be an autonomous entity. It also can be assumed that the cognitive capacity of the human specialist as to the learning capability and the a-priori knowledge as a whole by far exceeds that which is demanded for the specialist function. It just is inherently given and cannot be cut down to what is actually demanded. The ACUs, on the other hand, can be and usually are designed to fit exactly the demands as needed.

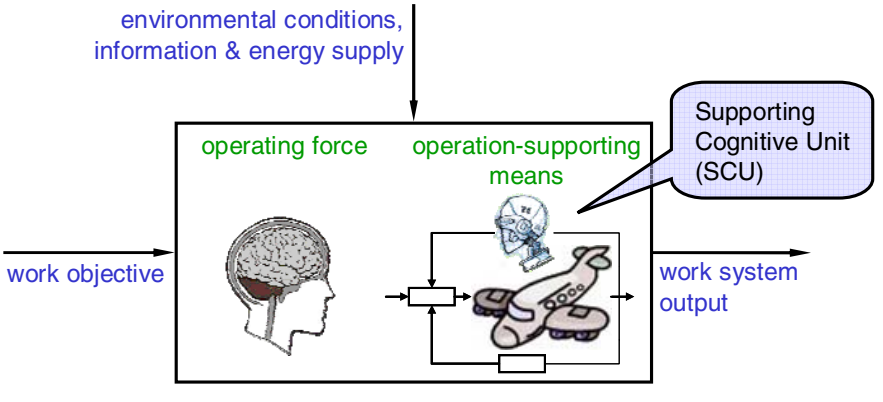


Fig. 35 Work system with supporting artificial cognitive units (SCUs) as part of the operation-supporting means (mode 1)

As a result, Figure 35 can be drawn in more detail making explicit the possible incorporation of ACUs as part of the operation-supporting means in the work system (see Figure 10). We call such a component ***"Supporting Cognitive Unit (SCU)"***. As being a part of the operation-supporting means, we call an SCU a component of *mode 1 cognitive automation*. In that sense, an SCU might be a software agent as a stand-alone component, or a software agent as part of an operation-supporting means like an aircraft which at the same time accommodates the human operator.

Now one might conclude that from thereon all automated functions which are part of the operation-supporting means are to be designed as SCUs. This would go too far, though. Simplicity is always a design goal, not only for the purpose of keeping the cost down. Therefore, the conventional might still be very useful and effective, indeed, for functions with low complexity. This was proven oftentimes in the past. The operation-supporting means of a more complex work system will therefore probably consist of a composition of both conventional automation and possibly mode 1 cognitive automation as depicted in Figure 35.

Anyway, the question is now, alluding to Figure 35: What have we really gained in terms of productivity (including safety) by introducing SCUs as part of the operation-supporting means? No question, there can be achieved a great enhancement of work system performance. In addition, referring to the experiences with conventional automation, we can affirm that the identified deficiencies of conventional automation will be partially overcome by the introduction of cognitive automation in terms of SCUs. For instance, brittleness, literalism and clumsiness, as described for conventional automation in Chapter 4.2, can be at least partially avoided, i.e. within the realm of its function as operation-supporting means only, not with respect to the work system as a whole.

In the context of automation in the automotive domain, [Kopf, 1997] has established 6 steps to go through in order to avoid this kind of brittleness:

(1) Step 1 is devoted to the specification of proposed functions to be automated.
(2) Step 2 has to consider any situational constraints such that they are made known and easily recognised by the human operator (driver).
(3) Step 3 is to make sure that the preliminarily specified functions are clearly delineated from other adjacent or closely related ones. The design of the relevant automation component should take care that the human operator is actively informed about the situation whether he is currently supported by an automated function or not.
(4) Step 4 identifies the operational limitations of the proposed automated functions. Again it is important that these limits are easy to recognise and to be taken care of by the human operator.
(5) Step 5 identifies the technical causes for the limitations just mentioned. The designer can decide on the basis of this information, whether the function has to be dropped, because it is not possible to make the functional limits sufficiently apparent for the driver.

(6) Finally, with step 6 a more detailed consistency check might be carried out for the proposed functions. Usually, a proposed function should be dropped, if it is not feasible that the functional limitations are not

- intuitively apparent
- easy to recognise or
- sufficiently easy to learn.

The typical deficiency of opacity, though, as also described for conventional automation can be fully avoided by introducing SCUs. These are able, in principle, to give an explanation of what has been done by them and for what reason. On the other hand, because they are still under supervisory control of the operating force (human operator), the process of work migration from humans to machines in terms of operation-supporting means is even being put further by the introduction of SCUs with its considerably higher degree of complexity. Unfortunately, this again fits the characteristics of the vicious circle as depicted in Figure 33. Therefore, it makes sense not to rely on cognitive automation of mode 1 only, if there is something else one can take advantage of.

4.3.2 Cognitive Automation: Mode 2

Mode 1 cognitive automation can be of great effect as to the enhancement of work system performance, however there is even more growth potential, if ACUs are co-operating with the human operator(s) as part of the operating force in the work system. In essence, this is not a new idea. It was being intuitively promoted for a long time already (see for instance [Hoc et al., 1995]), although barely realised so far in operational systems. These ACUs work on a level that their motivational contexts include in the first place the motivation to do one's best for the achievement of the work objective in co-operation with the other team members of the operating force. In order to be able to do that in a satisfying manner they must have sufficient knowledge of the work objective, too. Correspondingly, all what they are assigned to do is co-ordinated with the other members of the operating force (including human operators) and cannot but being in compliance with the work objective.

Once this becomes feasible, the solution looks like a system as depicted in Figure 36 which shows a human-ACU team as operating force. Moreover, it also should be noted at this point that it will not be excluded at this point that in case of several humans being involved as members of the team possibly (but not necessarily) one or the other of the human team members might be replaced by the cognitive automation of mode 2. We call this type of ACU ***"Operating Cognitive Unit (OCU)"*** which always is part of the operating force in a work system. An OCU might also be a software agent as a stand-alone team member of the operating force, or as part of a special type of robot as a team member.

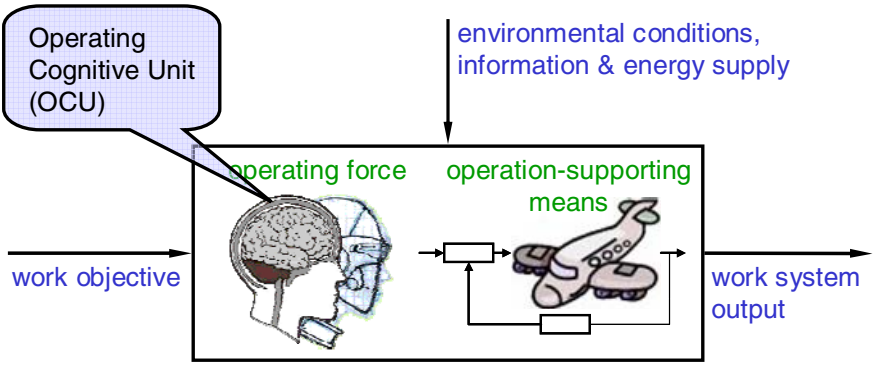


Fig. 36 Work system with co-operation between human operators and OCUs as operating force (mode 2)

As long as there is no effect on the work system environment, mode 2 cognitive automation can take over task commitments in parallel to the human operator. This is in particular convenient for the subtask of situation assessment, regarding the state of the work system components (including the operator) and the work system environment. In the extreme case even intervention in the commitment of the human operator and his actions is principally possible, for instance in case of secure recognition of safety-critical human failures.

The concept of OCUs, being part of the operating force in a work system, is supposed to break with the conventional design philosophy of increasing replacement of human work by increasingly complex automation as operation-supporting means only and the resulting shift of human work to more detached and more burdening supervision as a consequence (cf. Figure 33). In contrast to the kind of automation aiming at taking over human responsibilities, no matter how dumb or intelligent that kind of automation might be, an OCU in the sense of human-machine co-agency enables

- to collaborate in a close-partner work relationship with the human operator similar to what we are used to among humans,
- to negotiate within the close partnership the allocation of tasks adapted to the needs of the current situation, and
- to jointly supervise under the consideration of the overall work objective the performance of both highly automated (semi-autonomous) systems and less highly automated systems as part of the operation-supporting means.

However, to be very clear, this kind of automation should not work really autonomously, although possibly being capable to do so.

Figure 37 provides a survey of the prominent properties of OCUs as part of the working force in a work system example of the aviation application domain, i.e. a pilot or a pilot crew as human operator(s).

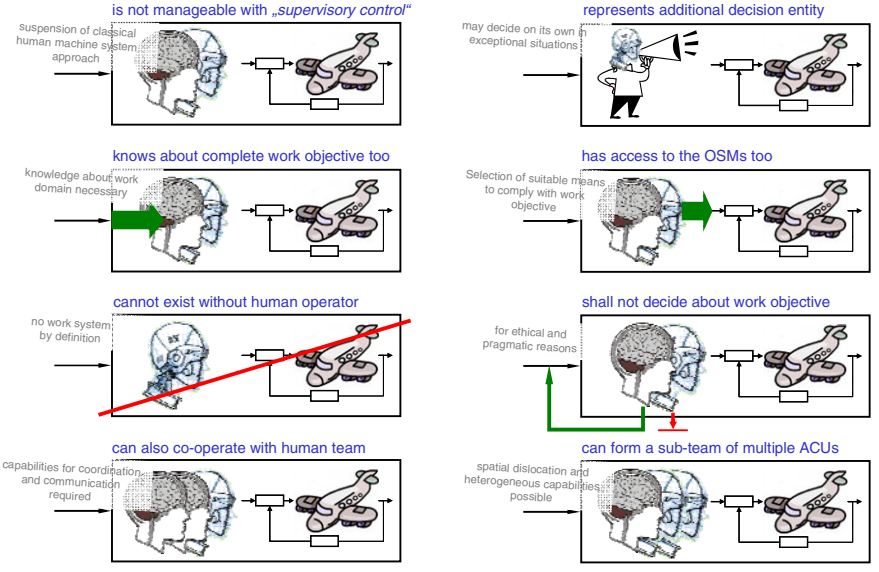


Fig. 37 Properties of OCUs as mode 2 cognitive automation

In fact, based on these properties, cognitive automation, especially by means of OCUs is capable to make up for all deficiencies which were experienced with conventional automation, i.e. brittleness, opacity, literalism, and clumsiness. One can expect that an OCU behaves like a human in a sensible manner no matter which situation will be encountered. The behaviour is based on the *a-priori knowledge about the motivational contexts.* Part of the motivational contexts is the explicit motive to behave according to the also known *work objective* of the pertinent work system. Therefore, as opposed to SCUs, brittleness as well as literalism with respect to warrant compliance with the work objective can be avoided by virtue. If there occurs a situation which was not accounted for by the designer of the OCU by corresponding a-priori knowledge about task situations according to the procedure-based behaviour level (see Figure 27), the also implemented concept-based behaviour level will deal with it by deriving a correspondingly adapted task agenda. The problem of clumsiness is dealt with in a similar manner. This works the better the more emphasis is put on aspects which are likely to be difficult to deal with by human operators. Whenever wanted and if designed for that, an OCU (as well as a SCU), based on its explicit a-priori knowledge, can also offer an explanation of why it is doing what it does. Therefore, the problem of opacity can also be mitigated in a systematic way. The same is true for the problem of literalism concerning conventional automation as long as the situation assessment comprises a-priori knowledge adequate to reveal conflicts with the known work objective. Avoiding clumsiness, finally, is a matter of how much a-priori knowledge is implemented in an OCU or SCU, thereby putting emphasis on aspects which are likely difficult to deal with by human operators.

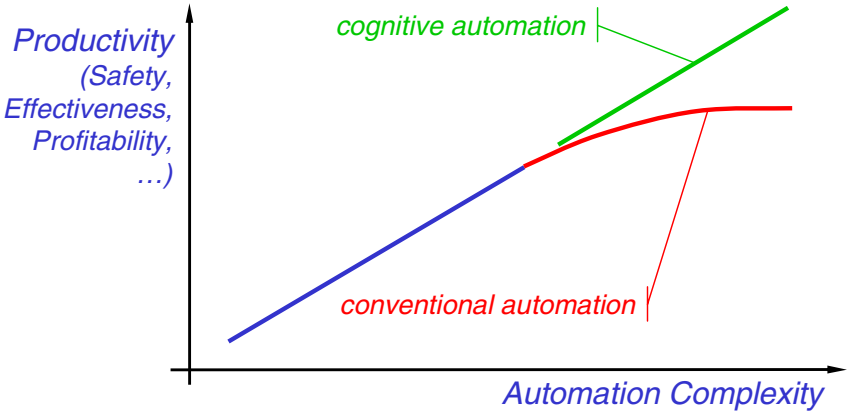


Fig. 38 Effect of conventional and cognitive automation on productivity as a function of automation complexity

At this point, it should be noted, though, that cognitive automation in the work system design is not meant in order to totally replace conventional automation. Since conventional automation as established today does a very good job in many cases, there is no compelling reason for that. Why should we want to inconsiderately break with well-proven technology of conventional automation? In most practical cases a work system design will not be made from scratch, anyway. Only in cases of higher system complexity, as discussed previously, conventional automation might be just not good enough. Therefore, we offer a way to overcome known shortcomings by the supplement of what we call cognitive automation, in particular by means of OCUs (see Figure 38).

Obviously, compared to conventional automation, complexity is a much less limiting factor for cognitive automation by way of OCUs. This is best illustrated by taking up the metaphor of an ant which is on its way to the place where food can be collected [Simon, 1969]. The way it takes is not like a straight line to the place it is targeting for food. Instead, it can be very complex. Because we know of the *objective* of the ant to reach the food with least impact by obstacles, we understand that deviating from the straight line is only because of the grains of sand and other small or larger obstacles in its way. The globally observable behaviour of the ant is no more that complex, as it appeared in the first place, but follows rather simple guidelines. As a result, this analogy tells us, that complexity in behaviours driven by the complexity of the world can be reduced to some more orderliness by the use of goals for action. Therefore, although higher automation complexity is the price to pay for introducing cognitive automation, it still offers the potential to pave the way towards the gains in productivity of work systems we want.

4.3.3 Conceptual Conclusions

Summing up, automation typically can be broken down in two main levels concerning the degree of cognitive capabilities. We call these levels *conventional*

automation and *cognitive automation* as being distinguished earlier in more detail. Conventional automation is what we are used to in almost all operational systems. Cognitive automation is being on its way to penetrate work systems without necessarily replacing conventional automation, but at least supplementing it wherever it is adequate.

In addition, cognitive automation can be integrated into the work system in two distinct, but not exclusive ways, i.e. by making use of cognitive automation in mode 1 and/or mode 2. Consequently, the work system designer has to consider a *dual-mode cognitive design* comprising ACUs as

- SCUs as part of the operation-supporting means in the work system at a rather low decision authority level, and/or as
- OCUs as part of the operating force in a team formed together with the human operator (human operator/ACU team) at the high-end authority level for decisions in the work system.

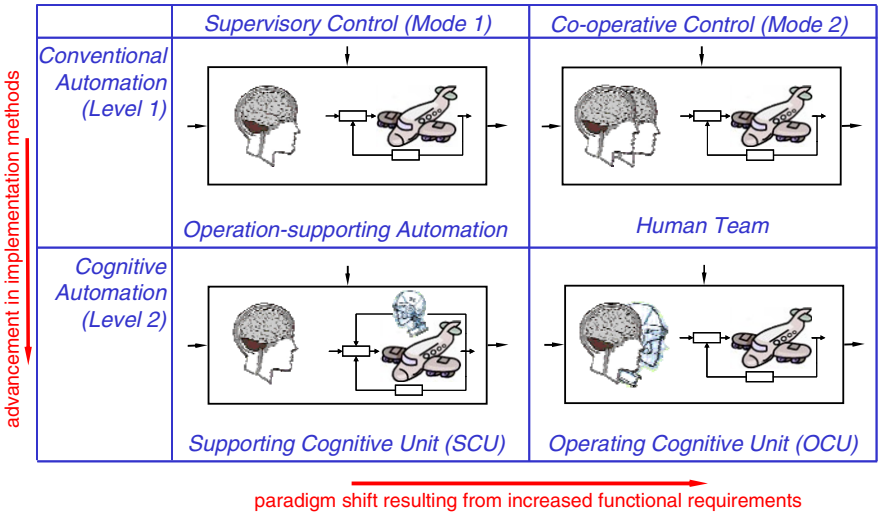


Fig. 39 Changes in work system structure with dual-mode cognitive automation, resulting from technological advances and increases of functional requirements

Apparently, these two modes of automation show great differences regarding design rational and resulting features and capabilities. Correspondingly, they are very different with respect to what the human operator can and will expect from them and what the interaction with the human operator looks like. As shown in Figure 39, mode 1 cognitive automation actually appears as, *supervisory control* and mode 2 cognitive automation as *co-operative control*. As mentioned earlier, in the mode of supervisory control the human operator is placed over the automation like we know from our experience with automatic speed control in our cars. Here, it is exemplified for a work system of the aviation application domain. In

distinction to supervisory control, co-operative control is characterised by the fact that the operating force of the work system consists of more than one operating unit, i.e. a team of at least one human operator and in case of co-operative cognitive automation at least one ACU in addition. We call this ACU an operating cognitive unit (OCU). The team members are **co-operatively** pursuing a common objective which is the work objective of the underlying work system they are operating in.

This is a new and exciting engineering perspective on the one hand, and a new, more complete conception of automation on the other hand. It is accounting for both, to be able to further increase the functional requirements without suffering from the ensuing increase of system complexity on the one hand, as well as the sensible use of these advances in technology on the other hand, i.e. not overdoing it. Herewith, we have got the means to systematically answer the two fundamental questions regarding the introduction of automation into the work system (see [Rouse et al., 1987]):

1) when to automate?
2) how to automate?

Now, the question of "when to automate" can be answered in the way of systematically pursuing both, to overcome human limitations and, at the same time, to let the human operator's strengths become fully effective by a co-operative approach (see also [Rouse et al., 1987] and [Hoc, 2000]). Thus, automation is certainly to be introduced as a provision of capabilities complementary to those of the human operator. This includes, though, **that in turn, the designer also has to take care of the always existing intrinsic limitations on the side of the automation components in the work system. These should be clear-cut in a way such that the human operator can be easily made prepared to cope with these limitations, if he is principally able to do so. In particular, this is the case for limitations of sensing capabilities of automation where the human operator could be superior.** Otherwise, new automation pitfalls will be generated. Thus, the design has to account for both the intrinsic human limitations and the automation limitations as well. It is of supreme importance to keep that in mind. In case of the OCU design, this is a crucial issue for defining its role as a team member of the operating force. Otherwise we have to deal with brittleness of the operating force. This is similar to the design task for a work system when defining roles for the team members of a purely human team with different qualifications of the team members. Of course, the design decision about the introduction of a SCU, similar to that for a component of conventional automation, has to account for similar brittleness, too, regarding its function as part of the operation-supporting means as was already alluded to in the section for cognitive automation of mode 1.

Regarding the question of "how to automate" the design goals mentioned can only be realised to a satisfying extent by the introduction of cognitive automation. Think of the deficiencies of conventional automation, concerning brittleness, opacity, etc., which are incompatible with the design goals just mentioned.

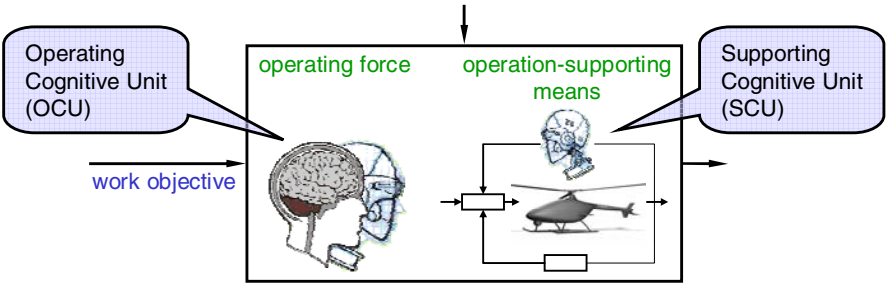


Fig. 40 Introducing SCUs and OCUs in the work system of UAV guidance and control

Designers of work systems have yet to get used to that. As an exemplification for a work system structure with implementations of both mode 1 and mode 2 cognitive automation, Figure 40 is shown. This figure depicts the work system of guidance and control of a helicopter as an UAV, where it becomes obvious that a dual-mode cognitive design is very appropriate, including both a SCU for mode 1 cognitive automation to control the helicopter in a semi-autonomous fashion and an OCU to assist the human operator in the control station on the ground in a co-operative manner as appropriate for the mode 2 cognitive automation.

The introduction of the OCU is the basis for human-machine symbiosis in its real sense and guides the way out of the vicious circle (see Figure 33) as shown in Figure 41. Although cognitive automation does not break the general trend of raising complexity as such, the introduction of the co-operative control mode counteracts the problems with supervisory control of increasingly complex automation by definitely mitigating the system monitoring and management task load through human-automation co-operation.

As an important conclusion from the preceding discussion about SCUs and OCUs (assistant systems) we can present a couple of general recommendations for the ACU configuration in a work system design.

1. Look first whether it is possible to set up an OCU rather than a SCU for the enhancement of the work system. The reason is that OCUs as opposed to SCUs, based on sound knowledge about the work process, undertake every effort they are able to that the known work objective will be achieved. This can considerably reduce the supervisory load on the human operator towards a much lower level, in particular with regard to the monitoring of the operation-supporting means.
2. Keep the number of ACUs, whether as OCUs or as SCUs, as small as possible. From a theoretical point of view, the ideal solution would be a single OCU. This leads to easier and more error-free interaction in the course of the work process, because the number of independent situation representations and different setups of knowledge bases is kept on a minimal level. Other aspects, however, may lead to dual-mode solutions or even a number of SCUs or OCUs to be defined.

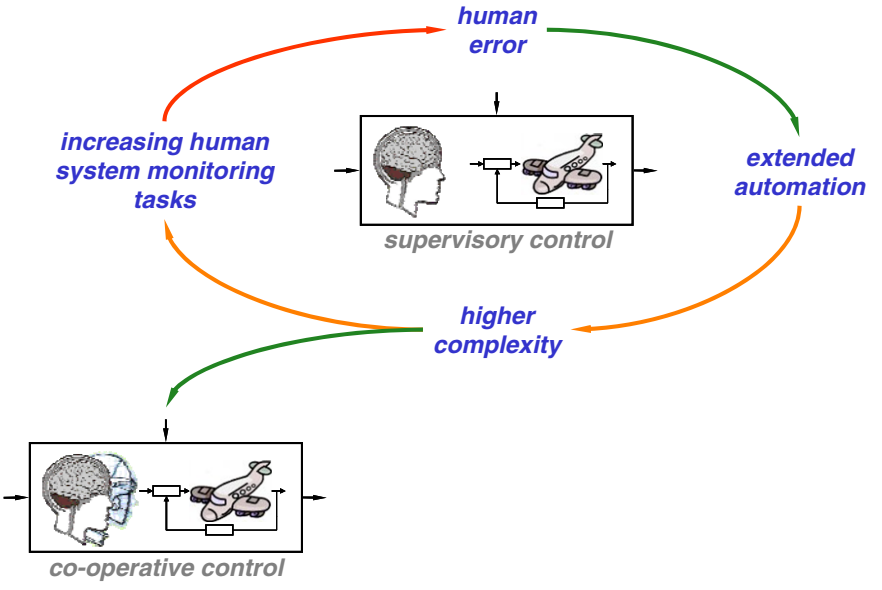


Fig. 41 The vicious circle of progressing automation and the escape from it

We often can take advantage of the little less extreme case that already one ACU in both the operating force and the operation-supporting means will do the job. In fact, to keep the number of ACUs as small as possible is an important design recommendation. As already pointed out, it is because we want to take advantage of the central ACU situation representation. Then, it will be ensured that as many functions as possible can rely on exactly the same situational information. Thereby, concept-based behaviours concerning the full functional spectrum shall be facilitated. In addition, if co-operation is involved, the co-ordination effort can be kept as low as possible.

By the way, proliferation of independent situation representations residing in separate cognitive entities in work systems is one of the inevitable weaknesses of human teams. This is a principal problem, because the number of team members might not be reduced to the same degree as with ACUs. Let us think in this context of the team in the surgery room with several “assistants” of the chief surgeon. There, for instance, it is necessary to have more than one “assistant” as part of the operating force, because humans are more limited regarding the task load they can bear. Just think of the load on attention or the simple fact that our eyes can only look in one direction at one gaze and that this gaze can only lead to detailed perception of objects within an angle of just a couple of degrees, if there is a busy environmental scene. Often, already the lack of resources at disposal, like lack of time, makes it necessary to rely on another team member which can offer certain simple skills at the right moment. Summing up, in the following we

shall focus in the first place on the case that there is only one OCU, such that there only remains the co-ordination effort between one assistant system and the remaining human team members.

The two preceding recommendations are very general ones. They are good to think of in the first place before considering other aspects. They may not be applicable or realistic, though, because application-related aspects might overrule them. For instance, there might be the aspect of cost effectiveness which may postulate not to make costly redesigns of existing components of operation-supporting means. It results that these components are to be integrated into work system designs with a new OCU or SCU by keeping them as they are.

Mostly, the reason to make use of more than one unit in a work system is an application-specific necessity to have units working at different locations. This might be illustrated by the following possible options of configuring a work system with a work objective of carrying out a reconnaissance mission by use of a UAV as part of the operation-supporting means (see Figure 42). The human operator (at least one) is the central part of the operating force and is hosted in a control station which is dislocated from the UAV. He or she is assisted by at least one OCU. The UAV is being piloted by another ACU.

Figure 42 illustrates that there are four options of where to place the ACU piloting the UAV:

(1) as SCU at the operator control station
(2) as OCU at the operator control station
(3) as SCU (and part of the operation supporting means) aboard the UAV
(4) as OCU (and part of the operating force) aboard the UAV

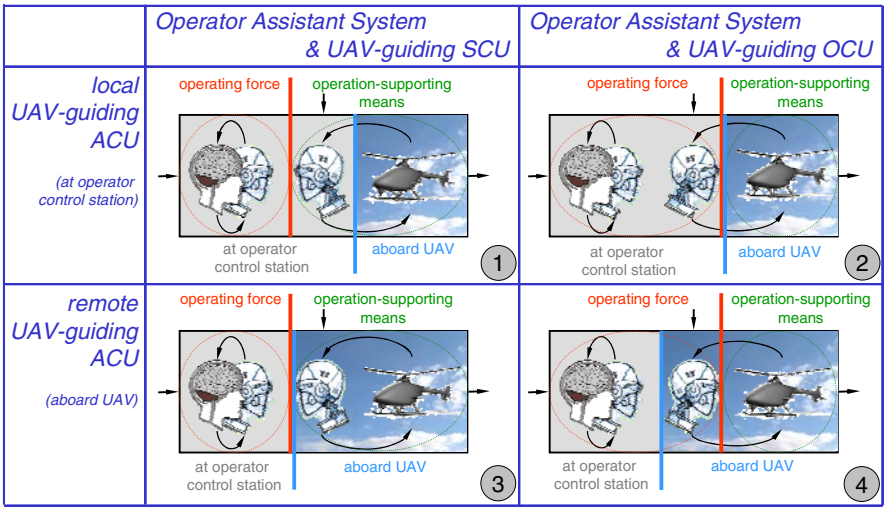


Fig. 42 Options within the work system of placing the ACU piloting a UAV

The decision which option to choose heavily depends on the role of the data link communication between ground station and UAV. If there are no technical or operational constraints on the data link, which in practice is usually not a case we can consider options (1) and (2) and assume, that option (2) probably would be the one chosen. The remaining two OCUs located at the operator control station could be joined into a single one which would lead to the ideal solution corresponding to the general reommendations being made earlier.

If there are major constraints on the data link, option (1) and option (2) immediately have to be dropped, because there would be no decision component (i.e. ACU) being able to maintain connection to the UAV in order to manage contingencies if necessary. Option (3) represents a dual-mode cognitive design with one OCU and one SCU. Option (4) incorporates two OCUs, one at the operator control station and the other one aboard the UAV. Then, option (4) would be probably the most favoured solution, because the airborne OCU has got knowledge about the work objective. Thus, it has got the capacity, in principal, to operate more independently than it can be achieved for option (3). The more knowledge about the work objective the more independence is possible for the airborne ACU when operating the UAV. The more this capacity is made use of in the work system design, the more effort is needed for the cognitive design of the airborne ACU, though. This eventually might lead to the question whether the design can be accomplished within the cost limit.

In Chapter 5 examples of realisations of both SCUs and OCUs are described in order to illustrate the theoretical foundations from the preceding chapters.

4.4 Cognitive Teaming in the Work System

So far, we have become acquainted with the fact that cognition in the work system is not confined to human individuals as part of the operating force. In addition, the designer can make use of artificial cognition in terms of ACUs which may appear as part of the operating force or the operation-suporting means. This brings up the issue of cognitive teaming between these different kinds of cognitive units in whatever configuration. With the advent of multi-agent systems the issue of cognitive teaming has become a major design issue as opposed to the analysis issue which dominated the work on human team structures in the past. There is a great deal of literature in that field which is pushed by the necessities in agent technology, although this segment in agent technology is still not fully consolidated. This becomes evident in the terminology which is still not a unitary one. In essence, we follow the elaboration on this topic of [Meitinger, 2008] in the following.

The underlying idea of cognitive teaming in the work system is *co-operation*. We as humans are used to take advantage of co-operation in our daily life. From experience we know we can accomplish more in a co-operating group as one could accomplish with the same number of people, who refrain from co-operation. This can be exploited in the context of work systems by means of cognitive teaming. Co-operation can appear in different forms, but will always be based on a common goal or at least a compatible one under all circumstances. Goals are

considered to be incompatible, if achieving one means that another one, which should also be pursued at the same time, cannot be achieved [Ferber, 1999]. Co-operation within a group can be made explicit by the formation of *teams*. The team members are cognitive units, humans or ACUs. The co-operation of team members can be very simple with only some form of task dependency, however in many cases there are a great deal of activity *interdependencies* among the team members which can make co-operation rather complex. Then, co-operation might demand for supplementary activities of the team members to account for activity interdependencies. This can be subsumed by what we call *co-ordination* [Malone & Crowston, 1994]. Finally, *communication* is a means for co-ordinating by exchanging information among team members.

Based on these introductory remarks, in the following we get acquainted with further details on different forms of co-operation, team structures, the necessity of co-ordination and ensuing demand for communication, and how this can be achieved.

4.4.1 Forms of Co-operation

In the context of multi-agent systems, [Ferber, 1999] has defined that a co-operation situation exists, if one of the following two conditions holds:

1. The addition of a new agent makes it possible to differentially increase the performance levels of the group.
2. The action of the agents serves to avoid or to solve potential or actual conflicts.

The latter condition might need some explanation. It defines the situation that despite of the addition of a new agent no differential increase of the group performance is achieved in situations free of conflicts, but that the potential of avoiding or resolving conflicts has been increased. In this sense co-operation is not increasing performance, it rather moderates the reduction of performance in case of conflicts.

We also follow [Ferber, 1999] as to his nomenclature for the classification of co-operation between work system players in different interaction situations where immediate actions are demanded. Co-operation assumes compatible or common goals, although the kind of co-operation depends on the degree of activity interdependencies and on whether there are sufficient resources and team member capabilities (knowledge and/or skills). Since sufficiency in both resources and capabilities of a single team member makes co-operation pointless, [Ferber, 1999] distinguishes between three types of co-operation, namely simple collaboration, obstruction, and co-ordinated collaboration. In this order the level of activity interdependencies involved among the co-operating team members is reflected, thereby beginning with the lowest level. A team may exercise more than one type of co-operation in the course of work. *Simple collaboration* is featured by sufficient resources and insufficient capabilities. This corresponds to the case that tasks are allocated to team members, that none of the team members is skillful enough to do the work alone, and that there is very little interdependency among

the co-operating parties, because there is about no potential of resource and authorization conflicts. This kind of collaboration is typical for a team of specialists who can work rather independently in parallel or in series. This can include that the team members allocate tasks among themselves and share information in the course of work. As an appropriate example of the domain of vehicle guidance and control we refer here to the rescue of shipwrecked mariners, for instance, including both a crew of a specially equipped search airplane on the one hand and a rescue helicopter crew on the other hand. Each of these parties has got its own capabilities. They exchange information with each other to do the job, in particular, when the crew of the search plane informs the rescue helicopter crew about the co-ordinates of the find when it is discovered. Since there is no overlap in capabilities and no coincidence in the demand of not sharable resources, there is almost no need for co-ordination. *Obstruction* stands for a type of co-operation situation which is featured by insufficient resources, sufficient capabilities, and less organized co-ordination. This is usually the case, if there are team members who are dependent on the same resource. Then, part of the co-operation within the team is the endeavour to commonly find a solution that the team members concerned can proceed with least loss of performance. Here, co-operation is typified by the use of techniques of reactive online co-ordination while observing the other's actions. Obstruction is characteristic of situations which are pertinent to the second of the two conditions mentioned earlier according to [Ferber, 1999], which define a co-operation situation. Finally, *co-ordinated collaboration* usually has to be considered, if there are insufficient resources. The capabilities of the individual work team members might but do not have to be insufficient, too. Possibly there might even changes of their roles, if necessary. It works on the basis of pre-organized or/and online managed co-ordination in the light of the limited resources. That can include the allocation of roles among the team members to avoid resource conflicts by organisational design. A typical example of a team which carries out co-ordinated collaboration to a great extent is that of a two-man pilot crew in a commercial transport aircraft. The two pilots have different roles as pilot flying and pilot not flying. Only the pilot flying can act on aircraft controls. The pilot not flying is responsible of an assisting role with tasks like flight plan preparation and communicating with agencies outside the aircraft like air traffic control. The capabilities of both pilots are about the same, such that they can change their roles with each other in certain situations, if appropriate.

A team may exercise more than one type of co-operation in the course of work. Even with only one task being executed the form of co-operation may switch from type to type, also from mode to mode like from co-operation to independence and vice versa. Thereby only work situations with compatible goals of the team members are being discussed. In this context, for instance, [Schmidt, 1991] identifies an additional co-operation situation of so-called *debative co-operation* with partially independent work of the team members. This can happen when they are to work on the same task without resource shortages and have about equivalent levels of capabilities. In the first place they are working independently to provide a proposal for the solution of the given task. In due time, they compare the results and decide on the best proposal. This will be executed. This type of co-operation

makes only sense for human- or human-machine teams, if the task is a rather complex one, if poor solutions are not sufficiently improbable, and if there is enough time for this debative procedure. We can conclude that debative co-operation is characteristic for problem solving in a team with temporary independence of the team members, for instance. This kind of co-operation might also evolve in the airliner cockpit in certain rare situations.

Having introduced ACUs into the work system both as part of the operating force and as operation-supporting means opens up the potential of co-operation within the work system in terms of cognitive teaming in principally two ways in addition to human-human co-operation:

(1) ACUs being part of the operating force could deliberately co-operate with the human operators for the sake of best work process performance. This kind of ACUs take initiatives on their own in order to have the work system comply with the work objective.
(2) There might be several ACUs as semi-autonomous systems as part of the operation-supporting means in a work system, which could accomplish in co-operation with each other a task given to them by the operating force. It should be noted at this point that also co-operation with or between other given cognitive entities might be possible.

In conclusion, the type of co-operation under index (1) is **the only possible version of human-machine co-operation.** Although there might be ACUs as part of the operation supporting means, there cannot be any co-operation between these ACUs and the human operator. Instead, the interaction between the human operator and these ACUs is characteristic for supervisory control. The type of co-operation under index (2) is representative for machine-machine co-operation. Both types of co-operation may amplify the capacity of the work system and/or improve the work system performance, of course based on the existence of a team.

Considering human-machine co-operation, an ACU is allowed to co-operate with the human operator as an assistant system like it would be the case if there was a human assistant. The only principal constraint on the assistant system is the fact that it must not make any alterations of the work objective. A typical example for a comparable purely human team on this level of co-operation is the pilot team in the two-man cockpit of a commercial transport aircraft. Just like them a team consisting of a pilot as human operator and an ACU as an assistant system could form a team to operate the same work process in a co-ordinated way in order to accomplish a common given work objective. This kind of co-operation might go as far as considering an assistant system which takes over full authority in case of incapacity of the human operator. Not all assistant systems on that co-operation level as part of the team constituting the operating force are of that high-end capacity. We can think of others which can be considerably constrained in their authority level and might have got no capacity at all to directly operate the work process, although knowing everything that is known by all other team members of the operating element, including the externally given work objective. Their assistance can be like that of a harbour pilot who advises the captain of a ship,

thereby supporting him to bring his vessel safely to the dock, but who never would have his hands on the rudder. Furthermore, there might be assistant systems which are even much more confined in their scope of action taking, only being specialists for certain simple tasks which are necessary from time to time in compliance with the work objective and the pertinent task agenda. This makes clear, that such an assistant system does not have to be equivalent in its capabilities to human co-members. However, at least it has to know the common work objective and the role it has to play within the team subject to the common work objective.

Figure 43 shows a work system which exemplifies both internal human-machine and machine-machine co-operation for the application domain of UAV guidance and control. A human operator, located in the ground station on the one hand, is assisted by an ACU, which stands for human-machine *vertical* co-operation (see [Millot, 1988]). On the other hand, a group of 2 ACUs as operation-supporting means controlling UAVs form a *horizontally* co- operating machine team (see also [Millot, 1988]) in order to comply with an instruction by the human operator, for instance for a search task. Each of the co-operating team members of the work system depicted in Figure 43 needs a certain motivational context for co-operation. Especially for ACUs, this has to be formulated explicitly and could be as outlined in Figure 44.

Figure 43 also indicates another type of interaction within the work system of concern. This is supervision as opposed to co-operation, which takes place between the operating force and the operation-supporting means. Note that double-sided vertical arrows in Figure 43 designate the interaction type of co-operation and horizontal arrows indicate the direction of supervising interaction. From the engineering point of view, it is important to distinguish carefully between supervision and co-operation. In particular, the requirements and challenges for the realisation are considerably different. The main difference between supervision and co-operation lies in the fact that the goals of both the supervisor and the supervised unit are not necessarily the same. It is even not warranted that the goals are compatible under all circumstances. In that sense, the

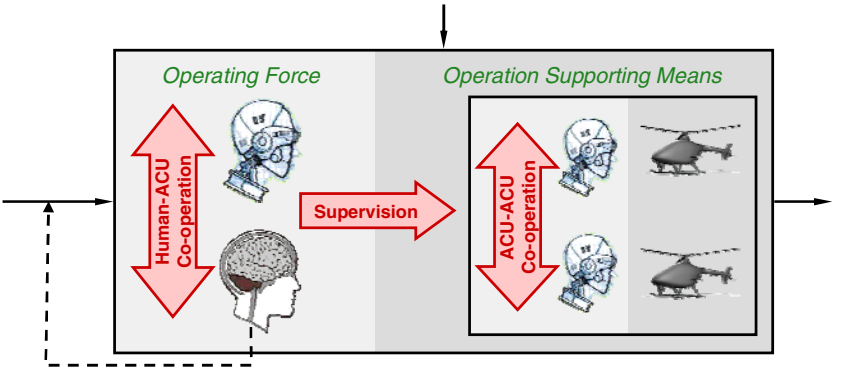


Fig. 43 Forms of human-machine and machine-machine co-operation in the work system. OF-OSM interaction (left to right): supervision

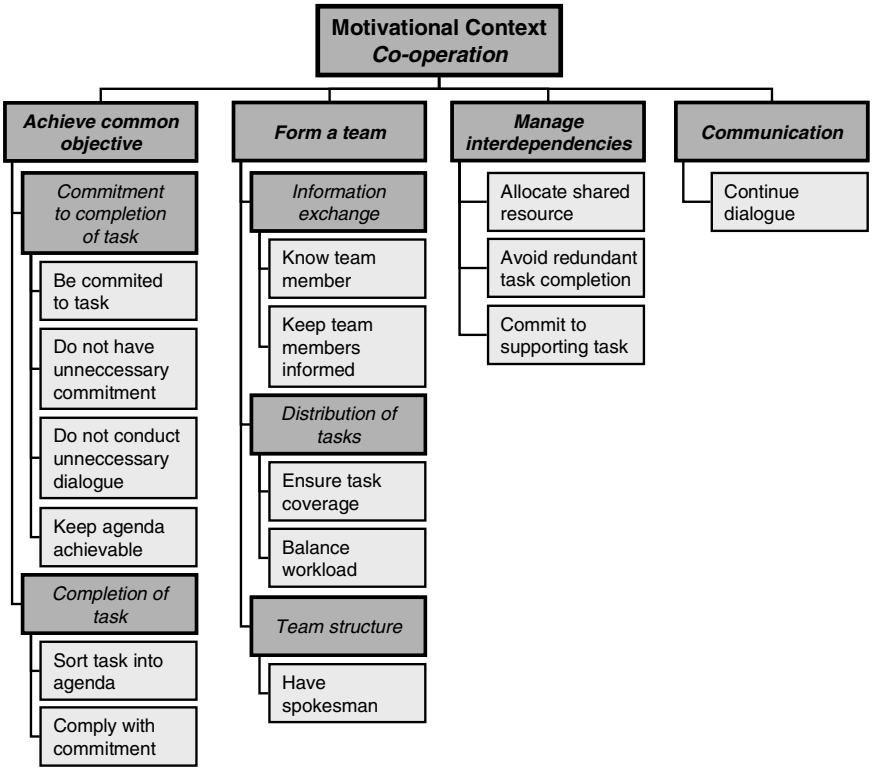


Fig. 44 Motivational contexts for co-operation [Meitinger, 2008]

term manned-unmanned teaming (MUM-T) which has become known from the literature (see for instance [Schulte & Meitinger, 2009a] and [Schulte & Meitinger, 2009b]), in particular in the context of airborne-controlled UAVs, is not necessarily the same as what we have defined here as teaming of co-operating units. There, teaming is used for team structures which can be associated with both supervision and co-operation like it is depicted in Figure 43. Also teaming including co-operation among units in different work system may be considered in MUM-T formations.

What are the forms of co-operation we can figure out in this work system? The co-operation of the team, which forms the operating force, can stand for both the type of simple collaboration and co-ordinated collaboration. This depends on whether the team members have to make use of the same resources which are insufficient to comply with multiple demands. There will be little chance of obstruction, though. One can expect that there are situations and corresponding tasks with simple collaboration and other situations where co-ordinated co-operation is appropriate. Similar conclusions can be made about the team of ACUs controlling the UAVs, although here obstruction situations might be more likely because of resource conflicts.

4.4.2 Team Structuring

Organisational team structuring is an important issue in order to warrant good team performance. We will consider teams where the team members are identified by their *role* and pertinent *authority domain*, which might change in the course of the work process under certain conditions. Roles translate different functions and pertinent activities to be covered by the individual team members and are relatively long-term structural elements. These may be reassessed and reorganized under certain circumstances. This is for instance the case for the two-man pilot crew of an airliner. The roles designated as pilot-flying and pilot-not-flying may change among the two crew members in flight for certain known reasons. The authority domain marks the range of decisions which can be made independently subject to a certain role. Usually, the human operator's authority domain might still include that one of ACUs, but this has to be carefully investigated for the particular work system case being considered. The principal of as much comprehensive authority domain as possible on the human operator's side is not necessarily the best solution in all cases. Therefore, flexibility of authority allocation in that respect is of great interest, but not easy to implement. There are not only complementary authority domains, there may be overlapping ones, too, in certain cases. Role assignments have to take account of the capabilities of the team members. These are often not the same for each team member. In order to have a great wealth of capabilities in the team, it often is of great benefit to have team members with different capabilities.

[Biggers & Ioerger, 2001] describe teams by categorising them within a range of structures from instruction-driven (hierarchical), in which the instructions are generated by a decision-making team member, usually the team leader, to peer-like distributed, in which each team member makes decisions for itself. The concrete team structure chosen by the work system designer within this range is heavily dependent on the given or designated capabilities of the individual team members, in particular regarding work domain skills and knowledge as well as social skills. Typical examples of human teams for the extremes which span the range of possible team structures are the following:

Example 1: Team structure with role (authority) hierarchy
A surgery team works as operating force in a surgery work system. It is characteristic for a team structure with role (authority) hierarchy. The chief surgeon holds the role as a team leader with the highest authority level in the team, being responsible to make the top-level decisions. This warrants a situation to be flexible enough in case of situations where a change in the task agenda becomes necessary. At that point the chief surgeon would react with a new work plan. The roles and pertinent authority levels of the other team members are established depending on their individual capabilities. The chief surgeon, for instance, has got completely different capabilities than, for instance, the theatre nurse, who also belongs to the team. Yet, the nurse, like all other team members, knows about the work objective, too. Furthermore, she knows about the normative work plan of the team and the pertinent task agenda which complies with the work objective. Thus, without being explicitly asked by the surgeon she passes the scalpel when it is

needed by the surgeon. However, she is by no means able to replace the chief surgeon in his role. In summary, all team members know very well what their role is and, as a consequence, what their tasks are in whatever situations in order to serve the common work objective. All this is valid, too, for a team which consists of both humans and ACUs. ACUs may also adopt the full range of possible roles, if it is technically feasible. Only the team leader cannot be replaced by an ACU. This is a very interesting conclusion. It postulates that an ACU has not necessarily to be equivalent to the human co-member with highest level capabilities. It has just to be equivalent to a human who would take the role which is allocated to the ACU by the work system designer.

Example 2: Distributed team structure
Tennis players in a double team are forming a distributed team of peers. They decide individually on what they are going to do in order to comply with the common work objective, based on their individual capabilities. They also can temporarily take on the tasks of their partner, if necessary, although there is a certain task sharing in terms of roles they have agreed upon. The same is true for a soccer team or the pilot crew in an airliner cockpit. Again, all this is valid, too, for a distributed team which consists of both humans and ACUs.

Although these structures are relatively long-term structures, it has been shown that different organisations are appropriate for different problem situations and performance requirements [Malone, 1987]. Hence as a situation evolves, the community may need to periodically reassess its structure to determine whether it is still appropriate or whether a rearrangement would be beneficial – see the work of [Ishida et al., 1990] for an illustration of the dynamic reorganisation of a group of cooperating agents in response to changes in the environment. In the electricity management scenario, for example, the community may decide that it is best to replace the agent carrying out high-voltage diagnosis with several spatially distributed agents so that the load and the reliance upon any one individual are reduced. This evaluation corresponds to a convention for the organisational structure.

The capabilities of the individual team members for the work as such are crucial determinants for the team structure to be chosen. This includes skills and knowledge, as well as the capability to manage the demanded work physically and timely. If all capabilities needed for the work to be done within an individual work process, can be found with a single human operator, this would be the best solution. There would be no reason to form a team. Accomplishing a work objective in co-operation only makes sense, if it is actually possible and useful to work on the given task with several cognitive units, due to the following reasons (see [Jennings, 1996,] and [Wooldridge, 2002]):

- No individual team member has sufficient capabilities, resources or information to solve the entire problem.
- Activities of team members depend on each other with respect to the usage of shared resources or the outcome of previously completed tasks.

- The efficiency of the work process or the utility of the outcome is increased e.g.
  - by avoiding the completion of tasks in unnecessary redundancy or
  - by informing team members about relevant situational changes

  so that task completion will be optimised.

Some of these reasons were already alluded to. As a consequence it has to be concluded that the size of a team, if it is necessary to have any, has to be kept as small as possible for the sake of productivity.

Thus, also in the context of team size, the work system designer has to make two fundamental decisions about the work system structure which are not independent from each other:

1. to decide, for which tasks operation-supporting means are to be made available
2. to decide about teaming, thereby considering teams in both the operating force and the operation-supporting means as main work system components

Productivity, including safety, effectiveness and profitability (see also Chapter 4.2), is the main criterion which is to be used when making these decisions.

In summary, it turns out that team productivity is highly dependent upon the effort and performance in co-ordination activity. Co-ordination and the resulting need for communication within the team create inevitable additional tasks which have to be accounted for. The more teams and the greater the team size, the more effort has to be placed on the co-ordination process. Therefore, in the following we will discuss in more detail what has to be considered with respect to team co-ordination and pertinent communication between team members. These findings are equally valid concerning both, human-machine and machine-machine teaming.

4.4.3 Team Management

If the human operator is working within a team, possibly as the team leader, the performance of the team is heavily dependent on how it is controlled by means of co-ordination and pertinent communication. Team control, though, is not only a matter of human-human and human-machine co-operation, but also of machine-machine co-operation. The latter case has great impact on the development of co-ordination and communication procedures and routines, since such methods have to be formalised in case of multi-agent systems. This becomes obvious in the following two sections.

4.4.3.1 Co-ordination

In order to achieve the desired positive effects of co-operative work in a team, the activities of the team members have often to be co-ordinated, i.e. interdependencies between these activities have to be managed [Malone & Crowston, 1994] to achieve coherent team behaviour. That is mainly the case for co-ordinated collaboration but also for obstruction to some extent (see Chapter 4.3.1).

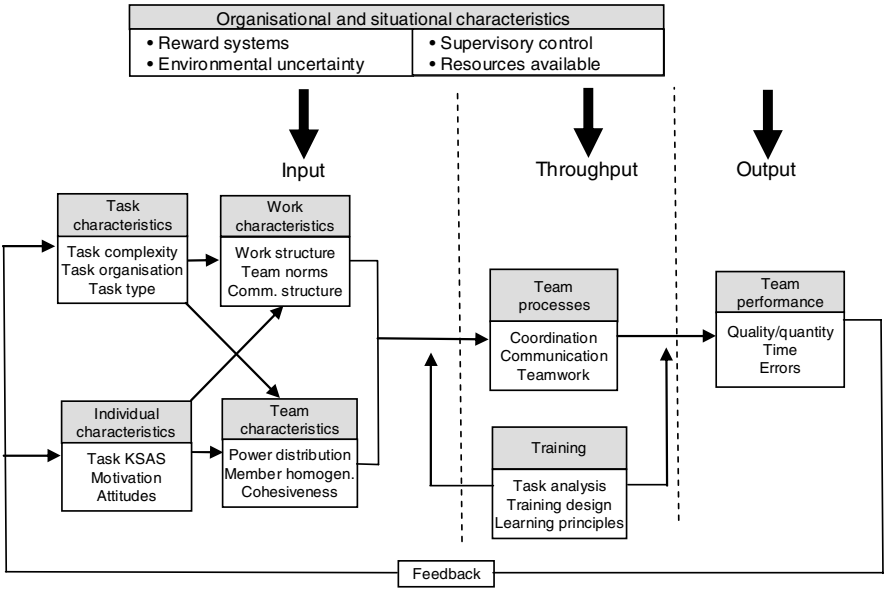


Fig. 45 A framework integrating existing models of human team performance and training (cf. [Salas et al., 1992])

Interdependencies include not only negative ones such as the usage of shared resources and the allocation of tasks among the team members, but also positive relationships such as demands for equal activities [von Martial, 1992] from more than one team member which opens the opportunity of deliberate redundancy or reduction in commitment for at least one team member. The complexity co-ordination can take on is illustrated by [Salas et al., 1992] by means of a framework that integrates a number of models about team performance and team training (see Figure 45). It becomes obvious from this framework that the process of co-ordination is dependent on the knowledge about the characteristics of the work including the task, the team, and the individual team member. Above all, there are external organizational and situational influences.

In hierarchically structured human teams as considered by [Swezey & Salas, 1992], the team leader gives instructions about which tasks are to be carried out by the team members and what is the pertinent time horizon of action, subject to a work plan. The work plan may be generated by the team leader himself or by the team under his leadership. This is the classic approach based on supervision. However, usually there still remains much room for co-ordination as becomes obvious by the following teamwork guidelines as published in [Swezey & Salas, 1992]. With emphasis on designing training programs, [Swezey & Salas, 1992] have listed and clustered desired capabilities of team members to ease interaction for the sake of team performance. Here, subsequently, only a selectively shortened version of the list is shown which hopefully comprises the most important items:

1. *Team mission and goals*
   - *Every team member should be able to state the title and purpose of the team mission.*
   - *Every team member should be able to describe how his or her own special (or subsidiary) team(s) will satisfy its task(s) in the execution of the team mission.*
   - *Every team member should be able to describe the general approach the team will employ to accomplish mission objectives.*
2. *Environment and operating situation*
   - *Teams should be trained under conditions that approximate the operational environment.*
3. *Organization, size, and interaction*
   - *Make use of graphed representations of timeline and critical paths to demonstrate interdependence.*
   - *All task responsibilities for team members should be specific and free from ambiguity.*
4. *Motivation, attitudes, and cohesion*
   - *Team members should be supportive of teammates when the latter make mistakes.*
   - *Every team member should be able to recognize and list all members of the team by position.*
   - *Every team member should be able to describe the consequences of team dependency failure(s).*
   - *Accurate expectations of the contributions of other team members to the overall team performance should be developed, both for the overall team and for special or subsidiary teams.*
5. *Leadership*
   - *Every team member should be able to recognize when he or she is the team leader or is expected to assume the team leader's position.*
   - *Every team member should be able to recognize when the team leader is unable to lead the team.*
   - *Team leaders should be trained to acquaint themselves with the details of the team's operation, as well as with the individual tasks required of each team member.*
   - *Every team member should recognize the authority of the team leader, both for the entire team and for the relevant subsidiary and special teams.*
   - *Team leaders should verbalize their plans for achieving the team goal.*
   - *Team leaders should be good communicators; they must keep the team informed about matters that affect team performance.*
6. *Communication: General, conveying information, feedback*
   - *Team members need to communicate information to team members in proper sequence.*

- *Proper information exchange facilitates the development or co-ordination of teamwork strategies by assuring that all team members have access to relevant information.*
- *Team members should use proper terminology when communicating information to other team members.*
- *Unnecessary interruptions of individual team member's tasks by other team members can lead to overall team performance failures.*
- *Feedback of performance information among team members is especially important in task conditions that ar changing rapidly and where direct equipment or instrument feedback is unavailable.*
- *Asking for help when in need of assistance is positively related to successful team performance.*

7. *Adaptability*
   - *Team members should be able to adapt relevant information to the task at hand.*
   - *Adaptable team members are able to change the way they perform a task when necessary or when asked to do so.*
8. *Co-ordination and cooperation*
   - *Members of well-co-ordinated teams are often able to anticipate when teammates are going to need specific information for the completion of a task. They also provide such information.*
   - *Effective team members typically help other team members who are having difficulty with a task.*
   - *Effective team members co-ordinate information gathering activities in a structured fashion.*

Work system designers should take into account as much as possible of this list. It is thought for training of purely human teams only, though, but about all of it can also be applied to human-machine and machine-machine teams. In order to comply with all these design recommendations there might be a danger of ending up with a level of complexity too hard to handle. However, this is not a principal dilemma. One can rank the recommendations for a particular team in terms of their effectiveness with regard to the overall team performance, taking the safety-relevant recommendations on top, for instance, and leaving out those from realisation which appear at the bottom of the ranking list.

In particular for machine-machine teams, however, the team structure often becomes a distributed one which makes the co-ordination issue by far more a central one in team design as for hierarchically structured teams. Among the multitude of approaches to realise co-ordination (cf. e.g. [Wooldridge, 2002]) in distributed artificial intelligence like multi-agent systems there is one which should be discussed briefly here, as it lends itself to be applied rather straightforward. With this approach [Jennings, 1996] proposes a distributed goal search formalism and reduces co-ordination to the combination of *commitments*, *conventions*, *social conventions*, and *local reasoning* (cf. also [Cohen & Levesque, 1990]).

Therein, commitments of an artificial team member are described as

> *"[…] pledges [of agents] about both actions and beliefs." [Jennings, 1996]*

These pledges are made subject to the role assignment and can refer to the past (with respect to beliefs), the present state of affairs or about the future, possibly dependent on certain conditions. In essence, these pledges are commitments for certain tasks which possibly may comprise the full amount of cognitive sub-functions from perceptual processing up to action control (see Chapter 3.2.2.2). Because of resource limitations they may lead to certain constraints, if there are proposals to take on additional commitments. Accordingly, there might be *joint commitments*, if not a single team member but a group of teammates or the team as such is concerned.

Conventions are explained as

> *"[…] describing circumstances under which an agent should reconsider its commitments." [Jennings, 1996]*

This is necessary, because there might evolve a situation that a commitment has to be withdrawn because of unforeseen changes and uncertainties in the context the team member is situated in. Conventions define the circumstances, when this should be the case. They are a means to pump some flexibility in the team behaviour when operating in dynamic environments. They also may be the reason for misinterpretations of contextual changes and losses in coherence of the team behaviour. Too many conventions can contaminate the co-operation in the team. Also how frequently the commitments are to be reviewed as a consequence of conventions has to be carefully sorted out:

> *"When specifying conventions a balance needs to be reached between constantly reconsidering all commitments (which will enable the agent to respond rapidly to changing circumstances, but will mean that it spends a significant percentage of its time reasoning about action rather than actually carrying out useful tasks) and never reconsidering commitments (which means agents spend most of their time acting, but what they are actually doing may not be particularly relevant in the light of subsequent changes)." [Jennings, 1996]*

This underlines that the specification of the conventions is a critical design issue. In addition, there are social conventions which

> *"[…] specify how to behave with respect to the other community members when their commitments alter" [Jennings, 1996]*

Here, the earliest possible opportunity has to be taken to inform those team members about commitment changes which could be affected by them. At the same time it has to be accounted for the effect of that message by considering the

state of affairs the message receivers are just going through. In the ideal case both sides become mutually aware of the convention as it becomes active.

Finally, local reasoning stands for the capability to think about the own actions and related ones of others.

In summary, by choosing conventions and social conventions appropriately, the distribution of commitments in a team can be adapted to the current situation and can consider aspects such as the workload of the human operator, opportunities of team members, or unexpected changes of the work conditions.

It should be noted at this point that these characteristics typical of co-operation in machine-machine teams of distributed structure can also be often valid in purely human teams or even in human-machine teams, if partial or complete distributed team structure is considered for the design of such teams.

Furthermore, the way co-ordination is designed, if it is necessary, can be decisive regarding success or breakdown of the pertinent work process. In case of complex task and team characteristics co-ordination may even amount to an online problem solving task. The human operator(s) and ACU(s) as team members involved in co-ordination have to take it as one of their main co-operation sub-goals. If the cognitive process of co-ordination works allright, there is still the task of communication to bring co-ordination in action. This can also amount to a pitfall. A survey about communication as part of co-ordination will be covered in the following section.

4.4.3.2 Communication

As noted previously, the prevailing means for co-ordinating actions is *communication*, which can be either explicit or implicit. *Explicit communication* in the area of multi-agent research is usually based on the speech act theory founded by [Austin, 1962] and further developed by [Searle, 1969]. It states that communicative acts can change the state of the environment as much as other actions, i.e. to communicate explicitly means to send messages to team members, usually in order to achieve certain desired states. *Implicit communication* in contrast is based on the observation of team members and an inference of their intentions in order to be able to conclude, which content messages could have had, if explicitly sent.

In order to enable machines to communicate with team members explicitly, formalized agent communication languages can be used. One of them is the FIPA ACL (Foundation of Intelligent Physical Agents – Agent Communication Language). It specifies

- a message protocol [FIPA, 2002a],
- communicative acts (performatives) [FIPA, 2002b],
- content languages, and
- interaction protocols.

Whereas the former three are focused on single messages, interaction protocols describe possible successions of messages. In Figure 46 for instance a typical request interaction based on the FIPA specification [FIPA, 2002c] is shown in a representation similar to state automaton with messages effecting transitions between states (see [Winograd & Flores, 1986]).

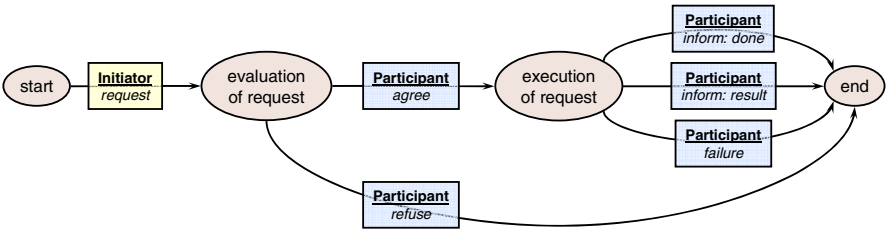


Fig. 46 Request Interaction (cf. [FIPA, 2002c])

A request interaction is started because an actor ("initiator") wants another one ("participant") to accomplish a certain task. Therefore, initially a message is sent detailing the request. The participant has to decide whether the request can be accepted and in the following can either refuse or accept the request. In case of acceptance, the requested task is being executed and in the end the participant informs the initiator about the outcome of the task. Such a formalisation of interactions involves that – no matter how the participant decides – the initiator always receives feedback on the request and can use this as a basis for selecting future actions. More complex message interactions of machine-machine communication using FIPA ACL will be exemplified in Chapter 5.1.2, regarding the communication between several co-operating UAVs.

Although humans mostly tend to stick to such protocols, they may in contrast to machines send redundant messages or use short-cuts not being intended. Thus, in case machines are supposed to communicate with humans, either the humans have to stick to protocols or – being a much more human-friendly alternative – the machines have to be enabled to cope with such imperfect human behaviour. In a first step, this might be a notification if a message has not been understood (a possibility already been intended by the FIPA ACL) and information about the expected messages. Depending on the degree of lack of understanding on the side of the machine, only additional explanations might be specified and requested from the human team member. Furthermore, if there is only little uncertainty about the semantic content of the message, just a confirmation of what seems to be understood by the machine might be requested. The latter case, in particular, lends itself to a more advanced concept-based approach by means of a dialogue manager which could imply an interpretation of the derivation and an inference of the underlying intention of the human transmitter with an appropriate adaptation of the message or interaction respectively. This could include that the dialogue manager adapts to the sequence of tasks which the human operator has chosen, and the appropriate time when the human operator is prepared to receive the message.

These facets of implicit communication, i.e. the capability to derive information from the situation and the human behaviour, demand for a model of the dialogue partner from which intentions can be deduced. Whereas for machine partners the development of suchlike models is quite straightforward due to the fact that their internal mechanisms are usually well known, many human factors engineering research issues regarding human behaviours in dialogues remain.

4.5 Engineering Approach for Artificial Cognitive Units

Considering system design in work environments, engineers are interested to build adequate tools and support systems for human operators for the sake of good, satisfying work results. The engineers' focus lies on the issue of work system synthesis rather than system analysis, thereby considering the human operator as a given, sufficiently known system element, taking into account the advanced findings about the cognitive behaviour of humans. The knowledge about human cognition is of great benefit, too, to promote the right ideas in the course of the endeavours of computer science people and engineers to achieve artificial cognition with capabilities equivalent or even better than those of the human species.

With the advances in computer science, engineers are more and more looking at designing systems which are relying heavily on artificial cognitive functions similar to those of humans. [Braitenberg, 1984], having a background in biological cybernetics, describes a fascinating series of experiments with toy vehicles, steadily being amended step by step until a kind of cognitive vehicle is achieved which behaves as it were controlled in a human-like fashion when moving around in its little toy world. He claims that by introducing two particular types of electronic core elements simulating certain neural properties he knows from his neuroscience experiments with animal brains, networking them in certain structures and adding sensors and motors, human-like behaviour in vehicle guidance and control is possible. This includes functional capabilities of cognition such as knowledge storage capability, knowledge acquisition capability about objects and concepts pertinent to the vehicle environment (static and dynamic), including also the mapping of the environment, thereby achieving the capability of process prediction. Among the vehicles also those are outlined which can cope with more rare situations when the prediction does not correspond with what actually happens. Here, the additional function of a short-term memory is introduced in the toy vehicle design new solutions, for instance for the identification of new situations, yielding the capability of exploring knowledge already incorporated in the brain. Eventually, in the last one of the succession of steadily upgraded vehicle designs, the capability of goal-directed behaviour based on explicitly stored desires is incorporated. [Braitenberg, 1984] has already perfectly illustrated by means of fascinatingly simple featuring of the crucial design steps how to accomplish the synthetic task of providing artificial cognition. His book is bridging thereby the disciplines of neuroscience, computer science, and engineering. It is strongly encouraging and stimulating for researchers of computer science and engineers to work for the realisation of artificial cognition in vehicles in real world applications.

4.5.1 Background Aspects

We can call the Braitenberg vehicles robots, if we look at them from the robotic community's point of view. We also can call them cognitive vehicles, if we look at them from the point of view of designing artificial cognition as such, which

addresses not the whole vehicle system but just the part which is responsible of information processing in the vehicle by means of computers. Both perspectives can be subsumed under the theory of agents, then distinguishing between hardware agents (e.g. robots) and software agents (e.g. internet machines). Both kinds of agents can be part of work systems. Software agents need communication links to their environment within and outside the work system, whereas hardware agents also observe the environment in a more self-reliant fashion by means of sensing capabilities before acting on it. A typical example for a hardware agent might be a semi-autonomous unmanned vehicle. From the system engineer's point of view, agents in work systems are supposed to be a kind of artificial equivalent to human operators. In that sense, we follow the characterisation of agents as stated by [Jennings & Wooldridge, 1998]: Agents *"must be*

- *Semi-autonomous: given a vague and imprecise specification, it must determine how the problem is best solved and then solve it, without constant guidance from the user.*
- *proactive: it should not wait to be told what to do next, rather it should make suggestions to the user*
- *responsive: it should take account of changing user needs and changes in the task environment*
- *adaptive: it should come to know user's preferences and tailor interactions to reflect these"*

In order to ensure these characteristics it has to be made use of capabilities like cognition, co-operation, communication, mobility, and learning at the extent needed.

It has to be noted though at this point that, by virtue, there are differences between the humans and artificial cognitive systems, mostly even huge differences. These differences are caused, in the first place, by the obvious differences regarding the physical implementation of the cognitive functions (especially the low-level functions according to the framework in Figure 30). As a consequence, depending on the technical implementation and the task requirements, the behavioural results of artificial cognitive systems might show both, better or worse performance than those of humans when encountering certain work situations. This has to be accounted for when designing artificial cognitive systems for use in work systems. Thus, agents are not necessarily designed as copies of humans. They rather are designed to do their job most effectively, regarding the function they have to carry out within a work system. There are applications, where the cognitive behaviour of an agent is deliberately designed to differ from that one of humans. At the other extreme there are applications, though, where the design goal is to achieve cognitive behaviour as similar as possible to humans.

So far, agents are deployed in work systems as part of the operation-supporting means only, often as software agents like BDI agents [Rao & Georgeff, 1995]. In this book, we lay our focus on a special kind of agents which are not only used as operation-supporting means in work systems, but also as part of the operating force. This aspect has been discussed extensively in Chapter 4.3. To co-operate as part of the operating force with the human operator on a given common work

objective demands that both the human and this kind of agents are able to understand the behaviour of each other the same way as humans understand each other. This often takes a-priori knowledge of human cognition on the agent's side, explicit and implicit (knowledge of designer), in order to generate beliefs about the mental and physical state of the human operator. As a prerequisite for that cognitive behaviour both human and agent must have got a similar understanding of how to tackle a given work objective and the explicitly existing mutual desire to comply with it.

Therefore, in distinction to other known kinds of agents, we call these software agents *artificial cognitive units* (ACU). An ACU features its own independently derived central situation representation, thereby forming its own situational picture about all what is of interest regarding the work process at hand as it might be also available to a human operator who may work next to it in the work process. Based on that and other functions needed it acts as a kind of "one-brain artificial individual" similar to a human person. This leads to a work system architecture embodying both natural and artificial individuals as distributed cognitive entities. This is a powerful architectural feature which we will take advantage of. To make it even clearer, the characteristics of an ACU as proposed here are in strong contradiction to other approaches like a Multi-Agent System (MAS) [Ferber, 1999] or the Real-Time Control System (RCS) as described in [Albus & Meystel, 2001]. An ACU is not composed of functional specialists with only a partial situational picture like we experience it as a recommended feature of individual agents in a multi-agent system. An ACU cannot be compared with an individual component of the RCS architecture. Concept-based performance according to the understanding we developed in Chapter 3.2.2.2, i.e. the capability of the generation of solutions to problems not previously anticipated by the system designer, cannot be easily achieved by a system reflecting a task decomposition of its application domain within its architecture. Concept-based performance in principle needs the freedom of taking any a-priori and situational knowledge into account, no matter whether it has been previously acquired in a context which is not the same as the one actually given. What we can learn from human mastering of unknown situations and human creativity is the voluntary use of knowledge from various, probably not even directly related domains while solving a particular problem. This at least requires the architecture of a central knowledge representation within the ACU. Every design decision which breaks down the system in separate units according to task decomposition hinders the emergence of concept-based behaviours. Therefore, we advocate a design policy, which concentrates as much functionality within one ACU as possible. As an analogy from daily life, it is certainly more result-oriented to ask a well educated general practitioner for a medical diagnosis on an unclear set of symptoms first, instead of consulting various specialists, each of them having a heavy focus and pre-occupation due to their special discipline.

As a last remark, before then going into more details of the design of ACUs, we are focusing on software agents as ACUs in this book. Thus, the hardware design will not play a major role in this book compared with the software design.

4.5.2 The Cognitive Process of an Artificial Cognitive Unit (ACU)

Similar to a human representing a cognitive unit in the work process we will talk in the following about artificial cognitive units (ACUs), as already mentioned. The ACU has got cognitive capabilities which potentially are of great similarity to those of humans regarding rational reasoning and decision making in order to recognise and identify the encountered situation and to make action plans in order to react properly to given work objectives and the individual desires pertinent to the ACU. An ACU, though, is not to mimic the functions of the human brain in all functional levels. It rather should be able to generate the kind of outputs which could be generated as well by human rational thinking or which, at least, are intelligible to humans.

In general, the ACU represents a software agent based on a so-called cognitive process (CP), exploiting the computer technology available in order to achieve a cognitive performance level as demanded. This includes incorporating relevant knowledge available about human cognition, but it does not necessarily include mimicking the physiology of the human brain as closely as possible. This would be too specific and limiting regarding the ACU potentials in general. At the time being there is, for instance, a framework of ACU software available under the name COSA (Cognitive System Architecture) (see Chapter 7.1 and [Putzer & Onken, 2003] and [Putzer, 2004]). This has already been used for certain applications. But before coming back to this specific implementation later on (see Chapter 7.1), let us first consider a rational of how to deduce the architecture of an ACU from what we learned from human cognition. There can be summarised three top-level functional design criteria already from what has been discussed so far:

- The ACU shall be able in the first place to provide the kinds of high-level cognitive functions as we know from human cognition, not necessarily on an equivalent performance level, and to provide the capability to expose **behaviour on all behavioural levels**, i.e. the concept-based, the procedure-based, and the skill-based behavioural level if necessary.
- The ACU shall provide purely **rational behaviour**. Therefore, the ACU features:
  - *Explicit and well balanced motivational contexts*, i.e. the motivational contexts are defined and weighted by the designer team subject to the work objective. This includes rational decision-making, i.e. the ACU shall take all available knowledge into consideration when it comes to a selection of action. Thereby, unbalanced behaviour, like human unbalanced emotional behaviour, can be avoided.
  - *Comprehensive situation understanding*, i.e. the ACU shall be enabled to create a mental picture of all situational aspects available and relevant for the performance in the context of the work process.
  - *Goal-oriented action*, i.e. the ACU shall align its selection of action with relevant goals and underlying motivational contexts.

  - *Problem-solving capabilities*, i.e. the ACU shall provide mechanisms to select from given options for the course of action with the aim to transfer the current situation into a desired one.

  As opposed to human behaviour, the ACU architecture shall not imply models of human performance limitations (e.g. attention bottlenecks, memory capacity limitations) or other psychological or psychophysiological factors (e.g. fatigue) pertinent to human cognition. The ACU stays being "artificial", and therefore the design shall highlight the inherent strengths of the computer compared to the human. If an application requires the modelling of human performance as part of the role the ACU is playing in the work system, this needs to be modelled explicitly later on as part of the application development. This might be the case for an adaptive assistant system, which needs e.g. to anticipate high workload situations on behalf of the human team member. For the moment, specifying the architecture of the ACU, we claim: Knowing about humans is different from designing the ACU like a human.

- The ACU shall be **interactive**, i.e. it shall be able to communicate with its environment in order to receive information from the environmental world and to manipulate objects in the environmental world.

Hence, three non-functional design criteria follow concerning the architecture of ACUs:

- The ACU shall make provisions for a **central representation of the situation** and in principal unlimited access to **all relevant knowledge**, i.e. the a-priori knowledge and the situational knowledge in order to facilitate concept-based behaviour as stated earlier.
- The architecture of the ACU shall **not encourage** any **preconception for the implementation of the** necessary **functional levels of cognition** (cf. Chapter 3.2.2.1. Emerging technologies in computer science and computational cognitive modelling might offer great opportunities for performance improvements in the future.
- The ACU shall provide **deterministic behaviour**, i.e. the ACU shall behave in an identical manner when encountering identical situations, given by its picture of the situation. Although it is anticipated to be very unlikely for a complex system in a complex work environment that an identical situation ever will re-occur, **predictability** is an important feature, though, in the context of validation and verification processes. In practice, this requirement will probably be shifted more towards the demand for adequate **robustness**, i.e. the characteristic of a system to maintain specified behaviours even despite variations of situational circumstances. Although, this might be hard to quantify and even harder to achieve for real applications, at least the underlying architecture of the ACU should comply with this requirement. This obviously permits learning as mechanism of influencing future behaviours by past and present experiences being a

feature of the architecture per se. However, if an application requires learning for any purpose, then the necessary mechanisms have to be part of the application. If you want a system that learns, teach it how to learn.

Figure 47 shows the elementary building blocks of the proposed architecture of an ACU. The central component is the *body*, the oval core of the ACU in Figure 47, which hosts all knowledge used and produced by the ACU. The ACU is linked to the external work environment via dedicated input and output information links in order to enable interaction. According to the understanding on the embodiment of cognition we dwelt on in Chapter 3.2, physical sensors, effectors, and communication means shall be seen as part of the work environment. The work environment represents the real world the ACU is interacting with, usually as part of a work system. Therefore, the environment includes the other components of the work system the ACU belongs to, including human operators and operation-supporting means, as well as environmental objects outside the work system including also other work systems of relevance. The relation between the output and input information determines the observable *behaviour* of the ACU as seen from the work environment.

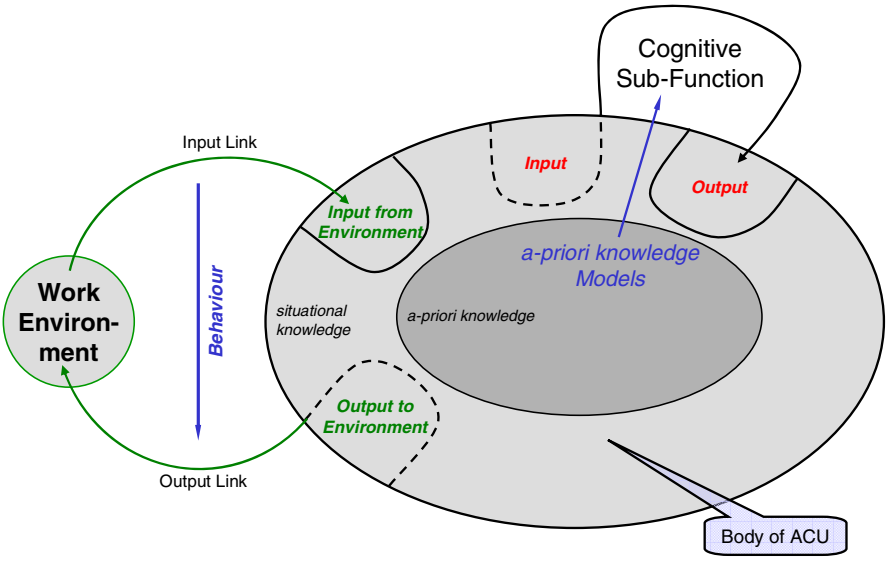


Fig. 47 Building blocks of the architecture of an ACU

Let us come back to the body of the ACU, first to the inner darker part of it. It contains the *a-priori knowledge* which is fed into the ACU during the application development process, or might as well be learned, if this is a characteristic of the application. Thus, the a-priori knowledge is application-dependent and it determines the behaviour of the ACU in the work process. It contains all kinds of knowledge, which we know from as part of the long-term memory of humans, i.e. it comprises

both categories of knowledge as it was defined in Chapter 3.2.1.3 for human cognition: the explicit as well as the implicit knowledge. According to the design criteria stated earlier, there is no decision made yet for any ontological commitments regarding the knowledge representation, no decision about implicit or explicit representation or any specific representation technique. There is one exception, though, as to the decision to represent the a-priori knowledge in terms of *models*, i.e. objects which are described by attributes and behaviours.

The outer part of the body contains the *situational knowledge* as time-varying information the ACU can draw on at the actual point of time during runtime. One particular portion of the situational knowledge is specified by the dynamic input information available from the work environment. In principal, all situational knowledge might be open to be fed back into the work environment as output of the ACU, in order to facilitate other agent's understanding of what the ACU is doing. In fact, there shall be one specific portion of the situational knowledge representing the dedicated output to the environment by the ACU. Except for the environmental input all situational knowledge is dynamically generated and updated resulting from the *cognitive sub-functions*. Figure 47 shows the principal design of one cognitive sub-function. What will be generated by the cognitive sub-functions is what we call the *cognitive yield* which reminds to some extent of what was said about human consciousness and working memory contents in Chapter 3.2.1.3. This element represents the *central representation* of the knowledge about the current internal and external situation. If wanted, information about situational knowledge as being generated by any cognitive sub-function in the past may become part of the a-priori knowledge as a kind of learning process. In that sense, also the a-priori knowledge is time-variant to some extent. Similar to the definition of the situation awareness, also extrapolations of the situation into the future are part of the situational knowledge, if the appropriate a-priori knowledge is implemented to generate them. It also might include the belief about its own situational status as internal situation, for instance, as also pointed out by [Grau & Menu, 1994]. In that sense it seems that the situational knowledge of ACUs represents more than that what is described by [Endsley, 1997].

Every cognitive sub-function, in principle, has full access to the entirety of situational knowledge, although it primarily draws on particular dedicated portions of the cognitive yield (zone encircled by the dashed line in Figure 47). The thereby implemented concept of the central representation of situational knowledge is a very crucial feature which implies that the ACU processing is not distributable over multiple independent systems. This is one reason why we do not take it as being built up by a customary multi-agent system [Ferber, 1999]. It has got one single processor like it is the case for the human brain, thereby providing the basis for being free for as much inter-netting as wanted. This makes it possible, in principle, that a cognitive sub-function may represent a full package of lower level cognitive sub-functions as well. The outputs of the cognitive sub-functions are supposed to be written into the designated areas at the arrowhead. These outputs consist of nothing but instantiations of models which are part of the a-priori knowledge. This brings up another important feature of the cognitive sub-functions: Their input-output behaviour is exclusively determined by the a-priori knowledge.

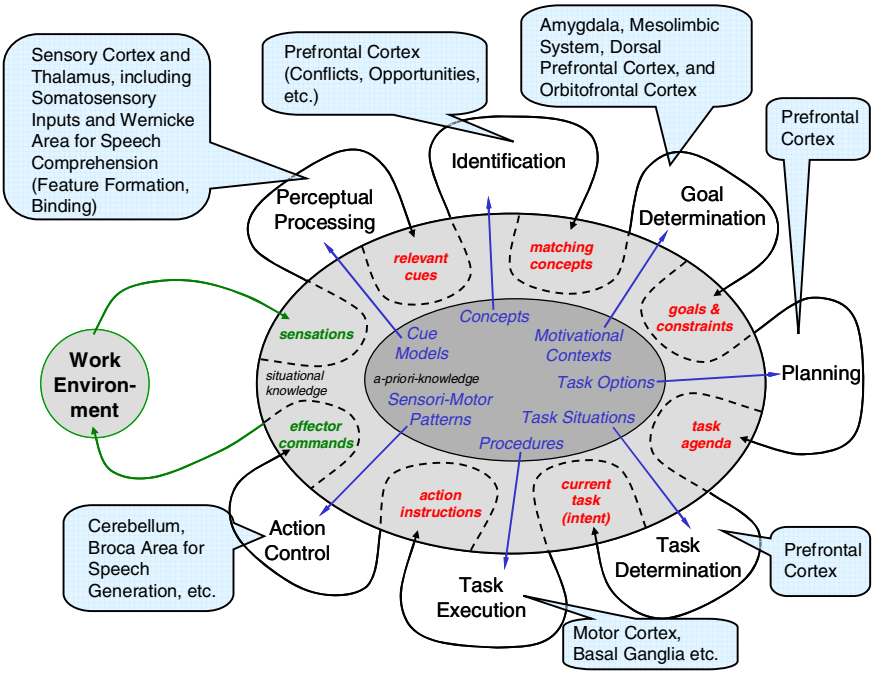


Fig. 48 ACU-architecture with a-priori knowledge and pertinent cognitive sub-functions. Also brain regions are depicted which are involved in both processing and pertinent memory of the corresponding high-level sub-functions in human cognition

Now, we can build up an ACU-architecture with a formation of cognitive sub-functions as shown in Figure 48. This figure might suggest that there is a kind of loop of consecutively working sub-functions, but this is not the case. The concept is that all sub-functions logically are working in parallel. Referring to the findings about human cognition in Chapter 3.2.1 we can associate the corresponding human brain structures involved in respective sub-functions. Deliberately, we also make total use of the terms used for the behavioural functions and associated a-priori knowledge in Figure 27 in order to indicate the similarities with human cognition.

One can think of it as superimposed partial functional cognitive ensembles corresponding to the behavioural levels of concept-based, procedure-based, and skill-based behaviour as discussed in section 3.2.2.2 (see Figure 27). The behavioural functions of the concept-based, procedure-based, and skill-based levels are taken as cognitive sub-functions with each of them keeping their particular a-priori knowledge. The corresponding three layers as shown in Figure 49 are interconnected via the body of situational knowledge forming a total of knowledge which is accessible from all cognitive sub-functions in all layers. The cognitive sub-functions as such, though, are working separately.

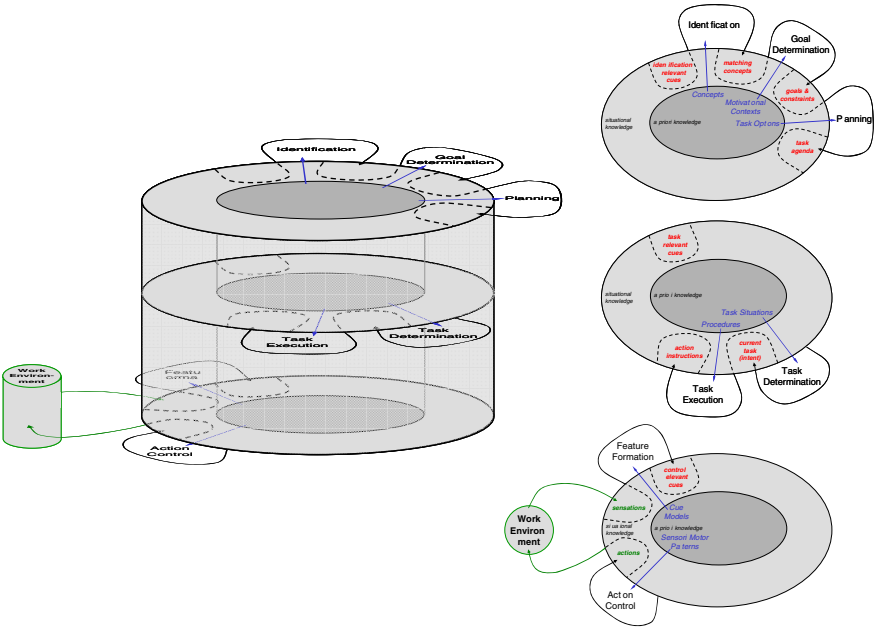


Fig. 49 Layered cognitive process including skill-, procedure- and concept-based behaviour

Depending on the extent of the a-priori knowledge, whether being of the explicit or the implicit type (see Chapter 3.2.1.3), the full range of involvement of cognitive sub-functions might occur, from a complete configuration, as shown in Figure 48, down to a minimal configuration of just one single cognitive sub-function, possibly also implicitly comprising a greater number of sub-functions in one. The latter is for instance the case, if no reasoning efforts are demanded and the activated a-priori knowledge comprises all aspects necessary to form a direct action. This helps a lot to warrant efficiency of cognitive processing in ACUs, which also corresponds to the lower levels of the behavioural scheme for human cognition (see Figure 27 in Chapter 3.2.2.2). A cognitive sub-function may also be broken down into lower-level sub-functions in order to form additional internal layers. An example is the sub-level of cognitive sub-functions which form the perception process under the cognitive sub-function of feature formation.

Chapter 5
Examples of Realisations of Cognitive Automation in Work Systems

The preceding chapters have established a kind of foundation for a systematic approach to achieve a system-ergonomic design of work systems. It was pointed out that there are two levels of automation which can be made use of, conventional and cognitive automation. As to cognitive automation, the artificial cognitive unit (ACU) was introduced as a particular kind of agents, based on a conception of the artificial cognitive process in some analogy to human cognition. Again, there are two possible modes of ACUs to account for in a work system design (dual-mode design), supporting cognitive units (SCUs) as part of the operation-supporting means in the work system, and operating cognitive units (OCUs) as members of a team making up the operating force of the work system.

This chapter shall contribute to illustrate that there are already implementations of either type of ACUs, SCUs and OCUs, as well as their integration in work systems of vehicle guidance and control. In the following sections a number of them will be described in some detail, starting with SCUs.

5.1 Realisations of Supporting Cognitive Units

This chapter does not deal with realisations of SCUs in general, because there are too many of them. Instead, it deals with cognitive systems which *are actually incorporated in work systems* as supporting cognitive units (SCUs) with the particular system architecture as elaborated on in Chapter 4.5.2. This is a very heavy constraint, but there is no room in this book to provide a general guide through the widespread world of agents, including that of robots. Therefore, also agent system developments are excluded, which are the product of engineering endeaours to create artificial systems as close as possible to autonomous ones, but without giving much attention to aspects of user requirements concerning their integration in operational work systems. The design goal of the work system designer is what we are giving much emphasis here. In addition, we account only for work systems associated to guidance and control of air and road vehicles.

As pointed out in Chapter 4.3, supporting cognitive units (SCUs) are in the first place distinct from OCUs (assistant systems) because of their insufficient knowledge about the work objective and what is related to the task of accomplishing this objective. There is no difference, though, regarding the pertinent cognitive process of SCUs. From the design point of view an even more

R. Onken and A. Schulte: System-Ergonomic Design of Cognitive Automation, SCI 235, pp. 129–211.
springerlink.com © Springer-Verlag Berlin Heidelberg 2010

significant distinction between SCUs and OCUs is the fact that the designer of SCUs is much freer regarding the use of enabling technologies. There is much less demand to account for the characteristics of human cognition. In essence, only the performance counts of providing the results which are requested from the supervising counterparts of the operating force in the work system.

In the following, a brief insight is given in an already operational SCU design known from the automotive application domain of work systems, followed by describing a prototype integration of SCUs in a work system of the aviation domain.

5.1.1 ACC System

As pointed out in Chapter 2, in the meantime there are a number of support functions at the disposal of the driver of road vehicles. One of them is the ACC system (Adaptive Cruise Control or Active Cruise Control). Many drivers are already used to it since years. ACC systems are based on former systems for plain *cruise control*. These former systems were typical examples of what we have defined as conventional automation. In extension to the plain cruise control, ACC systems have got an additional functionality, introducing on top of the set-speed control the *follow* and *stop and go* control function. These functions automatically adapt the relative speed to preceding vehicles in order to keep the headway distance at a safe range. This is the main step towards cognitive automation, making possible a safe ride in dense road traffic while staying in a lane. The motivational contexts of these systems can be specified as the implicitly implemented, prevailing desire of keeping safety while driving as close as possible to the set speed. If there is no longer a slower vehicle in its way, the system accelerates up to the set speed again. All this is not possible without the task determination capability of type 2 (see Chapter 3.2.2.2). At any time, the driving situation is evaluated against this desire, i.e. associated task situations as part of the a-priori knowledge of the system, in order to determine whether a new task should be initiated. The work objective is not completely known, though, what is one of the undeniable characteristics of SCUs (see Chapter 4.3.1).

Taking a closer look, the ACC function, including stop and go, requires the following capabilities [MOTIV ACC consortium, 1999]:

- remain safe distance from preceding vehicle
- safely slow down behind decelerating vehicle, eventually make a full stop
- automatic “go” when stopped behind vehicle
- “go” when initiated by driver in case no preceding vehicles are present
- control vehicle speed according to set speed when no preceding vehicles are present
- manage standstill condition, even on slopes
- comfortably manage near cut-ins from adjacent lanes
- recognise and manage lane changes initiated by the driver
- harmonise perturbed traffic flow
- inform driver when system limits are reached

- switch off when brake pedal is activated by the driver
- limit vehicle speed when set speed has been reached
- adjust headway according to driver preference,

The actual task is dependent upon the traffic situation and the actions of the driver. In analogy to Figure 48, Figure 50 illustrates the functional structure of the cognitive process of an ACC system.

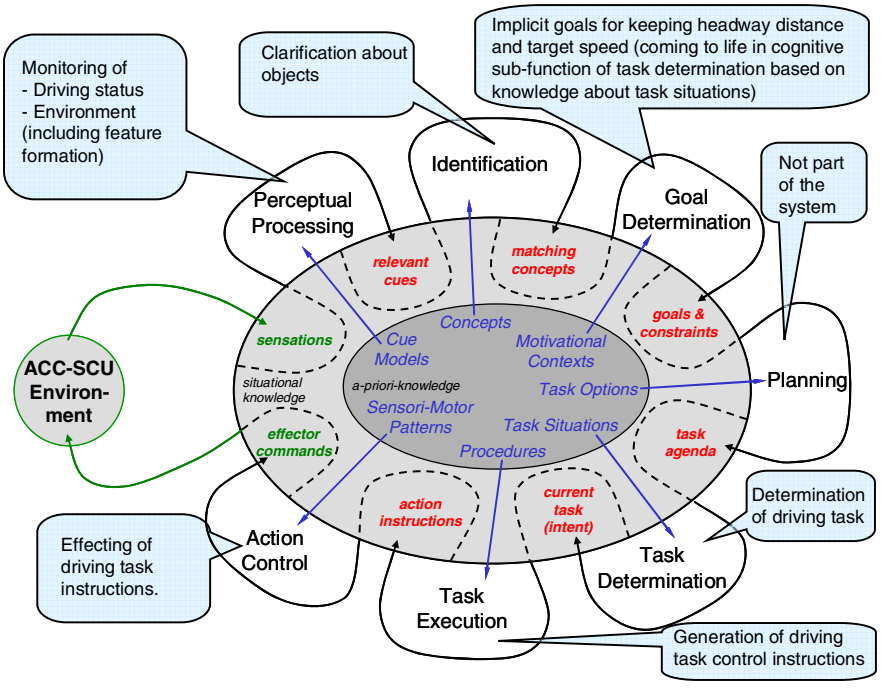


Fig. 50 The cognitive process of an ACC system and its functional structure

A great amount of a-priori knowledge is required about task situations to make sure that the traffic scene encountered leads to the determination of the correct task. All relevant environmental objects are modelled which can be perceived based on the sensor equipment. Also a driver model might include such that the system behaviour is coming close to that of the driver. [Schreiner, 1999] developed a prototype system which is able to adapt its behaviour to the individual driver. The sensory system should have the ablity to capture surrounding vehicles of relevance like cars, trucks, motor cycles, and bicycles. Usually there are several radar sensors for long and short range headway measurementwith a wide field of view. Some systems make use of infra-red sensing to avoid that pedestrians are overlooked. Future systems might include cameras for computer vision purposes.

5.1.2 Support by Co-operating UAVs

This project, as described in [Meitinger & Schulte, 2006a],[Meitinger & Schulte, 2006b], [Schulte & Meitinger, 2009 a,b] and much more detailed in [Meitinger, 2008], was focused on the activity of a team of UAVs as main operation-supporting means in the work system setup as shown in Figure 51. [Trouvain & Schlick, 2007] for instance investigate the human-machine interface for this kind of work system, confining the work system setup to conventional supervisory control of multiple ground-based unmanned vehicles. In the case to be described in the following, all UAVs are performing by means of cognitive automation. They are supposed to team up and to co-operatively pursue a common objective, i.e. a simplified air-to-ground attack mission as instructed by the operating force (human operator). Hence, each UAV of the team is piloted by an SCU. Part of the project is the development of an SCU which is the same for all UAVs. The co-operation between the fellow team members is eased by communication dialogues. Within the scope of this project the SCUs were in the first place to behave on the concept-based behavioural level, i.e. processing situational cues (identification), determining goals, and planning which results in a task agenda. Thus, the design of the a-priori knowledge is deliberately done without knowledge about task situations on the procedure-based behavioural level. As a consequence, all UAV actions can directly be tracked back to goals as being determined on the concept-based level. Obviously, requirements for computing efficiency were not of highest priority in this project, but the implementation of concept-based behaviour. As a result, it is an excellent example of high-level cognitive automation as part of the operation-supporting means. From a capability point of view, the focus of the project was co-operative mission management rather than ordinary flight planning. The latter and the coverage of the skill-based action control, were assumed to be given as well-established conventional automation like autopilot and flight management systems.

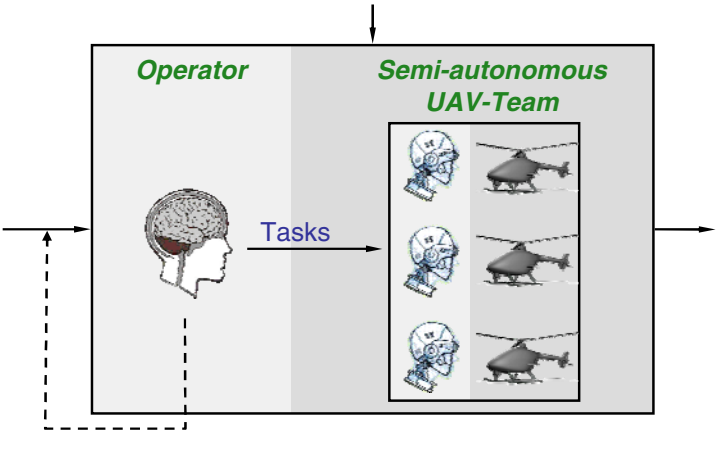


Fig. 51 Work system setup with the work support by a team of co-operating UAVs

The scenario that has been considered consists of a high level hostile target at a certain location which has to be attacked and some threats which have to be avoided or suppressed in order to create a safe corridor to the target. Some of the threat areas are known prior to the mission while others may pop up unexpectedly during the course of the mission. One of the UAV team members ("attack UAV") is equipped with the means to attack the target. However, this UAV cannot protect itself against threats. Therefore, there should at least be one additional UAV in the team which has got the sensor equipment for on-board detection of threats as well as appropriate means to suppress them. While the UAVs are equipped differently, they are piloted by an SCU of same cognitive capabilities as was earlier alluded to. These SCUs are co-operating on peer-level but are not all the same as to their roles. For instance, one team member is selected by the team by means of a certain procedure to act as spokesman. Thereby, it takes care of the communication with the operating force.

In order to accomplish the sketched mission, the participating UAVs have to cover the following capabilities [Platts et al., 2007]:

- Use of other UAV equipment of conventional automation, such as an auto-pilot, a flight management system and a flight planner minimizing threat exposure.
- Safe flight, i.e. ensuring that the UAV can fly within 3D-space without colliding with other UAVs or terrain.
- The capability of a single UAV to actually accomplish a task such as "suppress threat".
- Co-operative mission accomplishment, i.e. the ability to co-operate with other UAVs in order to achieve the common objective, namely the mission assigned to the team by the operating force

All of these aspects had to be considered when implementing a prototype of the SCUs. Since the focus of the project was on co-operative behaviour, the first three aspects were only covered as much as needed by the co-operative capability. In the following, the development of the prototype with respect to the last aspect, i.e. co-operation, will be described in some more detail. First, the most important models to be part of the a-priori knowledge relevant to co-operation will be explained. Then, an idea of how different models interact with each other will be given, using the distribution of workload within the team and an interaction between team members as an example. Finally, some results regarding functionality, behaviour on concept-based behaviour level, and human-machine co-operation will be shown.

The SCU prototype is designed according to the cognitive process as described in section 4.5.2 and implemented by the use of COSA (COgnitive System Architecture) (see section 7.1). Its capabilities regarding the different aspects of the cognitive process are depicted in Figure 52. So far, it is the most complex system implemented in COSA so far with 144 classes and 1139 Soar rules being modelled. The behaviour of this SCU is driven by the implemented a-priori knowledge, in particular that about the motivational contexts with emphasis on those for co-operative behaviour. In addition, a-priori knowledge about task options and task

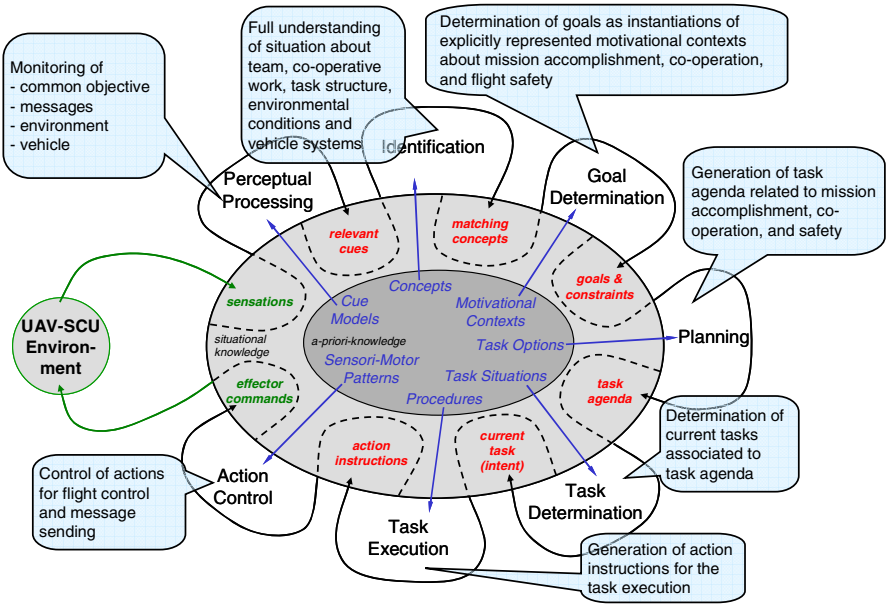


Fig. 52 The cognitive process of a UAV-SCU and its functional structure

execution procedures is included as much as required. Regarding the development process of defining the models of the a-priori knowledge of the SCU, the motivational contexts have to be modelled at first. In a second step, the task options (action alternatives), which the SCU should have at its disposal in order to achieve the potential goals, must be defined. As it should be possible to execute any task as soon it has been chosen as an element of a task agenda, appropriate procedure models have to be generated next. Subsequently, environment models have to be generated, which are necessary for activation of goals, selection and instantiation of task options or instruction models. After the declaration of the models and the specification of their attributes, the behaviour of the models and interactions among them have to be stated and integrated. In reality, this procedure of knowledge acquisition and representation is not as linear as described, but cyclic, because there might be identified additional potential system goals in the course of the design process which are likely to be covered by additional motivational contexts. This approach to implement the a-priori knowledge into the cognitive system is called the "CP method" (see [Putzer & Onken, 2003] and [Putzer, 2004]). The number of classes of this development amounts to 144 and the number of Soar rules implemented amounts to 1139. This might provide an idea about the size of the system. Some more details about the implementation of the a-priori knowledge of the SCU prototype can be looked up in section 6.2.3.1.

The main motivational contexts that underlie co-operative behaviour of the SCUs guiding UAVs are to

- form a team
- achieve the common objective

- manage interdependencies and
- communicate

They form four sub-groups representing to a great extent the capabilities which have been listed above. In case of possibly conflicting motivational contexts, normative rules of priorisation are introduced. Future extensions of the system might need to associate relative weights to the motivational contexts in order to avoid that the system gets stuck or is pursueing the wrong goal.

In order to achieve the common mission objective, commitments have to be managed and the associated tasks accomplished. Commitment management hereby refers to both accepting and dropping commitments under certain circumstances. Usually, a commitment is accepted by an individual team member or a team if somebody (e.g. operator) requests the accomplishment of a certain task and the accomplishment of this task is judged as being feasible. A commitment should be dropped if it is achieved, irrelevant or unachievable ([Cohen & Levesque, 1990] and [Jennings, 1994]). As this does not only refer to own commitments, but also to commitments which other team members have accepted due to own requests, irrelevant dialogs should not be continued. Moreover, a team member should not only consider each commitment separately but should also consider whether all its commitments are achievable as a whole. Team members as individuals are responsible for the actual accomplishment of tasks associated with commitments, therefore, commitments have to be arranged in a certain order in which they shall be considered (so-called 'agenda') and actually be complied with at a certain point in time.

Following the requirements derived by [Billings, 1997] to be imposed upon co-operative agents, co-operative behaviour includes appropriate information exchange within the team driven by the motivational context to have all necessary information about the other team members and to keep them informed with respect to e.g. capabilities, resources, and commitments. The distribution of tasks within a team should be organised in a way that all tasks which should be carried out are assigned to a team member and at the same time workload is balanced among the team members. Finally, in the project described here, the team is structured such that all team members are peers, but that communication with actors not belonging to the team is performed by a spokesman. Therefore, a motivational context "have spokesman" has been formulated.

Within the scenario described earlier three kinds of interdependencies between team members are relevant. First of all, shared resources (here a corridor which has to be used by all UAVs to fly to the area of interest) have to be assigned to individual team members. Secondly, redundant task assignment shall be avoided in order to maximize team effectiveness. Thirdly, team members shall support fellow team members by carrying out tasks that facilitate the task accomplishment of the fellow team members. The management of these interdependencies is triggered by appropriate goals.

Communication among the SCUs is structured according to the specific dialogue representation. In order to keep dialogues running, the goal 'continue dialogue' becomes active, if the current state of a dialogue requires an SCU to send a message next.

In order to be able to achieve these goals, the SCUs may choose among various *action alternatives*, namely to accept or drop commitments, send messages, and initiate dialogues. Within this context, the following different types of dialogues can be initiated all of them being based on FIPA specifications (see Chapter 4.4.3.2):

- ***Request*** – The UAV initiating the dialogue requests that the UAV receiving the message executes a certain action, usually to accept or drop a commitment about performing a certain task.
- ***Propose*** – The initiator proposes to the UAV receiving the message to execute a certain action on its own, usually to accept or drop a commitment about performing a certain task.
- ***Query*** – The initiator asks the the UAV receiving the message to provide him with some specified information.
- ***Subscribe*** – The initiator asks the the UAV receiving the message to provide him with some specified information and to repeat this every time the value of the specified information changes.
- ***Inform*** – The initiator provides some information to the UAV receiving the message without being requested to give that kind of information.
- ***Cancel*** – The initiator wants the UAV receiving the message to stop the execution of the activities begun due to a prior dialogue. Typical examples are to have the UAV receiving the message stop carrying out a task or providing the information as requested by the prior dialogue.

As mentioned earlier, appropriate *instruction models* are necessary in order to put selected action alternatives into effect. Unlike "typical" instruction models, most of the instructions within the context of co-operation do not directly influence the environment, but change the situational knowledge first. For instance, the execution of the action alternative "commit" creates a commitment within the belief section or the instantiation of a dialogue actually creates that dialogue in the belief. The only instruction being sent to the environment is sending a message to other UAVs which in most cases follows a change in situational knowledge.

Within this context, various *environment models* are required. First of all, a representation of actors is needed, which is instantiated for each actor in the setup, here for each SCU piloting a UAV and for the human operator. Closely related to these models are the ones which describe the various characteristics about capabilities and resources of the actors. Since the actors are organized in a team, also a model is necessary about how a team is to be defined. This model is for example attributed with the members of the team pointing at instantiations of actors. With respect to the common objective of a team, commitments have to be represented. These are further described by the task which is related to the commitment and the actor or team, which is responsible for the commitment. As commitments refer to tasks, tasks have to be modelled. Furthermore, a representation of dialogues, their states and transitions are needed, which again are related to actors and/or teams involved in an interaction.

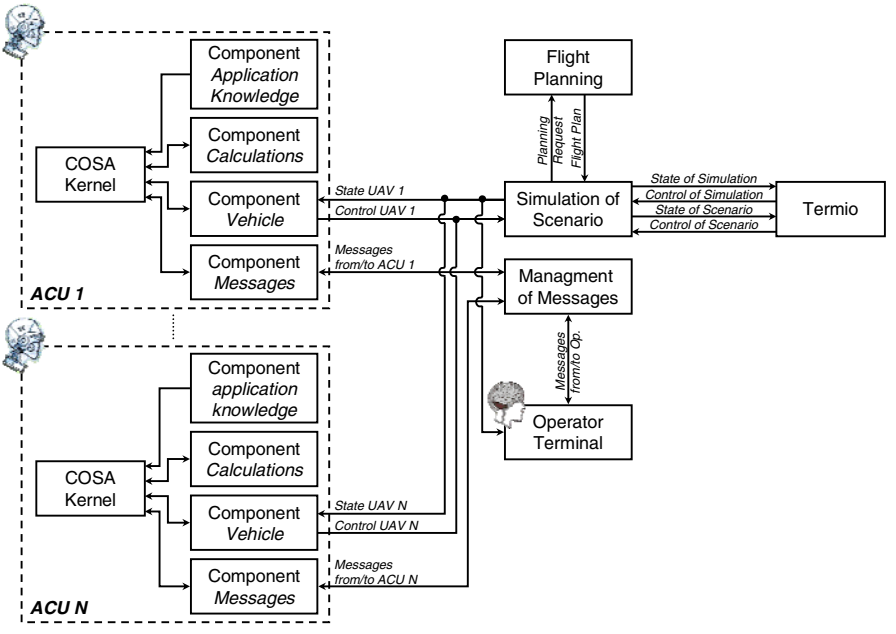


Fig. 53 Simulation Environment [Meitinger, 2008]

The functionality of the system was successfully tested in a simulation setup as shown in Figure 53. A great number of test runs were conducted. Hereby, different scenarios were used, some of which considering only parts of the overall setup and therefore demanding for particular key capabilities of the SCUs, while others comprised the complete mission scenario in order to test the overall capabilities of the SCUs. Figure 54 shows the operator terminal of the simulation setup.

In Figure 55 a sequence of pictures is shown of a simulation test run with a reduced scenario considering mission accomplishment in a heterogeneous UAV team. It consisted of a team of 2 UAVs, one attack UAV (ID 4) and another one (ID 0) which is able to suppress any of the threats being around. One threat area covers the location of the high value target to be attacked as shown in Figure 55 for the time 00:00. Figure 56 depicts some of the messages exchanged and dialogues conducted during this simplified mission. While dialogues with type "inform" consist of just one message with the performative "inform", request dialogues are composed of several messages which are connected with a line. At first, the operator requests to attack the target which is accepted by the UAV team. The attack UAV commits itself to attack the target as there is now a team commitment to do it. Moreover, it informs the fellow team member about this change in its commitments (dialog-inform) and asks it to suppress the threat being in the way when approaching the target (dialog-request). Due to the goal to keep team members informed, the ACU guiding UAV 0 also informs about its commitments



Fig. 54 Operator terminal of the simulation setup

(dialog-inform), namely to suppress known threats, to protect the attack UAV from possible pop-up-threats, and finally to return to the homebase.

In the following, both UAVs comply with their commitments, i.e. the attack UAV flies towards the target, while the other UAV (ID 0) starts flying towards the threat (see Figure 55, 02:07). After the UAV (ID 0) has successfully suppressed the threat (Figure 55, 03:53), it informs the ACU guiding UAV 4 and the attack UAV adjusts its flightplan and continues its way to the target, now being escorted by the other UAV (see Figure 55, 04:31). After having successfully attacked the target, the team continues the request dialogue initiated by the operator and informs him about the successful outcome of the mission. Finally, both UAVs start returning to the homebase (see Figure 55, 06:29 & 11:49).

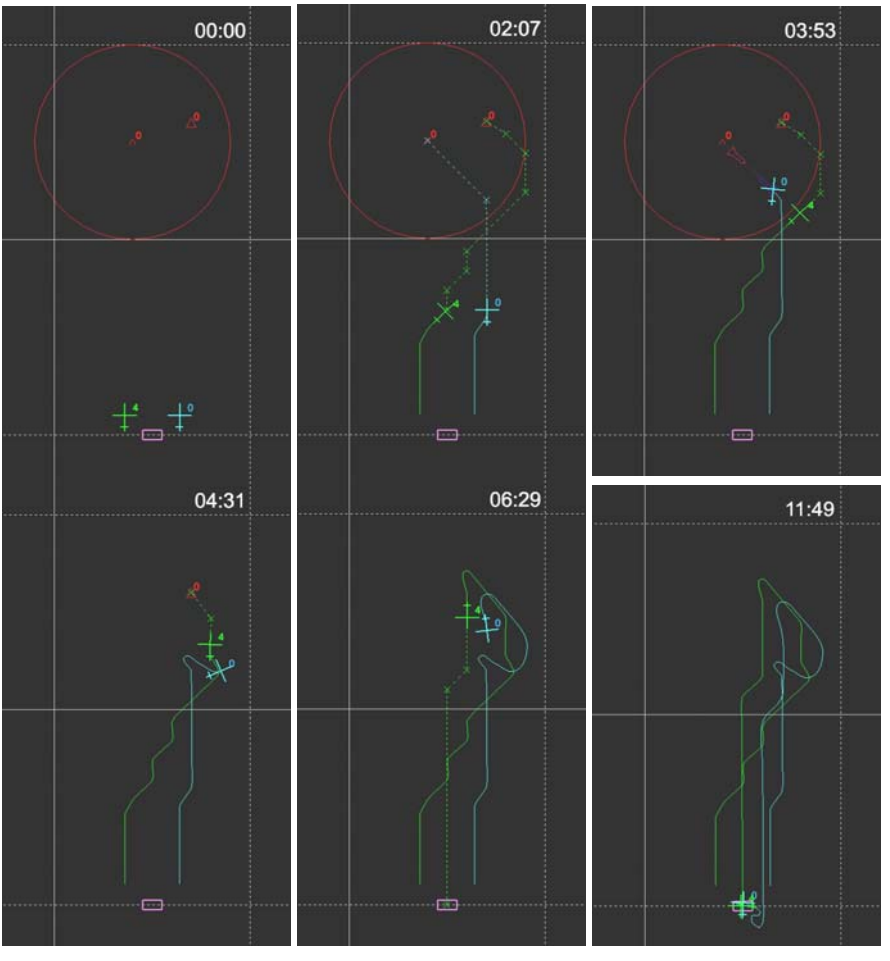


Fig. 55 Example of mission accomplishment of a team of 2 UAVs in a test run [Meitinger, 2008]

Within the project described here, scenarios consisting of up to 10 known and unknown threats and 5 UAVs as well as a corridor for penetrating into the action theater have successfully been tested in a simulated environment with the SCU prototype used for UAV guidance as described.

Besides the functionality as such also the capability of the prototype to behave on the concept-based level of performance was successfully evaluated (see [Meitinger, 2008] and [Meitinger & Schulte, 2009]).

Finally, this work also provides the basis for human-machine co-operation although its main focus on machine-machine co-operation aspects. This add-on could be achieved because of a consequent provision of models of goals for co-operation, of coordination techniques, and of dialogue management related

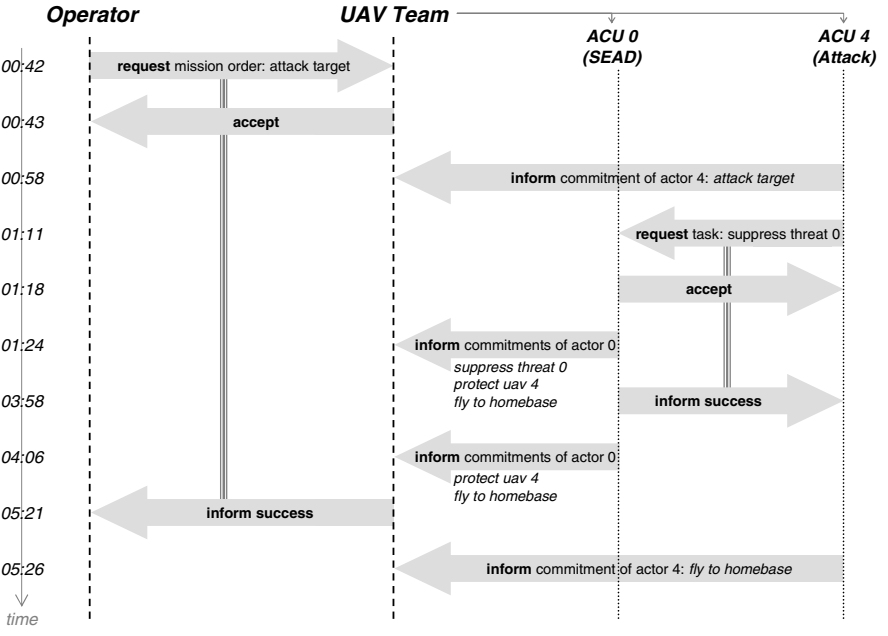


Fig. 56 Excerpt of communication between UAVs in test run [Meitinger, 2008]

knowledge. Within this context, an experiment was conducted in order to gain first insight to problems arising in human-ACU co-operation and to derive requirements for future ACU development. There, human pilots had to control up to three UAVs equipped with ACUs to accomplish a mission as described at the beginning of this section (see [Meitinger, 2008],[Rauschert et al., 2008], and [Meitinger & Schulte, 2009]). It was expected, that humans would be able to manage co-operation with one UAV but workload would exceed an acceptable level when increasing the number of UAVs.

To investigate this, five experienced military air crew members were asked to work together with one or three UAVs in different roles (attack or SEAD, see Figure 57, bottom) and scenarios. The manned aircraft could be controlled on autopilot level. Co-operation within the human-ACU team was based on the exchange of information on capabilities, resources, i.e. weapons, and commitments via a graphical user interface. Moreover, task requests could be sent to either individual UAV or the whole UAV team. After each run workload was measured using the NASA-TLX method [Hart & Staveland, 1988] and the subjects were interviewed.

The performance of the human-machine team was very good. More than 90% of all missions could be completed successfully. Regarding the workload of the subjects, it was on a medium level in all configurations that have been investigated (see Figure 57 for the average workload of humans in different configurations). Interestingly, there is only a small increase in workload when changing the number of UAVs within the team from one to three. Moreover, the increase in

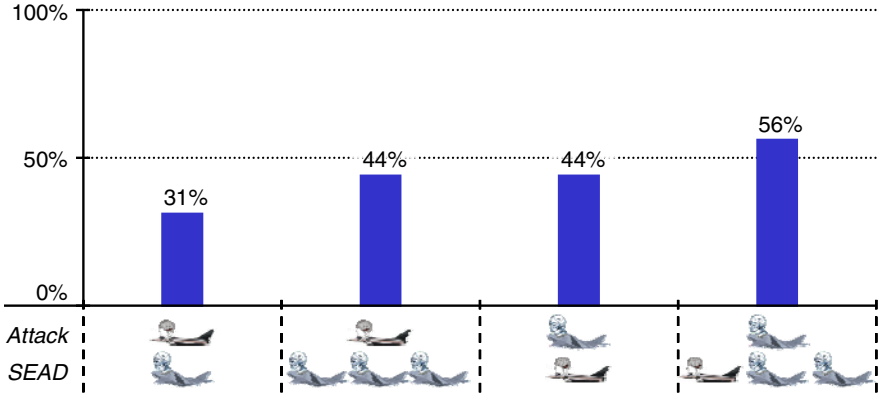


Fig. 57 Average workload of subjects in different configurations [Meitinger, 2008]

workload when changing the human role from attack to SEAD is as large as when adding two UAVs, which could be related to different performance of UAVs in the SEAD and attack role.

These results show that in principle human-ACU co-operation is possible with the approach of cognitive and co-operative automation and that workload is on a medium level even though ACUs were not designed to work together with humans. Still, further improvement potential was identified in the following areas after having analyzed interview protocols:

- *Team Structure.* The introduction of a hierarchical team structure is encouraged, although team members shall be capable of situation dependent deviation from leader input.
- *Abstraction of Interaction.* Interaction with ACUs shall be as abstract as possible (e.g. specification of tasks for UAV team rather than detailed instructions for single UAVs). In particular it shall be possible to give instructions on different abstraction levels.
- *Level of Automation.* ACUs shall be able to act on a high level of automation but the actual level shall be adaptable or self-adapted to the current situation and task.
- *Teamwork.* Co-operation of humans and ACUs shall be based on a common agenda and anticipate future actions of team members.
- *Task completion.* The capability of ACUs to actually accomplish tasks has to be improved.
- *Communication within team.* Vocabulary shall be adapted for more intuitive understanding of humans. Moreover, the number of dialogs and messages shall be reduced to a minimum in order to avoid overload of humans.
- *Assistant System.* The human team member shall be supported by a electronic assistant providing him or her with situation awareness concerning team members, task assignment and information flow within team as well as accomplishment of his or her primary mission related task, i.e. aircraft guidance.

Ongoing work is intended to account for these findings and bring together the different aspects of human-machine and machine-machine cognitive co-operation in manned-unmanned multi-agent scenarios. In the field of supervisory control of UAVs the question of how to reduce the operator-to-vehicle ratio will be predominant. Hereby, cognitive and co-operative automation offer solutions for many of the upcoming questions.

5.2 Operating Cognitive Units (Assistant Systems)

Referring to Chapter 4.3, SCUs and OCUs as part of a dual-mode cognitive design are work system components as artificial cognitive systems (see Chapter 3.1.2). In distinction to SCUs, OCUs as mode 2 cognitive automation play a special role by way of cognitive teaming of directly co-operating with the human operator(s) as part of the operating force in a work system. Since the human operator(s) naturally functions (function) as team leader(s) the OCUs are assisting the human operator(s). We are using the term "assistance" to intentionally separate it from the term "support" which is occupied for automated sub-systems of the operation-supporting means. We started the development of OCUs as so-called *assistant systems* about 20 years ago, at a time, when this term had not been used yet in the field of automation. It was chosen in analogy to human assistants in a human work team like that for instance in a surgery room with surgeons, anaesthesists and nurses as cooperating team members. In such a team the assisting team members may have qualifications which may be similar in skills and knowledge to that of the chief surgeon, but also quite different. One thing is for sure, though, that they all know that the chief surgeon is the team leader with certain unique competences. Here, we consider similar kinds of teams as operating force which consist of a mix of human operators and OCUs with one of the human team members being the team leader, Unfortunately, in the meantime the term "assistant" became a kind of synonym for various kinds of innovative systems in automation technology, which does not necessarily match with our definition, mainly because this term seemed to be an attractive one. Despite of that fact we want to keep this term and its definition, hoping, that this book will contribute in the future to have the term more and more understood again by its original meaning.

Assistant systems are introduced into work systems as co-operating team members of the operating force in addition to the human operator(s). All team members, including the assistant systems, are working in parallel according to their roles, which usually are different for each team member, but not exclusively different in all aspects and not independent. Assistant systems range widely. On the one side there are those which are highly specialised and fitting an unpretentious co-operating role, and on the other side there are those which may be more or less equivalent to the human operator regarding their a-priori knowledge about the work domain concerned and the corresponding breadth of activities in information management. A human operator as a co-ordinating agent for the team takes care for an overall work plan which is agreed upon by all team members and which specifies the tasks and the sequencing order of carrying them

out. All team members are interpreting the environmental situation in parallel and are deliberating individually on what their goals and actions ought to be, subject to their roles, their a-priori knowledge, and sufficient knowledge about the work objective of the work system. Let us repeat what was pointed out in Chapter 4.3.2: This takes a motivation as part of the a-priori knowledge on motivational contexts to do one's best for the achievement of the work objective.

In principle, all sub-functions of the cognitive process of any team member can be assisted no matter whether it is done by humans or OCUs. In order to keep the human operator as much involved as possible without overcharging him, usually only a limited selection of the human operator's cognitive sub-functions is assisted by OCUs, though. There are three design specifications which have to be decided on concerning every cognitive sub-function of the human operator:

a) whether it is to be assisted at all
b) whether the assistance is to be done by a human team member or by an OCU, and
c) the way of interaction with the human team member.

Regarding the latter specification, there is a great variety of how an assistant system might co-operatively interact with the human team member, ranging from whether the assistant system has got a commitment to display certain pieces of information which were figured out by it about something which possibly could be of interest for the human operator, up to whether it has got a commitment to act as a substitute for the human team member without taking care of informing anyone about its action. In summary, there are basically three characteristic styles of co-operative assistance which can be distinguished, i.e.:

(1) **associative assistance**,
(2) **alerting assistance** and
(3) **substituting assistance**.

These styles will be described in the following in some more detail. They may appear in a mix when designing the operating force of a work system.

5.2.1 Associative Assistance

Associative assistance is rather independently working on its commitments, which, in essence, are also belonging to those of the human team member it is assisting. The basis for the performance of the assistant system is its knowledge about the work domain, also including the knowledge about the work objective and possibly additional knowledge in order to be able to inference the intents of the human operator. It is continuously offering its results as proposals to be possibly utilised by the human team member, but not drawing the human team member's attention on them in an undeniable way like it would be done in case of an alert. The human team member may completely ignore the assistant's results for any reason. He also may look at them, thereby

- being free to take the results of the assistant system for granted without further checks and making use of them in whatever way, also by automatically having the assistant carry out the corresponding tasks or
- accepting the results of the assistant system after verifying them, or
- performing by himself the tasks which are necessary to achieve this kind of results and comparing them with those of the assistant system. This possibly can lead to a mismatch situation where the human operator has to decide about whose results are eventually to be used.

Associative assistance is well-suited for work system designs, where it is required for any reasons to leave all initiatives of action decisions with the human operator, including those of active communication between assistant system and human operator. Then it is only up to the human operator, whether to look for any information which is derived by the assistant system in the background and presented as an offered proposal without actively drawing the operator's attention towards it. This is the style of assistance with least amount of demands on the design to prevent inadequate assistance. Inadequate assistance might happen, though, because of insufficient harmonisation of the beliefs about the situation between both sides, that of the human operator and that of the assistant system.

Typically, the cognitive sub-function of planning is a candidate for associative assistance, in particular, if there are to be solved complex problems. However, also the area of the cognitive sub-functions of situation assessment and goal determination might be considered in that context. This is particularly effective when there are commitments of the associative assistant system to generate proposals for decisions which

- are due for decisions regarding concurrent urgent tasks of the human team member,
- are very laborious and involve time consuming computations, and
- can easily be assessed by the human operator regarding their effect on the work objective.

Automatic planning systems as such are already available as operation-supporting means. In distinction, cognitive planning assistants work on the basis of own knowledge about the work objective and a-priori knowledge, i.e. the a-priori knowledge regarding the cognitive sub-functions of situation assessment and the a-priori knowledge about motivational contexts and task options to select from in order to determine ranked alternatives of action agendas (plans) as the resulting proposals. Consequently, the cognitive planning assistant has to be capable to take care on its own, just like the operator, of being aware of the situation and to provide the proposals based on that.

The situation awareness of an ideal associative assistant should include what is considered as an important characteristic of a team member of the operating force, i.e. that it has continuous information about the activities and intents of the human counterpart. Therefore, the associative assistant is trying to infer on its own this kind of information, ideally not interrogating the operator. As a consequence, it has to deal with the problem that its own situation awareness might not be

harmonised with that of the human operator. This may lead to situations where the human operator does not understand what the assistant is doing, for instance when proposing a new plan or task.

In case of associative assistance it should also be taken care about not unloading the human operator of too much work, for instance by allowing the assistant to take over at a high authority level. This can lead to a particular kind of human overtaxing, the so-called out-of-the-loop phenomenon. Then there might be a situation that he is not prepared to perform in due time a task which is urgently demanded and which he is supposed to carry out.

In essence, *associative assistance* is capable of continuously proposing solutions for how to proceed, thereby taking the overall situation into consideration, the work objective included.

5.2.2 Alerting Assistance

The alerting assistant system is working in parallel to the human team member it is assisting. As a dissimilarly redundant agent it is working on certain same cognitive work commitments and pertinent cognitive sub-functions. The purpose of this style of assistance is to make sure that in case of discrepancies between the results of both the assistant and the assisted human team member these discrepancies will be actively communicated to the assisted human team member by drawing his attention towards this fact in an undeniable way, possibly also giving advices how to harmonise again. Thereby, the assistant might bring valuable monitoring information into the work process like it could also be the case if the same function would be occupied by an additional human team member. Alerting assistance is only intervening in case of situations which the human operator does not seem to be completely aware of. It is never acting as a substitute.

Alerting assistance as a kind of artificial cognitive redundancy is capable of making key contributions, if a high degree of work system performance is to be achieved and integrity is required. As to the performance aspect good candidates for this kind of cognitive assistance are tutoring work systems. Integrity is in particular of interest in case of safety-critical work systems and in case of work systems where human errors or failures might cause high amounts of cost. As a side product, this style of assistance is also a contribution to warrant high performance standards. The way of actively communicating with the human operator plays an important role for the success of this style of assistance (see [Gerlach, 1996]).

Alerting assistance can be an option for the enhancement of the human operator's situation awareness, for instance. Usually, the most disturbing cognitive limitations of humans are those of the mechanisms of attention control (see Chapter 3.2.1.4) which can also interfere, if there are no task load problems at all. Assistance of this kind would work in parallel to the human operator on cognitive sub-functions such as feature formation, task determination, and identification. Think for instance of the work process which includes driving a car. We all have experienced instances of overtaxing as drivers, regarding the cognitive

sub-functions to ensure situation awareness. Just think of the load on attention or the simple fact that our eyes can only look in one direction at one gaze and that this gaze can only lead to accurate perception of details within an angle of just a couple of degrees, if there is a busy environmental scenary (see Appendix 1.2). Often, already the lack of resources at disposal, like time, makes you wish that you could rely on another team member which can offer certain additional resources when they are needed. Another example is a modern two-man airliner cockpit with operation-supporting means including complex conventional automation. The second crew member in the cockpit is mainly there, because a single pilot is supposed to be overtaxed in terms of the amount of concurrent tasks to be managed in the work process. Yet cognitive overcharging might still occur. Apart of the role to fly the aircraft according to the flight plan there is also the fundamental task for the two crew members to monitor the work process as a whole from their own perspective subject to the work objective and the work plan agreed upon. This also includes monitoring the behaviour of the team mate. Two pairs of eyes see more then one pair. However, there is a question how effective that really is. There are situations when humans are not aware of an occurrence of inadequate behaviour of one of their team mates because of suffering from the same cognitive deficiencies. One drastic example is the famous Everglades crash of a Lockheed L 1011 when executing a missed approach at Miami. According to the National Transportation Safety Board (NTSB), it happened, because the whole cockpit crew (there were even four crew members at that time) was distracted by a problem with the landing gear. Flying at 2000 ft altitude, **none of them** realised in time that the autopilot altitude hold function was disengaged by a light force on the yoke and that the aircraft was going to descend. Inadequate behaviour is due to overtaxing in one or the other way, in particular resulting from principal weaknesses and limitations of human cognition (see Chapter 3). Also incapacity of the human operator because of a physical problem like an injury is a case of overtaxing, although this is occurring very rarely.

Alerting assistance for the cognitive sub-functions of ensuring situation awareness can make up for that drawback of pure human teams by monitoring the work process and the team members' behaviour from its own perspective. Thereby it might recognise inadequate human behaviour and might take care to have the process restored back to normal. The main reason for making use of alerting assistance to ensure situation awareness is the fact that it ensures that the human operator's attention is drawn to inadequacies without any time lag. This is essential, because a quick reaction of the human team member might be saving high costs as long as he is capable to carry it out. Therefore, alerting assistance for enhancing the human operator's situation awareness is the most important application, although it is not confined to it.

One other crucial point in this respect is the fact that artificial and natural cognitive systems are complementary in many aspects of cognitive capabilities. Hence, the assistant system should feature as much strengths as possible where the human operator is inferior. In other words, it should represent a complement in behavioural strengths, but still present as much overlap in capabilities as possible. Attention assistance is a supreme opportunity in that respect. This can make a key

contribution to ensure situation awareness which in turn is fundamental for the successful performance of the human operator. In order to closely monitor the human operator the alerting assistant system is best located at the work site of the human team members. Then, one can also take advantage of implementing only one OCU such that the design recommendation of keeping the number of OCUs as small as possible is optimally met.

In essence, *alerting assistance* is capable of recognising inadequate human behaviour and is drawing his attention towards that incident.

Fundamental for the development of alerting assistance are the available artificial capabilities of perception, not only to derive the cues necessary to artificially correctly understand the situationof the environment, but also to provide valid cues about whether the human operator has in fact become aware of this situation, too. The most important perceptual modality in this respect is the visual one. Therefore, modelling of human perceptual behaviour, including the knowledge about which are the visual cues the human operator is looking for, as well as gaze measurement (see [Findlay et al., 1995]) play a predominant role in this context. For instance in [Schulte, 1996] and [Schulte & Onken, 1995] some encouraging experiences are described as to experimenting on visual routines of pilots in low level flight.

5.2.3 Substituting Assistance

Substituting assistance is named that way, because it is substituting for work commitments of human team members. Thereby, the substituting assistant system is unloading the assisted human part of the operating force by participating as a full authority team member in order to accomplish the commonly known work objective, i.e. participating to negotiate among themselves the work plan and work commitments, and to carry out the tasks according to their commitments with great independence and mostly direct impingement on the real world environment (work object). Although operating with great independence, a substituting assistant system may also participate when the work plan and work commitments are negotiated among the team members. In certain extreme cases, this can lead to replacements of human team members, if there are several humans involved. Still, an assistant system filling this kind of role has to respect specific intermediate instructions of the human operator as the team leader. There can be distinguished two types of substituting assistance commitments:

- temporarily substituting
- permanently substituting

which will be discussed in the following.

The *temporarily substituting assistance* is of great effect, if unforeseen events of human overtaxing have to be counteracted. A typical example is the so-called emergency braking as an automotive application. The human driver tends to start braking by putting too low pressure on the brakes, also in cases where a collision with a vehicle in front can only be prevented, if maximum deceleration is achieved

with very hard braking right from the very beginning. An emergency braking assistant could warrant this. It takes the human operator out of the loop for a limited amount of time to warrant maximum deceleration as demanded. This may happen without indication, because it has to react as swiftly as possible before it is too late. Since this function bluntly overrides the human's activity once in a sudden in a safety-critical situation, it should be ensured that its intervention is based on perfect situation awareness and that it is exactly what is demanded in that situation. This is one of the greatest challenges for the designer of such an assistant system.

Temporarily substituting assistance is about a synonym with the cognitive functionality which is well-known in the meantime under the term *adaptive automation* (see [Scerbo, 2001], [Parasuraman, 2003], and [Di Nocera et al., 2003]). Adaptive automation is working on certain commitments to intervene without waiting for being authorised in case of rare events of overtaxing or incapacity, because the human operator fails to work on his commitments in the work process in a minimally appropriate manner. [Parasuraman, 2003] refers to four major categories of identification of the kind of overtaxing event which could be associated to adaptive automation, thereby using various techniques in order to recognise:

- critical events (environmental sensors)
- critical operator performance deterioration (behavioural models)
- critical operator drawbacks of physiological resources (physiological sensors, physiological models)
- critical operator drawbacks of cognitive resources (neurophysiological sensors, cognitive models)

Sub-systems of the category of environmental sensors are already in operative use. A prominent example is the safety function of having an aircraft automatically pop up in case it violates safety margins concerning the proximity to the terrain.

For the remaining categories modelling the situation-dependent state of human resources is a major design requirement [Donath & Schulte, 2006]. Sensors cannot be easily implanted into the brain, though. Even neurophysiological sensing devices like EEG receivers which have to be fastened on the body, are hardly acceptable for civil applications in the domain of vehicle guidance and control work processes. In addition, if it is achieved to identify a work situation of potential overtaxing because of too little load on the human operator, what should the assistant system do about it? Many engineering questions are arising at this point and there is still no final answer. One should bear in mind, here, that all functions being part of an assistant system as mode 2 automation are based on sharing the knowledge about the work objective with the human operator and having available the a-priori knowledge needed to co-operate with him effectively in pursueing the work objective. Otherwise, what was considered at the first glance to be an assistant system, would turn out to be just part of the operation-supporting means of the work system concerned, implemented as mode 1 cognitive automation.

Temporarily substituting assistance may also be an option of assistance to be authorised by the human operator in order to take over certain commitments from him for a limited period of time. This may occur, when the human operator envisions a considerable increase of workload which might go beyond his resources. In this case it is up to him to decide about when the assistant shall take over and to which extent. To be prepared for that it is necessary that the assistant is permanently following the human operator's course of action and the changing environment, subject to its knowledge about the work objective, thereby being perfectly informed about the situation when the authorisation will be received. This kind of assistance may have been originally a supporting cognitive unit under supervision of the human operator, more or less doing the same thing, but needing a lot more instruction and monitoring by the human operator when activated because of insufficient knowledge about the work objective. Corresponding design provisions are called *adaptable automation* [Opperman, 1994]. This approach takes it for granted that, in essence, the human operator can warrant the adequacy of his online-adjustments of the level of automation. In Chapter 5.3.1.3 a simulator project is described which demonstrates this feature for a fighter cockpit work environment on a task-by-task basis [Taylor et al., 2000]. Since it is not always certain that the human operator has got the time to carefully decide on the type and level of automation, a combination of both adaptive and adaptable automation is also considered in that program. [Flemisch et al., 2003] advocates a solution of haptic interface for vehicle control. By taking a tighter grip, the human operator is announcing that the substituting assistance for control is to withdraw (only intervening for the sake of safety), while taking a looser grip the operator is announcing that the substituting assistant system is to take over to a greater extent, possibly up to completely taking over with the operator being completely out of the loop, if no grip forces can be sensed anymore.

In the extreme case temporarily substituting assistance becomes permanently substituting assistance. Permanently substituting assistance may be authorised by the human operator similar to the temporarily substituting assistance, but now in advance of the beginning of the work process, or it acts as a function permanently built in by design. It actually works by rigidly substituting (partially or completely) for work commitments which otherwise were those of human team members of the operating force. This style of assistance becomes a compelling solution, if the human team member to be substituted by an assistant system would principally not be capable to do the job, or would have to work under conditions of high risk or would be the cause of too high costs. A typical example is an assistant system which is to replace a human pilot on purpose, because the air vehicle's size cannot accommodate a human pilot. Also a UAV which has to operate in a dangerous environment would be an example.

Oftentimes, permanently substituting assistant systems may have originally been SCUs under supervision of the operating force (see [Meitinger, 2008]). They are upgraded to become an OCU and to be part of the operating force. For that purpose they are provided with additional knowledge in both:

- knowledge about the work objective **and**
- a-priori knowledge which enables to make use of the knowledge about the work objective to co-operate effectively as a team member of the operating force in the work system concerned.

For new developments of work systems, which are designed from scratch or which are redesigned under no limiting preconditions in that respect, one may freely specify the formation of the operating force subject to criteria like safety or cost effectiveness, or a combination of them. Then, substituting assistant systems are introduced into the work system design in order to sensibly compose the human force as well as to keep the number of SCUs as small as reasonably possible.

This includes the specification of the combined formation of humans and assisting systems in the operating force on the one hand, and on the other hand, according to a dual-mode cognitive design, the specification of SCUs and other operation-supporting means. As a valid example we pick the highly relevant and actually pursued development of manned-unmanned teaming (MUM-T) for controlling UAVs. In this case, there are various options for the formation of the cognitive units as part of the operating force and the operation-supporting means, as depicted in Figure 58 by a selected number of possible design alternatives.

One of these structural alternatives in Figure 58 has got a configuration extremely relying on the human component, i.e. without any OCU and SCU involved. Each UAV is supervised by a human team being the operating force. This may include a varying operator vehicle ratio, depending on the application demands. The other extreme shown is the configuration with only one human operator left in the control station, forming the operating force together with permanently substituting assistants, dislocated in each UAV, and no SCU involved. In between an alternative is shown as a dual-mode configuration which comprises an operating force of a small human team, i.e. possibly only one single operator, together with a combination of category 1 through 3 assistance located in the control station, and one piloting SCU in each UAV as part of the operation-supporting means. If sensible according to the motivation for permanently substituting assistants, the extreme version with little participation of humans in the operating force, correspondingly more involvement of OCUs, and no SCUs, would be that one favoured at the first glance. We will dwell a little on this extreme case in the following, although it might not necessarily be the best solution after all, because the necessary communication between the different team members at various locations, the control station and the spatially dislocated uninhabited vehicles (e.g. UAVs), may cause considerable effort, if not even problems.

The main characteristic of such an operating force is the fact that the participating permanently substituting assistants are indispensable to eventually achieve the work objective. In that way, they are limiting the range of accomplishments the human force in the work system can achieve by soly relying on the own capabilities. This is obvious for the particular application of controlling UAVs as considered so far, where the assistant systems as team

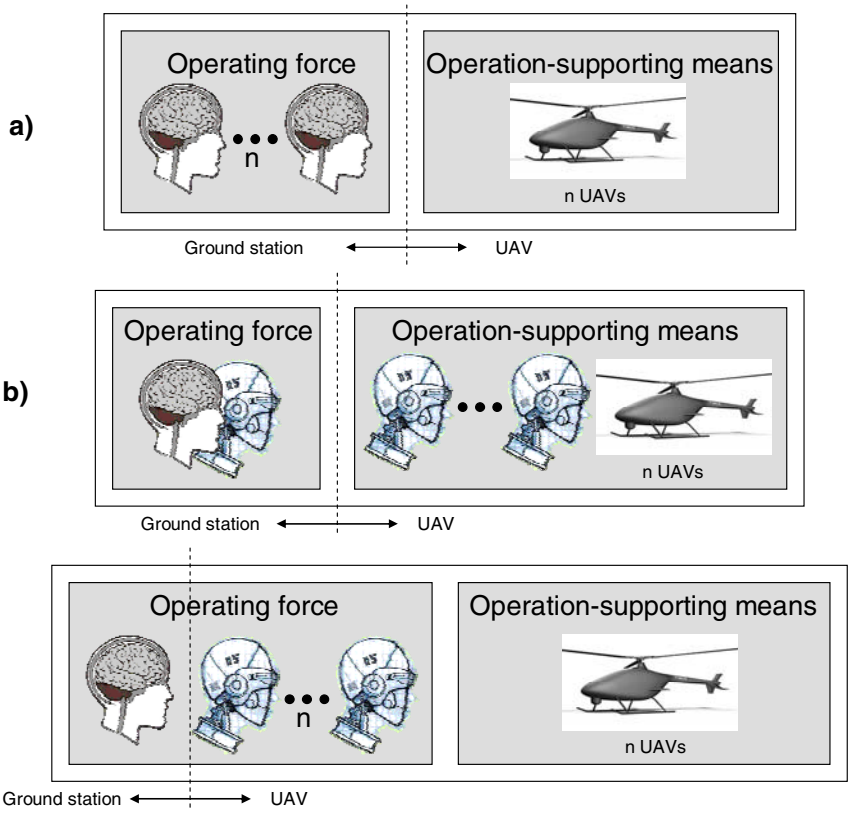


Fig. 58 Structural alternatives of integrating cognitive units in a work system for controlling n UAVs in a team formation: a) purely human, b) dual-mode cognitive automation, c) mode 2 cognitive automation by n permanently substituting assistants, one for each UAV

members are dislocated in their UAVs, but it would also promptly lead to disaster for other applications with the operating force located at one single work station. In essence, this means that the human operator is only defining or passing on the work objective to the assisting OCUs and is monitoring the process of negotiation and the execution of the commitments in order to intervene, if necessary, only having at his disposal to change the work objective or to define new constraints for the negotiation of commitments among the available assistant systems. This is the ultimate way of integrating robots into a work system.

Permanently substituting assistants may range from very simple ones and pertinent commitments to very comprehensive ones like that one which might completely operate an UAV, for instance. As was pointed out in Chapter 4.3, a cognitive unit might be capable to behave according to all three behavioural levels of the Rasmussen scheme. This is also true for a permanently substituting assistant. Depending on the actually allocated role within the work process

emphasis may be put on either the skill-based, or the procedure-based, or the concept-based level, but there is always a combination of the three. In a quite simple form, assistance is offered similar to that what we wanted to illuminate by use of the example of a theatre nurse's normal duties in the surgery room earlier in this book.

In that case, the corresponding assistance, based on its knowledge about the work objective together with the agenda of tasks the work process goes through, and the a-priori knowledge how to pursue it, would take the initiative on its own to do its job when the pertinent capabilities are demanded. It observes the work process going on and independently starts its action when it is needed. This should not be mixed up with an operation-supporting means which in fact may exhibit the similar basic control capabilities, but which has not the knowledge needed about the work objective to be effective and which therefore always needs to be explicitly instructed by a supervisor. As an example from the application domain of aircraft guidance and control, think of the autopilot which is a typical automated capability available in cockpits today as operation-supporting means and activated by the pilot by means of supervisory control. If we think of an autopilot as an assistant system, it would have to have the knowledge to be able to actively pursue the work objective, i.e. to intervene at the right time, independent from any other team member, in order to hold the altitude, for instance, thereby recognising possible hazards popping up, and reacting adequately on them. This autopilot assistant system would for instance recognise a mountain as a hazard in the aircraft's way and would fly around it.

It is important to realise that sufficient knowledge about the work objective is not enough to comply with the other important characteristic of a team member of the operating force, i.e. to assess the effect of other team mates' activities, for instance. This aspect is similar to what leads to the delineation of work systems as has been outlined in Chapter 3.1.3 in the context of the interaction among work systems. The work objective of the higher-level work system may be known in a work system, but this is not sufficient to be able to assess the effectiveness of the upper-level work system because of the lack of knowledge about the work process in the upper work system level.

If the assistance is not distributed over different locations, we can again take advantage of the case that already one OCU can do this style of assistance, again in accordance with the recommendation to keep the number of OCUs as small as possible.

With the integration of permanently substituting assistants in the operating force, overtaxing due to human task overload becomes more and more obsolete, but again not overtaxing due to underload and loss of situation assessment. Also permanently substituting assistance, if overdoing it, might lead to too little load on the human operator. Obviously, this has to be carefully investigated before considering permanently substituting assistance for whatever given application.

5.2.4 Summary on Characteristic Styles of Assistants

In the following table the differences between the characteristic styles of assistance are summarised. By combining these styles of assistance the designer

Table 3 Condensed resume of the properties of the characteristic styles of assistance

	Associative assistance	Alerting assistance	Temporarily substituting assistance	Permanently substituting assistance
Task commitments				
working in parallel	yes	yes	yes	no
own exclusive task commitments	no	no	no	yes
temporarily exclusive task commitment	no	no	yes	no
Kind of co-operation				
low degree of co-ordination	yes	no	no	no
co-ordinated	no	yes	no	no
debative	no	possibly	no	no
simple	no	no	yes	yes
obstructive	no	no	possibly	possibly
Assisting actions				
continuous presentation of results	yes	no	no	possibly
presentation of results, if necessary	no	yes	yes	yes
drawing attention to results in an undeniable way	no	yes	possibly	possibly
Action with direct impingement on team environment	no	no	possibly	possibly

can cover all requirements concerning aspects like those for task commitments, kinds of co-operation, and assisting actions as mentioned in Table 3. The kinds of co-operation as mentioned in the table are described in Chapter 4.4.1.

5.2.5 Conclusions for General Design Guidelines

With the following guidelines for the introduction of assistance in this section, support shall be given for the decision when to make use of associative, alerting or substituting assistance, or maybe combinations of them in the work system design. In the extreme case, there could be combinations with assistance of one out of the three styles of assistance for each cognitive sub-function respectively.

There are two main design criteria to be considered, work system productivity (including safety) and human operator satisfaction. Each criterion can become the prevailing one, depending on the application. Apparently, for a work process of private activity, the satisfaction of the human operator is the prevailing one. In this

case, productivity aspects are included in the satisfaction criterion. This is also true for self-determined professional work. For other professional work processes, on the other hand, productivity is mostly the dominant criterion, although the satisfaction of the human operator will presumably not be neglected.

From the system-ergonomic point of view, two main design goals usually have to be accounted for which serve both criteria, productivity and the satisfaction of the human operator:

- keeping the load of the human operator on a normal level, and
- contributing to ensure situation awareness.

Regarding these design goals, from what we learned about ways of co-operative interaction with the human operator by way of associative, alerting, and substituting assistance, the combination of alerting and substituting assistance is very promising for a work system with a given human team as members of the operating force. This combination of assistance, if technically feasible, typically leads to a very satisfying compliance with both work system performance in the sense of productivity and human operator satisfaction. It should be realised by the smallest number of OCUs as possible. Then, priority is given for alerting assistance regarding the cognitive sub-functions of situation assessment, and the substituting assistance is deployed in case of overtaxing of human team members regarding the other cognitive sub-functions. The substituting assistance can be a temporary one for commitments of the human team member when temporary overtaxing can occur. On the other hand, also permanently substituting assistance might be deployed, if there are commitments to be allocated which the human team member is principally not capable to accomplish, or which are of too high risk or likely a cause of too high costs. Therefore, this combination is particularly recommended for safety-critical work systems.

The design of the corresponding operating force of a work system can be specified by founding it on the compliance with the following basic requirements (see also [Onken, 1994a] and [Onken, 1994b]):

Requirement 1:
The assistant system has to be able to present the full picture of the work situation from its own perspective and has to do its best by own initiatives to ensure that the attention of the assisted human operator(s) is placed with priority on the objectively most urgent task or subtask.

Requirement 2:
If according to requirement 1 the assistant system can securely identify as part of the situation interpretation that the human operator(s) cannot carry out the objectively most urgent task because of overtaxing, then the assistant system has to do its best by own initiatives to automatically transfer this situation into another one which can be handled normally by the assisted human operator(s).

Requirement 3:
If there are cognitive tasks, the human operator(s) is(are) principally not capable to accomplish, or which are of too high a risk or likely a cause of too high costs, these tasks are to be allocated to the assistant system or operation-supporting means, possibly a supporting cognitive unit.

Comments on requirement 1: In the ideal case, requirement 1 can be fulfilled for all possible situations which could be encountered in the course of work. By virtue, it lends itself to be transformed into technical design specifications which the system engineers are looking for. The resulting design will be also in accordance with the more fundamental requirements of the so-called human-centred design (see Chapter 4.2).

Requirement 1 leads to functions to become active only at certain, normally rare occasions for contacting the human operator(s) with advisory messages when overtaxing of the human operator(s) seems possible in the task category of situation assessment and interpretation. This takes place in a way such that the human operator(s) can make up his (her/their) mind to accept the advice or not. This function is entirely advisory and can be reactive or proactive. Requirement 1 takes care that in most cases inadequate human behaviour is discovered at a state early enough that it can be easily corrected by the human operator himself before a situation with unacceptable detriment of behaviour might develop.

The requirement 1 is most important in case of safety-critical work processes. There are work processes of that kind where this is relatively easy to realise. Those are the favoured candidates work system designers might begin with. An objectively comprehensive picture of the situation might still not be altogether possible, but the information accessible to the assistant system can already be sufficient to provide the crucial advice to the human operator. Referring to the accident of Nagoya (see Chapter 4.2) the range of information the assistant system can rely on is probably the key issue to accomplish a successful design. Candidates of that type are, in particular, those where perception capabilities have not to be realised to be fully equivalent with those of the human operator. A good example is the work system of commercial air transport under instrument flight rules, where the most relevant information about the work system environment is provided via communication and navigation. Not only information about the situation of the work system environment is of interest in that respect, although it is the most relevant, but also information about the situation within the work system, including that about the human operator. This can be translated in technical terms by specifying the sensing and communication capabilities needed.

The way of presenting the work situation to the human operator(s) is crucial when striving for compliance with requirement 1. Drawing the attention of the human operator towards the most urgent task means that the delivery of advices cannot be accomplished by visual displays only. Today, the display technology is very advanced for both human perception modalities the visual and the auditory one. Furthermore, it is noteworthy at this point that there is good experience with additional modalities for certain applications, like the modality of tactile feeling. Also combinations of modalities are to be considered. Advancements are also made with regard to the methods of designing display layouts (see [Vicente, 1999] about ecological displays) to make them most efficient in the sense of requirement 1.

Although this issue of efficient information presentation deserves great attention from the design point of view, it would go beyond the practicable scope of this book. For those interested in that topic some publications like [Vicente & Rasmussen, 1992], [Wickens & Carswell, 1995], [Sachs et al., 1995], [Schulte et al., 1996], [Verly, 1996], [Theunissen, 1997], [Kubbat et al., 1998], [Vicente, 1999], [[Hecker et al., 1999], and [Kaiser et al., 2001] may alleviate the access to that subject. The display technology must not be the design segment which prevents a successful design. There can be pitfalls, though, if designing carelessly.

At this point, we also have to refer to the possibility that the human operator might have difficulties to understand the advice given by the assistant system. This might be caused by performance degradation of the human operator, of the assistant system, or of both of them as it also may happen between humans, by the way. This is a situation, where the one-way communication as discussed so far would no more be appropriate. Instead, means for dialogues between the assistant and the human operator are to be provided. Dialogues might be independently initiated on both sides. Therefore, part of the assistant system should be a solution of efficient dialogue communication, favourably by speech. Thus, it turns out that a dialogue manager might be specified to cover the following functions:

- to receive instructions from the human operator
- to display advice messages and to take care of good timing of these messages
- to manage dialogues with the human operator, if necessary.

There is a newly specified field of research, mainly emerged in the context of associative assistance for the cognitive sub-functional segment of achieving situation awareness and called *augmented cognition (AugCog)* ([http://www.augmentedcognition.org, 2006], [Miller & Dorneich, 2006). Currently, this is strongly promoted by the Defense Advanced Research Projects Agency (DARPA) in the US and is essentially addressing how to take online account of the human operator's state of cognitive processing and to adapt accordingly the presentation of information by online display modifications. This is not completely complying with requirement 1 as formulated in the preceding section, but it could expand the human operator's abilities by proactively avoiding occurrences of potentially poor operator behaviour. Augmented cognition explicitly includes online physiological gauging and neurophysiological sensors for cognitive state information. The measurement equipment on the operator's head might confine the range of application domains, but certain applications might still take great advantage from it. If it is possible to keep track with the human cognitive processes, there is the option to offer additional explicit information to the human team member about details concerning the situation which might not be immediately available to the operator. One should note, though, that augmented cognition can only be a kind of assisting function, if its operation is based on sharing the knowledge of the work objective with the human operator and pursuing it in co-operation with him (i.e. having a high degree of commonality with his a-priori knowledge on motivational contexts). Otherwise, it can be considered as a part of the operation-supporting means of the work system concerned. The latter would make a significant difference with often underestimated effect on human-machine interaction.

Comment on requirement 2: In order to catch up with that what human assistants are capable of in terms of temporary substituting assistance in rare occurrences of overtaxing of other team members, machine assistants should also be designed for temporary substitutive behaviour in order to intervene under very special conditions (in the extreme case out of the blue). Thereby, it should **keep the demand on the human resources on a normal level**. Therefore, in case that a situation of overtaxing of a human team member (possibly even of the team leader) can be reliably assessed by the assistant system, whether there is a case of task overload or, for instance, of attention overtaxing or any other human limitation or error, there will be a hint to the human operator according to requirement 1 and, in case of imminent danger of anything awful to happen, the assistant system shall take an action to heal the situation. At this point, the assistant system is temporarily replacing a human team member.

The necessity of an intervention of the assistant system according to requirement 2 is an extremely rare event. This is because usually the operation-supporting means of the work system concerned are already built up to an impressive extent like in case of the work process in an airliner cockpit. Thus, an assistant system without this additional capability usually makes already much sense, if requirement 1 alone is fulfilled to a great extent, or even accompanied by ordinary associative assistance.

Since requirement 2 is based on sufficient fulfilment of requirement 1, the technical realisation of the assisting function according to requirement 2 consists of two elements:

1. to realise reliable identification of situations with overtaxing of the human operator, albeit he or she has got full situation awareness, and
2. to realise reliable automated functionality to overcome this situation.

To design for requirement 2 according to these two realisation aspects is probably the greatest challenge to the work system designer so far. It leads to assisting functions which may comprise all cognitive functions necessary in addition to input interpretation, goal determination, planning, plan execution, and control actions. The second realisation aspect pointed out usually is the easier one to be settled. In essence, the corresponding functions are directly impinging on the work product by automatically carrying out the due tasks which otherwise the human operator was supposed to do. Depending on the situation, explaining information might be given to the human operator that he is enabled to take over again as soon as possible. Accordingly, the resulting action of the assistant system can be both

- completely taking over in case of overload, up to a point where control can safely be given back to the human operator, or
- taking other kinds of measures in case of underload such that the human operator is again becoming fully engaged into the work process.

As mentioned earlier in the context of describing the essence of temporarily substituting assistance, this problem is being tackled by the work on *adaptive automation* ([Scerbo, 2001], [Parasuraman, 2003], [Di Nocera et al., 2003] and [Donath & Schulte, 2006]).

In this context, as long as perfect fulfilment of requirement 2 cannot be achieved with acceptable cost, associative assistance automatisms have to be considered for decision aiding with laborious computations or certain operation-supporting means supervised by the assistant system. This is a good solution, if it can be assumed that sufficient (albeit not comprehensive) knowledge about the human capabilities exists. Still, this should be carefully studied, because there is the potential that the assistant system could unload the human operator too heavily, eventually taking him out of the loop. Therefore, of course, there has to be kept the option for the human operator, that he is doing all this on his own, possibly also making use of functions already implemented as part of the assistant system.

At this point, an important aspect of implementation should be mentioned. Assistant systems as part of the operating force in the work system not necessarily consist of all sub-functions of the cognitive process as shown in Figure 48. The sub-functions of planning and plan execution, in particular, or parts of them, might be better placed as part of the operation-supporting means. This could also be a favourable solution for certain execution aids pertinent to the sub-functions of situation interpretation and goal determination. In this case, the assistant system is making use of these operation-supporting means by supervisory control. It can possibly be implemented in a way that also the human operator can make use of these means in parallel. There should be noted, though, that the assistant system should rather have the complete a-priori knowledge about the function of these operation-suporting means such that it is able to correctly interpret the results provided by these means.

The difficulty to perfectly realise the fulfilment of requirement 2 has led to another idea. This suggests that the commitments of the assistant system mentioned above and the level of automation might be adaptively assigned online by the human operator. This reminds to what is mentioned in Chapter 5.2.3 by the approach of temporarily substituting assistance in terms of adaptable automation (see [Opperman, 1994] and [Taylor et al., 2000]).

After all, it turns out that at its best an assistant system should be a combination of all three categories 1 through 3. Anyway, the combination chosen should end up, if possible, in a single assistant system implemented as a single OCU. It seems that the category 1 assistance should be a mandatory element for the combinations, since thereby the crucial issue of harmonisation of the situation awareness of both the human operator and the assistant system for the sake of safety is best covered. As a general final remark, it should be noted, though, that an assistant system usually neither fulfils requirement 1 nor requirement 2 hundred percent, due to principle limitations. Despite of this, it is nevertheless extremely worth the effort. This has already been demonstrated, in particular in the work domain of vehicle guidance and control, in particular, where safety is a constraining factor. In Chapter 5.3 there will be some more details on that.

Comment on requirement 3: Requirement 3 refers to assistant systems (OCUs) carrying out tasks which in distinction to substituting assistance principally should not or cannot be taken over by a human operator. They are based on an a-priori design decision to permanently assign certain cognitive tasks within a given work

context to the assistant system. This may also include that the assistant decides to utilise certain operation-supporting means in order to fulfil the task. Moreover, the assistant system can have its own exclusive operation-supporting means at its disposal. Whereas assistant functions derived from requirement 1 or 2 most typically will be carried out at the operator's work station, assistant functions derived from requirement 3 might as well be dislocated, e.g. aboard a UAV. In this case the assistant system is guiding the UAV at a high degree of autonomy forming the operating force together with the human operator located somewhere else. In a MUM-T set-up, for instance, this will usually be onboard of manned aircraft.

5.3 OCU Prototypes

At the time being, we can refer to quite a number of programs worldwide which are working on prototype developments of OCUs in various application domains for vehicle guidance and control. These application domains range across aviation applications with OCUs onboard of aircraft and on the ground for air traffic control (see for instance [Erzberger, 1995] and [Völckers & Böhme, 1995]), automotive applications, and even applications on ships like the NARIDAS (Navigational Risk Detection and Assessment System) (see for instance [Kersandt & Gauss, 2006]). There are also applications in educational domains related to vehicle control like tutoring systems for truck driving. Some of these programs will be described in some more detail which might help to make the theoretical discourse of the preceding chapters more concrete.

5.3.1 Prototypes of Cockpit Assistant Systems

In this chapter we will report on a selected number of prototypes of assistant systems which were designed to assist pilots as human operators in the work process of aircraft guidance and control, and which have been developed within a period of about 15-20 years, beginning in the 1980s. [Banbury et al., 2008] list many more in a literature research and give brief descriptions which are also partially used in the following. The selected prototypes for this chapter, developed in different countries worldwide, often also referred to as electronic crew members, are the following:

- Pilot's Associate (PA) and Rotorcraft Pilot's Associate (RPA) (USA)
- Copilote Electronique (France)
- Cogpit (United Kingdom)
- ASPIO, CASSY, and CAMA (Germany).

These prototypes have been demonstrated in simulator experiments. Some of them have even been fielded in flight trials. They all have in common to belong to work systems of the kind as depicted in Figure 59.

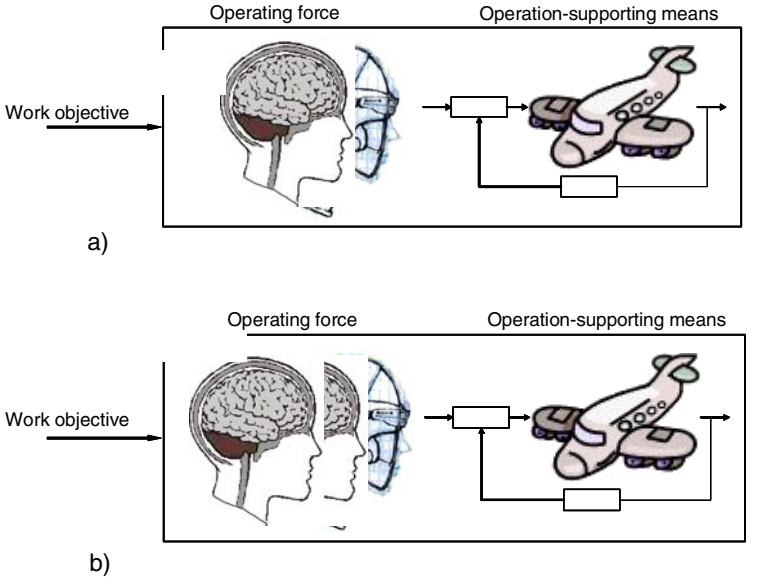


Fig. 59 Work system with cockpit assistant system: (a) one-man cockpit; (b) two-man cockpit

The prototype systems as cited above are dealing with work systems of both civil and military domains. ASPIO and CASSY belong to the civil domains of private and commercial transport flight. The Pilot's Associate, Copilote Electronique, Cogpit platform, and CAMA belong to military domains. Of course, the corresponding work objectives are not of the same nature. Therefore, in essence, when describing the approaches of the different prototype systems, we are focusing on the aspects which are related to that part of the respective work objective which leads to aircraft guidance and control tasks in the course of the work process. Unsurprisingly, the own developments are described in some more detail than those of other development teams, because of more detailed material available.

5.3.1.1 PA and RPA (USA)

The *Pilot's Associate (PA)* program was the first major program for the prototyping of a cockpit assistant system. This program was launched as a joint effort of the Defense Advanced Research Projects Agency (DARPA) and the US Air Force in 1986. In the framework of this program two contracts were awarded, one for Lockheed Aeronautical Systems Company and the other one for McDonnell Douglas Aircraft Company, each of them specialising on cockpit assistance for a different type of fighter aircraft mission. The development of Lockheed became the better known part of the program and will be described in some more detail in the following, more or less based on publications at an intermediate state of the program like that of [Banks & Lizza, 1991].

In the context of a demonstration after the first two years of the program by way of a simulation in non-real time the following situations in the course of a typical flight mission were reported [Banks & Lizza, 1991]:

> *"The first incident highlighting the systems pilot-aiding capabilities occurred while still in the cruise phase of the mission. A stuck fuel valve required correction. Given prior authority, Pilot's Associate corrected the problem and advised the pilot. After crossing the forward edge of that battle area, the aircraft was advised of a target assignment to need to intercept the four-bomber strike force protected by four fighter aircraft. Pilot's Associate created the new mission plan, including a probably intercept point consisting with pre-mission attack strategies.*
>
> *As threatening aircraft entered sensor range, Pilot's Associate allocated an onboard passive sensor to monitor the most critical threats. The system detected a combat patrol plane breaking orbit, but decided it was not an immediate threat to the mission. As the pilot flew the mission route, he could see his aircraft signature being managed with respect to known surface-to-air missile sites … the pilot was making his turn for egress and monitored the missile flyout when an unknown surface-to-air missile launched. Pilot's Associate advised the pilot of an effective defensive maneuver while it dispensed chaff and flares at appropriate intervals … the evasive maneuver and countermeasures enabled the aircraft to survive the surface-to-air-missile explosion; however explosion debris damaged control surfaces and the remaining good engine. The system planned a new egress route to an alternate recovery base, which was consistent with the aircraft's remaining capabilities while minimizing exposure to further threats." [Banks & Lizza, 1991]*

This example demonstrates that the Pilot's Associate provides a variety of decision support services for the pilot similar to the functions performed by the Weapons Systems Operator in a two-seat tactical aircraft [Winter, 1995]. The assisting nature of the system is derived from modelling the case of crew interaction if there were two crew members in such aircraft. The development is based on a number of prior conceptual research studies (see for instance [Rouse et al., 1987]. The system is active in the sense that it tends to anticipate the needs of the pilot subject to the known work objective (thereby relying also on a sub-function of pilot intent recognition [Geddes, 1992]) and to provide answers at the time and in the form to be most useful to the pilot. It does not, however, attempt to dictate solutions to the pilot.

Although the pilot can express his preferences and mandate action by the system, his direct input is generally not required for most system functions. Thus, the system collects its operational data directly from the aircraft sensors, forms its

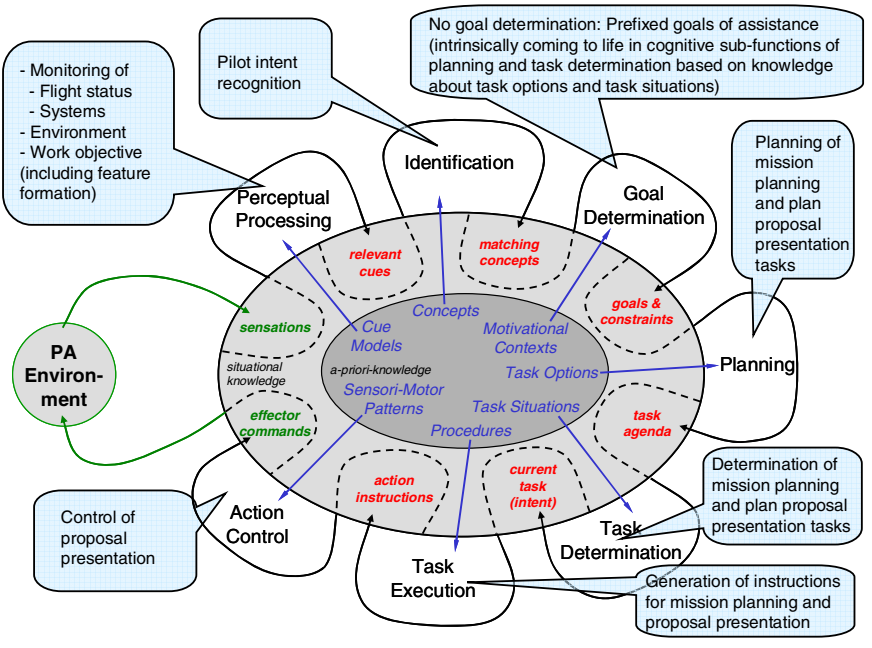


Fig. 60 Functional structure of the PA's cognitive process

own problem definitions, develops solutions, and, in certain special cases – authorised by the pilot prior to mission execution – carries out the plans upon explicit direction from the pilot. In every case, however, the plans must conform to the pilots own plan, and the pilot must be able to recognise that fact. If not, he can override or reject any recommendation either explicitly by merely pursuing his own different plan. This action will be recognised by the Pilot's Associate entailed by adjusting its plan accordingly.

Obviously, the Pilot's Associate is a good example of associative assistance as described in Chapter 5.2.1. Its main functional components are the assessor for the situation, also comprising the assessment of the status of the aircraft system, the tactics and mission planners, and the pilot vehicle interface.

This merges into a structure of the pertinent cognitive process, as shown in Figure 60, where

- the situation assessment is associated to the situation interpretation sub-functions
- the assistance goals are prefixed and permanently given. There is no active goal determination. The goals are intrinsically coming to life in the sub-functions of planning and task determination.
- the planning sub-function is associated to the generation of plans about how to assist. The mission planning of the PA is not part of this sub-function. Instead, it is a permanent task executed by the task execution sub-function.

- the plan execution is associated to the task determination, task execution and action control sub-functions.

The top goal of the PA, broken down in a number of sub-goals, was to do the best that the flight operation closely complies with the given mission order (work objective), i.e. that the mission will be successfully carried out. Mission success for military operations is not as clear-cut in all aspects as one might expect. Unfortunately, there is not a comprehensive set of generally accepted rules or regulations which ensure mission success. Therefore, it was up to the designer to define the goals from his point of view. As a result, there was much room left for mismatches between the goals of the individual fighter pilot and those of the PA. This turned out to be a weakness of the system. It was proved, though, that the principal idea of an associate of that kind was a good one.

The system has been flown by test pilots in a nonreal-time simulator facility. There were no plans for flight trials. Quantitative measures of success are not openly available. It can be assumed that the extraordinary complexity of the missions investigated was too extreme for the technological state-of-the-art at that time to achieve a system performance which would justify airborne trials.

The U.S. Army's *Rotorcraft Pilot's Associate* (RPA) program can be considered as being based on the earlier work in the development program of the PA. The RPA is the second development worldwide for a cockpit assistant prototype which led to a flight demonstration. It was aimed at becoming a cockpit assistant for the next-generation attack/ scout helicopter of the US army [Miller & Funk, 2001] with associative assistance capabilities. In essence, the functional structure of the RPA is very similar to that of the PA in Figure 60, although the application domain is different in many aspects.

The main modules of the RPA are shown in Figure 61. These include the functions of data fusion, data distribution, data fusion, situation assessment, planning, information management, controls, displays, and integrated mission processing (see [Miller & Hannen, 1999]).

Great emphasis is taken regarding the management of the information in future helicopters for the intended missions so that the relevant information as a portion of the total amount of available information can be figured out and presented to the crew in a way the crew can attend it without being overloaded. A cockpit information manager (CIM) has been developed to perform the following five major functions according to [Miller & Hannen, 1999]:

- Page (or format) selection – selection of a complete page or format to present on any of the aircraft's presentation devices
- Symbol selection/declutter – turning on or off specific symbols on a selected page
- Window placement – control of the location for pop-up windows which would overlay some other visual imagery on the multi-function displays
- Pan and zoom – control of centering and field of view of map and sensor display
- Task allocation – the assigning of tasks to various pre-defined, legal combinations of the two human pilots and automation.

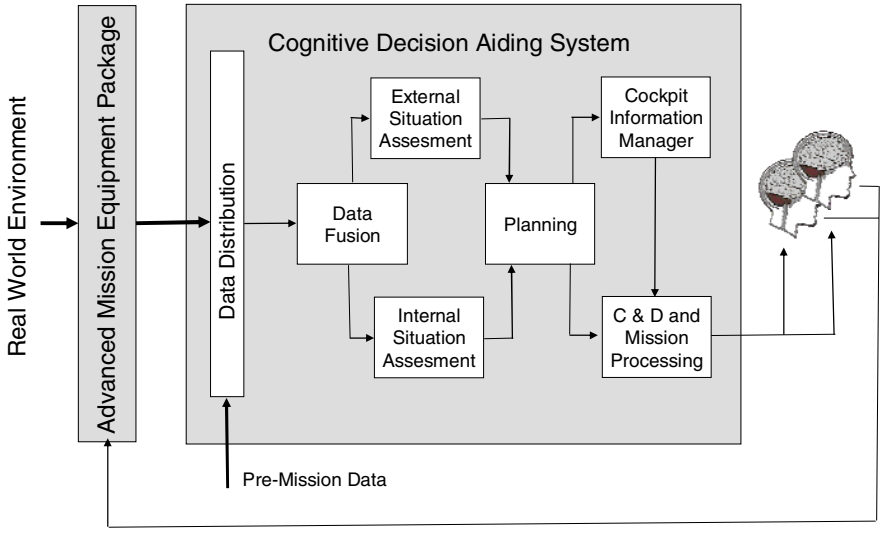


Fig. 61 Main modules of the RPA (after [Miller & Hannen, 1999]*)*

The last-mentioned function is certainly the most crucial one. In [Miller & Hannen, 1999] this was described in the following terms:

> *"Each behavior was adaptive and made use of an inferred task context to determine which information should be presented or which tasks should be allocated in what way. In addition, to relying solely on inferred pilot tasks, however, we also implemented several mechanisms to allow the pilots to control and interact with the adaptive behaviours decribed bove. Pilots could apply preference weights to those options. Pilots could also, during the mission, individually turn each of the options on or off. … In this sense, the pilot retained control over the behaviours that the CIM could, and was likely to perform."*

Extensive simulations in the full mission simulator of Boeing were carried out to evaluate the cockpit information management of the RPA. The four crews involved flew as they do in actual field operations. Each crew flew 14 part mission and four full-mission test scenarios, half of them with full RPA cognitive decision aiding system and the other half with an advanced mission equipment package alone, i. e. without the integrating support of the RPA.

Figure 62 shows the pilot's average overall ratings of the performance of the RPA cockpit information management. The cockpit information management was seen as very predictable. The capability to interact directly with the associate's assumptions about active tasks was very welcomed by the pilots. This is related to the important property of the RPA to facilitate interactions of the pilot to mitigate, correct, or just override its mistakes.

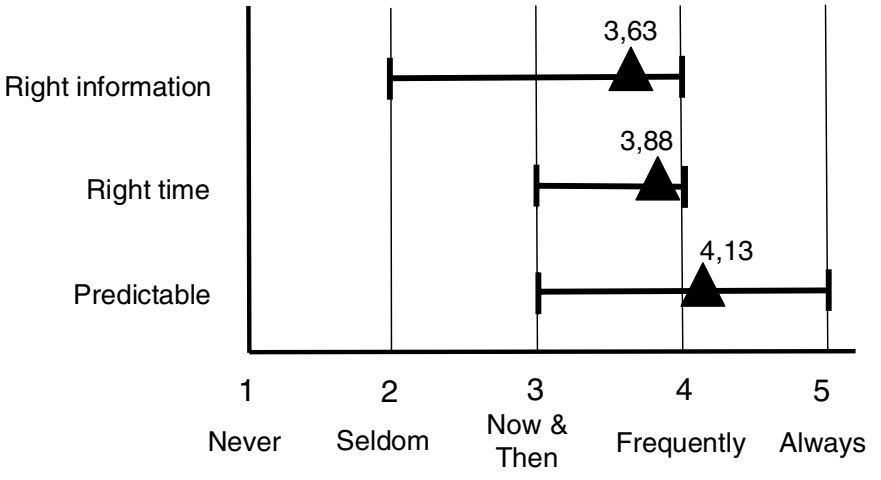


Fig. 62 Pilot's average overall ratings of RPA CIM performance [Miller & Hannen, 1999]

In October 1998, first flight demonstrations of the RPA were carried out in a modified AH-64D Apache longbow prototype. In the course of two months of flight testing more than 91 successful sorties were completed.

5.3.1.2 Copilote Electronique (France)

Almost in parallel to the PA-program in the US the French explorative development project of the *Copilote Electronique* was launched. It was also labelled as an in-flight mission replanning decision aid for an advanced combat aircraft. The program was led by the company Dassault Aviation with the support of several French industrial and scientific partners. It is meant to be part of a work system with a pertinent work objective to carry out a low altitude high speed penetration or an air-to-air escort flight mission of a single-pilot combat aircraft. From what is published about the program (see for instance [Champigneux & Joubert, 1997]), it was intended to become a decision aid in the sense of associative assistance as depicted in Figure 60 with correspondingly sufficient knowledge about the work objective. In cooperation with IMASSA (Institut de Médecine Aérospatiale - Service de Santé des Armées) it was also intended to take account of human centred design principles (see [Amalberti & Deblon, 1992]). In this context requirements were mentioned for instance in [Champigneux & Joubert, 1997] such that

- optimality of proposed plans should not be pursued if the pilot has not got sufficient time to understand,
- assistance must be adapted to the pilot's skills,
- assistance must be coherent to avoid surprises on the side of the pilot,
- assistance must know and respect its own limits,
- messages of the assistant system must be adapted to the situational context, pilot intents and pilot load.

In accordance to the PA, the assistance is focused on the generation of plan proposals, not going much further down regarding the resulting tasks or actions to take according to the plan in a particular situation. The motivational contexts which drive the system are not explicitly represented as it was the case for about all developments of that period.

The system was implemented in parts, in particular regarding the planning function. There have been laboratory tests, but unfortunately there is no publication about details on these tests and the final development result.

5.3.1.3 COGPIT (United Kingdom)

The Cognitive Cockpit project (COGPIT), led by DERA, aimed at the development of a laboratory demonstrator for an assistant in a fast military jet aircraft. The COGPIT system can be considered as an assistant system as long as the pertinent work objective (mission order) can be considered as being known by the system. COGPIT has got four main components. These are the following [Bonner et al., 2000]:

- a situation assessment support system (SASS) that recommends actions based on the status of the aircraft and the environment, both internal and external to the aircraft
- a cognition monitor (COGMON) that monitors the pilot's state
- a tasking interface manager (TIM) that tracks goals and plans and manages the pilot/vehicle interface and system automation, and
- a cockpit (COGPIT) layout that interprets and initiates display and automation modifications upon request.

Both the COGMON and the TIM are probably the most salient components of this design. The COGMON has 32 inputs of physiological, behavioural, and situational data. These permit analysis of both low and high frequency measures in near real time. The data provide information about

- overall load
- visual load
- auditory load
- somatosensory load
- motor load
- executive load
- time pressure
- stress level
- alertness level
- current tasks and
- current intents
- among others.

This probably most comprehensive component for workload assessment as part of an assistant system seems to be good basis for further studies in this respect.

Table 4 PACT levels [Bonner et al., 2000]

Automation pact level	Pilot Authority	Computer authority
5 (automatic)	interrupt	full authority
4 (direct support)	revoking action	action unless revoked
3 (in support)	acceptance of advice and authorising action	advice, and if authorised action
2 (advisory)	acceptance of advice	advice
1 (at call)	advice, only if requested	advice, only if requested
0 (command)	full authority	none

The TIM component utilises information from the monitoring and analysis of the mission tasks and from pilot state monitoring which is derived by the SASS and the COGMON component, respectively. Based on this information it drives the task and timeline management in accordance with the requirements of the thereby known mission plan. In addition, it drives the COGPIT component to adapt the degree of automation and the information presentation to the pilot.

Based on the results of the Rotorcraft Pilot's Associate program (RPA, see Chapter 5.3.1.1) the TIM is making use of a task model shared by the pilot and the COGPIT system. This shared model is used for task tracking. It affords a high level of co-ordination between the pilot and the system, enabling to determine pilot information and automation needs.

Part of TIM is the so-called Pilot Authorisation and Control of Tasks system (PACT). It uses military terminology for the five implemented levels of aiding by automation, going from fully under command up to fully automatic (see Table 4). By means of PACT the pilot is able to control the task allocation through the following:

- pre-set pilot-preferred defaults
- pilot selection during pre-flight planning
- changed by the pilot during i-flight re-planning, and
- automatically changed according to pilot-agreed, context-sensitive adaptive rules.

The COGPIT assistant mainly addresses a combination of associative and substituting assistance. In that sense, the functional structure is similar to that of the PA, RPA, and Copilote Electronique, with the exception of offering a kind of temporal substituting functions for task execution. In the sense of adaptable automation, it is up to the pilot, according to the PACT system, how far this kind of temporal substitution may go.

5.3.1.4 ASPIO, CASSY, CAMA, and TIMMS (Germany)

These prototypes as own developments are related to the work environment of guidance and control of transport aircraft. ASPIO (Assistant for Single Pilot IFR Operation) and CASSY (Cockpit ASsistant System) are civil applications for private and commercial flight, whereas CAMA (Crew Assistant Military Aircraft) is a corresponding application of military transport flight. ASPIO was the first development, started in the late eighties, followed by CASSY and CAMA. ASPIO [Dudek, 1990] and CASSY were developed at the University of German Armed Forces in Munich. In order to provide a life demonstration of typical features of a cockpit assistant system a video clip about CASSY is available to the reader on the CD attached to this book. The development of CASSY was partially sponsored by the company Dornier Luftfahrt GmbH and the DLR. CAMA was sponsored by the German DOD and developed in co-operation by the University of German Armed Forces in Munich together with the companies DASA (Daimler-Benz Aerospace AG) and ESG (Elektroniksystem- und Logistik-GmbH) as well as with the DLR.

The first significant results have been gained in extensive flight simulator experiments with the CASSY predecessor system ASPIO (Assistant for Single Pilot IFR- Operation) in 1990 at the German Armed Forces University in Munich [Dudek, 1990]. This system has been prototyped primarily for the approach and landing phases of IFR flights in the terminal area of Munich airport. Although the hardware was hardly powerful enough at that time, the results were astonishingly promising. The design philosophy proved to be well accepted by the pilots. The significant improvement of flight accuracy and acceleration of planning and decision-making tasks where further aspects of encouragement, which gave way for continued projects and industrial support [Onken, 1992]. This resulted in the next prototype CASSY. Since CASSY was the first cockpit assistant system worldwide, which was successfully demonstrated in-flight, we mainly concentrate on describing this prototype in more detail in the following.

General characteristics of CASSY

CASSY corresponds to an assistant system which represents a combination of alerting and associative assistance in a work system of the kind as depicted in Figure 59b. The work objective to operate on is a civil air transport objective by means of a given aircraft operating under IFR constraints. According to the basic requirement 1 for assistant systems (see Chapter 5.2.5), CASSY assesses/ interprets the work situation on its own and does its best by own initiatives to ensure that the attention of the assisted human operator(s) is placed with priority on the objectively most urgent task or subtask within the IFR-fligt regime, in particular that which might turn out to be safety-crucial. CASSY is exclusively advisory. Nevertheless, it offers certain services on its own initiative in reaction to certain situations. One of these services focuses on the laborious subtask of generating flight plan alternatives which otherwise would have to be worked out by the cockpit crew by use of certain operation-supporting means at their disposal. These services account for requirement 2 to a certain degree (see Chapter 5.2.5).

According to this assignment, CASSY acts like an additional, i.e. third crewmember, which stays passive as long as everything is working normally, and comes forward to contact the cockpit crew and to give advices and to offer service functions, if urgent necessities turn up which the crew has not yet taken care of. The goal is a situation-dependent flexible, cooperative distribution of tasking between the electronic and the human crewmembers.

Assessing and interpreting the work situation requires a wide-ranging knowledge base, consisting of static and dynamic knowledge about all elements, which possibly influence the cockpit crew's behaviour and its environment. This includes some basic functionalities, such as

- a data base for static aircraft and environment data (such as aircraft performance, equipment and navigation data),
- monitoring functions for health status of aircraft systems,
- functions providing dynamic environment data, such as air traffic, airline data and weather information
- planning and decision aiding functions, able to generate and update the global mission plan and its subgoals to react on adverse situations,
- a sophisticated model of the expected crew behaviour in order to pursue the mission plan,
- functionalities to recognise and identify the current crew behaviour and intention, and
- an interface to gather information from the own aircraft, i.e. aircraft state and aircraft subsystems.

The way this knowledge is being interpreted makes a difference between the human crew members and CASSY. CASSY is capable of taking all accessible information into consideration, whereas the human has to decide on fragmented excerpts of situational information due to information processing limits. Capacity restrictions also lead to a much higher amount of time in complex decision-making for humans. Furthermore, a computer system will always draw exactly the same conclusion in exactly the same situation, it cannot be grumpy.

On the other hand, ACUs only warrant correct behaviour, if there is sufficient a-priori knowledge. Therefore, application experts should be involved as much as possible into the system design. Just like a human the system may succeed or fail in handling unforeseen and/or unknown situations. Thus, as long as the human operator has more experience than the system and human decision-making in an unknown situation is usually superior to machine decision-making, the human operator should be the decisive part. Thus, we have kept CASSY far from holding the final decision authority in about any critical situation. It rather assists the cockpit crew by giving advisory and warning messages.

Communication plays an important role in assisting the human operator in order to realise the basic requirements. The need for a centralised communication management becomes quite obvious when the necessary communication amount between the human crew members and the assistant system is regarded. To achieve a most effective communication between the cockpit crew and CASSY, the crew interface is designed very carefully. Since the present state of HCI

(human-computer interface) in modern aircraft is not totally sufficient, e.g. [Dyck et al., 1993], new technologies are used to a much greater extent.

In order to avoid information overflow in the visual channel, although it must be used as the channel with the highest processing capacity, the auditory channel is exploited for use of new technologies for ATC communication (data link) and speech communication. Speech can be used in either direction. The addition of speech input to the already existing speech output in the cockpit makes communication close to natural. If speech recognition is sufficiently reliable, it will be very effective.

To accomplish the stated basic requirements, information transfer, in particular information ouput to the human operator, is centrally controlled and managed in an information managing functionality, based on the knowledge about the global work situation. This actually presents the voluntary action of CASSY as its impact on the work process. Therefore, all other modules of CASSY can be considered as service modules for that communication manager.

Functional structure of CASSY

As to the motivational contexts of CASSY to assist the cockpit crew, these are prefixed in a similar way as it was described for the PA in Chapter 5.3.1.1. Therefore, there is no goal determination sub-function. Essentially, the corresponding goals are intrinsically coming to life in the sub-functions of planning and task determination. Thereby, CASSY is doing its best that the flight as IFR (Instrument flight rules) flight operation closely complies with the given flight objective, i.e. that the flight to a given destination will be successfully carried out. Goal conflicts are not considered, because they are very unlikely, since, in distinction to the PA (see Chapter 5.3.1.1), the definition of success is backed by a tremendous amount of regulations for civil IFR flight operations. Therefore, there is almost no room for mismatches between the goals of the individual pilot and those of CASSY.

Therefore, referring to the structure of the cognitive process (see Chapter 4.5.2) the transformer for the goal determination was excluded in the CASSY implementation without causing inconvenience, such that there are remaining the following cognitive sub-functions (see Figure 63):

- perceptual processing, identification, and task determination for the situation interpretation,
- planning and
- task execution and action contol for the plan execution.

As to the *situation interpretation*, there are four sub-modules which are entirely devoted to that aspect, i.e. the sub-modules to monitor *flight status, systems, and environment*, as well as the *Pilot Intent and Error Recognition (PIER)* sub-module which is based to a great extent on a-priori knowledge about the pilot behaviour as explicitly available by the so-called *Piloting Expert (PE)*, using an explicit knowledge base of behavioural rules. All other a-priori knowledge involved in the CASSY modules is stored were it is needed. The three sub-modules monitoring flight status, systems, and environment are very similar to already developed and

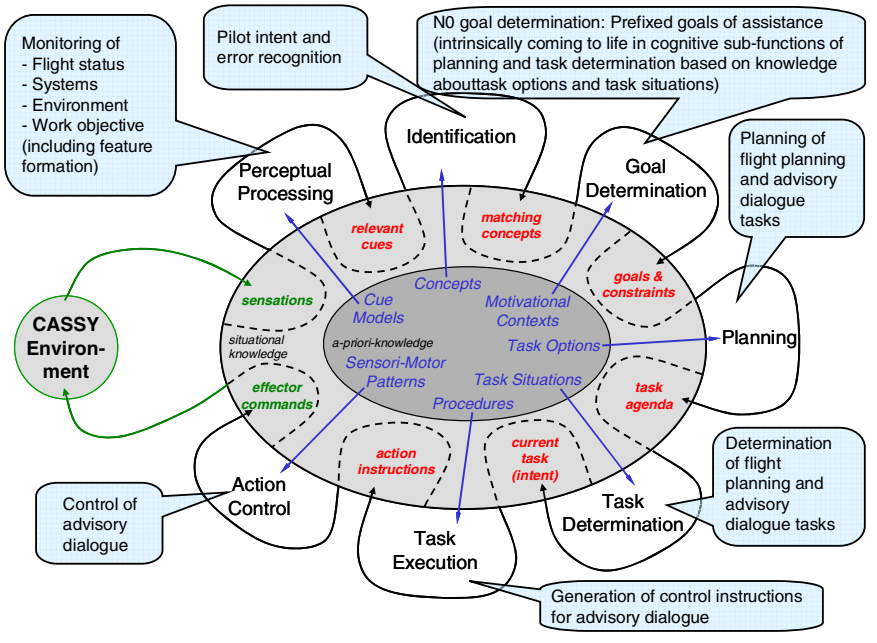


Fig. 63 Functional structure of CASSY's cognitive process

applied modules in modern aircraft. Therefore, the CASSY-modules for monitoring can be considered as sub-systems well-known from modern aircraft automation. The monitoring modules are an absolute need for the assistant system to serve the basic requirements. As the last of the five sub-modules entirely devoted to the situation interpretation, there is the *Automatic Flight Planner (AFP)* which interprets the situation by generating beliefs about alternatives of flight plans for the remaining portion of flight. Also the *Dialogue Manager (DM)* is partially involved in the situation interpretation by receiving the pilot inputs into CASSY. Both the assistance goals and the assistance plan are permanently given and are not explicitly represented. Therefore, the determination of assistance goals and assistance plan according to the goal determination and planning sub-functions of the cognitive process are not explicitly included. *Plan execution* is represented by the cognitive sub-functions of task determination, task execution, and control as shown in Figure 63. In the following, the functions of the CASSY-modules for the situation interpretation, planning, and plan execution are described in some more detail.

Situation interpretation:
Monitoring sub-modules
Besides the monitoring of the pilot crew by CASSY, three main areas of interest for situation monitoring can be figured out. The *Monitor of Flight Status* provides the crew and electronic assistant with data about the present flight state and flight progress. It reports the arrival at any waypoint of interest during the flight.

The status of the aircraft systems is monitored by the *Monitor of Systems*. Information about failures in aircraft systems is communicated to the crew and provided to other CASSY core elements. Based upon this information CASSY may conclude that re-planning might be necessary. The expected crew behaviour could be different subsequent to a modified system configuration, too. A comprehensive knowledge base about the aircraft system functions is not available so far in CASSY. Although it would be very desirable, this is not a major drawback in CASSY as a demonstrator prototype, because it does not present any principal design problems. By the way, this component is available as operative component in modern commercial transport aircraft, for instance known as the so-called EICAS (Engine Indication and Crew Alerting System) and ECAM (Electronic Centralized Aircraft Monitoring) systems in modern civil air transport aircraft.

The *Monitor of Environment* gathers information for the assessment and evaluation of surrounding traffic and weather conditions. When deviations to the stored data are identified, they are reported to the pilot crew and the CASSY modules.

Pilot intent and error recognition (PIER)

The comparison between the expected and the actual pilot behaviour is carried out in the *Pilot Intent and Error Recognition* (PIER) (see [Wittig, 1993] and [Wittig & Onken, 1993]). In case of a deviation, the PIER has to figure out, if this deviation is erroneous or intentional. In the first case, warnings and advisory hints are issued to the cockpit crew by means of the *Dialogue Manager* with the aim to make them correct the error. In the second case, the classification process has identified one out of various possible crew intentions. The PIER module is making use of the information about the work objective.

The intent recognition is performed by use of an inference algorithm based on known intent hypotheses. A pilot model of behaviour is available for each hypothesis. Over time more and more hypotheses are rejected until eventually one of them may remain as determined pilot intent. If no hypothesis for the crew's behaviour deviation from the expected behaviour could be verified by this method, only then a crew error will be assumed. A further advanced approach is described in Chapter 6.2.3.2 (see also [Strohal, 1998]).

The recognised intent will be first communicated to the pilot crew to ensure that it has been recognised correctly, and then passed to the *Automatic Flight Planner* in order to trigger the necessary flight plan modification. The PIER supports the pilot's situation awareness and therefore mainly supports the basic requirement 1.

Plan execution:

The plan execution combines the processing of both the *Automatic Flight Planner (AFP)* and the *Dialogue Manager (DM)*.

Automatic Flight Planner (AFP)

The Automatic Flight Planner (AFP) (see [Prevot & Onken, 1993] and [Prevot, 1996]) is a typical decision support component (see [Hollnagel et al., 1988]),

associated with the genuine assistant function. As a result of a detected conflict or on crew request, it automatically provides a ranked choice of alternative proposals for the

- selection of the best alternate airports,
- selection of the best emergency fields,
- lateral route,
- vertical flight path profile, and
- mission timing.

The AFP comprises the entire flight planning knowledge of CASSY for generating a complete global flight plan. This flight plan includes a detailed 3-D/4-D trajectory plan as well as alternate airports and emergency fields and possible conflicts along the intended flight. The basis for the planning tasks is the access to the knowledge about the global situation and to navigational and aircraft data bases. The destination for the flight is the only information which is needed from the pilot. This is part of the flight status and represents the work objective. This data and all other information can be extracted from the situation representation.

The extensive planning knowledge can be used in different ways. The initial flight plan for the given flight mission can be generated by the AFP. During the flight mission the AFP is activated automatically when an adverse situation demands any modification of the flight plan. Therefore, the available situation information is checked whether there arises a conflict with the current flight plan. Such conflicts can result from e.g. new ATC instructions, changing weather conditions, or system failures.

In the re-planning process one or more ranked proposals will be presented to the crew, which can be accepted, modified, or neglected. Only a plan proposal agreed upon by the crew will be instantiated as further flight plan. Besides the automatic activation of the re-planning functions, the pilot crew can make use of the planning knowledge for decision-making purposes at any time. Requests for routing and trajectory alternatives or alternate airports can be made with any number of directives and/or constraints. The generated recommendations have no influence on the actual flight plan, unless they are activated explicitly by the human crew.

In parallel to these functions, the AFP periodically updates the flight plan with respect to changing usable radio aids, wind conditions and the actual flight progress. This is a function which is a common feature of modern Flight Management Systems (FMS).

The automatic presentation of the resulting situation-dependent flight plan proposals to the crew by the Dialogue Manager is an example of dealing with basic requirement 2 by conventional automation, alleviating the cockpit crew from laborious tasks.

Dialogue Manager (DM)

The Dialogue Manager (DM) executes the communication of CASSY with the human cockpit crew. It relies on the information provided by the situation interpretation and the Automatic Flight Planner. It manages this information for a well-timed presentation to the crew. It is responsible to manage advising messages

to the cockpit crew [Gerlach, 1993], thereby essentially co-ordinating the information flow from CASSY to the human crew by accounting for priorities. In addition, it represents the interface component for the crew inputs into CASSY, for instance by speech. The design of the corresponding speech recognition system was accounting for the relevancy of the message subject to the knowledge about the pilot behaviour as provided by the PIER module. Besides other internal flow of information as well as that to and from separated service modules, the DM gathers the resulting outputs of the CASSY-modules. These are the outputs of the modules providing the CASSY-belief, like the monitoring sub-modules and the PIER on the one side, and the AFP as task execution module. These outputs provide information packages to be considered by the DM as drivers for potential messages to the crew. The DM is evaluating the potential messages for the purpose of priority ranking before sending them out and it is formatting these messages correspondingly. This ensures that the most important and most urgent information is always presented first and with the necessary emphasis.

The DM has got access to two components to present the situation to the crew: a speech synthesizer which allows varying voices, and a graphical colour display. Warnings and advisory messages are mainly issued via speech output. More complex information, like the flight progress situation, is primarily presented on the display. Both devices using different sensory modalities of the human pilot are complementing each other.

The major input channel to CASSY uses speech. Therefore, a speaker independent speech recognition system, which allows online switching of active syntaxes, is used. Thus, context-dependent speech recognition is enabled, which improves the recognition rates. Context-dependent in this case implies that the DM of CASSY is able to predict the entirety of speech input commands, which potentially could be given in the current situation characterised by the current flight phase, flight plan, autopilot and configuration settings an so forth. They outline the commands, which have to be recognised by the system. Therefore, the complexity of the language model to be processed by the speech recognition system can be reduced significantly with respect to the actual situation.

If no data link to ATC (Air Traffic Control) is available, all ATC information is accessed by the DM by other means. This is done via the speech recognition channel, picking up the obligatory acknowledgments of the ATC instructions by the pilot.

A very promising idea for the improvement of the communication management is the development and integration of a crew resource model the DM might have access to. Later developments have taken account for this aspect [Flemisch, 2001]. As one result it should be possible to adapt the information flow to the crew to their actual state of information processing resources (adaptive automation).

Obviously, the DM mainly supports the alerting assistance, but its assistance in flight planning is a typically associative one. Therefore, both kinds of subroles are integrated in the CASSY design. It is noteworthy that **harmonisation problems between associative assistance and the human operator can be overcome by this combination of associative and tak monitoring assistance.**

Execution Aids (EA)
CASSY comprises a lot of functions which should not necessarily be part of an assistant system as part of the operating force in the pertinent work system. The Execution Aids of CASSY are a good example. In an operative design this module would probably be part of the operation-supporting means. It offers a variety of optional service functions which can be controlled by the crew by the use of speech. The available services include configuration management, radio management, autopilot settings, and calculations regarding navigation and aircraft performance.

A-priori knowledge
As already mentioned, most of the a-priori knowledge is not explicitly represented. It is intrinsic to in the modules of CASSY whereever it is needed. There is one exception, the behaviour model of the pilot crew. This is explicitly available by the knowledge module called the *Piloting Expert (PE)*. In order to be able to present behavioural advices to the pilot crew, the actual crew behaviour must be assessed and monitored. A reference for pilot behaviour must be established for this purpose, possibly in connection with an estimation of the actual state of remaining unused pilot resources. This is achieved by processing a behavioural model of flying the specific aircraft type as represented by the PE. This module of a-priori knowledge covers the normative procedure-based behaviour of the crew regarding situation assessment and plan execution [Ruckdeschel, 1994] (see later Chapter 6.2.2.1). The model describes what is documented in pilot handbooks and air traffic regulations. The knowledge representation is based on Petri nets [Reisig, 1991].

The flight plan as generated and administrated by the AFP is a reference for both the pilot crew and the PE. The PE uses it with regard to the current flight progress to elaborate the currently expected crew action patterns and the remaining resources as well as admissible tolerances for executing these actions.

The comparison of the expected behaviour to the actual behaviour, as done in the PIER module, indicates, whether a situation with overcharge has already occurred. On the other hand relating the expected behaviour to the resources of the crew enables the assistant system to decide, whether a situation with overcharge is likely to occur. Thus, the PE is a precondition to be able to serve both requirements 1 and 2. The latter property for the sake of designing for requirement 2 is not exploited in CASSY, though. In a later development as part of the prototype CAMA [Stütz, 1999] added a component modelling individual behaviours.

Simulation experiments
For simulator test runs CASSY was implemented into two different flight simulators, at the University of German Armed Forces Munich and at DASA-Dornier, the company which sponsored parts of the development of CASSY. The simulator in Munich is a fixed-base one-seat simulator with artificial stick force and artificial outside vision capabilities. The DO 328 flight simulator experiments at DASA took place in 1993. This simulator represents a modern glass cockpit. Flights in the Frankfurt area have been simulated with different pilots [Onken & Prevot, 1994]. The good pilot scores for the crew monitoring and alerting functions and for the support in flight planning were confirming those already experienced with

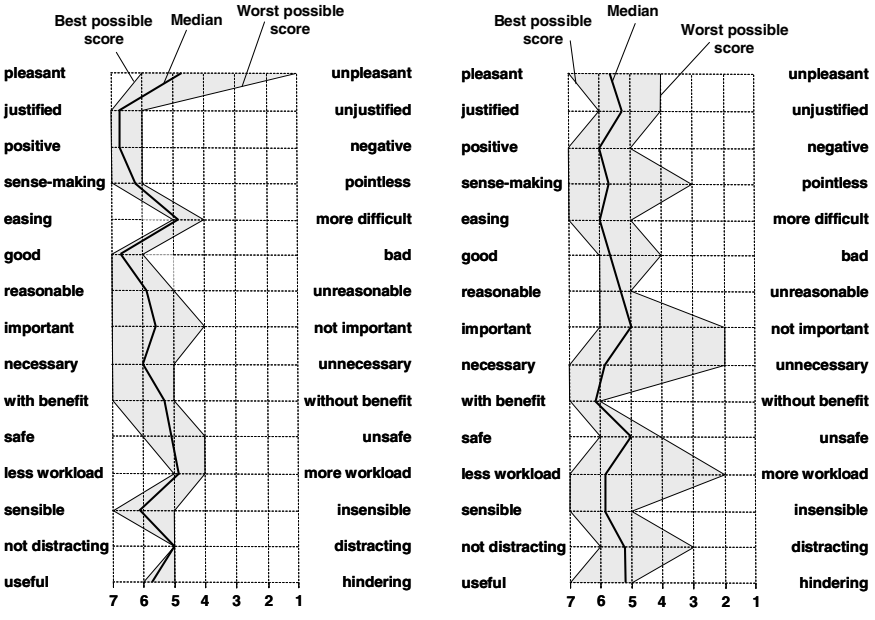


Fig. 64 Results of simulator evaluation of CASSY. Left: Evaluation of crew monitoring. Right: Evaluation of flight planning support. [Onken & Prevot, 1994]

ASPIO. The evaluation took place by way of questionnaires answered by the test pilots. For the questionnaires it was made use of the method of semantic differentials. The results are summarised in an orderly manner in Figure 64.

With respect to the monitoring of crew behaviour the test pilots were impressed with the fact that only in case the crew actually deviates from the nominal values beyond pertinent tolerances in an unacceptable fashion an advisory message is given for correction. When the pilot performs a control movement into the right direction, i.e. the tendency is correct, the system would be quiet again. When the error is not being corrected after a certain time, the intent recognition module tries to figure out this intention and reports it to the flight planning module to modify the flight plan, accordingly.

One major driving factor for developing CASSY with the structure as described has been to improve the present situation of working conditions of the aircrew. Therefore, it seems reasonable to give a more concrete description of the resulting situation with CASSY in the cockpit by giving a flavour of flying with CASSY as experienced with the simulator flights:

During normal flight, i.e. there are no occurrences of problems, the pilot crew will hardly become aware of the assistant system's activities unless a support function is requested. The flight situation is presented on the display. This means that the flight plan as generated by the AFP is displayed, necessary frequencies are indicated, possible conflicts endangering the flight segments ahead like thunderstorm

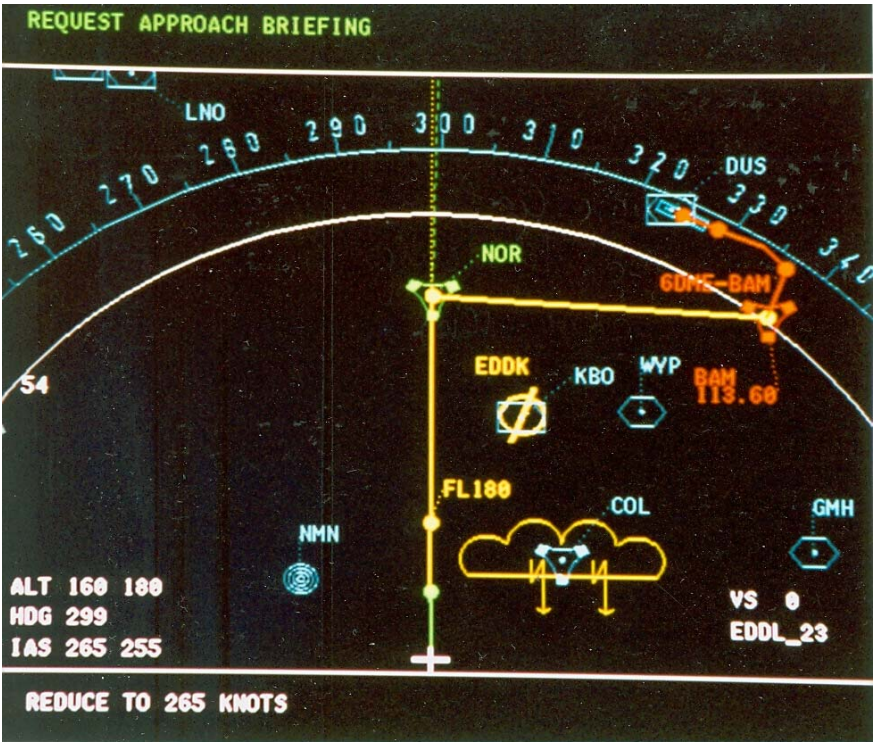


Fig. 65 CASSY map display with
- redundantly visualised pilot speech input at the top and verbal CASSY output at the bottom (possibly by speech, here visualising ATC instruction)
- heading-up and north-up option (here set on heading-up)
- compass card
- depiction of environment with navaids, weather peculiarities (here location of thunderstorm), next best emergency field, set navaid (NOR) with set radial, flight path with waypoints up to the destination airport, colour-coded differing types of clearances along the flight path, and nominal values for altitude, heading and speed for the actual and following flight path leg (bottom left) [Gerlach, 1996]

areas or missing clearances are illustrated and the next best emergency field is permanently displayed. Figure 65 shows the CASSY map display for a specific flight situation. This is about the same function which we are used to nowadays with the flight management system (FMS) in commercial airliners.

In addition to that, the existing flight plan is permanently adapted to the flight progress and ATC instructions. This distinctively exceeds the FMS functionality and represents the associative part of assistance CASSY is performing. Every new ATC instruction will be automatically inserted into the flight plan. In case of altitude or speed constraints the trajectory profile is updated with regard to the new clearance. Thus, resulting actual nominal values for indicated airspeed, magnetic

heading, altitude, and vertical velocity are indicated prominently permanently to serve as a reference for the crew.

The following essentials are monitored regarding the cockpit crew behaviour:

- flight path deviations, using primary flight data such as altitude, course, airspeed, power setting, climb/descent rate, pitch attitude;
- configuration: operation of flaps, landing gear, speed breaks;
- radio navigation settings.

The data are also checked for violations of danger limits, e.g.:

- descent below the minimum safe altitude and
- violation of minimum or maximum speed for the current aircraft configuration.

The planning and decision support, automatically given by CASSY as a result of a detected conflict or on crew request, includes:

- selection of the best alternate airports,
- selection of the best emergency fields,
- best lateral route,
- best vertical profile,
- best mission timing.

All this support is based on the current global situation, including atmospheric data, aircraft systems data, etc. and requires minimal crew inputs, although the planning process might be performed quite interactively, if wanted by the crew.

An example for a complex re-routing process is given in Figure 66. The aircraft (callsign CASSY1) approaches Cologne airport in critical weather conditions. During arrival the monitor of systems detects a failure of the onboard ILS-system. This information is transferred to the crew by CASSY using speech output. Also the consequences are automatically evaluated, which leads to a complex re-routing process, since the weather conditions at Cologne are insufficient for a non-precision approach.

After the selection of the best alternate airport a re-routing is requested. CASSY offers different alternatives, a routing via the standard arrival route, and a more direct routing using suitable radio aids (example in Figure 65). In this case, the crew activates the standard routing without modification. CASSY generates the detailed flight plan, but reminds the crew of the missing clearance. When the crew requests the clearance, ATC instructs the pilots to proceed directly to Frankfurt VOR. This instruction is integrated by CASSY into the flight plan and the execution is monitored. In this example the crew forgot to enter the frequency of Frankfurt VOR into the RMU (Radio Management Unit). The advice to select this frequency is only given after a certain period of time, since the crew needs some time to cope with the new situation. The crew might respond by assigning this action of data insertion to CASSY by using the “Do IT” command.

Before finally integrating the system into the test aircraft, flights have been simulated with the hardware in the loop in the simulation facilities of the German

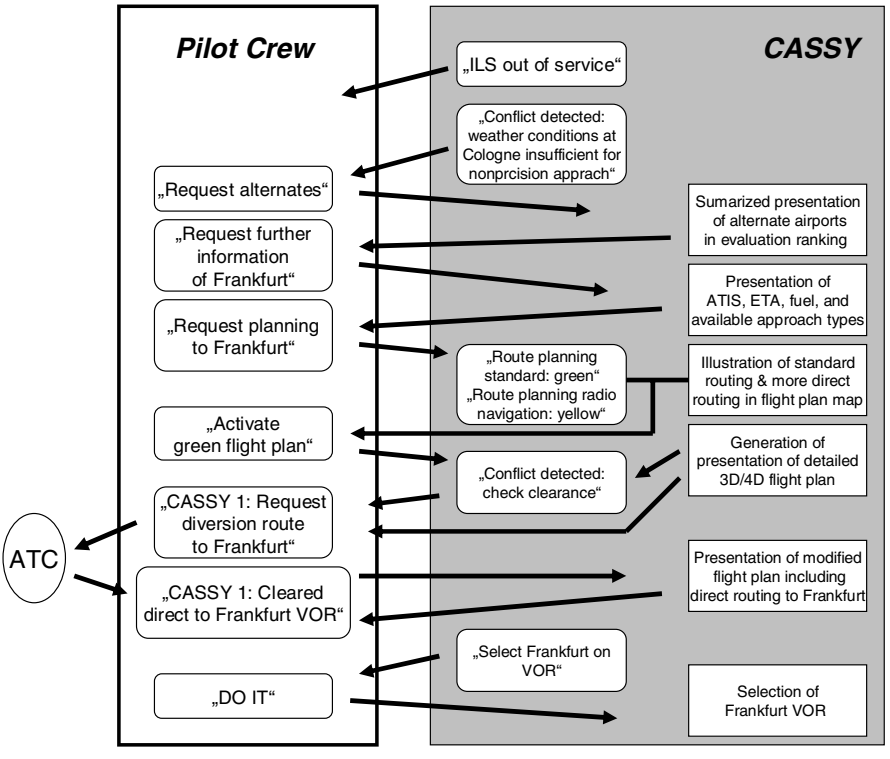


Fig. 66 Interaction with CASSY for complex rerouting process, including speech communication. [Onken & Prevot, 1994]

Aerospace Research Establishment (DLR) in Braunschweig. In these simulator runs the CASSY software version was verified to an extent considered as reasonable for the flight trials, and the test pilots for the flight experiments were introduced to the functionalities of CASSY and the experimental environment. After the successful completion of these simulator trials the flight experiments were performed in June 1994 (see attached DVD).

Flight tests

The flight experiments were performed with the aim to evaluate the performance of CASSY in a real aviation environment. The system was integrated into the experimental cockpit of the *Advanced Technologies Testing Aircraft System* (ATTAS) of the DLR, as shown in Figure 67 and typical regional flights in high traffic areas where performed.

The ATTAS test aircraft, an especially developed modification of the 44-seat commuter jet VFW614, was equipped with an experimental fly-by-wire flight control system and a versatile computer and sensor system. Beyond many other test programs it is used as the airborne segment in the air traffic management



Fig. 67 Advanced Technologies Testing Aircraft System (ATTAS) of the DLR: (left) view on aircraft; (right) view into the experimental cockpit in the cabin, right behind the regular cockpit (this picture was made in the context of the CAMA flight experiments)

demonstration program [Adam et al., 1993] of DLR and it is equipped with very good facilities for testing complex on-board systems in instrument flight scenarios. The CASSY hard- and software was integrated into ATTAS using its additional experimental cockpit in the rear cabin directly behind the front cockpit. The experimental cockpit is a generic flight deck (one seat) with side stick, Airbus-like display and autopilot control panel, and other ARINC (Aeronautical Radio Incorporated) control panels. Therefore, it represents a sufficiently realistic pilot work environment for IFR operation.

The hardware of the assistant system consisted of

- an off-the-shelf SGI (Silicon Graphics Inc.) Indigo (R4000) workstation to run the core modules of the assistant system connected to the ATTAS experimental system via Ethernet,
- a PC equipped with a Marconi MR8 PC-card providing speaker dependent continuous speech recognition activated by a push-to-talk button on the side stick,
- a DECtalk speech synthesizer providing various voices for speech output connected to the ATTAS intercommunication facilities, and
- a BARCO monitor (about 25cm) connected to the graphics channel over the SGI-Indigo and additionally built into the experimental cockpit.

The computers were located in the rear of the main cabin close to the experimenter work stations of the aircraft, one of which equipped with a laptop for starting and maintaining the CASSY software.

During the flight tests CASSY was running throughout the complete flights from taxi-out to taxi-in. All data, which CASSY received via the avionics data bus, have been recorded with a frequency of 10 Hz. All in- and output messages have also been recorded and every time the flight plan had changed by a major planning activity or when a checkpoint has been passed the entire situation representation has been stored. These data enabled a replay of all flights and a reproduction of all situations.

Table 5 Flight test scenarios

Flight	T/O T/D	Flight time [hours]	G/A in	after	ATC instructions	Pilot
1	EDVE	1:03	EDDH	0:33	26	1
2	EDDF	0:50	In-flight	simulated	13	1
3	EDVE	1:27	EDDF	0:43	27	1
4	EDVE	0:50	EDDH	0:09	24	2
5	EDVE	1:32	EDDF	0:41	32	1
6	EDVE	0:57	EDDH	0:32	27	2
7	EDVE	0:57	EDDH	0:31	24	1
8	EDVE	0:58	EDDH	0:31	21	1
9	EDVE	1:31	EDVV	1:14	42	1

The results to be presented in the following have been gained by online-observing the behaviour of the pilot and the intelligent assistant during the flights, by off-line evaluating the collected data and in the debriefings immediately after the flights. Two professional pilots served as experimental pilots and additional pilots from Lufthansa German Airlines were participating as observers.

A total amount of 10 flight hours has been flown, comprising eight flights from the regional airport Braunschweig (EDVE) to the international airports of Frankfurt (EDDF), Hamburg (EDDH), and Hannover (EDVV) at which a missed approach procedures were conducted before returning back to Braunschweig (see Table 5).

Test flight number two has been an in-flight simulation of departure and approach to Frankfurt, which was necessary to investigate certain incidents, which would have been safety critical in the real Frankfurt area, e.g. descending below the minimum safe altitude. In all other flights nothing has been simulated and no special situations have been provoked, since the system should be evaluated in the real environment, which includes coping with all events which occur during IFR flight in a high density area.

One important result held true throughout the complete test program: There was no significant difference in system performance between the flight tests and the simulation trials. Consistently, the following discussion of results concentrates on the major questions concerning the real environment rather than system performance.

Situation assessment with respect to pilot behaviour

The basic requirements described in Chapter 5.2.5 emphasize the necessity for a complete understanding of the global flight situation. To get an impression of the situation assessment capabilities the duration of discrepancies between the actual and expected pilot behaviour has been evaluated, related to the total flight time. This has been done on the basis of the stored data for the six flights 2, 3, 5, 6, 7, and 8.

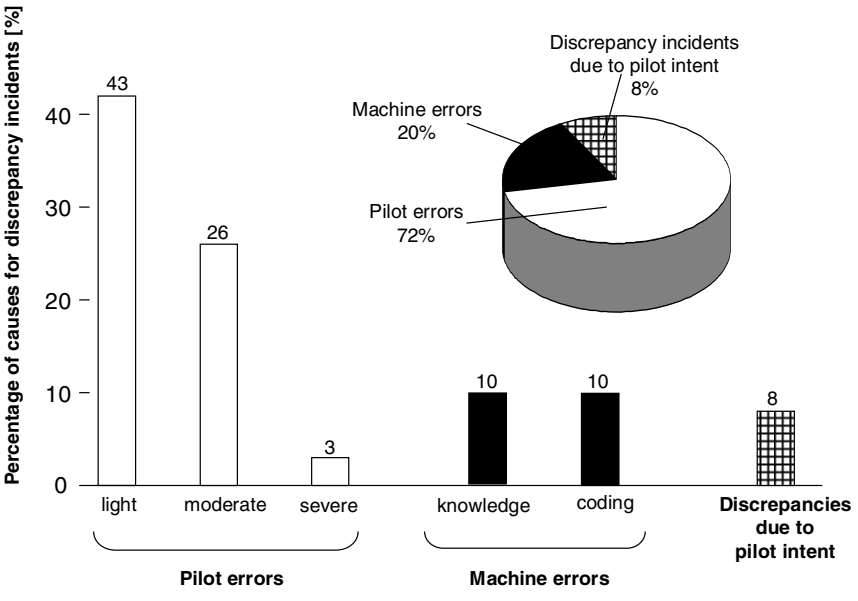


Fig. 68 Percentage of causes for discrepancy incidents (pilot errors, machine errors, and pilot intent discrepancies) during the flight trials. [Prevot et al., 1995]

For almost 94% of the time flying in a high density environment the pilot and the machine assessed the situation equally, because otherwise they would not expect or perform the same action patterns.

A total amount of 100 incidents have led to warnings. They have been evaluated to find out the reasons for the warnings and the consequences they had. All incidents have been related to one of the following three categories: pilot error, pilot intent, and machine error (i.e. CASSY errors in this case).

In five cases of the intentional deviations from the flight plan the intention was automatically figured out by the assistant system and the flight plan has been adapted, accordingly. In three cases the pilot had to inform CASSY about his intention.

Half of the machine errors were caused by an incompletely implemented knowledge, the other half by coding errors of the experimental system. In all cases of machine errors the pilot realised that a wrong warning was issued by CASSY. No negative influence on the pilot's situation assessment could be observed. In one case, the CASSY system had to be restarted in flight. That took about 15 seconds. The only pilot input needed to recover was the flight destination.

Concerning the pilot errors, light errors are considered to reside in an inaccurate or uneconomical but safe manoeuvre. Moderate errors probably would lead to a safety critical situation, and severe errors surely would lead to a dangerous safety hazard unless an immediate correction is made. All pilot errors which occurred during the flight tests were detected by CASSY. All moderate and severe errors as

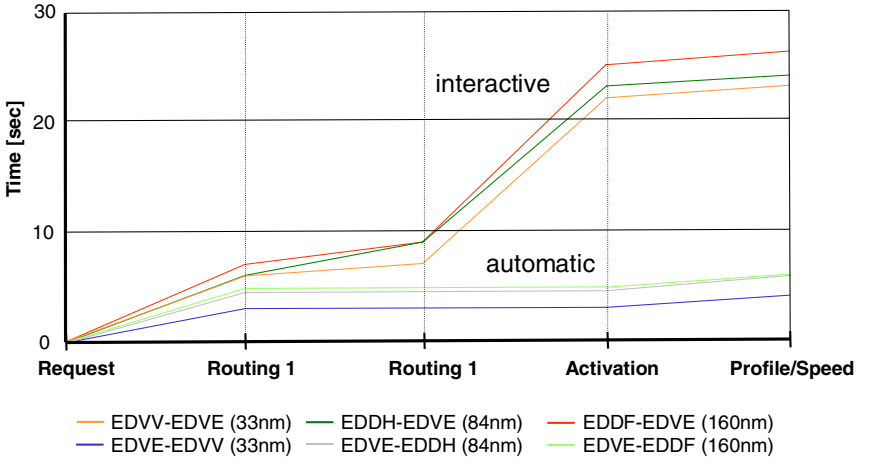


Fig. 69 Duration of flight planning, comparing interactive use of the AFP and automatic proposal of the AFP. [Prevot et al., 1995]

well as about 70% of the light errors were immediately corrected by the pilot after having received the warning or hint.

Flight planning and decision aiding
CASSY's flight planning capabilities have been stated by the experimental pilots and the observers as very impressive. As a matter of fact, all planning proposals have been accepted and none of the automatic radar vectoring predictions have been modified or caused any doubt from the pilot. The time needed for planning a complete flight from one airport to the other is illustrated in Figure 69.

Before every flight, the flight destination and the departure runway were entered into CASSY and an automatic planning of the complete flight was initiated. After the go around procedure at this destination the pilot initiated an interactive planning to return to the airport from which he had departed by entering its name. CASSY presented two routing proposals in parallel. The pilot could select or modify one of them. After this selection the trajectory profile was planned in detail and recommended speeds, times of over-flight, radio aids etc. were inserted.

The distance to the destination had only little impact on the duration of planning. The automatic planning took between four and six seconds, the interactive re-planning up to 26 seconds of which the pilot needed about 16 seconds to decide for a proposal. This confirms the approach to re-plan automatically when the flight plan must be generated very fast. When there is more time available, re-planning can be done interactively, too, in order to keep the pilot more involved.

Operation of pilot-CASSY speech communication
To evaluate the speech recognition performance three different speakers made speech inputs during the flight tests, summarized in Table 6.

Table 6 Speech inputs

Speaker	Time [flight hours]	Inputs
Pilot 1	8:18	324
Pilot 2	0:50	36
Experimenter	0:57	56

During their first flight pilot one and pilot two were not very familiar with the speech recognition system and the specific syntax to be used. In-flight number six a CASSY experimenter made the complete speech input for the pilot. He was familiar with the syntax and the speech recognizer from simulation experiments. In Figure 70 the results are shown with regard to the recognition performance.

Obviously, speech recognition inside the noisy aircraft is possible. It takes some time for the pilot to become familiar with the recognizer and the syntax to be used. This learning process can also be done in the simulator as the flight with the experimenter has shown. The achieved percentage of recognised speech commands is almost of the same level as could be achieved in ATTAS simulator runs with the same recognition system.

For entering the ATC commands into the system two different experiments have been made throughout the flights. 92 ATC commands of a total of 236 have been fed into the system by the pilot using speech. The remaining 144 commands have been keyed into the system by one of the experimenters onboard the aircraft

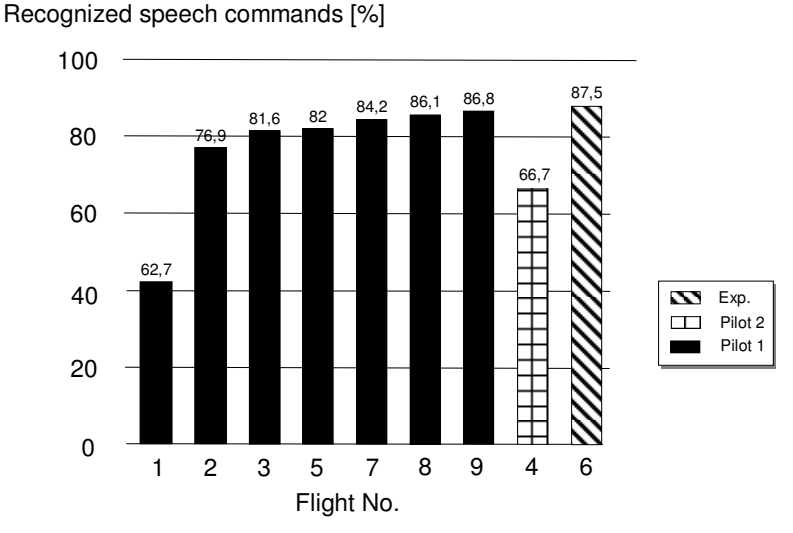


Fig. 70 Percentage of recognized speech commands by the speech recognition system. [Prevot et al., 1995]

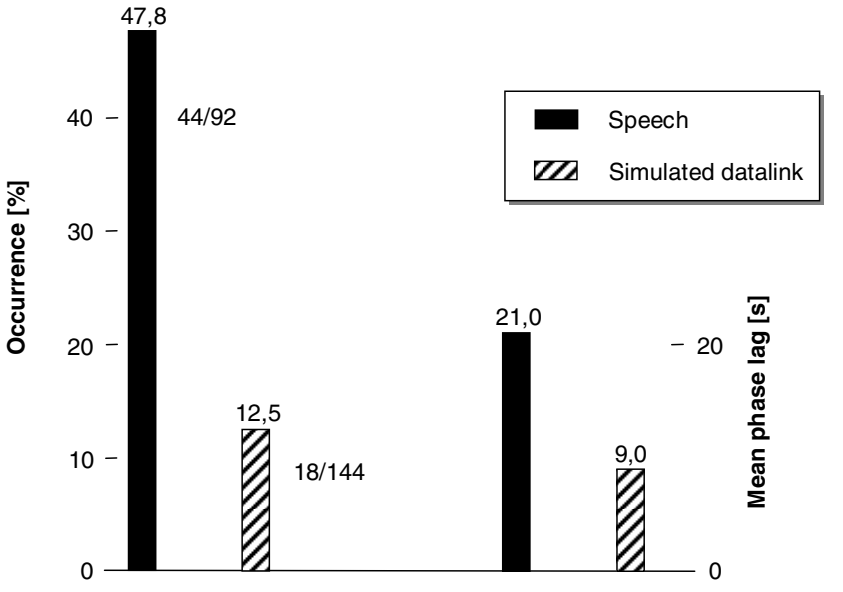


Fig. 71 Unnecessary warnings or hints resulting from time lags when entering instructions from air traffic control. [Prevot et al., 1995]

immediately after receiving the message to simulate a data link from the ground into the aircraft. This took some seconds. The pilot reacted to the commands at the same moment he received the ATC message, but acknowledged the commands with some time lag. This sometimes led to unnecessary warnings and hints. The percentage of occurrence of these incidents compared to the respective number of ATC messages and the mean phase lag is illustrated in Figure 71.

This effect was typical for the one-pilot configuration of the experimental cockpit. The figure illustrates the importance of a fast and powerful ATC interface. Optimal system performance can only be achieved with a digital data link.

Pilot acceptance

The acceptance of the planning and monitoring functions was at least as good as in the previous simulator trials [Onken & Prevot, 1994]. All participants in the evaluation attested CASSY a nearly operational performance and a very promising concept. It was noted that situation assessment, monitoring and good planning capabilities are completely in-line with human centred design.

As a conclusion, the successful flight tests of the Cockpit ASsistant System (CASSY) in real IFR flights have demonstrated that it is possible to integrate intelligent onboard systems into modern aircraft. Optimal performance can be achieved with a digital data link between ground and airborne segments. The amount of detected and avoided pilot errors, the availability as well as the power demonstrated in complex planning indicate the performance level of CASSY. Speech recognition proved to be a powerful device as pilot interface, but other

devices should also be considered. This kind of system can be considered as a solution to overcome existing criticism with respect to current flight management systems representing the counterpart as operation-supporting means.

As mentioned earlier, an operative aircraft cockpit of modern airliners represents a great number of highly developed automatic systems which all serve a separate specific task. The approach of integrating a cockpit assistant is not to replace all existing systems by the 'one and only' system. It neither should be a simple addition of another black box. As pointed out as a guideline in Chapter 5.2.5, it should be part of a cognitive work system design with possibly partial shift of former operation-suporting means into the assistant system, or possibly a dual-mode cognitive design.

Add-ons in CAMA and TIMMS, and PILAS

As mentioned earlier, CAMA as a follow-on of CASSY was intended as a cockpit assistant for military transport aircraft (see [Strohal & Onken, 1998], [Schulte & Stütz, 1998], [Onken & Walsdorf, 2001], and [Stütz & Schulte, 2000]). This implies regular IFR flight and in addition tactical flight phases, possibly close to terrain for longer periods, thereby minimising the exposure to unknown threats.

The principle functional structure of the cognitive process of CAMA corresponds to that of CASSY according to Figure 63, including the implementation of the central situation representation. Of course, the a-priori knowledge is taking account of the military aspects as well. Ground proximity is continuously monitored. Therefore, corridors of flight trajectories, achievable within the aircraft's performance capabilities are determined for safe terrain avoidance by using a DTED (Digital Terrain Elevation Data). A warning, by visual display and by voice, is issued in case of violations of safety margins by the pilot. CAMA also generates proposals to solve the conflict.

Results of simulator and flight trials

In November 1997 and May 1998 first flight simulator test runs of CAMA were conducted. 10 German Airforce transport pilots (Airlifter Wing 61, Landsberg) were participating as test subjects.

In addition, there were two periods of flight trials with the ATTAS (Advanced Technology and Testing Aircraft System) test aircraft of the DLR (see Figure 72) which amounted to a total of 15:50 hours of flight testing with CAMA in operation. A number of modifications have been made after the first period of flight trials. Therefore, representative results mainly were gained from the second period of flight trials in October 2000 with 5 test flights (7:15 hours in total), flown by 4 German Airforce transport pilots (two of them were test pilots). The flight trials were the first successful ones worldwide of a cognitive crew assistant for a work system with a work objective of military air transport into a potentially hostile area.

In both test environments, the flight simulator and the flight trials, the pilots were tasked with full scale military air transport missions. This comprised the preflight mission briefing, takeoff from base, an IFR flight segment to the ingress corridor, a minimum-risk low level flight through a mountain area to a drop zone. The low level flight over the potentially hostile area represented a dynamic tactical

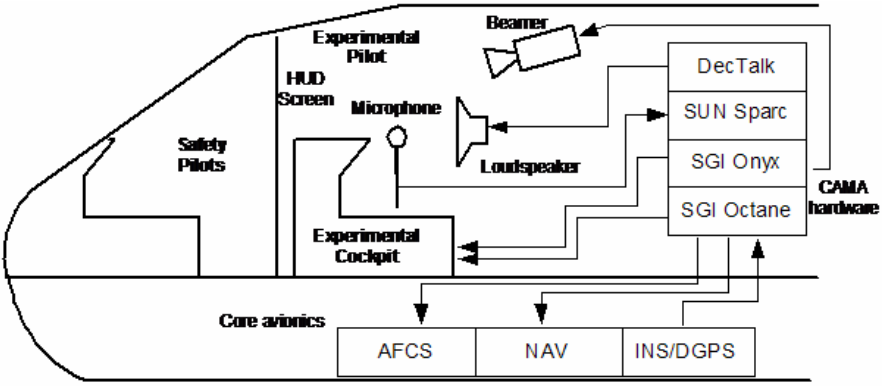


Fig. 72 Integration of CAMA system into ATTAS [Stütz & Schulte, 2001]

scenario with multiple SAM stations (Surface to Air Missiles). After executing the drop, the mission consisted of recovering from drop manoeuvre, an egress flight segment on a minimum-risk-route (MMR), and a subsequent IFR flight segment to the home base.

The IFR segment incorporated:

- Adverse weather conditions
- High density airspace
- Changing availability of landing sites
- ATC communication (e.g. clearances, radar-vectoring, redirection)

The tactical segment incorporated:

- Varying SAM sites
- Drop procedure
- Changed egress corridor
- Redirect to new destination

Each test flight was entailed by a debriefing. Here, the pilot's overall acceptance of the cognitive assistant system was documented through a questionnaire with the following topics to be rated:

- Test environment
- Situation awareness
- Assistance quality
- Pilot acceptance.

All ratings were given within a range from 1 to 7, where 1 represented the best and 7 the worst score. A choice of the results is shown in Figure 73, Figure 74, and Figure 75, showing the means of ratings. The ratings of the flight simulator test runs are marked by an S and the flight test results are marked by an F. These results were extremely encouraging. They also confirmed that the pilot interface, including the efficient pilot-CAMA speech communication, was quite adequate. A more detailed documentation of the test runs and its results is given in [Lenz & Onken, 2000], [Stütz & Schulte, 2001] and [Frey et al, 2001].

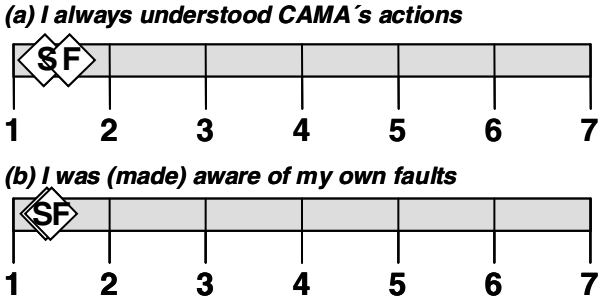


Fig. 73 Evaluation of the co-operative approach of CAMA (S = Simulator test runs, F = Flight trials)

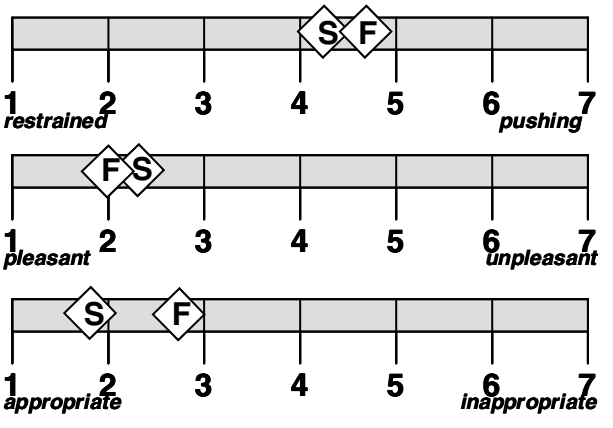


Fig. 74 Acceptance of CAMA by pilots (S = Simulator test runs, F = Flight trials)

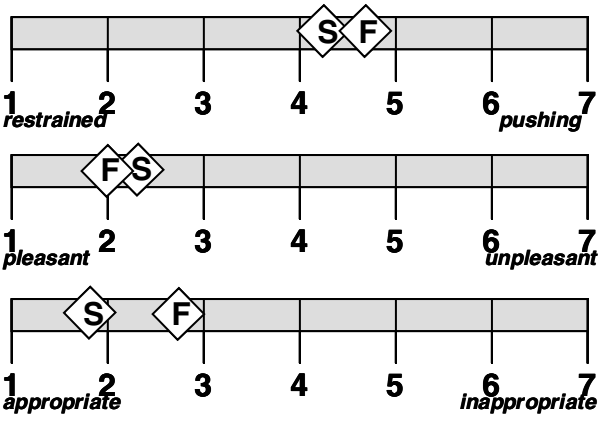


Fig. 75 Overall evaluation of CAMA (S = Simulator test runs, F = Flight trials). [Frey et al, 2001]

Along with the CAMA project new results were obtained for design methods as to explicitly representing motivational contexts (see [Walsdorf, 2002] and [Onken & Walsdorf, 2001], modelling of individual pilot behaviour (see [Stütz, 1999]), and recognising pilot errors and the pilot's intent (see [Strohal & Onken, 1998] and [Strohal, 1998]). The latter two approaches are described in some more detail in Chapter 6.2.2.1 and Chapter 6.2.3.2.

TIMMS (Tactical Information and Mission Management System) was a first German simulator study to also investigate an OCU approach for a fighter cockpit. This system was structured in a similar way as CAMA (see [Schulte, 2002] and [Schulte, 2003]).

5.3.2 Prototypes of Driver Assistant Systems

In this chapter we will report on two developments of prototypes of assistant systems which were to assist drivers of motor vehicles as single human operators in the work process of vehicle guidance and control, and which have been developed within a period of about 10 years, beginning in the nineties. These prototypes, developed in different countries are the following:

- Generic Intelligent Driver Support (GIDS)
- Driver Assistant system (DAISY).

The driver assistant system DAISY is an own development which will be described in more detail, unsurprisingly.

These prototypes have been demonstrated in simulator experiments and even in field trials. They have in common to belong to work systems of the kind as depicted in Figure 76.

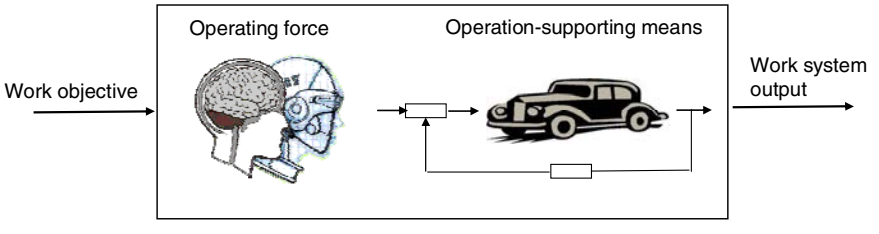


Fig. 76 Work system with driver assistant system

Again, these systems are of the kind of alerting assistant systems and comply with the main characteristic for a member of the operating force in the work system, i.e. to have sufficient knowledge about the work objective which is pertinent to the respective work system. That means that in addition to the work objective as such sufficient a-priori knowledge is available to understand the work objective in the sense of what that means to serve this objective in the framework of the assigned role by taking into account aspects concerning the driver, the motor vehicle, environmental factors, and mission constraints.

Driver assistant systems mainly differ from those of aviation applications in the requirement to react on fast-changing environment and to account for great behavioural differences of individual drivers.

5.3.2.1 Generic Intelligent Driver Support (GIDS)

To our knowledge, the GIDS project was the first major programme for the prototyping of an assistant system for the driver of road vehicles. This project was launched in 1988 as DRIVE project V 1041. It was a joint effort of altogether 13 partners from 6 European countries to stimulate the introduction of modern RTI (Road Transport Informatics). It was carried out in collaboration of engineers and behavioural scientists, where, interestingly enough, behavioural scientists took the responsibility for the realisation of the system.

[Michon, 1993] provides a comprehensive survey on that project. GIDS stands for the development of a prototype for a driver assistant system, being both associative and alerting. The system architecture as specified at the outset of the project is shown in Figure 77. The a-priori knowledge needed is represented in rule bases. The core functions for knowledge processing in order to assist the driver are the

- reference driver model
- actual driver model
- analyst and the
- dialogue controller

The *reference driver model* represents the knowledge about the behaviour of a driver who can be considered to perform the driving tasks safely, effectively, and efficiently. The *actual driver model* models the way the actual driver performs the driving tasks. At its best, it also includes knowledge about systematic error forms due to limitations of the driver's behavioural capacity. The analyst detects discrepancies between the behaviour of the reference driver model and the actual driver model. Depending on the amount of discrepancies, it infers what ought to be done about them within the range of available options, mainly by alerting the driver in one or the other way or to derive a proposal with regard to navigation. This leads to a functional structure of the resulting cognitive process which is very much alike that of CASSY.

Based on this architecture a prototype system was specified as shown in Figure 78. The manoeuvring and control support module comprises most of the core functions for driver assistance as mentioned above. This prototype system was integrated with its major components in a "small world" simulator and a test vehicle. Regarding the simulator version, the inputs from sensors were simulated and the functional modules of active controls and workload estimation were left out. The integration into the test vehicle was based on the same modules. It suffered, though, from lacking appropriate sensor hardware.

The experiments, in particular those in the simulator for evaluation purposes, provided the following results. There were 7 scenarios selected for the functional test of GIDS. All of them led to dialogue actions of GIDS through displays, i.e. a changing counterforce on the accelerator, a discrete pulse on the steering wheel feedback, or a route guidance message.

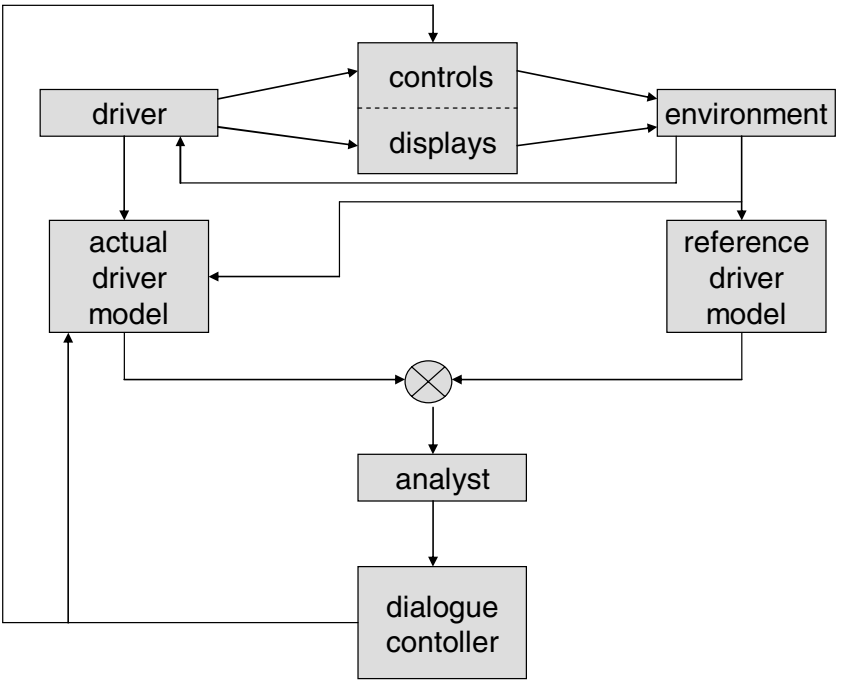


Fig. 77 Architecture of GIDS (cf. [Smiley & Michon, 1989])

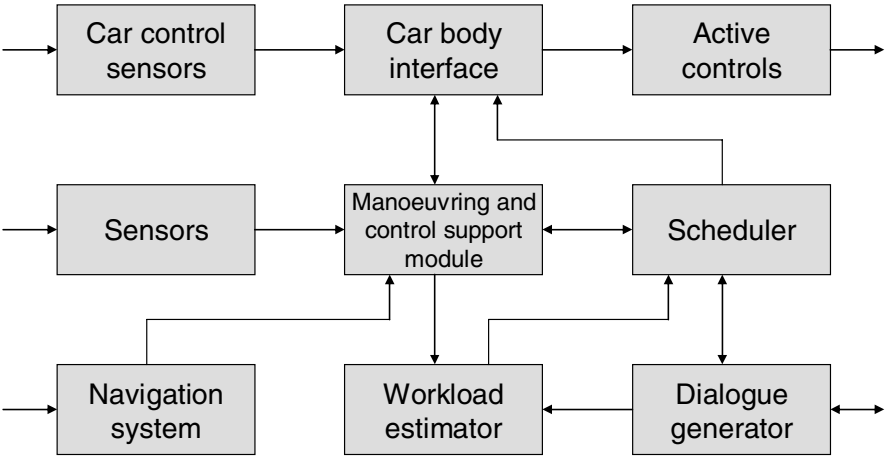


Fig. 78 GIDS prototype system (cf. [Smiley & Michon, 1989])

Without question, route guidance support can be principally considered as a very effective function for driver support. It was highly justified to take this for granted. Therefore, apparently there was no specific evaluation of this functional

aspect. As to the remaining dialogue actions of GIDS it turned out in the course of the experiments that most of the scenarios which were selected for the system evaluation were not sufficiently demanding for the test subjects to lead to significant performance increases through GIDS assistance. Only one of the scenarios led to assistance activity through accelerator feedback which resulted in a main effect on driver performance. In this scenario, the own vehicle negotiates a curve, beyond which it encounters another stationary vehicle ahead. It cannot overtake the vehicle as other vehicles are approaching. Compared with the condition of no assistance the test subjects in the condition with assistance had

- a lower speed when entering the curve
- a longer headway in terms of both time and distance
- a smaller deceleration and
- a larger accepted gap size.

Considering the time this project was launched, it was a very successful study which included the successful development of prototype hardware. This endeavour represented a great encouragement for further study, eventually merging into the development efforts which are presently under way in the automotive industry.

5.3.2.2 Driver Assistant System (DAISY)

This prototype as own development is related to the work environment of guidance and control of land vehicles to be driven in German road traffic. The first version, as decribed in [Kopf, 1994] and [Kopf & Onken, 1992], was designed to be in operation when driving on German motorways. It was developed in the framework of the Prometheus program, which was launched as part of EUREKA of the EU. There was a later version which, in essence, differed in how the adaptive modelling of the behaviour of the driver individual was accomplished. This is a crucial component of the system in order to achieve good work system performance and full acceptance of the driver [Feraric & Onken, 1995] and [Onken & Feraric, 1997]. Approaches of adaptive modelling were successfully further investigated in [Grashey & Onken, 1998], [Grashey, 1998], and [von Garrel et al., 2000], also extending the driving domain from motorway driving towards driving in the streets of cities.

Subsequent to extensive investigations in simulator experiments also results were achieved in a field experiment [Kopf, 1994] about

- verifying results from simulator campaigns
- implementing the software in a given test vehicle
- testing a haptic warning device
- gains in safety
- ratings of subjects on the system performance

Further results on own advanced approaches about adaptive modelling of driver behaviour were gained later on in additional simulator experiments (see [Feraric, 96], [Feraric & Onken, 1995], [Grashey, 1998], [Grashey & Onken, 1998], [Schreiner, 1999], [von Garrel, 2003], and [von Garrel et al., 2000].

General characteristics of DAISY

DAISY corresponds to an alerting assistant system in a work system of the kind as depicted in Figure 76. The pertinent work objective to operate on is that of road transportation by means of a given road vehicle like a car, a bus, or a truck. The assistant function is confined to certain sections of a transportation ride for transportation. For its original version it assists the driver when driving on a motorway, starting to be active after entering a motorway and terminating its activity when an exit is approached to leave the motorway or to turn into another motorway at a motorway junction. The development of DAISY focuses on the avoidance of accidents by monitoring the skill-based and partially procedure-based driver behaviour. The potential accidents considered are

- collision with leading vehicle
- crossing of lane borderline because of inadequate lateral control, and
- crossing of lane borderline because of violation of maximum speed in turns.

This corresponds to alerting assistance defined in Chapter 5.2.2. As its output, DAISY conveys a warning to the driver, if necessary. According to the basic requirement 1 for assistant systems (see Chapter 5.2.5), DAISY assesses/interprets the work situation on its own and does its best by own initiatives to ensure that the attention of the assisted human operator(s) is placed with priority on the objectively most urgent task or subtask in safety-relevant situations. DAISY is exclusively advisory as typical for assisting functions in the subrole (1). Similar to the assistant system CASSY (see Chapter 5.3.1.4), the functionality of DAISY could be extended by certain services like for instance adding navigational route planning as an automatic service or at the request of the driver. For the latter case a speech interface for input and output would be a very sensible feature. This kind of service would account for requirement 2 (see Chapter 5.2.5). Also other characteristics of CASSY to assist the human operator, like concept-based assistant behaviour not acconted for in the context of DAISY, can be considered feasible in a similar way as it is realized in the CASSY design (see Chapter 5.3.1.4).

According to this assignment, DAISY is to act like an ideal co-driver, who is quiet as everything is working normally, and who comes forward on its own initiative to contact the driver in an adequate way to give the necessary advice.

The design of DAISY, in particular the development of the a-priori knowledge, is considerably eased, because one can presume that all designers are very knowledgeable experts themselves about driving and traffic environment.

The efficacy of communication, in particular when considering warnings to the driver, plays an even more important role for this application rather than in the aviation domain. There are situations where the driver has to react safely within an extremely short time span. This demands special displays which may trigger reflexive, correct behaviour by the driver. With DAISY certain proposals are made in this respect.

Functional structure of DAISY

The motivation for voluntary actions of DAISY is only an advisory one in reaction to dangerous situations coming up. This motivational context is prefixed, but not

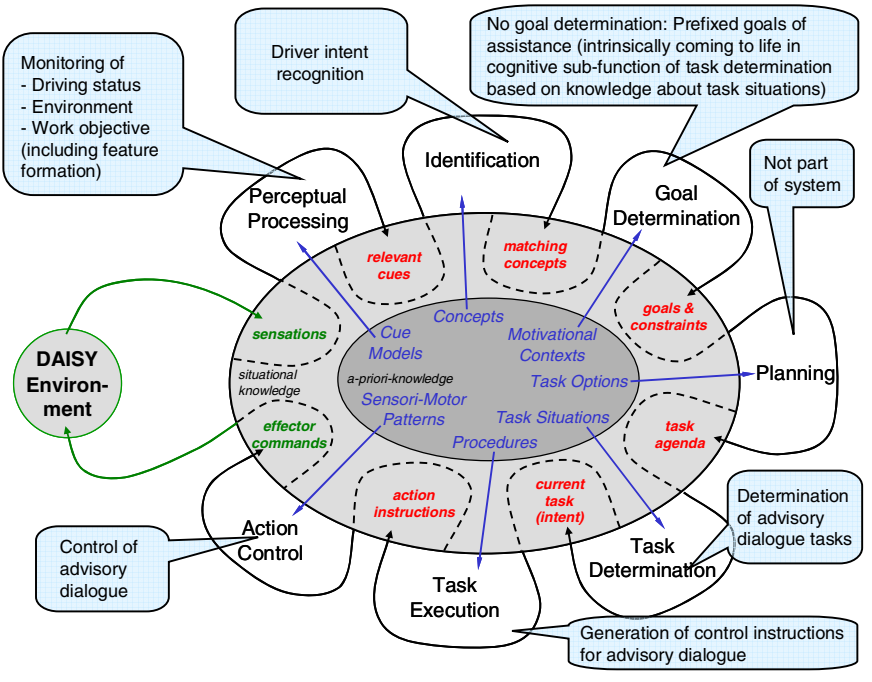


Fig. 79 Functional structure of DAISY as cognitive process

explicitly represented. It is intrinsically coming to life in all modules of the functional structure, similar to human procedure- and skill-based behaviour. Therefore, referring to the structure of the cognitive process (see Chapter 4.5.2), only the superposition of both procedure-based and skill-based cognitive processing is relevant for DAISY. The corresponding transformers to be covered are (see Figure 79):

- Situation interpretation, including perceptual processing, identification, and task determination
- Task execution, and
- Action control.

As to the *situation interpretation*, there are three main modules, the IRD and the IMAD as instantiations of the *reference driver* and the *model of actual driver*, and the *discrepancy interpretation* (see Figure 80). The task execution and action control is part of the fourth main module, the *warning device*. There is some resemblance of this structure with other driver support systems, in particular the GIDS (Generic Intelligent Driver Support) which is described in Chapter 5.3.2.1.

As becoming obvious from Figure 80, the main modules comprise several submodules. The *IRD* module extracts from the incoming data a symbolic description of the current driving situation. This information is used by all functional submodules of DAISY. Within the IRD a situation analysis takes place by use of existing a-priori knowledge about potential danger, hindrance by other vehicles, and

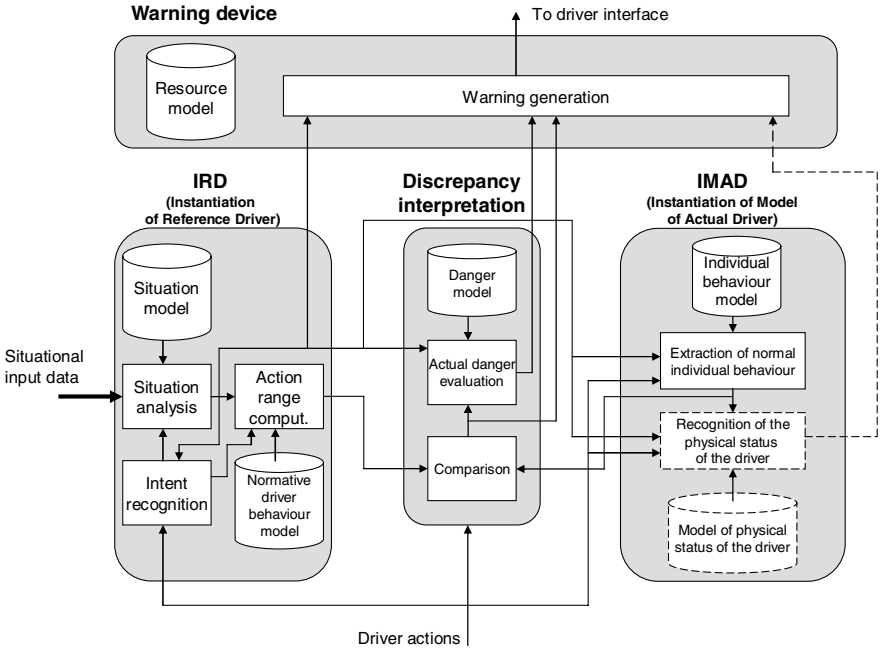


Fig. 80 Functional module structure of DAISY [Kopf, 1994]

normative driver behaviour. In addition, adherence to road traffic rules is evaluated. The IRD also comprises the recognition of explicit intent parameters like the desired speed which is a driving parameter for the driver's actions. In addition, it calculates the range of possible situation-specific actions, making use of a normative model of driver behaviour and the intent recognition sub-module.

The *IMAD* module is making use of the situation analysis of the IRD module in order to model how the actually driving individual probably behaves in terms of adherence to so-called time reserves for longitudinal and lateral control. As depicted by using dashed lines in Figure 80, also the modelling of deviations of the normal physical status of the driver was conjectured, although not realised in a verified function yet.

Both the normative reference and the behaviour modelled as that of the actual driver which accounts for individual deviating traits, in particular concerning time reserves as safety margins, are set against the actual driver action in the *discrepancy interpretation* module. Discrepancies are evaluated by use of the danger model which is essentially based on the so-called *time reserve*. The time reserve is an objective measure reciprocal to the danger, i.e. is a measure of how far the situation is still apart of a boundary surface as actual danger boundary (ADB) in the situational parameter space beyond that in the so-called uncontrollable region (UR) an accident will undoubtedly occur unless there are some benign circumstances making the accident not happen. In distinction to the UR, there is a safe

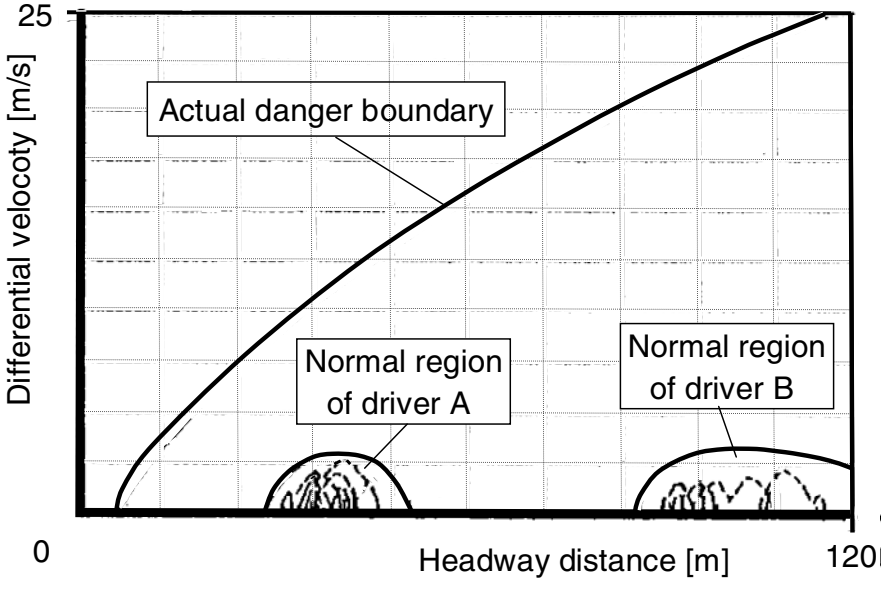


Fig. 81 Normal region of car following behaviour of two different drivers [Kopf, 1994]

region normally used by the driver, the so-called normal region (NR) and a transition region between both the NR and the UR. Figure 81 shows an example taken from a car following situation where the NR of two different drivers and the driver-independent ADB is depicted [Kopf & Onken, 1992].

In case of an actual danger coming up, i.e. a situation outside the normal region, the *Warning device* is triggered to issue an appropriate warning to the driver through both graphical and haptic displays. In case that the time reserve is going to fall below the driver-specific minimum reaction time, a point very close to the UR, an acoustic alarm will be triggered. Obviously, if initialized at the first time, DAISY has to be operated in a learning phase for a certain amount of time in order to establish the individual behaviour model as part of the a-priori knowledge.

A-priori knowledge of DAISY

The a-priori knowledge which is accumulated within DAISY is allocated in smaller units to the pertinent module in terms of models. The most essential and model is the model of the objective situation. The situation mortar comprises the external situation, including all elements describing the environment of the system driver/vehicle and of the vehicle itself, the internal situation, i.e. the situation of the driver, and altogether the system situation of the complete system driver/vehicle/environment. The model of the external situation also includes an anticipation of its future development. Other models are used as an extension to the internal and external situation model to describe the degree of danger of different kinds of potential accidents (danger model) or to describe the degree of

hindrance. Furthermore, there is also model of the stationary speed as targeted by the driver.

In distinction to the objective model with a normative model of the driver a model of the actual driver is used containing all available knowledge about the actual driver's individual condition and behavioural traits.

Situation model

The situation is modelled by normatively describing an external and an internal situation. Thus, the condition and geometrical configuration of the road, the static markings and traffic signs, the other surrounding dynamic entities participating in road traffic, and their time-varying relative positioning are typical elements of the external situation. On top of the external situation the internal situation comprises a corresponding normative set of possible driving tasks to be selected from by the driver. For example, there might be the situation of an obstacle vehicle on the right lane such that the driver has to choose between an immediate lane change from right to left, or to follow the obstacle vehicle because of occupied target lane, or deliberately following the vehicle without the intent to pass it. Correspondingly, the process of task determination has to take into account the intent to change the lane inorder to keep or to regain a desired speed. It also has to consider for instance the time needed for the manoeuvre of lane change by accounting for parameters like current speed, lane width, time gap between vehicles on the target lane and elements of situational evolution like closing up at an obstacle vehicle on the right lane before overtaking and changing to the free target lane. Hence, the total situation model is an overall normative behaviour model of the total system driver/vehicle/environment, including the situational elements of the external and internal situation.

For the modelling purpose, a vector of situational parameters and a corresponding situation space is established in the DAISY software. Each elementary type of situation corresponds with an elementary cell in the situation space. Furthermore, subspaces can be defined. A subspace contains all elementary types of situations, which can be described by vectors of same situational parameters but different ranges of parameter values. For the normative situation model in the reference driver module the granularity of the elementary cells can be rather coarse. For the driver actions solely situation-specific ranges of danger-free actions are considered. If the situation model with the individual behaviour of the actual driver is of interest for adaptation purposes, which is the case in the model of actual driver module, much higher degrees of granularity are required.

In its first version of implementation, decision trees and finite automata have been used for the normative model to represent the generic knowledge of the situation, including the driver behaviour. Decision trees were used to represent rules or static situational aspects, automata were useful for stereotype sequences of events or actions (see also Chapter 6). The concept "lane change state", for example, contains the geometric parameters of road and vehicle as well as the driver's actions in terms of steering wheel activity and the resulting parameters. Here, a behavioural model has been integrated to detect as early as possible whether the driver intends to initiate a lane change or is willing to keep the actual lane.

The version of an upgraded implementation which is being developed makes use of Petri Nets as the most suitable representation for both sequential and concurrent events.

Danger model
The actual degree of danger is explicitly evaluated by use of a danger model. For our purpose, three principal types of danger with respect to potential accidents are distinguished:

- collision with a vehicle in front because of inadequate deceleration behaviour
- violating nominal lane boundaries because of inadequate steering behaviour
- violating road boundary lines in curves because of inadequate velocity.

For these types of danger, a general danger function can be defined, which is reciprocal of the time reserve which is left in a given situation to initiate a danger-suppressing action in order to just avoid an accident. During normal driving this measure will be far above the human reaction time, because the driver normally drives safely. The time reserve is determined by

- the dynamic state of the system environment/vehicle
- the physical safety margins (road condition)
- the maximum possible action to avoid danger.

It should be taken notice of the fact that the time reserve is closely related to the definitions of time-to-collision (TTC) and the time to line crossing (TLC) (see [Godthelp & Konings, 1991]) except that the aspect of danger-suppressing actions was not included. For the simple case of lateral control subject to the lane keeping task, the a-posteriori time reserve T_{res} can be formulated by the following relation (see [Kopf, 94] and [Onken, 94]):

$$T_{res} = -(v_c / a_c) \pm \left\{ (v_c / a_c) 2 - \left[v_c^2 - 2v^2 (W - y)(c_r - c_{p,pot}) \right] / \left[v^2 a_c (c_p - c_{p,pot}) \right] \right\}^{1/2},$$

if $a_c \neq 0$,

$$T_{res} = (W - y) / v_c - v_c / \left[2v^2 (c_r - c_{p,pot}) \right], \text{ if } a_c = 0 \text{ and } v, v_c \neq 0,$$

and

$T_{res} = \infty$, if $a_c = 0$ and $v_c = 0$, using the following notation:

v:	vehicle speed
c_p:	curvature of vehicle path
$c_{p,pot}$:	potential curvature of vehicle path for danger suppression
c_r:	curvature of road boundary line
W:	available lane width
y:	cross track position

v_c : cross track speed

a_c : cross track acceleration

Depending on the parameter values not known for certain by the driver but to be used for the computation, the time reserve can be considered as subjectively assessed by the driver (a-priori time reserve) as opposed to an objectively measured quantity (a-posteriori time reserve). Both the subjective assessment and the objective result are of interest for the danger model.

If the a-posteriori time reserve drops under the minimum human reaction time, the accident is inevitable unless the obstacle is cooperatively moving away or the driver manages to circumvene the obstacle.

This concept of a-priori and a-posteriori time reserves will be exemplified in the following for the danger case of a potential collision with a vehicle in front. The time reserve is a function of the actual accelerations and the velocities of both the own vehicle and the one in front, the actual distance between the two vehicles, the road condition coefficients, and the potential decelerations as a result of the danger-suppressing braking action of the driver. To compute the a-priori time reserve, the value of the deceleration of the leading vehicle is determined by the 95% percentile of situation-specific driver braking actions, for instance corresponding to probability distributions as given in [Diepold, 1991] and [Winzer, 1980]. The a-posteriori time reserve is computed by use of the actually measured acceleration of the vehicle in front. The deceleration of the vehicle in front is set at the maximum value depending on the road condition.

Model of driver target speed

The knowledge about the target speed desired by the driver is a basis for the recognition of the driver's intentions.

The modelling process of the target speed is an estimation process based on observation. The basic idea is that the target speed can be estimated in a sufficiently accurate way from the observed situation data, if influencing factors like curves and road gradients, for instance, are not present. Also driver actions of vehicle acceleration are not allowed during this phase.

Individual behaviour model

Driver models have been developed within a wide range of paradigms and methods for many different applications. [Jürgensohn, 1997] gives an overview about the actual state of the art in this respect. The model of the actual driver as individual behaviour model as part of the IMAD module is to describe the individual style of the actual driver in performing certain driving tasks. Two approaches are used to implement the individual driver model, a statistical and a neural net approach [Feraric et al., 1992]. The statistical approach will be further outlined in the following.

In case of the statistical approach, the knowledge of the situation and driver-specific behaviour is represented by the empirical probability distribution of task-relevant time reserves. Hence, the individual behaviour model does not explicitly deal with instantiations of driver actions as such. The learning procedure

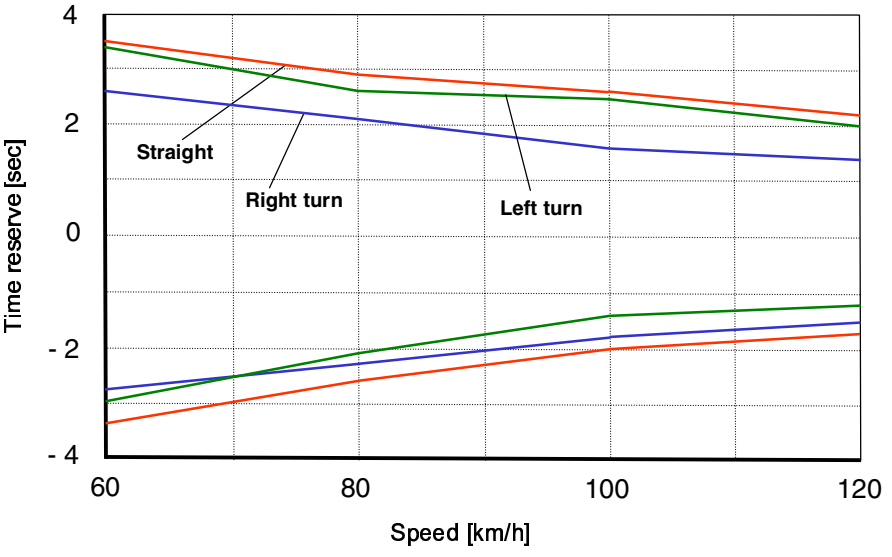


Fig. 82 5% time reserve for lateral control [Kopf, 1994]

essentially consists of recording the situation-specific empirical probability density. When a predefined number of data has been collected the learning phase terminates and the distribution will be computed. With regard to the use of the model for driver-adaptive warning, an algorithm computes a situation- and driver-specific threshold value (about 5% percentile), taking into account the particular shape of the distribution. This kind of distribution is to be established for all elementary cells of relevance in the situation space. The transition from learning to operation occurs independently for each situation element. In order to minimize the number of elementary cells to be covered, it is necessary to derive a model of the situational dependence of the behavioural parameter of concern. Figure 82 shows the influence of speed and road course on the lateral time reserve, as could be evidenced by simulator trials with 10 test subjects [Diepold, 1991].

Preferably, the model should provide the a-priori knowledge for all relevant situations about procedures individuals are normally using. Apparently, this is a difficult task on the basis of the statistical approach. Therefore, some alternatives of neural net approaches have been investigated. Neural nets seem to be a natural approach to implement such a model for different reasons:

- a neural net has the ability to process inputs from different sensors simultaneously, therefore complying with the requirement of a situation-specific description of the driver behaviour
- a neural net can learn from examples, being able to model each driver individually
- once trained a neural net can be retrained in order to adapt to changes in the individual driving style.

Using the back propagation algorithm it was taken advantage of a combination of nets, in order to get also information about the deviations from the mean behaviour of the actual driver [Feraric et al., 1992]:

Net one is trained first on the normal steering action. Its input is formed by the situation parameters. The output of the net is compared with the actual steering action and the difference is used for training. Subsequent of the training of net one net two is trained on the deviations of the actual driver actions from what is expected by net one. During the operation phase both the output of net one and net two can be used for adaptive warning. Results of test runs on a driving simulator were very encouraging. As part of the model of the actual driver, also some information about the driver condition with regard to his driving capacity can be derived by this approach. Subsequent to learning, the characteristics of observable deviations from the normal behaviour can be exploited for this purpose. For instance, the situation-specific ratio of the actual frequency of violations of the individually minimal time reserve to that one when the driver is behaving normally is an interesting measure. It should be close to 1 under normal conditions. Values distinctly greater than 1 are indicating a change towards unsafe behaviour.

As DAISY in its original version only relied on an individual behaviour model based on the time reserve, without considering a model of the individual driver action as such, later versions endeavoured to develop a driver-adaptive modelling of the action, as for instance the longitudinal acceleration for longitudinal control or the steering wheel deflection for lateral control (see [Feraric, 1996], [Grashey, 1998], [Schreiner, 1999], and [von Garrel, 2003] and Chapter 6).

Implementation and experiments

DAISY is implemented in an SGI computer and integrated in a fixed-base driving simulator facility of the University of German Armed Forces Munich. The vehicle model and the DAISY software are computed within 0.05 second timeframe. All software code is written in C. Part of the DAISY software and functionality, i.e. the monitoring and warning for lateral control, was ported in the VITA (computer Vision Technology Application) test vehicle of Daimler-Benz for field tests. In this vehicle, a computer vision and control system was installed which was developed in the first place for automatic driving by the University of German Armed Forces Munich [Dickmanns et al., 1993] in cooperation with Daimler-Benz. The DAISY software for lateral control was implemented into this system in place of the software module for automatic control, making use of the built-in computer vision capability for the road geometry perception.

The experiments in the simulator were aimed at investigations concerning the adaptation aspect of DAISY and the determination of the driver's conditional status. Also the investigation on estimating the driver's target speed was fostered by various simulator experiments [Kopf, 1994]. The experiment with VITA was intended for an overall system validation with respect to lateral control.

Experiments for driver-adaptive warning

The adaptivity experiments will be described in more detail in the following. They were aimed at testing the process of determination of the situation-specific normal

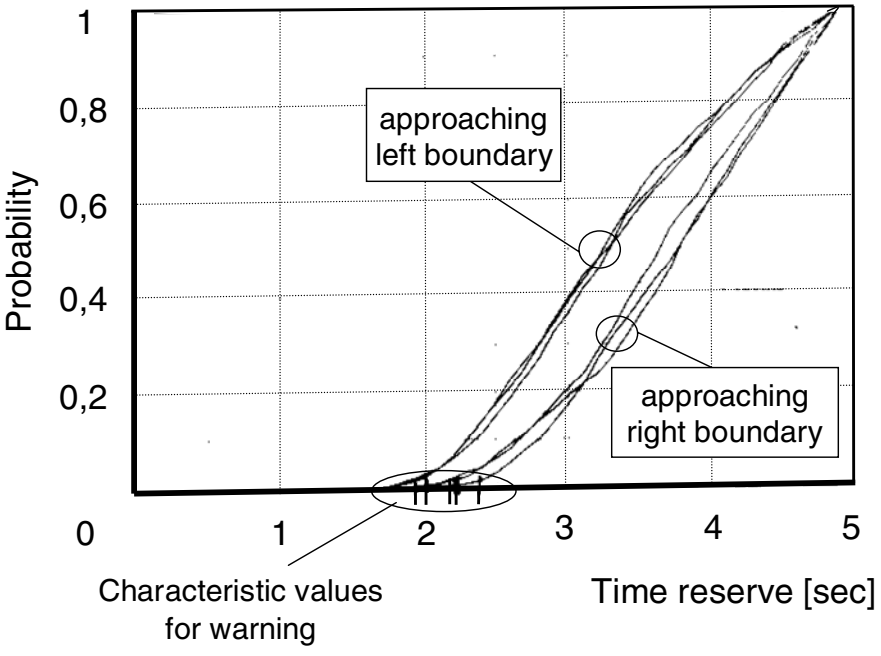


Fig. 83 Distribution of time reserve for lateral control. Three test runs at 75 km/h on straight road segments with one test subject were evaluated. [Kopf & Onken, 1992]

individual behavioural traits of the actual driver and its effect on actual warnings. A basic hypothesis in order to pursue this adaptive approach was to assume consistency of the driver behaviour over time. Thus, the experiments were aimed at proving that this hypothesis is valid, in essence. Another aspect is to demonstrate the necessity of the adaptive approach, i.e. that different drivers can be so much apart with respect to their behaviour in certain situations such that a warning strategy based on an average driver is no longer acceptable. For these experiments the statistical adaptivity approach was used, only.

First, some experimental results will be given with respect to adaptivity in lateral control. 11 test persons were involved in this experiment. Each of them carried out three test runs of 10 minutes each. The task was to drive on the right lane and to stay abreast with a truck which is driving on the adjacent lane on the left with constant speed (75 km/h). No other traffic was involved. Through this set-up of the task the test person was not aware of the fact that the test was solely focused on lateral control. Disturbances from road surface and wind gusts were simulated by coloured noise in the relative azimuth with standard deviation of 0.01 rad. The course to drive on was 13 km long, straight and curved portions to the right and left were equally distributed. The radius of the curvatures was between 500 and 3000 m.

Figure 83 shows the empirical distributions of the time reserve in three test runs of one test person, taking into account the situational elementary cells one and two

Table 7 Tested elementary cells of situation subspace of lateral control [Kopf, 1994]

Situational elementary cell	Dangerous obstacle	Course of road
1	right	straight
2	left	straight
3	right	right turn
4	left	right turn
5	right	left turn
6	left	left turn

for driving straight ahead and watching the danger of violating the lane boundaries on the left and on the right (Table 7).

The 0.5, 5, 10 and 20% percentile values were calculated for each distribution. With respect to warning adaptation only the lower limits are of interest. A characteristic value CV for this lower limit was geometrically determined by drawing a straight line through the points of the distribution curve for the values for the 5 and 10% percentiles. The point of intersection of that straight line with the time reserve axis indicates the CV. The condensated results for all test persons are listed in Table 8.

Table 8 Evaluation of overall characteristic values (CV) [Kopf, 1994]

	Situational elementary cell					
	1	2	3	4	5	6
Minimum CV	0.87	1.67	0.63	1.33	1.17	0.90
Maximum CV	2.40	2.33	2.00	2.07	2.30	2.57
Mean CV	1.91	1.94	1.33	1.82	1.86	1.79
CV standard deviation	0.42	0.26	0.46	0.27	0.37	0.42

The mean values over all test persons with a minimum and maximum CV in each situational elementary cell and the overall mean and standard deviation in each of the elementary cells are enumerated. A variance analysis showed that behavioural differences between test persons were highly significant on a 0.1% level. Thus, adaptation to the individual driver of warning messages makes sense. Considering a single test person, also a variance analysis is carried out in order to evaluate the differences of the CVs with respect to different situational elementary cells. For one of the 11 test persons no significant difference could be determined, for two of them significant differences could be determined on the 5% level and for the rest on the 2.5% level at least. Therefore, it can be concluded that it is necessary to break down the situation space for the purpose of situation adaptation in about the way it was done. For DAISY the mean characteristic value minus a constant of 0.2 seconds was used as a time reserve threshold to trigger a warning message. Then, less than 1% of all data collected for the time reserve behaviour of the single test person lies below that value.

With regard to longitudinal control results of similar effect has been achieved [Kopf, 1994].

System evaluation with test vehicle
As mentioned earlier, DAISY was ported into a test vehicle of Daimler-Benz AG in order to validate the design concept. This was exemplified on the basis of assisting the driver in lateral control. The validation was focused on aspects like portability of simulator test results into real world environment including the aspect of making use of real data from computer vision, haptic driver interface, and effect of warning strategy on safety and acceptance by the driver. The advantages of this kind of display were obvious:

- less mental load on the driver
- possibly shorter driver reaction time
- vehicle passengers would not become aware of the warning message.

The haptic warning message primarily consists of a continuous torque signal on the steering wheel (1.5 km), which is activated when the situation-specific, driver-adapted time reserve value threshold is exceeded. The torque signal by itself already produces an effect to reduce danger. In order to make this warning signal discernible from other torque stimuli on the steering wheel which might be caused by low frequency road impacts, for instance, a vibration signal is superimposed.

Two elementary cells of the situation space for lateral control were evaluated by these trials, corresponding to that of driving straight ahead in Table 8. Road markings corresponding to a one-way piece of motorway with two lanes were drawn on the taxiway of the former airport of Neubiberg for a stretch of about 2 km in order to have a realistic environment on a perfectly reproducible basis. The outer lane was used only in both directions and no other traffic was involved. The test runs started from standstill. The test vehicle was to be accelerated until a constant speed of 60 kph was reached. This speed was to be kept as long as possible up to the end of the test. Twelve test persons were involved between 21 and 27 years old, with driving experience of more than three years and more than 15,000 km/year.

Each test person carried out nine test runs in three test blocks. The first block consisted of four runs in order to test the learning capability for the warning threshold adaptation. The test person had three runs of normal driving without DAISY to collect data for the learning algorithm and one test run with DAISY. The second block consisted of two test runs in order to evaluate the warning signal with regard to recognizability of the warning direction. The warning was enforced by effecting visual occlusion for at most 10 seconds in pseudorandom time intervals. The last test block again consisted of two runs in order to evaluate the effect of the warning mechanism with respect to safety. The test person was driving once with DAISY and once without, again with enforced distraction from the driving task. Prior to the second test block the test person carried out an extra run in order to become familiar with the occlusion device.

The following results were achieved. The individual warning thresholds learned from the first three test runs were again quite different between the test persons,

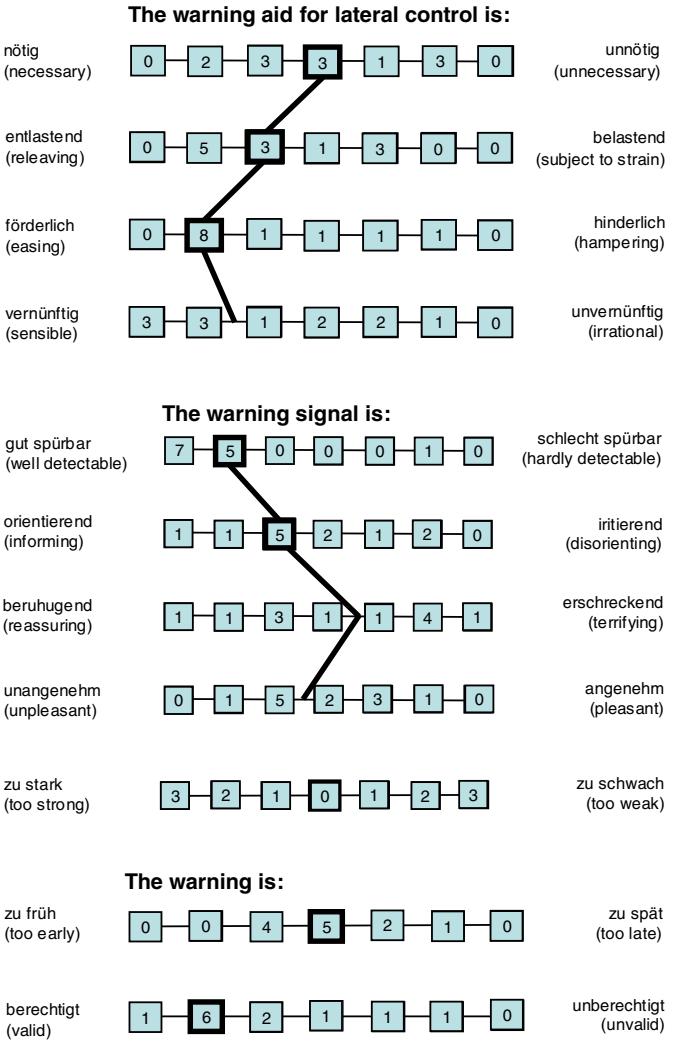


Fig. 84 Results of questionnaire with semantic differentials after completion of test runs. The numbers in the blocks represent the number of test drivers giving that score. The median is highlighted [Kopf, 1994]

similar to simulator tests. However, different to the simulator tests, the values did not differ significantly for the two cases of danger corresponding to violation of right and left lane boundaries. The reason might be that, as opposed to the simulator trials, there was no vehicle aside on the adjacent lane. With regard to the second test block, recognizability of the warning direction was not completely unambiguous for all test persons. 92.8% of the driver reaction was in the correct direction. Only one test person had some difficulty with the correct interpretation

of the torque signal. Regarding the third test block this evaluation showed a highly significant difference between the test runs with DAISY (with warning) and those without regarding the effect of distraction on driving safety. It can be concluded that (at least under the conditions of the test runs) driving safety could be secured with DAISY despite significant distraction.

Each test person filled in a questionnaire after completion of the test runs giving a subjective assessment about DAISY. The questions were mainly formulated in the format of semantic differentials. Some of the more important results are summed up in Figure 84.

Of course, the results should be treated with caution because of the relatively small number of test persons. The English translation of the semantic attributes used is given in parentheses. As the results demonstrate, the assessment resulted mainly in favour with DAISY. Interestingly, the question whether there is a need for DAISY or not did not lead to a clear answer despite the other closely related general questions were answered clearly in favour with DAISY. One reason for this might be the psychological aspect that the danger situations experienced by the drivers were not due to evident driver mistakes, since they were prompted by a forced obstruction of the driver's situation awareness.

Conclusions

This section presented the concept, the realization, and the validation of DAISY, a situation-specific, driver-adaptive monitoring and warning aid for the driver. The more outstanding functionalities are based on the use of computer vision data about the driver environment and the modelling of the driving situation, including the situation with respect to normative driver behaviour, as well as modelling of the types of danger concerned, types of hindrance, and driver objectives. In addition, and learning algorithm is providing a model of the individual behavioural traits of the actual driver to be warned.

DAISY was tested in a driving simulator and a test vehicle. These tests led to following results:

- the realization of a monitoring and warning aid is possible on the basis of computer vision data
- the modelling of the individual behavioural traits of the actual driver is possible, presuming that a certain amount of time for the learning process is acceptable
- the concept of driver-adaptive warning is well accepted by the test drivers, including the haptic display
- DAISY enhances driving safety in situations susceptible for distractions
- longterm effects like risk compensation effects can be avoided by countermeasures which are generated on the basis of knowledge about the development of this effect.

5.3.3 Assistant for Driver Tutoring (ADT)

An emerging application domain for assistant systems is that in work systems for operator tutoring in vehicle guidance and control. These kinds of work systems by

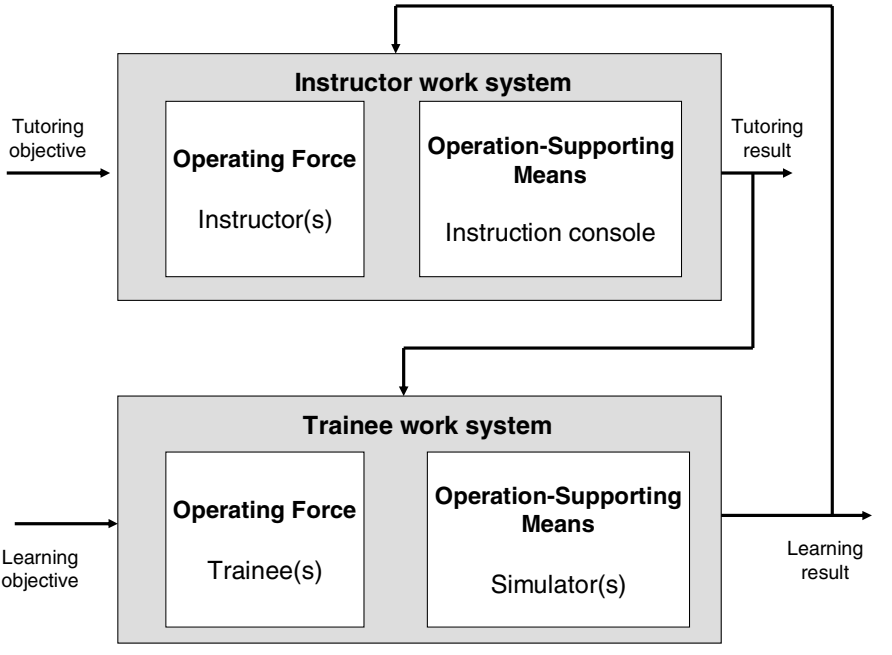


Fig. 85 Configuration of work systems in the process of tutoring

use of simulators are operational for aircraft pilot tutoring as well as for driver tutoring of a structure of work systems as depicted in Figure 85.

Today, operational systems of that kind usually consist of a purely human operating force. The operating force of the instructor work system comprises one instructor person or an instructor team, and similarly the operating force of the trainee work system comprises one trainee or a team of trainee persons. Both work sytems the instructor work system and the trainee work system make use of more or less complex operation-supporting technical means, in particular an instruction console and a simulator respectively. The simulator provides a virtual reality environment about the work environment the trainee is to be trained for. It includes the work station of the trainee. The instruction console makes up the work station of the instructor(s). It is separate of the simulator, possibly remotely located from the simulator. The instruction console provides all kinds of automation components to facilitate the job of the instructor, including means of communication with the trainee, indicators and displays which inform about the state of the training process and the well-being of the trainee. For economical reasons there are investigations to have several independently or dependently working trainees instructed by one instructor at the same time. Apparently, then a correspondingly increased number of simulators are being run simultaneously.

Since there are two separate work systems in the process of tutoring, there is the possibility of two assistant systems, one for the driver work sytem and the other one for the instructor work system as shown in Figure 86.

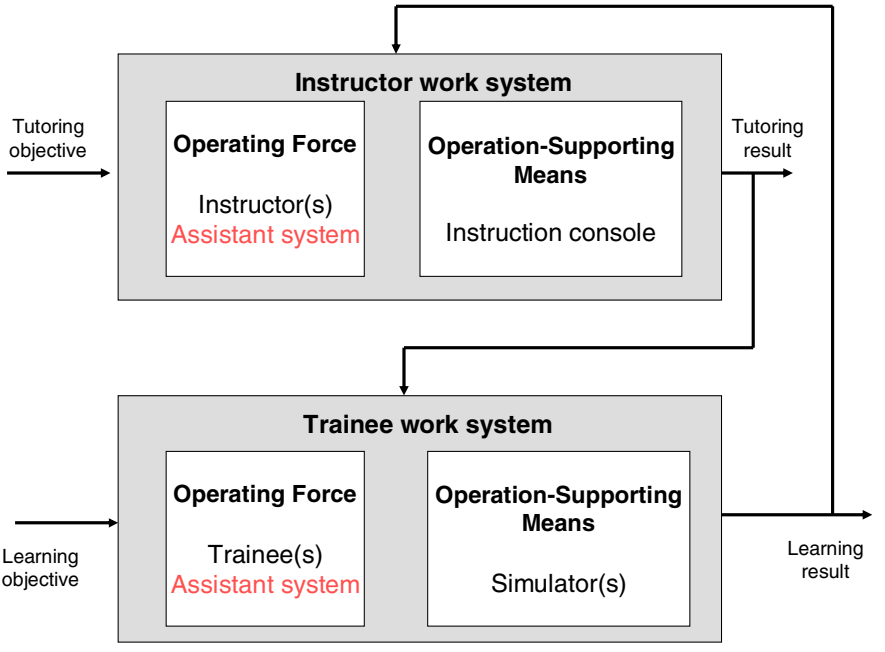


Fig. 86 Introduction of cognitive assistant systems inconventional tutoring work systems

The trainee assistant certainly is an alerting assistant system, whereas the instructor assistant can well be a substituting assistant system, then reducing an instructor team to possibly only a single instructing person. In case of full automation of the instruction function with no instructor left by means of a cognitive system, the instructor work system ends to be a separate work system. The instructor function is subsumed under the trainee work system, leading to a single tutoring work system. Then, the instructor function becomes either part of the operation-supporting means of the tutoring work system or of the operating force as part of the trainee assistant system as shown in Figure 87.

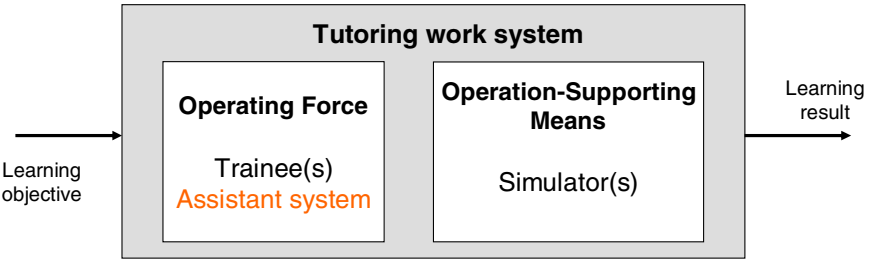


Fig. 87 Compacted tutoring work system with only one ACU as cognitive assistant system containing all necessary tutoring functions



Fig. 88 Example of visualisation of optimal driving path for turning right at the intersection in front

The trainee assistant system essentially monitors the trainee with respect to the work objective of learning and issues situation-dependent advices to the trainee if necessary. A typical exmple may be the type of visualisation as depicted in Figure 88. Furthermore, because of the knowledge about the learning objective and the learning progress, the assistant system can provide proposals for appropriate learning lessons to go through. It also may be authorized to automatically control the sequence of lessons on its own. Also the analysis of the trainee performance can be part of the assistant system (possibly of the trainee as well). It receives information about the learning progress and the situations where performance deterioration has occurred, possibly repeatedly. Based on that, it can carry out further analysis investigations and simulations in order to provide the trainee with background knowledge and detailed explanations about the causes of poor behaviour after each lesson. In addition it can take over the scheduling of the simulator operation. This is particularly of interest, if several simulators are operated in parallel.

Generally speaking, one great constraint of cognitive engineering is the relatively sparse sensing capability of machines, so far. This is particularly true for automotive applications. Although the advances in the field of computer vision, for example, are impressive (see for instance [Dickmanns, 2007]), it is still a long way that computer vision becomes equivalent to human vision. This is the reason why tutoring under machine assistance does not take place on the job in the real world. The tutoring system according to Figure 87 does not suffer from this

drawback, though, because it represents the real world of the tutoring process as a virtual reality application a virtual sensory system. The process of generating the virtual environment the trainee has to deal with automatically also generates as an output what wold have to be gained by real sensors in the real world environment. The design of simulators used for tutoring sytems can take this aspect into account. Another reason for operating tutoring systems by use of simulators is the safety aspect.

Again, both the assistant system and the human operator as depicted in Figure 87 has to rely on a considerable amount of a-priori knowledge about the application domain, i.e. the trainee in this case. The assistant system also knows about the learning objective.

The assistant system

A typical functional structure of the cognitive process of an assistant sytem as part of the operating force in a tutoring work system (see Figure 87) could be like that depicted in Figure 89.

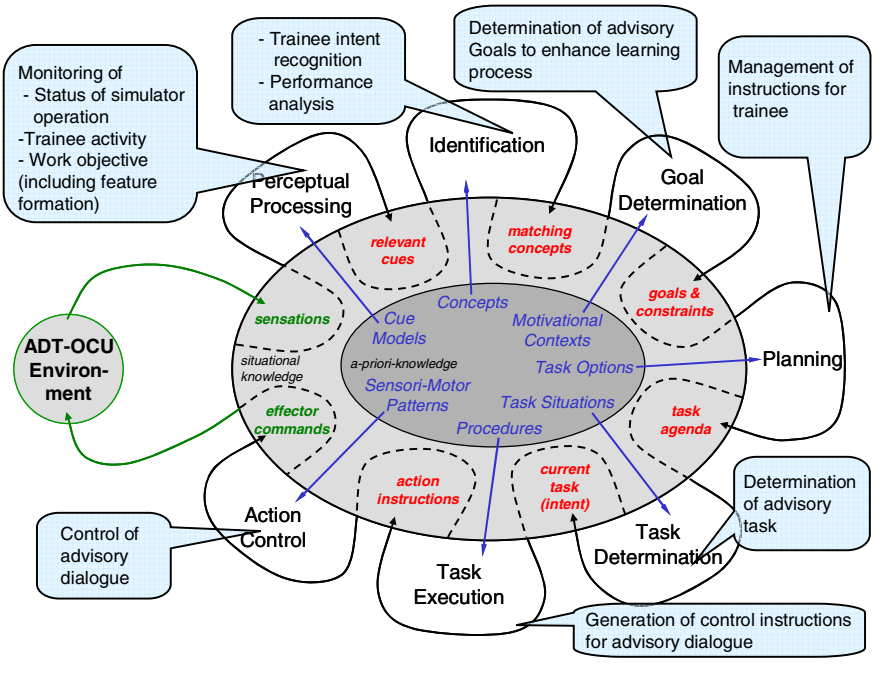


Fig. 89 Structure of the cognitive tutoring process

The cognitive process of the assistant system is driven by the a-priori knowledge regarding a motivational context such as doing its best to co-operate with the trainee to enhance his/her performance. This entails the situation-dependent determination of advisory goals to enhance learning. There is also a

functional structure of the cognitive tutoring unit as part of the operation-supporting means which, in essence, receives the results of the situation interpretation of the assistant system for analysis purposes. This functional structure may also contain the full range of cognitive behaviour as that of the assistant system, i.e. concept-based, procedure-based, and skill-based behaviour.

In case of a driver tutoring system, the a-priori knowledge required for the situation interpretation is similar to that of the IRD module of DAISY (see Chapter 5.3.2.2). In addition it contains the knowledge about incorporating the analysis results of the cognitive analysis unit. The crucial part of the a-priori knowledge is the model about the normative driver behaviour as a reference which can be established similar to the corresponding model as part of the IRD module of DAISY (see Chapter 5.3.2.2). Again, the reference is defined by way of normative borderlines between performance levels, i.e. for instance the levels of *good*, *acceptable*, and *unacceptable* performance.

In [Otto, 2005] an approach is verified to establish a reference model of that kind. This was done in two steps. The first step comprises the determinantion of the envelope of "good driving" by collecting data in simulator runs pertinent to this category of good driver behaviour. These data are assessed by professional instructors when observing the drivers. The borderline between good driving and acceptable driving might be defined by the 95% percentile, for instance. The second step comprised the collection of data, again in simulator runs, pertinent to the category of unacceptable driving. These data came about by the corrective comments of professional instructors when observing the drivers. Here, the borderline between acceptable and unacceptable driving behaviour was established by use of support vector machines (see [Burges, 1998], [Witten & Frank, 2000], [Kecman, 2001], and [Bousquet et al., 2003]).

Chapter 6
Implementation Examples of Crucial Functional Components of Cognitive Automation

Referring to the functional framework in Figure 30 for human cognition shown in Chapter 3.2.4 and the process of artificial cognition in an ACU described in Chapter 4.2.1 some implementation examples of crucial functional components of cognitive automation will be presented in more detail in the following. The choice of these implementation examples is made in a way to

1. include, from the historical point of view, some milestones about developments of models of human cognition which can be considered as important ones recent or current developments are based on in one or the other way.
2. exemplifying that about all functional aspects, as indicated in the functional framework of Figure 30, are included in these implementation examples. The low-level functions will not be covered, since we focus on software approaches in this book. This does not imply any constraint, though. We hope that the reader may appreciate at this point that we rely chiefly on implementations which were carried out in our own laboratories.
3. make sure that the realisation of an implementation is in fact feasible and powerful enough to achieve work system enhancements as theoretically claimed. In that sense the examples presented have not necessarily to belong to the status of latest achievements in implementation methods. These are anyway permanently in progress.
4. mainly focus on *knowledge management* as the central process of cognition. Methods and techniques which might also be important for the design of cognitive automation like those for perceptual processing, also including sensing and data fusion, or management of data bases are not be covered in the following.

The order of presenting the examples of component implementations, which will be presented in the following, is essentially chosen from a bottom-up perspective. About all of them are covering more than one cell of the array shown in the functional framework of Figure 30. That means that lower-level functions as enablers and behaviourally lower levels come prior to higher-level functions and higher-level behaviour. It turns out that there is at least one implementation example associated to each cell of functional and behavioural levels concerned. Furthermore, it should be kept in mind that, in the first place, all high-level

R. Onken and A. Schulte: System-Ergonomic Design of Cognitive Automation, SCI 235, pp. 213–310.
springerlink.com © Springer-Verlag Berlin Heidelberg 2010

functions from skill-based through concept-based behaviour, which are essentially identical to those of ACUs, are determined by the a-priori knowledge to be represented and implemented. Therefore, the knowledge management is the main issue of the following implementation examples.

As opposed to humans, the world with its dimensions of work environment and entailed work objectives a single ACU is exposed to in current designs, is extremely limited. This book, for instance, is deliberately confined to particular applications only of the work domain of vehicle guidance and control. Therefore, for an ACU in such a confined domain, attention control does not play a role as it does for human cognition. Looking at it from the perspective of the layout of the cognitive process it is the motivational contexts of such an ACU which is very limited so far. In that case, the designer of an ACU can rather easily take into account all situational cues right away which might be associated to the motivational contexts. The resulting ACU will monitor all the situational cues in parallel. There is no limitation of computational resources in this respect. As a consequence, also the weighting between motivations can usually be achieved rather easily. If future developments include ACUs with much enlarged work responsibilities, attention control might become necessary, at last. Then, a more complex management of the a-priori knowledge about motivational contexts and priority control of motivational contexts will be necessary, too. Possibly, the knowledge management could be of a kind similar to the theoretical basis of what is outlined in Chapter 6.1.2.2. Following those basic ideas of [Landauer, 2002], there would be a vector in a semantic space of motivational contexts which would trigger actions as appropriate ones for the actual work situation. It is still a long way to get there, but surely there are coming developments of that kind some day.

6.1 Knowledge Management

The knowledge which is processed in a cognitive system of a particular domain, i.e. which we are interested in, is the knowledge to make intelligent functions work in the respective domain. Here, an intelligent function means that it produces an appropriate output when faced with stimuli which indicate that there is a situation with problems to be solved. Thereby, it is not specified beforehand in explict mathematical terms how the outputs depend on the inputs. In [Kosko, 1992] this is called *model-free estimation of functions* from data. Based on this, he presents an illustrative taxonomy of model-free estimators as shown in Figure 90.

Structured rules of a production system are considerably different from those of a fuzzy system. Production rules are acquired, stored ad processed as bivalent symbols as opposed to fuzzy rules which are treated as numerical entities. In the numerical framework one deals with continua instead of discretes. This becomes even more obvious in case of neural systems. These are not structured beforehand, though. The patterns neural networks can easily recognize cannot even be defined by us. A car, for instance, can only be defined by examples.

In the following when we will discuss concrete approaches of knowledge management, in particular knowledge representation, we will first cover the

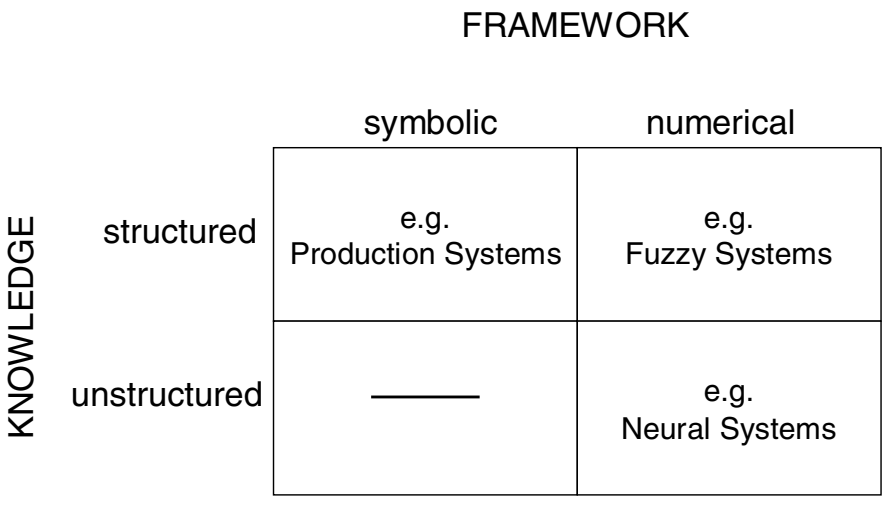


Fig. 90 Taxonomy of model-free estimators [Kosko, 1992]

symbolic approaches before dealing with the numerical ones, and discussing structured knowledge representation prior to the unstructured one.

6.1.1 General Discussion on Knowledge Representation

The medium-level artificial cognitive functions are background functions to enable the high-level functions. Their main purpose is to manage the a-priori knowledge when carrying out the high-level functions. Knowledge management comprises knowledge acquisition, representation, and retrieval functions. As already stated, the a-priori knowledge determines the behaviour of the ACU. Once being acquired and formally incorporated, this knowledge therefore determines how effective the the ACU will perform. Taking for granted that the necessary knowledge is available, the way this knowledge is represented within the ACU, predominantly determines the efficacy of the knowledge processing and whether it is well-accustomed to the demands of the work process. Therefore, when regarding medium-level artificial cognitive functions we shall dwell in the following on the knowledge representation issue in the first place.

Interestingly enough, it is a fact that the research subject of modelling how the knowledge is being represented in the human brain has been mainly advanced by research people working in the artificial intelligence realm. These people, who are working since the 60ties on intelligent machines of a performance level similar to humans, had to figure out how knowledge representation can be accomplished in a sufficiently efficient way. It seemed to be the easiest way to try to mimic the way as it was assumed at that time how it takes place in the human brain. Typical examples for this kind of approach are the early investigations on knowledge representation of [Newell & Simon, 1972] for the purpose of modelling human problem solving and [Schank, 1975] for the purpose of machine translation of

languages. In essence, since that time the way of tackling the development task for the function of knowledge representation has not changed considerably. More recent work, though, indicates that these approaches have taken insufficient account of the knowledge representation in the human brain by way of semantic coding. That means, this issue of knowledge management seems to become in the future again a more interdisciplinary one by benefitting from the advances in biopsychological research about the binding problem and the way in more detail how the global work space works (see Chapters 3.2.1.2 and 3.2.1.4). Before entering into some illustrations of representative approaches on knowledge representation, also including more recent work, a more general analysis will be taken up as published in [Davis et al., 1993].

What is knowledge representation really about? [Davis et al., 1993] try to answer just that question which, surprisingly enough, has been rarely answered directly before. They argue that the notion of knowledge representation

"can be best understood in terms of five distinctly different roles it plays each of which places different and at times conflicting demands on the properties a representation should have".

These are:

- A knowledge representation is most fundamentally a surrogate, a substitute for something internal or external. An intelligent entity (e.g. person or computer) represents knowledge in its memory in order to keep it ready for use while reasoning about the world, for instance in the context of a work process. The correspondence in view of the intended identity between the surrogate and the thing it substitutes is the semantics. For instance, a semantic net is a corresponding representation in symbolic terms and an artificial neural net directly complies with this in a connectionistic way. The latter corresponds to the human semantic coding.

  Another issue concerning a surrogate is the fidelity which determines how close the surrogate is to the real thing? The real thing might be something tangible or intangible like an oak tree or a mathematical expression, respectively. Perfect fidelity might be achieved regarding the mathematical expression mentioned, whereas it is impossible in the case of the oak tree. This leads to the conclusion that one has to live with imperfect representations which, in turn, can potentially cause incorrect inferences when used in the reasoning process using the representations.
- A knowledge representation is a set of ontological commitments. This alludes to the commitments to be made about how and what to see due to the fact that the surrogate cannot be more than an imperfect approximation of the real thing. Selecting one or another ontology can produce a very different view on the problem to be reasoned about. Through these commitments some part of the real thing is brought "into sharp focus, at the expense of blurring other parts" as stated by [Davis et al., 1993]. These ontological commitments comprise the most essential part of the process of representing knowledge, beginning with the selection of the representation technique like mathematical representational means, or rules, or frames, or

neural systems etc. (see Figure 90 on Kosko's taxonomy). Other commitments in accumulating layers have to follow. A commitment to frame-based representation lends itself to form collections of prototypes and taxonomies but other commitments have to follow to decide which real things actually are to be selected as prototypes. In turn, "rules suggest thinking in plausible inferences, but do not tell us which plausible inferences to attend to". Considering human semantic coding on the baisis of connectionism, this is most convenient in this respect, because it **efficiently** comprises on the whole the relevant amount of information which is important for potential reasoning processes (see also Chapter 3.2.1.3).

- A knowledge representation is a fragmentary theory of intelligent reasoning, since all possible inferences can only be made based on what is represented as knowledge and how it is represented. The theory of intelligent reasoning, underlying the way the knowledge representation is carried out, can be made evident by examining its three components [Davis et al., 1993]:]:
    - The representation's fundamental conception of intelligent reasoning
    - the set of inferences the representation sanctions, and
    - the set of inferences the representation recommends."

  [Davis et al., 1993] state:

  "While sanctioned inferences tell us what conclusions we are permitted to make, that set (of these inferences) is invariably very large and hence provides insufficient constraint. Any automated system attempting to reason, guided only by knowing what inferences are sanctioned, will soon find itself overwhelmed by choices. Hence we need more than an indication of which inferences we can legally make, we also need some indication of which inferences are appropriate to make, i.e., intelligent. That indication is supplied by the set of recommended inferences. Once we admit that there is a need for recommended inferences we also admit that the representation has to tell something about how to specify the recommended inferences, i.e. how to reason intelligently. One way is to have the human operator tell what to do, to recommend the set of inferences to be represented, thereby expressing their reasoning strategies. Thus, knowledge representation and intelligent reasoning are closely intertwined."

  [Sowa, 2000] sums up while alluding to this particular role of knowledge representation:

  "To support reasoning about the things in a domain, a knowledge representation must also describe their behaviour and interactions. The description constitutes a theory of the application domain. The theory must be stated in explicit axioms or it might be compiled in

executable programs by what means and how the knowledge is manipulated through inferences."

- A knowledge representation is a medium for efficient computation. This role covers the inevitable limitation in computing power which will be present forever despite the permanent progress in computing equipment. It has always to be kept in mind that the way knowledge is represented is of great impact on computing efficiency. Again considering human semantic coding on the baisis of connectionism in this respect, the computing efficiency as to the knowledge representation is very high as mentioned earlier, whereas the computing power with regard to conscious reasoning processes in less familiar domains is rather low.
- A knowledge representation is a medium of human expression. Communication between the humans involved and possibly the ACUs, in particular when incorporating knowledge in the ACU, should take place on the level of facilitating mutual understanding. [Davis et al., 1993] states:

  "The representation is the language in which we communicate, hence we must speak it without heroic effort."

  This is also alluding to the concept of work with at least one human being involved in a work process. Mutual understanding on the cognitive level is of crucial importance to accomplish satisfying work results.

Each of these roles that a knowledge representation places different and possibly conflicting demands on the choices one has to take when building up the representation. These choices are of great impact on both the reasoning and the computational architecture. If there is clear awareness of these above mentioned roles and if there are some conflicts in demands, one might succeed at the end by combining different representations. [Yen et al., 89] have given an example on the combination of rules and frames, both contrasting schemes for representation, more or less focusing on the reasoning aspect:

"Rules are appropriate for representing logical implications, or for associating actions with conditions under which an action should be taken. … Frames (or semantic nets) are appropriate for defining terms and for describing objects and the taxonomic class/membership relationships among them. An important reasoning capability of frame systems with well-defined semantics is that they can infer the class/membership relations between frames based on their definitions.
Since the strengths and weaknesses of rules and frames are complementary to each other, a system that integrates the two will benefit from advantages of both techniques. This paper describes a hybrid architecture called classification based programming which extends the production system architecture using automatic classification capabilities within frame representations. In doing so, the system enhances the power of a pattern matcher in a production system from symbolic matching to semantic matching, organizes rules into rule classes based on their functionalities,

and infers the various relationships among rules that facilitate explicit representation of control knowledge."

In general, oftentimes symbolism and connectionism are considered as contrasting schemes of knowledge representation. However, they indeed lend themselves for combining both for the sake of representational power. Both kinds of approaches have certain advantages and disadvantages. For instance, founding cognition and the incorporated knowledge representation on explicit symbols being used by the cognitive process, for example, as symbols representing a-priori knowledge or as former process states, certainly provides the desired efficient means to track back the process of information transformation. This is indeed an important feature to alleviate the design process. When applying the principles of connectionism, though, this is not easy to realise. Connectionism approaches, on the other hand, as opposed to the explicitness of symbolism, directly deal with semantics by way of semantic coding. In the first place, the meaning of something is what is represented, also under conditions of uncertain information. Neural structures lend themselves to semantic coding just by unique activation patterns. Thereby, they can make use of the capability of *generalisation.* In spite of these different properties, symbolism and connectionism have not to exclude themselves. This is crucially important to be realised. The human brain is a good example for how the cognitive process can benefit from combining both symbolic and connectionistic processing in one design (see Chapter 3.2.1.4), although the human system is solely implemented on the basis of neural structures, regarding the underlying low-level cognitive functions. This was established for many good reasons in the course of evolution as part of the basic general design of human physiology.

This illustrates that purity in design may have its drawbacks regarding the functional design of knowledge representation. As a result, agent systems may benefit from combinations of approaches, although this will possibly end up in a higher degree of complexity, sometimes possibly with too great an expense. If it is not possible for any reason to combine different ACUs, each of them working under different representational schemes and pertinent architectures, design purity might be still a good option, though. This is the case in nature to some extent, too.

To be a little bit more concrete on possible means of knowledge representation, in Table 9, thereby following [Ruckdeschel, 1997], a choice of representation techniques is shown which are often used in work system components for vehicle guidance and control, whether for ACUs as operation-supporting means or as part of the operating force. These techniques are originated by disciplines like

- Control theory
- Logic
- Modelling of distributed systems
- Artificial intelligence
- Learning from (experimental) data
- Software engineering, and
- Stochastic processes,

and they are classified by way of these disciplines.

Table 9 A selection of the representation methods used in vehicle guidance work systems (Legend: "x" stands for "feature holds true", " –" stands for "feature does not hold true", and "blank" stands for "feature is not suited in that classification") (cf. [Ruckdeschel, 1997])

Disciplines	Methods	Features												Potentials for cognitive sub-functions		
		Quasilinear	Nonlinear	Time discrete	Continuous	Sequential	Parallel	Crisp	Fuzzy	Certain	Uncertain	Procedural	Declarative	Skill-based	Procedure-based	Concept-based
Control theory	*Quasilinear control*	x			x	–	–	x		x		x		x		
	Nonlinear control		x	x		–	–	-/x	x/-	x/-	-/x	x		x	x	
Formal grammars		–	–	x		x		x		x		x		x	x	
Finite automata		–	–	x		x		x		x		x			x	
Logic	*Propositional and predicate logic*	–	–	–	–	–	–	x		x			x		(x)	
	Temporal logic	–	–	–	–	x	x	x	x	x	x		x		x	
Modelling of distributed systems	*Petri nets*			x			x	x		x		x			x	
	Process algebras/Parallel languages			x		x	x	x		x		x			(x)	
	Statecharts			x		x	x	x		x		x			x	
„Artificial Intelligence"	*Frames/Semantic nets*	–	–	–	–	–	–	x	x	x	x	x	x		x	x
	Production systems	–	–	x		x		x		x		x		x	x	x
Learning from experimental data	*Neural nets,*		x	x					x		x			x	(x)	
	Fuzzy modelling		x	x		–	–		x		x	x		x	x	
	Statistical modelling techniques		x	x					x		x			x	x	
Software engineering	*Decomposition diagrams*	–	–	–	–	–	–	x		x			x		(x)	
	Decision tables/trees	–	–	x		x		x		x		x			x	
	Data flow diagrams	–	–	–	–	–	–	–	–	–	–	–	x		(x)	
	Structural analysis/RT	–	–	x		x	x	x		x		x			(x)	
Stochastic processes	*Markoff chains*	–	–	x		x		x		x		x		x	x	
	Stochastic Petri nets	–	–	x			x		x	x		x			x	
	Queues	–	–	x		x						x		x		

Significant features pertinent to the representation methods in Table 9 are marked, although there might be some vagueness in the feature membership. Anyway, this can be a first guide when looking for candidates of representation or combinations of them. For instance combining fuzzy logic and neural nets had become very popular at times in the soft computing community.

In the following, a selection of approaches for the knowledge representation is discussed in more detail. Thereby, it is referred to the distinction made with respect to human cognition about *explicit* and *implicit* knowledge.

6.1.2 Management of Explicit Knowledge

By far most endeavours to aquire, represent, and process knowledge subject to artificial cognitive systems were focused on explicit knowledge, so far. This is probably because the representation is easily readable by the designer. It can be built up by starting with simple facts and constantly supplementing these by additions as far as it is needed in order to achieve a sufficiently satisfying quantitative model of any object to be represented. Although the demands on artificial cognition have come to a point that the value of representing knowledge

implicitly becomes more and more appreciated the representation and processing of explicit knowledge is still that what is mostly applied in actual developments. Therefore, some of the best known approaches for the management of explicit knowledge are being briefly presented in the following. **The selection is by far not a comprehensive one, but it sufficiently serves the purpose to get a feeling that enough experience is available to accomplish the development task of knowledge representation for a given case of ACU design to significantly enhance the performance of the work system concerned.** Most of these approaches are of the kind describing the semantics by symbolic terms only. A couple of them, however, are referring to the issue of semantic coding, too.

6.1.2.1 Some Milestones Concerning Management of Explicit Knowledge

Quantitative models of the human cognitive functions, which determine human behaviour in a work process, are of particular interest for the work system designer. The application of these models is twofold. At first, it can be made use of these models as background knowledge when designing assisting ACUs for the human operator which are required to co-operate on a cognitive level and act in the right manner at the right time and place when needed, not interfering with the tasks and actions of the human operator. Secondly, these models can be incorporated in ACUs to simply let them behave in a human-like manner, possibly adapted even to the traits of the individual operator concerned. They also can be used for analysis purposes, for instance for the analysis or assessment of the human performance when operating in a work system. In summary, the capacity of these models is crucial for the design success. A great number of quantitative models of the human cognitive functions and resulting behaviour have been developed in the past, mostly for certain sub-functions in much confined worlds and work domains. In the following we shall briefly allude to some of the most outstanding contributions of that kind in the past like those of [Petri, 1962], [Minsky, 1975], [Newell & Simon, 1972], [Newell, 1990], [Anderson, 1983]. A different architecture, but similar to Newell's and Anderson's work, was EPIC (Executive-Process/Interactive Control) in [Meyer & Kieras, 1997]. [Rumelhart & Mc Clelland, 1986] stands for the best known work on the connectionistic approach, and the Fuzzy logic approach was proposed at first by [Zadeh, 1975]. Amazingly, most of them have come into being within a rather short period of time. About all current designs of intelligent agents in general and ACUs in particular are somehow inspired by the earlier fundamental developments of that kind. Although there have been much more similar endeavours which deserve great appreciation, too, we have to leave it to other dedicated surveys which are specially devoted to the historical background in this field. As to the application background we are addressing in this book, a simulation study of [Döring, 1983] should not left unmentioned as one of the first more elaborated implementations of a procedure-based pilot behaviour model. By far most of these models put emphasis on behaviour models and the representation of the pertinent explicit knowledge. Some developments, though, are also expanding the represented knowledge on concept-based behaviour. Referring to Figure 27, this cannot be all, though, since we also need quantitative models for skill-based behaviour and

corresponding representation of implicit knowledge. As compared to those of procedure-based models these are still very rare. Most of them are only very recent developments like for instance those of [Feraric, 1996], [Grashey, 1998] and [von Garrel, 2003] (see Chapter 6.2.1). Before getting to these developments, let us first discuss one by one in the following some of the earlier milestone developments of explicit knowledge management by means of quantitative modelling, starting with Minsky's frame system approach:

Frames

The idea about representing explicit knowledge in frames is intimately related to [Minsky, 1975] with the title: A framework for representing knowledge. A frame is a data structure which represents a "stereotyped situation, like being in a certain kind of living room, or going to a child's birthday party. Attached to each frame are several kinds of information. Some of this information is about how to use the frame. Some is about what one can expect to happen next. Some is about what to do if these expectations are not confirmed.

We can think of a frame as a network of nodes and relations. The "top levels" of a frame are fixed, and represent things that are always true about the supposed situation. The lower levels have many terminals – "slots" that must be filled with specific instances of data. Each terminal can specify conditions its assignments must meet. (The assignments themselves are usually smaller "sub-frames.") Simple conditions are specified by markers that might require a terminal assignment to be a person, an object of sufficient value, or a pointer to a sub-frame of a certain type. More complex conditions can specify relations among the things assigned to several terminals."

In essence, [Minsky, 1975] is dealing with " … a remembered framework to be adapted to fit reality by changing details as necessary." Thus, a frame rather stands for a category of real world entities than only for the uniquely featured one of that category which has just been experienced.

Undoubtfully, this representation language from the early days of AI is a symbolic one with all its advantages and disadvantages, but it apparently already accounted for the necessity of a generalisation potential of the representation language.

No question, the frame idea has inspired about all following developments of symbolic knowledge representation. Ensuing his publication, there have been implemented a great number of versions of frame systems, although these days it is no longer as such a serious candidate for knowledge representation.

State, operator, and result (Soar)

Newell's work probably marks the starting point of model development with reasonable value for application in work system design. At that time there have been also other rather popular frameworks distinct from that of production systems like the schema architectures [Schank & Abelson, 1977]. However, none of those achieved the degree of comprehensiveness and the potential of psychological interpretation as it has become the case based on the production system framework, since rules apparently play a major for the central cognitive behaviour. With the background from computer science, in particular by use of

techniques of artificial intelligence for knowledge representation, Newell and Simon found the way to computer modelling of human procedure-based behaviour based on a production system framework (first proposed in [Post, 1943]). This became further advanced and merged into an architecture called *Soar* [Laird et al., 1987] and [Newell, 1990]. Soar already repesents an agent architecture. The operative system consists of a so-called *working memory* (in imitation of the working memory as described by [Baddeley, 1990]) (see also Chapter3.2.1.4), a so-called *long-term memory* which contains the productions, and a *control mechanism* to make sure that the appropriate productions will be selected.

As already mentioned, the production constitutes a rule, i.e. a condition-action pair. The condition specifies a data pattern. If there is a situation with elements in the working memory that match the pattern, then the production can apply and the action will be executed. The content of the working memory is represented by a context graph (see Figure 91). The working memory elements have the form of a triplet (**object ^attribute value**). **object** stands for the name of an object like for instance the object A4 or the relation object R1. **attribute** represents an attribute (callsigns, doors, distance, etc.) of the object as named in the triplet, and **value** means the value of the attribute which can be a symbolic constant, a number, a string or also an object. Thus, objects and relations of objects can be represented this way.

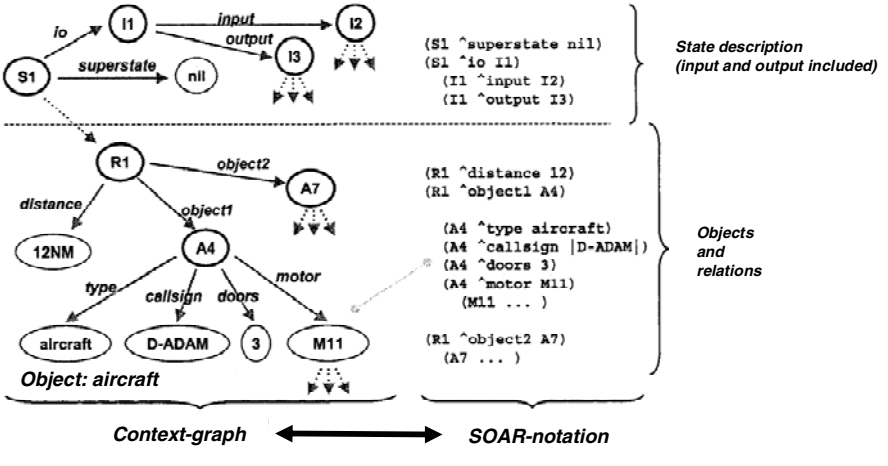


Fig. 91 Representation of objects and relations in Soar [Putzer, 2004]

Regarding the situational context as depicted in Figure 91, this might be a snapshot of the content of working memory, after two rules (productions) have been fired. There might be a first rule whose condition is testing on two objects which have the attribute ^type and value aircraft. If these objects are found in the working memory, the action part of the rule generates the relation R1 with two edges each of which point to one of the aircraft objects as found in working

memory beforehand. In a next step another rule might have matched this relation with the working element elements of the aircraft's positions. The action part of this second rule, then might determine the distance between the two aircraft from the position data and extends the relation by the attribute ^distance 12nm.

The cyclic knowledge processing is depicted in Figure 92. Usually, a number of rules are packed into an operator. Operators, in turn, might be structured in hierarchies where a number of operators are forming an upper-level operator. With each cycle certain operators are proposed and one of them is selected by the control mechanism which is making use of some kind of background knowledge. Once selected it is applied by firing all pertinent rules simultaneously. Thereby, the system state is transformed by operators into a new subsequent state. This goes on continuously. The input and output in Figure 92 stand for the interface with the environment by means of sensors and effectors.

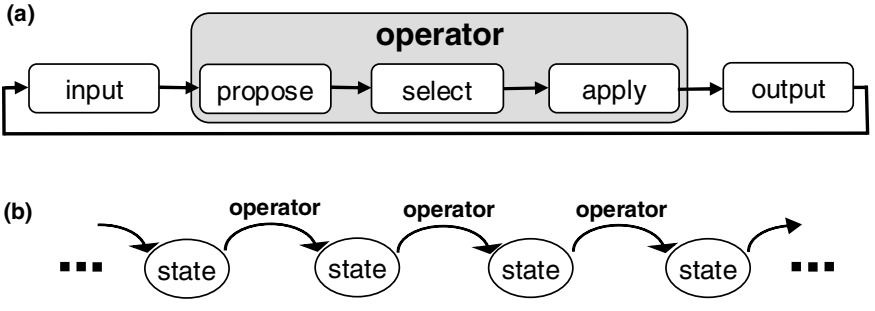


Fig. 92 Cyclic knowledge processing in Soar

Soar as the implementation of what is described in [Newell, 1990] as *unified theory of cognition*, is deliberately confined to symbolic knowledge representation. The first major application was the modelling of human problem-solving [Newell, Simon, 1972], which is best suited to that type of representation as a cognitive function being considered as part of the central executor function in human cognition (see Chapter 3.2.1.4). Although this application was very successful as others too in the procedure-based level of cognitive system behaviour, this is only a part of human cognition as a whole as we know. Anyway, Soar has proved a valid cognitive processing machine as long as rule-based systems are concerned.

Adaptive Control and Thought (ACT)

Quantitative models of the human cognitive functions subject to certain behaviour are of particular interest for the analysis or assessment of the human performance when behaving in the working environment. This is the information needed to design technical means which are appropriate to enhance the work process. Quantitative models of the human cognitive functions and resulting behaviour have been developed, each for certain sub-functions. Those dealing with the procedure-based and concept-based behavioural level, e.g. the developments of [Newell & Simon, 1972], [Newell, 1990] and [Anderson, 1983] are the best

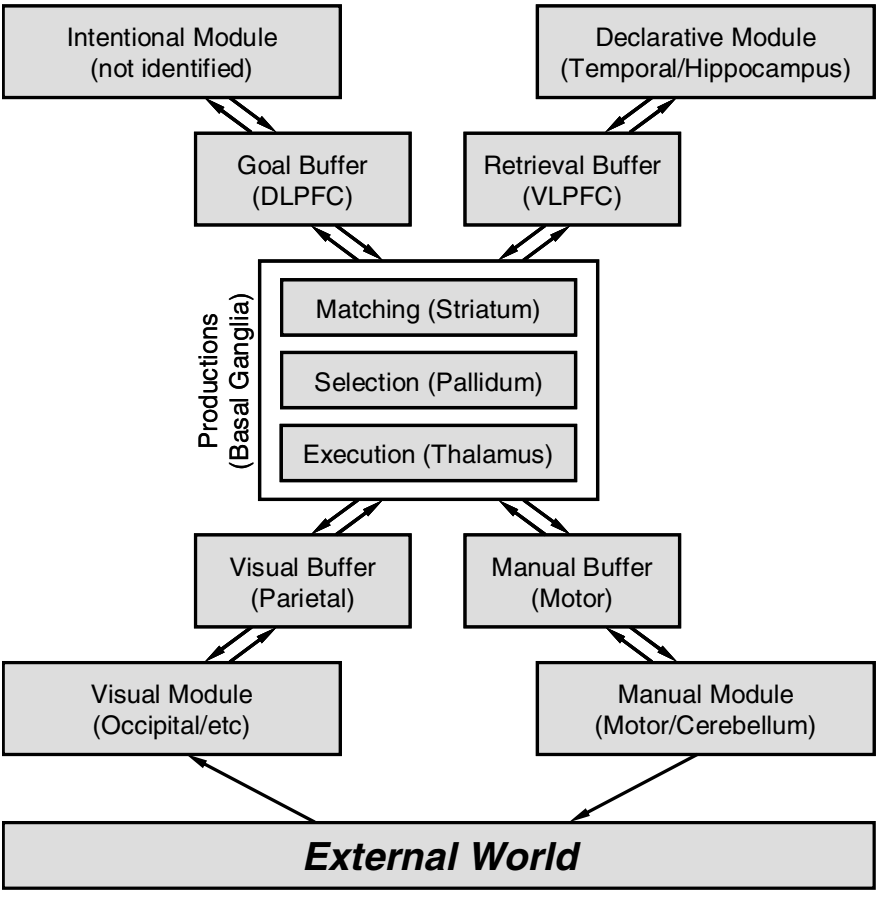


Fig. 93 The organization of information in ACT-R 5.0. Information in the buffers (short-term memory) associated with modules is responded to and changed by production rules. (Legend: DLPFC = dorsolateral prefrontal cortex, VLPFC = ventrolateral prefrontal cortex) [Anderson et al., 2004]

known. There are others, too, though. Some of them will also be briefly described in the following. Procedure-based models have also been expanded to concept-based behaviour.

Anderson also started in the seventies with somewhat similar to Newell's ideas. He developed the ACT (Adaptive Control and Thought) system resulting in a number of subsequent derivatives of the system over the years. An intermediate state is described in [Anderson, 1983]. Later, Anderson arrived at a status of probably the most refined model of human procedure-based and concept-based behaviour, also including learning [Anderson et al., 2004]. Since it seems to be the most advanced and comprehensive model so far with a psychology background, ACT will be described at some more length in the following. ACT is modelling

both the high level and medium level cognitive functions. At this point, there will be most emphasis on the high level cognitive functions.

In [Anderson et al., 2004], one of the latest publications, the architecture ACT-R (**A**daptive **C**ontrol and **T**hought - **R**ational) is proposed which will be described in some detail in the following. This work accomplishes with a high degree of validity to portray the way human cognition works. The strength of this theory is the endeavour to account for typical traits and facets of human cognition which are well known by the psychologists on a phenomenal basis. The basic architecture is shown in Figure 93, which shows some similarity with Rasmussen's model and its interpretation, here for the sake of lucidity only highlighting perceptual-motor functions, the visual perception and the manual motor modalities for controlling the hands. There are also some cues in the figure about which parts of the brain are involved, thereby referring to work in modern neuroscience results. It is tried that each of the components is contributing to an integrated architecture of cognition.

Besides the visual and the manual module (see Figure 93) representing the interface to the external world, ACT-R comprises a module to retrieve information from explicit long-term memories, a goal module, and a central production system which co-ordinates the behaviour of the modules mentioned. Distinct from the explicit memory, the implicit memory of productions is not explicitly depicted. It is implicit part of the production system. The various buffers of the modules act as working memory. They hold the limited information that the production system can respond to:

> *"The contents of the visual buffer represent the products of complex processes of the visual perception and attention systems. Similarly, the contents of the retrieval buffer are determined by complex memory processes. ACT-R includes a theory of how these buffers interact to determine" human "cognition. The basal ganglia … and associated connections are thought to implement production rules. The cortical areas corresponding to these buffers project to the striatum, part of the basal ganglia, which we hypothesize performs a pattern-recognition function (in line with proposals ; e.g. [Amos, 2000; Frank, Loughry, & O'Reilly, 2000; Houk & Wise, 1995, Wise, Murray, & Gerfen, 1996]). This portion of the basal ganglia projects to a number of small regions known collectively as the pallidum, serving a conflict resolution function. The projections to the pallidum are substantially inhibitory, and these regions in turn inhibit cells in the thalamus, which projects to select production actions in the cortex. The organization of the brain into segregated cortico-striatal-thalamic loops is consistent with the described hypothesized functional specialization." In the architecture shown in figure 3.1* (see Figure 93 in this book)*, it is also not accounted for the fact that not all procedural processing has to involve the basal ganglia, since there are also*

> *direct connections between cortical brain areas. In order to account for this, it would be another extension of ACT-R.*
>
> *When running ACT-R, within a processing cycle, which takes about 50 ms, "buffers hold representations determined by the external world and internal modules, patterns in these buffers are recognized, a production fires, and the buffers are then updated for another cycle." [Anderson et al., 2004]*

The visual module separates into the visual-location sub-module to model the anatomically neural dorsal where system and the visual-object sub-module to model the ventral what system. The EMMA (Eye Movements and Movement of Attention) system of [Salucci, 2001] uses ACT-R to provide a more detailed theory of visual encoding, modelling eye movement control in reading. Regarding the perceptual-motor system, ACT-R takes into account how the different perceptual-motor modalities work together in parallel tasks [Byrne & Anderson, 2001].

The goal module keeps track of the steps of actions and partial results in order to warrant that the behaviour will serve the pre-specified goals. This corresponds to the problem-solving task and thereby, might be extended to that what is meant by concept-based behaviour in the qualitative Rasmussen model. It seems that the generation of the goals is not considered, though.

The explicit memory module represents a typical medium-level function. It forms the central part of ACT-R together with the production system (procedural system) which appears as a combined high-level function with production matching, selection, and execution. By the explicit memory module, ACT-R executes a number of formulas like for activating a chunk in the explicit memory, for probability and latency of retrieval, and for production utility based on a computed probability of production success. The latter is used for learning of appropriate use of productions in certain contexts. ACT-R might even learn new productions in that manner. This is reflecting known phenomena of changes of human behaviour in a given context.

As mentioned earlier, ACT is not only dealing with procedure-based but also with concept-based behaviour like it is defined by Rasmussen. ACT models inferential processes in problem solving. Productions to generate an inference are acquired to encode the type of contingencies that have tended to happen in past experience. It is tried to model the human way of reasoning, which means that the inference needs not be complete and sound in the sense of mathematical logic.

Therefore, usually humans are not as good in generating the inference as in recognising whether the inference is valid. ACT implements rules for recognition of correctness. This might be useful to check our own inference, before applying it.

Real-world applications of ACT-R have been proposed in training and education [Anderson, 2002], human-computer interaction [Byrne, 2003], and synthetic agents [Freed, 2000].

Figure 93 has given evidence about the crucial role of knowledge in the human cognitive process. Each functional block in this abstract model can be defined with its own knowledge base of a-priori explicit and implicit knowledge. The accumulated knowledge is the true enabling factor. A central issue in modelling

human cognitive behaviour is therefore to model the enabling medium-level functions for knowledge encoding, representation, retrieval, and retention. For the discussion of these medium-level functions we refer to ACT* [Anderson, 1983]. ACT* describes human cognition based on the following assumptions, thereby mainly regarding medium-level functions about making available the knowledge for the mainstream knowledge processing. These assumptions are basic for the ACT theory and, in essence, are also the assumptions of the subsequent versions of ACT.

Assumptions of ACT* (quoted from [Anderson, 1983])

1. Technical time assumption: Time is continuous.
2. Basic architectural assumption: There is a production system as a component which operates on the explicit knowledge representation.
3. Explicit representation: Explicit knowledge can be composed into a tangled hierarchy of cognitive units. Each cognitive unit consists of a set of no more than five elements in a specified relation.
4. Activation of explicit memory: At any time t, any cognitive unit or element i has a nonnegative level of activation $a_i(t)$ associated with it.
5. Strength in explicit memory: Each memory node i (cognitive unit or element) has a strength s_i. The relative strength r_{ij} of a link between node i and node j is defined as $s_j/\Sigma_k s_k$ where the summation is over nodes connected to i.
6. Spread of activation: The change in activation at a node i is described by the following differential equation
   $da_i(t)/dt = Bn_i(t) - p*a_i(t)$
   where $n_i(t)$ is the input to the node i at time t and is defined as
   $n_i(t) = c_i(t) + \Sigma_j r_{ji} a_j(t)$
   where r_{ji} is the relative strength of connection from node j to i and $c_i(t)$ is zero unless i is a source node. If i is a source node, $c_i(t)$ is a function of the strength of i.
7. Maintenance of activation: Each element that enters into working memory is a source of activation for Δt time units. There is a single goal element which can serve as a permanent source of activation.
8. Decay of activation: ACT*'s assumptions about decay are already implied in 6 and 7.
9. Structure of productions: All productions are condition-action pairs. The condition specifies a conjunction of features that must be true of explicit memory. The action specifies a set of temporary structures to be added to memory.
10. Storage of temporary structures: When a temporary cognitive unit is created and there is not a permanent copy of it, there is probability p that a permanent copy will be created. If there is a permanent copy, its strength will be increased one unit.

11. Strength of productions: Each production has a strength associated with it. That strength increases one unit with every successful application of the production.
12. Production selection: When the condition of a production achieves a satisfactory match to a set of explicit structures, the production is selected to apply. The pattern matcher is represented as a data-flow network of (sub)pattern tests. The rate at which these tests are performed is a function of level of activation of the pattern node that performs the tests. The level of activation of that node is a positive function of the strength of the node, the level of activation of the data structures being matched, and the degree of match to these structures. It is a negative function of the level of activation of competing patterns matching to the same data.
13. Goal-directed processing: Productions can specify a goal in their condition. If their goal specification matches the current goal, these productions are given special precedence over productions that do not.
14. Production compilation: New productions can be created from the trace of production application. *Composition* collapses a sequence of productions into a new one. *Proceduralization* builds new productions that eliminate the long-term memory retrievals of old ones.
15. Production tuning: New productions can be created by generalizing or discriminating the conditions of existing productions. *Generalization* works from pairs of more specific productions to create more general ones. *Discrimination* works from feedback about erroneous production application to create more specific productions that avoid the mistakes.

It might be useful to have some further remarks on these assumptions:

The activation process defines the current contents of the working memory. Working memory refers to explicit knowledge that is in an active state. Productions can match only the knowledge that is currently active. The most active one will be preferred in a matching process, if there are matched more structures than one. The activation level specifies the likelihood that a particular piece of knowledge, a cognitive unit (or chunk), will be useful in a particular moment. If two structures in explicit memory might be matched to the same input pattern, the more active one will be preferred. The activation process helps to avoid ambiguity.

Activation is spread in the explicit memory network from certain source nodes which are created by environmental encoding or elements deposited by a production's action. Each source node supports a pattern of activation over the network. The total activation is the sum of the patterns pertinent to the source nodes. The momentary change of activation of a node is a function of the input n to the node, controlled by the parameter B, and a spontaneous rate of decay at the node. This rate is controlled by the parameter p*. Input n is possible source

activation plus the sum of activation from associated nodes weighted by relative strength.

Temporary cognitive units enter working memory through the encoding process by placing descriptions of the environment in working memory or through actions of productions to record the results of internal computations. Encoding refers to the process by which cognitive units become permanent long-term memory traces. There is a probability then, that a copy of the temporary cognitive unit will be created in the explicit long-term memory. Storage processes in explicit memory act on whole cognitive units in the representational form of strings, images, and propositions.

Production selection is performed in rather unlimited parallel manner. That is in good compliance with the neuro-physiological fact of parallel processing in the human brain. Otherwise, there would be a serious computation time problem in testing the total of thousands of productions in long-term memory against the content of short-term memory. Production selection also allows that the pattern in short-term memory is only partially matched within certain defined limits. If no complete match can be accomplished, the production with the condition of best interpretation of the available data is chosen, provided the match is sufficiently close to complete. This is also a prerequisite for modelling typical human behavioural performance degradations because of partial matching, for instance, if we have difficulty to find the appropriate word and use another one which is not quite correct but sufficiently related to the right one.

Production tuning by the generalization process constitutes a gradual procedural learning mechanism as opposed to the direct learning of explicit knowledge. Production tuning in this way is similar to what is described in [Rasmussen, 1983] by the behaviour level downshift resulting from repetitive practise from knowledge-based (concept-based) to rule-based (procedure-based) behaviour.

ACT* proposes a so-called tri-code knowledge representation for explicit knowledge with three representational types: temporal strings, which encode the order of a set of items, spatial images encoding spatial configuration, and abstract propositions. These representational types are defined in terms of the processes that operate upon them like encoding, storing, retrieval, matching, and constructing new structures based on old ones, for instance by combination, deletion or insertion of elements. The production system framework is easing to communicate among these three representational types. Conditions and actions can specify different types. Because of limits on how much can be encoded in one cognitive unit, large structures are encoded hierarchically. For a unit to be encoded into long-term memory all of the elements must be in working memory and the system must be able to address each element.

The representation of the procedural knowledge in ACT is given by the production system resulting in a production knowledge base. Productions are somewhat simplified compared to Newell's production system, omitting such things as control nodes and global variables. Both the explicit knowledge and the production knowledge base constitute distinct parts of the long-term memory and form together the basis of cognitive behaviour. There are certain representational

features in structuring the production knowledge base like associative networking in order to alleviate the matching process.

Another interesting aspect, however, is the featuring of procedural learning. Productions are not – i.e. cannot be – directly learned like explicit knowledge. Production acquisition takes place by a kind of adaptation process. ACT supports the following distinct ways of forming new procedural knowledge based on already existing ones:

- Procedures to interpret explicit declarative information
- Production compilation
- Production tuning

At first, explicit declarative information can be used in several ways to guide behaviour, for instance in the form of directly presented instructions to be followed step by step, by using known general problem-solving procedures to interpret explicit knowledge in a very flexible but resource consuming way, or simply by exploiting analogy.

The development of procedures through the interpretation of explicit knowledge, as mentioned, is usually preceding both production compilation and production tuning.

The process of production compilation can be a process of composition or, if going further, of proceduralisation. Production composition reformulates a number of subsequently used productions in a single one in order to make the procedure more time-effective because of less load on the working memory. In order to achieve further reduction of working memory load, the process of proceduralisation leads to specialisations of productions by specifying the production condition and action in terms of concrete values instead of variables. Thereby, retrieval operations in explicit memory will be saved. This makes sense, if this production will mostly be used in this specialised version.

Searching of the problem space becomes more selective with increased problem solving experience in a work domain. The corresponding change of the underlying productions is called production tuning. One way is to broaden the range of applicability of productions in order to have a better chance to make the problem solving process more effective in work situations having never encountered before. This corresponds to the method of generalisation. One obvious way of generalisation is the application of variables instead of constants. Also changes of the structure with use of general constraints might lead to a similar effect. On the other hand, restriction of the range of application might be appropriate for certain contexts after experiencing both incorrect and correct applications. This process of looking for more specific productions well-targeted to a certain application context in the work process is called discrimination. The application-adaptive learning process has to come up with the appropriate balance between generalisation and discrimination.

Petri nets

State transition diagrams are being used in design work for rule bases to represent *discrete processes*. The simplest version is the *Finite state machine*. Petri nets, though, as proposed by [Petri, 1962], are a much more elaborate version of state

transition diagrams which represent distributed systems. They lend themselves to represent concurrent processes and to account for normative preconditions of sequencing situation-dependent actions (transitions). All Petri nets are characterized by two kinds of net elements and the relations between them. One of them stands for situations and the other for events. The situations can be interpreted as states, conditions, or properties of objects, often designated as *S-elements*. They are represented in the net graph as circles. The events can be interpreted as state transitions, events, or operations, often designated as *T-elements*. They are represented in the net graph as rectangles.

A general (more formal) definition of Petri nets is the following (see [Desel & Juhas, 2001]:

The Petri net structure is a four-tuple (S, T, F, M_0)

- S is a set of S-elements
- T is a set of T-elements
- F is a set of arcs known as flow relation. The set F is subject to the constraint that no arc may connect two situations or two events, or more formally: F is a subset of (S x T) unified with (T x S)
- M_0 is an initial marking of the possible S-elements in the net by means of a certain number of tokens

We are referring to two basic kinds of Petri nets, the

- *condition/event net* and
- *place/transition net.*

The condition/event nets constitute the basic class of Petri nets. The S-elements of these nets are called conditions and the T-elements are called events. The marking of the S-elements can only be of a binary type which indicates whether the condition is met or is not met. An event is enabled, if all input conditions are met and all output conditions are not met. Along with the occurrence of an event the markings of the input conditions are removed and all output conditions are marked. This is shown in Figure 94.

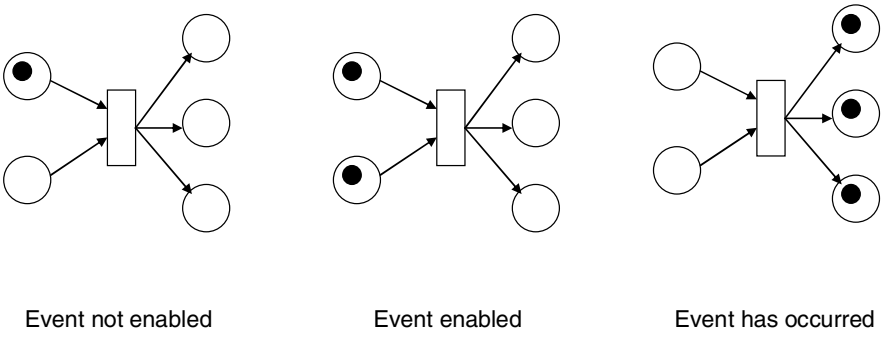


Fig. 94 Switching from input to output conditions because of an enabled event in a condition/event net

Figure 94 also suggests one of the most important characteristics of Petri nets that they can represent concurrency of discrete processes, for instance, if there are arcs from the output conditions which have arcs to events in parallel.

The place/transition nets are an extension of the condition/event nets. The S-elements of these nets are called places and the T-elements are called transitions. The marking is no more a binary one. The number of markings of a place is only limited by the marking capacity of the place. Each arc is labelled by a so-called weight which indicates how many markings will be transferred when the transition fires. In that sense the place/transition net is defined by the 5-tuple (*S, T, F, W, M_0*), whereby *W* is the set of arc weights. In case the transitition is activated (fires), markings are removed from the input places according to the associated input arc weights and markings are added to the output places, again according to the associated output arc weights. An example of transition firing is shown in Figure 95.

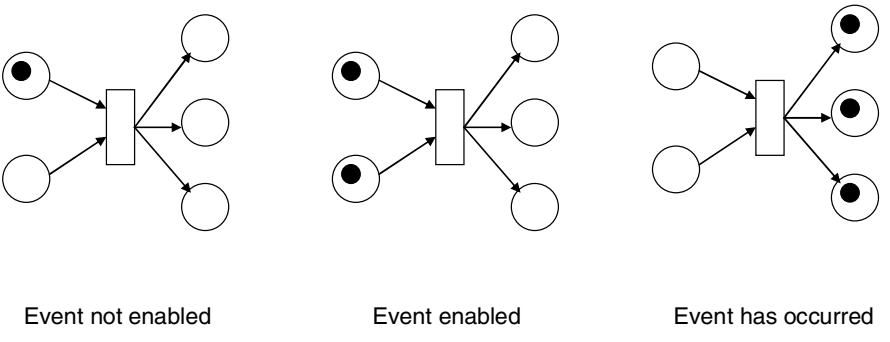


Fig. 95 Switching from input to output places concerning one transition in a place/transition net

Petri nets have specific properties which can be exploited by designers for analysis purposes. This is very useful to warrant errorfree nets. At first, the graphical representation is very comprehensible, although it can be very extensive. It also provides a representation of order of succession of situations and events, also in case of concurrent processes. There are graphical tools available. Figure 96 gives an example for a model of driver-copilot interaction.

An often used analysis technique is the *reachability* test. All states (markings), which can emerge from any initial marking M_0 in the any course of transition switching through the net, are called reachable and form a set of reachability. If this set is finite, it can be represented in the so-called reachability graph. Each node of the graph contains a complete marking state of the net. Arcs in the reachability graph represent transitions from one state to the following in the course of the underlying process.

In computer programming and other applcations, the property of *liveliness* of a Petri net is used to model that the system can never lock up. The property of

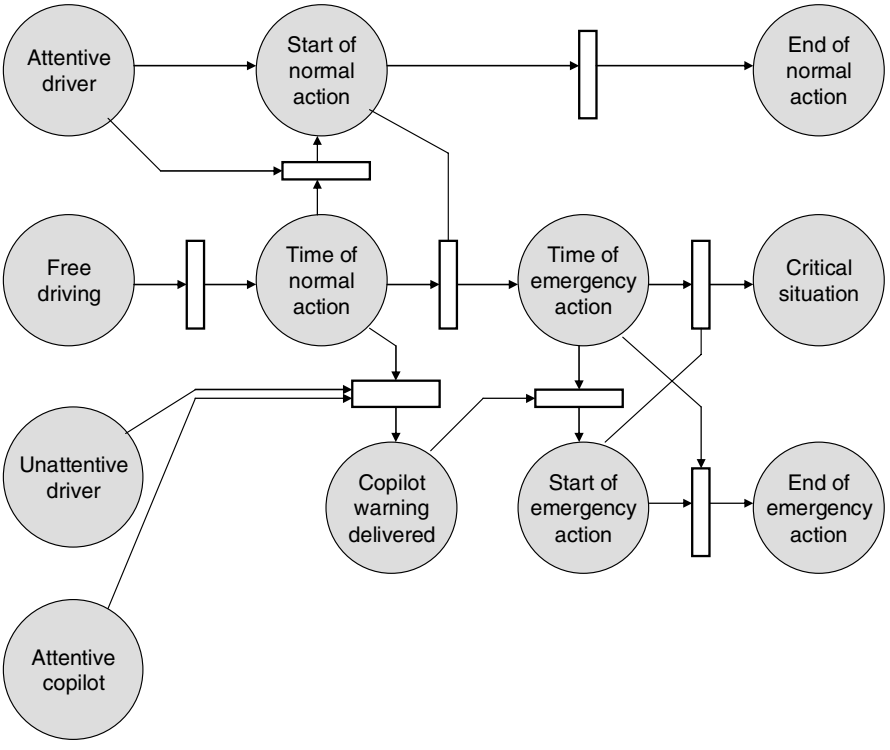


Fig. 96 Petri net for a model of driver-copilot interaction (cf. [Blum et al., 1989])

boundedness of a Petri net describes that the number of reachable markings at any place will not go beyond a given maximum. A net is k-bounded, if the maximum number of reachable markings is k. A Petri net is called *safe*, if it is 1-bounded. For other possible properties of Petri nets we refer to the literature, for instance ([Peterson, 1981], [Reisig, 1992]).

There are many extensions to the original Petri net [Russel & Norvig, 1994]. One particular extension is focused on modelling the human operator behaviour by a process to adapt to his individual traits [Stütz, 2000]. It should be taken into account, though, that the extension of properties usually makes it harder for the designer to use standard tools.

There is another valid alternative to the Petri nets for many applications, the so-called *statecharts* [Harel, 1987]. They are based on finite state machines, extending them by properties like providing hierarchical and concurrent structures. Statecharts are well introduced into designs of consumer products and can be applied by using commercial tools (e.g. Statemate [Harel et al., 1990]).

Fuzzy systems

Fuzzy systems belong to the category of structured model-free estimators and are part of the low-level cognitive functions in the functional framework (see Figure 30). The term “fuzziness” was introduced in the paper [Zadeh, 1965] on “fuzzy

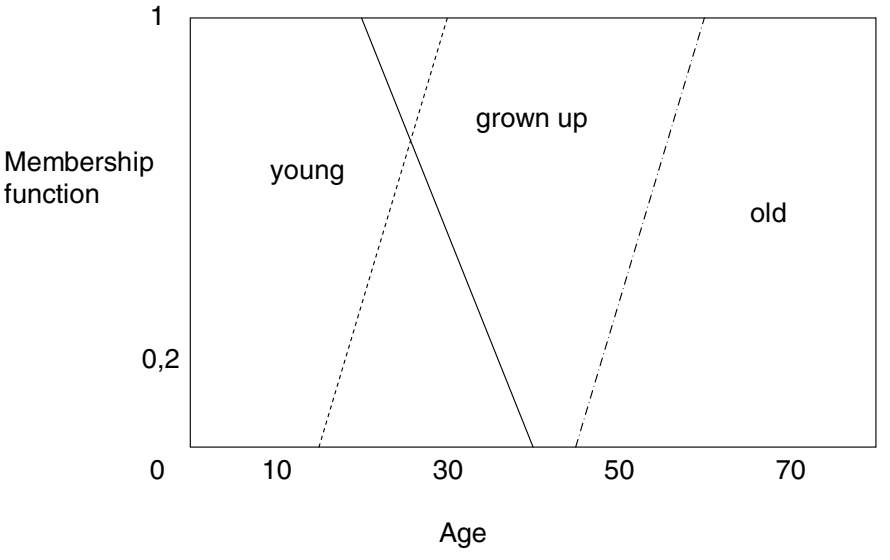


Fig. 97 Fuzzy sets for different phases of age

sets". Instead of the bivalent *indicator function* of a nonfuzzy set in the classic set theory of 1 or 0, indicating whether element x is part of the set or not, [Zadeh, 1965] defines the so-called *membership function* which can take values between 0 an 1. The membership value mü indicates the degree to which element x belongs to a set *S*. Similarly, a so-called subsethood can also be measured by values between 0 and 1. When being operationalised fuzzy sets linguistic variables are favourably used which corresponds to the fuzzyness of human language. An example of fuzzy sets is given in Figure 97, from which the membership of a person to sets for different categories of age can be derived.

Relations between membership functions can be dealt with by logical operators similar to those in classical logic. The operators Fuzzy-OR, -AND, and -NOT can be operationalised in the following way:

The membership function $\mu(A \cup B)$ of the union of the fuzzy sets A and B for all elements x of the set X of values of a fuzzy variable is:

$$\mu(x)(A \cup B) = \mu(x)\,A\ \mathrm{OR}\ \mu(x)\,B = \mathrm{MAX}\left(\mu(x)\,A, \mu(x)\,B\right).$$

The membership function $\mu(x)(A \cap B)$ of the intersection of the fuzzy sets A and B for all elements x of the set X of values of a fuzzy variable is:

$$\mu(x)(A \cap B) = \mu A(x)\ \mathrm{AND}\ \mu B(x) = \mathrm{MIN}\left(\mu A(x), \mu B(x)\right)$$

and the membership function $\mu(x)(\overline{A})$ of the complement of a fuzzy set A for all elements x of the set X of values of a fuzzy variable is:

$$\mu(x)(\overline{A}) = 1 - \mu(x)(A) \ .$$

This will be illustrated in the following by a simple example for a fuzzy rule which is part of the error detection module in the CAMA system. During landing approach a more serious error occurs, if the pilot does not recognize that the aircraft becomes too high and too fast. The rule says, if it is verified that the aircraft altitude is too high and the speed is too high, too, then there is an error which has quickly to be worked on. For the sake of simplicity, we will only concentrate here on simultaneous deviations in both altitude and speed from the appropriate ones and neglect errors which might occur in only one of the two variables speed or altitude. Anyway, this kind of error usually can be compensated for rather easily. Figure 98 shows the following:

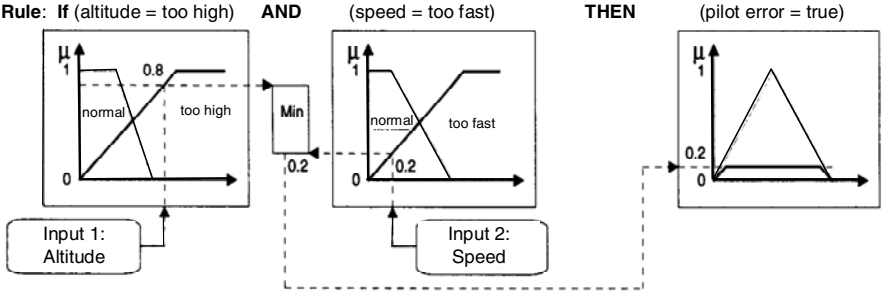


Fig. 98 Example of a simple Fuzzy-AND relation in a fuzzy rule

There are two queries in respective fuzzy set diagram concerning the variables altitude and speed. The value for the membership function of the input for the aircraft altitude can be read from the fuzzy set diagram for the altitude as 0.8 and the membership value for the speed is read as 0.2 only. Here, we have to use both fuzzy sets in an AND relation in the antecedent of the rule such that we take the minimum of both values which is 0.2 what will not make a significant contribution in the rule consequent.

In Chapter 6.2.3.2 we will refer to fuzzy systems when discussing the knowledge components of the pilot intent recognition module of the CAMA system [Strohal, 1998].

There have been other approaches, too, to deal with uncertain knowledge. Best known is the so-called Bayes theorem [Kosko, 1992]:

$$P(H_i|E) = \frac{P(E/H_i) \cdot P(H_i)}{\sum_{j=1}^{k} P(E/H_j) \cdot P(H_j)}$$

It uses available uncertainty information in the description of uncertain phenomena by expanding the a-posteriori conditional probability $P(H_i|E)$, the

probability that H_i, the ith of k-many disjoint hypotheses $\{H_j\}$, is true given observed evidence E,

A generalisation of the approach of Bayes is the theory of Dempster-Shafer [Shafer, 1976] which deals with sets of hypotheses.

Unfortunately, these mathematically valid theories usually are not of great value in the design process of operational systems, because usually many of the presumptions cannot be met in a satisfying manner.

PDP (Parallel distributed processing)

The PDP approach is the attempt to get as close as possible to that what is known about management of explicit knowledge in the human brain ([Rumelhart & Mc Clelland, 1986] and [Mc Clelland & Rumelhart, 1986]). It is also called a *connectionistic* approach, since it relies on the internetting of a great number of elementary *processing units* and certain network formations of them, corresponding to the internetting of millions of neurons within a brain nucleus and the interconnections within and between the various nuclei in the brain. Thus, similar to human cognition, these units can activate or deactivate eachother through respective exhibitory and inhibitory connections. Each connection represents a "synaptic input weight" (*strength*) of the signal which enters the receiving unit. The weights vary in the course of the learning process until they reach a stable state, thereby complying with what is called the *multiple constraint satisfaction*. A weight determines how much the pertinent input contributes to the *netto input* of a processing unit. The netto input as the sum of the weighted inputs from all incoming connections is transformed by an *activation function* into the output of the unit. The activation function mostly is of a non-linear type.

An individual processing unit stands for a kind of elementary semantic feature which usually cannot be described in explicit terms of natural language. It might exhibit a strong activation when the pertinent elementary feature is one of the features looked for in a retrieval act. More precisely, the activation of a unit encodes the degree of confidence that its preferred elementary feature is present. Explicit declarative knowledge items like a word or an object are represented by an activation pattern of a network formation of processing units which work in parallel. The processing units of a network are classified according to where they can receive inputs from, and how they can influence other units. Therefore, it is distinguished between so-called input units, output units, and hidden units. Input units can receive external inputs, output units can transmit information to the external world, and hidden units exclusively interact with other units of the pertinent network.

As already alluded to above networks of that kind are able to learn by changing the weights of the connections during a training phase. There are certain learning rules which are used for PDP networks. The simplest one is that proposed by [Hebb, 1949]. Hebb has claimed that the strength (weight) of a connection between two units will be amplified to a certain extent, if both units are firing. Among others often used learning rules are the Widrow-Hoff rule and the Generalised Delta rule [Rumelhart & Mc Clelland, 1986]. All these lead to the storage of knowledge in a network by changing the pattern of connection weights.

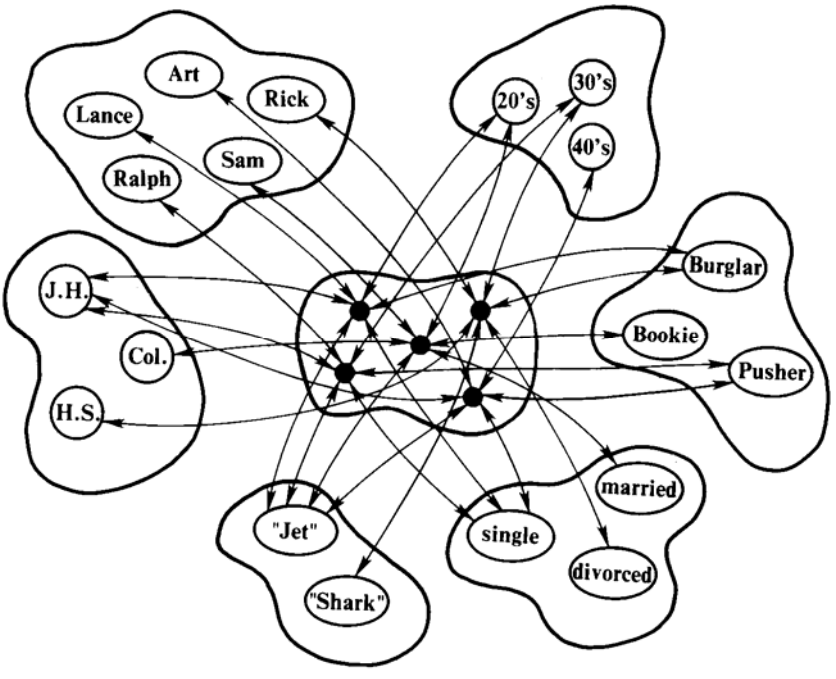


Fig. 99 Associations in a PDP network [Mc Clelland & Rumelhart, 1981]

While exhibiting a single pattern of connection weights, the trained network still may have simultaneously stored a great number of different associations. This may be illustrated by Figure 99 which is taken from [Mc Clelland & Rumelhart, 1981]. Different partial networks represent classes of features of individuals and the individuals. Connections between units within a partial network (not depicted) are inhibitory. If certain features are activated by any input, then the activation is spreading out from there to associated units and networks by means of excitatory connections in terms of a stable state of activation (*attractor*). For instance, if the feature "married" is activated several associations become fired from there, like Sam, Col., Bookie, Jet, and 20's.

An important property of the PDP approach is the fact that it is founded on semantic coding. The activated associations are semanticly related. The internetting and appointed spread of associations also supports that incomplete or partially erroneous input features do not necessarily result in a retrieval failure. However, there are also drawbacks which have to be mentioned. It is often the case that the number of needed training sets becomes unrealistic and that stability of the network behaviour is not warranted. It might happen that certain associations learned are being destroyed because of inferences which might take place in the course of learning of new semantic variants. This can be an indication for a deficiency of the network structure, for instance to be not sufficiently powerful to accommodate the full amount of the knowledge as being specified.

This is possibly caused by the fact that large scale structures like those in the human brain are still not feasible.

6.1.2.2 Semantic Vector Space

More recently models for explicit memory functionality, including knowledge acquisition, representation, and retrieval were being developed, which differ considerably from those used for symbolic knowledge bases, like those of Soar or ACT, and which seem to be at least good candidates for dealing with human language knowledge. As a first example of semantic coding for knowledge representation these models of explicit knowledge are based on a semantic vector representation. For instance, the representation of words, i.e. the meaning of words, is taking place in terms of vectors in a high-dimensional space and thereby is categorising words and text passages semantically and even grammatically, too. These approaches can deal with synonymous words and ambiguity in word meaning. We refer here to HAL (Hyperspace Analogue to Language) (see [Burgess & Lund, 1997]), the work published in [Finch & Chater, 1992], and LSA (Latent Semantic Analysis) as published in [Deerwester et al., 1990] and [Landauer & Dumais, 1997]. All these approaches are independently developed but similar in the sense that they capture explicit knowledge as a semantically coded representation in highly dimensional vector spaces. The LSA approach has been further pursued with vigour and has been successfully applied in several ways in the language domain. Therefore, in the following of this section our focus will be on describing the LSA approach in more detail.

The LSA approach is of the type of performing empirical associations somewhat like human knowledge acquisition (see Chapter 3.2.1.3) as a general purpose analytic device, thereby exhibiting humanlike generalisation characteristics. "Empirical" means that the association inputs are based on vast amounts of observations of real world data, whereas the resulting output is a condensed one of semantically coded representation of the real world in a highly dimensional vector space. In that sense, it is more like a full-scale bottom-up approach as opposed to a top-down one by explicitly describing and representing real world objects feature by feature in symbolic terms from a top view. It becomes immediately obvious that this kind of symbolic representation has to build up a knowledge base with lots of layers of level of detail. Breaking down describing details by these layers can become about endless. Therefore, the symbolic representation is crucially limited in comprehensiveness as opposed to the semantically coded one, never being able to ensure whether really all is captured from what is necessary to be captured in order to get the semantics sufficiently accurate with no ambiguities left. This is also the cause for ontological problems being a typical feature of symbolic representations.

Since most experience with LSA has been gained in the field of verbal semantics, in the following we will stay with this application domain when describing LSA in more detail. [Landauer, 2002] claims while exploring this approach by means of computational simulation which is necessary for validation purposes, that its application for verbal semantics has been easier and more revealing. He has given evidence, though, that this approach LSA is relying on

can be applied for semantically coded representations of knowledge in other domains as well.

The fundamental secret for success in the choice of going for a semantic vector space seems to be the idea about how the semantic vector space is derived, i.e. how co-occurrences of verbal expressions are dealt with. In addition, it is a crucial decision in this process which kind of co-occurrence data and which form of computation is chosen. In case of verbal semantics, the co-occurrence data chosen in LSA is not of events and events as elementary components (like words and words in the language application domain), but of events and contexts (like words and text passages). Staying from now on with the language application,, successful execution, in addition, relies on huge amounts of source data (tens of thousands or more) for both words and related text passages. The size of the passages is also a control variable to be accounted for in the sense of achieving a good trade off concerning generalisation versus precision. Moreover, as is stated in [Landauer, 2002]:

"The computation is not estimation of probability of sequential contingencies between words, but rather the use of the algebra of simultaneous equations to induce meaning relations between words from all the contexts (not only those shared), in which they do or do ***not*** appear."

Thus, the solution is how words and text passages are computationally combined, thereby keeping in mind that it is the relationship between meanings of words with those of available text passages we are interested in. Since the meaning of a passage is contained in its words, in [Landauer & Dumais, 1997] a relationship is assumed in that the meaning of a passage is simply defined as ***the algebraic sum of the meanings of its words***. Although everybody knows that this is not the full truth (it is not just the words, but also the structuring of the words in the passage which provides the meaning of a passage), it turns out that this simple approach is a surprisingly effective one. It seems to be a brilliant way to achieve sufficient performance at rather low cost. [Landauer, 2002] gives a simple example for illustration:

"Given this way of representing verbal meaning, how would a learner go about using data on how words are used in passages to infer how word meanings and their combinations (in passages) are related to each other? Just assuming that words that often occur in the same passages have the same meaning won't do at all. For one thing, it is usually false; it is the combination of words of different meanings that makes a passage meaning different from a word meaning. Consider the following passages, which are represented as equations of sums of word meanings as specified above:

System 1:

$$\text{ecks} + \text{wye} + \text{aye} = \text{foo}$$

$$\text{ecks} + \text{wye} + \text{bie} = \text{foo}$$

ecks and wye always co-occur in the same passages, aye and bie never. Together the two equations imply that aye and bie must have the same meaning, but nothing

at all about the relation between ecks and wye. Thus, the way to use empirical association data to learn word meanings is clearly not just to assume that words that have similar meanings to the extent that they tend to appear together. Now add two more equations.

System 2:

$$\text{ecks} + \text{wye} + \text{aye} = \text{foo}$$

$$\text{ecks} + \text{wye} + \text{bie} = \text{foo}$$

$$\text{ecks} + \text{wye} + \text{cee} = \text{bar}$$

$$\text{ecks} + \text{wye} + \text{dee} = \text{bar}$$

We know that cee and dee are also synonymous. Finally consider

System 3:

$$\text{aye} + \text{cee} = \text{oof}$$

$$\text{bie} + \text{dee} = \text{rab}$$

To be consistent with the previous passages, from which aye = bie and cee = dee, these two passages must have the same meaning (oof = rab) even though they have ***no*** words in common.

Here we have the makings of a computation for using observed combinations of elements that appears more subtle and promising than simple classical conditioning in which one stimulus comes to act much like another, if and only if it occurs soon before it, or by which two passages are similar just to the extent that they contain the same words or even the same base forms."

The example nicely points out how meanings can be filtered out through this type of mathematical approach. By the way, system 3 in the preceding example is a good illustration of what is meant by "latency" when talking about latent semantic analysis. Semantic information is not only explicitly but also implicitly contained in the source data.

Following the mathematical approach of providing a set of equations from all passages available as source data, this set probably will not be exactly solvable even if there might be more equations than variables of meanings. A known computational method to overcome this is the so-called SVD (Singular Value Decomposition) which is applied to a special formation of the source data matrix. [Gorrell, 2007] illustrates this, using the following set of short passages:

"The man walked the dog"

"The man took the dog to the park"

"The dog went to the park",

and the pertinent source data matrix:

Table 10 Source data matrix for a simple example

	Passage 1	Passage 2	Passage 3
the	2	3	2
man	1	1	0
walked	1	0	0
dog	1	1	1
took	0	1	0
to	0	1	1
park	0	1	1
went	0	0	1

As shown in Table 10, the matrix formation is done in the following way. Each row of the matrix stands for the meaning of a unique word and each column stands for the meaning in terms of a semantic profile of a source data passage. That implies that there is a semantic word-passage correlation which can be exploited by this setup of the matrix. (Notice the similarity to the chunking phenomenon of human memory.) Each cell of the matrix represents the frequency of the occurrence of the word (meaning), which belongs to the pertinent row, in the passage of the pertinent column. The frequency values will be weighted by a particular function to counteract against the effect that the words contribute by varying extents to the semantic profile of the used passages. The word "the", for instance, is a typical example of this adverse effect, because its frequency value is usually relatively high, although its impact on the semantic profile of the respective passage is likely very low. In addition, a word which is evenly distributed over the passages used is of very little value to tell apart the semantic profiles of the passages. [Dumais, 1990] has provided a global weighting function gw_i of the word count at column i of a matrix like that in Table 10, that expresses both the word's importance for each passage and the degree to which the particular word carries information in terms of the semantic profile of each passage:

$$c_{ij_{new}} = gw_i \left(c_{ij_{old}} + 1 \right)$$

$$gw_i = 1 + \sum_j \frac{p_{ij} \log\left(p_{ij}\right)}{\log(n)} \text{ with } p_{ij} = \frac{t_{f_{ij}}}{g_{f_i}},$$

where cij is the frequency value as the content of the cell at column i and row j of the source matrix to be modified by the global weighting function gw_i, n is the number of passages (documents), t_f is the term frequency, i.e. the original word count, and g_f is the global frequency, i.e. the total count for that word across all passages (documents).

The matrix M resulting from this transformation is then to be applied to SVD:

$$M = U \cdot S \cdot V',$$

where U and V are orthonormal matrices and contain the eigenvectors of MM' and $M'M$ respectively. Both MM' and $M'M$ have the same non-zero eigenvalues. S is a diagonal matrix and contains the so-called singular values s_i, which determine the rank of M and are ordered on the diagonal from large to small. U contains the eigenvectors of MM', while V contains the eigenvectors of $M'M$. The singular value decomposition is shown in more detail as follows:

$$M = U \cdot S \cdot V'$$

$$\begin{bmatrix} m_{1,1} & \cdots & m_{1,n} \\ \vdots & & \vdots \\ m_{m,1} & \cdots & m_{m,n} \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} \\ u_1 \\ \\ \end{bmatrix} \cdots \begin{bmatrix} \\ u_l \\ \\ \end{bmatrix} \end{bmatrix} \cdot \begin{bmatrix} s_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & s_1 \end{bmatrix} \cdot \begin{bmatrix} [& v_1 &] \\ & & \\ [& v_l &] \end{bmatrix}$$

The way of presentation is taken from http://en.wikipedia.org/wiki/Latent_semantic_analysis. The vectors u_1 through u_l and v_1 through v_l are called left and right singular vectors. Let t_i be the row vector of the ith row of M *with*

$$t_i' = \left[m_{i,1} \ldots m_{i,n} \right],$$

and d_j the jth column vector of M *with*

$$d_j' = \left[m_{1,j} \ldots m_{m,j} \right],$$

then we can sort out from the equation when M is applied to SVD that V' is contributing to d_j only with its jth column vector $d_{v,j}$:

$$d_{v,j}' = \left[v_{1,j} \ldots v_{1,j} \right].$$

Correspondingly, U is contributing to t_i only with its ith row vector $t_{u,i}$:

$$t_{u,i}' = \left[u_{i,1} \ldots u_{i,1} \right].$$

Since LSA is based on a matrix M of extremely high dimensionality, SVD offers the possibility to drastically reduce the dimensionality of the matrices U, S, and V by only accounting for the k largest singular values and setting those to zero

which are of smaller value. [Landauer & Dumais, 1997] are calling this an inductive process in analogy to inductive human learning. This is of great importance, because it extremely alleviates the computational effort. More dramatic is its effect on semantic learning performance. If choosing k with some care, there is the surprising effect that the resulting rank k approximation may yield even better results as when using the original matrices. At the same time it translates the vectors of word meanings and passage semantic profiles in a semantic space with dimension k. Then, the vector $t_{u,i}$ has k entries, each giving the relation between word i and each of the k semantic concepts. Correspondingly, the vector $d_{v,j}$ yields the relation between passage j and each of the k semantic concepts. The decomposition translates then into

$$M_k = U_k \, S_k \, V_k^{'}$$

and the word and passage vectors in the semantic space are

$$d_{v,j} = S_k^{-1} \, U_k^{'} \, d_i$$

$$t_{u,i} = S_k^{-1} \, V_k^{'} \, t_i$$

To compare passages to words, the matrix *S* is to be accounted for, typically done by moving both passages and words through an additional sqrt (*S*) transform.

We can now see in the semantic space how certain word and passage meanings are related, for instance by evaluating the corresponding dot product. This measure of similarity becomes 1, if the two vectors to be compared are perfectly matching. One characteristic of LSA with highly dimensional vector representations seems to be the fact that there emerge clusters which are more or less sparsely distributed over the entire range of the vector space. That leads to a situation that it usually proves sufficient to choose the cosine of the angle between the two vectors to be compared as the semantic similarity measure.

If a query is wanted about a passage, for instance “the dog walked” when taking the example from above, it can be taken as a pseudo-passage and be compared to the given passage vectors in the semantic space by using the same transformation. Figure 100 taken from [Gorrell, 2007] shows how it compares in reduced 2-dimensional semantic space with the other passages given in the above example. The co-ordinates in Figure 100 stand for abstract semantic features which cannot be named by any presented word or passage. It should be noted here that this figure is only given to show by a simple example how the algorithms work and the resulting semantic vectors may be placed in semantic space. However, any expectations about semantic accuracy should not be checked in this figure, because there are too few source data in this example to make it a test case for accuracy. It also becomes obvious at this point that not only verbal semantics are represented by LSA, but also the words as symbols associated to the corresponding vectors of their meanings. The symbolic coding of the words,

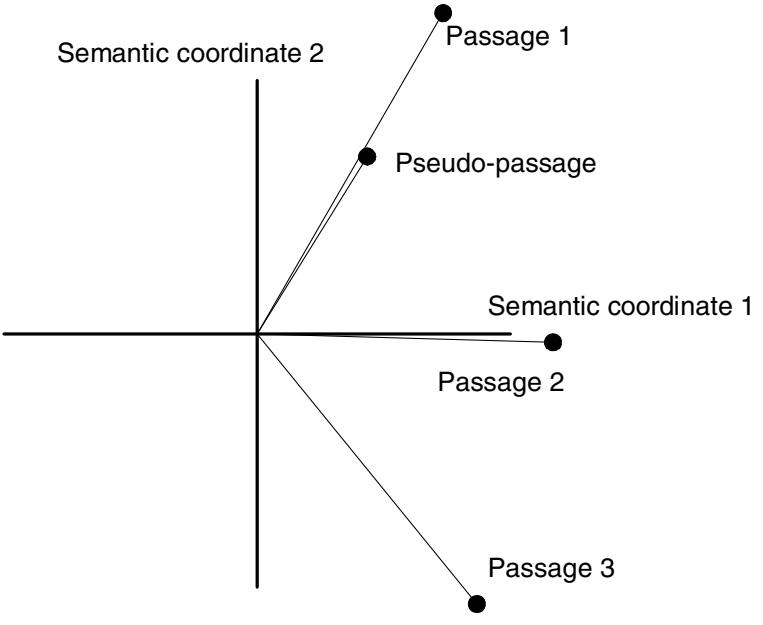


Fig. 100 Comparison of source data passages and query pseudo-passage in semantic space for a primitive example [Gorrell, 2007]

depending on the way they enter the system, whether through image or acoustic sensing or through communication links, is the transformed in a vector presentation. **Highly important, both semantic and symbolic representation is achieved at the same time** by LSA, very alike to human cognition. By the way, it also seems to model what we refer to as chunking in human memory management.

Once learning of a word is achieved by associating a vector to the word regarding it as a symbolic item and its meaning, the corresponding co-ordinates are stored and can be called upon for retrieval purposes.

Applications

From [Landauer & Dumais, 1997] and [Landauer, 2002] we conclude that LSA has been applied in various segments of language research. It "has been used to improve automatic information retrieval, where it produces 15-30% gains in standard accuracy measures over otherwise identical methods by allowing user's queries to documents with the desired conceptual meaning but expressed in different words." This is certainly based on a large-scale experiment when LSA was applied for test purposes to 30.473 articles as text samples with a total of 4.6 million words. A sample of each article was taken with a mean of n 150 words. In total a number of 60.768 unique words were filtered out of the text samples which appeared at least in two samples. Thus the corresponding matrix M contained more than 60.000 rows and more than 30.000 columns. The dimensionality reduction for the SVD procedure went as far as to 300 dimensions of the semantic

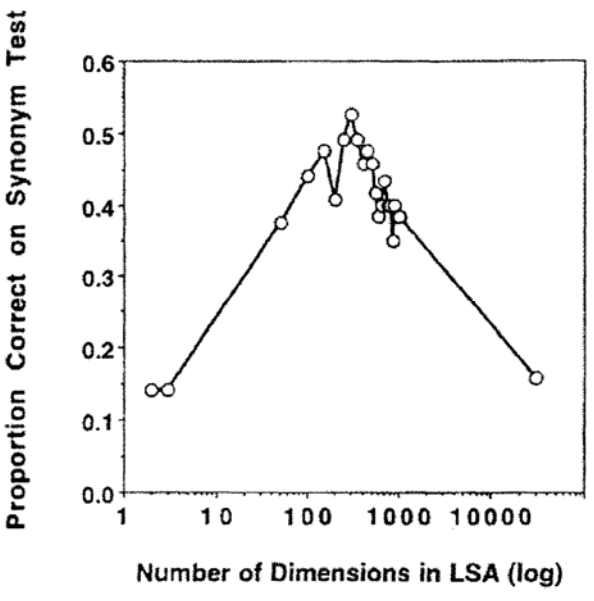


Fig. 101 Effect of number of dimensions retained in LSA test simulations of word meaning similarities. The ideal value for the proportion correct on synonym test would be the value 1 [Landauer & Dumais, 1997]

space. A semantic space of this dimensionality already shows a degree of sparseness, as claimed by [Landauer & Dumais, 1997], that the word and passage meanings can easily being told apart from each other as isolated entities. According to Figure 101, this procedure of optimizing dimensionality was derived using tests with 80 retired synonym items from the multiple-choice test of English as a foreign language (TOEFL). LSA has also provided good retrieval performance when queries and documents are in different languages. Moreover, LSA was applied to measure the similarity of student essays on a topic of instructional texts in order to predict how much an individual student will learn from a particular text. Among further applications, LSA was also used to assess the psychiatric status of depressed people in interviews compared to normal interview behaviour.

[Landauer, 2002] also discusses application possibilities outside of verbal semantics. This is to support the idea that LSA is essentially a model for the management of explicit knowledge as such, including knowledge about objects, episodic events, and rules. In particular, he sketches how to apply LSA to objects to be learned by visual recognition, probably the most complex category of knowledge about objects. He claims that there can be established analogies to the treatment of verbal semantics, like visual saccadic scenes might correspond to text passages and retina ganglion outputs correspond to words. While the semantic space of word meanings must be established by tens of thousands of words and tens of thousands of passages the visual semantic space may possibly need

empirical associations of tens of millions of visual percepts, though, or more which is presently out of reach with current computational power. Landauer conjectures about an incremental derivative of LSA which could lead to a feasible visual semantic space with 1000 dimensions. He makes the point that success of that kind of approach is predominantly dependent on the number of empirical associations taking place. The realisation of an incremental LSA approach for online learning is proposed by [Gorrell, 2006]. Another variant of LSA is a probabilistic approach as published by [Hofmann, 1999]. [Landauer, 2002] has also alluded to that problem, looking for instance at neural net solutions, but for his explorations and experiments to gain a maximum of insights, mathematical traceability was a too important aspect, so far.

In summary, semantic along with symbolic representation in artificial cognition obviously is becoming a realistic way of amendment in order to take advantage of unique strengths in memory management humans have. Perhaps, these ideas are pointing to a way, too, how to model what is known as the binding problem of human cognition. A representation of that kind can deal with synonyms and ambiguity of word meaning. Since there is no experience yet with this way of knowledge representation for the application domain of cognitive systems in work systems like those of vehicle guidance and control, this approach has still to be considered somewhat premature. However, we take the risk of a prognosis that, in general, this is the way to proceed in explicit knowledge representation for future designs of ACUs in work systems. In many cases, online knowledge acquisition is not a requirement, such that offline learning by use of the mathematical learning theory, as outlined, is not a principal drawback. Still the online knowledge acquisition and retrieval by means of LSA does not seem possible in a dynamic situation of vehicle guidance and control, unless a parallel computing approach like PDP of very high dimensions in processing units and interconnections is available.

6.1.2.3 Concluding Remarks

It seems that the knowledge management in artificial cognitive systems like the ACUs will arrive at a really humanlike powerful one only until approaches for semantic memory coding like that of LSA on the one side and PDP on the other side are being combined in the process of knowledge acquisition and retrieval. This takes a move to bravely undertake a big step from a much perforated symbolic representation to a large scale semantically coded representation in a highly dimensional vector space presenting a semantic continuum similar to the LSA approach, implemented in a PDP-like connectionistic structure which exhibits a human-like hardly feasible abundance of interconnections.

6.1.3 Management of Implicit Knowledge

Another issue of knowledge management is the acquisition, representation and retrieval of implicit knowledge as a-priori-knowledge of skill-based cognitive high-level functions according to Figure 30. Modelling of this behavioural level had been one of the greatest challenges so far. Similar to the problem of representing explicit knowledge including semantics, this field was not satisfyingly dealt with

for a long time. Here, there was a principle problem, because there is no way of making use of verbal protocols of subjects about their skill-based behaviour, since this is based on unconscious automatisms which a subject cannot reliably report on. To rely on subjects' assertions about their skill-based behaviour may even become dangerous, because they might try to talk about what they imagine they might have done, but what in fact may be totally different from what they really were doing. Therefore, modelling approaches are being pursued in this context, which differ considerably from those used for rule-based cognitive behaviour. Mostly, they were launched for the purpose of modelling the behaviour of humans when driving a road vehicle. In early approaches, models of the driver behaviour were mainly developed in the form of linear continuous systems. However, during the last two decades, hybrid approaches with the combination of various nonlinear modelling techniques have become more and more important. In particular, researchers working on cognitive systems for the application in road vehicle guidance and control have achieved great advancements in this respect [Jürgensohn, 1997]. They are striving for more and more precise models, taking into account the specific human behavioural traits, including skill-based sensori-motor behaviour [Rasmussen, 1983].

Staying with the application domain of road vehicle driving, quantitative modelling of the cognitive driver behaviour has to account for the fact that the driver behaviour is characterised by variability, flexibility and great inter-individual differences. Quantitative modelling of the driver behaviour also has to be situation-adaptive, because the driver is deriving his/her actions based on wide-ranging perceptions entailed by the assessment of the full picture of the driving situation he/she is encountering at each instant of time. So far, no analytical equations have been found to cover all this, although this was tried again and again [Mc Ruer & Weir, 1969], [Gazis et al., 1963], [Wiedemann, 1972], [Fritsche, 1994], and [Jürgensohn, 1997]. Normative non-linear modelling has also been tried, though (see [Protzel et al., 1993], [Rekersbrink, 1995], and [Henn, 1995]).

Therefore, alternative data-based approaches had to be developed which rely on observing the driver while he/she operates and which learn from these data. Thereby, the driver is able to adapt his/her behaviour to the situational reality, which might be different from what he/she expected. The aspect of inter-individual differences leads to the demand for driver-adaptive models as opposed to a normative driver model. The more complex the driver's behaviour, the more challenging are the requirements for the method to learn the driver's behaviour by a machine. This leads to the domain of neuro-computing and classical statistics. In [Pommerleau, 1995] a 3-layer feedforward neural net, based on the backpropagation algorithm for offline training, is presented for automatic lateral control of agricultural vehicles. [Fritz & Franke, 1993] used a similar approach for the lateral control of road vehicles. The input vector comprises the speed, the yaw angle of own vehicle relative to the road, the road parameters like curvature and width, the lateral deviations from the lane centre line and the target lane. The output of the neural net is the steering wheel deflection rate. The net is trained to model the combined driver and vehicle behaviour. The system was demonstrated

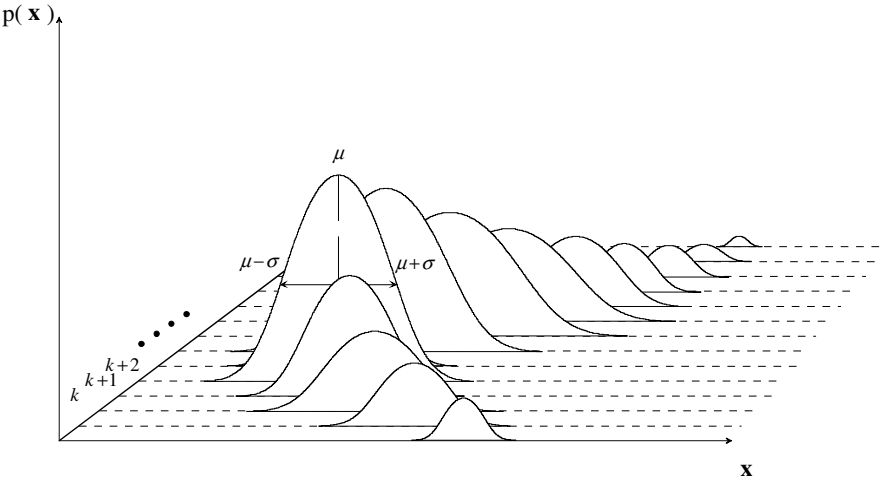


Fig. 102 Principle of the likelihood classification [Grashey, 1998]

on a ride from Stuttgart to Ulm in 1991. Also [Neusser et al., 1991] designed a similar neuro-model, which was fielded as well. On the basis of Fuzzy Art [Carpenter et al., 1991] a neuro-model for longitudinal control was presented by [Feraric, 1996]. The training of this model is online and in quasi-real time. An important feature of this approach was the ability to take care of rarely occurring situations.

Numerous other approaches based on neural nets can be found in the literature like [Shepanski & Macy, 1991], [Narendra & Parthasarathy, 1990], [Kornhauser, 1991], [Kehtarnavaz, & Sohn, 1991], [Fujioka & Takabo, 1991], [Mecklenburg, 1992], [Chen & Billings, 1992], [Kraiss & Küttelwesch, 1992], and [Kraiss, 1995].

Besides the neuro-approaches, there also are several ones based on statistical modelling (see [Grashey, 1998] and [Schreiner, 1999]). That one described in [Grashey, 1998] generates a model online which is based on the likelihood classification. It maps the longitudinal and lateral control for driving situations on the German “autobahn”. Thereby, the possible driver actions are quantised in terms of k classes. It is assumed that for all k action classes the feature values are distributed according to the Gaussian distribution. Figure 102 shows an example for a one-dimensional feature space.

While the model is trained online, the class-specific distributions are developed by observing the actions and corresponding situation features. The application (recall) of the model selects the actually valid action class by looking for the feature value corresponding with the greatest density value across all possible action classes. Driver actions, which are not complying with the assumption of the Gaussian distribution, have to be treated separately. For this purpose it becomes necessary to partition the feature space correspondingly.

In summary, it becomes obvious from the preceding discourse that the systematic investigation of the online modelling of individual, i.e. driver-adaptive skill-based behaviour has not got very much attention so far. Most approaches

were focused on normative behaviour modelling. These models are not suited to be extended for individual skill-based models.

Numerical learning algorithms can, in principal, map the linear and non-linear individual traits of driving behaviour without the necessity of formulating parameterised analytical expressions. Therefore, some of the approaches mentioned favour numerical learning algorithms, namely through statistical and neural net approaches. However, all of these former approaches have still some drawbacks. The backpropagation algorithm has got the problem of slow learning performance and the tendency of neglecting rare but important situational events. The main drawback of the Fuzzy Art approach is the hardly controllable demand for memory. The statistical approach based on the maximum-likelihood classification is suffering from the principal assumption of a Gaussian type distribution for the situation feature values with respect to each of the classes of actions.

A workable combined modelling theory for skill-based cognitive behaviour in terms of an adaptive driver modelling is successfully elaborated in the more recent years in, [von Garrel et al., 2000], and [von Garrel, 2003]. This modelling approach was also developed in the context of applications based on driver cognitive behaviour modelling in the road vehicle guidance and control work domain. The driver behaviour is characterised to a great extent on skill-based behaviour in terms of unconscious automatisms. This approach will be discussed in more detail in Chapter 6.2.1.

6.2 A-Priori Knowledge Components in ACUs

The framework in Figure 30 indicates that there is to be acquired and represented knowledge about all three behaviour levels, skill-based, procedure-based, and concept-based behaviour, in order to realise a complete cognitive system which exhibits a similar performance than humans. Therefore, some implementation examples of a-priori knowledge components of these three levels are described in some more detail in the following. This may alleviate the process of getting into the subject for those who have no experience in designing cognitive systems. Therefore, it makes sense at this point that implementation examples are presented which were developed by our own.

6.2.1 Components with Emphasis on Skill-Based Behaviour

The implementation of knowledge representation about skill-based behaviour in cognitive automation has been more or less neglected for a long period of time in favour of that about procedure-based and concept-based behaviour. Obviously, one had looked for applications first where one could do without and in fact there are a great number of applications of that kind. In particular, the applications of cognitive automation in work systems of road vehicle control can hardly do without. The main reason for prioritising applications for a long time which did not need knowledge representation about skill-based behaviour lies in the fact that skill-based behaviour is subconscious (automatic). Questionnaires and interviews are mostly not worth the effort. To get access to that kind of knowledge therefore is a highly demanding task. For these reasons, there are not many productive

approaches currently around for the acquisition and representation of knowledge about skill-based behaviour. Therefore, the following examples are probably of particular interest for the reader. Either example is addressing adaptive modelling of driving behaviour. As was already alluded to in Chapter 6.1.3, in this case of application the acquisition of implicit knowledge about skills is needed in addition to that of explicit knowledge. In particular, the implicit knowledge of human drivers is different from person to person. It is represented by semantic coding. Therefore, in order to model human driving behaviour with respect to implicit knowledge, we can establish the model by observing the human driver actions and the pertinent driving situations. Still, either example is making use of quite different methods for its realisation. In that sense, one of the examples can be considered as a more classical approach, whereas the other one is making use of soft-computing methods.

6.2.1.1 A Classical Approach

In [Schreiner, 1999] a classical approach was pursued which was based on a control-theoretical realisation of an automatic guidance and control system for road vehicles. If this system is turned on by the driver, it is to automatically drive the vehicle in a driver-adaptive fashion for acceptance reasons, i.e. to drive in a style close to that human driver actually driving. Therefore, part of this system should be a model of the individual driving behaviour of the actual driver. This system comprised both, explicit a-priori knowledge as a rule base about how to determine the current task as part of the procedure-based behaviour level as well as a combination of explicit and implicit knowledge for the determination of control instructions and the action control in the skill-based level. In this section the latter will be of main interest.

For this approach certain application-related basic requirements were taken into account. Very likely, these requirements could also be valid for other applications, where models of individual operator cognitive behaviour are of interest. These basic requirements are:

- Driver adaptivity
- Situation adaptivity
- Intrinsic knowledge of model validity

Driver adaptivity:
The modelling process has to take place, while the driver to be modelled is operating the vehicle on the road. The skill-based cognitive behaviour (see Chapter 3.2.2.2) of this person is learned online, thereby including the individual behavioural traits. Switching from learning (even if not complete) to operational use of the model has to be possible at all time like it is known from human learning. The learning process should be as short as possible to make the model sufficiently useful.

Situation adaptivity:
The driver behaviour is dependent on the driving situation. The driving situation is given by a vector in the situation feature space. The situation has got elements,

which control the sensori-motor part of the skill-based behaviour, but there are other elements which have to be analysed by a certain structure of rules, too. Suitable representation of both the sensori-motor part and the rules is to be provided.

Intrinsic knowledge of model validity:
Since the model is not complete from the beginning of the online learning process, but possibly already useful for certain regions of the situation feature space at the actual state of learning, the useful part of the model should be ready for usage and the other part should be excluded from usage.

This leads to the following more detailed requirements for the learning algorithm:
Sequential learning: The training phase has to be online (sequential) and in real time. As opposed to batch-learning, where the data processing takes place in epochs, sequential learning is based on data from observations of the relation "Relevant situation features → Action". These data have to be processed continuously right at the time when they are coming in. There is no access to the data again at any later time. This leads to

- efficiency with respect to computation performance and memory demand
- ability to track stochastic and dynamic fluctuations of parameter values
- ability to process any quantity of data (continuous learning lifelong)

Plasticity and stability:
The learning algorithm must be able to take into account new facts without getting saturated (plasticity). At the same time it must be able to keep what was already learned. Many algorithms lack that capability to permanently store knowledge learned and, at the same time, to refine this knowledge.

Modelling of non-linear behaviour:
It must be possible to approximate non-linear behaviour (e.g. step function transitions) with sufficient accuracy.

Rare situational events:
The probability of occurrence of the potential situational events differs to a great extent in the domain of driving. Dangerous situations, for instance, can be considered as rare events. However, it is most important that they are covered by the model.

Processing of obstructed data:
There is always the possibility of obstructed data. The learning algorithm has to deal with these data appropriately.

The modelling approach
The basic concept of the adaptive driver modelling is motivated by the need of modelling behaviour that is similar to the main stream of behavioural characteristics of driver individuals (see Figure 27). The similarity should be achieved

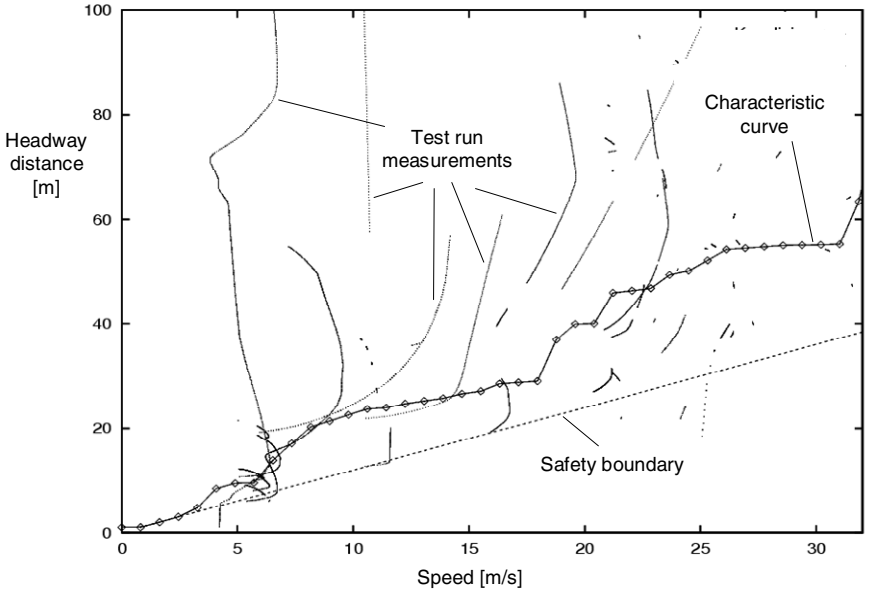


Fig. 103 Characteristic curve for the headway distance as adopted by an individual driver in a simulator study [Schreiner & Onken, 1997]

to an extent that systems which apply the model can predict the behaviour of the individual with sufficient accuracy and that these individuals can predict and understand the model outputs as well. The main focus of the model concept is to map the driver behaviour when following a predefined route plan on German freeways.

One particular feature of this modelling approach is the use of characteristic curves and parameters about the individual driving style of the driver. They are being determined by continuously observing the driver when being in action, resulting in statistical histograms. For instance, there results a histogram which shows the values of headway distance to a preceding vehicle a certain individual driver is striving for when coping at certain speeds with that vehicle appearing in front. In order to exclude behavioural outliers, in this case the 15. percentile of all values collected for the distance adopted at a certain speed is the corresponding one taken for the characteristic curve. Figure 103 shows an example of that kind of characteristic curve as determined in a simulator study. Among several others another histogram is to be developed for the target speed being adopted as a function of the curvature of the road or for the target lane chosen as a function of the existing number of lanes.

Since usually there are several task alternatives to choose from in a given driving situation, an evaluation process is modelled which results in selecting one of the alternatives as the representative one for the driver behaviour. Since this part of the overall model is associated to the procedure-based driver behaviour, it

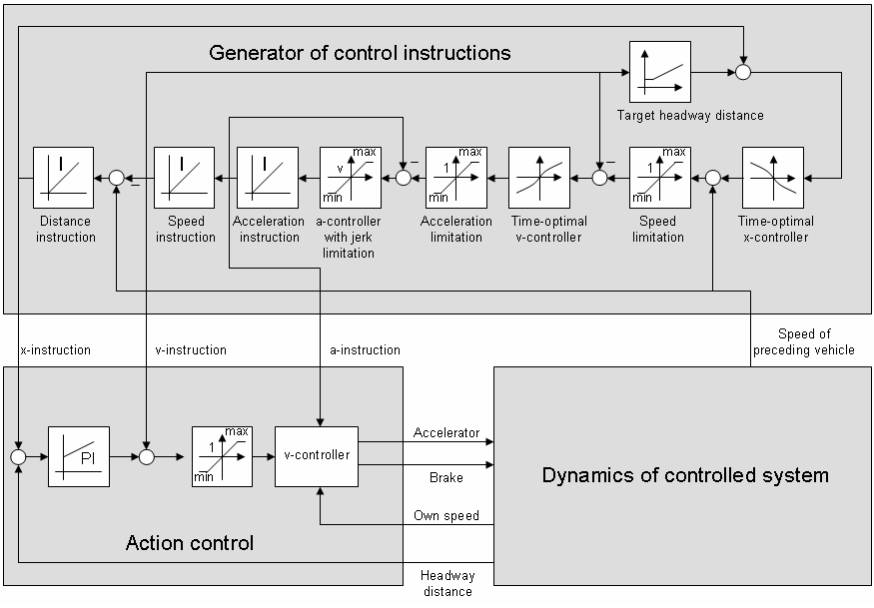


Fig. 104 Automatic longitudinal control (headway distance) [Schreiner, 1999]

will not be further commented. At this point it will only be mentioned that use is made of a catalogue of criteria with driver-adaptive weighting factors.

Once a task alternative is determined, control instructions are to be generated by means of the task execution function. This is modelled as skill-based action control for the control of the longitudinal and lateral vehicle loops of action control via accelerator (see Figure 104 and Figure 105). These were successfully implemented in a simulator study.

Regarding the control instructions for speed control, both instructions for the speed v and the acceleration a are generated. As shown in fig, limitations for the stationary speed as a function of the anticipated curvature of the road are accounted for as well as the limitations of dynamic performance of the vehicle. The control instruction for the vehicle acceleration (commanded acceleration) results from integration of a physical jerk which is limited, too. The control instruction for the speed (speed command) results from integrating the control instruction for the acceleration. Two nonlinear control loops take care that the target value of the control instructions for the speed are achieved in a time-optimal manner (see [Leonhard, 1990]). The corresponding “controlled system” consists of the two integrators. It should be noticed how the characteristic curve for the target speed is incorporated.

Similarly, the control instructions for the headway distance, including the pertinent instructions for the acceleration and the speed, are generated. Accordingly, there are three nonlinear time-optimal control loops for a “controlled system” of

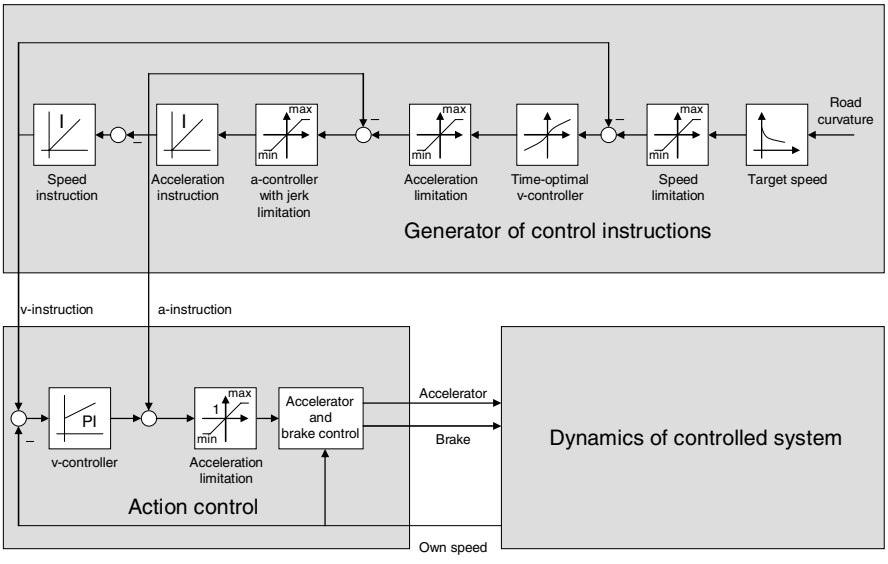


Fig. 105 Automatic longitudinal control (speed) [Schreiner, 1999]

three integrators. The control instructions for the headway distance result from integrating the difference between the speed of the preceding vehicle and the speed command for the own vehicle.

Example of simulator study
[Schreiner, 1999] carried out simulator experiments for the validation of this kind of model. A typical example of a test run is shown in Figure 106.

As can be identified from this figure with great evidence, the behaviour of the model is rather well adapted to that of the test subject after a period of familiarisation when driving through the first 4 kilometers of the test course. The main deviations in speed control occur in road turns, in particular in S-turns. They result from the very cautious driving style of the test person which could not sufficiently modelled by the characteristic curve for the target speed as a function of the road curvature. The main cause for this deficiency of the characteristic curve lies in the fact that only one fixed distance ahead was simply taken into account for the anticipation of the curvature. This distance parameter was not adapted to the test subject.

6.2.1.2 A Soft-Computing Approach

Inspired by earlier work such as [Feraric, 1996], [Grashey, 1998], [von Garrel, 2003] pursued another quite different approach which was based on a particular combination of statistical techniques and neuro-computing. It proved to provide a way to capture the individual non-linear driver behaviour when controlling a vehicle on the road. This soft-computing approach was developed to be applied for the driver tutoring assistant system as described in Chapter 5.3.3, providing a

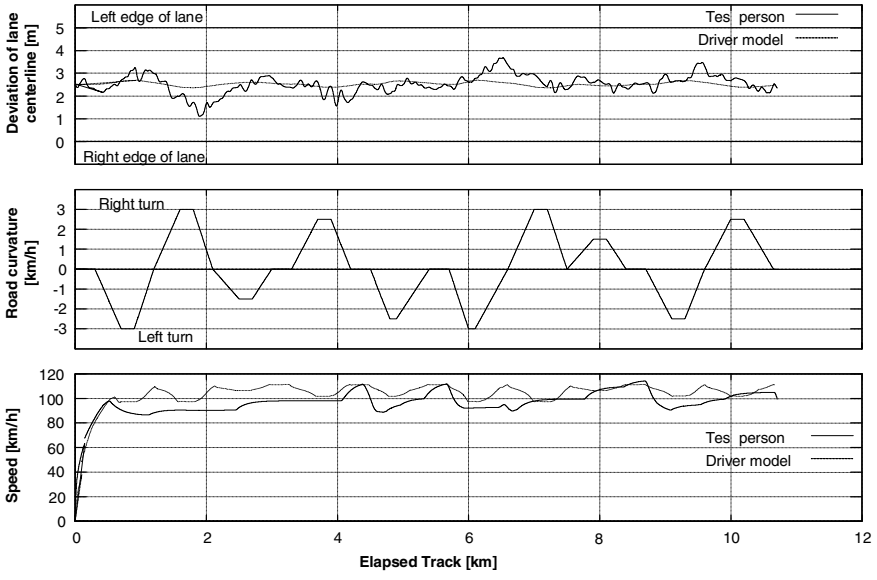


Fig. 106 Behaviour for the control of the lateral deviation off the road centreline and of the vehicle speed, looking at both the behaviour of a human driver as test subject and the model of the subject's driving behaviour [Schreiner, 1999]

crucial part of the knowledge to be used in the context of the cognitive sub-function of situation interpretation according to Figure 89. Again, the same application-related basic requirements were taken into account as already described for the classical approach in Chapter 6.2.1.1. They are based on the functional needs for applications like driver assistant systems in road vehicles and in simulator systems for driver tutoring.

The modelling approach

Again similar to the classical approach, we usually have to consider several potential elementary driving tasks to choose from in a particular traffic situation. In this case, there are many situation-dependent decision rules to be modelled. Although these decision rules oftentimes can be considered as "explicit", conscious procedure-based ones of task determination (like the decision to overtake a car in front), they cannot be conscious ones in all cases. We rather have also to account for "implicit", unconscious ones in terms of semantic coding (see Chapter 3.2.1.3). There can be considerable differences between drivers in that respect. Assistant systems have to account for that, if we want their hints and warnings being accepted by the driver. In [Otto, 2006] an approach is shown to approximately model this implicit decision behaviour of the driver when driving in urban streets. As was already proposed in [Schreiner, 1999], a discretisation of the potential

Table 11 Main potential elementary driving tasks of longitudinal and lateral control in an urban environment (cf. [Otto, 2006])

Lateral control → / Longitudinal control ↓	Lane driving					Changing lane		Keeping lateral safety distance	
	Deliberate left lateral deviation from lane centerline	*Straight driving*	*Deliberate right lateral deviation from lane centerline*	*Turning right*	*Turning left*	*To right*	*To left*	*Traffic in opposite direction*	*Traffic in same direction*
Straight driving									
Driving at target speed (no interferences)	X	X	X					X	X
Driving interferences (road junctions, traffic lights, etc.)	X	X	X					X	X
Turning at road junction									
Turning right at road junction		X	X	X				X	X
Turning left at road junction	X	X			X			X	X
Turning right with right to go		X	X	X				X	X
Turning left with right to go	X	X			X			X	X
Keeping safety distance to vehicle ahead									
Stop and go	X	X	X	X	X			X	X
Following	X	X	X	X	X			X	X
Closing up	X	X	X	X	X			X	X
Speeding off	X	X	X	X	X			X	X
Stopping	X	X	X	X	X			X	X
Coming to a standstill									
Decelerating	X	X	X	X	X			X	X
Stopping	X	X	X	X	X			X	X
Overtaking									
Closing up	X	X	X			X	X		
Following	X	X	X			X	X		X
Passing	X	X	X			X	X		X

situation-dependent elementary driving tasks is used rather than a discretisation of potential situations. In a matrix of some main potential elementary driving tasks of longitudinal and lateral control is depicted for the case of driving in an urban environment (see also [Fastenmeier, 1995]).

We can see that various elements of the matrix are empty, but that several potential candidates can exist which have to be chosen from in a particular driving situation and others which are concurrently valid. For instance, elementary driving tasks for longitudinal control are usually to be performed concurrently with someone of lateral control.

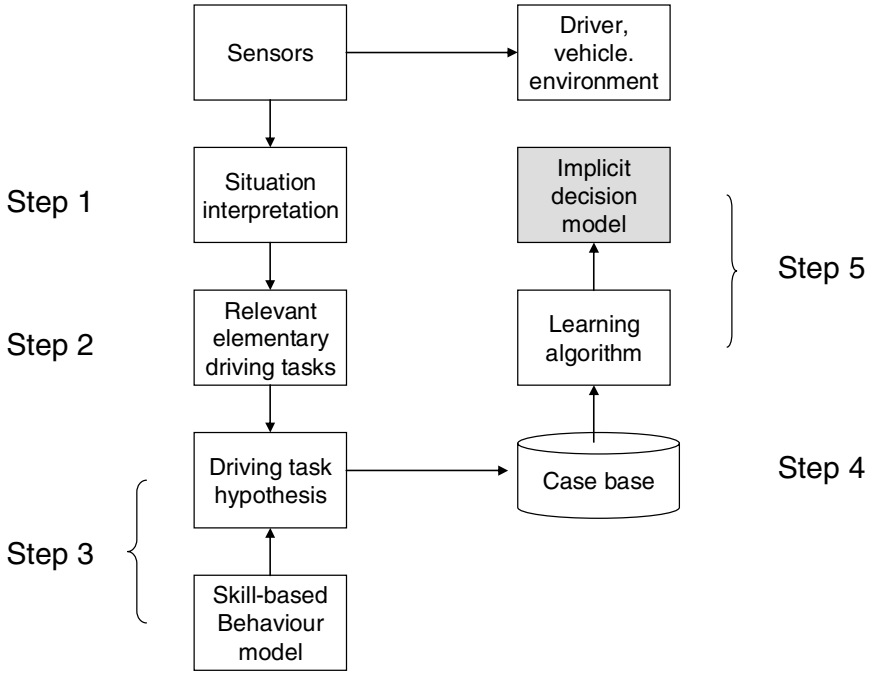


Fig. 107 Process of learning of a driver-adaptive, behaviour model for skill-based (implicit) decisions, deciding on the valid situation-dependent elementary driving task (cf. [Otto, 2006])

Figure 106 shows the sequence of steps, to go through in order to learn rules of the kind:

"if situation A, then elementary driving task and pertinent action pattern B",

which approximately models the "implicit", unconscious rule applied by the individual who drives in that situation [Otto, 2006]. This stepwise process of knowledge acquisition will be started with step 1, the dynamic situation interpretation, and ends with step 4, the generation of a case base for driver-selected, situation-dependent elementary driving tasks:

Step 1 (situation interpretation): The step of interpreting the actual driving situation concerned goes as far as to ensure that the representation is unambiguously described. It contains information about the *external situation* regarding the relevant objects of the environment, the road, the own vehicle, those other relevant vehicles nearby as well as other participants in road traffic which are to be observed by the driver. It possibly may also include the *internal situation* regarding information about the status of the driver himself.

Step 2 (generation of situation-dependent alternatives of elementary driving tasks, i.e. the task situation): This step is about the same as has been already described in the context of Figure 27. In case of procedure-based behaviour the task determination function compellingly leads to a single elementary driving task. This is achieved by use of the task situation knowledge which includes a normative rule base to evaluate the incoming situation-dependent cues. Figure 131, coming up later in Chapter 6.2.2.2, will exemplify that. In case of skill-based behaviour, though, the process of feature formation is first of all usually left with a set of task alternatives. Then, use is made of knowledge about the decision to implicitly determine the valid alternative. The following steps are to acquire this knowledge ending up in an implicit decision model.

Step 3 (learning of sensori-motor action patterns associated to elementary driving tasks): This step is crucial for the adaptive modelling of individual behaviour. An implementation example, developed in [von Garrel, 2003], will be described lateron in this chapter. It is a machine learning process based on observing the driver behaviour.

Step 4 (generation of case base for driver-selected, situation-dependent elementary driving tasks): Here, it is made use of the results in step 3. In a given situation Si of machine learning runs which might be even online, the resulting simulated hypothetical action pattern outputs (available through the knowledge derived in step 3) associated to the alternatives of elementary driving tasks (determined in step 2) are compared with the action of the individual who is actually driving at the same time (see Figure 109). The comparison results for as many as necessary situations Si, one for each alternative, are evaluated by taking that one as a sample of the case base which is matching best with the action of the driver. The steps of the evaluation process are the following:

(1) Evaluation of the differences d_{k_j} between hypothetical action values a_{i,k_j} of alternatives k_j, and the driver action a_D:

$$d_{k_j} = \left| a_{i,k_j} - a_D \right| \text{ for all } i = 1, 2, 3, \ldots N_s \text{ and } j = 1, \ldots N_a$$

with N_s as the total of evaluated situation samples and N_a as the total of alternatives of elementary driving tasks, and

the differences s_{k_j,k_k} between hypothetical action values a_{i,k_j} of alternatives k_j and hypothetical action values a_{i,k_l} of alternatives k_l:

$s_{k_j,k_l} = \left| a_{i,k_j} - a_{i,k_k} \right|$ for all $i = 1, 2, 3, \ldots N_s$, $j = 1, \ldots N_a$, and $1 = 1, \ldots N_a$ with $k_j \neq k_l$.

(2) Determination of the best match: The alternative with the best match $d^*_{k_j}$ is determined by the smallest value of the differences d_{k_j}, given that this value lies in between a predetermined range of the maximally allowed d_{k_j} and the minimally allowed s_{k_j,k_l}.

The alternative with the best match is taken as the valid case to become part of the case base. In Figure 108 the evaluation parameters d_{k_j} and the s_{k_j,k_l} are depicted to be used for the evaluation steps (1) and (2) for the traffic situation S_i at t_i. This corresponding traffic scenario is characterized by the own vehicle approaching a red traffic light with another vehicle ahead. The corresponding alternatives of elementary driving tasks are:

- driving at desired speed (DDS)
- keeping safety distance to vehicle ahead (KSD), and
- coming to a standstill (CS).

It turns out that the acceleration a as the control variable, associated to the model for driving at desired speed, is the hypothetical skill-based action which is closest to the actual driver action. If the supplementary conditions about the amount of the corresponding d associated to driving at desired speed and the values of s_{k_j,k_l} regarding the other alternatives (for keeping safety distance to vehicle ahead and for coming to a standstill) are met, the elementary driving task of driving at desired speed becomes associated to the situation S_i as a case to be kept in the case base.

Figure 109 shows a sample of time histories of a driving simulator run for the same type of traffic situation. When approaching a traffic light with another vehicle ahead, it has to be dynamically decided all along until the standstill which elementary driving task of longitudinal control has to be selected out of the three potential ones, driving at desired speed, keeping safety distance to vehicle ahead, and coming to a standstill. The driver action is plotted, and in addition to it the simulated skill-based actions corresponding to the sensori-motor action patterns associated to the situation-dependent elementary driving tasks.

For each computation cycle, the evaluation steps (1) and (2) are performed, providing the time history of actually best matching elementary driving task alternatives. This is plotted at the bottom of the diagram in Figure 109.

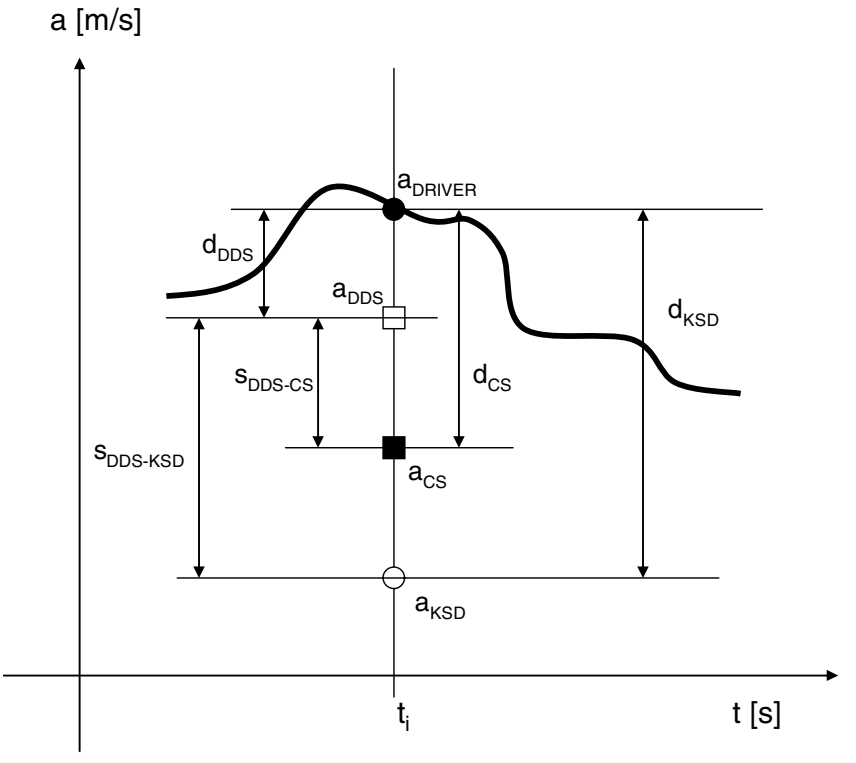


Fig. 108 Matching parameters d and s for a traffic situation S_i with the potential elementary driving tasks DDS, KSD, and CS [Otto, 2006]

Step 5 (learning of decision model based on the content of the case base): When a sufficient number of cases is gathered, the case base eventually can be used to model by an appropriate learning algorithm the situation-dependent implicit driver decision about the elementary driving task to be executed (see Figure 110). It is an adaptive model, as it models the individual driver who is observed during the learning process while being in action.

The driver model as such, when being in operation, makes use of both

- the combined decision model of implicit and explicit decisions on the actual elementary driving task and
- the model of sensori-motor action patterns (see Figure 110).

This combines the procedure-based and skill-based driver behaviour.

The modelling of skill-based sensori-motor action patterns

The main difficulty in developing driver behaviour models is to find a mapping function of the skill-based behaviour in terms of sensori-motor action patterns. This behaviour must be viewed as a quasi-stochastic process and it is in principle

- discontinuous,
- nonlinear and
- varying in time.

Due to the great inter-individual differences of driver behaviour the adaptation to an individual behaviour with a high degree of detail can only be achieved if data-based modelling is used. Data-based modelling takes place in the sense of use of online observation data, concerning the vehicle, the environment and the driver (mainly his actions). These data are sequentially presented for each observation cycle to a learning component. The learning component produces the behaviour model by mapping the situation characteristics (vehicle and environment) on the driver action.

For modelling skill-based driver behaviour, complying with the requirements as described earlier in this chapter, the so-called Constructive and Evaluating Learning Method (CEL-Method) has been developed [von Garrel, 2003]. This new kind of algorithm combines local and global learning strategies by a net of local units and works according to the principle of supervised learning.

The algorithm is based on a constructive learning strategy. New units are inserted successively in regions, where the model does not yet work appropriately. The region of influence of the new units decreases as the learning process moves on, so that the approximation function is being approached in a strategy from-coarse-to-fine.

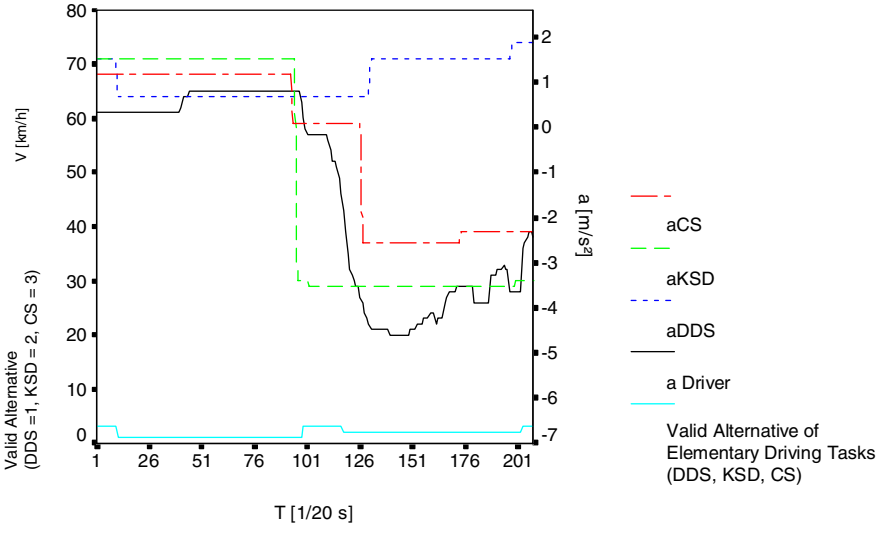


Fig. 109 Simulator test run result, showing the time histories of the model outputs for the the situation-dependent alternative elementary driving tasks DDS, KSD, and CS in comparison to the driver action in longitudinal control (vehicle acceleration a). In addition, the time history of the hypothesis for the actually valid elementary driving task alternative is shown as the result of evaluating the relevant alternatives DDS, KSD, and CS. [Otto, 2006]

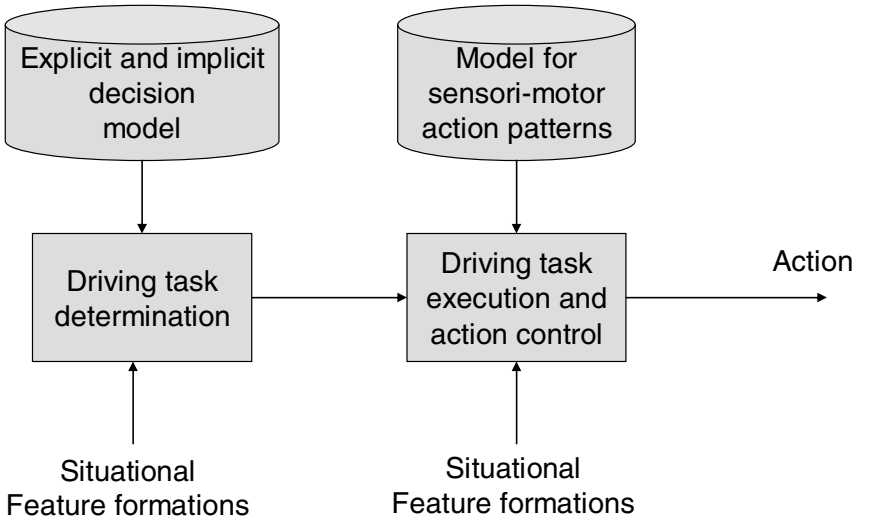


Fig. 110 Driver-adaptive model of procedure-based and skill-based driver behaviour

With this kind of processing, the CEL-algorithm keeps the size of the model at a moderate level and thereby makes it possible to process large amounts of training samples. Furthermore, the algorithm adapts quickly to new relevant data without "forgetting", which is still useful information (stability vs. plasticity dilemma). After completion of the learning process the CEL-method also warrants appropriate processing of data in situations which have not been accounted for among the training patterns (generalisation). Accordingly, the method can interpolate as well as extrapolate.

In general, to approximate a continuous non-linear function in great detail, a compromise has to be found between the quality of approximation and an over-adaptation to special data. The CEL-algorithm follows this aim by a successive update, where new parameters are inserted into the model dependent on the quality of the current approximation. Thereby, the amount of parameters is being optimized in correspondence to the target function inherent to the data. This way the stability of the learning process is warranted, and it offers the possibility to make the progress of learning transparent.

In order to reduce the amount of data and increase the ability of generalization this algorithm is equipped with a special version of vector quantization. The vector quantization connected to the learning method serves the purpose to classify often occuring examples which are within the region of influence of a local unit.

CEL-Model Architecture

The hybrid modelling approach of the CEL-algorithm is based in the first place on locally placed, radially symmetric functions as kernels in the (input) feature space, which are optimized in their location in the feature space by means of a special version of vector quantization. These functions are specified as normal kernels of second order and correspond to the Gauss function:

$$\phi(\mathbf{x}) = \exp\left(-\frac{1}{2} d_{ma}^2(\mathbf{x}, \mathbf{c}_i)\right)$$

This type of kernel is monotonically decreasing with increasing distance d from the kernel center point **c**. According to the theory of neural networks the area of influence of such a kernel is called a receptive field. In order to be able to consider that the statistical characteristics of the situational features observed may be different, the Mahalanobis distance d_{ma} is used. One of the special characteristics of this kind of distance is that it is invariant to translation, scale change and rotation. Furthermore, it is independent from potentially existing correlation between features. This leads to the approach of the generalized radial basic function. The Mahalanobis distance is calculated from the inverse of the empirical covariance matrix as follows:

$$d_{ma}^2(\mathbf{x}, \mathbf{c}_i) = (\mathbf{x} - \mathbf{c}_i)^T \, \boldsymbol{\Sigma}_i^{-1} (\mathbf{x} - \mathbf{c}_i)$$

Here $\mathbf{c}_i$ is the center of the kernel with index i, $\boldsymbol{\Sigma}_i$ is the corresponding covariance matrix.

Furthermore, a local model of action y_{li} is assigned to each kernel function. It is based on a linearly parameterized polynomial function g(**x**). This model is determined sequentially by using the recursive least squares method (RLS). Then, the output $f(\mathbf{x})$ of the model can be calculated as locally weighted regression for the input vector x:

$$f(\mathbf{x}) = \frac{\sum_{o=1}^{m} \phi_i(\mathbf{x}) y_{li}(\mathbf{x})}{\sum_{o=1}^{m} \phi_i(\mathbf{x})}, \text{ with } y_{1i}(\mathbf{x}) = \mathbf{w_i} \cdot g(\mathbf{x})$$

$\mathbf{w}_i \in \Re^m$ is the corresponding weighting vector of the polynomial function g(**x**) in the local model of action y_{li}. Using a general vectorial transformation **g**: $\Re^n \rightarrow \Re^m$ the vector **x** is transformed into the functions $g_i(\mathbf{x})$

$$g_i(\mathbf{x}): \begin{cases} \Re^n \rightarrow \Re \\ \mathbf{x} \rightarrow g_i(\mathbf{x}) \end{cases}, \quad i=1,...,m$$

As a result, an extraordinarily flexible modelling becomes possible for the regression function. In order to reach the highest possible flexibility for the regression function on the one hand, but not to consume too high an amount of computer resources on the other hand, polynomials of third degree have turned out to be most suitable. Figure 111 shows the principle of this locally weighted regression based on normal kernels distributed across the input feature space receptive fields. The global model results from assembling all local models by way of the regression function. This can be implemented by a specific type of artificial neural

network, combining the contributing local components, the kernels and local models, in hidden units. In this type of network nonlinear functions determine the weight with which the local approximation contributes to the total result.

Thus, using the CEL-algorithm, local and global methods of approximation are combined to a semi-parametric approach. Using this concept, high generalization capability and efficient learning is possible.

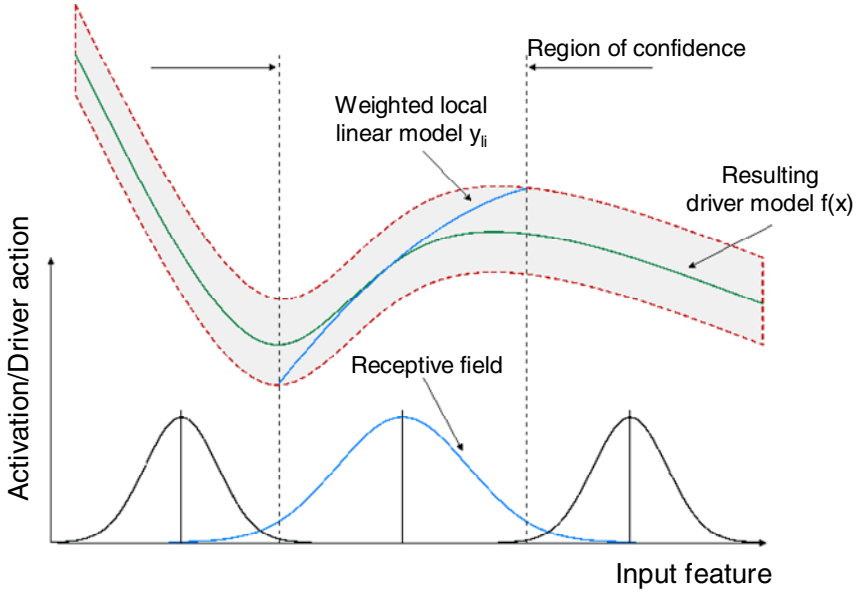


Fig. 111 Principle of the local regression associated to one input feature dimension [von Garrel, 2003]

Adaptation of local linear models

The local adaptation of model parameters is made in a way that during the presentation of a new observation solely the local linear model is adapted, which is associated with the receptive field which shows the greatest amount of activation. As a consequence, after each presentation of a new observation the corresponding parameter vector **w** is recalculated, using the recursive least square algorithm (RLS-algorithm).

The RLS algorithm [Moschytz, 2000] is an online extension of the classical pseudo-inverse-method and is based on the sequential inversion of the input correlation (covariance) matrix. The inverse of the input correlation matrix is calculated iteratively, so no explicit matrix inversion is necessary. Furthermore, this algorithm provides very quick convergence.

Adaptation of the receptive fields

For a good approximation performance the placement of receptive fields (support vectors) in the (input) feature space is of great importance. In order to be able to

include the desired model output into the learning process, it is not enough to simply connect two learning methods in series like it is realised with RBF nets [Moody & Darken, 1988]. Another independent learning method is needed for that purpose. The CEL-algorithm is an independent learning approach, where this is achieved by combining two different methods. First, the model is built up constructively by placing new local units exactly in that region of the feature space, where the mapping performance is particularly poor. Next, a vector quantization algorithm is initialized in a way, that a relatively quick convergence is achieved and that, for the most part, so-called dead units can be avoided, which are not useful, because they are far out of the meaningful area in the feature space to be captured by the model. The vector quantization is realized by an adaptive version of the so-called EM-algorithm [Bishop, 1995]. This method leads to an adaptation of the center and the width of the receptive fields.

Constructive Learning

As already mentioned, the CEL-algorithm is based on a constructive approach. In this case no fixed size of the model is given. Rather, according to a certain growth criterion, new units are created successively, only if necessary. The goal of the constructive model expansion is to find a model structure which is best suiting the learning problem. In addition, the growth criterion offers a potential to avoid the problem of stability while still warranting plasticity of the learning process. This approach has a higher learning speed than most classical approaches.

The constructive approach of learning can be described in more detail as follows: When the learning process is started the model does not yet have any hidden units. The growth criterion decides whether a new unit shall be added. This criterion reflects the *modelling performance* on the one hand and the *level of membership* subject to the model on the other hand. With respect to the modelling performance two different versions are distinguished. The *current modelling performance* of the model is defined as

$$E_A = \left| f\left(\mathbf{x}_n\right) - y_{sn} \right|,$$

where (x_n, y_{sn}) represents the current observation. The *accumulated local modelling performance* of a unit j is based on all samples $(\mathbf{x}_{j1}, \mathbf{x}_{j2}, \ldots, \mathbf{x}_{jn})$ observed so far for which the center $\mathbf{c}_j$ is the nearest neighbour to the vector $\mathbf{x}_{ji.}$ It is recursively defined as follows (λ is an adaptation factor and n_j is the amount of samples for center $\mathbf{c}_j$):

$$E(j) = \left(1 - \frac{1}{n_j}\right) E(j) + \frac{1}{n_j} w(n_j) E_A \quad \text{with} \quad w(n_j) = 1 - \exp(-\lambda n_j)$$

Here a combination of both versions is used as a criterion for the recruitment of a new unit. The current modelling performance serves to achieve a fast but

possibly rather rough placement of support vectors. The growth criterion for a new unit requires:

- The actual *modelling performance* is greater than a threshold E_{max} and the *level of membership* $\phi_j(\mathbf{x})$ is less than a threshold θ_-.
- The *accumulated modelling performance* of unit j is greater than a threshold E_{max} and *level of membership* $\phi_j(\mathbf{x})$ is less than a threshold θ_+.

Thereby, according to Figure 112, three levels of membership (little, medium, strong) are defined by $\arg\max \phi_j(\mathbf{x})$. For each new unit, the area of the receptive field is determined dependent on the neighbouring units and a factor of influence $\delta(t)$. The factor of influence is initialized at the beginning of the learning process at a high value in order to achieve a good placement of the receptive fields. During the training phase this factor is reduced with a time constant τ until it has reached a minimum value, according to

$$\delta(t) = \max\left(\delta_{max} \exp(-t/\tau), \delta_{min}\right).$$

The initialization of the learning process is aiming at suitably setting the width parameters of the receptive fields based on a neighbourhood heuristic only effective for this purpose (see Figure 112).

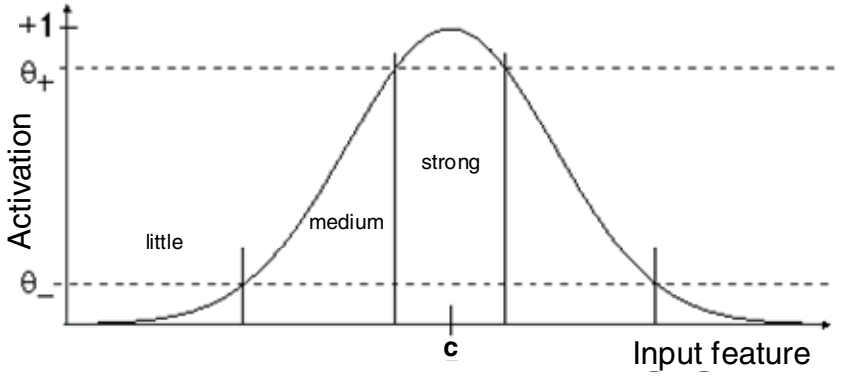


Fig. 112 Parameters of the constructive learning process

Further refinement of these parameters happens while the learning process is running. This differs from [Moody, 1988]. This method of adaptation has some advantage, not only because the number and the positions of the receptive fields change continously during the learning process, but also because the choice of widths in accordance with the neighbourhood leads to extraordinarily variable distances [Sarle, 1994]. Thereby, undesirably large overlappings can appear,

though. For this reason it is advantageous to use a clustering-algorithm in a way as it is realized in this approach.

Examples of Applications

In the following example a representative result is shown for the learning algorithm described (see Figure 113). It focuses on the action pattern of "car following" in an urban environment. This pattern is learned, when there is a vehicle in front on the own traffic lane, which prevents from driving at the desired speed.

For this situation the learning process uses as input variables: the distance d to the vehicle driving ahead, the relative speed Δv and the velocity of the front vehicle v_f. These are presented to the learning component at a time delay of T seconds, thereby accounting for the reaction time of the driver.

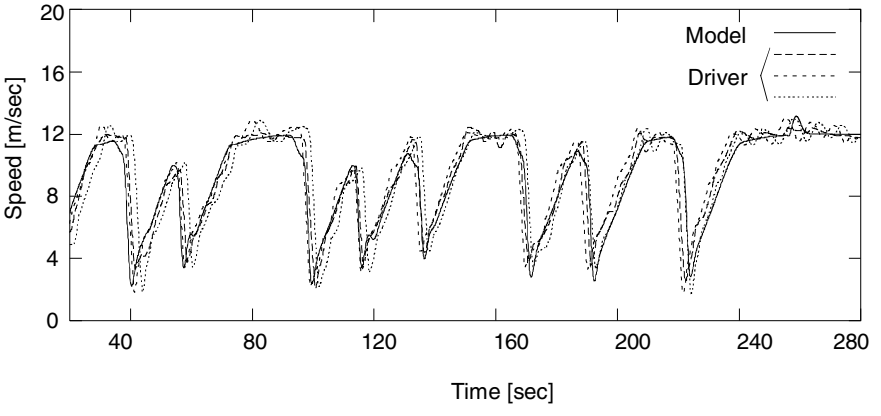


Fig. 113 Behaviour of the driver model in the closed loop simulation [von Garrel, 2003]

In order to demonstrate the quality of the dynamic model behaviour in a closed loop (for instance to be used for automatic vehicle control based on the model learned), Figure 114 shows the speed and distance behaviour of the model of a test driver after a certain period of learning (adaptation) in comparison to the actual behaviour of this driver himself. There is no previous classification. The output of the model as well as of the driver is the acceleration of the own vehicle. The CEL-algorithm processes these data in real time. After a training phase of about 20 minutes the resulting behaviour of the driver model becomes very similar to that of the driver. Only 30 units have been used for modelling.

We have to take into account that the driver's behaviour is deviating within a certain range. Therefore, there cannot be complete identity between the model and the driver behaviour.

Using a section of the test drive, the left part of Figure 114 shows the speed chosen by the driver in comparison to the driver-adapted model. In the right part of Figure 114, for comparison, the complete trajectories of the test run are

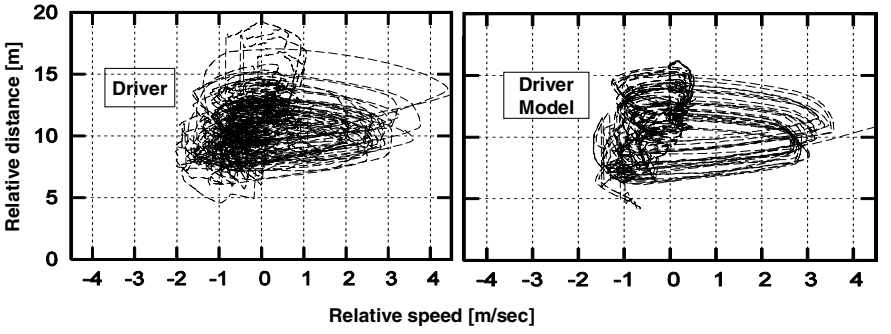


Fig. 114 Behaviour of the driver model in the closed loop simulation [von Garrel, 2003]

depicted in the Δv-Δs – phase diagram. The high similarity between the driver behaviour and the model is obvious. The driver and the model mainly stay within the same region of the feature space. Maxima and minima with reference to the relative speed and relative distance are essentially identical.

6.2.2 Components with Emphasis on Procedure-Based Behaviour

The knowledge representation about procedure-based behaviour is the most commonly known by designers of cognitive automation. This knowledge can most easily be aquired from experts of the application domain. In the following, examples of own implementations are described. Of course, the methods used for these examples are not mandatory ones. There are a great number of methods available to choose from. Here, the main emphasis is laid on the issue to demonstrate that the knowledge representation about procedure-based behaviour is very straightforward and feasible with rather little cost.

6.2.2.1 Piloting Expert Implementation in CASSY and CAMA

The cockpit assistant systems CASSY and CAMA represent examples of cockpit assistant systems for co-operating with the 2-man pilot crew of a transport aircraft on both the procedure-based and concept-based pilot behaviour level (see Figure 27). Together with the pilot crew they are part of the operating force of the pertinent work system which also comprises the aircraft with all its equipment as operation-supporting means. The work objective is a typical flight objective for a transport aircraft flying under instrument flight rules (IFR), for instance on a flight from Frankfurt to New York.

For the development of such a system as an example of mode 2 cognitive automation as much knowledge as possible has to be implemented as a-priori knowledge concerning the human operator(s) to cooperate with. This includes a model of pilot behaviour when flying under instrument flight rules. This section will focus on the model for the pilot's procedure-based behaviour resulting in a model of normative procedure-based behaviour and his individual deviations from that (see [Ruckdeschel, 1997] and [Stütz, 1999]). By the way, a model of skill-based pilot

behaviour was not needed for that application in CASSY and CAMA, since the pilot actions as such were considered as discrete ones by pushing buttons and inputting setups of the autopilot as direct consequences when applying the rules. The cognitive behaviour concerned is that of situation assessment, goal determination, problem solving/planning, and plan execution. In the following, we give a survey on the procedure-based pilot behaviour model of plan execution as part of this kind of a-priori knowledge. Since the pilot actions as such can be considered as discrete skill-based ones by pushing buttons and inputting setups of the autopilot as direct consequences of rule applications, the model is implemented by use of Petri nets in a module called Piloting Expert (PE). With CASSY only the normative behaviour model was covered, CAMA provided both the normative and the individual behaviour model.

Knowledge base
Pilot behaviour in plan execution can be separated into the situation assessment and action processing components. Some other special functions are also part of the knowledge base. Behaviour modelling is done for all flight segments (taxi, takeoff, departure, cruise, approach, and landing) and concerns the following tasks:

a) modelling of situation assessment:
   - recognition of actual flight segment
   - recognition of process of plan execution related to flight plan and procedures

b) modelling of pilot actions/pilot performance:
   - primary flight guidance (altitude, course, airspeed, power setting, climb/descent rate, and pitch attitude)
   - operation of flaps, landing gear, speed brakes
   - radio navigation
   - communication with air traffic control

c) modelling of special functions:
   - callouts (often important checkpoints, e.g. altitudes)
   - checklist processing (normal, abnormal, emergency)

Analysis of knowledge base
To choose adequate modelling formalism, the pilot tasks were analyzed with regard to causal, temporal and structural relations. This analysis gave the following characteristics:

- piloting tasks are *strongly concurrent*. This can be stated in the domain of situation assessment as well as in parallel processing off several tasks (e.g. maintaining altitude, reducing airspeed, radial tracking, ATC communication).
- processing of pilot tasks (e.g. radio navigation) is driven by situation-dependent choices of different rule domains (e.g. cruise navigation or approach navigation), this is *a choice between (excluding) alternatives*.
- the basic element within the considered tasks is always a causal relation, which can be formulated by a *production rule* (if …, then…).

- the situation space as well as the pilot's action space can be described by discrete states (e.g. flight segments, flaps setting) and *state transitions* (e.g. flight segment transition, flaps setting transition).
- state transitions are driven by *discrete events* (e.g. "passing station X", "reaching altitude Y", "system Z breakdown").
- pilot behaviour can be broken down into *several levels of abstraction*, like flight segments and their decomposition into sub-segments in the domain of situation assessment as well as holding procedure and its decomposition into single actions in the domain of pilot actions.

Representation of knowledge

One of the most important objectives of this modelling activity was to obtain a homogeneous representation of the considered pilot behaviour. Homogenity should be required with respect to low expense for software tools and - if available - to enable formal analysis methods. It is obvious that the knowledge representation method to be chosen must be adequate to the system characteristics named above.

In former systems knowledge was often represented by production rules and so-called production systems. However, production systems become difficult to control, if the number of rules increases. Reasons must be seen in the lack off methods for structuring and decomposition. Finally, concurrency cannot be represented explicitly by production rules. Thus, solely modelling pilot behaviour by production systems is no longer adequate.

Another alternative was the use of finite automata. However, the number of states which had to be modelled explicitly is enormous in view of the concurrencies to be considered.

These considerations led to the choice of Petri nets by making use of different Petri net classes, adequately matched to the particular properties of the knowledge domains considered [Reisig, 1992]:

- a considerable part of the knowledge is well representable by condition/event-nets.
- another part of knowledge, even for modelling of multiple resources, requires at least use of place/transition-nets.
- finally, a further part can in principle also be formalized by place/transition nets, however for multiple identical model structures individual tokens are demanded and thus the application of high-level nets is suggested.

To keep the model complexity and the expense for net tools (primarily of the real-time tools) within limits, at first the class of place/transition nets was chosen and used extensively for modelling.

Recently several Petri net applications in the domains of aviation and aerospace arose (see for instance [Lloret et al., 1992]). Modelling is done for simulation as well as for analysis purposes, partly by high-level nets. Regarding the criticality of aviation/aerospace software and with respect to safety and the resulting certification processes, further - even industrial - applications should be expected in the future.

Model design process

When applying Petri net theory to a concrete technical process there will be encountered the problem of missing general rules concerning the formulation of application knowledge by Petri net constructs. In general, the question is: how does the transformation real world → model look like and which rules have to be applied?

Typical questions arising in the modelling process are:

- which real world components have to/must not/may be formulated as places/transitions?
- which levels of net modularization are suitable?
- how can local testability of a large net construct be secured?

In the following, some characteristics of the Petri net application for the PE will be summarized. Especially, we try to illustrate the design process, beginning with single production rules and rounding up with a hierarchically structured net system.

Semantic of net primitives

Places:

Discrete states in the field of situation assessment and during pilot action procedures are represented by places (C/E-nets). Examples are flight segments ("final approach"), conditions for subsequent actions ("turn right after the passing altitude A") and states of discrete aircraft systems ("flaps 20 degrees"). Multiple resources, e.g. redundant navigation devices, are represented by multiple marked places (S/T-nets). Within the scope of modelling pilot workload, limited pilot resources are also modelled by multiple marked places.

Transitions:

Transitions are used to represent situation state transitions, e.g. between flight segments ("final approach → landing") and discrete aircraft systems ("landing gear up → down). In the domain of pilot actions transitions represent for instance changes between basic tasks ("cruise → descent"), navigation instrument settings and callouts of checklist items (" landing gear down?).

Because transitions are typically used to model state transitions, their firing time is assumed to be zero. In case a state transition time unequal to zero is to be modelled the transition is decomposed into a place and timeless firing transitions.

Tokens:

The distribution of tokens in the places of the net represents the state of the net. They indicate which places are active at the time being. The flow of the tokens illustrates the dynamic behaviour of the net.

Transfer of production rules into net constructs

The knowledge base to start with usually consists of production rules which had to be transformed into net constructs. In the following example the transfer of two simple rules is shown. The rules are:

- IF (flight segment = "Final Approach") AND (altitude < 50 feet over ground) THEN new flight segment: "Landing"

- IF (flight segment = “Final Approach”) AND (recognized crew intent = “Missed approach”) THEN new flight segment: “Missed approach”

Either rule can be represented as place-transition-place construct. The transitions are attributed by external conditions (altitude and crew intent criteria), see Figure 115 left. It is evident that the identical net pre-condition of both transitions (“Final Approach”) lead to a joint net construct (see Figure 115 right). Analogously, if pre-and post-states of different rules are identical, they are connected sequentially.

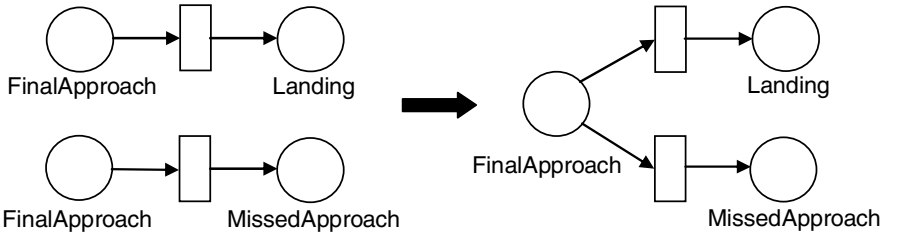


Fig. 115 Petri net representation of two rules as formulated in the text

Modular construction
Size and complexity over the knowledge to be represented by Petri nets for the application domain considered require extensive use of modularization techniques. Several requirements have to be satisfied:

1) subsystems must be testable and analyzable on the local level. For this reason, activation and deactivation of coupling mechanisms is needed.
2) because no tokens may be inserted or removed dynamically, or subsystems have to be strongly connected and marked.
3) token flow between subsystems is not allowed. Nevertheless, implicit token flow is realized by the activation of subsystems.
4) requirement 3) implies that access to places of other subsystems is only permitted as read-only.

The literature names different kinds of modular net construction, e.g. place and transition refinements, place and transition fusion sets, invocation transitions etc. [Huber et al., 1990] and [Vogler, 1992]. According to the mentioned requirements the place and transition fusion sets are chosen for the application at hand. This method allows multiple graphical representations for a place or a transition, even in different nets.

Modular construction often requires access to state or state transition information established in other subsystems. This information must be accessed in a read-only way. Thus, we use two coupling mechanisms, both construed of place and transition fusion sets.

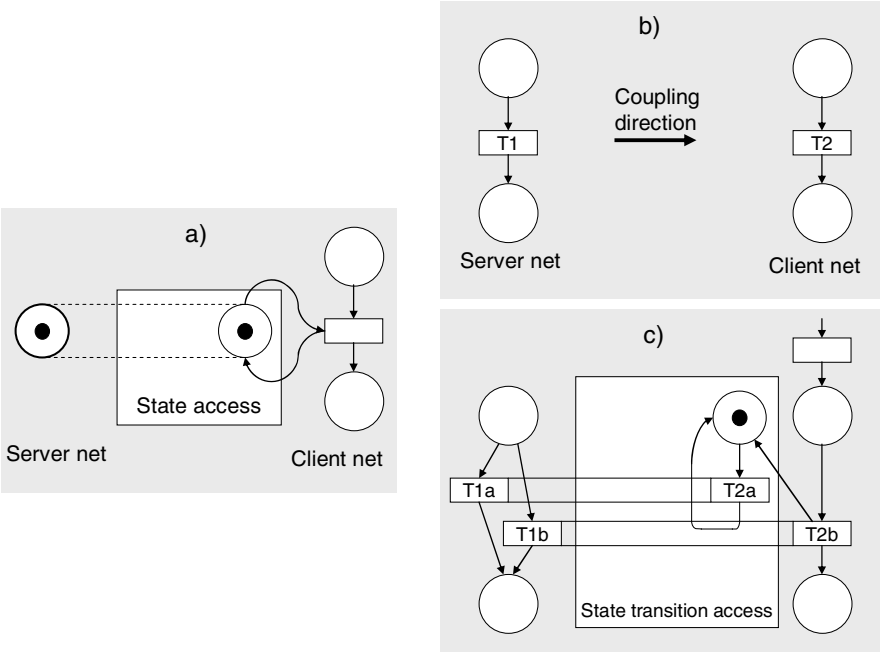


Fig. 116 a): Read-only access to state: The required state information is imported into the client net by place fusion. The place can now be accessed by test (double) arcs. No token flow between the nets is allowed. b) and c): Read-only access to state transition: A coupling mechanism is used to model a unidirectional dependence between two transitions (see b)): Firing of T1 should be a pre-condition for firing T2, but T1 should fire independently. The construct is realised by splitting transition T1 into T1a and T1b and by importing them into the client net via transition fusion (T1a, T1b). A place complement is needed to guarantee firing of T1a and T1b. The state transition of the server net (firing of T1a and T1b) is not restricted by the state of the client net, while firing of T2b is coupled with the state transition of the server net (firing of T1b) (see c). Such coupling is often applied for reset purposes. [Ruckdeschel, 1997]

Hierarchical construction

The design over the net model is done in a top-down way. In many cases already modelled behaviour aspects have to be expanded by a more detailed model. Of course, net models cannot be extended boundlessly (graphical representation, testability etc.). Thus, it has to be decided which parts of the net model are suited to be located in a subsystem and which coupling mechanisms are applied. In many cases it is desired to refine a state which is represented in the coarse net by a single place. A direct replacement of the coarse state by a subsystem does not comply with the modularization requirements mentioned above (primarily requirement 1)). Since the coarse state carries semantic information (e.g. access by other nets), it is essential not to substitute the coarse state (as done by place refinement). For these reasons we refine states by duplicating their “interface” transitions into a

subsystem, where the coarse state is modelled in more detail. In case the coupling mechanisms are deactivated, this construction preserves the behaviour of the properties of the coarse net. To fulfil requirements 1) and 2), a marked complement place is added to the subnet (see Figure 116). Coupling of the two nets is done by transition fusion sets of the interface transitions. This construction is an extension of pure place refinement (see [Vogler, 1992]).

Figure 118 illustrates this technique using the example of Figure 115: we consider the place "Finalapproach", i.e. the decision between "Landing" and "MissedApproach" has to be modelled in more detail.

The upper path of the subnet in Figure 117 represents a natural decomposition of the flight segment "Final Approach". This is done to pay regard to the outer marker (OM) beacon. A recognized crew intent "MissedApproach" is told to the net by a message from the PIER module (see Chapter 5.3.1.4). This message has to be received concurrently to the described flight segment decomposition. This is modelled by the places "WaitForCrewIntent", "CrewIntent" and the transition "recv message" (lower path of the subnet). In case no crew intent message is received, the flight segment "FinalApproach – BehindOM" terminates regularly by firing of transition T2. In case of a crew intent the subnet terminates via transition T3. Different actions are performed dependent on the actual flight segment (transition T4, T5).

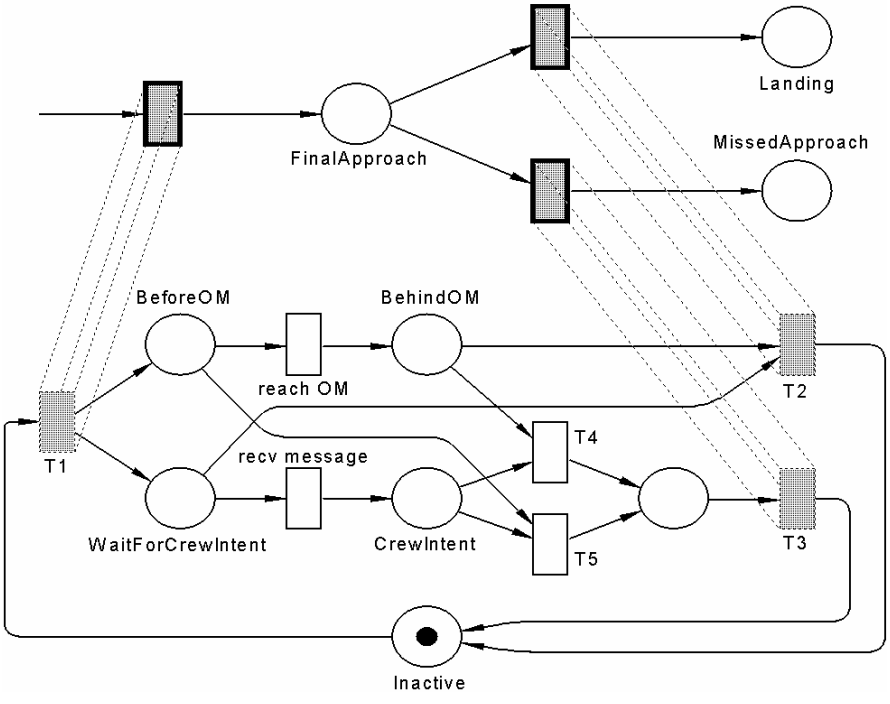


Fig. 117 Refinement of place "Final Approach" [Ruckdeschel, 1997]

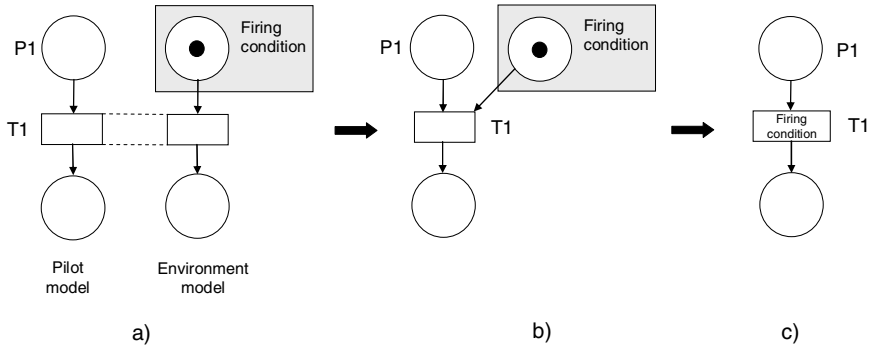


Fig. 118 Compressing the representation of a firing condition from a) to c) [Ruckdeschel, 1997]

Process interface

Modelling of pilot behaviour in plan execution can be separated into a situation assessment and action components. As a basis for a rule-based situation assessment model a discrete situation space with well-defined state transitions has to be established.

The procedure-based situation assessment process is characterized by a permanent consideration of all possible state transitions with regard to the actual situation state vector. The state transitions are typically defined as discrete limits within the - more or less - continuous state space of aircraft and environment (e.g. “passing station X”, “reaching altitude Y”).

For the assessment of the actual situation the state transition itself is sufficient. Nevertheless, the processes leading to the state transition influence the dynamics of situation assessment. Obvious questions like “what is reached earlier – station X or altitude Y?” show that the causal structure of the underlying processes has the essential effects on the assessment results.

Relative to real-time situation assessment this “why?” of state transitions can be neglected. But for an overall investigation of the dynamics of situation assessment, the causal structure of aircraft and environment has to be made accessible to analysis methods. This means these systems have to be modelled by Petri nets, including qualities like “x happens before y”. This extended modelling could be done at a later development stage.

For real-time situation assessment state transitions within the net model have to be executed dependent on external (aircraft/environment) conditions, in the following called “firing conditions”. These firing conditions can be understood as states within a - not realized – aircraft/environment net model (Figure 118a). These two net models can be connected by a common transition. The marking of the places P1 and ’firing condition’ enables the firing of transition T1. With regard to the firing of T1 the net structure containing ‘firing condition’ can be neglected. This leads to a compressed representation (see Figure 118b). A disadvantage of this representation is that a token has to be inserted when the condition occurs. Besides, if the condition is left without having fired T1, the token has to be removed to avoid

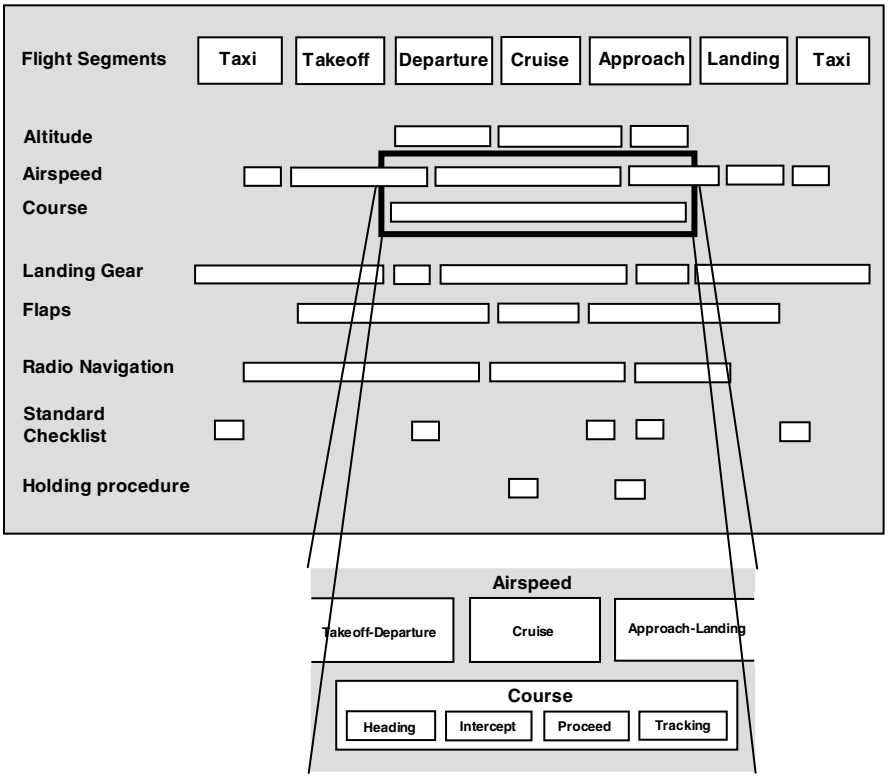


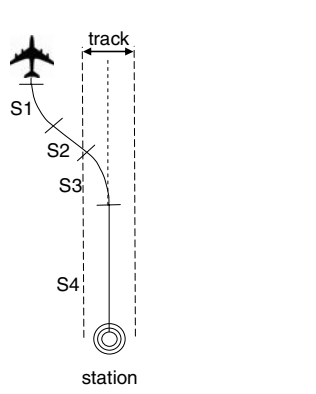
Fig. 119 Model structure [Ruckdeschel, 1997]

subsequent, faulty firing. For this reason we formulate this firing condition briefly as transition attribute (guard) (see Figure 118c). A transformation back to the other representations, e.g. for analysis purposes, can easily be done.

Model structure

Figure 119 gives an overview of the strongly simplified model structure. This structuring of the net model was done according to knowledge structures as far as possible. The primary determinants of structuring are the pilot tasks within plan execution: recognition of flight segment, primary flight guidance (altitude, course, airspeed), system operation (gear, flaps, radio navigation) etc. These are typical examples for concurrent tasks.

To come up with subnets of handy size and complexity (not more then 10 to 15 places, reset constructs excluded), most of the tasks need further subdivision. For this purpose subclasses of behaviour within the main tasks had to be identified. An efficient structuring was done by separating behaviour with regard to different situation characteristics. The resulting behaviour classes are always related to excluding situation elements. Therefore, they are exclusive alternatives. The situation characteristics can mainly be attached to two groups: flight segment subsets



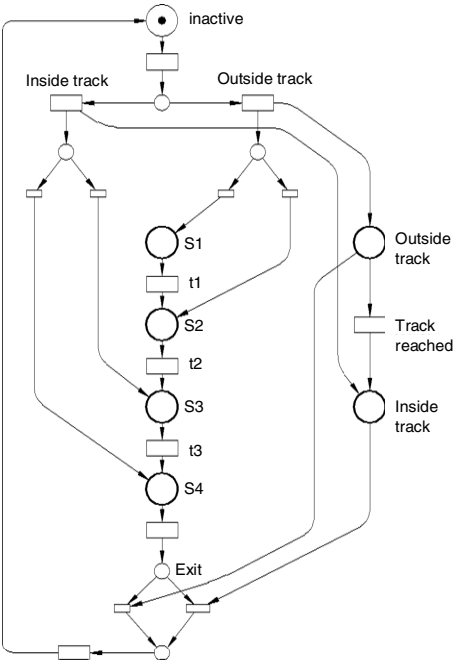


Fig. 120 Petri net for the "intercept procedure" [Ruckdeschel, 1997]

(e.g. departure airspeed behaviour) and instructions derived from the flight plan (e.g. course behaviour for "proceed to station X").

Figure 119 shows the concurrent task models in the vertical axis. Alternative (exclusive) sub-models related to flight segments are drawn in horizontal axis. Sub-models are represented by white boxes; best size, complexity, and hierarchical depth (i.e. number of subnets) differs widely. Nets for co-ordination purposes and their couplings with task model nets are also neglected in this illustration.

A small part of the model structure is zoomed out and discussed in more detail. Figure 119 shows the course model and a part of airspeed sub-models. Subdivision of airspeed behaviour is done related to flight segments (takeoff-departure, en route, approach-landing subnets). The actual course selection behaviour class is derived from the flight plan and is a choice between "turn to heading H", "intercept course C of station S", "proceed to station S", and "tracking from station A to station B". A simplified "Intercept" subnet is described in the following as an example (see Figure 119).

Example

An intercept is carried out to reach a given (magnetic) course to a given station (e.g. a radio navaid). This can be required within published departure or approach procedures or can be caused by an instruction of air traffic control. In the general case, an intercept covers four sections (see Figure 120 left): turning to a special intercept heading (S1), maintaining on intercept heading (S2), turning to given

course (S3), tracking on given course (S4). Sections are skipped if the aircraft fulfils the characteristics of a following section, e.g. if the aircraft is already on intercept heading at the time the procedure is started.

In addition to this heading behaviour, the given course to the station relative to the ground and an admitted lateral deviation specify a track. After having reached this track, the aircraft should not leave the track until a new lateral procedure is started (see Figure 120 right). In this example it is assumed that the transition point between section S2 and S3 (start of second turn) is always placed outside the track.

The intercept net is mainly characterized by two concurrent constructs:

- a sequence of four places ("S1" to "S4") represents the four subsequent behaviour sections described above. The transitions connecting these states are attributed with heading conditions or other lateral conditions.
- the places "OutsideTrack", "InsideTrack", and the transition "track reached" are used for the modelling of the tracking performance mentioned.

When the net becomes active several conditions have to be considered within the initial section of the intercept procedure (not discussed here in detail). A further concurrent construct is needed to enable a net reset from all (stable) states, for instance in case of a changed flight plan (not shown in Figure 120).

As final result of the modelling process the model comprises more than 2000 places and nearly 3000 transitions in 175 subnets.

Modelling of individual pilot behaviour

As already pointed out in Chapter 5.3.1.4 the Piloting Expert module in CAMA consists of two kinds of models, the normative and the individual model. In the following we will focus on the modifications of the normative model which are to be taken care of to provide the individual version. The normative model essentially describes deterministic pilot behaviour as documented in pilot handbooks and air traffic regulations [Ruckdeschel, 1997]. Modelling is done primarily within the domain of procedure-based behaviour. Skill-based behaviour is coarsely modelled as step function control actions, thereby also accounting for admissible tolerances. The individual model adapts to the procedure-based behavioural parameters of the individual pilot. It is developed as a real-time adaptive component [Stütz, 1999], aiming at a customised model. This is achieved by learning from observed behaviour examples, making use of Case-Based Reasoning (CBR) as inductive machine learning method [Kolodner, 1993]. The Piloting Expert module therefore has become a hybrid Petri net/CBR system. Fundamental for the realisation of this adaptive model is the assumption that normative regulations and procedures are still the guideline, even when they are slightly modified by the individual pilot. This process of customising can be done in three stages:

- varying the state parameters of the Petri net (online)
- varying the transition parameters of the Petri net (online)
- varying the state-transition structure within a Petri net (offline)

Parameter variations from person to person may be rather frequently, whereas the variation of the state transition structure is a very rare case for this application

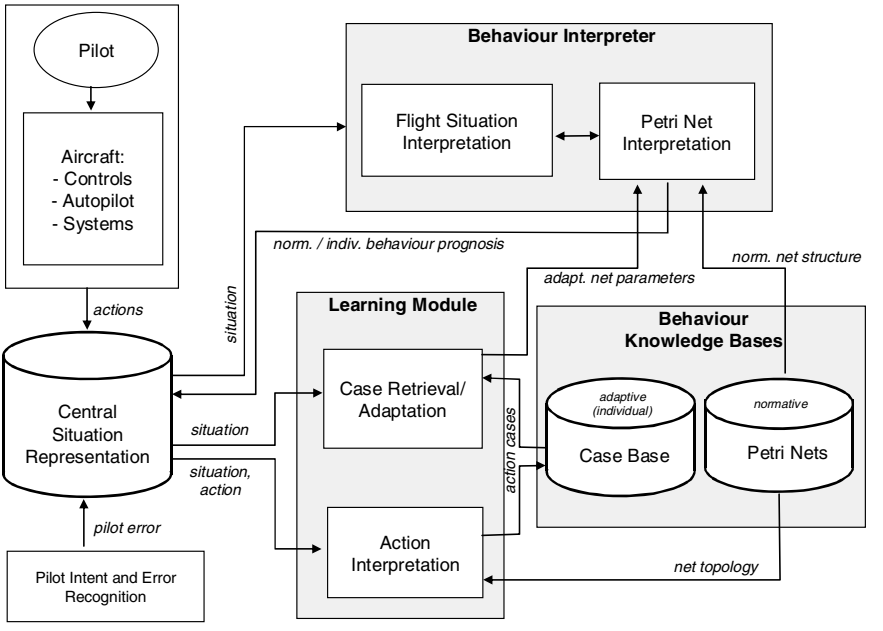


Fig. 121 Modules of normative and adaptive piloting model [Stütz & Onken, 2001]

of IFR-flight. Figure 121 shows the modules of the piloting model covering both an adaptive and a normative model.

The action interpretation module is the basis for supplying the case base with a constant stream of discrete pilot action events. Detected actions are classified as state transitions and attributed according to their storage format (Figure 122). A commercial SQL-database-tool is used for case storage management. Erroneous actions are eliminated using the output of the assistant system's pilot error recognition subsystem (PIER).

During the problem analysis the problem to be solved has to be described in a characteristic way to distinguish it from others. This leads to its comparability and enables the search process. The "feature extraction" is a part of almost all problem solving concepts. Applying similarity metrics to these features, *case retrieval* (or initially match) isolates these cases in the *case base* which are considered to be "similar" to the actual problem description. Emphasis is given on fast retrieval. The result is normally a whole bunch of cases. *Case selection* then figures out the case that is most promising for further processing. *Case adaption* aims at making the selected case reusable for the current problem. Either the past solution (transformal reuse) or the past method that constructed the solution is copied or adapted (derivational reuse) [Aamodt, 94]. Typically, *storage* successfully adds solved problems as new cases to the case base.

The *retrieval and adaptation* module in Figure 121 provides the online access of the Petri net system to case base. This comes about in a way to preserve the

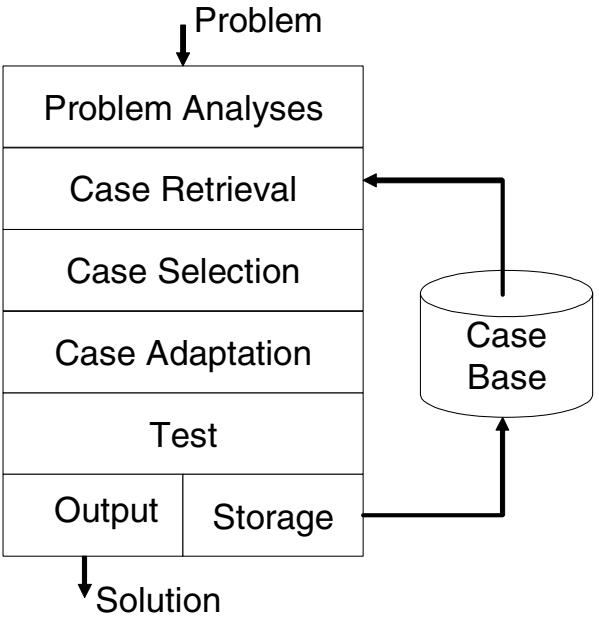


Fig. 122 Processing steps within CBR [Stütz & Onken, 2001]

overall normative task sequence, but also to take into account individual, admissible deviations. For illustration, it is considered that the Petri net pre-conditions of the transition concerned (e.g. flaps0 → flaps15, fly straight → turn) are fulfilled. When the pilot model is running, the firing condition can now be acquired during runtime from the examples in the case base given for the individual pilot. Thereby, it is making use of observations which were just recently collected.

Figure 123a shows a flight situation for a typical decision task. Given a lateral deviation from the current leg of the planned track, the pilot has to decide whether to intercept the current leg (*Intercept*), to directly proceed to the next waypoint (*Proceed*) or to disregard the current leg and to steer towards the following leg (*Exit*). Relevant situational information (Figure 123b) to be considered for this decision is the aircraft's relative position to the current flight plan leg as defined by the cartesian values *dist_basis* and *dist_track*, the aircraft's speed (*ias*), the aircraft's current heading relative to the actual track (angle β), and the change of the track course from the current leg to the following one represented by the angle α.

Figure 124 shows the respective Petri net indicating that the pilot is most likely to proceed directly to the next waypoint. A normative functional relation between the situational attributes and the manoeuvre decision is hard to achieve, as objective rules and recommendations do not exist. This freedom typically favours the formation of individual pilot customs such that a rigid normative model looses its validity.

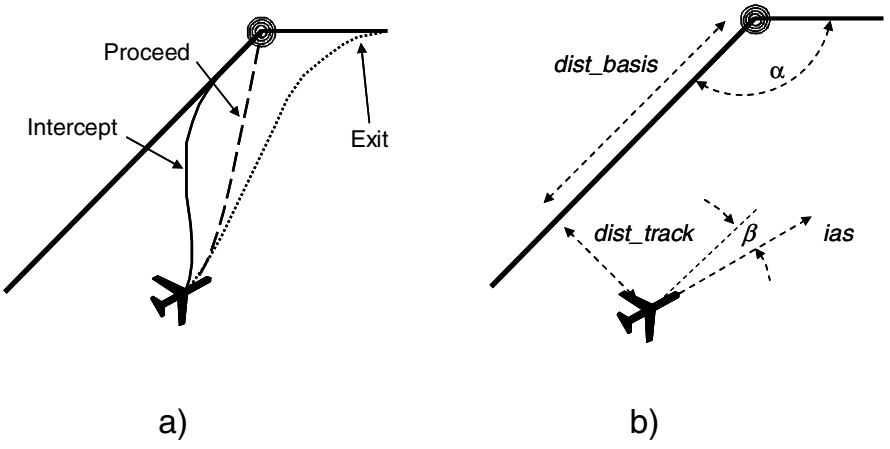
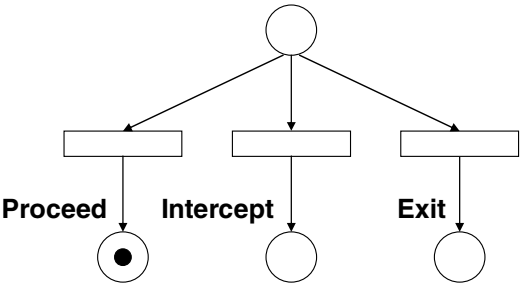


Fig. 123 Typical example of a flight situation with considerable cross track deviation requiring a decision for one of the manoeuvre alternatives "Intercept", "Proceed", or "Exit": a) Typical flight paths associated to the respective manoeuvre alternatives. b) Situational attributes designating the flight situation. [Stütz & Onken, 2001]

Fig. 124 Petri net fragment from net Basic_Tracking_Off_Track for the decision between the manoeuvres "Proceed", "Intercept", and "Exit".[Stütz & Onken, 2001]



The way to tackle this problem in the sense of case-based reasoning is to see in which situation (described by the above mentioned attributes) a pilot made these decisions in the past and to conclude that he will come to the same decision again in a similar situation coming up.

Past manoeuvre cases are being stored in the case base by the manoeuvre interpretation function in the format as shown in Table 12:

Table 12 Case format and inserted sample

Manoeuvre description		**Situational attributes**		
		required		supplementary
pre (heading):	337°	*dist_track* (cross track deviation):		*alt (altitude)*: 12180 ft
post (heading):	300°	0.66 nm		...
type:	Intercept	*dist_basis* (distance to waypoint):	10.21 nm	
		α (angle between current and next leg):	-56°	
		β (angle between planned track and A/C headg.)	14°	
		ias (speed):	200 kts	
		fph (flight phase):	Enroute	

In order to compare stored case attributes with the current situation, local similarity values are computed individually for all attributes. These can be of metrical, ordinal or nominal nature. They are combined to a global similarity *SIM*. For the given example the global similarity is made up from the local similarities *sim* and corresponding weights *w* for the attributes position, alpha, airspeed, beta and flight phase:

$$SIM = \frac{w_{pos} sim_{pos} + w_{\alpha} sim_{\alpha} + w_{ias} sim_{ias} + w_{\beta} sim_{\beta} + w_{fph} sim_{fph}}{w_{pos} + w_{\alpha} + w_{ias} + w_{\beta} + w_{fph}}$$

Note that the flight phase attribute (fph) is not a metrical one. Coming back to the example of the situation depicted in Figure 123 one can assume that the aircraft for some reason deviated from the planned track beyond a certain threshold. This causes the Petri net interpreter to first invoke the aforementioned net Basic_Tracking_Off_Track and its fragment as shown in Figure 124, and then to issue a request to the retrieval and adaptation function together with a description of the actual situation in order to conclude for the most likely pilot reaction. This function now tries to retrieve suitable course manoeuvre cases and sorts them according to their situational similarity. The manoeuvre type of the most similar case (e.g. Intercept) is than passed back to the interpreter and the respective transition is allowed to fire.

Differences in individual behaviour

Figure 125 clearly shows the differences between the manoeuvre type solutions produced by the adaptive models for individual pilots. Obviously, the model for pilot B indicates a much stronger tendency compared to pilot A to directly proceed to the waypoint when positions on the outside are considered. Even at quite large dist_basis values the model does not favour the Intercept option. Another peculiarity is that the model does not foresee at all the *Exit* manoeuvre for pilot A when being on an outside position.

To get more information on the modelling validity, the pilots were asked to prepare a rough subjective drawing indicating the areas of their manoeuvre preferences. Overlaid on the results of the pilot model, the drawings indeed confirm the effects mentioned. Pilot B admits himself a quite large area where he would rather choose the *Proceed* option than to intercept the planned track. His decision is different to that of pilot A, who prefers the *Intercept* option up to a distance of about 8 miles to the waypoint. This threshold value was stated identically by the pilot model when considering small *dist_basis* values. Likewise pilot A ruled out the *Exit*-option for his course behaviour on an *outside* position and the pilot model was able to reproduce the *Exit* area for pilot A on an *inside* positon almost identically.

Analysis - goals and results

When the development was started, model testing was done mainly by simulation. Only few simple properties like connectedness were checked automatically after passing the net declaration. Simulation tests have to be carried out anyway, even to check numerical results and the interfacing of the pilot model. However, the

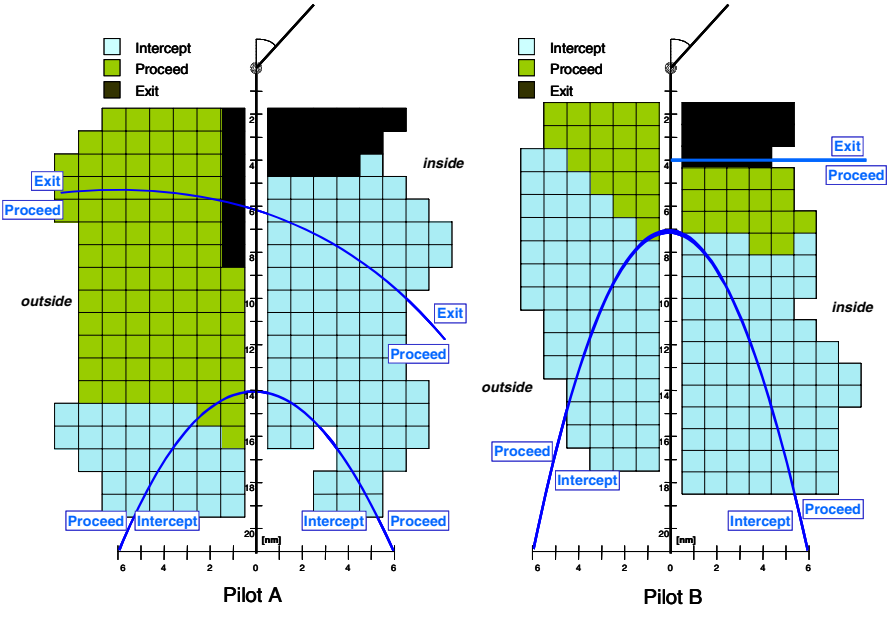


Fig. 125 A typical result of simulator experiments for modelling of individual pilot decision behaviour regarding Intercept (light blue), Proceed (green), and Exit (black) manoeuvres of the two test subjects, pilot A and pilot B. The corresponding flight situation is given by the $|\alpha| = 45°$, $\beta = 0°$, ias = 200 kts, fph = Enroute. The position left and right off the track is labelled in this case as outside and inside respectively. The areas separated by blue lines and labelled in blue colour with Proceed, Intercept, and Exit depict the pilots' subjective estimate of manoeuvre preferences as stated by them independent from the simulator trial. [Stütz & Onken, 2001]

main problem of testing only by simulation runs is that it is impossible to reach all (critical) net states and state transitions within the test run. With respect to reversibility all net states are critical, because if a reset condition occurs and the net is unable to perform the reset successfully, this must lead to deadlock or at least to a malfunction of the net and the parent nets.

For this reason formal analysis methods are applied to the net model. The analysis strategy is bottom-up, thus in a first step all subnets are checked to satisfy some obligatory qualities. These are at least:

- strong connectedness
- boundedness
- reversibility
- liveliness and safeness

Strong connectedness characterises a subnet which is clearly delineated from the other subnets, but internally causally coherent. Token flow between subnets is not allowed. A subnet exhibits boundedness, if it represents a limited number of rules in a limited state space. Reversibility of a subnet designates the property to

get back to an initial state of inactivity after having been active. Liveliness warrants that there is no situation modelled by the subnet which can occur only once. Finally, safeness describes that the number of tokens in a place does not exceed a certain maximum value. For 1-safeness the maximum number of tokens is 1.

Analysis was done for all subnets by the analysis tool called INA [Starke, 1990, 1993]. Net analysis proved about one essential defect in every tenth net, mainly incomplete reset structures.

The next analysis step is to combine (already checked) subnets into more complex net systems, step-by-step, and to prove the required properties again for the more complex net. This has not been done yet.

Besides these general properties there are other special properties which can be derived from the pilot model specification. Examples are:

- Exclusive states (e.g. exclusive flight segments, exclusive aircraft system states)
- State sequences: “states S1, S2, …, Sn have to occur/may never occur subsequently” (e.g. flight segments)
- State refinements: “activity of refined state Sr requires activity of coarse state Sc”
- Predetermined reset procedures: ”firing of transition Tr carries over every net marking to the initial marking” (e.g. subnet reset)

These properties can be proved using facts, invariants or special evaluations of the reachability graph (critical with large models). Some of the listed properties have already been investigated on the basis of the reachability graph generated by INA.

For extensive verification a checking tool enabling the formulation of logical terms (numerous and/or operations, negations) and avoiding the calculation of the reachability graph is desirable.

Tools

As a main requirement, the net model, as presented, is to be used not only in the design phase but also as final implementation of the PE-module. For this purpose, a real-time Petri net interpreter was needed. This central role of highly integrated real-time net interpretation is an atypical aspect of this application and has some unfavourable effects on the suitability of commercial Petri net tools. In the following, the main tool requirements are summarized:

- availability of real-time tools (interpreter and monitor) on the hardware platform (Silicon Graphics workstation)
- interpreter interface to program in language C for integration or transition guard/action functions (process interface)
- strict separation of interpreter (simulator) and graphical user interface
- graphical and textual net declaration (especially important to large nets and declaration of transition attributes).

Because of these requirements only a fraction of the commercial tools could be applied.

A description language for place/transition-nets was defined in enabling the declaration of modular constructs and process interfacing by transition attributes (guard and action functions).

By use of the commercial tool Design/OA [Ruckdeschel & Onken, 1993] a graphical editor was developed which supports the required coupling and refinement methods and the local treatment of sub-systems.

The net interpreter is integrated in the real-time data processing of the assistant system and does not have any graphical interfaces. The central requirement for interpreter development was to gain response times not dependent on net size and nearly independent of the number of actually acting transitions. The process couplings are achieved by use of an open interface to programming language C. Transition guard and action functions implemented in C are automatically linked to the net data structures and to the interpreter kernel.

Debugging of net simulations is supported by a graphical monitor system using OSF/Motif. The monitor receives actual marking information from the interpreter. Transition firing (overwriting of transition guard functions) can interactively be done by the monitor. Special attention was paid to a net activity dependent choice of presented information. Because of system size this information reduction is indispensable.

As presented in the preceding section, the commercial tool INA is used for net analysis.

Use of high-level nets

It was stated earlier that parts of the model gained greater compactness and expressiveness by use of high-level nets. Individual tokens can reduce net external data flow and enable folding of identical net structures. By use of different interpretations (algebras) of the same net structure a very compact model can be construed. An example for the re-modelling of a pilot behaviour aspect by use of high-level nets is presented in the following.

Checklist processing of pilots was modelled by place/transition-nets. The nets have been simplified for better clearness of description (see Figure 126 through 100):

Checklists can be disabled, for instance, within inadequate flight segments. A checklist is enabled as soon as the checklist processing can be allowed regarding the flight segment and further situation elements. The checklist processing itself is started when the pilot requests the check or when a timeout condition occurs, e.g. the latest point within flight progress to do the check (see Figure 126). After the check has been started, a sequence of n checklist items is treated. This is done for each item in the same way and modelled in n structurally identical subnets (“Item N” in Figure 127). These subnets hold the complementary places “ItemSet” and ”ItemNotSet” as an initial state (the reset of these states is not described here). After subnet activation, the concrete check is done or skipped dependent on the state of the item. If the item has to be checked and the pilot confirms the check (e.g. by speech input), it is made sure that the check has been done correctly (e.g. by checking the aircraft system state). If the check was successful, a state transition to “ItemSet” is performed. Subsequently, the subnet “$Item_{i+1}$” is activated simultaneously.

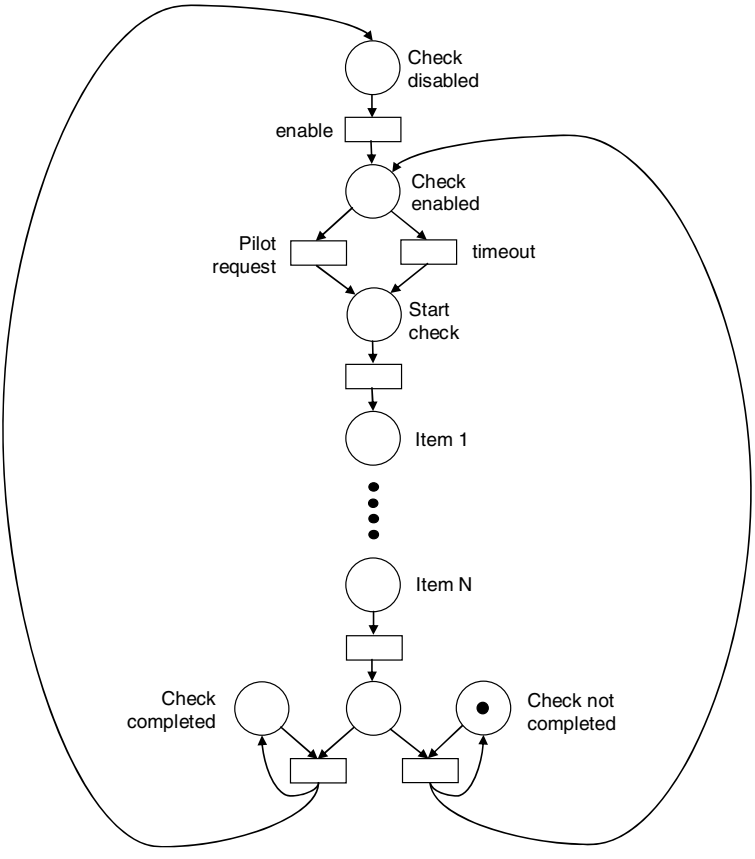


Fig. 126 Checklist subnet "Frame" [Ruckdeschel, 1997]

It is essential to know whether all checklist items have been checked successfully or if there are items remaining. These states are represented by the complementary places "CheckCompleted" and "CheckNotCompleted" and are evaluated in the "ItemState" subnet in Figure 128. They are made accessible to the "Frame" net by place fusion.

After the last item has been processed, the checklist is enabled again (for later processing over the remaining items) or finally disabled (if the check is complete).

As presented, checklist models contain identical net structures for the treatment of each checklist item. By use of a high-level net class, a first folding step reduces n item-subnets to one item-subnet used for n items. Figure 129 shows the high-level net construct which replaces the highlighted area of the S/T-net in Figure 126. The sequence of n items is replaced by a cyclic structure; variable tokens ($item_i$… $item_n$) represented different checklist items. The item sequence is realized by use of the successor function $SUC(item_k) = item_{k+1}$.

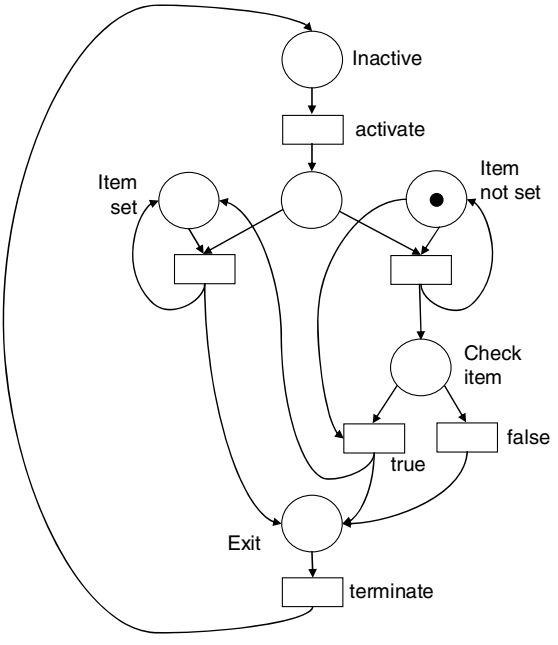


Fig. 127 Checklist subnet"Item N" [Ruckdeschel, 1997]

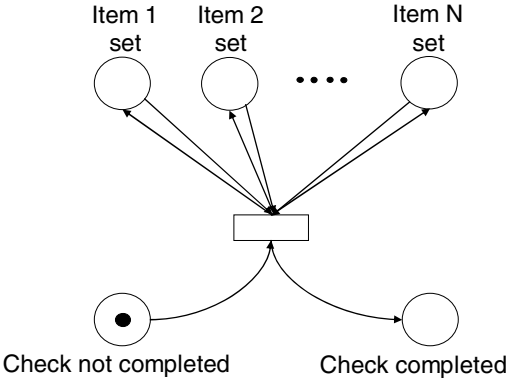


Fig. 128 Checklist subnet "ItemState" [Ruckdeschel, 1997]

After this step all checklist items are modelled by the same subnet, i.e. different transition attributes representing concrete aircraft systems are folded into a single transition. Thus, a transition interpretation (i.e. choice of transition guard and action functions) dependent on firing modes is needed.

Having done this first folding step, all checklist models are structurally identical, they still differ in the number of items and the transition attributes. Thus, a second folding step can be done to reduce m checklist nets to one checklist net used for m checklists. To perform this step, different interpretations of one net structure are required.

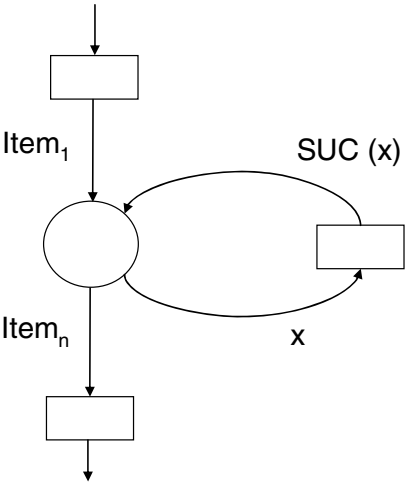


Fig. 129 High-level net construct for checklist modelling [Ruckdeschel, 1997]

To allow these folding steps, the high-level net class must enable variable cyclic structures (e.g., n subsequent checklist items) and variable semantics of a single net structure (e.g., m different checklists).

Another example for the use of high-level nets is the modelling of pilot behaviour in case of navigation instrument breakdowns: this requires a distinction between multiple, redundant devices. Use of individual tokens enables a folding of identical net structures and thus improves model compactness.

The choice of net class and specification algebra was done in co-operation with Reisig.

Conclusions

To judge the actual behaviour of the human operator as part of an overall situation assessment process a reference is needed. In case of the application in CASSY and CAMA this reference is provided by a model combining normative behaviour and individual pilot behaviour.

Resulting from an examination of crew behaviour, Petri nets are considered to be an adequate knowledge representation method. By use of different – in part high-level - net classes model expressiveness can be maximized. Application of high-level nets to parts of the knowledge base leads to considerable reduction of net extent compared to pure place/transition-net modelling.

The great size of the rule knowledge base requires a modular and hierarchical model structure. Some frequently applicable standard coupling and refinement constructs could be established.

First analysis steps detected model faults which could not be located in many simulation runs before. The analysis of connected subsystems enables further model improvements.

The knowledge base is used not only in the design phase but also as part of the implementation. Net data processing is done by a package of - partially commercial - tools consisting of editor, parser, real-time interpreter, monitor, and analyzer.

6.2.2.2 Implementation of Rule Bases for DAISY

In the preceding section a model of human procedure-based behaviour for vehicle guidance is presented which is based to a very great extent on regulations and standard procdures. This is a typical characteristic of the application of commercial flight. For safety reasons there is almost no room for the pilot to individually select a procedure accounting for his own potentially possible ideas and intents. Opportunities for deviations from the rule-base are extremely rare.

A completely different situation arises for the designer of work systems for guidance and control of road vehicles. Firstly, really professional drivers are not representing the majority of driver population as a whole. Secondly, the individual driving style allows for great variations in the decision on the next task to be carried out in a given situation. As to the procedure-based behaviour, individual drivers differ considerably regarding the structure of the underlying rules. For example, the decision to overtake a vehicle or to stay behind in a situation when the own cruise speed is higher, is very different from person to person, depending on their experience and their attitude to take risks. Therefore, modelling of the individual behaviour is a predominant requirement for this application.

Another distinction to the implementation of the pilot model is the fact that for a model of the driver behaviour the skill-based behaviour plays a central role. This is dealt with primarily in Chapter 6.2.1.2 in detail. Thus, for a model of lower level driver behaviour both are needed the procedure-based and the skill-based part. In this section we are dealing with the procedure-based part of a driver model in the first place, leading to the specific alternatives of elementary driving tasks to select from in any driving situation (see also Chapter 6.2.1.2. Again taking the example of being faster than the preceding vehicle, the choice has to be made, for instance, between the alternatives of taking over or staying behind. Whereas the selection process of one of these alternatives is a subjective one of the driver individual, the derivation of these alternatives can be considered as normative, based on rules.

The modelling of the driver behaviour for the assistant system DAISY in its first version was not going as far as to model the generation of situation-dependent driving task alternatives, because it was focused on particular situations of potential accident danger. There, account was made only for the aspect of individual differences in time to collision (TTC) or time to lane crossing TLC), because only the range of allowed actions was of interest in order to provide adequate warnings [Kopf, 1994]. Actions considered for the particular potential accidents to be avoided were simply braking and steering manoeuvres. Consequently, representations of rules were only necessary for the purpose of representing task situations as part of the situation analysis. This knowledge is used to determine the actual elementary driving task on a normative basis, all being part of the cognitive sub-function of situation interpretation in Figure 79, and to generate corresponding alerting advices by way of the remaining sub-functions in Figure 79.

For the implementation of the corresponding rule-bases the representation in decision trees and finite, deterministic automata was applied [Kopf, 1994]. Decision trees are used for the situation representation. According to [Morik, 1990] a

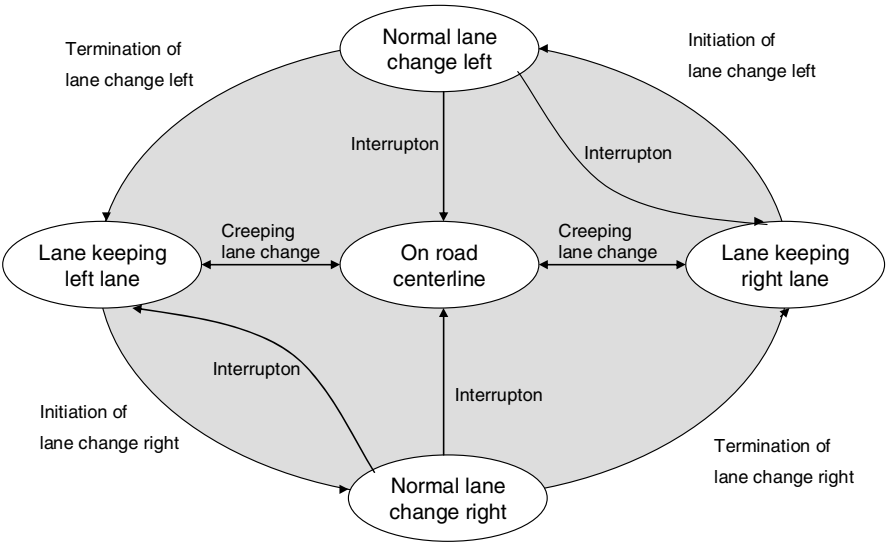


Fig. 130 State transition diagram (finite automat) for lane change/ lane keeping situations when driving on a two-lane Autobahn [Kopf, 1994]

decision tree consists of nodes, ordered in node levels, and arcs. The nodes correspond with attributes, one in each node level, and the arcs correspond with the respective values of the attributes. In our case, the attributes are situation features, simple as well as more complex ones. The values of the attributes may be of metric, nominal, ordinal, or Boolean type. In one of the terminal nodes of the decision tree the situational class is determined which corresponds with the current driving situation. Finite, deterministic automata as state transition diagrams (directed graphs) were mainly applied for the representation of transitions between elementary situational subspaces which are considered to be associated with particular elementary driving tasks. Elementary situational subspaces are considered as low-dimensional subspaces of the high-dimensional space of potential driving situations. An example for a simple graph of that kind is shown in Figure 130.

Later versions of DAISY refined the driver model regarding the situation-dependent elementary driving tasks (see [Feraric, 1996], [Grashey, 1999], and [Otto, 2006]). A simple normative rule-base depicted as a decision tree is shown for longitudinal control when driving on the "Autobahn" in [Grashey, 1999].

The corresponding rules used in [Grashey, 1999] are the following:

Free driving:
Free driving is defined by [Kopf, 1994] for situations when the headway distance to a leading vehicle is greater than the sum of 150 [m] added by the product of the actual speed times 0.5 seconds. If a situation of free driving is true, then the wanted target speed v_{target} is being adopted.

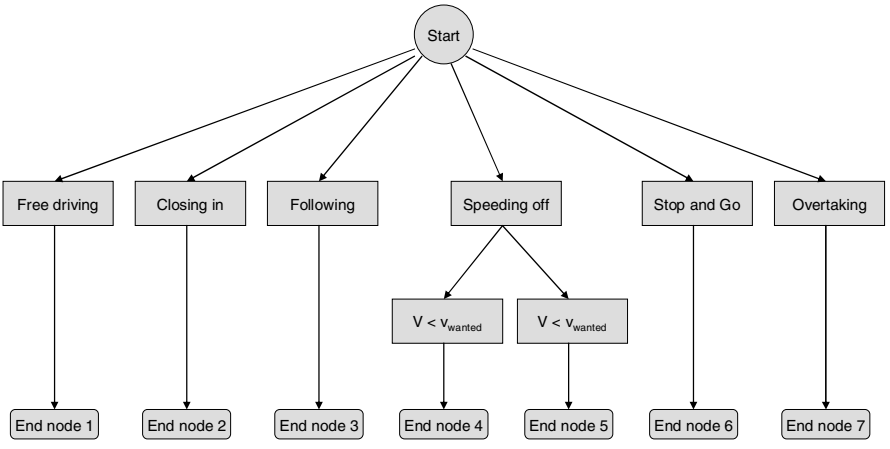


Fig. 131 Decision tree for normative rules in longitudinal conrol on a two-lane Autobahn [Grashey, 1999]

Closing-in:
Closing-in is given by situations where there is no free driving, nor stop and go, and if the speed difference Δv between the leading vehicle and the own one on the same lane is negative. The driver has to reduce the speed in order to avoid a collision. Oftentimes there is a transition to the driving task of following the leading vehicle in safe distance.

Following:
Following is given by situations where Δv as defined for the closing-in driving task is about zero. The driver is typically keeping a certain distance to the leading vehicle. Again, situations pertinent to free driving and stop and go are excluded.

Speeding off:
Speeding off becomes relevant as a driving task, if there is a situation of closing-in or following and if the leading vehicle accelerates, causing Δv becoming positive. As depicted in, Figure 131 there are two modes, depending on whether the target speed v_{target} is smaller or equal and greater than the speed of the own vehicle.

Stop and Go:
Stop and go can be characterized by a situation when both the own vehicle and the vehicle ahead are driving on a speed level below a given low speed.

Overtaking:
The overtaking manoeuvre signifies the situation when changing the lane to pass a vehicle ahead. It is associated with the action of setting the direction indicator and speeding up.

6.2.3 Components with Emphasis on Concept-Based Behaviour

Concept-based behaviour stands for the most complex human behavioural level. Although this kind of behaviour is founded on procedure-based behaviour to a great extent, the representation of the knowledge about relevant concepts is highly demanding. The way the concepts are represented decides about whether the matching of any cues being developed by the feature formation process and pertinent concepts will work satisfyingly. It is of great value for an artificial cognitive system to be integrated in a work system, if adequate concept-based behaviour of the artificial cognitive system can be accomplished. Again, because of this very demanding task there are not many developments of that kind which could be verified as being as useful as wanted. Therefore, the following examples may possibly be of a great yield for the unexperienced reader.

6.2.3.1 Co-operation of UAVs

In Chapter 5.1.2 a project on co-operating UAVs as operation-supporting means was introduced (see [Schulte & Meitinger, in press] and [Meitinger, 2008]). In this project each UAV team member was piloted by an SCU. This SCU prototype has got the explicit a-priori knowledge necessary to participate in achieving the common objective, i.e. a mission order instructed by the human operator as operating force of the respective work system. In the following, more insight is given into the modelling of this a-priori knowledge.

In total, the following capabilities were considered and modelled as packages:

- *Co-operation* in order to ensure co-ordinated team work
- *Communication* in order to carry out certain types of dialogues
- *Problem solving* in order to generate the task agenda
- *Mission task execution*
- *Environment recognition*
- *Safe manoeuvring* in order to avoid the total loss of the UAV
- *Priorisation* of goals, and
- *Operator instruction handling*

Based on the cognitive programming language CPL and the cognitive process library of COSA (see Chapter 7.1) these packages represent the top abstraction level of modelling of the a-priori knowledge. They are usually linked to several of the sub-functions of the cognitive process. Each of them contains pertinent model classes which are associated with a particular cognitive sub-function, though, representing the lowest abstraction level of model representation. A class is structured in a way that one part contains the knowledge about static items, i.e. attributes, and that another part contains dynamical properties, i.e. the behaviour by which an instantiation is established or eliminated, the concrete values of the attributes are set, and relevant interactions with other instantiations are brought about. Modelling of application-specific knowledge is developed separately from that one for use in possibly several different applications like that to be able to communicate with other agencies. Certain models are connected to external server components which are for instance needed to execute numerical calculations and to link to components in the SCU environment, e.g. sensors and vehicle systems.

The total of relevant instantiations of models as part of whatever model packages forms the situational knowledge of the SCU.

The interplay of a-priori knowledge models and their dynamic behaviour resulting in observable ACU behaviour in the end will be explained using two examples from the packages co-operation and communication, respectively. These two packages comprise more than half of the knowledge implemented in all packages mentioned above and are the central packages as far as co-operative capabilities of UAVs are concerned. Each of the following figures shows models of the a-priori knowledge (solid line) as well as instances of these models within the situational knowledge (dotted line). The allocation of classes to model categories (environment models, desires, action alternatives, instruction models) can be derived from the colour of the boxes (yellow, blue, green, red) as well as from the tag at the top of the class (<belief>, <goal>, <plan (task agenda)>, <schedule (control instruction)>). Attributes and their values are shown in italics while the dynamic behaviour of the models will be described in the text.

The first example refers to the distribution of workload within an ACU team. To begin with, Figure 132 shows two instances of the environment model **actor** (*actor-self* and *actor-1*) as part of the situational knowledge of the considered ACU which is associated with *actor-self*. Moreover, there are two instances of **task-destroy** (*task-destroy-0* and *task-destroy-1*). While *actor-1* is committed to both tasks (cf. two instances of **commitment**), there is no commitment of *actor-self* to any task. This results in the conclusion, that workload of *actor-self* is low (cf. attribute *is* of instance *workload-0* of **workload**) while the workload of *actor-1* is high (cf. attribute *is* of instance *workload-1* of **workload**).

So far, the prevailing situation has been interpreted, i.e. instances of environment models have been created and values have been assigned to their attributes.

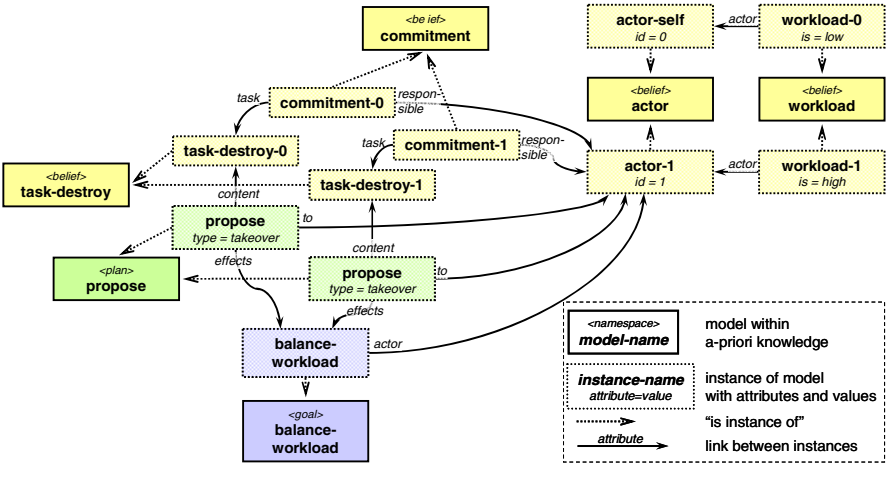


Fig. 132 Interplay of models (a-priori knowledge) and instances (situational knowledge) [Meitinger, 2008]

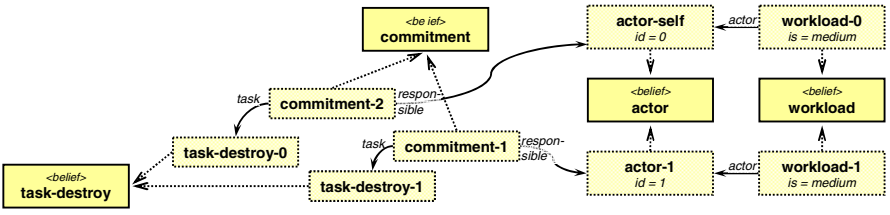


Fig. 133 Models and instances thereof after appropriate actions have been successfully carried out in order to achieve goal "balance workload" (cf. [Meitinger, 2008])

Activation criteria of the desire **balance-workload** comprise the knowledge, that **balance-workload** shall be pursued as active goal in **case** the workload of one actor is low while the workload of another actor is high. As this is the case, an instance of **balance-workload** is created within the situational knowledge. To achieve this goal, several actions are possible from the perspective of *actor-self*, namely to propose to take over *task-destroy-0* or to propose to take over *task-destroy-1*. Both are instances of the action alternative **propose**.

As there are more than one possible ways to achieve the active goal, appropriate selection knowledge has to be used to choose from the action alternatives being available. Assuming, that `actor-self` selects to propose taking over `task-destroy-0` from `actor-1`, an appropriate dialogue will be initiated and conducted, again leading to the instantiation of a-priori knowledge models including the activation of desires.

This interaction results in a change in commitments, which is also communicated to other team members and reflected within the situational knowledge (see Figure 133). In concrete terms, there is no commitment of `actor-1` to `task-destroy-0` any more (previously `commitment-0`), but instead, `actor-self` has committed to `task-destroy-0` (`commitment-2`) and the workload of both actors is assumed to be "medium" now. Thus, the desired situation has been reached and there is no reason for an ongoing activation of the desire **balance-workload** any more.

As a second example, we consider the representation of a dialogue of the request type as depicted in Figure 134 (see Chapter 5.1.2). It refers to the dialogue taking place when the human operator requests the completion of a certain task ('mission order') of the UAV team, which will be answered by the SCU of the spokesman UAV team member. Hereby, Figure 134 shows both knowledge models (classes) of the a-priori knowledge of the SCU of the spokesman UAV team member as well as instantiations of these classes which are involved in the representation of such a dialogue. To begin with, this type of dialogue could not take place unless the existence of an environment model **dialog-request** in the a-priori knowledge of the SCU. There is also the class **actor** as part of the environment models which has instantiations referring to the human operator (*actor-operator*), two team members, which are represented by the instantiations *actor-self* and *actor 4*, and the team as a whole represented by the instantiation *team-all*. As *actor-self* is spokesman of the team, there is an instantiation *spokesman* of the environment model **spokesman** within the

situational knowledge, which refers to the actor actually being the spokesman (*actor-self*) as well as to the respective team.

Upon receiving a message from the operator, which requests to perform the mission to attack the target, the representation of an appropriate dialogue is created within the situational knowledge. Here, several models are involved, first of all an instantiation of the environment model **dialog-request**, which is inherited from the class **dialog** which encapsulates all knowledge relevant to all types of dialogues. The attributes of that instantiation are the

- "protocol" used
- identification code of the dialogue ("conversation id")
- dialogue "initiator" (*actor-operator*)
- dialogue "participant" (*team-all*) and the
- "subject" of the dialogue, in this case the *mission order* as an instantiation of the class **mission order**.

Besides, several models are needed to represent the states and transitions of the dialogue. Directly after the creation of *dialog-request*, the **dialog-state** *start* is created, pointing to the **dialog-transition** *request*, which indicates that there is only one possible message, which can be sent when being in state *start* and which points to *actor-operator*, who is responsible for sending that message. As the operator did already send a message of type ("performative") request, the appropriate transition is attributed with "done" = t (the value t means "true"), and a new state

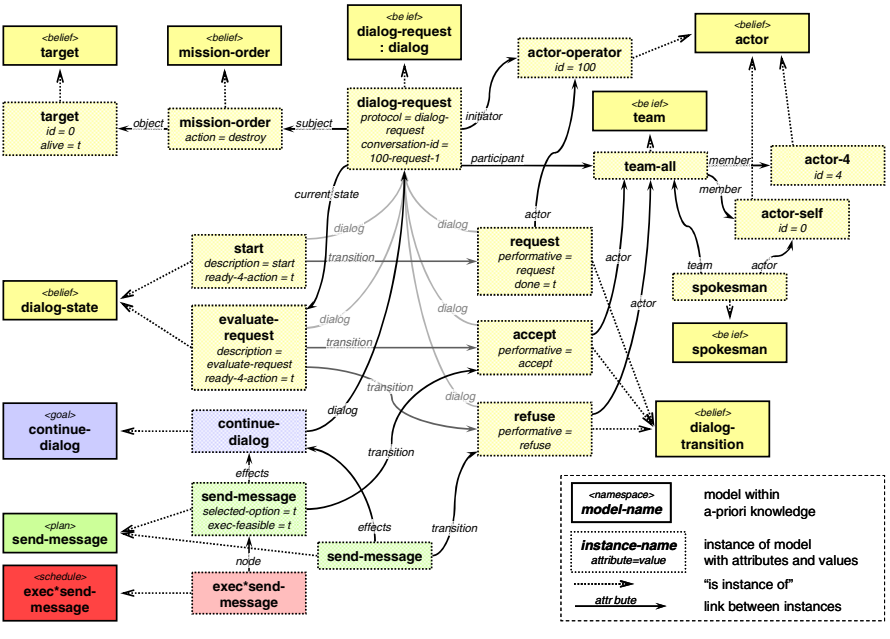


Fig. 134 Representation of dialogue 'Request' [Meitinger, 2008]

evaluate-request is created as the state of **dialog-**request. In Figure 134 the process of the dialogue is frozen at a point in time where this is just the current state. Here, two different messages can be sent, which is indicated by the possible transitions *accept* and *refuse*, both of which *team-all* can initiate.

As the *actor-self* is spokesman of *team-all*, the motivational context **continue-dialog** is activated, which can be achieved by sending a message – either accepting or refusing the request. In this example, the option to accept the request is selected (instantiation *send-message*, pointing to transition *accept,* is attributed with "selected-option" = t). Subsequently, this option will be executed by the appropriate instantiation of an instruction model, which will send a message to the operator.

6.2.3.2 Example for the Identification Function: Pilot Intent and Error Recognition

Knowledge about the environment of an ACU may include the knowedge about the human operator. This is of particular interest regarding an OCU as part of the operating force in a work system, because the quality of co-operation with the human operator depends on that knowledge to a great extent. Part of it is to know the actual intent(s) of the human operator.

Unless the human operator would be communicating it, the intent of the human operator is not always an obvious one as a consequence of the work plan agreed upon. Usually, this is the case when the human operator is deviating from the work plan for particular reasons. One reason may be that he/she is simply mistaking. Another reason could be that the actual situation was not considered when creating the work plan which requires a decision about how to proceed further without exactly taking up the due task according to the work plan. Then, the OCU (assistant system) has to "make up its mind" about the human operator's possible intent. In order to accomplish this it draws on what is observable of the human operator's behaviour to derive satisfactory conclusions on what the operator's intent might be. This is what we consider as a typical example of what is meant by the identification function as part of concept-based-based behaviour according to Figure 27. In the following it will be outlined how the necessary knowledge is derived for the identification (recognition) of the intent of an aircraft pilot while conducting a transport mission under instrument flight rules. This function was developed in the framework of the CAMA project which is described in Chapter 5.3.1.4. Also CASSY, as described in Chapter 5.3.1.4, already contained a similar module (see [Wittig, 1994]) to recognize the pilot's intent for the same reason. This is shown as part of the cognitive sub-function of situation interpretation in Figure 63, but it will not be discussed further at this point.

Scenario

As has been described in Chapter 5.3.1.4, most of the time in the course of a transport flight under instrument flight rules the pilot is almost never free to select from choices for his piloting behaviour. About everything is regulated in terms of procedures based on a given flight plan and what the air traffic clearance is providing as limitations on the flight path of the aircraft. Consequently, the pilot behaviour is almost completely normative with very little tolerances and exceptions. If everything goes normal, the normative model of pilot behaviour

called piloting expert (PE) is perfectly valid as outlined in Chapter 5.3.1.4 on the CASSY and CAMA module. If things are not going normal because of an exception not captured by the normative model or because of pilot errors, the identification process of pilot error and intent recognition has to be activated. This process is considerably facilitated by the fact that one has to take only care of rare exceptional events where it is up to the pilot to decide on his own how to proceed. These events can be considered as known. They can be listed and described in terms of certain situational circumstances associated to each of them. One typical event of that kind is for instance the encounter with a thunderstorm right on track. At this situation, the pilot is usually free to decide whether he is going to fly right through it or to circumvene it (weather diversion). Then, the task of the OCU would be to recognize as early as possible which decision has been made, i.e. which intent is pursued by the pilot. Based on this it has to propose an adjustment of the flight plan if necessary. The complete list of intent hypotheses as being considered by the intent recognition module in CAMA was the following:

- change of course not complying with the flight plan
- speed change not complying with the flight plan
- landing intent not complying with the flight plan
- go around intent
- turn onto a new waypoint not complying with the flight plan
- flight path shortcut not complying with the flight plan
- weather diversion
- escape manoeuvre for collision avoidance
- escape manoeuvre for threat avoidance

The main task for the system designer is to implement the knowledge for identification needed in that case. In [Strohal, 1998] and [Strohal & Onken, 1998] a co-operative neuro-fuzzy approach is published concerning this task which we will elaborate on in the following.

Concept of knowledge acquisition

We know of three types of co-operative neuro-fuzzy approaches which are [Nauk, 1994]:

1. Based on training data, a neural network determines weight factors for fuzzy rules online to conrol their impact on the overall conclusion.
2. Based on training data, a neural network models the membership function parameters of a fuzzy rulebase which is predetermined in its structure by experts. This can be done offline or online in a preliminary phase prior to the system operation.
3. Based on training data, a neural network determines the rulebase of a fuzzy system offline, while the parameters of the member functions are provided by experts.

[Strohal, 1998] pursued a solution to the problem by making use of the approach under 2. Essentially, he is following the approach of [Pedricz & Card, 1992]. The neural computing part as the essential part of learning consists of the

application of the self-organizing maps (SOMs) as introduced by [Kohonen, 1990]. These maps preserve the topology of the knowledge, i.e. patterns which are learned are represented in the map close to other patterns which are similar ones. In that sense they are characterized by a kind of clustering, although there is no information about the cluster semantics and there are no instructions how to delineate the clustering in the map in order to be able to directly use the learning result for classification purposes. The point is to find an interpretation of the resulting SOMs which symbolically captures the semantics and to achieve delineated clusters. For the semantic interpretation [Pedricz & Card, 1992] propose the employment of fuzzy sets and linguistic variables. Delineated clusters are achieved by [Pedricz & Card, 1992] in terms of deriving so-called fuzzy feature maps as nonlinear transformations from the original SOMs and making use of α-cuts in these maps for delineation purposes. These three steps of this approach will be outlined in more detail in the following.

The self-organizing map consists of n_1 x n_2 neurons. All of them are connected to all input nodes. The input is an n-dimensional vector **x** of input features. An input element x_i is multiplied by a numerical weight $w_{i1\ i2\ i}$ regarding the connection of the input node and a neuron on the map with the co-ordinates i_1 and i_2. Thus, the whole of the weights is represented by the 3-dimensional connection matrix $\mathbf{W} = [w_{\ i1\ i2\ i}]$ with $i_1 = 1,2,\ldots, n_1$, $i_2 = 1,2,\ldots, n_2$, and $i = 1,2,\ldots,n$. The learning algorithm is the following:

1. Input patterns $\mathbf{x_k}$ from a training data set $\mathbf{x_1}$, $\mathbf{x_2}$, …, $\mathbf{x_N}$ are consecutively presented to the SOM network. Thereby, an input pattern is stochastically selected from a training data base. The input vector is normalized such that it is lying within a [0, 1] hypercube (x is an element of $[0, 1]^n$).
2. In order to achieve a mapping of similar input patterns into close neighbourhoods in the SOM network modifications to the weight matrix **W** are brought about by competitive learning between the various neural elements of the map. The winner is determined by the minimal distance between the input vector $\mathbf{x_k}$ and the weight vector associated with the neural element (i_1, i_2), i.e. the minimum of $\|\mathbf{w}_{\ i1\ i2} - \mathbf{x_k}\|$ for all elements (i_1, i_2). The winner's weight is modified to become closer to the input vector. Also the weights of the near neighours are modified, for instance according to the so-called Mexican-hat function [Kohonen, 1990].
3. The learning algorithm is terminated when there is hardly any plasticity left in the neural network.

The second step concerns the interpretation of the learned map by use of fuzzy sets, since the SOM as such does not offer much for the interpretation, as was alluded to earlier. We first rather break down the SOM into n maps of each element i (feature i) of the input vector **x**. Thereby the input feature i corresponds to the n_1 x n_2 matrix $\mathbf{W_i}$ which represents the part of **W** pertinent to the feature i. Each matrix $\mathbf{W_i}$ can be considered as a map on its own as depicted at the bottom left of Figure 135 as feature map i. For the n input features we get n corresponding feature maps.

Now, the human expert is involved who has some experience with the domain (concept) of concern. He takes each input feature i, i.e. x_i, as a linguistic variable and defines a number of linguistic labels for the range of values of the input feature and the shapes of the corresponding fuzzy sets. Figure 135, for instance, exhibits the labels A_1, A_2, and A_3 for ranges of small, medium, and large values, respectively. The corresponding fuzzy sets, which where defined for the feature x_i based on the linguistic labels A_1 through A_3, can now be applied to the feature map corresponding to W_i, yielding a set of transformed maps which we call A_1 (W_i), A_2 (W_i), and A_3 (W_i). For the linguistic label A_1, for instance, this transformation enhances those regions of the feature map which correspond to small feature values. The transformation for the linguistic labels A_2 and A_3 works correspondingly (see Figure 135 on the right).

The maps resulting from this transformation can be viewed as fuzzy relations in terms of a fuzzification of the feature maps, thereby introducing semantics according to the linguistic labels.

As a next step one can look for combinations of such fuzzy relations which might exhibit good compatibility with an expert's class description of input patterns, i.e. a potential input cluster of interest. Thus, the compatibility with such a class description can be described by combining fuzzy relations of the type as described above, thereby for instance forming a fuzzy-AND decision relation of intersections like the following one:

$$D = A_{j1}(W_1) \cap A_{j2}(W_2) \cap \ldots \cap A_{jn}(W_n),$$

where each feature map is transformed by use of the A_j's associated to it and processed according to the decision relation by use of the MIN-operator for intersection relations. In this case, the elements $D_{i1\,i2}$ of D become

$$D_{i1\,i2} = \mathrm{MIN}\left[A_{j1}(w_{i1\,i2\,1}),\, A_{j2}(w_{i1\,i2\,2}), \ldots, A_{jn}(w_{i1\,i2\,n}) \right].$$

This is exemplified in Figure 137 by taking a relation between transformed (fuzzyfied) maps of two features and combining them. Note that one of them is feature i of Figure 135, here only accounting for certain combinations of the linguistic labels small and large. Thereby, $D_{i1\,i2}$ describes the degree to which the node (i_1, i_2) supports the compatibility of decision D. The maximum of $D_{i1\,i2}$ over all nodes (i_1, i_2) gives a measure for the validity of the relation D as such. If it is sufficiently high the corresponding node represents a kind of cluster center. If it is too low, the relation D might be dropped as a description of a potential cluster.

This is used for the further detailed evaluation as the third step and as the second step of the interpretation process of this approach. This is to develop a solution for which relations can be considered as reasonable concepts and how they are clustered in the self-organizing map, i.e. where are the cluster boundaries. [Pedricz & Card, 1992] are making use of α-cuts in the maps of fuzzy relations.

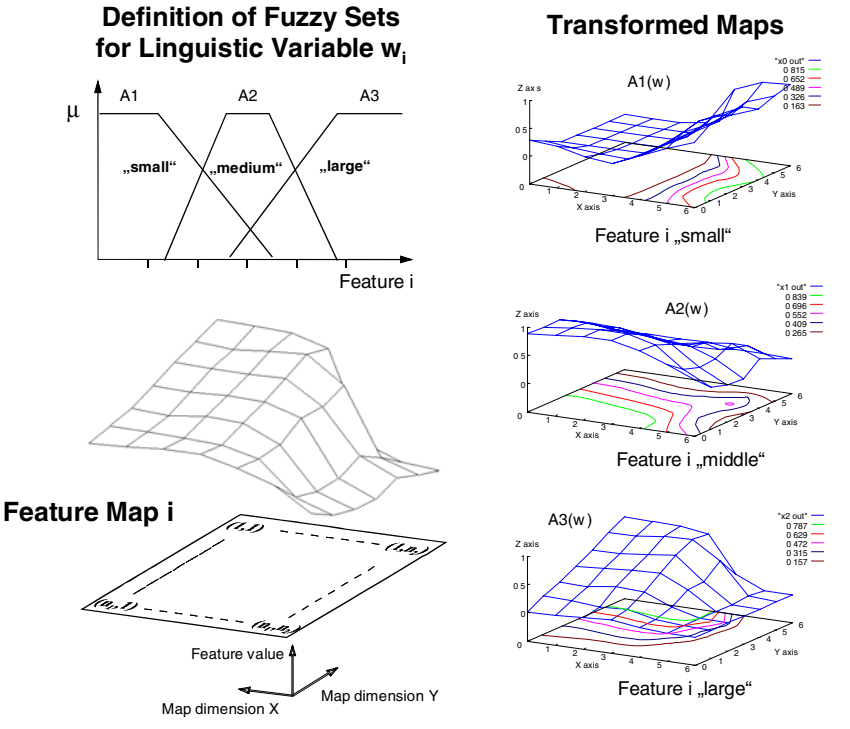


Fig. 135 Transformation of a feature map i for linguistic interpretation and the definition of fuzzy sets for the corresponding linguistic variable w_i [Strohal, 1998]

Then, the relation is represented by a nested series of projections of α-cuts (see Figure 136). These are encircling regions of the map and corresponding nodes with grades of membership to a relation not lower than α. Apparantly, there is less credibility of relations which turn out to be associated with low α-values. The α-values can also be used as a measure for the delineation of fuzzy relations occupying certain areas on the Kohonen-map. Another aspect is to find out whether certain areas of the Kohonen-map cannot be associated to one of the possible relations. In this case, the layout of the predefined fuzzy sets should be reconsidered and appropriately modified by the knowledge engineer.

Still this approach hardly provides a solution for the difficult decision about the cluster boundaries and about the appropriate number of clusters to achieve the best trade-off regarding generalisation and precision. [Strohal, 1998] has found a way to fill this gap by applying the hierarchical cluster analysis to the SOM resulting from the training phase, i.e. its n_1 x n_2 feature vectors. A cluster tree is showing all possible clusters of the situations concerned, called dendrogram, having on one side a cluster which includes all feature vectors and stepwise breaking down this cluster into smaller ones until eventually on the other side each feature vector of the SOM represents a cluster as a tree leaf (see Figure 140) or object. The best

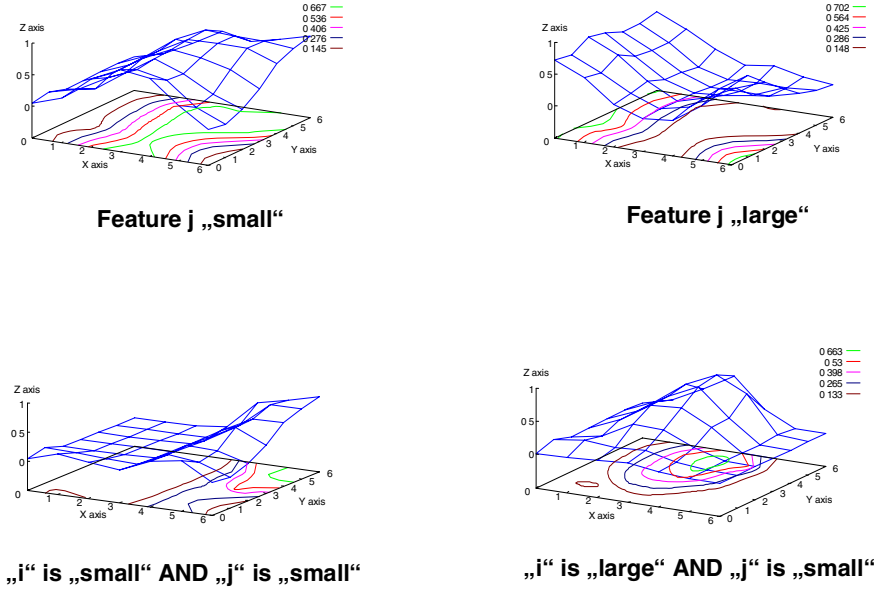


Fig. 136 Fuzzy-AND combinations of features i (according to Figure 135) and j by the fuzzy MIN-operator [Strohal, 1998]

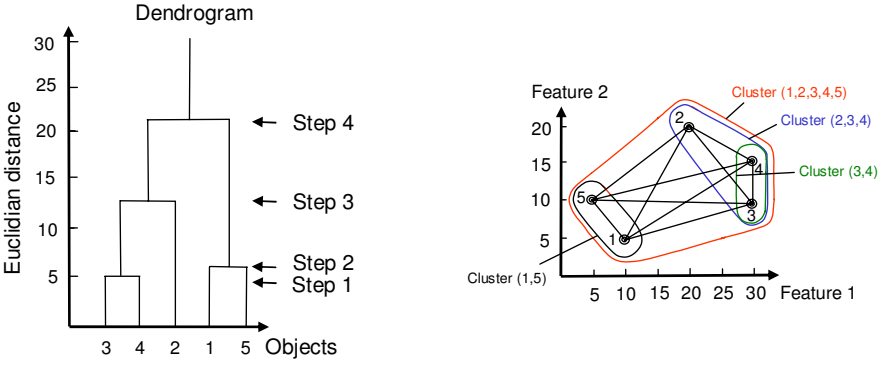


Fig. 137 Dendrogram presentation of hierarchical clustering

way to establish the dendrogram is to start with the leaves on the abscissa and to proceed upwards by forming the next new cluster by combining that pair of clusters which are the closest to each other, i.e. the most similar ones. This requires a decision about an appropriate metric for distance measurement. In this case, the same metric is to be used as that in the Kohonen algorithm. Consequently, the ordinate of the dendrogram indicates the distance between clusters, i.e. how far apart they are semantically.

In Figure 137 one can place a so-called ß-cut of the dendrogram which indicates which number of clusters might be considered as appropriate by the knowledge engineer as a first choice. These clusters can be positioned in the Kohonen-map by extracting the feature vectors (objects) from the dendrogram which are included in these clusters. In addition, for each feature involved in a cluster, there can be determined the pertinent membership function. Using a symmetrical trapezoidal function for the sake of simplicity, the extreme feature values occuring in the cluster form the bottom corners of the trapezoid and the standard deviation of the feature values occuring in the cluster determines the upper corners (see Figure 138).

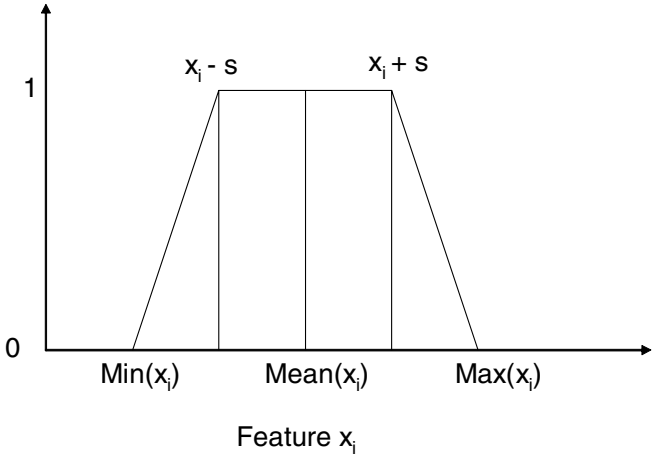


Fig. 138 Fuzzy set of continuous feature [Strohal, 1998]

The corresponding membership functions for all features and clusters can be used to follow the approach of [Pedricz & Card, 1992] again by providing the corresponding transformed feature maps on a refined basis and the fuzzy relations which correspond to the clusters.

Application

In the following, an exemplification of this approach is shown to some extent by the application as carried out for the pilot intent recognition of the CAMA system. Figure 139 shows a bock diagram of the main components of the intent and error recognition (PIER) module of CAMA as a fuzzy system.

The error recognition component contains a task monitor and a safety monitor. The task monitor contains all knowledge in terms of fuzzy sets and rules which enables a fuzzy recognition of whether the flight plan is carried out as planned or not. This knowledge is arranged in subdivisions for the total of n tasks which might be as part of a flight plan. The safety monitor contains all knowledge in terms of fuzzy sets and rules which enables the recognition of violations of safety constraints.

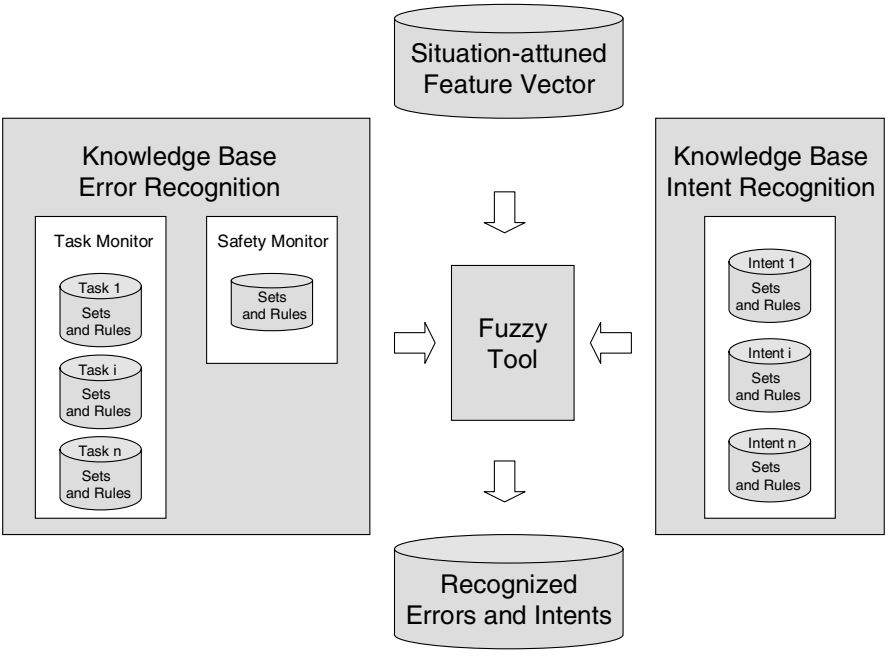


Fig. 139 Main components of the intent and error recognition (PIER) module of CAMA [Strohal, 1998]

The intent recognition component contains all knowledge in terms of fuzzy sets and rules which enables a fuzzy recognition of a reasonable intent in case of pilot behaviour deviating from the flight plan and not being deliberately communicated to CAMA.

The recognition of errors and intents as such is carried out by the fuzzy tool software which works on the inputs of intent and error recognition components and a situation-attuned feature vector ([Stein, 1994], [Schreiner, 1996]). This feature vector allows the fuzzy tool to selectively draw on the pertinent knowledge components concerning task performance measures, safety measures, and hypotheses about pilot intents. Then, in case of a deviation from the flight plan and no violation of a safety constraint the system might embark on an intent hypothesis.

The training data base is given from 71 flights which are carried out by 10 pilots in a flight simulator. Filtering algorithms can be used to search for flight segments with certain feature combinations which are characteristic for particular behaviour in case of an error or an unscheduled intent. Based on these data we will now follow the procedure of semi-automatized knowledge acquisition for the recognition of the intent “weather avoidance” as an example.

Figure 140 shows a result of data processing in order to filter out and depict all segments of the simulator flights where the pilot has developed the intent not to

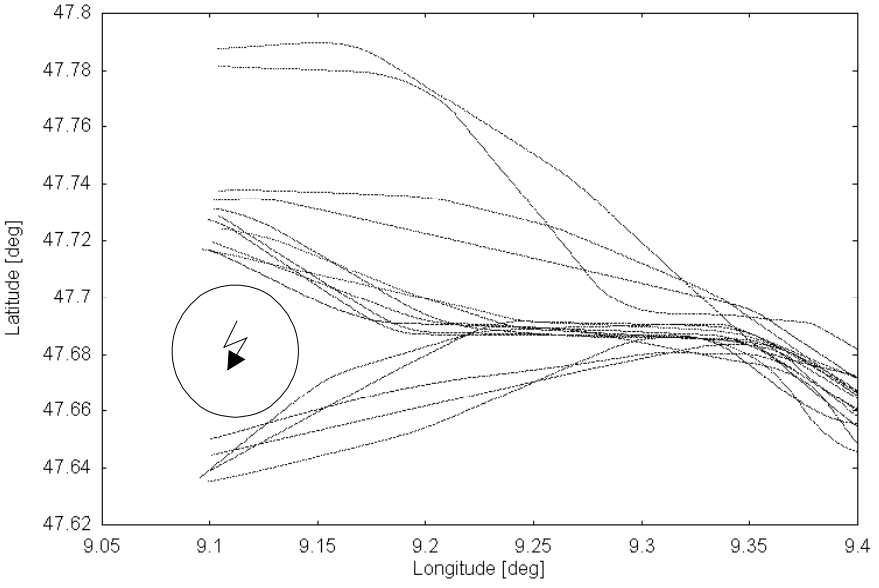


Fig. 140 Depiction of all segments of flights with active avoidance of a thunderstorm conducted by transport aircraft pilots in flight simulator trials [Strohal, 1998]

follow the flight plan right through the thunderstorm but rather to avoid a thunderstorm by means of circumvening it.

At first, the engineer, who applies this knowledge acquisition tool, has to define the features which he considers as relevant. In this respect he can rely on expert knowledge which has been gathered beforehand. The following features are chosen as potential components of the input feature vector for the first flight phase of weather avoidance when turning off to move away from the track (compare with Figure 141):

Feature 1: Rate of turning off the track according to the flight plan (ranging widely)
Feature 2: Distance from the flight track (small values near the track)
Feature 3: Course deviation from track (large values)
Feature 4: Tendency of approaching the track (negative tendency)
Feature 5: State of being on course into the thunderstorm (true/false)
Feature 6: State of being positioned within the thunderstorm (false)
Feature 7: Minimal distance between aircraft and thunderstorm (rather small values)
Feature 8: State of flying to or away from the thunderstorm (flying towards the thunderstorm)

This flight phase is the most important for recognzing the intent as early as possible. The range of feature values which are expected by the knowledge engineer, are attached to the feature names in parentheses. The SOM which results

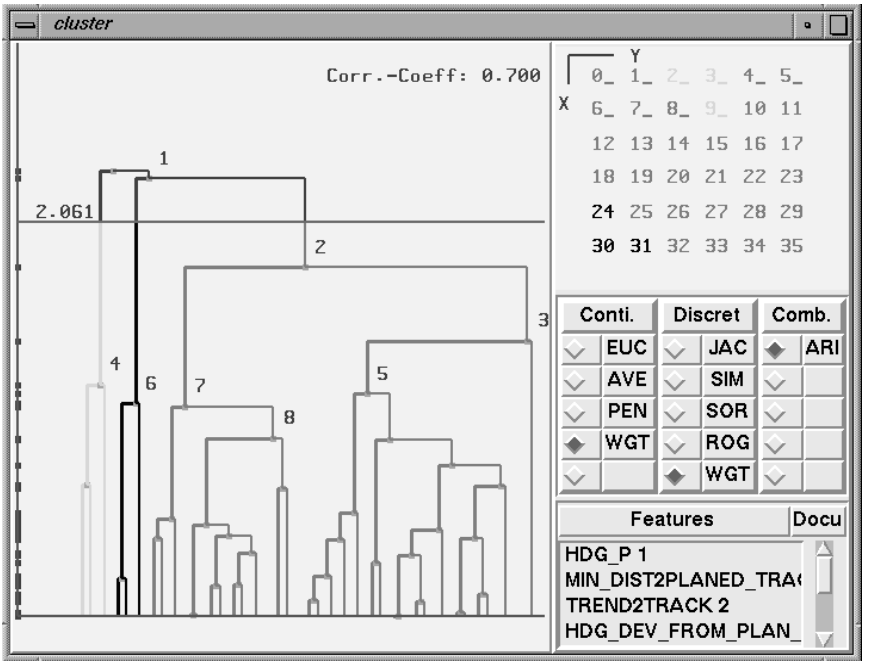


Fig. 141 Screen-shot including the dendrogram to be used for the cluster analysis [Strohal, 1998]

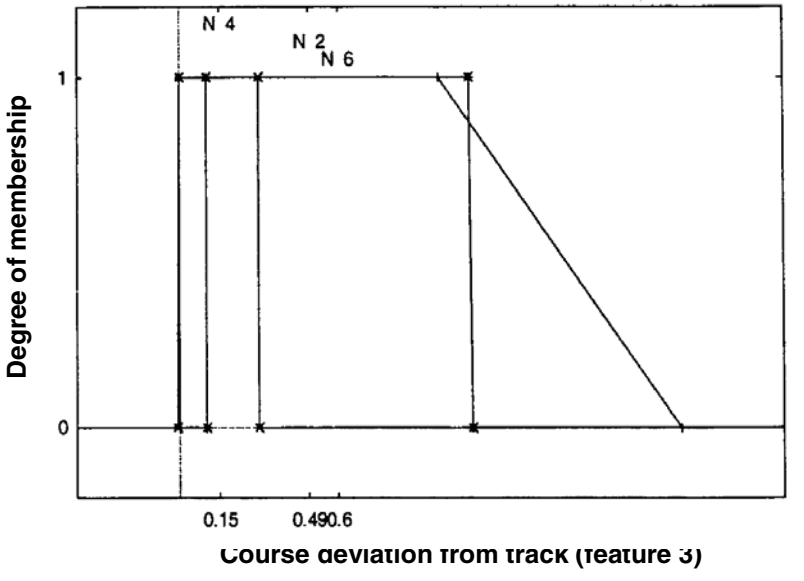
from the Kohonen algorithm consists of a 6x6 matrix of weight vectors (feature vectors). If all features are taken as components of the input, there can be derived eight feature maps. Figure 141 shows, for instance, a screen shot for a choice of the first four features on the list above. It shows the dendrogram as cluster analysis diagram, the SOM pattern with 36 cells and their cluster membership, and a window designating the features chosen for the investigation. Note that a couple of weight vectors are very close to each other such that the corresponding cluster nodes are almost not visible because they are located extremely close to the abscissa. At last, a window on the right indicates which distance measures are set depending on whether the features are metric or nominally scaled variables or a combination of both. One can choose between the Euclidean distance, the mean Euclidean distance, the Penrose distance, and the so-called weighted distance for metric variables. For nominal variables the distance measures to select from are the Jaccard coefficient, the simple-matching coefficient, the Sorensen coefficient, the Rogers and Tanimoto coefficient, and the weighted coefficient. For combinations of metric and nominal features the arithmetical mean is used.

Next, the dendrogram is evaluated for investigations about the semantic interpretation of the clusters and about the reasonable number as well as the extent

Table 13 SOM cells associated to cluster nodes next to the ß-cut above cluster node 2

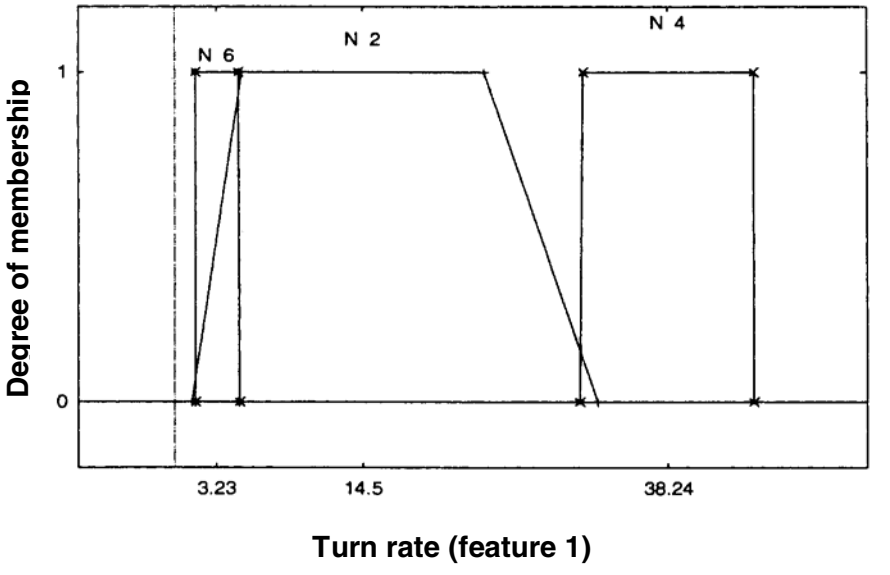
Cluster	Associated SOM cells
4	2, 3, 9
6	24, 30, 31
2	0, 1, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 32, 33, 34, 35

of the clusters [Strohal, 1998]. The dendrogram shown in Figure 141, which depicts the resulting clustering for the case of evaluating an input vector with four components, suggests a ß-cut above cluster node 2 or node 3, thereby advocating for a trade-off of gaining sufficient of both generalisation and accuracy. Staying with a ß-cut above cluster node 2, Table 13 indicates the resulting clustering and which SOM cells are taken by which cluster.



	Left corner ↓	Left corner ↑	Mean	Right corner ↑	Right corner ↓
Set of Node 2	1.300	5.113	14.497	23.881	32.800
Set of Node 4	31.300	31.580	38.400	44.900	44.910
Set of Node 6	1.590	1.600	3.133	4.858	5.000

Fig. 142 Membership functions for the feature 3 [Strohal, 1998]



	Left corner ↓	Left corner ↑	Mean	Right corner ↑	Right corner ↓
Set of Node 2	0.000	0.000	0.443	0.985	1.900
Set of Node 4	0.000	0.000	0.200	0.300	0.300
Set of Node 6	0.100	0.100	0.500	1.100	1.110

Fig. 143 Membership functions for the feature 1[Strohal, 1998]

When having decided about the ß-cut one can also call for the membership functions for each feature. Figure 142 shows the membership functions for the feature 3 (course deviation from track) concerning the impact of the cluster nodes 2, 4, and 6.

It becomes obvious from Figure 142 that cluster 4 consists of vectors whose components corresponding to feature 3 have all large values. In distinction, feature 3 is varying over a large range of values for the cluster 2 with a tendency of smaller values than those for the cluster 4. This already gives enough evidence that cluster 4 is not a relevant cluster for the first phase of the weather avoidance manoeuvre when turning off to move away from the track. This is confirmed by the membership functions for feature 1 (turn rate) (see Figure 143).

The turn rate for the situations covered by cluster 4 consistently shows rather small values, whereas the cluster 2 stands for a large range of values of the turn rate. Also cluster 6 can be excluded right away for similar reasons. A further refinement can be accomplished by looking at the membership functions of other features and by stepwise cutting at lower levels and thereby generating a higher number of clusters for a greater granularity. Also the depiction of transformed feature maps and those of fuzzy relations is very helpful for the final refinement

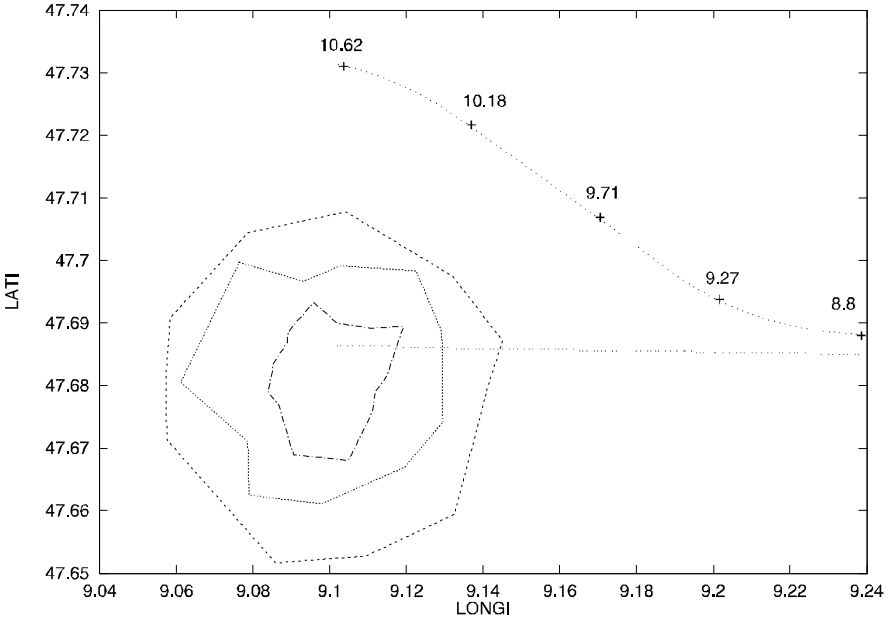


Fig. 144 Test result of weather avoidance intent recognition [Strohal, 1998]

process. In [Strohal, 1998] this example ends up with the finding that cluster 9 stands for the turn off phase of the weather avoidance manoeuvre and cluster 14 stands for the consecutive phase of relatively high values of feature 3 (difference between track course and actual course). The corresponding ß-cut is right between cluster nodes 5 and 6.

It becomes obvious that by means of the cluster analysis an abundance of fuzzy relations can be found. Those which make much sense for the knowledge engineer, capturing that what he is after in terms of situations like those associated to unplanned changes of pilot intent, can be added to the knowledge base at any time.

In the framework of the CAMA project the error and intent recognition module was successfully flight tested by 10 transport aircraft pilots in the course of more than sixty simulator flights. No weather avoidance maneouvre remained undiscovered. Figure 147 shows an example of successful recognition at flight time of 8.8 minutes right at the beginning of the turn off manoeuvre.

Concluding remarks

This approach has proved to be a very powerful observer which sensibly monitors the behaviour of the human operator all through the work process and generates information which can be crucial for the performance of an OCU as an assistant system. In particular, it is able to judge continuous behavioural patterns of skill-based behaviour in a concept-based identification process.

What makes this approach even more attractive is the fact that it combines sub-symbolic semantic knowledge coding and a symbolic interpretation of the semantic code. This is the only piece of knowledge we have developed so far which meets a requirement like that.

There is some potential for improvement, though. The fuzzy relations being investigated so far are of the fuzzy-AND type. Certainly, the possibility of fuzzy-OR and fuzzy-AND relations with negated features or conditional relations would be an interesting extension. Also the Kohonen-algorithm could be enhanced by further pre-processing the training data in order to avoid predominantly using data of most frequently occurring situations in the course of data mining. This is important because usually one is interested in data patterns of rarely occurring situations. Certain filter algorithms could do the job. Finally, one can think of a greater variety of fuzzy set patterns different from the trapezoidal one.

Nevertheless, this approach has proved to be sufficiently robust to perform successfully in field tests for an important function of a cognitive system like that of CAMA.

Chapter 7
Operationalisation of Cognitive Automation in Work Systems

The operationalisation of cognitive automation in work systems is probably the most controversial topic of this book. This is because there are hardly any automated functions of that kind in operational use today. Because of the lack of engineering experience there is great hesitation in most industrial design teams, in particular as to the software certification of cognitive automation, since this software seems to them even more complex than that known for conventional automation. This concern, however, is definitely not justified. The strongly praised reliability of human individuals when acting is mainly founded on the fact that their motivational contexts and their knowledge about the work objective provide the criteria needed to appropriately carry out and monitor his activities. The same is valid for artificial cognition in work systems, if it works on the same basis of motivational contexts and knowledge about the work objective. In order to illustrate this point, an example for an implementation of cognitive automation is given in Chapter 7.2.1 which is dealing with self-monitoring on that basis. Another issue is the lack of development standards. This is still a rather open field. There is not much application literature around. Therefore, at first a development framework, called COSA (COgnitive System Architecture), will be described in the following to give some guidance for those who otherwise would hesitate to take care of that and to make the start on their own.

7.1 Cognitive System Architecture (COSA)

COSA (Cognitive System Architecture) (see [Putzer & Onken, 2003] and [Putzer, 2004]) is an architecture for the development of cognitive systems. It offers a framework to implement applications according to the theory of the cognitive process of ACUs and supports the developer in two ways: Firstly, COSA provides an implementation of the application independent inference mechanism, so that the development of a cognitive system is reduced to the implementation of interfaces and the acquisition and modelling of a-priori knowledge. Secondly, knowledge modelling is supported by the provision of a front end language, the programming paradigm based on the theoretical approach of the cognitive process. This will be further outlined in the following.

R. Onken and A. Schulte: System-Ergonomic Design of Cognitive Automation, SCI 235, pp. 311–331.
springerlink.com © Springer-Verlag Berlin Heidelberg 2010

7.1.1 Design Goals of COSA

In most cases of current research, special designs are used to model certain isolated aspects of cognitive systems and solve problems of the respective application. With COSA [Putzer, 2004] we try to design a framework which uses results from several sources and combines them to get an improved, holistic and unified architecture for cognitive systems. This section will give a survey on the design goals of COSA in relation with the design decisions.

The first step towards COSA was a specification phase which included the analysis of former cognitive systems and other state-of-the-art systems. Moreover, the requirements of three different groups of users were analyzed:

System developers add functionality or external components to the actual architecture. For this task they need good documentation, simple interfaces for internal and external sub-systems, good modularity and support to easily extend and maintain the system. Design decisions regarding these requirements are accounted for in the following sections referring to the 'architecture' and 'implementation'.

Knowledge engineers design *a-priori knowledge* implementing the behaviour of the cognitive system. They need an interface to potentially any knowledge modelling language in order to be able to choose the one which is best suited for their particular application domain. Moreover, they need a design methodology for knowledge modelling and a structured concept to partition the knowledge. These aspects are supported by the 'language front end' being described in the next section.

Users of the resulting cognitive application put some more general requirements on the system. These are described in the following paragraphs along with other features derived from our own experience with artificial cognition.

The COSA framework is designed to ensure a unified architecture of cognitive systems. It is based on the model of the cognitive process as described in Chapter 4.5.2. The cognitive process as the core element of the system makes all its properties available for the resulting framework, i.e. domain and application independence through general knowledge processing, reusability, comprehensible processing to humans and support for cognitive operations.

The kernel of COSA, which establishes the cognitive process, is designed to carry out high level knowledge processing. It is not designed to implement number-crunching algorithms, high frequency control loops, or low-level cognitive functions. As it will turn out, this is no restriction for an application because there are interfaces to external components which can cope with such requirements.

Certification is eased by COSA. It can be identified, that COSA supports approaches to strengthen system integrity. One approach of that kind is described in Chapter 7.2.1.

7.1.2 Architecture

After having sketched some design goals of COSA, this section describes their realisation in the architecture of COSA already taking into account some aspects of the implementation.

7.1.2.1 Overview

The COSA framework is based on a modular architecture in order to ensure clear internal interfaces and decoupling of functional components. The single sub-systems of the architecture are marked on the left-hand side in Figure 145 and are built on a distribution layer based on CORBA [Römer et al., 2001], which is an industry standard for distributed systems.

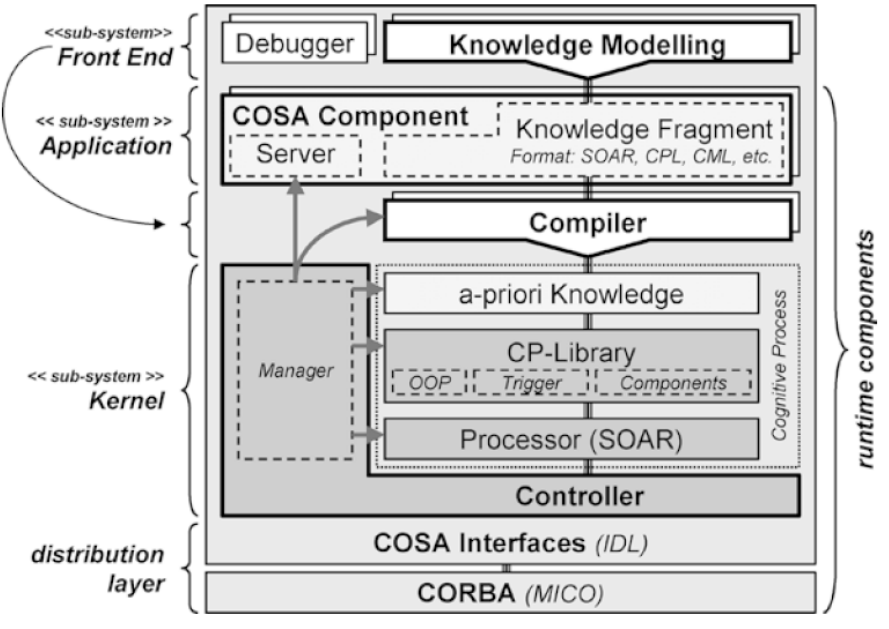


Fig. 145 Block diagram of COSA's modular architecture [Putzer & Onken, 2003]

To begin with, the *kernel* encapsulates the knowledge processing engine by which the cognitive process is implemented. Here, no knowledge specific for a particular application is included. This part of the architecture only covers the conversion of knowledge into behaviour. Moreover, organisational tasks such as the integration of distributed components of the framework to an integrated system and the management of runtime components and compilers are accounted for by the controller and manager.

The *application* consists of at least one so-called "COSA component". Each COSA component usually covers a more or less encapsulated capability of the

resulting cognitive system and consists of application-specific knowledge and/or so-called "servers". The latter provide containers for the integration of functions which shall not be covered within the cognitive process, e.g. interfaces to external systems, number-crunching algorithms, human-machine interfaces or image processing algorithms. The knowledge in turn has to be transformed into a format which is understood by the kernel during system start-up by the use of adequate compilers.

For knowledge modelling and debugging various tools are necessary which are provided by the *front-end*. These tools are used by the knowledge engineer during application development.

With these sub-systems an important feature of the COSA framework is described, namely the separation of application and architecture. As basic architectural problems have already been solved, development time can be saved. Moreover, due to the provision of behaviour generation on the basis of explicit knowledge, great flexibility with respect to application development and reusability of components can be achieved.

7.1.2.2 Kernel

As indicated above, the kernel has two major responsibilities, namely the management of other components of the framework by the *controller* and the implementation of the cognitive process by means of the *Processor* and the *CP Library*.

The *controller* can be regarded as the central manager in the COSA architecture. At start-up, all components such as compilers provided by front-ends and COSA components register with the controller. After the start-up process, the controller takes control over all components, searches for the knowledge of COSA components, translates it with the appropriate compiler, and loads the result into the processor. During this starting phase (offline phase) dependencies of the components are checked and resolved. After the starting phase, the control is transferred to the processor which processes all loaded knowledge of procedure behaviour (runtime or online phase).

As the central element of the kernel a component is needed that can process knowledge to yield behaviour. Soar (see [Newell, 1990] and [Laird et al., 1987]), which is developed and maintained by the University of Michigan, is a good candidate because it meets all requirements. The main reasons for selecting Soar as *processor* are the following: it is easy to learn, the developing community is very active and reacts quickly to queries, and last but not least, Soar comes with portable source code and can be easily integrated in C / C++ environments. Additionally, Soar provides interfaces to integrate basic features that are not supported in its pure implementation.

Soar stores its 'knowledge' about the situation in its working memory which has a uniform and symbolic structure similar to conceptual graphs. The *behaviour* is uniformly stored in rules. Thus, Soar is a kind of production system but with a very special multi-stage processing loop so that even advanced features like learning are supported to some degree.

Like other production systems, Soar offers a very fine-grained interface: the rules. With this feature, extending existing knowledge models is as easy as writing new rules and loading them into Soar (even at runtime). In Soar, all loaded productions can fire in parallel so this supports the idea of Cognitive Automation: the application of *all* knowledge at hand at *any* situation.

It turns out that maintenance can be done as easy as extensions: the debug code is a set of rules that can be loaded into Soar at any time. This code can trace values, set breakpoints or just print out portions of the working memory. This is supported by symbolic representation of the working memory that is understandable by human beings.

Like many other architectures that build on production systems, Soar has deficiencies in implementing e.g. number crunching algorithms or image processing. But this is not the job of the core processor within COSA anyway. Instead, the processor is used to implement higher decision levels while the above-mentioned functionality is implemented in extensions outside the cognitive processes, namely within servers of COSA components.

Further on, some key features on the knowledge level are added to Soar by loading a set of basic productions called the *CP library*, thereby giving it, for examples, the means to organise knowledge in components and to implement the cognitive process.

The function of the CP library is to provide the abstract layer of the cognitive process by making a set of appropriate Soar rules available. These are combined with the a-priori knowledge of the application and executed in parallel by the processor. Three main features of the CP library lead to the required functionality:

1. **Timers/triggers** handle simulateous events and synchronisation throughout the whole system. They are needed internally to the CP library and not used in the actual application.
2. The **object-oriented approach** introduces object-oriented design philosophies to Soar. With this feature, “models” (= classes) are supported. These can be instantiated, which is a similar process to instantiating a class in C++. Instances contain data (attributes) describing their individual state. Models also contain the behaviour for all instances throughout the whole lifecycle: creation, behaviour during life time, and removal.
3. **Component management** enables the kernel to keep track of activated components and determine dependencies and prioritites. This is done in co-operation with the controller component.

Again, it is emphasised, that the cognitive process and its implementation within the CP library is a basic element for cognitive applications and does not include any domain-specific knowledge.

7.1.2.3 Distribution Layer

The distribution layer of COSA is based on CORBA, an industry standard for distributed systems that connects distributed software objects. It serves as middleware to connect the controller to other components containing knowledge, I/O interfaces, or providing compilers. These components can be distributed over

a network and may differ in programming language, operating system, and computer platform.

To use the Soar processor within the CORBA environment, the interfaces of the processor have to be mapped to the middleware layer. The main features of the middleware layer of COSA are as follows. The *client-server-structure* puts the controller into a central position. Components of the application register with the controller on start-up so that they can be used to retrieve a-priori knowledge or to access other interface functions. The *knowledge mapping* is the link between the different knowledge representations in Soar and in the transport layer CORBA. It defines the representation CORBA uses to transport any piece of knowledge via the network from the processor to other objects and vice versa. The *encapsulation of callbacks* connects I/O functions and internal callbacks of the knowledge processor to published member functions of distributed objects.

With these features COSA can dispatch tasks to processing units external to the kernel, which are servers as a part of COSA components (see Figure 145) with all the advantages described previously.

7.1.2.4 Front-End

The main purpose of the front-end is to support the knowledge engineer with the modelling of the knowledge required for the application components, i.e. the actual capabilities of the cognitive system. Moreover, the debugging of the application shall be supported. Within this context, the developer shall not have to cope with specific requirements of knowledge processing, but be able to concentrate on the contents and be allowed to use a modelling suite which is best suited for his or her particular application domain.

Therefore, the specific knowledge representation or format is separated from the process of generating the behaviour, which is delegated to the kernel while various front-ends provide different modelling facilities. This separation is used to reach a high abstraction for the knowledge modelling process. Modelling takes place in terms of “mental concepts”, i.e. cue models, concepts, motivational contexts, task options, task situations, procedures, and sensori-motor patterns, as they are defined by the cognitive process. This increase of abstraction as compared to traditional software engineering methods enables human beings to understand and cover an increasing complexity of applications as is necessary for cognitive systems.

Within COSA, a front-end is defined by the following five aspects: concept, language, compiler, method, and tools.

- The *concept* represents the basic idea of how a cognitive system should produce its behaviour. For example, the cognitive process can be used as concept. In fact, this is the only concept, we consider here.
- A *language* is needed to express the knowledge that is to be modelled in a formal syntax. Here, text-based as well as graphical notations are possible and will be discussed in some more detail later.
- The *compiler* translates the language with its underlying concept into the knowledge format of the processor such that it can be executed by the kernel. Hereby, the compiler does not only have to convert knowledge

formats but also underlying concepts. A fragment of knowledge and its compiled version both have to represent the same knowledge in terms of producing the same behaviour.

- For knowledge modelling a *method* is needed. This method defines a guideline for the knowledge engineer on how to create an application using the given concept and language of the front-end.
- *Tools* are optional elements of the front-end. They support the knowledge modelling process. The minimal tool is a text editor to produce a textual representation.

In more sophisticated systems as many front-ends as desired can be provided so that the knowledge engineer can choose whichever is best suited for his or her needs. In the current version of COSA, only few front-ends are available, all of which are based on the cognitive process as concept. Of course, native Soar representation is supported as language. To facilitate knowledge modelling on a higher level of abstraction and closer to the paradigm of the cognitive process, CPL (Cognitive Programming Language) has been developed which extends Soar with elements of the object-oriented paradigm and allows the declaration of a-priori knowledge models as they are defined within the cognitive process. To further abstract knowledge modelling, CML (Cognitive Modelling Language) is currently under development where the appropriate tools allow for the definition of a common ontology and the graphical declaration of behaviour so that the knowledge engineer does not have to cope with language syntax and can focus on actual modelling. As far as a method for knowledge modelling is concerned, the so-called "CP method" is part of the CPL front-end. It defines five steps to model a certain capability of the respective cognitive system, which can be repeated as often as requried. At first, the so-called static model is built, i.e. the necessary desires (motivational contexts), action alternatives (task options), instruction models (procedures), and environment models (cue models, task situations, and concepts) are identified and their attributes denominated. Here, the order in which models are specified is important, especially when considering concept-based behaviour. Here, behaviour is goal-driven, therefore the designer has to start with a definition of motivational contexts, then list all action alternatives (task options) at hand which can be used to achieve a desired state, afterwards specify how single action alternatives can be put into effect, and finally name all environment models that are needed to properly activate goals, plan task sequences and execute them. After this static model has been built, the behaviour of models has to be described, i.e. the so-called dynamic model has to be created. This comprises mainly the creation and destruction of instances of models as well as assigning values to attributes. As there are usually attributes linking instances of different models, the interplay of models has to be considered.

To be more concrete on the knowledge modelling aspect, a short example from the flight domain is presented using CPL. Figure 146 shows an excerpt from a package encapsulating knowledge relevant for flight safety for an aircraft. In concrete terms, there are two knowledge models, an environment model defining the concept of 'danger' and a desire denoting the potential goal to avoid collisions

```
package flight-safety
{
    ...
    class <belief> danger
    {
        attributes:
            link    aircraft;
            string  threat;
        behavior:
            sp {create
                (instance[belief::aircraft::*] <own> ^self true)
                (instance[belief::aircraft::*] <other> -^self true)
                (instance[belief::distance::*] <dist> ^between <own> ^between <other>
                                                      ^distance small)
             -->
                (elaborate <i> ^aircraft <other> ^threat high)
            }
    }; // end class 'danger'
    ...
    class <goal> avoid-collision
    {
        attributes:
            link    threat;
        behavior:
            sp {activate
               (instance[belief::danger::*] <danger> ^aircraft <ac> ^threat << medium high >>)
            -->
               (elaborate <i> ^threat <ac>)
            }
    }; // ende class 'avoid-collision'
}; // ende package 'flight-safety'
```

Fig. 146 Sample knowledge models referring to flight safety

with other aircraft. The static attributes of both models further detail the concepts of danger and collision avoidance and are modelled on a very shallow depth here. **danger** for instance is being described by a link to the (other) aircraft posing a potential threat to the own aircraft and the actual risk. The desire **avoid-collision** is attributed only with a link to the threatening aircraft.

The behaviour of the class **danger** only describes the circumstances under which an instance of this class shall be created. The appropriate rule "create" can be read as follows:

"If (as long as) there is an instance of the class aircraft which represents the own aircraft (i.e. there is an attribute 'self' which has the value 'true')

and there is an instance of the class aircraft which does not refer to the own aircraft (i.e. there is no attribute 'self' which has the value 'true')

and there is an instance of the class distance which represents the distance between the own and the other aircraft (there are two attributes 'between' referring to the above-mentioned instances of aircraft) and the distance is small (there is an attribute 'distance' with has the value 'small')

(then) have an instance of the class danger which refers to the other aircraft and indicates that the threat is high."

In a similar way, the behaviour of the desire **avoid-collision** describes the circumstances under which there should be an active goal, i.e. instance of this class:

"If (as long as) there is an instance of the class distance which refers to an aircraft and the threat is medium or high

(then) have an instance of the class avoid-collision which refers to the respective aircraft with the attribute threat."

Looking at the knowledge modelling task from the knowledge engineers' point of view, they have to make sure that adequate knowledge is stored at the right place. For the first design in a certain application, disregarding the knowledge representation task for a moment, it might be difficult to decide where to start and how to define the dimensions of the knowledge components. Here, some guidance will be given in that respect, taking the CP method as described above as process model.

Like it has been mentioned, we start with the knowledge insertion of motivational contexts, i.e. in particular those motivational contexts which may be relevant at some point in the course of the life cycle of the ACU concerned. Assuming, that knowledge engineers want to reproduce expert knowledge as good as possible, it will be feasible to extract knowledge about motivational contexts from experts in the application field concerned. As basic motivational contexts concerning vehicle guidance and control work processes experts would mention, for instance, to accomplish the vehicle mission order (work objective), to comply with economy standards, and to warrant safety. We immediately recognise that these motivational contexts may be conflicting in certain cases. Therefore, it should be possible to weigh these motivational contexts against each other, possibly by means of situation-dependent priority weights. In addition, the degree of actual compliance with these motivational contexts should be measurable in terms of enumerable parameters. This should be enabled by providing appropriate information within input data and environment models, in particular, but also of the other categories of world models. Here, it becomes evident that the definition of desires leads to requirements about the necessary models pertinent to the other cognitive sub-functions like those for the situation interpretation, that of planning, and those for plan execution.

The same applies when considering action alternatives necessary for planning. These models represent possible actions that can be used within the process of searching the best way to proceed from the present situation. A lot of information has to be accounted for in that process. First of all, the actually determined relevant goals as outputs of the cognitive sub-function of goal determination are forming the basis for this process. For the definition of the action alternatives (task options), of course, all possible goals are taken into account. In addition, the available means necessary for certain actions have to be taken into account as well as all other situational information about the world which is of interest in that respect. This necessary information has to be provided through the remaining cognitive sub-functions and corresponding models. There are models which are commonly required for both goal determination and the planning. The instantiations of these models as part of the situational knowledge are used then for both the goal determination and the planning.

We can move on with this for the definition of the instruction models, too. There is principally the same procedure. These models are mainly based on the possible plans of proceeding, but also the possible goals might have an impact on

the modelling. In addition, again these models possibly call for some additional models pertinent to other cognitive sub-functions.

Thus, the last step of a-priori knowledge incorporation into the cognitve process will be that all models will be represented which have been required in the context of the preceding definitions of models of the other categories. This procedure ensures that there is no proliferation of modelling.

Certain environmental models (cue models, task situations, and concepts) might be extended, though, to serve the purpose of processing efficiency. There should be a mechanism that the cognitive process has not to follow the basic procedure to carry out all cognitive sub-functions in all cases. If it can be figured out at an early state that there is no need for any goal or plan modification, the due actions may be effected by a short-cut procedure which is modelled as an extension of the pertinent environmental model. This is a similar design feature as described by the Rasmussen scheme for human cognitive behaviour.

To conclude this section, some comments on the capacity of the cognitive process shall be made. In the first place this is dependent on the richness of knowledge the system can rely on. Regarding the different cognitive sub-functions of the cognitive process, there are different dimensions of capacity levels. Without knowledge about the motivational contexts the cognitive process would exhibit no behaviour at all. Therefore, this knowledge is fundamental. Probably most important is the richness of knowledge to generate the belief (situational cues, current tasks, and matching concepts) from available internal and external stimuli in compliance with what is of interest with respect to the motivational contexts. This includes perceptional knowledge apart from sensor system capacities as well as task knowledge and knowledge needed for the concept-based function of identification. The richer the knowledge related to the cognitive sub-function the more powerful cognitive systems can be designed.

7.1.3 Implementation

Originally, the operating system used for the implementation of COSA was IRIX, a UNIX version from SGI. As programming environment the IRIX native tools were used along with the latest versions of the packages listed below. These are distributed with source code and are portable to Windows and many UNIX derivatives:

- **Soar** as the implementation of the Unified Theory of Cognition [Newell, 1990]
- **MICO** which is a free implementation of the CORBA standard (see [Römer et al., 2001]).
- **Qt** library for the (graphical) user interface (see [Trolltech, 2001]).
- **STL** the Standard Template Library, which provides basic types and containers and is part of the IRIX native C++ environment but also distributed freely for other platforms (see [SGI, 2001]).

The documentation for most parts was done with DOXYGEN (see [v. Heesch, 2001]) which generates documentation from the C and C++ structures by using code comments of a special format.

Some time ago, the COSA framework has been ported to the Linux operating system. This has led to a substantial increase of performance due to cheap computing power on the desktop computer market. With this increase in performance, even large applications such as the SCU prototype that has been described in section 5.1.2, which is also the most elaborated application on the basis of COSA so far, can be run in realtime.

7.1.4 Conclusion and Perspectives

COSA is the architecture currently used by us for the design of cognitive systems, both OCUs and SCUs. During development of COSA the research results gained in our and other projects on cognitive systems were taken into account. Especially the cognitive process of an ACU as described in Chapter 4.5 is used as the basis for the design of COSA.

Meeting the requirements of several user groups COSA became a highly flexible and usable framework with its implementation based on free libraries. Especially the implementation of the cognitive process as the kernel of the uniform architecture for cognitive systems as a set of reusable Soar rules fits very well. This is very plausible because Soar implements the Unified Theory of Cognition [Newell, 1990] which should be a good basis for the architecture of cognitive systems as ACUs.

Several applications have so far been implemented that prove that COSA is a valid framework to go on with. However, there is still a great amount of research needed and in progress to further verify the concept of the cognitive process and the design of COSA. Future work therefore ranges from work on performance improvements to the integration of enhancements of the cognitive process into the CP library and even further to the realisation of semantically coded representations of a-priori knowledge.

7.2 Integrity Amendments through Cognitive Automation

For safety-critical work systems, but also for work systems where failures might cause considerable cost, the integrity issue is a very important one for the sake of productivity. Cognitive automation offers a number of advantages compared to conventional automation to ensure integrity, in particular, if we make use of cognitive automation with explicitly represented motivational contexts. Generally speaking, bad experiences with conventional automation like brittleness, opacity, literalism, or clumsiness (see Chapter 4.2 and Chapter 4.3.2), can be excluded in the case of cognitive automation, if it is carefully designed. This will be further described in the following Chapter 7.2.1. [Jarasch & Schulte, 2008] are using this approach for the enhancement of the design process of operation-supporting means for automatic guidance and control of UAVs which consist of both conventional and cognitive automation. They propose a design approach based on an extension of the Vee-model [INCOSE, 2007], the core element for the systems engineering design process.

Furthermore, cognitive automation may be the key element to identify performance deficits of existing conventional automation and may include online-monitoring and amendment of its own system performance by a corresponding metafunction. The latter aspects will be further described in the following Chapter 7.2.2.

7.2.1 Metafunction for Online Monitoring and Control of System Performance (Application Example)

One of the outstanding capabilities of a human operator is the typical behavioural trait of deliberating on what he/she is thinking. There is a kind of metalevel of deliberations in order to monitor own cognitive processes, mostly by carrying out plausibility checks. This is known under metacognition (see for instance [Metcalfe & Shimamura, 1994] and [Paris, 2002]). In particular, impending actions are reconsidered subject to whether they will really comply with what is intended as resulting effect. Before actually acting or while acting there might be a "second thought" in terms of challenging the first approach of deliberations by a dissimilar approach and of comparing the results of both approaches. This is in particular true, if there are actions which can cause great harm in case of being not appropriate ones. Then, inappropriate behaviour might be possibly prevented before it is effecting. Other ways of complying with the goals may be found and incorrectness of a-priori knowledge which has caused inappropriate behavioural intentions may be repaired, if possible. This behavioural trait may be the cause, so far, for having much more confidence in the reliability of the human performance as opposed to that of an artificial device.

This might change in the future with the advent of cognitive automation. In the following, it will be shown that by means of cognitive automation in terms of introducing a metalevel of monitoring and control of system performance in the ACU a significant enhancement of system integrity can be established, similar to that of the human operator. For ACUs of that kind we simply have to extend the a-priori knowledge correspondingly for this functionality, i.e. to

- monitor the overall operation in order to identify possible inconsistencies or errors
- trace back these inconsistencies in the cognitive process in order to identify the true cause by an analyzing process. Therefore, this functionality involves all cognitive sub-functions in order to detect and trace also those errors in the course of the cognitive process that originally have occurred in one of the cognitive sub-functions, but that might show up in another cognitive subfunction somewhere in the course of the cognitive process.
- find a solution that keeps the system operational as long as possible despite of an error. In case that an error is identified but that it is not possible to immediately eliminate the error, the operation is continued by a work-around procedure which should not necessarily end up in a dissimilar back-up system (for instance manual human control).

There are constraints, too, resulting from the cognitive system architecture, that have to be considered for this functionality:

- It must be in line with the principles of the cognitive process. This includes that all behaviour is goal-driven and evolves from applying the a-priori knowledge.
- The knowledge for the mainstream cognitive functionality and that for the metacognition is executed in parallel by a single cognitive processor.

7.2.1.1 The Technical Concept

Keeping both in mind the functional requirement and the architectural constraints, leads to the following concept [Frey, 2005]. The productive mainstream process is extended by another process working in parallel on a-priori knowledge that enables to check decisions and behaviour of the productive mainstream process from a higher level perspective and to build up a belief about the system's integrity status. The meta-process mimics that of a human operator who would do it in case he/she would have the same job. It exhibits the following functionality:

- The belief of the monitoring process level is elaborated by means of separate and simple routines and crosschecks.
- There is to be explicitly established a-priori knowledge for all cognitive subfunctions related to the monitoring and control functionality, in the first place including motivational contexts. Goals emerging from the goal determination subfunction on the basis of the corresponding motivational contexts are the primary basis for all action planning in the light of identified inconsistencies.
- The action planning in the light of identified inconsistencies applies special strategy-models for dealing with the particular inconsistencies which have occurred. In essence, this enables flexible decision making and ensures to comply with the work objective and pertinent goals in the best possible way. This means, the planning process creates a task agenda (plan) that brings the system as close as possible back to normal operation. Possible resulting measures are:
  - Modifying the system's a-priori knowledge of the productive mainstream process to heal the system and eliminate the problem from thereon.
  - Modifying the system's cognitive yield by installing a work-around procedure, if possible. This does certainly not eliminate the problem, but it enables the system to sufficiently cope with the problem.
  - Prohibiting the operation of identified erroneous knowledge elements. As a consequence, for instance, the meta-process creates a new plan for the mainstream functionality that might not be optimal, but that ensures the highest possible level of safety.

7.2.1.2 A Functional Prototype

In order to prove the concept of this kind of functional extension for system self-monitoring and performance controlling it is introduced in a given system called COSY^{flight} which represents the mainstream as an onboard SCU for semi-autonomous guidance and control of a UAV. This extended cognitive system has been implemented in a simulation facility and is used as an operational system. The extended system is called COSY+^{flight}.

An introduction to COSY+^{flight}

COSY+^{flight} is a first and relatively simple application implemented on the basis of COSA. It models a system which semi-autonomously controls a UAV during a reconnaissance mission.

The focus of this application lies on the goal-based cognitive behaviour of the UAV that evolves from the a-priori knowledge implemented.

To better understand the environment of COSY+^{flight} along with the interfaces used, Figure 147 shows all communication paths internal and external to the UAV.

- an external data link for messages to and from the human operator to receive a mission order or any updates. The operator communicates with the UAV in the same way as if he/she would communicate with a human pilot sitting in the aircraft's cockpit (except for formal communication protocol to ease the UAV implementation).
- Internal connections to sensors (including TCAS sensors) to get information about the traffic situation and, in turn, to build an internal representation.
- Internal connections to the onboard radio unit to communicate with the operator on the ground or other entities in the environment.
- An internal connection to the flight control system to control the UAV.
- Internal connections to onboard databases.

7.2.1.3 The Scenario

In order to show the capabilities of the cognitive system a mission order is given to the UAV. The mission has been kept simple in order to be able to point out the essential elements. COSY+^{flight} develops a flight plan in accordance with the system goals. If no unforeseen event occurs, the flight will be carried out as shown in Figure 148.

- Take-off on airport Stuttgart (ICAO code: EDDS) on runway 25
- Departure via a standard instrument departure (SID) route
- Execution of action “photo” at point foto1
- Execution of action “photo” at point foto2
- Execution of action “photo” at point foto3
- Arrival and approach via standard arrival route (STAR)
- Landing on airport EDDS on runway according to air traffic control (ATC) instructions

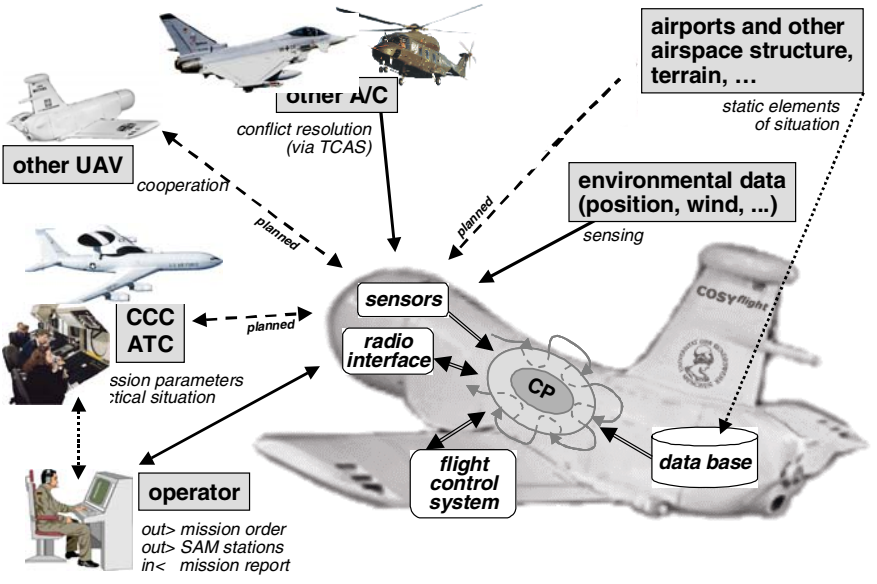


Fig. 147 Environment and communication links of COSY+^{flight} [Putzer, 2004]

Furtheron, an area of adverse environmental conditions (AEC) as a disturbing element is communicated to the UAV at start up. More disturbing elements can occur throughout the flight and will be passed to the UAV via data link.

As the UAV is flying in accordance with the flight plan as seen in Figure 148, unexpected traffic is reported by the simulated TCAS sensor on board of the UAV.

The following classification and calculation of an evasive route is handled according to the a-priori knowledge of COSY^{flight} mainstream sub-functions that a minimum distance is required to any other aircraft for full safety goal achievement.

Directly after finishing the planning and before the new flight plan is put into action, the cognitive sub-functions of feature formation and identification of the metalevel process detect that the planned evasion flight path would require a longer flight time than the remaining fuel quantity will allow. This means, an error of a-priori knowledge regarding the cognitive sub-function for flight planning in the mainstream system COSY^{flight} is detected. Therefore, the metalevel process determines a goal to cope with the knowledge error of the mainstream process. The metalevel cognitive sub-function for planning selects a strategy to make the system create a flight plan that needs less time. This is a typical example for a work-around procedure. The chosen strategy is to eliminate a waypoint, thereby searching for a plan where the overall achievement is still satisfying. This may include evaluating different routes according to the strategy. Still the mainstream process does not know about a fuel problem (as a result of the knowledge error which has not been corrected online), but creates a route that is sufficiently short to be satisfactory.

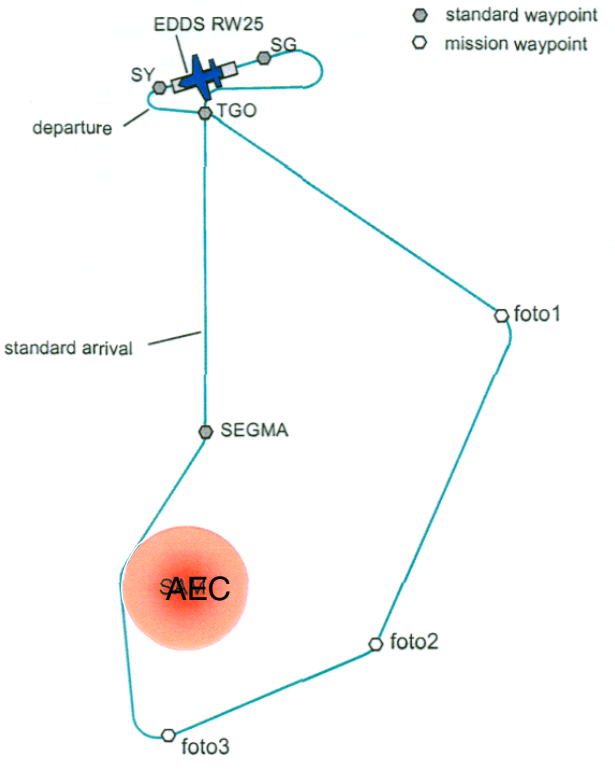


Fig. 148 Initial flight planning which is generated by the corresponding cognitive sub-function of COSY^{flight} [Frey, 2005]

The activated flight plan is shown in Figure 149, together with the intermediate plan (dotted line) that would require an unacceptable amount of fuel. If the metasystem builds up the belief that the automatic route calculation cannot produce a correct route at all, it is substituted by a deliberately simple algorithm to generate a new route which is not optimized at all but which ensures a sufficient level of safety (not shown here).

Figure 150 shows the general structure of motivational contexts for this application. It shows those motivational contexts which are accounted for here and those highlighted ones which are relevant for the actual planning error.

The motivational contexts in Figure 150 are described in Table 14. In this table the motivational contexts which are relevant for the planning error described are highlighted in colour. As shown these motivational contexts are not all permanently relevant. If relevant, there are to be certain criteria to decide about any error in the productive mainstream process. There are to be for instance criteria for the recognition of erroneous sensations, recognition of erroneous

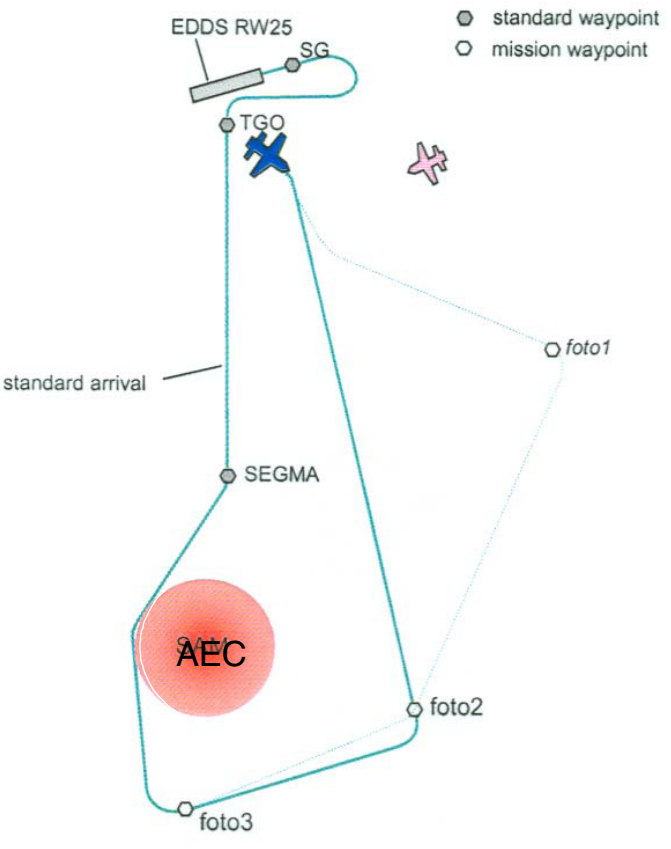


Fig. 149 New flight plan corrected by COSY+^{flight}[Frey, 2005]

determination of danger of collision with surrounding A/C (for instance by comparison with rough calculation in metafunction), recognition of erroneous a-priori knowledge about goal attributes, etc.

7.2.1.4 Conclusion

Simulator trials have shown that the extended cognitive system COSY+^{flight} has proven to be successful. The UAV communicates and interacts with its simulated environment to achieve its goals which are to fulfil a given reconnaissance mission under certain safety constraints. It is shown, too, that the semi-autonomous system is able to cope with internal knowledge errors in the a-priori knowledge by using the same strategies as a human operator would apply. Furthermore, the system would principally be able to correct its own a-priori knowledge online, but not necessarily in real time, as corresponding knowledge for knowledge acquisition (learning) would have to be part of the system (not shown in detail).

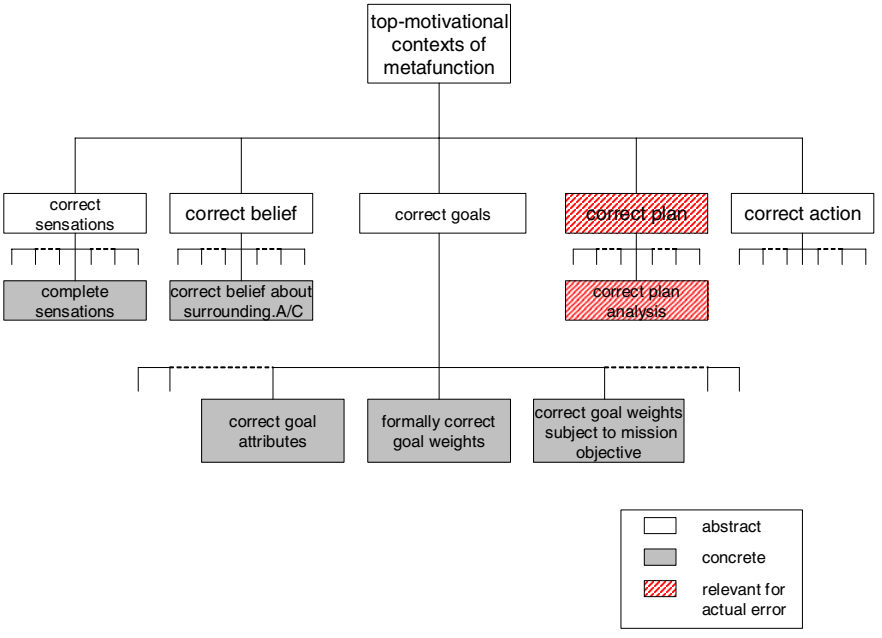


Fig. 150 Structure of motivational contexts with the choice of those accounted for in the application described and highlighting those which are relevant for actual error [Frey, 2005]

Table 14 Description of the motvational contexts which are accounted for in the application case discussed in this section. Those motivational contexts which are relevant for the planning error described are highlighted in colour.

Motivational context	Description
Top motivational context	Avoidance of compromising safety because of error in productive mainstream process Relevance: Permanently relevant
Correct sensations	Avoidance of compromising safety because of erroneous sensations in productive mainstream process Relevance: Permanently relevant
Complete sensations	Avoidance of compromising safety because of incompleteness of sensations in productive mainstream process Relevance criterion: Permanently relevant
Correct belief	Avoidance of compromising safety because of erroneos belief in productive mainstream process Relevance: Permanently relevant
Correct belief about surrounding A/C	Avoidance of compromising safety because of erroneous interpretation of sensations in productive mainstream process about surrounding traffic Relevance: In case of surrounding traffic

Table 14 (*continued*)

Correct goals	Avoidance of compromising safety because of erroneous goals in productive mainstream process Relevance: Permanently relevant
Correct goal attributes	Avoidance of compromising safety because of erroneous goal attributes in productive mainstream process Relevance: Permanently relevant
Formally correct goal weights	Avoidance of compromising safety because of formally incorrect distribution of goal weights in productive mainstream sub-function for goal determination Relevance: Permanently relevant
Correct goal weights subject to mission objective	Avoidance of compromising safety and poor compliance with the mission objective because of bad goals due to wrong goal weights Relevance: Permanently relevant
Correct plan	Avoidance of compromising safety because of planning errors Relevance: In case of planning necessity
Correct plan analysis	Avoidance of compromising safety because of a plan which cannot be carried out Relevance: In case of planning necessity
Correct action instruction	Avoidance of compromising safety because of erroneous action instruction Relevance: Permanently relevant

7.2.2 Identification of Performance Deficits of Work Systems

It is of great relevance to be able to measure the performance of new designs of work systems or just existing ones. Referring to Figure 32 in Chapter 4.2, unexpected overtaxing of the human operator is a typical occurrence experienced with conventional automation in work systems. This may happen with cognitive automation, too, quite easily in case of SCUs, if a situation arises which was not forseen by the designers and therefore not covered by the system design, but even with OCUs, too, as long as alerting assistance in combination with temporarily substituting assistance (see Chapter 5.2.2 and 5.2.3) is not sufficiently warranted in case of human overtaxing in task execution and action control (see Figure 27). The alerting assistance might work allright, providing alerts in case of violations as to the work objective. The substituting assistance, though, might lack sufficient a-priori knowledge for the assessment of the actual human resources, and therefore might be unable to figure out with sufficient accuracy that the human resources have been exhausted in the situation concerned. Research is going on in this field on resource modelling (see [Parasuraman, 2003] and [Donath & Schulte, 2006]), but completely satisfying results have not been achieved yet.

Airlines, for instance, are working with great effort on Flight data monitoring (FDM) programs to make records of all flight data of interest. This is done on a regular basis onboard the airliners to figure out from the huge amount of data in an

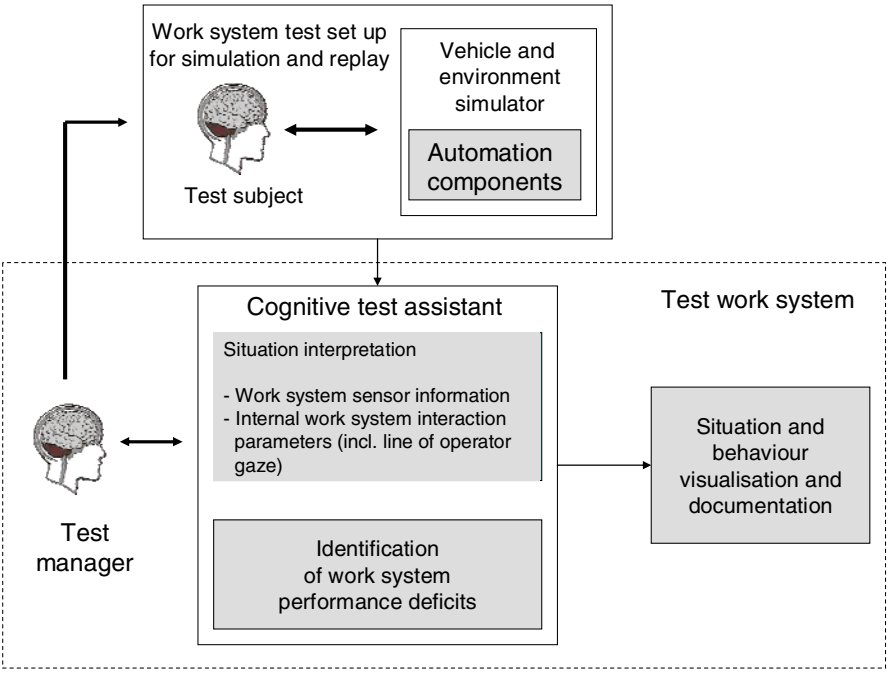


Fig. 151 Facility for performance monitoring of work systems (cf. [Flemisch, 2001])

offline and partially manual process based on considerable computation power when a critical situation of potential pilot overtaxing might have been occurred (see [Wright, 2005]).

Cognitive automation can be of great benefit to make this offline process much more effective by developing a software which would correspond to an onboard alerting assistant system and integrating this software into the offline process. This would make this process much more effective. The critical situations identified by this more automatised procedure could be possibly replayed in a flight simulator in order to present it to airline pilots for training purposes.

The longterm solution would be to have this software of alerting assistance working in real time onboard the aircraft. The output of alerts might be directly made use of by the pilots, or might be effected by a temporarily substituting assistant system, or might be saved for other offline purposes on the ground, for instance to trigger design changes of the work system for performance enhancement.

Another aspect of performance enhancement can be a comprehensive post-design process to test the effect of automation components on the work system performance. For instance, for testing certain components of cockpit automation, this test process, possibly as part of system validation, can take place in a flight simulator as a test rig. Such kind of test facility is depicted in Figure 151 (see [Flemisch & Onken, 2000] and [Flemisch, 2001]) for the general case of work systems for vehicle guidance and control. There is the work system which is to be

investigated by the test and the work system to manage the test. The operating force of the work system to manage the test comprises the test manager and a cognitive test assistant. The assistant has got the knowledge about the work objective for the test and the motivational contexts to identify performance deficits of the work system to be tested. These deficits and the situations during the test associated to the deficits are documented and may be visualised for the test manager and possibly also for the test subject after the test. There may be also initiated a replay of these situations. In the light of the experience of the replay the test subject can comment the identified deficits without any load on his memory.

The cognitive test assistant can be considered as an alerting assistant, not monitoring the test work system but the work system to be tested.

Chapter 8
Abbreviations

ABS - Anti-lock Braking System
A/C - Aircraft
ACAS - Airborne Collision Avoidance System
ACC Adaptive Cruise Control or Active Cruise Control
ACL - Agent Communication Language
ACT - Adaptive Control and Thought
ACT-R - Adaptive Control and Thought - Rational
ACU - Artificial Cognitive Unit
ADB – Actual Danger Boundary
ADS-B - Automatic Dependent Surveillance-Broadcast
AEC - Adverse Environmental Conditions
AFCS – Automated Flight Control System
AFP - Automatic Flight Planner
AI – Artificial Intelligence
ALT – ALTitude acquisition mode
APP - Approach
ARINC – Aeronautical Radio Incorporated
ASPIO - Assistant for Single Pilot IFR Operation
ATTAS - Advanced Technologies Testing Aircraft System
ATC – Air Traffic Control
A/THR – Auto THRust
ATIS – Automatic Terminal Information Service
AugCog - Augmented cognition
BARCO – Company for professional display and visualisation solutions
BDI – Beliefs, Desires, and Intentions
BMBF - German Bundesministerium für Bildung und Forschung
CAMA - Crew Assistant Military Aircraft
CAN – Controller Area Network
CASSY - Cockpit ASsistant System
CBR - Case-Based Reasoning
CCC – Comand, Control, and Communication
CDU - Control and Display Unit CEL - Constructive and Evaluating Learning
CFIT - Controlled Flight Into Terrain
CIM - Cockpit Information Manager
CLB - Climb
CML - CommonKADS Markup Language
COGMON - Cognition Monitor
COGPIT - Cognitive Cockpit project

CORBA – Common Object Request Broker Architecture
COSA - COgnitive System Architecture
COSY – Cognitive System
CP - Cognitive Process
CPDLC - Controller Pilot Data Link Communications
CPL - Cognitive Process Language
CS - Coming to a StandStill
CV - Characteristic Value
DAISY - Driver Assistant system
DARPA - Defense Advanced Research Projects Agency
DASA - Daimler-Benz Aerospace AG)
DDS - Driving at Desired Speed
DERA – Defence Evaluation and Research Agency
DES - Descend
Design/OA Seite233
DGPS – Differential Global Positioning System
DLH – Lufthansa
DLPFC – Dorsal-Lateral PreFrontal Cortex
DLR – Deutschs Zentrum für Luft- und Raumfahrt (German Aerospace Research Establishment)
DM - Dialogue Manager
DOD – Department of Defense
DOXYGEN Seite260
DTED - Digital Terrain Elevation Data
EA - Execution Aids
ECAM - Electronic Centralized Aircraft Monitoring
EDDF - Airport Frankfurt
EDDH – Airport Hamburg
EDDS – Airport Stuttgart
EDVE - Airport Braunschweig
EDVV – Airport Hannover
EEG – Electroencephalography
EFIS - Electronic Flight Instrumentation System
EICAS - Engine Indication and Crew Alerting System
EMMA - Eye Movements and Movement of Attention
EPIC - Executive-Process/Interactive Control
EPPC - Entorhinal, Perirhinal, and Parahippocampal Cortex
ESC - Electronic stability Control
ESG - Elektroniksystem- und Logistik-GmbH
ETA – Estimated Time of Arrival
EU – European Union
EUREKA – Acronym for a research program which is sponsored by the EU
FAA –Federal Aviation Authority
FAR – Federal Aviation Regulations
FFM – Airport Frankfurt
FIPA - Foundation of Intelligent Physical Agents

FLX-TOGA – FleXible TO/GA
fMRT - functional Magnetic Resonance Tomography
FMS - Flight Management System
FDM – Flight Data Monitoring
FPA – Flight Path Angle
GEMS - Generic Error Modeling System
GIDS - Generic Intelligent Driver Support
G/S – Glide Slope
HAL - Hyperspace Analogue to Language
HCI - Human-Computer Interface or Human-Computer Interaction
HDG - Heading
HOTAS - Hands on Throttle and Stick
OLG - Outer-Loop-Guidance
HUD – Head-Up Display
ICAO – International Civil Aviation Organisation
IFR – Instrument Flight Rules
ILS – Instrument Landing System
IMAD – Instantiation of Model of Actual Driver
IMASSA - Institut de Médecine Aérospatiale - Service de Santé des Armées
IMC - Instrument Meteorological Conditions
INA – Integrated Net Analyser
INS – Inertial Navigation System
IO – Input/Output
IRD - Instantiation of Reference Driver
IRIX - a UNIX version from SGI
KADS – Knowledge Acquisition and Documentation Structuring
KSD - Keeping Safety Distance to vehicle ahead
LAND
LSA - Latent Semantic Analysis
LOC - Localizer
LTP - Long-Term-Potentiation
MAS - Multi-Agent System
MCT – Maximum Continuous Thrust
MEG - Magneto-Encephalography
MFD - Multi Function Displays
MICO – Implementation of CORBA Standard
MIL-STD – Military Standard
MMR - MiniMum-Risk-route
Motif – User interface standard
MOTIV – Research program sponsored by BMBF
MUM-T - Manned-Unmanned Teaming
NARIDAS – Navigational Risk Detection and Assessment System
NAV - Navigation
ND - Navigation Display
NR – Normal Region
OA – Object-oriented Architecture

OCU - Operating Cognitive Unit
OF – Operating Force
OM – Outer Marker
OSF – Open Software Foundation
OSM – Operation-Supporting Means
QT - Library for the (graphical) user interface
PA - Pilot's Associate
PACT - Pilot Authorisation and Control of Tasks system
PC - Personal Computer
PDP - Parallel distributed processing
PE - Piloting Expert
PET- Positron Emission Tomography
PFD - Primary Flight Display
PIER - Pilot Intent and Error Recognition
RA - Resolution Advisory
RBF – Radial Basis Function
RCS - Real-Time Control System
REFA – Reichsausschuss für Arbeitszeitermittlung
RLS - Recursive Least Square
RMU - Radio Management Unit
RPA - Rotorcraft Pilot's Associate
RTI - Road Transport Informatics
RWY - Runway
SAM - Surface to Air Missile
SASS - Situation Assessment Support System
SCU - Supporting Cognitive Unit
SEAD – Suppression of Enemy Air Defenses
SEGMA - Standard waypint of IFR flight
SG - Standard waypint of IFR flight
SGI – Silicon Graphics Inc.
SID - Standard Instrument Departure
Soar - State, operator, and result
SOM - Self-Organizing Map
SOP - Standard Operating Procedures
SQL - Structured Query Language
SPD/MACH – Speed/Mach Hold
SRS – Speed Reference System
STAR - Standard Arrival Route
STL - Standard and Collision Avoidance System
TA - Traffic Advisory
TAWS - Awareness and Warning Systems
TCAS - Traffic Alert and Collision Avoidance System
TGO – Standard waypint of IFR flight
TIM - Tasking Interface Manager
TIS-B - Traffic Information Services-Broadcast
THR – auto THRottle

TOEFL – Test of English as a Foreign Language
TOGA – Takeoff-Go-Around: An aircraft automation mode that controls and displays information about the takeoff and go-around manoeuvres
TOGA LK – TOGA locked
TLC - Time to Line Crossing
TRK - Track
TTC - Time-To-Collision
UAV - Unmanned Aerial Vehicle
UML – Unified Modeling Language
UR – Uncontrollable Region
VFW – Vereinigte Flugwerke
VITA – computer Vision Technology Application
VLPFC – Ventral-Lateral PreFrontal Cortex
VMC - Visual Meteorological Conditions
VOR - Very High Frequency Omnidirectional Radio Range
V/S – Vertical Speed

Appendix

1 Supplementary Useful Facts about Human Factors

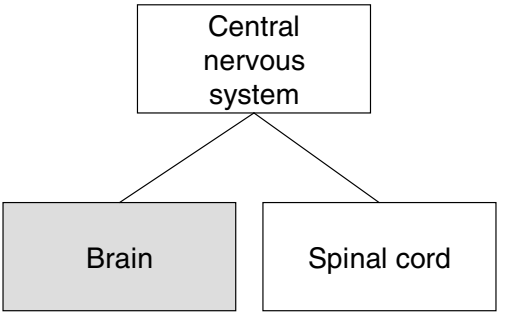
In this appendix a selection of facts about human factors are made available to the reader, mainly concerning those related to human cognition. They are in the first place referring to aspects which are discussed in the main part of the book, mainly in Chapter 3.2.

1.1 Main Brain Structures

1.1.1 Anatomical Aspects

The human brain is the place of the body where cognition as such takes place. The structure of the brain has evolved from the biological process of evolution over a time span of millions of years. It is the current end product of this evolutionary process. The brain is one of the two divisions of the human *central nervous system*. It forms the central nervous system together with the *spinal cord* (see Figure 1).

Fig. 1 Brain as part of the human central nervous system



The major divisions of the brain are the *forebrain*, *midbrain* and the *hindbrain*. Illustrative depictions of the main parts of these divisions are for instance given in [Pinel, 2007]. The forebrain with its neuronal formations (telencephalon and diencephalon) is of most interest for our discussions on cognition. The telencephalon as part of the forebrain takes the main volume of the brain and comprises the *cerebral cortex* with the *neocortex* and hippocampus, the sub-cortical structures such as the *amygdala* and the *basal ganglia* (see Figure 2) which in turn

consists of the *globus pallidus*, the *corpus striatum* (*nucl. accumbens*, *putamen*, and *nucl. caudatus*), the basal forebrain, and in the broader sense also the *nucleus subthalamus* and the *substantia nigra* of the midhirn. The neocortex is subdivided in pairs of *occipital lobes*, *temporal lobes*, *parietal lobes*, and *frontal lobes*.

The diencephalon as part of forebrain consists of the *hypothalamus*, the *epithalamus*, the *thalamus* and the *mammillary* bodies (see Figure 2).

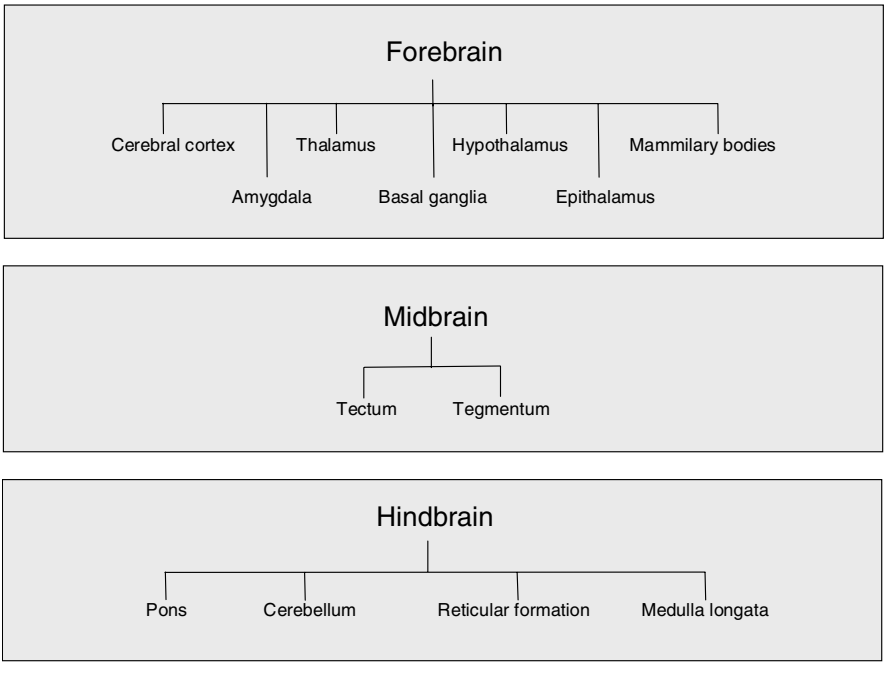


Fig. 2 Major brain structures

The midbrain (mesencephalon) and the hindbrain (*metencephalon* and *myelencephalon*) form the transition to the spinal cord. The midbrain has got a dorsal (see Figure 3) part with the *tectum* and a ventral part, called the *cerebral peduncle*, with the *tegmentum and the substantia nigra*. If looking from the front (rostral) towards the rear (caudal), the hindbrain consists of the metencephalon (*cerebellum, pons,* and *reticular formation*) and the myelencephalon (*medulla longata*) (see Figure 2). The reticular formation as a net-like structure above the pons from the medulla longata to the midbrain forms three longitudinally arranged strings of nuclei, the *median* string with the Raphe nuclei, the *medial* string, and finally the *lateral* string with the *PPT* (*nucl. tegmenti pedunculo-pontinus*), the *nucl. subcoeruleus* and in the broader sense the *locus coeruleus*.

Functional properties of main brain structures

All of the structures mentioned so far have their own functional properties most of which will be briefly outlined in the following.

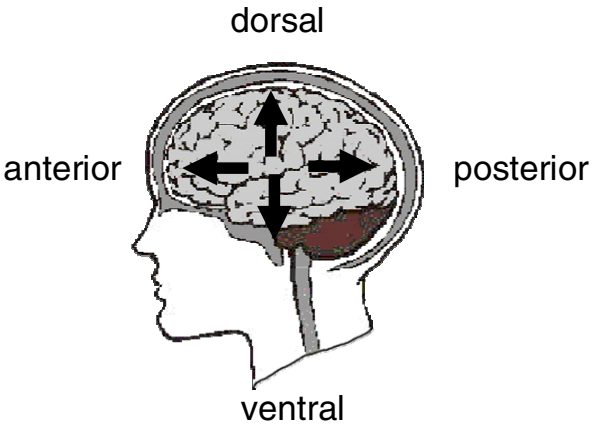


Fig. 3 Terms used to anatomically locate an area of interest in the human brain. In addition, the terms superior and inferior are used to distinguish between what is located more towards the top and what is located more towards the bottom, respectively, and the terms medial and lateral are used to describe what is located more inside and what is located more outside, respectively, if looking from the front

The neocortex mainly comprises the sensory and motor cortex areas as well as associative areas. According to the main interest in the context of this book, there are to be mentioned the

- Sensory cortices
  - primary and secondary somatosensory cortex
  - primary, secondary, and tertiary visual cortex
  - primary and secondary auditory cortex
- motor cortex (like primary motor cortex , premotor cortex, Broca speech centre, and frontal eye fields)
- cingulate cortex
- insular cortex and
- association cortices

The *somatosensory cortex* serves the function of perception of body states. It processes the sensory modalities of receptors of the skin for touch, pressure, vibration, temperature, and partially pain. It also processes the modalities of receptors inside the body (proprioception). The pertinent cortex areas for the different parts of the body are shown in.

As an example of the *perceptual processing* that of the visual modality is described in Chapter 3.2.1.2 of the main part of this book, including pictures of the locations of the pertinent cortices. In appendix 2.1 some more information about other modalities is given, including that of the motor system.

The cortical part associated with the *cingulate gyrus* is referred to as *cingulate cortex*. The anterior cingulate cortex (ACC), which is of most interest here, can be divided anatomically based on attributed functions into executive (anterior),

evaluative (posterior), cognitive (dorsal), and emotional (ventral) components [Bush et al., 2000]. The ACC is connected with the prefrontal cortex and parietal cortex as well as the motor system and the frontal eye fields [Posner & DiGirolamo, 1998] making it a central station for processing top-down and bottom-up stimuli and assigning appropriate control to other areas in the brain. The ACC seems to be especially involved when effort is needed to carry out a task such as in early learning and problem solving (Allman et al., 2001). Many studies attribute functions to it such as error detection, anticipation of tasks, motivation, and modulation of emotional responses to the ACC [Bush et al., 2000], [Nieuwenhuis et al., 2001], and [Posner & DiGirolamo, 1998].

The *insular cortex* represents and processes sensations of tastes besides others such as pain feelings.

The association cortices occupy the greatest part of the neocortex. They are the areas of the cortex which serve the more complex processing within the sensory system of one modality or between sensory systems of different modalities. From the association cortices projections go to other cortex areas for further processing, for instance to motor cortices. Essentially, there are

- the parietal association cortex
- the temporal association cortex
- the frontal or prefrontal association cortex.

The parietal association cortex serves the orientation in space by means of a spatial conception of the surrounding real world (including supporting means like human depictions of the real world (maps, drawings)). This includes the mental setup of a three-dimensional world (also in spatial perspective) and the localisation of the environmental stimuli, the own body and its motion relative to its environment. The functions of the parietal association cortex also comprise reading, calculating, and handling of symbols in general.

The temporal association cortex is in charge of functions like the integration and evaluation (recognition) of visual aspects of objects and processes as well as auditory information.

The prefrontal association cortex provides spatiotemporal structuring of sensory perceptions and is concerned with planning of goal- and context-dependent actions. The location of the working memory is considered to be within the prefrontal association cortex. The orbitofrontal cortex as part of the prefrontal cortex is thought to regulate planning of behaviour associated with sensitivity to reward and punishment. In that sense the orbitofrontal cortex accommodates the capacity to understand the context in a situation of social-communicative interaction with other people. It also plays a role for attention control.

Moving to the functions of the subcortical structures, that of the *amygdala* is probably the most prominent one. It plays a chief role in the process of generation and learning of emotions. It primarily concentrates on negative emotions like fear, anxiety, fury, anger, disgust. In that sense it is the central, top-level evaluator of incoming information and reacts by activating the other regions of the brain to act in accordance to its evaluation. The amygdala formation directly affects the hormone releasing and vegetative systems by its projections to the hypothalamus.

As to the functions of the *basal ganglia*, there are to be described those of the *corpus striatum,* the *globus pallidus*, the *nucleus subthalamus*, and the *substantia nigra*.

The corpus striatum (often simply "striatum") consists of the nucl. accumbens, the nucl. caudatus, and the putamen. The dorsal part of the striatum (most of nucl. caudatus and putamen), which constitutes the dorsal basal ganglia, plays a major role together with the globus pallidus in the process of controlling voluntary actions. The striatum is part of important processing loops of the brain to regulate execution functions (originated in the frontal lobe). These loops realise the interaction of motivation, emotion and rational cognition on the behaviour. The substantia nigra is also involved in these circuits. The striatum and the globus pallidus are functioning in an antagonistic way. In humans the striatum is activated by stimuli associated with reward which is partly associated with the ventral part of the striatum.

Among the functions of the components mentioned as part of the diencephalon are those of the *hypothalamus*, the *epithalamus*, the *thalamus*, the *nucl. subthalamus*, and the *mammillary bodies* only that of the thalamus will be described here in more detail.

The thalamus consists of a number of nuclei with correspondingly multiple functions. Deduced from the design of the isothalamus, it is generally believed to act as a translator by which various "prethalamic" inputs are processed into a form readable by the cerebral cortex. The thalamus processes and selectively **relays** sensory information to various parts of the cerebral cortex, as one thalamic point may reach one or several regions in the cortex. Thereby, it is also involved in the process of attention control, and it decides which of the information coming from sense organs and other parts of the body is important enough to be forwarded to respective cortical areas. Moreover, some of this information might eventually be forwarded to consciousness. A major role of the thalamus is also dedicated to the motor systems through connections to the basal ganglia. Certain nuclei of the thalamus have got strong connections to limbic structures (therefore also called limbic nuclei, see also following details of the limbic system) like the gyrus cinguli, the mammillar bodies, and the orbitofrontal cortex. It is believed that these connections are necessary to maintain attention. The thalamus also plays an important role in somatic functions like regulating states of sleep and wakefulness. Damage to the thalamus can lead to permanent coma.

Besides its participation for regulating vegetative functions like the medulla longata and the pons, the hypothalamus takes part to control basic affective behaviour in the heat of the moment by increasing the state of arousal and performance of the nervous system and associated body functions.

Being located in the middle of the brain, the epithalamus has got a function to take care of connections between the limbic system and other parts of the brain. In the context of visual performance it is also responsible of the pupillary reflex in reaction to changes in ambient light of a work site.

The functions of the nucl. subthalamus are much less known than for many of the other parts of the brain. It is strongly connected with the globus pallidus and is mainly responsible for gross movement.

The mammillary bodies along with the anterior and dorsomedial nuclei in the thalamus, are involved with the processing of recognition memory.

Regarding the hindbrain structures, the *cerebellum* should be briefly described in the first place. It is the important sensori-motor structure which assures the ability to precisely control body motions and to adapt them to changing conditions.

The *reticular formation* is a part of the brain that is involved in actions such as awaking/sleeping cycle. In that sense it is an important component to warrant vigilance and attention as prerequisites of consciousness. The lateral string of nuclei as part of the reticular formation accommodates the *tegmenti pedunculo-pontinus* (PPT) and the locus coeruleus as was mentioned before. The locus coeruleus announces to other parts of the brain that there are new external stimuli or any which stand out, whereas the PPT along with a network structure formed by the Nucleus basalis Meynert and others (*basal forebrain*) informs in a similar way about the relevance and significance of external stimuli. Evidently, the basal forebrain stands for the subcortical centre of attention control. The Raphe nuclei, however, provide an attenuating effect on those structures which are projected as target areas from both the PPT and the locus coeruleus.

From the functional point of view, there certainly should also be addressed what is the so-called *limbic system*. The limbic system is often called the centre of feelings. It is considered to be the central evaluator of the information entering the brain by regulating emotions, motivation, attention (as prerequisite of consciousness), associations, and memory. It more or less dictates the first cognitive reaction, and after ping-pong-like activations sweeping through numbers of brain structures, and it also has the final say. It is not easy to accurately delineate the limbic system. By many classifications it is composed of the components shown in Figure 4:

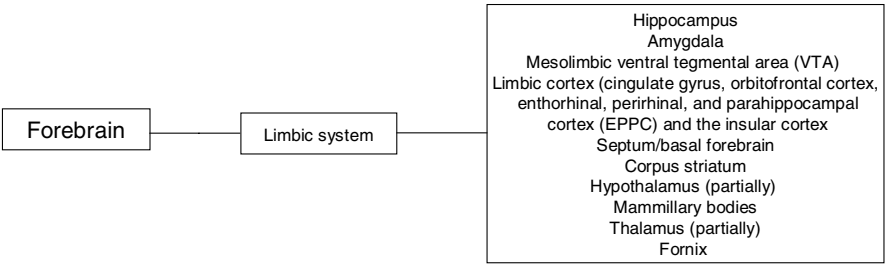


Fig. 4 Components of the limbic system

In the following the functional contributions of these components (the latter four components are not always recognized as part of the limbic system in the literature) will be briefly described as far as they are not covered before. For instance, the function of the amygdala was already described which is one of the central structures of the limbic system.

The *septum* participates to regulate emotions and the ability to learn and control impulses as well as such drives as sex, hunger, thirst, aggression, and fear.

The *hippocampus* is the main component for the organisation of the declarative memory (episodic in the first place) whose contents can get access to conscious experience, if relevant. The memory itself is not located in the hippocampus. It rather is stored in the different modality- and function-specific areas of the neocortex. The sematic memory is mainly controlled by the *enthorhinal, perirhinal, and parahippocampal cortex* (EPPC) which surrounds the hippocampus formation.

The functions of the *cingulate gyrus* have already been described in the context of cortical functions. Besides the insular cortex and the medial and orbitofrontal cortex it can be considered as the cortical part of the limbic system. It works as a mediator between the cortical-cognitive and the limbic-emotional functions.

The *ventral tegmental area* (VTA) is part of the *mesolimbic system* which is linked to expectancy of reward and the feeling of satisfaction.

The *fornix* stands for a bundle of axon fibres which carries signals from the hippocampus to the mammillary bodies and septal nuclei.

1.2 Perceptual Processing

The main part of the necortex is devoted to the perceptual processing, in particular the systems for the processing of visual, auditory, gustatory, and somatosensory stimuli, to mention the main ones.

The visual perceptual processing is described in the main part of the book. Here, the perceptual processing of other sensing modalities is being discussed. Whereas the main visual pathway is directly projecting into the neocortex, the main pathway of human audition from the ear to the association cortex is going all the way through the hindbrain and midbrain into the cortex, i.e. the geniculate nuclei of the thalamus and subsequently the primary and secondary auditory cortex in the temporal lobe. The main somatosensory pathway goes even all the way up from the spinal cord to the thalamus (ventral posterior nuclei) and further to the corresponding primary and secondary somatosensory cortex, The primary somatosensory cortex is somatotopic which means that it is organised according to a map of the body surface, the so-called somatosensory homunculus.

The most prominent property of human audition is the fact that it is omni-directional. The ears are always receptive, no matter where the stimuli are located around the body. In order to have this information processed further, though, attention is needed, too. This is of great importance, because there is no difficulty to trigger the operator's attention, if the stimulus has got a certain degree of relevance. This is best illustrated by the so-called cocktail-party effect. When one is focusing on a conversation at a cocktail-party, one is usually unaware of the conversations around, although one could hear what is talked about, if one wanted to. In case, however, your name was mentioned in one of the other conversations, this immediately would gain your attention. There is another aspect of audition which might lead to critical situations and errors, though: The stimuli are not lasting. This is in contrast to vision, where the stimuli usually stay for a while. A printed word does not disappear while you are reading.

1.3 Motor Processing

Similar to the perceptual processing the motor processing is hierarchically organised with so-called functional segregation [Pinel, 2007]. Each hierarchy level of the system has got different functions carried out by different neural structures. The neuronal pathways of motor processing start from the pertinent association cortex and go via the secondary motor cortex, the primary motor cortex, the brain stem motor nuclei and the spinal motor circuits. The lowest hierarchy level is represented by the muscles with their neuromuscular junctions.

The motor cortices are supported during both the programming and the execution of their tasks by control systems. Main players in this respect are the cerebellum and the basal ganglia. The cerebellum is involved in the initiation and proper timing of movements and the basal ganglia play a role for the speed of movements [Nieuwenhuys, 2001].

1.4 Language-Based Communication

Besides using senses and effecting by means of hands and feet, language-based communication certainly plays an important role in work processes when interacting with other collaborating members of the operating force as well as interacting with operation-supporting means and other work processes. Communication in this sense means exchange of language-based (coded) information and is carried out written and spoken. Verbal language is probably the human peculiarity as a result of the evolutionary process which has caused the human species standing far out from any other creatures on earth. It is the basis for the cultural achievements of mankind. It has to be noted here, though, that humans, as opposed to communication in technical systems, can only communicate via sensing (mainly visual and auditory system) and effecting, which results in time lags. This may generate an efficiency problem. It might happen, in particular, in the case of use of natural language in situations with time constraints, where both misunderstanding of short but ambiguous instructions or highly time-consuming utterances can be fatal. This is in particular true with regard to non-formal verbal commands for supporting machinery.

The classic theory of localisation of language in the brain is the so-called Wernicke-Geschwind model. It suggests the following:

The sounds of a spoken word by another person are sent to the primary auditory cortex and projected to *Wernicke's area*, an area of the left temporal cortex. There the meaning of the word is generated. In order to speak, Wernicke's area generates the neural representation of the word meaning which is transmitted to *Broca's area*, an area of the inferior prefrontal cortex. Broca's area holds a representation for articulating words. Thereby it drives the appropriate neurons of the primary motor cortex and ultimately the muscles of articulation. In order to read, the signal received by the primary visual cortex is transmitted to the left angular gyrus, the area of left temporal and parietal cortex just posterior to Wernicke's area, and from there to Wernicke's area. In order to read loudly, Wernicke's area then triggers the appropriate responses in Broca's area.

This model was derived from observations made in the context of brain surgeries. It is certainly a simplification as it is the case with all models, but it captures some main pathways. There is more recent evidence [Hinton et al, 1993], though, that there is parallel and interconnected processing involved to some extent as it is also known from the sensory and sensorimotor systems.

Language-based interaction between humans and machines is an important aspect for the design of work systems. There is great development endeavour for artificial speech systems which are able to understand human speech and to generate and speak out verbal utterances on their own. Great progress is already made in this field. Designers can already count on these techniques for a number of applications.

2 Input/Output Modalities

2.1 Sensing Modalities

There are plenty of sensing modalities of the human operator. First, there are the five senses: vision, hearing, tactile feeling, tasting, and smelling. In addition, we can sense heat and coldness, vibration, and pain and our motions. This is still not all. Our body is full of somatosensory receptors. The visual modality is by far the sensing resource mostly relied on by the human operator with the auditory one coming next. Thus, these are the sensing modalities which will be dwelt on somewhat further, because they are the ones being of predominant concern for the work system designer. It should be noted that sensing is not pertinent to cognition as such, but in the case of human cognition, there is surely no way to do without.

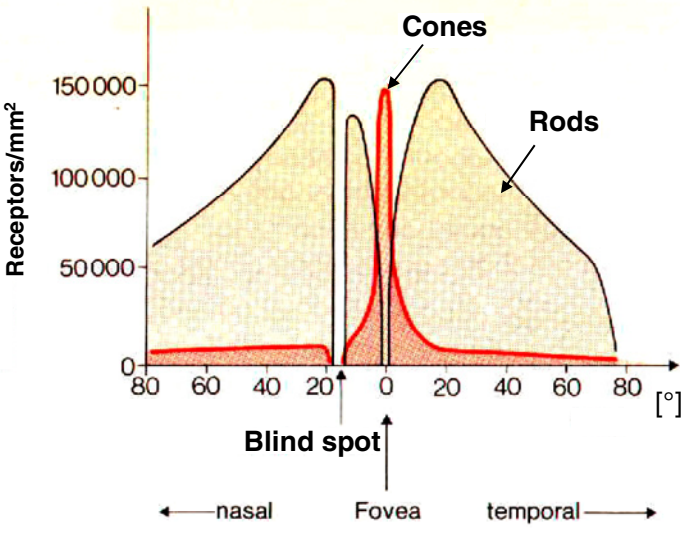


Fig. 5 Distribution of cones and rods across the retina (cf. [Schmidt & Thews, 1980])

For humans, taking in information and learning can only take place through the senses. Artificial cognition could possibly rely on communication links instead.

As to the human eyes as visual modality, the two kinds of receptors on the eye retina (see Figure 5), the so-called cones and rods, mediating two kinds of vision: *photopic* and *scotopic* vision, respectively. Almost all cones are located in the central part of the retina in a very confined area around the fovea. Thus, there is only a small region around the fovea of about two degrees of visual angle where we can make use of the special properties of this kind of visual receptors. For instance, by means of the cones we are able to perceive colours. There are specialised sub-groups of cones for the colours blue, green and red. Probably more important, in particular for the human operators of fast vehicles, is the fact that solely the cones let us perceive the imagery details with high acuity, a constraint very important to know for human operators of fast vehicles. This is even more a constraining factor, because this mode of vision demands for attention. In order to get objects of interest in this small region around the fovea, the eyes have to be moved towards them. There are automatisms of eye movements, for instance to catch a conspicuous object not too far from the line of sight. In particular in a stationary scene, there are rather short, discrete and swift unconsciously activated exploring saccadic movements of the line of sight onto so-called fixation points in the scene, which leaves us with the illusionary impression of a much larger field of fine-grained perception. However these automatisms do not work if the interfering object is not a conspicuous one and if the attention resource is exhausted in situations of complex scenes and fast ego-motion. In certain situations in the work process it may be a fatal weakness of the human operator when relying on these automatisms of eye movement. Therefore, in order to avoid that a threatening object or something dangerous escapes one's notice one has to allocate conscious endeavour to move the line of sight into the direction of a potential but unlikely threat. Since there is no automatism this has to be trained beforehand on the basis of knowledge about the threat potentials.

Scotopic vision relies on the rods as receptors. They are located in abundance about all over the retina (120 million as opposed to 6 million cones) except in the inner fovea region (see Figure 5). Therefore, rod-mediated vision is responsible for peripheral vision. It is almost completely driven by automatic (subconscious) processes and provides information about the gross impression of the scene along with human body's attitude relative to the environment, although with much less visual acuity as compared to photopic vision. Scotopic vision stays alive at night as opposed to photopic vision (500 times as sensitive to light), with some deficiencies, though, and it brings into focus objects which are fast moving or particularly conspicuous in shape and colour as was already mentioned above.

There are many other peculiarities concerning the human eye system. [Guyton, 1987], [Bruce et al., 1996], and [Schmidt & Thews, 2000] may be further helpful survey references for physiological and psychological issues. Other human factor references may be [Boff et al., 1986] and [Ernsting et al., 1999]. [Kraiss & Moraal, 1976] is an older reference for application-oriented issues.

Modelling of the visual behaviour of human operators depends heavily on the ability to observe which stimuli being sensed play a role in the process of the

generation of voluntary actions. This is probably the greatest challenge the work system designer has to face who wants to make use of behavioural models. Eye and head movement measurement alone is not sufficient; the semantic link to what is in fact attentively looked at in the environment is also important and might be even more crucial. This is a very lively field of research. The advances in eye gaze and head tracking equipment are encouraging (see [Jacob, 1993] and [Ohno et al., 2004]).

Also for the auditory modality, only a flavour of what is involved can be given here (see [Zwicker, 1982], [Boff et al., 1986], and [Kimble, 1988] for more detailed information). The cochlea, a long coiled structure of the inner ear, houses the auditory receptors. They are tiny hair cells, located between two membranes. Sound stimuli cause membrane oscillations with the highest amplitude at the location which corresponds to the frequency of the predominant tone of the stimulus. The resulting motion of the hair cells produces the action potential in the auditory nerve. The threshold of hearing is important which is dependent of both the frequency and the loudness of the sound (tone). There is also an upper bound of loudness up to which the sound can be sustained without harm.

The auditory modality plays an important role in the context of human-human and human-machine speech communication. Designing for speech inputs about environmental objects, which have to be seen in the context of a task in a work process, can be a good remedy for lacking measurements of visual inputs. Artificial speech recognition in parallel to human speech recognition will provide any artificial cognitive units for assistance of the human operator with important information. For instance, verbal instructions of air traffic controllers can be monitored in parallel to the pilot this way. Speech systems of good quality are available these days. Furthermore the auditory modality seems to be most appropriate, when it is desired to capture the human operator's attention, since other then the visual modality it is not physically preferring one direction as information source.

2.2 Effecting Modalities

All muscles in our body are effectors which are ready to be deployed in the course of an action. The human operator makes use of those, which are the adequate ones for a task when operating the work system. Within the scope of this book, we only touch this topic for the sake of describing the human body functions involved in the behavioural control process as a whole. We can easily imagine that the muscles of arms and hands on the one side and the legs and feet on the other side are those predominantly used for the control actions considered in the application domain of this book. However, we should not forget eye, head, and body movements to direct our senses and to generate speech. The main issues from the work system designer's point of view are the human factors about control accuracy, control rate, and achievable control power. The latter is quite relevant, since the control force is dependent on the time span the force has to be maintained in a certain task. This is important for the design of the controls and the anthroprometric design of the body position at the work site [Schmidtke, 1993]. Any human drawbacks in this respect can be rather easily overcome by

well-known technical aids. Control precision is dependent on the control force demanded (impact of force opposing the motion for wanted displacement of control lever). If there is no significant demand on the control force, like those for human-computer interaction, hand movement towards a target and keystrokes, the so-called Fitt's law is very relevant and provides a good estimate of the time needed for the move (see Figure 6) [Card et al., 1983]. The time of the hand movement towards the target area is given by

$T_{pos} = I_M \log_2 (2D/S)$, where $I_M = - (\tau_p + \tau_c + \tau_M)/\log_2 \varepsilon$, $\varepsilon = x_i/x_{i-1}$, and $i = 1 \ldots n$

The movement is not continuous. It may take some subsequent corrections which involve perception (τ_p), central cognitive processing (τ_c), and motoric action (τ_M). All together takes about 250 milliseconds for each corrective move. The total number n of corrective moves is dependent on D, S, and ε.

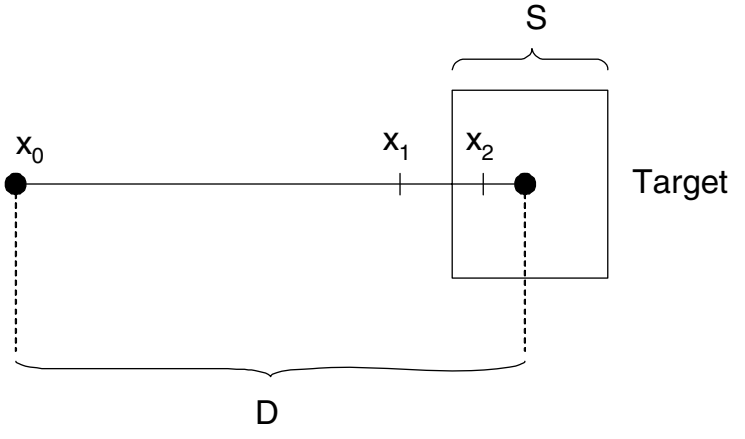


Fig. 6 Hand movement to a target area, xi is the remaining distance to the target centre point, x0 equals D [Fitts, 1954]

Regarding the application domain of vehicle guidance and control, it should be accounted for the dynamics involved in the control process. In this context, the neuro-muscular delay of the control movement might be something to be concerned about. Furthermore, there are other additional findings of interest to the work system designer about human effectors and the design of control levers which will not be discussed in this book in detail. For those who want to study this further we would recommend at this point for human factor issues [Boff et al., 1986] and [Kimble, 1988], for physiological issues [Guyton, 1987] and [Schmidt & Thews, 2000], and anthropometric issues [Tilley, 2002].

References

[Albus & Meystel, 2001] Albus, J.S., Meystel, A.M.: Engineering of mind, an introduction to the science of intelligent systems. Wiley & Sons, Chichester (2001)

[Allman et al., 2001] Allman, J.M., Hakeem, A., Erwin, J.M., Nimchinsky, E., Hof, P.: The anteriror cingulate cortex: the evolution of an interface between emotion and cognition. Annual New York Academic Science 935, 107–117 (2001)

[Amalberti & Deblon, 1992] Amalberti, R., Deblon, F.: Cognitive modeling of fighter aircraft process control: a step towards an intelligent onboard assistance. International Journal of man-machine studies (1992)

[Anderson, 1983] Anderson, J.R.: The architecture of cognition. Havard University Press, Cambridge (1983)

[Anderson, 2000] Anderson, J.R.: Cognitive psychology and its implications. Worth Publishers, New York (2000)

[Anderson, 2002] Anderson, J.R.: Spanning seven orders of magnitude: A challenge for cognitive modelling. Cognitive Science 26, 85–112 (2002)

[Anderson et al., 2004] Anderson, J.R., Byrne, M.D., Bothell, D., Douglas, S., Lebiere, C., Qin, Y.: An integrated theory of the mind. Psychological Review 111(4), 1036–1060 (2004)

[Augmented Cognition International, 2006] Augmented cognition International (retrieved, July 2006), `http://www.augmentedcognition.org/`

[Austin, 1962] Austin, J.L.: How to do things with words. Oxford University Press, Oxford (1962)

[Baddeley, 1990] Baddeley, A.D.: Human memory: theory and practise. Allyn & Bacon, Boston (1990)

[Baars, 1983] Baars, B.J.: Conscious contents provide the nervous system with coherent, global information. In: Davidson, R.J., Schwartz, G.E., Shapiro, D. (eds.) Consciousness and self-regulation. Plenum Press, New York (1983)

[Baars, 1993] Baars, B.J.: How does aserial, integrated and very limited stream of consciousness emerge from a nervous system that is mostly unconscious, distributed, parallel, and of enormous capacity. In: Bock, G.R., Marsh, J. (eds.) Experimental and theoretical studies of consciousness, Ciba Foundation Symposium, vol. 174. John Wiley & Sons, Chichester (1993)

[Baars, 1997] Baars, B.J.: In the theater of consciousness. Oxford University Press, Oxford (1997)

[Baberg, 2000] Baberg, T.W.: Man-machine interface in modern transprt systems from an aviation safety perspective. In: 4th IEE/IFP International Conference BASYS 2000, Berlin (2000)

[Bainbridge, 1983] Bainbridge, L.: Ironies of automation. Automatica 19(8) (1993)

[Banbury et al., 2008] Banbury, S., Gauthier, M., Scipione, A., Hou, M.: Intelligent adaptive systems. Defence R & D Canada (DRDC) Toronto, CR 2007-075 (2008)

[Banks & Lizza, 1991] Banks, S.B., Lizza, C.S.: Pilot's Associate: a cooperative, knowledge-based system application. IEEE Expert 6(3), 18–29 (1991)

[Bickhard & Terveen, 1995] Bickhard, M.H., Terveen, L.: Foundational issues in AI and cognitive science. In: Advances in Psychology, vol. 109. North Holland, Amsterdam (1995)

[Biggers & Ioerger, 2001] Biggers, K.E., Ioerger, T.R.: Atomatic generation of communication and teamwork within multi-agent teams. In: Applied Artificial Intelligence, vol. 15, pp. 875–916 (2001)

[Billings, 1997] Billings, C.E.: Aviation automation: the search for a human-centered approach. Lawrence Erlbaum Associates Inc., Mahwah (1997)

[Birbaumer & Schmidt, 2003] Birbaumer, N., Schmidt, R.F.: Biologische Psychologie. Springer, Heidelberg (2003)

[Bishop, 1995] Bishop, C.M.: Neural networks for pattern recognition. Oxford University Press, Birmingham (1995)

[Blum et al., 1989] Blum, E.J., Haller, R., Nirschl, G.: Driver-copilot interaction: modeling aspects and techniques. In: 2nd workshop of EUREKA project PROMETHEUS, Stockholm (1989)

[Bonner et al., 2000] Bonner, M., Taylor, R., Fletcher, K., Miller, C.: Adaptive automation and decision aiding in the military fast jet domain. In: Proceedings of the conference on human performance, situation awareness and automation: User centred design for the new millenium, Savannah, GA (October 2000)

[Boff et al., 1986] Boff, K.R., Kaufman, L., Thomas, J.P.: Handbook of perception and human performance, vol. I & II. John Wiley and Sons, Chichester (1986)

[Bousquet et al., 2003] Bousquet, O., van Luxburg, U., Rätsch, G. (eds.): Advanced lectures on machine learning. Springer, Heidelberg (2003)

[Brämer & Schulte, 2002] Brämer, E., Schulte, A.: Künstliche Außensicht für Privatpiloten zur Vermeidung von Geländekollisionen. In: Deutscher Luft- und Raumfahrtkongress 2002, Stuttgart (2002)

[Braitenberg, 1984] Braitenberg, V.: Vehicles. MIT Press, Cambridge (1984)

[Brockhaus, 2001] Brockhaus, R.: Flugregelung, 2nd edn. Springer, Berlin (2001)

[Brodmann, 1909] Brodmann, K.: Vergleichende Lokalisationslehre der Großhirnrinde, Leipzig (1909)

[Brooks, 1989] Brooks, R.: How to build complete creatures rather than isolated cognitive simulators. In: Van Lehn, K. (ed.) Architectures for Intelligence, pp. 225–239. Erlbaum, Hillsdale (1989)

[Bruce et al., 1996] Bruce, V., Green, P.R., Georgeson, M.A.: Visual perception, physiology, psychology, and ecology. Psychology Press, Hove (1996)

[Bubb, 2006] Bubb, H.: A consideration of the nature of work and the consequences for the human-oriented design of production and products. Applied Ergonomics 37(4), 401–407 (2006)

[Burges, 1998] Burges, C.J.C.: A tutorial on support vector machines for pattern recognition. In: Data Mining and Knowledge Discovery, vol. 2, pp. 121–167 (1998)

[Burgess & Lund, 1997] Burgess, C., Lund, K.: Modelling parsing constraints with high-dimensional context space. Language and Cognitive Processes 72(2/3), 177–210 (1997)

[Bush et al., 2000] Bush, G., Luu, P., Posner, M.: Cognitive and emotional inferences in anterior cingulated cortex. Trends in Cognitive Science 4(6), 215–222 (2000)

[Byrne & Anderson, 2001] Byrne, M.D., Anderson, J.R.: Serial modules in parallel: the psychological refractory period and perfect time-sharing. Psychological Review 108, 847–869 (2001)

[Byrne, 2003] Byrne, M.D.: Cognitive architecture. In: Jacko, Sears (eds.) The human-computer interaction handbook. Erlbaum, Mahwah (2003)

[Cacciabue, 2004] Cacciabue, P.C.: Guide to applying human factors methods. In: Human error and accident management in safety critical systems. Springer, London (2004)

[Card et al., 1983] Card, S.K., Moran, T.P., Newell, A.: The psychology of human-computer interaction. Lawrence Erlbaum, Hillsdale (1983)

[Carpenter et al., 1991] Carpenter, G.A., Grossberg, S., Rosen, D.B.: Fuzzy ART: fast stable learning and categorization of analog patterns by an adaptive resonance system. Neural Networks 4, S759–S771 (1991)

[Champigneux & Joubert, 1997] Champigneux, G., Joubert, T.: Copilote electronique project. In: The human-electronic crew: the right stuff? Proceedings on the 4th Joint GAF/RAF/USAF Workshop on Human-Computer Teamwork, Tegernsee, Germany (1997); printed and released by US Air Force Research Laboratory, AFRL-HE-WP-TR-1999-0235 (1999)

[Chen & Billings, 1992] Chen, S., Billings, S.A.: Neural networks for nonlinear dynamic system modelling and identification. Int. Journal of Control 56(2), 319–346 (1992)

[CML, 2001] CommonKADS, Engineering and Managing Knowledge (2001), `http://www.commonkads.usa.nl`

[Cohen & Levesque, 1990] Cohen, P.R., Levesque, H.J.: Intention is choice with commitment. Artificial Intelligence 42, 213–261 (1990)

[Collete & van Linden, 2002] Collete, F., van Linden, M.: Brain imaging of the central executive component of working memory. Neuroscience and Biobehavioural Reviews 24, 105–125 (2002)

[Corbetta et al. 1990] Corbetta, M., Miezin, F.M., Dobmeyer, S., Shulman, G.L., Petersen, S.E.: Attentional modulation of neural processing of shape, color, and velocity in humans. Science 248, 1556–1559 (1990)

[Davis et al., 1993] Davis, R., Shrobe, H., Szolovits, P.: What is a knowledge representation? AI Magazine 14(1), 17–33 (1993)

[Deerwester et al., 1990] Deerwester, S.C., Dumais, S.T., Landauer, T.K., Furnas, G.W., Harshman, R.A.: Indexing by latent semantic analysis. Journal of the American Society of Information Science 41(6), 391–407 (1990)

[Dehaene et al., 1998] Dehaene, S., Kerszberg, M., Changeux, J.-P.: A neuronal model of a global workspace in effortful cognitive tasks. Proceedings of the National Academy of Sciences of the USA (PNAS) 95(24), 14529—14534 (1998)

[Dehaene & Naccache, 2001] Dehaene, S., Naccache, L.: Toward a cognitive neuroscience of consciousness: basic evidence and a workspace framework. Cognition 79, 1–37 (2001)

[Dehaene et al., 2006] Dehaene, S., Changeux, J.-P., Naccache, L., Sackur, J., Sergent, C.: Conscious, preconscious, and subliminal processing: a testable taxonomy. In: Trends in cognitive Sciences, Article in press. Elsevier (2006)

[Desel & Juhas, 2001] Desel, J., Juhás, G.: What is a petri net? In: Ehrig, H., Juhás, G., Padberg, J., Rozenberg, G. (eds.) APN 2001. LNCS, vol. 2128, pp. 1–25. Springer, Heidelberg (2001)

[Dickmanns et al., 1993] Dickmanns, E.D., Behringer, R., Brüdigam, C., Dickmanns, D., Thomanek, F., von Holt, V.: An all-transputer visual autobahn-autopilot/copilot, Universität der Bundeswehr München (1993)

[Dickmanns, 2007] Dickmanns, E.D.: Dynamic vision for perception and control of motion. Springer, Heidelberg (2007)

[Diepold, 1991] Diepold, S.: Fahrsimulatorversuche zur Ermittlung situationsspezifischer Verhaltenparameter. Diplomarbeit Universität der Bundeswehr München (1991)

[Di Nocera et al., 2003] Di Nocera, F., Lorenz, B., Tattersall, A., Parasuraman, R.: New possibilities for adaptive automation and work design. In: Robert, G., Hockey, J., Gaillard, A.W.K., Burov, O. (eds.) Operator functional state. NATO Science Series, vol. 355. IOS Press, Amsterdam (2003)

[Döring, 1983] Döring, B.: Analyse des Arbeitsprozesses beider Fahrzeugführung am Beispel eines Landeanflugs. FGAN Forschungsinstitut für Anthropotechnik, Bericht Nr. 59 (1983)

[Döring, 2007] Döring, B.: Systemanalyse. In: Handbuch der Ergonomie, D – 1.3.2, herausgegeben von Bundesamt für Wehrtechnik und Beschaffung, Koblenz (2006)

[Donath & Schulte, 2006] Donath, D., Schulte, A.: Ein Konzeptansatz für modellbasierte adaptive Automation. 48. Fachausschusssitzung Anthropotechnik T5.4 Cognitive Systems Engineering in der Fahrzeug- und Prozessführung, 24–25 (2006)

[Driver, 1998] Driver, J.: The neuropsychology of spatial attention. In: Pashler, H. (ed.) Attention. Psychology Press, Francis & Taylor (1998)

[Dudek, 1990] Dudek, H.-L.: Wissensbasierte Pilotenunterstützung im Ein-Mann-Cockpit bei Instrumentenflug, Dissertation. University of German Armed Forces Munich (1990)

[Dumais, 1990] Dumais, S.: Enhancing performance in latent semantic indexing (1990), `http://citeseer.ist.psu.edu/dumais92enhancing.html`

[Dyck et al., 1993] Dyck, J.L., Abbot, D.W., Wise, J.A.: HCI in multi-crew aircraft. In: Salvendy, G., Smith, M.J. (eds.) Proceedings of the 5th International Conference on Human-Computer Interaction, Orlando (1993)

[Ebbinghaus, 1885] Ebbinghaus, H.: Über das Gedächtnis: Untersuchungen zur experimentellen Psychologie. Wissenschaftliche Buchgesellschaft Darmstadt (1992) (first published in 1885)

[Endsley, 1997] Endsley, M.R.: The role of situation awareness in naturalistic decision making. In: Zsambok, C.E., Klein, G. (eds.) Naturalistic decision making, pp. 269–283. Lawrence Erlbaum, Mahwah (1997)

[Engel, 1996] Engel, A.K.: Prinzipien der Wahrnehmung: Das visuelle System. In: Roth, G., Prinz, W. (eds.) Kopf-Arbeit. Spektrum Akademischer Verlag, Heidelberg (1996)

[Eriksen & Yeh, 1985] Eriksen, C.W., Yeh, Y.Y.: Allocation of attention in the visual field. Journal of Experimental Psychology, Perception and Performances 11, 583–597 (1985)

[Ernsting et al., 1999] Ernsting, J., Nicholson, A.N., Rainford, D.J.: Aviation medicine. Butterworth-Heinemann, Butterworths (1999)

[Erzberger, 1995] Erzberger, H.: Design principles and algorithms for automated air traffic management. In: Winter, H. (ed.) Knowledge-based functions n aerospace systems, AGARD LS 200, 7.1– 7.31 (1995)

[Fastenmeier, 1995] Fastenmeier, W.: Autofahrer und Verkehrssituation. Verlag TÜV Rheinland. Deutscher Psychologen-Verlag (1995)

[Feraric et al., 1992] Feraric, J.P., Kopf, M., Onken, R.: Statistical versus neural net approach for driver behaviour description and adaptive warning. In: 11th European Annual Manual, vol. F-9, pp. 429–436 (1992)

[Feraric & Onken, 1995] Feraric, J.P., Onken, R.: DAISY – A driver assisting system which adapts to the driver. In: Man-Machine Systems (MMS 1995), Cambridge (1995)

[Feraric, 1996] Feraric, J.P.: Echtzeitfähige Modellierung des individuellen Fahrerverhaltens zur Realisierung adaptiver Unterstützungsfunktionen in einem Monitor- und Warnsystem. Dissertation, University of German Armed Forces Munich (1996)

[Ferber, 1999] Ferber, J.: Multi-agent systems. Addison-Wesley, Reading (1999)

[Finch & Chater, 1992] Finch, S., Chater, N.: Bootstraping syntactic categories by unsupervised learning. In: Proceedings of the 14th annual meeting of the Cognitive Science Society, pp. 820–825. Lawrence Erlbaum Associates, Mahwah (1992)

[Findlay et al., 1995] Findlay, J.M., Kentridge, R.W., Walker, R. (eds.): Eye movement research: Mechanisms, processes, and applications. Elsevier, North Holland (1995)

[FIPA, 2002a] FIPA ACL Message Structure Specification. Document Number SC00061G, http://www.fipa.org/specs/fipa00061/SC00061G.pdf (retrieved December 7, 2005)

[FIPA, 2002b] FIPA ACL Communicative Act Library Specification. Document Number SC00037J, http://www.fipa.org/specs/fipa00037/SC00037J.pdf (retrieved December 7, 2005)

[FIPA, 2002c] FIPA Request Interaction Protocol Specification. Document Number SC00026H, http://www.fipa.org/specs/fipa00026/SC00026H.pdf (retrieved Decfember 7, 2005)

[Fitts, 1954] Fitts, P.M.: The information capacity of the human motor system in controlling the amplitude of movement. Journal of Experimental Psychology 47(6), 381–391 (1954)

[Flemisch, 2001] Flemisch, F.O.: Pointillistische Analyse der visuellen und nicht-visuellen Interaktionsressourcen am Beispiel Pilot-Assistenzsystem, Dissertation, University of German Armed Forces Munich (2001)

[Flemisch et al., 2003] Flemisch, F.O., Adams, C.A., Convay, S.R., Goodrich, K.H., Palmer, M.T., Schutte, P.C.: The H-metaphor as a guideline for vehicle automation and interaction. NASA/TM-2003-212672 (2003)

[Franklin & Graesser, 2001] Franklin, S., Graesser, A.: Modeling cognition with software agents. In: Moore, J.D., Stening, K. (eds.) Proceedings of the 23rd Annual Conference of the Cognitive Science Society (2001)

[Freed, 2000] Freed, M.: Simulating human agents. In: AAAI Fall Symposium, Menlo Park (2000)

[Frey, 2005] Frey, A.: Überwachung und Kontrolle in einem künstlichen kognitiven System zur autonomen Fahrzeugführung, Dissertation, University of German Armed Forces Munich (2005)

[Frey et al, 2001] Frey, A., Lenz, A., Putzer, H., Walsdorf, A., Onken, R.: In-flight evaluation of CAMA – the Crew Assistant Military Aircraft, DGLR-Jahreskongress (2001)

[Fritsche, 1994] Fritsche, H.-T.: A model for traffic simulation, Traffic Engineering + Control (1994)

[Fritz & Franke, 1993] Fritz, H., Franke, U.: Neuronale Netze in der autonomen Fahrzeugführung. Nachdruck, VDE Jubiläumskongress 93, Berlin, p. 8 (1993)

[Fujioka & Takabo, 1991] Fujioka, T., Takubo, N.: Driver model obtained by neural network system. JSAE Review 12(2), 82–85 (1991)

[Funk et al., 1996] Funk, K., Lyall, B., Riley, V.: Perceived human factors problems of flightdeck automation. FAA Grant 93-G-039 (1996)

[von Garrel, 2003] von Garrel, U.: Adaptive Modellierung des fertigbasierten Fahrerverhaltens durch konstruktives und evaluierendes Lernen, Dissertation, University of German Armed Forces Munich (2003)

[von Garrel et al., 2000] von Garrel, U., Otto, H.-J., Onken, R.: Adaptive modeling of driver behaviour in town environment. In: 4th IFIP/IEEE International conference on information technology for balanced automation systems in manufacture and transportation BASYS 2000, Berlin (2000)

[von Garrel & Onken, in press] von Garrel, U., Onken, R.: Modeling individual driver behaviour by a constructive and evaluating learning process. In: Jürgensen (ed.) (in press)

[Gazis et al., 1963] Gazis, D.C., Herman, R., Rothery, R.W.: Analytical methods in transportation: mathematical car-following theory of traffic flow. Journal of Engineering Mechanics Division, ASCE Proceedings Paper 3724 (Paper 372), 29–46 (1963)

[Geddes, 1992] Geddes, N.D., Hosmer, D.M.: Knowledge integration in the Pilot's Associate. In: 12th IFAC Symposium on Automatic Control in Aerospace (1992)

[Gerlach, 1996] Gerlach, M.: Schnittstellengestaltung für ein Cockpitassistenzsystem unter besonderer Berücksichtigung von Spracheingabe, Dissertation, University of German Armed Forces Munich (1996)

[Godthelp & Konings, 1991] Godthelp, H., Konings, H.: Levels of steering control; some notes on the time-to-line-crossing concept as related to driving strategy. In: 1st European annual Conference on Human Decision –Making and Manual Control, Delft (1981)

[Gorrell, 2007] Gorrell, G.: Latent semantic Analysis: How does it work, and what is t good for? (2007), www.dcs.shef.ac.uk/~genevieve/lsa_tutorial.html

[Goschke, 2003] Goschke, T.: Voluntary action and cognitive control from a cognitive neuroscience perspective. In: Maasen, S., Prinz, W., Roth, G. (eds.) Voluntary action-brains, minds, and society. Oxford University Press, Oxford (2003)

[Grafton et al., 1992] Grafton, S.T., Mazziotta, J.C., Presty, S., Friston, K.J., Frackowiak, R.S., Phelps, M.E.: Functional anatomy of human procedural learning determined with regional cerebral blood and PET. Journal of Neuroscience 12, 2542–2548 (1992)

[Grashey & Onken, 1998] Grashey, S., Onken, R.: Adaptive modeling of the individual driving behaviour of car drivers based on statistical classifiers. In: Global Ergonomics Conference, Cape Town (1998)

[Grashey, 1998] Grashey, S.: Ein Klassifikationsansatz zur fertigkeitsbasierten Verhaltensmodellierung beim Autofahren, Dissertation, University of German Armed Forces Munich (1999)

[Grau & Menu, 1994] Grau, J.Y., Menu, J.P.: Ergonomical implications of supermaneuverability in future combat aircraft. In: Workshop on human factors/future combat aircraft, Ottobrunn (1994)

[Guyton, 1987] Guyton, A.C.: Human physiology and mechanisms of disease. W.B. Saunders, Philadelphia (1987)

[Harel, 1987] Harel, D.: Statecharts: A visual formalism for complex systems. Science of Computer Programming 8, 231–274 (1987)

[Harel et al., 1990] Harel, D., Lachover, H., Naamad, A., Pnueli, A., Politi, M., Sherman, R., Shtull-Trauring, A., Trakhtenbrot, M.: Statemate: A working environment for the development of complex reactive systems. IEEE Transactions on Software Engineering 16, 403–414 (1990)

[Harris, 2002] Harris, D.: Engineering psychology and cognitive ergonomics, vol. 1 and 2. Ashgate Publishing Ltd. (2002)

[Hart & Staveland, 1988] Hart, S.G., Staveland, L.E.: Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research. In: Hancock, P.A., Meshkati, N. (eds.) Human Mental Workload, pp. 139–184. North Holland Press, Amsterdam (1988)

[Hebb, 1949] Hebb, D.O.: The organisation of behaviour. Wiley, Chichester (1949)
[Hecker et al., 1999] Hecker, P., Doehler, H.-U., Suikat, R.: Enhanced vision meets pilot assistance. In: SPIE International Symposioum on Aerospace/Defense Sensing, Simulation, and Controls, Orlando (1999)
[van Heesch, 2001] van Heesch, D.: DOXYGEN homepage (2001), http://www.stack.nl/~dimitri/doxygen
[Helander et al., 1997] Helander, M.G., Landauer, T.K., Prabhu, P.V. (eds.): Handbook of human-computer interaction. Elsevier, Amsterdam (1997)
[Henn, 1995] Henn, V.: Utilisation de la logique floue pour la modelisation microscopique du trafic routier. L'revue du Logique Floue (1995)
[Hinton et al, 1993] Hinton, G.E., Plaut, D.C., Shallice, T.: Simulating brain damage. Scientific American 269, 76–92 (1993)
[Hoc et al., 1995] Hoc, J.-M., Cacciabue, P.C., Hollnagel, E. (eds.): Expertise and technology: cognition and human-computer cooperation. Lawrence Erlbaum Associates, Hillsdale (1995)
[Hoc, 2000] Hoc, J.-M.: From human-machine interactions to human-machine cooperation. Ergonomics 43(7), 833—843 (2000)
[Hofmann, 1999] Hofmann, T.: Probabilistic latent semantic analysis. In: Uncertainty in artificial intelligence. In: UAI 1999, Stockholm (1999)
[Hollan et al., 1995] Hollan, J.D., Hutchins, E., Kirsh, D.: A Distributed Cognition Approach to Designing Digital Work Materials for Collaborative work places (1995), http://www.nsf.gov/cgi-bin/showaward?award=9873156
[Hollnagel et al., 1988] Hollnagel, E., Mancini, C., Woods, D.D.: Cognitive engineering in complex dynamic worlds. Academic Press, Harcourt Brace Jovanovich Publishers (1988)
[Hollnagel, 1995] Hollnagel, E.: Automation, coping, and control. In: Post HCI 1995 Conference Seminar on Human-Machine Interface in Process Control, Hieizan, Japan, July 17-19 (1995)
[Hollnagel, 2004] Hollnagel, E.: Barriers and accident prevention. Ashgate (2004), ISBN 0-7546-4301-8
[Hollnagel & Woods, 2005] Hollnagel, E., Woods, D.D.: Joint cognitive systems. Taylor & Francis, Abington (2005)
[Huber et al., 1990] Huber, P., Jensen, K., Shapiro, R.M.: Hierarchies in coloured Petri nets. In: Rozenberg, G. (ed.) APN 1990. LNCS, vol. 483, pp. 313–341. Springer, Heidelberg (1991)
[Hutchins, 1995] Hutchins, E.: Cognition in the wild. MIT Press, Cambridge (1995)
[IEA, 1964] International Ergonomics Association: Ergonomics system analysis checklist, 2nd International Congress in Ergonomics, Dortmund (1964)
[INCOSE, 2007] International Council on Systems Engineering: Systems Engineering Handbook (2007)
[Ishida et al., 1990] Ishida, T., Yokoo, M., Gasser, L.: An organisational approach to adaptive production systems. In: AAAI 1990, pp. 52–58 (1990)
[Jacob, 1993] Jacob, R.J.K.: Interaction styles and input/output devices. Behaviour and Information Technology 12(2), 69–79 (1993)
[James, 1890] James, W.: The principles of psychology. Havard University Press (1981) (first published in 1890)
[Jarasch & Schulte, 2008] Jarasch, G., Schulte, A.: Satisfying integrity requirements for highly automated UAV systems by a systems engineering approach to cognitive automation. In: 27th Digital Avionics Systems Conference (2008)

[Jennings, 1996] Jennings, N.R.: Co-ordination techniques for distributed artificial intelligence. In: O'Hare, G.M.P., Jennings, N.R. (eds.) Foundations of Distributed Artificial Intelligence, pp. 187–210. Wiley, Chichester (1996)

[Johannsen, 1993] Johannsen, G.: Mensch-Maschine Systeme. Springer, Heidelberg (1993)

[Jonas, 1989] Jonas, H.: Das Prinzip Verantwortung: Versuch einer Ethik für die technologische Zivilisation. VerlagInsel-Verlag (1989)

[Jürgensohn, 1997] Jürgensohn, T.: Hybride Fahrermodelle, Dissertation, ZMMS Spektrum Band 4 (1997)

[Kaiser et al., 2001] Kaiser, J., Mayer, U., Helmetag, A.: Stereoscopic head-up display for aviation. In: SPIE proceedings, vol. 4297, pp. 117–126 (2001)

[Karg & Staehle, 1982] Karg, P.W., Staehle, W.H.: Analyse der Arbeitssituation. Haufe Freiburg (1982)

[Karwoski, 2001] Karwoski, W. (ed.): International encyclopedia of ergonomics and human factors. Taylor and Francis, London (2001)

[Kecman, 2001] Kecman, V.: Learning and soft computing. MIT Press, Cambridge (2001)

[Kersandt & Gauss, 2006] Kersandt, D., Gauss, B.: NARIDAS – Ergebnisse einer zweiten Expertenbefragung. HANSA International Maritime Journal 143(10), 51–56 (2006)

[Kehtarnavaz & Sohn, 1991] Kehtarnavaz, N., Sohn, W.: Steering control of autonomous vehicles by neural networks. In: Proceedings of the American Control Conference (1991)

[Kim & Shadlen, 1999] Kim, J.-N., Shadlen, M.N.: Neuronal correlates of a decision in the dorso-lateral pre-frontal cortex of the macaque. Nature Neuroscience 2, 176–185 (1999)

[Kimberg et al., 1998] Kimberg, D.Y., D'Esposito, M., Farah, M.J.: Cognitive functions in the pre-frontal cortex – working memory and executive control. Current Directions in Psychological Science 6, 185–192 (1998)

[Kimble, 1988] Kimble, D.P.: Biological psychology. Rinehart and Winston, Inc., Holt (1988)

[Kirchner & Rohmert, 1974] Kirchner, J.H., Rohmert, W.: Ergonomische Leitregeln zur menschengerechten Arbeitsgestaltung, Katalog arbeitswissenschaftlicher Richtlinien über die menschengerechte Gestaltung der Arbeit (BVG §§ 90, 91). Carl Hansen Verlag (1974)

[Kohonen, 1982] Kohonen, T.: Self-organizing formation of topologically correct feature maps. Biological Cybernetics 43, 39 (1982)

[Kolodner, 1993] Kolodner, J.L.: Case-based reasoning. Morgan-Kaufmann Publications, San Francisco (1993)

[Kopf & Onken, 1992] Kopf, M., Onken, R.: DAISY, a knowledgeable monitoring and warning aid for drivers on German motorways. In: 5th IFAC Man-machine Symposium, The Hague (1992)

[Kopf, 1994] Kopf, M.: Ein Beitrag zur modellbasierten, adaptiven Fahrerunterstützung für das Fahren auf deutsche Autobahnen, Dissertation, University of German Armed Forces Munich (1994)

[Kopf, 1997] Kopf, M.: Vorschlag eines Entwurfs- und Bewertungsschemas für aus Nutzersicht konsistente Assistenzsysteme, 2. Berliner Werkstatt Mensch-Maschine-Systeme, Berlin (1997)

[Kornhauser, 1991] Kornhauser, A.L.: Neural network approaches for lateral control of autonomous highway vehicles. In: Proceedings of Vehicle Navigation & Information Systems, vol. 91, pp. S.1143–S.1151 (1991)

[Kosko, 1992] Kosko, B.: Neural networks and fuzzy systems. Prentice Hall, Englewood Cliffs (1994)

[Kraiss & Moraal, 1976] Kraiss, K.-F., Moraal, J.: Introduction to human engineering. TÜV Rheinland, Köln (1976)

[Kraiss & Küttelwesch, 1992] Kraiss, K.F., Küttelwesch, H.: Identification and application of neural operator models in a car driving situation. In: 5th IFAC Symposium on Analysis, Design and Evaluation of Man-Machine Systems, The Hague, The Netherlands (1992)

[Kraiss, 1995] Kraiss, K.F.: Implementation of user-adaptive assistants with neural operator model. Control Engineering Practise 3(2), 249–256 (1995)

[Krüger et al, 2001] Krüger, H.-P., Neukum, A., Hargutt, V.: Effort management: a necessary concept to understand driving. In: 8th Confrence on Cognitive science Approaches to Process Control, CSAPC 2001, pp. 245–254 (2001)

[Kubbat et al., 1998] Kubbat, W., Lenhart, P.M., von Viebahn, H.: 4D flight guidance displays: a gate to gate solution. In: Digital Systems Conference 17, Bellevue, Washington (1998)

[Laird et al., 1987] Laird, J.E., Newell, A., Rosenbloom, P.S.: Soar: An architecture for general intelligence. Artificial Intelligence 33(1), 1–64 (1987)

[Landauer & Dumais, 1997] Landauer, T.K., Dumais, S.T.: A solution to Plato's problem: The latent semantic analysis theory of aquisition, induction, and representation of knowledge. Psychological Review 104(2), 211–240 (1997)

[Landauer, 2002] Landauer, T.K.: On the computational basis of learning and cognition: arguments from LSA. In: Ross, N. (ed.) The psychology of learning and motivation, vol. 41, pp. 43–84 (2002)

[Lenz & Onken, 2000] Lenz, A., Onken, R.: Pilot's assistant in tactical transport missions – Crew Assistant Military Aircraft CAMA. In: NATO RTO/HFM Symposium, Oslo (2000)

[Libet et al., 1983]Libet, B., Gleason, C.A., Wright, E.W., Pearl, D.K.: Time of conscious intention to act in relation to onset of cerebral activity (readiness-potential). The Behavioural and Brain Sciences 106, 623–642 (1983)

[Lizza et al., 1990] Lizza, C.S., Rouse, D.M., Small, R.L., Zenyuh, J.P.: Pilot's Associate: an evolving philosophy. In: Proc. of the 2nd Joint GAF/RAF/USAF Workshop on Human-Electronic Crew Teamwork, Ingolstadt (1990)

[Lloret et al., 1992] Lloret, J.C., Roux, J.L., Algayres, B., Chamontin, M.: Modelling and evaluation of a satellite system using EVAL, a Petri net based industrial tool. In: Jensen, K. (ed.) ICATPN 1992. LNCS, vol. 616, pp. 379–383. Springer, Heidelberg (1992)

[Maasen et al., 2003] Maasen, S., Prinz, W., Roth, G. (eds.): Voluntary action-brains, minds, and society. Oxford University Press, Oxford (2003)

[Malone & Crowston, 1994] Malone, T.W., Crowston, K.: The Interdisciplinary study of co-ordination. ACM Computing Surveys 26(1), 87–119 (1994)

[von Martial, 1992] von Martial, F.: Coordinating Plans of Autonomous Agents. LNCS (LNAI), vol. 610. Springer, Heidelberg (1992)

[MATLAB, 1995] MATLAB version 4: The student edition of MATLAB. Prentice Hall, Englewood Cliffs (1995)

[Mc Clelland, 1981] Mc Clelland, J.L.: Retrieving general and specific knowledge from stored knowledge of specifics. In: Proc. of the 3rd Annual Conference of the Cognitive Science Society, Berkeley (1981)

[Mc Clelland & Rumelhart, 1981] Mc Clelland, J.L., Rumelhart, D.E.: An Interactive model of context effects in letter perception: Pt. I, an Account of Basic Findings. Psychological Review 88, 375–407 (1981)

[Mc Clelland & Rumelhart, 1986] Mc Clelland, J.L., Rumelhart, D.E. (eds.): Parallel distributed processing, vol. 2. The MIT Press, Cambridge (1986)

[Mc Ruer & Weir, 1969] Mc Ruer, D.T., Weir, D.H.: Theory of manual vehicular control. Ergonomics 12, 599–633 (1969)

[Mecklenburg, 1992] Mecklenburg, K.: Neural control of autonomous vehicles. In: IEEE Vehicular Technology Conference, pp. S.205–S.212 (1992)

[Meitinger & Schulte, 2006a] Meitinger, C., Schulte, A.: Human-centred automation for UAV guidance: Oxymoron or tautology? – the potential of cognitive and co-operative systems. In: Proceedings of the 1st Moving Autonomy Forward Conference, Lincoln, UK (2006)

[Meitinger & Schulte, 2006b] Meitinger, C., Schulte, A.: Cognitive machine co-operation as basis for guidance of multiple UAVs. In: Proceedings of NATO RTO meeting on Human Factors of Uninhabited Military Vehicles as Force Multipliers, HFM-135, Biarritz, France (2006)

[Meitinger, 2008] Meitinger, C.: Kognitive Automation ur kooperativen UAV-Flugführung, Dissertation, University of German Armed Forces Munich (2008)

[Menzel & Roth, 1996] Menzel, R., Roth, G.: Verhaltensbiologische und neuronale Grundlagen des Lernens und des Gedächtnisses. In: Roth, G., Prinz, W. (eds.) Kopf-Arbeit. Spektrum Akademischer Verlag, Heidelberg (1996)

[Metcalfe & Shimamura, 1994] Metcalfe, J., Shimamura, A.P. (eds.): Metacognition – knowing about knowing. MIT Press, Cambridge (1994)

[Meyer & Kieras, 1997] Meyer, D.E., Kieras, D.E.: A computational theory of executive cognitive processes and multiple task performance, Pt. 1, basic mechanisms. Psychological Review 104, 2–65 (1997)

[Michon, 1993] Michon, J.A. (ed.): Generic intelligent driver support. Taylor &Francis, London (1993)

[Miller & Hannen, 1999] Miller, C.A., Hannen, M.D.: The Rotorcraft Pilot's Associate: design and evaluation of an intelligent user interface for cockpit information management. Knowledge-based Systems 12(8), 443–456 (1999)

[Miller & Funk, 2001] Miller, C.A., Funk, H.B.: Associates with etiquette: meta-communication to make human-automation interaction more natural, productive, and polite. In: 8th Conference on Cognitive Science Approaches to Process Control (CSAPC 2001), Germany (2001)

[Miller & Dorneich, 2006] Miller, C.A., Dorneich, M.C.: From associate systems to augmented cognition: 25 years of user adaptation in high criticality systems (2006)

[Millot, 1988] Millot, P.: Automated systems control and ergonomics, Hermes, Paris (1988)

[Minsky, 1975] Minsky, M.L.: A framework for representing knowledge. In: Winston (ed.) The psychology of computer vision. Mc Graw Hill, New York (1975)

[Moody & Darken, 1988] Moody, J., Darken, C.: Learning with localized receptive fields. In: Proceeding of the 1988 Connectionist Models Summer School, pp. S.133–S.143. Morgan Kaufmann, San Mateo (1988)

[Moschytz, 2000] Moschytz, G.: Adaptive Filter. Springer, Zürich (2000)

[Motiv ACC consortium, 1999] Motiv ACC consortium: ACC im Ballungsraum, Abschlussbericht (1999)

[Narendra & Parthasarathy, 90] Narendra, K.S., Parthasarathy, K.: Identification and control of dynamical systems using neural networks. IEEE Trans. Neural Networks 1(1), 4–27 (1990)

[Nauk, 1994] Nauk, D., Klawonn, F., Kruse, R.: Neuronale Netze und Fuzzy systeme. Vieweg, Wiesbaden (1994)

[Neusser et al., 1991] Neusser, S., Hoefflinger, B., Nijhuls, J.: A case study in car control by neural networks. In: Proc. ISATA International Symposium, Florence, Italy (1991)

[Newell & Simon, 1972] Newell, A., Simon, H.: Human problem solving. Prentice Hall, Englewood Cliffs (1972)

[Newell & Simon, 1976] Newell, A., Simon, H.: Computer science as empirical inquiry: symbols and search. Communications of the ACM 19, 113–126 (1976)

[Newell, 1990] Newell, A.: Unified theories of cognition. Havard University Press, Cambridge (1990)

[Nieuwenhuys et al., 2001] Nieuwenhuys, R., Ridderinkhof, K.R., Blom, J., Band, G.P., Kok, A.: Error-related brain potentials are differentially related to awareness of response errors. Evidence from an antisaccade task. Psycophysiology 38(5), 752–760 (2001)

[Nieuwenhuys, 2001] Nieuwenhuys, R.: Neocortical macrocircuits. In: Roth, G., Wullimann, M.F. (eds.) Brain, Evolution and Cognition. Wiley & Sons and Spectrum Publication (2001)

[Ohno et al., 2002] Ohno, T., Mukawa, N., Yoshikawa, A.: Freegaze: A gaze tracking system for everyday gaze interaction. In: Proceedings of Eye Tracking Research & Application (ETRA 2002), pp. 125–132 (2002)

[OMG, 2001] Object Management Group: The OMG's CORBA Website (2001), `http://www.corba.org`

[Onken, 1994] Onken, R.: Funktionsverteilung Pilot-Maschine: Umsetzung von Grundforderungen im Cockpitassistenzsystem CASSY, DGLR-Bericht 94-01 (1994)

[Onken & Prevot, 1994] Onken, R., Prevot, T.: CASSY – cockpit assistant system for IFR operation. In: 19th ICAS Congress, Anaheim (1994)

[Onken, 94] Onken, R.: DAISY, an adaptive, knowledge-based driver monitoring and warning system. In: Vehicle Navigation & Information Systems Conference (VNIS 1994), Yokohama (1994)

[Onken, 1995] Onken, R.: Functional development and field test of CASSY – a knowledge-based cockpit assistant, AGARD LS 200 (1995)

[Onken & Feraric, 1996] Onken, R., Feraric, J.P.: Adaptation to the driver as part of a driver monitoring and warning system. In: 2nd International conference on fatigue and transportation: Engineering, enforcement, and education solutions, Fremantle (1996)

[Onken & Walsdorf, 2001] Onken, R., Walsdorf, A.: Assistant systems for aircraft guidance: cognitive man-machine co-operation. Aerospace Science Technology 5, 511–520 (2001)

[Onken, 2002]Onken, R.: Cognitive Cooperation for the Sake of the Human-Machine Team Effectiveness. In: RTO-HFM Symposium on The Role of Humans in Intelligent and Automated Systems, Warsaw, Poland (2002)

[Opperman, 1994] Opperman, R.: Adaptive user support. Erlbaum, Hillsdale (1994)

[Otto, 2006] Otto, H.-J.: Ein Konzeptbeitrag zur Realisierung eines kognitiven Tutorsystems für die simulatorgestützte Fahrausbildung im Stadtverkehr, Dissertation, University of German Armed Forces Munich (2006)

[Parasuraman, 1986] Boff, K., Kaufman, L., Thomas, J.P.: Handbook of perception and human performance. In: Parasuraman, R., Davies (eds.) Vigilance, monitoring, an search. John Wiley and Sons, Chichester (1986)

[Parasuraman, 1998] Parasuranam, R. (ed.): The attentive brain. MIT Press, Cambridge (1998)

[Parasuraman, 2003] Parasuraman, R.: Adaptive automation matched to human mental workload. In: Robert, G., Hockey, J., Gaillard, A.W.K., Burov, O. (eds.) Operator functional state. NATO Science Series, vol. 355. IOS Press, Amsterdam (2003)

[Paris, 2002] Paris, S.G.: When is metacognition helpful, debilitating, or benign. In: Chambres, P., Izaute, M., Marescaux, P.-J. (eds.) Metacognition – process, function and use, pp. 106–120. Kluwer Academic Publishers, Boston (2002)

[Pashler, 1998] Pashler, H. (ed.): Attention. Psychology Press, Francis & Taylor (1998)

[Passingham et al., 2000] Passingham, R.E., Toni, I., Rushworth, M.F.S.: Specialization within the pre-frontal cortex: the ventral pre-frontal cortex and learning. Experimental Brain Research 133, 103–113 (2000)

[Pedricz & Card, 1992] Pedricz, W., Card, H.C.: Linguistic interpretation of self-organizing maps. In: Proc. IEEE International Conference on Fuzzy Systems, San Diego, pp. 371–378 (1992)

[Peterson, 1981] Peterson, J.L.: Petri net theory and the modelling of systems. Prentice Hall Inc., Englewood Cliffs (1981)

[Petri, 1962] Petri, C.A.: Kommunikation mit Automaten, Ph.D. Dissertation, University of Bonn, English translation in Technical Report RADC-TR-65-377, Griffiss AFB (1966)

[Petrides, 2000] Petrides, M.: The Role of the mid-dorso-lateral pre-frontal cortex in working memory. Experimental Brain Research 133, 44–54 (2000)

[Pinel, 2007] Pinel, J.P.J.: Biopsychology. Pearson Education Inc., London (2007)

[Platts et al., 2007] Platts, J., Ögren, P., Fabiani, P., di Vito, V., Schmidt, R.: Final Report of GARTEUR FM AG14; GARTEUR/TP-157 (2007)

[Pommerleau, 1995] Pommerleau, D.: Ralph: Rapidly adapting lateral position handler. In: Proceedings of the Intelligent Vehicles 1995 Symposium, Detroit, pp. 506–511 (1995)

[Posner, 1989] Posner, M.I.: Foundations of cognitive science. MIT Press, Cambridge (1989)

[Posner & DiGirolamo, 1998] Posner, M.I., DiGirolamo, G.J.: Executive attention: conflict, target detection, and cognitive control. In: Parasuranam, R. (ed.) The attentive brain. MIT Press, Cambridge (1998)

[Post, 1943] Post, E.L.: Formal reductions of the general combinatorial decision problem. American Journal of Mathematics 65, 197–268 (1943)

[Prevot & Onken, 1993] Prevot, T., Onken, R.: On-board interactive flight planning and decision making with the cockpit assistant system CASSY. In: 4th HMI-AI-AS Conference, Toulouse (1993)

[Prevot et al., 1995] Prevot, T., Gerlach, M., Ruckdeschel, W., Wittig, T., Onken, R.: Evaluation of intelligent on-board pilot assistance in in-flight trials. IFAC, Man-Machine systems, Cambridge (1995)

[Prevot, 1996] Prevot, T.: Maschineller Flugplaner für Verkehrsflugzeuge als autonomes und cooperatives System, Dissertation, University of German Armed Forces Munich (1996)

[Protzel et al., 1993] Protzel, P., Holve, R., Bernasch, J., Naab, K.: Abstandsregelung von Fahrzeugen mit Fuzzy Control. In: Reusch, B. (ed.) Proceedings of 3. Dortmunder Fuzzy Tage, Reihe Informatik Aktuell. Springer, Heidelberg (1993)

[Putzer & Onken, 2003] Putzer, H.J., Onken, R.: COSA – a generic cognitive system architecture based on a cognitive model of human behaviour. In: Cognition, Technology & Work, pp. 140–151. Springer, London (2003)

[Putzer, 2004] Putzer, H.J.: COSA – ein uniformer Architekturansatz für kognitive Systeme und seine Umsetzung in einen operativen Framework, Dissertation, University of German Armed Forces Munich (2004)

[Rämä et al., 2001] Rämä, P., Sala, J.B., Gillen, J.S., Pekar, J.J., Courtney, S.M.: Dissociation of the neural systems for working memory maintainance of verbal and nonspatial visual information. Cognitive, Affective, & Behavioral Neuroscience 1, 161–171 (2001)

[Rao & Georgeff, 1995] Rao, A.S., Georgeff, M.P.: BDI agents: theory and practise. In: Proc. of 1st International Conference on Multi-Agent Systems, ICMAS (1995)

[Rasmussen, 1983] Rasmussen, J.: Skills, rules and knowledge, signals, signs and symbols, and other distinctions in human performance models. IEEE Transactions on Systems, Man, and Cybernetics SMC-13, 257–266 (1983)

[Rasmussen et al., 1988] Rasmussen, J., Duncan, K., Leplat, J. (eds.): New technology and human errors. J. Wiley & Sons, Chichester (1988)

[Rasmussen & Vicente, 1989] Rasmussen, J., Vicente, K.J.: Coping with human errors through system design: Implications for the ecological interface design. International Journal of Man-Machine Studies (1989)

[Rational, 2001] Rational Software GmbH: Rational Rose (2001),
http://www.rational.com/products/rose/index.jsp

[Rauschert et al., 2008] Rauschert, A., Meitinger, C., Schulte, A.: Experimentally discovered operator assistance needs in the guidance of cognitive and cooperative UAVs. In: First International Conference on Humans Operating Unmanned Systems. HUMOUS 2008, Brest, France (2008)

[Reason, 1988] Reason, J.: Generic error-modelling system (GEMS): A cognitive framework for locating common human error forms. In: Rasmussen, J., Duncan, K., Leplat, J. (eds.) New technology and human errors. J. Wiley & Sons, Chichester (1988)

[Reisig, 1991] Reisig, W.: Petri nets and algebraic specifications. Theoretical Computer Science 80, 1–34 (1991)

[Reisig, 1992] Reisig, W.: A Primer in Petri net design. Springer, Heidelberg (1992)

[Rekersbrink, 1995] Rekersbrink, A.: Mikroskopische Verkehrssimulation mit Hilfe der Fuzzy Logik. Straßenverkehrstechnik 2(95), S.68–S.74 (1995)

[Robert et al., 2003] Robert, G., Hockey, J., Gaillard, A.W.K., Burov, O. (eds.): Operator functional state. NATO Science Series, vol. 355. IOS Press, Amsterdam (2003) [Römer et al., 2001] Römer, K., Puder, A., Pilhofer, F.: MiCO – Mico is COrba (2001), http://www.mico.org

[Roth, 1996] Roth, G.: Das Gehirn des Menschen. In: Roth, G., Prinz, W. (eds.) Kopf-Arbeit, Spektrum Akademischer Verlag, Heidelberg (1996)

[Roth, 2001a] Roth, G.: Die neurobiologischen Grundlagen von Geist und Bewußtsein. In: Pauen, M., Roth, G. (eds.) Neurowissenschaften und Philosophie. Fink-Verlag (2001)

[Roth, 2001b] Roth, G.: Wie das Gehirn die Seele macht, 51. Lindauer Psychotherapiewochen (2001)

[Roth, 2003] Roth, G.: The interaction of cortex and basa Ganglia in the Control of Voluntary Actions. In: Maasen, S., Prinz, W., Roth, G. (eds.) Voluntary action-brains, minds, and society. Oxford University Press, Oxford (2003)

[Roth & Prinz, 1996] Roth, G., Prinz, W. (eds.): Kopf-Arbeit. Spektrum Akademischer Verlag, Heidelberg (1996)

[Roth & Wullimann, 2001] Roth, G., Wullimann, M.F. (eds.): Brain, evolution and cognition. Wiley & Sons and Spectrum Publication (2001)

[Roth & Rudnick, 2008] Roth, G., Rudnick, H.-J.: Antriebe des Unbewussten. Süddeutsche Zeitung, Nr. 158 (2008)

[Rouse et al., 1987] Rouse, W.B., Geddes, N.D., Curry, R.E.: An architecture for intelligent interfaces: outline of an approach to supporting operators of complex systems. In: Human-Computer Interaction, vol. 3, pp. 87–122. Lawrence Erlbaum Inc., Mahwah (1987/1988)

[Ruckdeschel & Onken, 1993] Ruckdeschel, W., Onken, R.: Petrinetz-basierte Pilotenmodellierung. In: Scheschonk, G., Reisig, W. (eds.) Petri-Netze im Einsatz für Entwurf und Entwicklung von Informationssystemen. Springer, Heidelberg (1993)

[Ruckdeschel & Onken, 1994] Ruckdeschel, W., Onken, R.: Modelling of pilot behaviour using Petri nets. In: 15th International Conference on Application of Petri nets, Zaragoza (1994)

[Ruckdeschel, 1997] Ruckdeschel, W.: Modellierung regelbasierten Pilotenverhaltens mit Petrinetzen, Dissertation, University of German Armed Forces, Munich (1995)

[Rumelhart & Mc Clelland, 1986] Rumelhart, D.E., Mc Clelland, J.L. (eds.): Parallel distributed processing, vol.1. The MIT Press, Cambridge (1986)

[Russel & Norvig, 1994] Russel, S., Norvig, P.: Artificial intelligence: a modern approach. Prentice Hall, Englewood Cliffs (1994)

[Sachs et al., 1995] Sachs, G., Möller, H., Dobler, K.: Synthetic vision experiments for approach and taxiing in poor visibility. In: Verly, J.G. (ed.) Enhanced and Synthetic Vision 1995, Orlando. SPIE Proceedings, vol. 2463, pp. 128–136 (1995)

[Salas et al., 1992] Salas, E., Dickinson, T.L., Converse, S.A., Tannenbaum, S.I.: Toward an understanding of team performance an training. In: Swezey, R.W., Salas, E. (eds.) Teams: their training and performance, pp. 3–29. Ablex Publishing Corporation (1992)

[Salucci, 2001] Salucci, D.D.: An integrated model of eye movements and visual encoding. Cognitive Systems Research I, 201–220 (2001)

[Salvendy, 1997] Salvendy, G. (ed.): Handbook of human factors and ergonomics. Wiley, Chichester (1997)

[Salvendy & Smith, 1993] Salvendy, G., Smith, M.J. (eds.): Proceedings of the 5th International Conference on Human-Computer Interaction, Orlando (1993)

[Sanders, 1983] Sanders, A.F.: Toward a model of stress and human performance. Acta Psychologica 52, 61–97 (1983)

[Sarle, 1994] Sarle, W.S.: Neural networks and statistical models. In: Proceedings of the Nineteenth Annual SAS Users Group International Conference (April 1994)

[Sarter & Woods, 1994] Sarter, N.B., Woods, D.D.: Decomposing automation: autonomy, authority, observability and perceived animacy. In: 1st Automation and Human Performance Conference, Washington, DC (1994)

[Sarter & Woods, 1995] Sarter, N.B., Woods, D.D.: How in the world did we ever get into that mode? Mode error and awareness in supervisory control. Human Factors 37(1), 5–19 (1995)

[Scerbo, 2001] Scerbo, M.W.: Adaptive automation. In: Karwoski, W. (ed.) International Encyclopedia of Ergonomics and Human factors. Taylor and Francis, London (2001)

[Schank, 1975] Schank, R.C.: Conceptual information processing. North-Holland Publishing Company, Amsterdam (1975)

[Schank & Abelson, 1977] Schank, R.C., Abelson, R.P.: Scripts, plans, goals, and understanding: an inquiry into human knowledge structures. Erlbaum Associates, Hillsdale (1977)

[Schmidt, 1994] Schmidt, K.: Cooperative work and its articulation requirements for computer support, Le Travail Humain. Tome 57(4), 345–366 (1994)

[Schmidt & Thews, 1980] Schmidt, R.F., Thews, G.: Physiologie des Menschen. Springer, Berlin (1980)

[Schmidt & Thews, 2000] Schmidt, R.F., Thews, G.: Physiologie des Menschen. Springer, Berlin (2000)

[Schmidtke, 1993] Schmidtke, H.: Ergonomie, Hauser (1993)

[Schreiner, 1996] Schreiner, F.: Bedienhandbuch ISF-Fuzzy, Programmdokumentation, IB Institut für Systemdynamik und Flugmechanik, University of German Armed Forces Munich (1996)

[Schreiner & Onken, 1997] Schreiner, F., Onken, R.: Ein Konzept zum fahreradaptiven, autonomen Führen eines KFZ, 2. Berliner Werkstatt Mensch-Maschine-Systeme, Berlin (1997)

[Schreiner, 1999] Schreiner, F.: Automatische Führung des Kraftfahrzeugs mit fahreradaptiven Eigenschaften, Dissertation, University of German Armed Forces Munich (1999)

[Schüz, 2001] Schüz, A.: What can the cerebral cortex do better than other parts of the brain. In: Roth, G., Wullimann, M.F. (eds.) Brain, Evolution and Cognition. Wiley & Sons and Spectrum Publication (2001)

[Schulte & Stütz, 1998] Schulte, A., Stütz, P.: Evaluation of the crew assistant military aircraft CAMA in simulator trials. In: RTO System Concepts and Integration Panel Symposium, Canada (1998)

[Schulte & Onken, 1995] Schulte, A., Onken, R.: Empirical modelling of pilot's visual behaviour at low altitudes. In: IFAC/IFIP/IFORS/IEA Symposium on Analysis, Design, and Evaluation of Man-Machine Systems, pp. 169–174. MIT, Cambridge (1995)

[Schulte, 1996] Schulte, A.: Visualle Routinen von Piloten, Dissertation, University of German Armed Forces Munich (1996)

[Schulte et al., 1996] Schulte, A., Rogozik, J., Klöckner, W.: Visualizing of planning and decision-aiding functions for crew assistance in military aircraft operations. In: Verly, J.G. (ed.) Enhanced and Synthetic Vision 1996, Orlando. SPIE Proceedings, vol. 2736, pp. 164–175 (1996)

[Schulte, 2002] Schulte, A.: Cognitive automation for attack aircraft: concept and prototype evaluation in flight simulator trials. International Journal of Cognition, Technology, and Work 4(94), 146–159 (2002)

[Schulte, 2003] Schulte, A.: Kognitive Assistenzsysteme für Transport- und Militärflugzeuge, Fortschritt-Berichte VDI Reihe 12, Automatisierungs- und Assistenzsysteme für Transportmittel: Möglichkeiten, Grenzen, Risiken (2003)

[Schulte et al., 2008] Schulte, A., Meitinger, C., Onken, R.: Human factors in the guidance of uninhabited vehicles: oxymoron or tautology? The potential of cognitive and co-operative automation, Cognition, Technology & Work (2008), http://dx.doi.org/10.10111-008-0123-2

[Schulte & Meitinger, 2009a] Schulte, A., Meitinger, C.: Introducing cognitive and co-operative automation into UAV guidance work systems. In: Proceedings of Human-Robot Interactions in future Military Operations (2009)

[Schulte & Meitinger, 2009b] Schulte, A., Meitinger, C.: Cognitive and cooperative automation for manned-unmanned teaming missions. In: NATO RTO Lecture Series SCI-208 on /Advanced Automation Issues for Supervisory Control in Manned-unmanned Teaming Missions (2009)

[Searle, 1969] Searle, J.R.: Speech acts: an essay in the philosophy of language. Cambridge University Press, Cambridge (1969)

[Seitz, 2003] Seitz, R.J.: How do we control action. In: Maasen, S., Prinz, W., Roth, G. (eds.) Voluntary action - brains, minds, and society. Oxford University Press, Oxford (2003)

[SGI, 2001] SGI AG: Standard Template Library Programmer's Guide (2001), http://www.sgi.com/tech/stl

[Shafer, 1976] Shafer, G.: A mathematical theory of evidence. Princeton University Press, Princeton (1976)

[Shepanski & Macy, 1987] Shepanski, J.F., Macy, S.A.: Manual training techniques of autonomous systems based on artificial neural networks. In: IEEE - First International Conference on Neural Netzworks in San Diego, vol. 4, pp. 697–704 (1987)

[Sheridan, 1987] Sheridan, T.B.: Supervisory control. In: Salvendy, G. (ed.) Handbook of Human Factors, ch. 9.6, pp. 1243–1268. John Wiley & Sons, New York (1987)

[Sheridan, 1992] Sheridan, T.B.: Telerobotics, automation and human supervisory control. MIT Press, Cambridge (1992)

[Sheridan, 2002] Sheridan, T.B.: Humans and automation: system design and research issues. John Wiley & Sons, Inc., Chichester (2002)

[Shortliffe, 1976] Shortliffe, E.: Computer-based medical consultations: MYCIN. Elsevier Publishing, New York (1976)

[Simon, 1969] Simon, Herbert, A.: The sciences of the artificial. MIT Press, Cambridge (1969)

[Singer 2003] Singer, W.: Synchronization, binding and expectancy. In: Arib, M.A. (ed.) The handbook of brain theory and neural networks, 2nd edn., pp. 1136–1143. The MIT Press, Cambridge (2003)

[Singer, 2007] Singer, W.: Understanding the brain, EMBO reports, European Molecular Biology Organisation. Special issue 8, 16–19 (2007)

[Sloman, 1999] Sloman, A.: What sort of architecture is required for a human-like agent? In: Wooldridge, M., Rao, A. (eds.) Foundations of rational agency. Kluwer Academic Publishers, Dordrecht (1999)

[Smiley & Michon, 1989] Smiley, A., Michon, J.A.: Conceptual framework for generic intelligent driver support. In: Deliverable Report DRIVE V1041/GIDS-GEN03, Traffic Research Center. University of Groningen, The Netherlands (1989)

[Solso, 2001] Solso, R.L.: Cognitive psychology. Pearson Education Inc. & Bacon (2001)

[Sommerhoff, 1990] Sommerhoff, G.: Life, Brain and consciousness. North-Holland, Amsterdam (1990)

[Sowa, 2000] Sowa, J.F.: Knowledge representation, Brooks/Cole (2000)

[Stanney, 2002] Stanney, K.M. (ed.): Handbook for virtual environments: design, implementation, and applications. Lawrence Erlbaum Associates, Inc., Mahwah (2002)

[Starke, 1990] Starke, P.: Analyse von Petri-Netz-Modellen, B.G. Teubner, Stuttgart (1990)

[Starke, 1993] Starke, P.: INA-Integrated net analyzer, version 1.3, Berlin (1993)

[Stein, 1994] Stein, A.: Programmierung eines Software-Tools für die Entwicklung von Fuzzy-Controllern, Diplomarbeit LRT/WE13/D/94-10, University of German Armed Forces Munich (1994)

[Strohal & Onken, 1998] Strohal, M., Onken, R.: Intent and error recognition as part of a knowledge-based cockpit assistant. In: International Society for Optical Engineering (SPIE), Orlando (1998)

[Strohal, 1998] Strohal, M.: Pilotenfehler- und Absichtserkennung als Baustein für ein Cockpitassistenzsystem mittels eines halbautomatischen Verfahrens zur Situationsklassifikation, Dissertation, University of German Armed Forces Munich (1998)

[Stütz, 1999] Stütz, P.: Adaptive Modellierung des regelbasierten Pilotenverhaltens in Cockpitassistenzsystemen, Dissertation, University of German Armed Forces Munich (2000)

[Stütz & Schulte, 2000] Stütz, P., Schulte, A.: Evaluation of the crew assistant military aircraft CAMA in flight trials. In: 3rd International Conference on Engineering Psychology and Cognitive Ergonomics (2000)

[Stütz & Onken, 2001] Stütz, P., Onken, R.: Adaptive pilot modelling for cockpit crew assistance: Concept, realisation, and results. In: 8th Conference on Cognitive Science Approaches to Process control CSAPC 2001, Neubiberg, Germany (2001)

[Swezey & Salas, 1992] Swezey, R.W., Salas, E.: Guidelines for use in team-training development. In: Swezey, R.W., Salas, E. (eds.) Teams: their training and performance. Ablex Publishing Corporation, Norwood (1992)

[Taylor et al., 2000] Taylor, R.M., Bonner, M.C., Dickson, B.T., Howells, H., Miller, C., Milton, N., Pleydell-Pearce, C., Shadbolt, N., Tennison, J., Whitecross, S.: Cognitive cockpit engineering: Coupling functional state assessment, task knowledge management and decision support for context sensitive aiding, TTCP Technical Panel 7 Human Factors in Aircraft Environments, Workshop on Cognitive Engineering, Dayton, Ohio (2000)

[Timpe et al., 2002] Timpe, K.P., Jürgensohn, T., Kolrep, H. (eds.): Mensch-Maschine-Systemtechnik - Konzepte, Modellierung, Gestaltung, Evaluation, 2nd edn. Symposion Publishing GmbH, Düsseldorf (2002)

[Theunissen, 1997] Theunissen, E.: Integrated design of a man-machine interface for 4D navigation. Thesis Delft University of Technology. Delft University Press (1997)

[Thurston & Carraher, 1966] Thurston, J.B., Carraher, R.G.: Optical illusions and the visual arts. Reinhold Publication Corporation, New York (1966)

[Treisman & Gelade, 1980] Treisman, A., Gelade, G.: A feature integration theory of attention. Cognitive Psychology 12, 97–136 (1980)

[Trolltech, 2001] Trolltech Software Company: Qt Reference Documentation (2001), `http://doc.trolltech.no`

[Trouvain & Schlick, 2007] Schlick, C., Trouvain, B.: A comparative study of multimodal displays for multirobot supervisory control. In: Harris, D. (ed.) HCII 2007 and EPCE 2007. LNCS (LNAI), vol. 4562, pp. 184–193. Springer, Heidelberg (2007)

[Tulving, 1993] Tulving, E.: What is episodic memory? Current Directions in Psychological Science 2, 67–70 (1993)

[Varela et al., 2001] Varela, F., Lachaux, J.-P., Rodriguez, E., Martinerie, J.: The brainweb: phase synchronization and large-scale integration. Nature Reviews-Neuroscience (2001)

[Venhovens et al., 2000] Venhovens, P., Naab, K., Adiprasito, B.: Stop and go cruise control. International Journal of Automotive Technolgies 1(2), 61–69 (2000)

[Verly, 1996] Verly, J.G. (ed.): Proceedings of SPIE Conference on Enhanced and Synthetic Vision as part of SPIE' s International Symposium on Aerospace/Defense Sensing and Control (Aerosense), Orlando, vol. 2736 (1996)

[Vicente & Rasmussen, 1992] Vicente, K.J., Rasmussen, J.: Ecological interface design: theoretical foundations. IEEE Transactions on Systems, Man and cybernetics SMC-22, 589–606 (1992)

[Vicente, 1999] Vicente, K.J.: Cognitive work analysis: towards safe, productive and healthy computer-based work. Lawrence Erlbaum, New Jersey (1999)

[Völckers & Böhme, 1995] Völckers, U., Böhme, D.: Structures, architectures, and design principles for dynamic planning functions in ATM. In: Winter, H. (ed.) Knowledge-based functions n aerospace systems, AGARD LS 200, pp. 7.1–7.31 (1995)

[Vogler, 1992] Vogler, W. (ed.): Modular Construction and Partial Order Semantics of Petri Nets. LNCS, vol. 625. Springer, Heidelberg (1992); Rozenberg, G. (ed.): APN 1990. LNCS, vol. 483. Springer, Heidelberg (1991)

[Wallentowitz & Reif, 2006] Wallentowitz, H., Reif, K.: Handbuch Kraftfahrzeugelektronik. Fr. Vieweg-Verlag/GWV Fachverlage GmbH (2006)
[Walsdorf, 2002] Walsdorf, A.: Zentrale, objektorientierte Situationsrepräsentation angewandt auf die Handlungsziele eines Cockpitasasistenzsystems, Dissertation, University of German Armed Forces Munich (2002)
[Wickens, 1984] Wickens, C.D.: Processing resources in attention. In: Parasuraman, D. (ed.) Varieties of attention. Academic Press, New York (1984)
[Wickens, 1992] Wickens, C.D.: Engineering psychology and human performance. Harper Collins Publishers (1992)
[Wickens & Carswell, 1995] Wickens, C.D., Carswell, M.: The proximity compatibility principal: 1st psychological foundation and relevance to display design. Human Factors 37, 473–494 (1995)
[Wiedemann, 1972] Wiedemann, R.: Simulation des Straßenverkehrsflusses, Schriftenreihe des Instituts für Verkehrswesen der Universität Karlsruhe (1974)
[Wiener & Curry, 1980] Wiener, E.L., Curry, R.E.: Flight deck automation: promises and problems, NASA Tech. Mem. No. 81206 (1980)
[Wiener, 1989] Wiener, E.L.: Human factors of advanced technology (glass cockpit) transport aircraft, NASA Tech. Rep. No. 177528 (1989)
[Winograd & Flores, 1986] Winograd, T., Flores, F.: Understanding computers and cognition. Ablex Publishing Corporation, Norwood (1986)
[Winter, 1995] Winter, H. (ed.): Knowledge-based guidance and control functions, AGARD-AR-325 (1995)
[Winzer, 1980] Winzer, T.: Messung von Beschleunigungsverteilungen, Forschung, Straßenbau und Straßenverkehrstechnik, Heft 319, BMV (1980)
[Witten & Frank, 2000] Witten, I.H., Frank, E.: Data mining. Academic Press, London (2000)
[Wittig & Onken, 1993] Wittig, T., Onken, R.: Inferring pilot intent and error as a basis for electronic crew assistance. In: Proceedings of the Fifth International Conference on Human-Computer Interaction (HCI International 1993), Orlando (1993)
[Wittig, 1994] Wittig, T.: Maschinelle Erkennung von Pilotenabsichten und Pilotenfehlern über heuristische Klassifikation, Dissertation, University of German Armed Forces Munich (1994)
[Woods et al., 1994] Woods, D.D., Johannesen, L.J., Cook, R.I., Sarter, N.B.: Behind human error: cognitive systems, computers and hindsight, State-of-the-art-report for CSERIAC, Dayton, Ohio (1994)
[Woods, 1996] Woods, D.D.: Decomposing automation: apparent simplicity, real complexity. In: Parasuraman, R., Mouloua, M. (eds.) Automation and human performance: theory and application. Lawrence & Erlbaum Associates, Mahwah (1996)
[Wooldridge, 2002] Wooldridge, M.: An introduction to multi-agent systems. Wiley, Chichester (2002)
[Wright, 2005] Wright, D.: Flight data monitoring: more than just data. In: Teledyne Controls 2005 Users Conference, Aviation Data Management Solutions, Today and Tomorrow (2005)
[Yen et al., 1993] Yen, J., Neches, R., MacGregor, R.: Classification-based programming: a deep integration of frames and rules, Rep. ISI/RR-88-213, USC, Information Sciences Institute, Marina del Rey (1989)
[Zadeh, 1965] Zadeh, L.: Fuzzy sets. Information and control 8, 338–353 (1965)
[Zwicker, 1982] Zwicker, E.: Psychoakustik. Springer, Berlin (1982)

Author Index

Abelson 222
Abbot 170
Adams 149
Adiprasito 16
Albus 121
Algayres 271
Allman 342
Amalberti 165
Amos 226
Anderson 50, 68, 221, 224-228
Arib 40
Austin 117

Baars 53, 56
Baberg 81
Baddeley 53, 223
Bainbridge 3, 91
Banbury 160
Band 342
Banks 160, 161
Behringer 201
Bernasch 248
Bickhard 40
Biggers 110
Billings 2, 3, 7, 25, 27, 80, 87-91, 135
Billings S.A. 249
Birbaumer 36, 50, 51
Bishop 266
Blom 342
Blum 234
Böhme 159
Boff 348, 349, 350
Bonner 149, 158, 166, 167
Bothell 226, 227
Bousquet 211
Braitenberg 47, 119
Brockhaus 7
Brodmann 38
Brooks 3
Bruce 348
Brüdigam 201
Bubb 17
Burges 211
Burgess 239
Burov 148, 158
Bush 342
Byrne 226, 227

Cacciabue 74, 95
Card, H.C. 298, 299, 300, 303
Card, S.K. 46, 49, 54, 55, 350
Carpenter 249
Carswell 156
Chambres 322
Chamontin 271
Champigneux 165
Changeux 53, 55, 56
Chater 239
Chen 249
Cohen 115, 135
Collete 57
Convay 149
Converse 113
Cook 3
Corbetta 51
Courtney 57
Crowston 105, 112
Curry 3, 100, 162

Darken 266
Davidson 55
Davis 216, 217, 218
Deblon 165
D'Esposito 53
Deerwester 239
Dehaene 53-57, 61
Desel 232

Dickmanns, D. 201
Dickmanns, E.D 44, 201, 209
Dickinson 113
Dickson 149, 158
Diepold 199, 200
DiGirolamo 50, 342
Di Nocera 148, 158
Di Vito 133
Dobler 156
Dobmeyer 51
Doehler 156
Döring 19, 221
Donath 148, 157, 329
Dorneich 156
Douglas 226, 227
Driver 50
Dudek 4, 168
Dumais 239, 240, 242, 244, 246
Duncan 52, 74
Dyck 170

Ebbinghaus 43
Ehrig 232
Endsley 125
Engel 41, 42
Eriksen 75
Ernsting 342
Erwin 347
Erzberger 159

Fabiani 133
Farah 53
Fastenmeier 257
Feraric 192, 199, 201, 222, 249, 256, 291
Ferber 105, 106, 121, 125
Finch 239
Findlay 147
Fitts 350
Flemisch 149, 174, 330
Fletcher 166, 167
Flores 118
Frackowiak 49
Frank E. 211
Frank 226
Franke 248
Franklin 61
Freed 227
Freegaze 349
Frey 187, 188, 323, 326-328
Friston 49
Fritsche 248
Fritz 248
Fujioka 249
Funk 90, 163
Furnas 239

Gaillard 53, 148, 158
Gasser 111
Gauss 159
Gauthier 160
Garrel, von 192, 201, 222, 250, 255-271
Gazis 248
Geddes 100, 161
Gelade 50, 71
Georgeff 120
Georgeson 348
Gerfen 226
Gerlach 146, 174, 177, 182-185
Gillen 57
Gleason 59
Godthelp 198
Goodrich 149
Gorrell 242, 244, 245, 247
Goschke 62
Grafton 49
Graesser 61
Grashey 192, 201, 222, 249, 255, 291, 292
Grau 125
Green 348
Grossberg 249
Guyton 348, 350

Hakeem 342
Haller 234
Hancock 140
Hannen 163, 164, 165
Harel 234
Hargutt 67
Harris 25
Harshman R.A. 21, 239
Hart 140
Hecker 156
Hebb 47, 237
Heesch, van 320
Helander 25
Helmetag 156
Henn 249
Herman 248

Hinton 347
Hoc 95, 100
Hockey 148, 158
Hof 342
Hofmann 247
Hoefflinger 249
Hollan 32
Hollnagel 32, 85, 90, 172
Holt, von 209
Holve 248
Hou 160
Houk 226
Howells 149, 158
Huber 273
Hutchins 32

INCOSE 30, 321
Ioerger 110
Ishida 111
Izaute 322

Jacob 349
James 43
Jarasch 321
Jennings 111, 115, 116, 120
Jensen 273
Johannesen 3
Johannsen 25
Jonas 24
Joubert 165
Jürgensohn 25, 199, 248
Juhas 232

Kaiser 156
Karg 19
Karwoski 148, 157
Kaufman 348, 349, 350
Kecman 211
Kehtarnavaz 249
Kentridge 147
Kersandt 159
Kerszberg 55, 56
Kieras 221
Kimberg 53
Kimble 349, 350
Kirchner 25
Kirsh 32
Klawonn 298
Klein 125
Klöckner 156
Kohonen 299
Kok 342
Kolodner 279
Kolrep 25
Konings 198
Kopf 4, 192-205, 290, 291
Kornhauser 249
Kosko 214, 215, 217, 236
Kraiss 249, 348
Krüger 67
Kruse 298
Kubbat 156
Küttelwesch 249

Lachaux 55
Lachover 234
Laird 223, 314
Landauer 214, 239, 240, 244- 247
Lebiere 226, 227
Lenhart 156
Lenz 187, 188
Leonhard 254
Leplat 52, 74
Levesque 115, 135
Libet 59
Linden, van 57
Lizza 4, 160, 161
Lloret 271
Lorenz 148, 158
Loughry 226
Lund 239
Luu 342
Luxburg 211
Lyall 90

Maasen 62, 66
MacGregor 218
Macy 249
Malone 105, 111, 112
Mancini 172
Marescaux 322
Martial, von 113
Martinerie 55
Mayer 156
Mazziotta 49
Mc Clelland 221, 237, 238
Mc Ruer 248
Mecklenburg 249
Meitinger 5, 104, 109, 132, 137-141, 149, 293-296

Menu 125
Meshkati 140
Metcalfe 322
Meyer 221
Meystel 121
Michon 190, 191
Miezin 51
Miller 149, 156, 158, 163, 164, 165
Millot 108
Milton 149, 158
Minsky 221, 222
Möller 156
Moody 266, 267
Mooral 348
Moore 61
Moran 46, 48, 54, 350
Moschytz 265
Mouloua 3
Mukawa 349
Murray 226

Naab 16, 248
Naamad 234
Naccache 53, 55-57, 59, 61
Narendra 249
Nauk 298
Neches 218
Neukum 67
Neusser 249
Newell 47, 48, 50, 56, 70, 215, 221–225, 230, 314, 320, 321, 350
Nicholson 342
Nieuwenhuys 342, 346
Nijhuls 249
Nimschinsky 342
Nirschl 234
Norvig 234

Ögren 133
O'Hare 111, 115, 116, 135
O'Reilly 226
Ohno 349
Onken 5, 122, 134, 147, 154, 168-202, 253, 280-286, 298, 313, 331
Opperman 149, 158
Otto 211, 256-258, 261, 262, 291

Page-Jones
Palmer 149
Parasuraman 3, 52, 148, 157, 329, 342
Paris 324
Pashler 50
Pauen 53
Pearce 149, 158
Pearl 59
Pedricz 298, 299, 300, 303
Pekar 57
Petersen 51
Peterson 234
Petri 223, 234
Petrides 54, 57
Phelps 49
Pilhofer 313, 320
Pinel 36, 65, 339, 346
Platts 135
Plaut 347
Pleydell-Pearce 149, 158
Pnueli 234
Politi 234
Pommerleau 248
Posner 50, 342
Post 223
Prabhu 25
Presty 49
Prevot 172,175, 176, 179, 182-185
Prinz 36, 62, 66
Protzel 248
Puder 313, 320
Putzer 122, 134, 187, 188, 223, 311–313, 325

Quin 226, 227

Rainford 342
Rämä 57
Rao 61, 120
Rasmussen 25, 52, 66, 67, 74, 151, 156, 230, 248
Rätsch 211
Rauschert 140
Reason 53, 76
Reif 14, 15
Reisig 175, 234, 271
Rekersbrink 248
Reusch 248
Ridderinkhof 342
Riley 90
Robert 148, 158
Rodriguez 55
Rogozik 156

Rohmert 25
Römer 313, 320
Rosen 249
Rosenbloom 223, 314
Ross 214, 239, 240, 246
Roth 36, 37, 46, 48, 53, 58, 59, 62, 64, 66
Rothery 248
Rouse 4, 100, 161
Roux 271
Rozenberg 273, 275
Ruckdeschel 175, 182-185, 220, 269, 274–279, 285-289
Rudnick 64
Rumelhart 222, 237, 238
Russel 234

Sachs 156
Sackur 54
Sala 57
Salas 113
Salucci 227
Salvendy 3, 25, 170
Sanders 670
Sarter 3, 86-92
Sarle 267
Scerbo 148, 157
Schank 215, 222
Scheschonk 286
Schlick 132
Schmidt, K. 106
Schmidt, R. 133
Schmidt, R.F. 36, 50, 51, 347, 348, 350
Schmidtke 349
Schreiner 131, 192, 201, 249, 251-256, 304
Schüz 36, 47, 48
Schulte 5, 109, 132, 139, 140, 147, 148, 156, 157, 186-189, 293, 321, 329
Schutte 149
Schwartz 55
Scipione 160
Searle 117
Seitz 38
Sergent 53
Shallice 347
Shadbolt 149, 158
Shafer 237
Shapiro 55
Shepanski 249
Sheridan 3, 25, 27, 69, 79, 82
Sherman 234
Shimamura 322
Shortliffe 84
Shrobe 216, 217, 218
Shulman 51
Shtull-Trauring 234
Simon H. 47, 215, 221, 223, 224
Simon, H.A. 98
Singer 1, 41, 42
Sloman 61
Small 4
Smiley 191
Smith 170
Sohn 249
Solso 44, 75
Sowa 217
Staehle 19
Stanney 47
Starke 285
Staveland 140
Stein 304
Stening 61
Strohal 172, 186, 189, 236, throughout chapter 6.2.3.2
Stütz 175, 186, 187, 189, 237, 271, 279–284
Suikat 156
Swezey 113
Szolovits 216, 217, 218

Takabo 249
Tannenbaum 113
Tattersall 148, 158
Taylor 149, 158, 166, 167
Tennison 149, 158
Terveen 40
Theunissen 156
Thews 347, 348, 350
Thomanek 201
Thomas 348, 349, 350
Tilley 350
Timpe 25
Trakhtenbrot 234
Treisman 50, 71
Trolltech 322
Trouvain 132
Tulving 44

VanLehn 3
Varela 55
Verly 156

Venhovens 16
Vicente 25, 156
Viebahn, von 156
Völckers 159
Vogler 273, 275

Walker 147
Wallentowitz 14, 15
Walsdorf 186, 187, 189
Weir 248
Whitecross 149, 158
Wickens 25, 46, 52, 73, 156
Wiedemann 248
Wiener 3, 90
Winograd 118
Winter 19, 159, 161
Winzer 199
Wise J.A. 170
Wise S.P. 226
Witten 211
Wittig 172, 182-185, 297
Woods 3, 32, 86-90, 172
Wooldridge 61, 112, 115, 120
Wright, D 330
Wright, E.W. 59
Wullimann 36

Yeh 73
Yen 218
Yokoo 111
Yoshikawa 349

Zadeh 221, 234, 235
Zenyuh 4
Zsambok 125
Zwicker 349

Subject Index

Adaptive Control and Thought (ACT) 224ff
Adaptive Control and Thought - Rational (ACT-R) 225, 226ff
ACT* 228-230
Adaptive cruise control (ACC) 15, 16, 130ff
Agent 22, 26, 43, 61, 92-95, 104-125, 129, 135, 142, 219, 221, 223, 227, also throughout Chapter 4
- cooperarting 111
- co-ordinating 142
- hardware 120
- physical 117
- software 62, 94, 95, 120, 121, 122
- synthetic 228

Aircraft guidance and control 13, 152, 159, 160
Amygdala 48, 58, 59, 64, 126, 340, 342-344ff
Anti-lock Braking System (ABS) 14, 15, 26, 27, 80, 82
Artificial cognitive process 69, 91, 119ff, 129
Artificial cognitive unit (ACU) 122ff, also throughout Chapters 4, 5, 6 and 7
ASPIO 4, 160, 168, 176
Assistant system 2, 25, 101, 103, 107, 108, 123, 129, 144ff, also throughout Chapters 4, 5, 6 and 7
Assistance 1-209
- alerting 143, 145-149, 154, 174, 193
- associative 143-145, 153–158, 162, 163, 165, 167, 174
- attention 147
- cognitive 145
- cockpit 161
- driver 191
- machine 209
- substituting 143, 147-150, 152–154, 157, 158, 167

Attention 6-252,
- covert 51
- divided 52
- focused 51
- overt 51
- selective 51

Attention allocation 74
Attention assistance 147
Attention control 35, 41, 43, 49ff, throughout Chapters 3 and 7, 145, 214
Attention overtaxing 157
Augmented cognition (AugCog) 156
Automatic Flight Planner (AFP) 171ff
Automation 2–333
- adaptable 149, 158, 167
- adaptive 148, 157, 174
- assisting 3
- built-in 82ff, 91
- cognitive 1, 72, 79ff, also throughout Chapters 4, 5, 6 and 7
- control 7
- conventional 72, 79ff, also throughout Chapters 4, 5 and 7, 311, 321, 322, 329
- cooperative 23, 100, 141, 142,
- degree of 167
- information 7, 27
- interactive 3
- human-centred 90
- level of 25, 141, 149, 158
- management 7, 27
- operation-supporting 99
- operator-controlled 4, 82ff

Autonomous system 23-24
- artificial 23, 24
- semi- 24

Basal ganglia 49, 50, 59, 126, 225, 226, 339, 343, 346
Behaviour 2–350
- adaptive 164, 258
- assistant 193
- automatic 54
- cognitive 35, 58, 63, 64, 66, 67, 211, also throughout Chapters 4, 6 and 7
- concept-based 70ff, 74-76, also throughout Chapters 4, 5 and 6, 319
- conscious 57, 68, 70
- co-operative 133-135
- crew 169, 172, 176-178, 289
- decision 256
- deterministic 123, 279
- driver 193-197, 200, 202, 206, 211, 248–269, 290
- dynamic 14, 272, 294
- emotional 57, 76, 122
- erroneous 51
- goal-directed 119
- human 2, 31-37, 60, 66ff, also throughout Chapters 3, 4, and 6
- human-like 119, 223, 239, 250
- individual 133, 175, 189, also throughout 195-206, 251, 259–270, and 283, 290
- intelligent 64, 217, 218
- intentional 57, 60
- knowledge-based 67
- learning 49
- mean 201
- normative 135, 175, 195, 197, 206, 211, 250, 269, 270, 279, 280, 289, 297
- observable 98, 124, 294, 297
- operator 44, 49, 66ff, 156, 234, 297
- perceptual 147
- pilot 221, 270, 271, 279, 286, 289, 297, 304, also throughout 170-189
- procedure-based 70ff, 82, 97, 122, 126, 127, 175, 193, 211, also throughout Chapter 6
- rational 58, 122
- routine 60, 61
- rule-based 67ff, 225, 230, 248
- skill-based 51, 67ff, 82, 123-127, 132, 152, 194, also throughout Chapter 6
- substitutive 157
- system 131, 224
- subconscious 70
- team 113, 116

Behaviourism 59
Behaviour level 68, 71-76
Behaviour model 60, 63, 67ff, also throughout Chapter 5 and 6
Behaviour visualisation 330
Binding problem 41, 216, 247
Boundedness 234, 284
Brittleness 89, 94, 97, 100, 321

CAMA 159, 160, 168, 175, 180, 186-189ff, 236, 269, 270, 279, 289, 297, 304, 305, 310
Camera-based sensing 16
CASSY 161, 169-188ff, 192, 195, 269, 270, 289, 297, 298
Cerebellum 44, 49, 225, 340, 344, 346
Chunk 46ff, 56, 61, 68-69, 71, 72
Clumsiness 89, 90ff, 94, 97, 321
Cluster analysis 302, 306, 309
Cockpit assistant system 2, 159, 160ff, 168, 269,
Cockpit information manager (CIM) 163ff
Cognition 1-322
- artificial 2, 3, 16, throughout Chapter 3, 79ff, 91, 104, 119, 213, 220, 247, 311, 312
- augmented 156
- conscious 50ff
- degree of 79
- distributed 32
- human 1, 32ff, throughout Chapters 3, 4, 5, and 6, 322
- machine 66
- natural 32,
- unconscious 60

Cognitive design 2, 6, 99ff, 101, 104, 142, 150, 186
Cognitive process 49-70ff, also throughout Chapters 4, 5, 6 and 7
Cognitive sub-function 124ff, 222, 293, 297, also throughout Chapters 5 and 7

Cognitive system 311-327
- artificial 2, 5, throughout Chapters 3 and 5, 79, 120ff, 220, 247, 293
- co-operating 5, 6
- extended 324, 327
- human 35,
- joint 32,

Cognitive Process Language (CPL) 293, 317
Cognitive System Architecture (COSA) 122, 133, 137, 293, 312ff, 324
Cognitive teaming 104, 107, 142
Collaboration 23, 105ff, 113, 190
CommonKADS Markup Language (CML) 317
Communication 6-349, 117ff
- explicit 117
- centralised 169
- implicit 117
- speech 170, 187, 349

Condition/event-net 271
Connectedness 284
Connectionism 45, 61, 217-219
Connectionistic information processing 36
Conscious experience 40, 48, 56-59, 62, 71, 79
Consciousness 35, 48, 53-59ff, 66, 68
Co-operation 3-299, 106-108ff
- co-ordinated 106
- cognitive 142
- debative 106
- horizontal 108
- human-automation 101
- human-machine 107, 112, 134, 139
- machine-machine 107, 108, 114, 139
- obstructive 105, 106, 109, 112
- simple 105, 108-110
- vertical 108

Co-operative control 1, 99ff
Co-ordination 103-106, 112, 113ff, 167
Copilote Electronique 159, 165ff, 167
Commitment 96, 109, 113, 115-117ff, 125
Connectedness 283ff
Constructive and Evaluating Learning Method (CEL-Method) 262ff, 265, 266, 268
CORBA 313, 315, 316, 320
Cortex 36-347
- association 38, 39, 71, 342, 345, 346
- auditory 345, 346
- cerebral 36, 39, 40, 59, 339, 340, 342
- cingulated 50, 57, 58, 76, 345, 347
- entorhinal 46, 344, 345
- frontal 342
- inferotemporal 39, 71
- insular 342, 344
- limbic 58, 344
- motor 49, 59, 126, 341, 346
- orbito-frontal 60, 61, 66, 78, 128, 342–345
- parahippocampal 46, 345
- parietal 36, 39, 71, 342, 346
- perirhinal 46, 344, 345
- pre-frontal 52, 55, 58, 60, 61, 65, 66, 128, 228, 342, 346
- pre-motor 341
- prestriate 39
- somatosensory 341, 345
- sensory 39, 41, 54, 71, 126, 341
- temporal 36, 340, 346
- visual 39, 346

COSY^{flight} 324-327
Cruise control 14, 15, 130
Cue 46, 51, 52, 70-76, 81, 126, 127

Decision model 258- 263
Dempster-Shafer, theory of 237
Dendrogram 301, 302, 303, 306, 307
Dialogue Manager (DM) 156, 171-173
Difference reduction 68
Discrepancy interpretation 194, 195
Display
- colour 174
- flight situation 176
- graphical 196
- haptic 196, 206
- ecological 155
- map 163, 177
- multi-function 165
- sensor 165
- visual 156, 163, 186

Display layout 155
Display modifications, online 156
Display technology 156

Driver adaptivity 251
Dual-mode cognitive design 99, 101, 104, 142, 150

Elementary driving tasks 256-262ff, 290, 291
Electroencephalography (EEG) 65
Electronic Stability Control (ESC) 14
EM-algorithm 266
Error recognition 280, 297ff, 304
Execution Aids (EA) 158, 175ff
Executive-Process/Interactive Control (EPIC) 221
Experimental cockpit 179, 180, 185
Expert system 84, 85

Fatigue problem 63
Feature formation 39, 67, 70, 71, 77
Finite automata 220, 271
Flight management system 8, 10, 11, 15, 132, 133, 173, 177, 186
Follow control 130
Fovea vision 41
Frame 218-220, 222ff
Frontal lobe 36, 50, 71, 340, 343
Functional magnetic resonance tomography (fMRT) 37, 65
Fuzzy ART 249
Fuzzy system 214, 215, 234ff, 236, 298, 305

Gaze measurement 147
Generalisation 40, 61, 62, 219, 222, 231, 237, 239, 240, 263, 301, 307
Generic error modelling system (GEMS) 52
Gestalt laws 41, 43
Global workspace 55-57, 60, 61
Global workspace hypothesis 55
Goal determination 70ff, 75-77, throughout Chapters 4 and 5, 270, 319, 323, 329
Ground proximity and warning system (GPWS) 13
Gyrus cingulated 51, 343

Hierarchical clustering 302
High-level cognitive function 33, 64ff, 65
Hippocampus 46, 48, 58, 339, 344, 345
Human-centred
- automation 91
- design 2, 25, 90, 135

Human-machine
- co-agency 96
- co-operation 107, 112, 133, 139
- interaction 2, 3, 4, 6, 156
- interface 6, 11, 132, 314
- symbiosis 101
- system 2-6, 25, 34, 97
- team 107, 112, 117, 140

Human
- assistant 107, 142, 157
- authority 24
- behaviour 2, throughout Chapters 3, 4, and 6, 146, 147, 155
- body 33, 34
- cognition 1, 17, 32ff, throughout Chapters 3, 4, and 6, 320
- component 79, 150
- crew 81, 173, 174
- cut-off-frequency 8
- effectors 34
- element 22
- engineering 34
- error 52, 81, 87, 88, 102, 147
- factors 6, 9, 35, 60, 67, 68, 79, 118
- failure 90, 96
- learning 37, 48, 244, 251
- limitations 63, 100, 157
- memory 36, 43ff, 242, 245
- operator 2-6, 22ff, then throughout Chapters 3, 4, 5, 6, and 7
- overtaxing 52, 145, 147, 329
- overload 86
- performance 23, 24, 56, 67, 70, 71, 123, 155, 221, 224, 322
- problem solving 58, 216, 224
- resources 23, 86, 148, 156, 329
- semantic coding 216, 217, 218
- senses 34
- team 5, 6, 95, 97-118, 141-150, 154-157
- team member 23, 95, 103, 118, 123, 141-149, 154-157, 169
- vision 39-43ff, 62, 209
- work 25, 86, 96

Identification 69ff, throughout Chapters 4, 5, and 7, also throughout 298-311
Individual pilot behaviour 191, 280ff

Intent recognition 163, 164, 174, 178, 196, 197, 213, 239, 299-312ff

Jaccard coefficient 306
Joint cognitive system 32

Knowledge 6-332
- application 137, 272
- a-priori 64ff, also throughout Chapters 4, 5, 6, and 7
- behavioural 68
- built-in 16
- declarative 15, 237
- domain 25
- dynamic 169
- embodied 37
- erroneous 323
- expert 319
- explicit 45ff, 69, 70, 85, 170, throughout Chapter 6, 314
- human language 239
- implicit 44, 48, 62, 69, 125, 220, 222, 227, 247, 251
- intrinsic 251
- procedural 15, 230
- procedure 71
- process 28, 220, 314
- processing 40
- relevant 122, 123
- situational 121ff, throughout Chapter 5, 294-296, 319
- task 320
- uncertain 236

Knowledge acquisition 49, 62, 69, 70, 119, 134, throughout Chapter 6, 328
Knowledge-based system 71, 84
Knowledge management 213-250ff
Knowledge representation 214ff, throughout Chapters 3 and 6, 122, 125, 175, 316, 319
- central 122
- explicit 228, 247
- structured 215
- semantic (connectionistic) 44, 45, 239-247
- symbolic 45, 222, 224
- tri-code 230

Lapse 74
Learning 35, 37, 46ff, 62-66, 70, 77, throughout Chapters 4, 5 and 6, 314, 327
- application-adaptive 231
- automatic 37
- batch 252
- competitive 300
- constructive 262, 266ff
- deliberate 66, 77
- human 37, 49, 244
- machine 259, 279
- offline 247
- online 247, 252
- procedural 230, 231
- semantic 244
- sensory 46
- sequential 252
- supervised 262
- unconscious 66

Learning algorithm 250, 252, 253, 258, 268, 299
Learning component 262, 268
Learning method 262, 263, 266, 279
Learning module 280
Learning performance 244, 250
Learning process 38, 47, 49, 66, 231, 237, 251, 252, 259-263, 266-268
Learning speed 266
Learning strategy 262
Level of automation (LoA) 27, 141, 148, 157
Limbic system 48, 50, 53, 57-59, 77, 344, 345
Literalism 89, 90, 94, 97, 321
Liveliness 233, 284
Logic 220, 227, 235
- fuzzy 220, 221
- predicate 220
- propositional 220
- temporal 220

Long-term-potentiation (LTP) 48
Low-level cognitive function 33, 65

Magneto-encephalography (MEG) 65
Mahalanobis distance 264
Means-ends analysis 68
Medium-level cognitive function 33, 65

Memory 46-352
- artificial 46
- declarative 44, 59, 77, 345
- emotional 59
- episodic 44, 48
- explicit 45, 61, 77
- external 47
- human 36, 43ff
- implicit 58, 61, 77
- long-term 44, 46, 47, 48, 54, 56, 69
- permanent 44
- procedural 44, 77
- semantic 44, 349
- sensory 47, 48
- short-term 44, 46, 47, 48, 58
- transient 46
- working 40, 46, 50, 52, 53ff, 58, 63, 66, 76, 77, 342

Memory access 47
Memory management 37, 57
Mesolimbic system 48, 58, 64, 345
Metacognition 322, 323
Mistake 74, 114, 164, 206, 229
Model 35-329
- ACT 227
- behavioural 66ff, 77, 82, 91, throughout Chapter 5, throughout 213-309
- cognitive 148
- computer 66, 123, 223
- cue 70, 71, 73, 131, 134, 162, 171, 194, 210, 316, 320
- concept 131, 134, 162, 171, 194, 210, 316, 320
- crew interaction 161
- danger 195, 196, 198ff
- dynamic 268, 317
- environment 134, 136, 276, 294–296, 317-319
- error 52
- fuzzy 220
- goal 139
- individual behaviour 175, 194–204, 237, throughout 251–290
- instruction 134, 138, 294-297, 317, 319
- interaction 234
- language 174
- learning 49
- memory 55
- neuro-fuzzy 298-310
- neural network 255, 256-269, 299
- normative 195, 197, 210, 269-279, 292, 298
- online 249
- perception 226
- performance 113, 121, 230
- physiological 148, 195
- procedure 131, 134, 162, 171, 194, 210, 222-224, 292
- process 319
- psychological 67
- qualitative 63, 66,
- quantitative 35, 62, 220, 221, 222, 224, 248
- reference 211
- resource 175, 195, 330
- static 317
- statistical 220, 249
- task 133, 134, 162, 166, 171, 194, 210, 278, 317, 319
- task situation 75, 133, 134, 162, 171, 194, 208, 210, 316, 320
- vee- 321
- vehicle 201
- world 71, 319

Model of distributed systems 219, 220
Model of motivational contexts 133, 134, 162, 171, 194, 210, 319
Model of sensori-motor patterns 133, 134, 162, 171, 194, 210
Model validity 251, 252
Modality 35-349
- auditory 349
- visual 38, 341, 347-349

Monitor of
- environment 172
- flight status 171
- systems 172

Motivational contexts 50, 57, 58, 62-75ff, throughout Chapters 4, 5 and 7, 214
Multi-Agent System (MAS) 26, 105, 112, 115, 121, 125, 142
Multiple constraint satisfaction 237

Navigation system 15, 16, 191
Neocortex 36, 38, 46, 50, 339, 340-342, 345

Network work structure 29
Neural system 214, 215, 217
Neuro-fuzzy approach 298
Neuron 36ff, 41, 48, 55-59, 66, 77, 346

Obstruction 105, 106, 109, 112
Occipital lobe 37, 39, 340
Ontological commitment 216
Opacity 89, 95, 97, 100, 321
Operating cognitive unit (OCU) 97-104ff, throughout Chapter 5, 297, 298, 309, 321, 329
Operating force (OF) 21-32ff, 43, 45, 49, throughout Chapters 4 and 5, 219, 269, 293, 297, 331
Operation-supporting means (OSM) 21-34ff, 47, 80, throughout Chapters 4 and 5, 219, 269, 293, 321

Parallel distributed processing (PDP) 237-239ff, 247
Parietal lobe 36, 50, 340
Penrose distance 306
Perceptual processing 38-43ff, 45, 213, 341, 345
Performance 2-333
- cognitive 36, 56, 122
- control 85
- human 3, 6, throughout Chapters 3 and 6, 123, 130, 147, 148, 156, 210-211
- human-machine system 5, 6, 122, 142
- learning 244, 250
- machine 215, 321, 322, 324
- memory 43, 46
- modelling 266, 267
- monitoring 85
- subsystem 5, 16, 28, 60, 85, 92, 121, 139-142, 164, 165, 179-186, 192, 250, 309
- team 110, 113-115
- visual 43
- work process 18, 26, 28, 48, 79, 107
- work system 2, 27, 48, throughout Chapters 4 and 5, 221, 329-331

Petri net 175, 198, 220, 231-234ff, 269-289
Pilot's associate (PA) 4, 160–163, 165-167, 170
Piloting Expert (PE) 170, 175, 269-289, 298
Pilot Intent and Error Recognition (PIER) 172-175, 275, 280, 303
Place/transition net 232, 233, 271, 286, 289
Plan execution 157, 158, 163, 170-172, 175
Plasticity versus stability dilemma 252, 263, 266, 299
Positron emission tomography (PET) 37, 48, 65
Production system 214, 215, 218, 220, 222-231, 271

RBF net 266
Reachability 233, 285
Recursive least squares method (RLS) 264, 265
Real-Time Control System (RCS) 121
Reference driver 190, 191, 194, 195, 197
Reticular formation 50, 53, 340, 344
Reversibility 284
Rogers coefficient 306
Rotorcraft pilot's associate (RPA) 160-165, 167

Safeness 285
Schema architecture 223
Selection versus monitoring dilemma 62
Semantic coding 37, 45, 47, 56, 61, 62, 216-221, 238, 239, 251, 256
Semi-autonomous system 24, 96, 101, 107, 120
Sensori-motor behaviour 248
Sensori-motor modality 35
Sensori-motor pattern 44, 67-72ff, 127, 129, 131, 134, 162, 171, 194, 210, 260, 261, 263
Sensori-motor processing 38
Sign (Rasmussen) 67, 72, 74
Simple-matching coefficient 307
Situation adaptivity 251
Situation awareness 81, 85, 125, 141, 145-148, 154, 156-158, 172, 187, 206
Slip 51, 74
Sorensen coefficient 306
Statechart 220, 234
State, operator, and result (Soar) 222-224ff, 239
State transition diagram 231, 291
Statistical modelling 220, 249
Stop and go control 130

Substantia nigra 58, 340, 343, 345
Supporting cognitive unit (SCU) 93-104ff, 129-139, 142, 149, 150, 293-295
Support vector 265, 267
Supervisory control 3-6, 10, throughout Chapter 4, 132, 142, 152, 158
Switching dilemma 63
Symbol (Rasmussen) 67, 72
Synapse 37, 47
System of work systems 30ff

Tanimoto coefficient 306
Task 2-331
- central 63
- cognitive 65, throughout 155-158, 221, 227
- concurrent 148, 278, 279
- control 7, 8, 54, 68, 70, 160
- current 70, 72, 73, 81, 83, 126, 127, 131, 134, 162, 194, 210, 251, 320
- dual 52
- due 66
- elementary driving 256-262, 290, 291
- guidance 7, 8
- information manipulating 27
- inspection 53
- management 89, 101
- monitoring 88, 102
- normative 281
- parallel 227
- planning 15, 80, 82, 106, 131, 134, 162, 171, 173, 194, 210
- problem solving 117, 227
- search 108
- senso-motoric 49
- sub- 27, 85, 96, 153, 167, 193
- supervisory control 10, 88
- supporting 109
- transportation 27
- work process 87

Task agenda 64, 70-76ff, throughout Chapters 4 and 5, 293, 294, 323
Task allocation 96, 113, 163, 167
Task analysis 113
Task assignment 135, 141
Task characteristic 113
Task command 81
Task commitment 98, 137, 155
Task complexity 113
Task control 131
Task decomposition 121
Task determination 72-77ff throughout Chapters 4 and 5, 256, 259, 263
Task distribution 109, 135
Task environment 120
Task execution 23, 27, 52, 70-77ff, throughout Chapters 4 and 5, 254, 263, 293, 329
Task instruction 131
Task load 101, 102, 145
Task model 167, 278
Task option 70, 76, 126, 127, 131-134, 144, 162, 171, 194, 210, 316, 319
Task organisation 113
Task overload 152, 157
Task-relevant cue 70, 72, 73
Task requirement 79, 121
Task responsibility 84-86, 114
Task situation 70-76ff, throughout Chapters 4 and 5, 259, 290, 316, 317, 320
Task structure 134
Task sequencing 143
Task sharing 111
Team 5-295
- ACU 107, 108, 109, 2947
- human 5, 6, 22, 97-102, 105, 107, 110, 113, 115, 118, 146, 150
- human-machine 95, 99, 107, 117, 141
- operating 22, 108, 142
- sub- 97
- UAV 132, 137, 140-141, 295

Team behaviour 113, 116
Team commitments 136
Team control 113
Team co-ordination 113, 294
Team co-operation 133, 294
Teaming
- cognitive 104, 107, 142
- unmanned 109, 150
- machine-machine 112, 115

Team leader 5, 110-114, 142, 147, 157
Team management 112
Teammate 114-116, 146, 152
Team member 23, 25, 95, throughout 100-123, 132-137, 141-157, 293, 295
Team mission 114
Team performance 110, 113-115

Team structure 104, 105, 109-111, 115, 117, 137, 141
- co-ordinated 115
- distributed 111, 117
- hierarchical 114, 141
- human 104
- subsidiary 114
Teamwork 113, 114, 115, 141
Temporal lobe 37, 39, 340, 345
Terrain awareness and warning system (TAWS) 13
Time reserve 195, 196, 198-202, 204
- a-posteriori 198, 199
- a-priori 199
- driver-adapted 204
- lateral 195, 198, 202
- longitudinal 195, 198, 199
Time-to-collision (TTC) 198
Time to line crossing (TLC) 198
Thalamic nucleus mediodorsalis 58
Thalamus 39, 42, 48, 50, 53, 58, 59, 340, 343ff, 345
Theatre metaphor 56
Traction control 15
Training 3, 23, 31, 37, 40, 68, 89, 113, 115, 201, 207, throughout Chapters 6 and 7
Tutoring 145, 159, 207-211

Unmanned aerial vehicle (UAV) 87, 101-109, 118, 132-142, 149-151, 159, 293-297, 321-329

Value system 58, 75
Vehicle guidance and control 6, 7-16ff, also throughout Chapters 3, 4, 5 and 6, 319, 331
Ventral loop 36, 57, 58, 59
Ventral pallidum 58
Ventral striatum 58
Ventral tegmental area 58, 344, 345
Vigilance 52, 66, 344
Voluntary action 92, 170, 193, throughout Chapter 3
Vicious circle 87, 88, 95, 101, 102
Virtual reality 47, 208, 210
Visual stream 39
- dorsal 39
- ventral 39

Warning 12, 14, 169, 172-187, 194-207, 234, 256, 290
- adaptive 201-206
- haptic 192, 204
Warning aid 205, 206
Warning device 192, 194-196
Warning message 170, 203, 204
Warning signal 204, 205
Warning strategy 202, 204
Warning system 12, 13, 14
Warning threshold 204, 206
Weighted distance 306
Work 1-333
Work
- human 24
- professional 17, 154
Work assistance 3
Work commitment 145, 147, 149
Work domain 2, 20, 29, 99, 112, 143, 159, 214, 221, 231, 250
Work end product 20
Work environment 3, 19ff, 28, 33, 36, 67, 70, throughout Chapters 4 and 5, 214
Work hierarchy 28, 29
Work object 19, 20, 21, 26, 28
Work objective 18-30ff, 65, 66, 76, throughout Chapters 4, 5 and 7, 214, 270
Work performance 6, 60
Work plan 22, 111, 113, 142, 146, 147, 297
Work process 2-6, 17ff, throughout Chapters 3, 4, 5 and 6, 310, 319
Work product 27, 158
Work site 21, 25, 26, 80, 85, 147
Work situation 66, 94, 108, 122, 150, 156, 157, 170, 171, 172, 195, 216, 234
Work state 20, 27
Work station 152, 159, 180, 207
Work structure 29, 30, 113
Work system 1-6, 19ff, throughout Chapters 3, 4, 5, 6 and 7
Work system component 30, 31, 79, 96, 112, 219
Work system design 30, 31, 32, 47, 49, 60, 63, 64, throughout Chapters 4 and 5, 221, 222
Workspace activity 55, 56
Workspace neuron 55, 56
Workspace state 57
Work system network 29, 30
Work task 2
Work team 106, 142
Workload 5, throughout Chapters 4 and 5 272, 294, 295